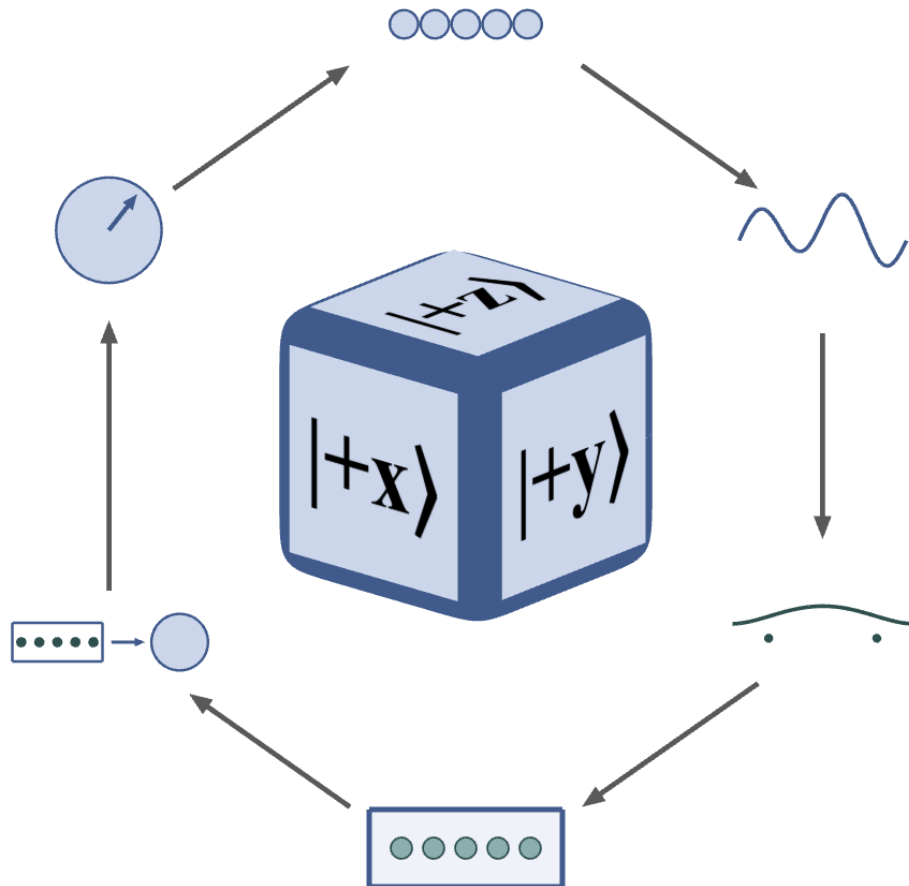


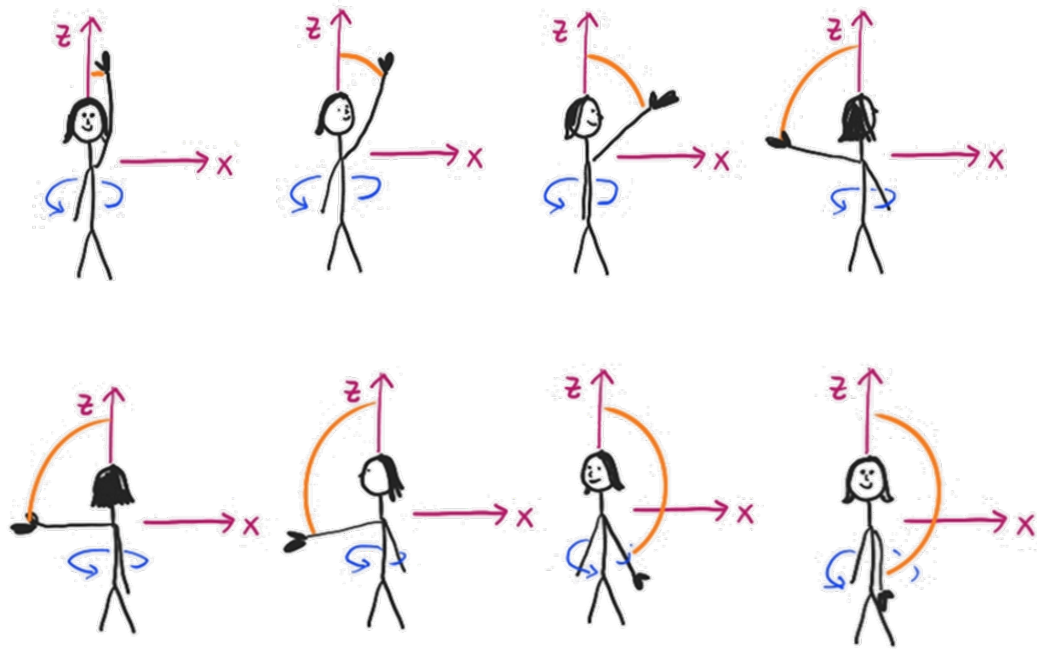
# From Qubit to Qubit

A Graduate Course in Quantum Mechanics

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*Illustration by Natalia Valderrama.*

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# Preface

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This book is the textbook for a two-semester graduate course in quantum mechanics taught at the University of Pittsburgh (PHYS 2565 and PHYS 2566). The notes have evolved over several years of teaching, and the present arrangement is the one I have settled on.

What distinguishes this treatment from other graduate texts is the order of presentation. The first two chapters develop quantum mechanics within the spin-1/2 system, with the Bloch cube, a six-sided physical object whose faces are the cardinal states of a single qubit, used as a concrete companion to the algebraic notation. Rotations of the cube enact unitary operations; pairs of cubes make entangled states visible; many of the basic calculations can be performed with a cube in hand. Quantum fields are introduced early on, in chapter 3, where the lattice version of a quantum field theory is the natural object once one has many qubits in a row. Three-dimensional angular momentum, the hydrogen atom, and perturbation theory occupy the middle chapters and are developed in the standard way. Bell's theorem, the Dirac equation, and the renormalization group complete the year. The book closes by returning to the qubit, now as a logical qubit protected by quantum error correction; the threshold for fault-tolerant computation appears as an unstable fixed point of the renormalization group, which is what the title points at.

The discrete-to-continuous transition is a recurring theme: in Hilbert space, when finite-dimensional qubits give way to infinite-dimensional state spaces; in real space, when lattice models give way to wave mechanics; in time, when discrete unitary steps give way to the Schrödinger equation; and finally in scale, when block-spin renormalization group transformations give way to continuous rescaling and to continuous variation in the number of spatial dimensions.

Viewed this way, the book differs from a standard graduate text in its global topology rather than its local content. Each section taken individually is conventional and could appear in any other textbook; this is by design. The order, however, closes into a loop: the qubit at the start identifies with the qubit at the end, the way coordinates on a circle wrap back to themselves rather than running off to infinity on a line. The path between them, through lattice fields to wave mechanics to relativistic dynamics to scaling, is the rearrangement, and many of the connections that this rearrangement makes visible are harder to see when the same material is presented linearly.

The treatment is largely self-contained. A reader with a working command of linear algebra, complex analysis, multivariable calculus, and the standard undergraduate sequence of mechanics and electromagnetism should be able to follow it. Many sections are accessible to advanced undergraduates; the later material assumes more mathematical maturity. Researchers in quantum information, condensed matter, or atomic physics may find the integrated treatment useful as a single reference for material that is usually scattered.

Each chapter uses a small set of pedagogical elements. Key Idea boxes mark the central conceptual moves. Examples work calculations end to end. Bloch Cube Activities are short physical

exercises meant to be done with a cube in hand. Mathematical Asides give technical derivations that can be skipped on a first reading. Physical Insights connect the formalism to experiment.

I am grateful to the students whose questions and frustrations have shaped this presentation, and to colleagues who have read earlier drafts. Special thanks are due to Chandralekha Singh for the collaborative development of the pedagogical approach, and to the American Journal of Physics for publishing our paper on the dot-vector formalism and the Bloch cube.

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# Chapter 1

## Foundations via Spin-1/2 Systems

### 1.1 Introduction: Quantum Foundations via Two-State Systems

#### Key Idea: The Power of Two-State Systems

The mathematical structure of quantum mechanics reveals itself most clearly through the study of two-state systems. These systems, despite their apparent simplicity, encode all the essential features of quantum theory: superposition, measurement, collapse, and unitary evolution. By developing a thorough understanding of two-state quantum mechanics, we establish a foundation that extends naturally to infinite-dimensional Hilbert spaces.

This course takes a pedagogically distinct approach to graduate quantum mechanics. Rather than beginning with wave mechanics and the Schrödinger equation, we start with finite-dimensional systems where the mathematical structure is transparent and the physical interpretation is unambiguous. One important pedagogical tool will be the “Bloch cube,” a physical representation of quantum states that makes abstract mathematical operations tangible and intuitive.

The Bloch cube is a six-sided object, similar to a die, where each face represents a distinct



Figure 1.1: Two Bloch Cubes.

quantum state of a two-level system. Through physical manipulation of these cubes, students can perform quantum operations, visualize superposition, and understand measurement in a concrete way before abstracting to the mathematical formalism. This approach bridges the gap between classical intuition and quantum formalism through what we call “dot-vec notation,” a mathematical framework that connects familiar vector operations to the Dirac notation of quantum mechanics.

The power of this approach lies in its ability to make the abstract concrete. When we say that quantum states can be “superposed,” what does this really mean? When we claim that measurement “collapses” a quantum state, how can we visualize this process? The Bloch cube provides immediate, tactile answers to these questions. By the end of this chapter, you will understand quantum mechanics not just as a mathematical formalism, but as a physical theory with clear operational meaning.

## 1.2 Mathematical Preliminaries: Vectors and Dot-Vectors

### 1.2.1 Review of Two-Dimensional Vector Spaces

We begin with the familiar setting of two-dimensional Euclidean space. Consider orthonormal basis vectors  $\hat{x}$  and  $\hat{y}$  satisfying the standard inner product relations:  $\hat{x} \cdot \hat{x} = 1$ ,  $\hat{x} \cdot \hat{y} = 0$ ,  $\hat{y} \cdot \hat{x} = 0$ , and  $\hat{y} \cdot \hat{y} = 1$ . Any vector in this space can be expressed as a linear combination  $\vec{V} = V_x \hat{x} + V_y \hat{y}$ , where the components are extracted through the inner product:  $V_x = \hat{x} \cdot \vec{V}$  and  $V_y = \hat{y} \cdot \vec{V}$ .

This elementary construction forms the basis for our approach to quantum mechanics. The key insight is that quantum states behave mathematically like vectors, while measurement processes behave like inner products. To make this connection explicit, we introduce a notational innovation that clarifies the dual roles played by vectors in quantum theory.

Let us explore this vector space more deeply. Consider a general vector  $\vec{V} = 3\hat{x} + 4\hat{y}$ . Its magnitude is  $|\vec{V}| = \sqrt{3^2 + 4^2} = 5$ , and we can normalize it to create a unit vector:  $\hat{V} = \frac{3}{5}\hat{x} + \frac{4}{5}\hat{y}$ . The angle this vector makes with the x-axis is  $\theta = \arctan(4/3) \approx 53.1$  degrees. These familiar calculations from classical mechanics will have direct quantum mechanical analogs.

### 1.2.2 The Dot-Vector Construction

We now introduce a formal distinction between vectors and what we term “dot-vectors.” For each basis vector, we define a corresponding dot-vector by symbolically attaching the dot operation:  $\hat{x} \rightarrow \hat{x} \cdot$  and  $\hat{y} \rightarrow \hat{y} \cdot$ . While this may appear to be merely a notational trick, it serves a profound pedagogical purpose by making explicit the dual nature of vectors in quantum mechanics.

A general dot-vector takes the form  $\vec{W} \cdot = W_x \hat{x} \cdot + W_y \hat{y} \cdot$ . The action of a dot-vector on a regular vector produces a scalar through the natural extension of the inner product:  $\vec{W} \cdot \vec{V} = W_x V_x + W_y V_y$ . This operation mirrors the action of a bra on a ket in quantum mechanics, providing students with a concrete mathematical bridge to the more abstract Dirac notation.

To illustrate the power of this notation, consider the dot-vector  $\vec{W} \cdot = 2\hat{x} \cdot + 3\hat{y} \cdot$  acting on our previous vector  $\vec{V} = 3\hat{x} + 4\hat{y}$ :

$$\vec{W} \cdot \vec{V} = (2\hat{x} \cdot + 3\hat{y} \cdot)(3\hat{x} + 4\hat{y}) \quad (1.1)$$

$$= 2 \cdot 3 + 3 \cdot 4 \quad (1.2)$$

$$= 6 + 12 = 18 \quad (1.3)$$

This calculation, while elementary, establishes the pattern for all quantum mechanical calculations involving state amplitudes and measurement probabilities.

### 1.2.3 Operator Construction from Vectors and Dot-Vectors

The true power of the dot-vector notation emerges when we consider products in the reverse order. The expression  $\hat{x}\hat{x}\cdot$  represents an operator that acts on vectors. To understand its action, consider its effect on an arbitrary vector  $\vec{V}$ :

$$(\hat{x}\hat{x}\cdot)\vec{V} = \hat{x}(\hat{x} \cdot \vec{V}) = \hat{x}V_x = V_x\hat{x} \quad (1.4)$$

This operator extracts the  $x$ -component of the vector and returns a vector pointing in the  $x$ -direction with that magnitude. Similarly,  $\hat{y}\hat{y}\cdot$  projects onto the  $y$ -direction.

Let's examine these projection operators more carefully. Consider their action on our example vector  $\vec{V} = 3\hat{x} + 4\hat{y}$ :

$$(\hat{x}\hat{x}\cdot)\vec{V} = \hat{x}(\hat{x} \cdot (3\hat{x} + 4\hat{y})) = \hat{x}(3) = 3\hat{x} \quad (1.5)$$

$$(\hat{y}\hat{y}\cdot)\vec{V} = \hat{y}(\hat{y} \cdot (3\hat{x} + 4\hat{y})) = \hat{y}(4) = 4\hat{y} \quad (1.6)$$

The sum of these projections recovers the original vector:

$$[(\hat{x}\hat{x}\cdot) + (\hat{y}\hat{y}\cdot)]\vec{V} = 3\hat{x} + 4\hat{y} = \vec{V} \quad (1.7)$$

This demonstrates that the sum of projection operators yields the identity operator:

$$\hat{I} = \hat{x}\hat{x}\cdot + \hat{y}\hat{y}\cdot \quad (1.8)$$

This decomposition of the identity will later correspond to the completeness relation in quantum mechanics. More generally, we can construct operators that transform between different directions. For instance, the operator  $\hat{y}\hat{x}\cdot$  maps the  $x$ -direction to the  $y$ -direction:  $(\hat{y}\hat{x}\cdot)\hat{x} = \hat{y}(\hat{x} \cdot \hat{x}) = \hat{y}$ .

## 1.3 The Quantum Correspondence

### 1.3.1 From Dot-Vec to Dirac Notation

We now establish the formal correspondence between our dot-vector notation and the Dirac notation of quantum mechanics. This mapping transforms our concrete two-dimensional vector space into the abstract two-dimensional Hilbert space of a quantum two-state system:

| Dot-Vec Notation        | Dirac Notation           | Physical Meaning         |
|-------------------------|--------------------------|--------------------------|
| $\hat{x}$               | $ +z\rangle$             | Spin up along z-axis     |
| $\hat{y}$               | $ -z\rangle$             | Spin down along z-axis   |
| $\hat{x}\cdot$          | $\langle +z $            | Measurement of spin up   |
| $\hat{y}\cdot$          | $\langle -z $            | Measurement of spin down |
| $\hat{x} \cdot \hat{y}$ | $\langle +z  - z\rangle$ | Transition amplitude     |
| $\hat{x}\hat{y}\cdot$   | $ +z\rangle\langle -z $  | Spin flip operator       |

This correspondence is not merely formal—it captures the deep mathematical structure shared by classical vectors and quantum states. The orthonormality of our basis vectors translates directly to the orthogonality of quantum basis states:  $\langle +z| + z\rangle = 1$ ,  $\langle -z| - z\rangle = 1$ , and  $\langle +z| - z\rangle = \langle -z| + z\rangle = 0$ .

### 1.3.2 Quantum Superposition

The principle of superposition, fundamental to quantum mechanics, emerges naturally in our framework. Just as any two-dimensional vector can be expressed as a linear combination of basis vectors, any quantum state in our two-dimensional Hilbert space can be written as:

$$|\psi\rangle = \alpha|+z\rangle + \beta|-z\rangle \quad (1.9)$$

Here,  $\alpha$  and  $\beta$  are complex numbers satisfying the normalization condition  $|\alpha|^2 + |\beta|^2 = 1$ . This condition ensures that the total probability of finding the system in some state equals unity. The complex nature of these coefficients, absent in our classical vector analogy, introduces the uniquely quantum phenomenon of phase.

To understand the role of complex coefficients, consider the state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|+z\rangle + \frac{e^{i\pi/4}}{\sqrt{2}}|-z\rangle \quad (1.10)$$

The magnitudes  $|\alpha| = |\beta| = 1/\sqrt{2}$  determine measurement probabilities, while the relative phase  $e^{i\pi/4}$  affects interference between the components. This phase has no classical analog in our dot-vector notation—it represents genuinely quantum behavior.

The coefficients can be extracted using the inner product, just as in the classical case:  $\alpha = \langle +z|\psi\rangle$  and  $\beta = \langle -z|\psi\rangle$ . These probability amplitudes encode all information about the quantum state and determine the probabilities of measurement outcomes through the Born rule.

## 1.4 The Bloch Cube: A Tangible Quantum System

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### 1.4.1 Construction and Concept

We now introduce the central pedagogical tool of our approach: the Bloch cube. This physical object serves as a concrete representation of a quantum two-state system, making abstract quantum concepts tangible and manipulable. The Bloch cube is a regular cube (such as a die) where each of the six faces represents a specific quantum state.

The full Bloch sphere, which represents all possible states of a two-level quantum system, forms a sphere in three-dimensional space. The general state  $|\psi\rangle = \cos(\theta/2)|+z\rangle + e^{i\phi}\sin(\theta/2)|-z\rangle$  corresponds to the point with spherical coordinates  $(\theta, \phi)$  on this sphere. While this continuum of states is mathematically elegant, it presents pedagogical challenges for beginning students.

The Bloch cube simplifies this picture by restricting attention to six special states that form the vertices of a regular octahedron inscribed within the Bloch sphere. These states correspond to the intersections of the sphere with the three coordinate axes. By limiting ourselves to these six states initially, we can perform all fundamental quantum operations while maintaining a clear physical picture.

### 1.4.2 The Six Fundamental States

The six faces of the Bloch cube represent the following quantum states:

Along the z-axis, we have  $|+z\rangle$  and  $|-z\rangle$ , which serve as our computational basis states. These represent definite values of the z-component of spin for a spin-1/2 particle.

Along the x-axis, we find  $|+x\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + |-z\rangle)$  and  $|-x\rangle = \frac{1}{\sqrt{2}}(|+z\rangle - |-z\rangle)$ . These states represent equal superpositions of the z-basis states, differing only in their relative phase.

Finally, along the y-axis, we have  $|+y\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + i|-z\rangle)$  and  $|-y\rangle = \frac{1}{\sqrt{2}}(|+z\rangle - i|-z\rangle)$ . The presence of the imaginary unit  $i$  in these superpositions introduces a phase difference of  $\pi/2$  between the components.

### 1.4.3 Complete Basis Representations

A crucial tool for quantum calculations is the ability to express any state in any basis. Here we provide the complete representation of all six Bloch cube states in all three bases:

#### Bloch Cube States in All Bases

##### Z-basis representations:

$$|+z\rangle = |+z\rangle \quad (1.11)$$

$$|-z\rangle = |-z\rangle \quad (1.12)$$

$$|+x\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + |-z\rangle) \quad (1.13)$$

$$|-x\rangle = \frac{1}{\sqrt{2}}(|+z\rangle - |-z\rangle) \quad (1.14)$$

$$|+y\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + i|-z\rangle) \quad (1.15)$$

$$|-y\rangle = \frac{1}{\sqrt{2}}(|+z\rangle - i|-z\rangle) \quad (1.16)$$

##### X-basis representations:

$$|+z\rangle = \frac{1}{\sqrt{2}}(|+x\rangle + |-x\rangle) \quad (1.17)$$

$$|-z\rangle = \frac{1}{\sqrt{2}}(|+x\rangle - |-x\rangle) \quad (1.18)$$

$$|+x\rangle = |+x\rangle \quad (1.19)$$

$$|-x\rangle = |-x\rangle \quad (1.20)$$

$$|+y\rangle = \frac{(1+i)|+x\rangle + (1-i)|-x\rangle}{2} \quad (1.21)$$

$$|-y\rangle = \frac{(1-i)|+x\rangle + (1+i)|-x\rangle}{2} \quad (1.22)$$

##### Y-basis representations:

$$|+z\rangle = \frac{1}{\sqrt{2}}(|+y\rangle + |-y\rangle) \quad (1.23)$$

$$|-z\rangle = \frac{1}{\sqrt{2}}(-i|+y\rangle + i|-y\rangle) \quad (1.24)$$

$$|+x\rangle = \frac{(1-i)|+y\rangle + (1+i)|-y\rangle}{2} \quad (1.25)$$

$$|-x\rangle = \frac{(1+i)|+y\rangle + (1-i)|-y\rangle}{2} \quad (1.26)$$

$$|+y\rangle = |+y\rangle \quad (1.27)$$

$$|-y\rangle = |-y\rangle \quad (1.28)$$

These representations are essential for calculating transition probabilities between different bases and understanding how the same quantum state appears different when viewed from different measurement perspectives.

#### 1.4.4 Using Cross-Basis Representations

Let's work through an example that demonstrates the utility of these cross-basis representations. Suppose we want to calculate  $\langle +x| + y \rangle$  directly using the Z-basis.

##### Example 1: Cross-Basis Inner Product

Calculate  $\langle +x| + y \rangle$  using Z-basis representations:

From our table:

$$| + x \rangle = \frac{1}{\sqrt{2}}(| + z \rangle + | - z \rangle) \quad (1.29)$$

$$| + y \rangle = \frac{1}{\sqrt{2}}(| + z \rangle + i| - z \rangle) \quad (1.30)$$

Therefore:

$$\langle +x| + y \rangle = \frac{1}{\sqrt{2}}(\langle +z| + \langle -z|) \cdot \frac{1}{\sqrt{2}}(| + z \rangle + i| - z \rangle) \quad (1.31)$$

$$= \frac{1}{2}[\langle +z| + z \rangle + i\langle +z| - z \rangle + \langle -z| + z \rangle + i\langle -z| - z \rangle] \quad (1.32)$$

$$= \frac{1}{2}[1 + 0 + 0 + i] \quad (1.33)$$

$$= \frac{1}{2}(1 + i) \quad (1.34)$$

We can verify this is correct since  $|\langle +x| + y \rangle|^2 = \frac{1}{2}$ , as expected for adjacent cube faces.

#### 1.4.5 Orthogonality and the Cube Geometry

The geometric structure of the cube encodes the orthogonality relations between quantum states. States on opposite faces of the cube are orthogonal, meaning their inner product vanishes. This can be verified through direct calculation:  $\langle +z| - z \rangle = 0$ ,  $\langle +x| - x \rangle = 0$ , and  $\langle +y| - y \rangle = 0$ . This orthogonality has profound physical meaning—orthogonal states can be perfectly distinguished by appropriate measurements.

States on adjacent faces of the cube are not orthogonal. The inner products between adjacent states all have magnitude  $1/\sqrt{2}$ :

$$|\langle +z| + x \rangle| = |\langle +z| + y \rangle| = \frac{1}{\sqrt{2}} \quad (1.35)$$

$$|\langle +x| + y \rangle| = |\langle +x| - y \rangle| = \frac{1}{\sqrt{2}} \quad (1.36)$$

$$|\langle +y| + z \rangle| = |\langle +y| - z \rangle| = \frac{1}{\sqrt{2}} \quad (1.37)$$

This uniform overlap between adjacent states reflects the high symmetry of the cube arrangement on the Bloch sphere.



**Bloch Cube Activity: 1.1: Constructing and Exploring the Bloch Cube**

To begin your hands-on exploration of quantum mechanics, first construct or obtain a Bloch cube. You can use a standard six-sided die or create a cube with labeled faces. Label the six faces according to the states described above:  $|+z\rangle$ ,  $|-z\rangle$ ,  $|+x\rangle$ ,  $|-x\rangle$ ,  $|+y\rangle$ , and  $|-y\rangle$ . Once you have your Bloch cube, verify the following properties through direct manipulation:

- (a) Opposite faces represent orthogonal states
- (b) Each state can be expressed as a superposition of any pair of orthogonal states
- (c) The cube can be oriented to place any face “up,” representing the current quantum state
- (d) Rotations of the cube correspond to unitary transformations of the quantum state
- (e) Use the basis representation table to verify that  $|+x\rangle$  is indeed an equal superposition of  $|+z\rangle$  and  $|-z\rangle$

## 1.5 Measurement Theory and the Born Rule

### 1.5.1 Measurement as Fundamental Process

Measurement in quantum mechanics differs fundamentally from classical measurement. While classical measurements merely reveal pre-existing properties, quantum measurements actively disturb the system and, in general, change its state. Our Bloch cube framework provides a concrete model for understanding this process.

A quantum measurement on our two-state system corresponds to asking a yes-no question about the state. For the Bloch cube, we have three natural measurement types, corresponding to the three spatial axes. The Z-measurement asks whether the system is in state  $|+z\rangle$  or  $|-z\rangle$ . Similarly, X-measurements distinguish between  $|+x\rangle$  and  $|-x\rangle$ , while Y-measurements distinguish between  $|+y\rangle$  and  $|-y\rangle$ .

To understand how measurement procedures translate into mathematical operators, let's construct them step by step from our concrete measurement outcomes. We've already encountered the concept of eigenstates and eigenvalues in the context of rotations—here, the eigenvalues will correspond to the possible measurement outcomes and eigenstates correspond to the states that yield definite results.

Consider the Z-measurement procedure: we ask whether the system is in state  $|+z\rangle$  or  $|-z\rangle$ . If we assign numerical scores to these outcomes—say +1 for finding  $|+z\rangle$  and -1 for finding  $|-z\rangle$ —then we can construct a mathematical operator that encodes this measurement:

$$\hat{\sigma}_z = (+1)|+z\rangle\langle+z| + (-1)|-z\rangle\langle-z| = |+z\rangle\langle+z| - |-z\rangle\langle-z| \quad (1.38)$$

This operator has the property that when it acts on the measurement eigenstates, it returns the corresponding measurement outcome:  $\hat{\sigma}_z|+z\rangle = (+1)|+z\rangle$  and  $\hat{\sigma}_z|-z\rangle = (-1)|-z\rangle$ .

Similarly, we can construct operators for X and Y measurements:

$$\hat{\sigma}_x = |+x\rangle\langle+x| - |-x\rangle\langle-x| \quad (1.39)$$

$$\hat{\sigma}_y = |+y\rangle\langle+y| - |-y\rangle\langle-y| \quad (1.40)$$

These measurement operators are known as the Pauli operators, which form the foundation of two-level quantum mechanics. Notice that the six Bloch cube states we introduced earlier are precisely the eigenstates of these three operators:  $|\pm z\rangle$  are eigenstates of  $\sigma_z$ ,  $|\pm x\rangle$  are eigenstates of  $\sigma_x$ , and  $|\pm y\rangle$  are eigenstates of  $\sigma_y$ .

Observe the pattern in our construction: each measurement operator was built by assigning real numerical values ( $\pm 1$ ) to the possible outcomes and weighting the corresponding projection operators. This systematic construction ensures that the eigenvalues—which represent the measurement outcomes we can observe—are always real numbers. This is a general feature of all quantum measurements.

### 1.5.2 The Born Rule and Probability

The Born rule provides the fundamental connection between the mathematical formalism of quantum mechanics and experimental predictions. According to this rule, if a system is in state  $|\psi\rangle$  and we measure an observable with eigenstates  $|\phi_i\rangle$ , the probability of obtaining the outcome corresponding to  $|\phi_i\rangle$  is:

$$P(|\psi\rangle \rightarrow |\phi_i\rangle) = |\langle\phi_i|\psi\rangle|^2 \quad (1.41)$$

This formula encodes one of the most profound aspects of quantum mechanics: even with complete knowledge of the quantum state, we can only predict probabilities of measurement outcomes, not the outcomes themselves.

Let's explore several examples to understand the Born rule deeply.

#### Example 2: Measurement Probabilities for $|+x\rangle$

Consider the state  $|+x\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + |-z\rangle)$ . Let's calculate the probabilities for all three measurement types.

**Z-measurement:**

$$P(|+x\rangle \rightarrow |+z\rangle) = |\langle+z|+x\rangle|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2} \quad (1.42)$$

$$P(|+x\rangle \rightarrow |-z\rangle) = |\langle-z|+x\rangle|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2} \quad (1.43)$$

**X-measurement:**

$$P(|+x\rangle \rightarrow |+x\rangle) = |\langle+x|+x\rangle|^2 = |1|^2 = 1 \quad (1.44)$$

$$P(|+x\rangle \rightarrow |-x\rangle) = |\langle-x|+x\rangle|^2 = |0|^2 = 0 \quad (1.45)$$

**Y-measurement:** Using  $|+x\rangle = \frac{1}{\sqrt{2}}(|+y\rangle + |-y\rangle)$  from our basis table:

$$P(|+x\rangle \rightarrow |+y\rangle) = |\langle+y|+x\rangle|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2} \quad (1.46)$$

$$P(|+x\rangle \rightarrow |-y\rangle) = |\langle-y|+x\rangle|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2} \quad (1.47)$$

### 1.5.3 State Collapse and Sequential Measurements

Following a measurement, the quantum state “collapses” to the eigenstate corresponding to the observed outcome. This collapse is instantaneous and irreversible within the standard quantum formalism. If we measure  $|+x\rangle$  along the z-axis and the system collapses to  $|+z\rangle$ , the post-measurement state is  $|+z\rangle$ , not  $|+x\rangle$ .

This collapse leads to one of the most striking features of quantum measurement: the order of measurements matters. Consider performing X and Z measurements in sequence. If we start with  $|+y\rangle$ , measure X (causing the system to collapse to, say,  $|+x\rangle$ ), then measure Z, we get random Z results. However, if we first measure Z, then X, we get random X results. The non-commutativity of measurements reflects the non-commutativity of the corresponding operators.

#### Example 3: Sequential Measurements

Start with  $|+y\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + i|-z\rangle)$ .

##### Path 1: Measure Z, then X

- (a) Initial state:  $|+y\rangle$
- (b) Measure Z: The system collapses to  $|+z\rangle$  with probability 1/2 or  $|-z\rangle$  with probability 1/2
- (c) Suppose the system collapses to  $|+z\rangle$
- (d) Now measure X on  $|+z\rangle = \frac{1}{\sqrt{2}}(|+x\rangle + |-x\rangle)$
- (e) The system collapses to  $|+x\rangle$  or  $|-x\rangle$  each with probability 1/2

##### Path 2: Measure X, then Z

- (a) Initial state:  $|+y\rangle = \frac{(1+i)|+x\rangle + (1-i)|-x\rangle}{2}$
- (b) Measure X: The system collapses to  $|+x\rangle$  with probability 1/2 or  $|-x\rangle$  with probability 1/2
- (c) Suppose the system collapses to  $|+x\rangle$
- (d) Now measure Z on  $|+x\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + |-z\rangle)$
- (e) The system collapses to  $|+z\rangle$  or  $|-z\rangle$  each with probability 1/2

The final probability distributions are identical, but the intermediate states differ!

#### Bloch Cube Activity: 1.2: Simulating Quantum Measurement

Using your Bloch cube, simulate the measurement process:

- (a) Set the cube to show  $|+x\rangle$  (facing up)
- (b) To simulate a Z-measurement, calculate the probabilities for collapsing to  $|\pm z\rangle$  states
- (c) Use a random process (coin flip) to determine the outcome

- (d) Rotate the cube to show the post-measurement state
- (e) Repeat multiple times to verify the statistical predictions
- (f) Try starting from  $|+y\rangle$  and perform measurements in different orders

#### 1.5.4 Measurement Outcomes and Expectation Values

So far, we have discussed measurements as producing states  $|\pm z\rangle$ ,  $|\pm x\rangle$ , or  $|\pm y\rangle$ . To connect with observable physical quantities, we assign numerical values to these outcomes. We adopt a standard “scoring” system:

- Measuring  $|+w\rangle$  yields outcome  $+1$
- Measuring  $|-w\rangle$  yields outcome  $-1$

where  $W \in \{X, Y, Z\}$  labels the measurement type. This assignment is not arbitrary—it corresponds to measuring the component of spin angular momentum along the specified axis (in units of  $\hbar/2$ ).

With this scoring system, each measurement operator can be written as:

$$\hat{\sigma}_z = |+z\rangle\langle +z| - |-z\rangle\langle -z| \quad (1.48)$$

$$\hat{\sigma}_x = |+x\rangle\langle +x| - |-x\rangle\langle -x| \quad (1.49)$$

$$\hat{\sigma}_y = |+y\rangle\langle +y| - |-y\rangle\langle -y| \quad (1.50)$$

These are the Pauli operators. Notice something important: unlike the rotation operators whose eigenvalues were complex phases (1 and  $i$  for  $\hat{Z}$ ), the Pauli operators have real eigenvalues ( $\pm 1$ ) that represent the possible measurement outcomes. This is not a coincidence—measurement outcomes must be real numbers that we can observe in experiments.

This pattern—that measurement operators constructed from real-valued outcomes always have real eigenvalues—holds generally in quantum mechanics. Operators with this property are called *Hermitian* operators. The requirement that measurement outcomes be real numbers forces measurement operators to be Hermitian, while the requirement that rotations preserve probability forces rotation operators to be *unitary* (ensuring eigenvalues with magnitude 1).

Interestingly, there’s a simple relationship between the rotation and Pauli operators:  $\hat{\sigma}_W = \hat{W}^2$  for  $W \in \{X, Y, Z\}$ . That is, the Pauli operators correspond to  $180^\circ$  rotations about their respective axes. This makes sense geometrically—a  $180^\circ$  rotation takes each Bloch cube face to its opposite face, which is precisely what the Pauli operators encode through their  $\pm 1$  eigenvalues.

For a system in state  $|\psi\rangle$ , the *expectation value* of a measurement is the average outcome we would obtain if we performed the measurement on many identically prepared systems:

$$\langle \hat{\sigma}_W \rangle = P(|\psi\rangle \rightarrow | +W \rangle) \cdot (+1) + P(|\psi\rangle \rightarrow | -W \rangle) \cdot (-1) = P(|\psi\rangle \rightarrow | +W \rangle) - P(|\psi\rangle \rightarrow | -W \rangle) \quad (1.51)$$

where  $P(|\psi\rangle \rightarrow | \pm W \rangle)$  are the probabilities calculated using the Born rule.

**Example 4: Expectation Values for  $|+x\rangle$** 

Calculate the expectation values of all three Pauli operators for the state  $|+x\rangle$ .  
From Example 1.2, we found:

- Z-measurement:  $P(|+x\rangle \rightarrow |+z\rangle) = P(|+x\rangle \rightarrow |-z\rangle) = 1/2$
- X-measurement:  $P(|+x\rangle \rightarrow |+x\rangle) = 1, P(|+x\rangle \rightarrow |-x\rangle) = 0$
- Y-measurement:  $P(|+x\rangle \rightarrow |+y\rangle) = P(|+x\rangle \rightarrow |-y\rangle) = 1/2$

Therefore:

$$\langle\hat{\sigma}_z\rangle = P(|+x\rangle \rightarrow |+z\rangle) - P(|+x\rangle \rightarrow |-z\rangle) = \frac{1}{2} - \frac{1}{2} = 0 \quad (1.52)$$

$$\langle\hat{\sigma}_x\rangle = P(|+x\rangle \rightarrow |+x\rangle) - P(|+x\rangle \rightarrow |-x\rangle) = 1 - 0 = 1 \quad (1.53)$$

$$\langle\hat{\sigma}_y\rangle = P(|+x\rangle \rightarrow |+y\rangle) - P(|+x\rangle \rightarrow |-y\rangle) = \frac{1}{2} - \frac{1}{2} = 0 \quad (1.54)$$

The state  $|+x\rangle$  has definite value +1 for X-measurements but gives random (zero average) results for Y and Z measurements.

Expectation values can also be calculated directly using the inner product:

$$\langle\hat{A}\rangle = \langle\psi|\hat{A}|\psi\rangle \quad (1.55)$$

This formula, fundamental to quantum mechanics, connects the abstract mathematical formalism to measurable physical quantities.

**Bloch Cube Activity: 1.3: Measuring Expectation Values with Bloch Cubes**

Using your Bloch cube:

- Prepare the state  $|+y\rangle$
- Perform 20 simulated Z-measurements, recording +1 for  $|+z\rangle$  and -1 for  $|-z\rangle$
- Calculate the average of your results
- Compare with the theoretical expectation value  $\langle\hat{\sigma}_z\rangle = 0$
- Repeat for X and Y measurements
- Discuss how many measurements would be needed to distinguish  $\langle\hat{\sigma}_z\rangle = 0$  from  $\langle\hat{\sigma}_z\rangle = 0.1$

## 1.6 Unitary Evolution and Quantum Dynamics

### 1.6.1 Discrete Rotations as Quantum Gates

The time evolution of isolated quantum systems is governed by unitary transformations. For our Bloch cube, these transformations are implemented as physical rotations of the cube. This provides

an immediate, tactile understanding of quantum dynamics that complements the mathematical formalism.

Consider rotations by 90 degrees about the three coordinate axes. These discrete rotations form a group and can be represented by unitary operators. The rotation by 90 degrees clockwise about the z-axis (when viewed from the positive z-direction) is represented by the operator:

$$\hat{Z} = | +z \rangle \langle +z | + i | -z \rangle \langle -z | \quad (1.56)$$

Let's trace through how this operator acts on all six Bloch cube states:

#### Example 5: Action of $\hat{Z}$ on All Cube States

$$\hat{Z}| +z \rangle = | +z \rangle \quad (1.57)$$

$$\hat{Z}| -z \rangle = i| -z \rangle \quad (1.58)$$

$$\hat{Z}| +x \rangle = \hat{Z} \frac{1}{\sqrt{2}}(| +z \rangle + | -z \rangle) = \frac{1}{\sqrt{2}}(| +z \rangle + i| -z \rangle) = | +y \rangle \quad (1.59)$$

$$\hat{Z}| -x \rangle = \hat{Z} \frac{1}{\sqrt{2}}(| +z \rangle - | -z \rangle) = \frac{1}{\sqrt{2}}(| +z \rangle - i| -z \rangle) = | -y \rangle \quad (1.60)$$

$$\hat{Z}| +y \rangle = \hat{Z} \frac{1}{\sqrt{2}}(| +z \rangle + i| -z \rangle) = \frac{1}{\sqrt{2}}(| +z \rangle - | -z \rangle) = | -x \rangle \quad (1.61)$$

$$\hat{Z}| -y \rangle = \hat{Z} \frac{1}{\sqrt{2}}(| +z \rangle - i| -z \rangle) = \frac{1}{\sqrt{2}}(| +z \rangle + | -z \rangle) = | +x \rangle \quad (1.62)$$

This confirms that  $\hat{Z}$  implements a 90° rotation:  $+x \rightarrow +y \rightarrow -x \rightarrow -y \rightarrow +x$ .

Notice something remarkable in the calculations above: when  $\hat{Z}$  acts on  $| +z \rangle$  and  $| -z \rangle$ , these states are only multiplied by phase factors (1 and  $i$  respectively). This illustrates a fundamental concept in quantum mechanics: *eigenstates* and *eigenvalues*.

#### Key Idea: Eigenstates and Eigenvalues

A state  $|\psi\rangle$  is called an eigenstate of an operator  $\hat{A}$  if the operator's action on that state merely multiplies it by a complex number:

$$\hat{A}|\psi\rangle = \lambda|\psi\rangle \quad (1.63)$$

The complex number  $\lambda$  is called the eigenvalue. For our rotation operator  $\hat{Z}$ :

- $| +z \rangle$  is an eigenstate with eigenvalue  $\lambda = 1$
- $| -z \rangle$  is an eigenstate with eigenvalue  $\lambda = i$

Physically, eigenstates represent states that are “stable” under the operation—they don't change direction, only acquire a phase. For rotations about the z-axis, it makes intuitive sense that states aligned with the z-axis ( $| \pm z \rangle$ ) would be special in this way.

The eigenvalues of rotation operators are always complex numbers with magnitude 1 (i.e., pure

phases), which ensures that the operators preserve the normalization of quantum states. This is a consequence of unitarity. For measurement operators, as we've seen earlier, the eigenvalues will be real numbers corresponding to the possible measurement outcomes.

Similarly, we can define the X and Y rotation operators:

$$\hat{X} = | +x \rangle \langle +x | + i | -x \rangle \langle -x | \quad (1.64)$$

$$\hat{Y} = | +y \rangle \langle +y | + i | -y \rangle \langle -y | \quad (1.65)$$

### 1.6.2 The Non-Commutativity of Rotations

A fundamental feature of quantum mechanics is that operations generally do not commute. This non-commutativity can be directly experienced through Bloch cube manipulations. Consider the unitary operators  $\hat{X}$  and  $\hat{Y}$ , representing 90-degree rotations about the x and y axes respectively.

To understand this rigorously, we need to determine how these operators act on the state  $| +z \rangle$ . Using the operator definitions:

$$\hat{X} = | +x \rangle \langle +x | + i | -x \rangle \langle -x | \quad (1.66)$$

$$\hat{Y} = | +y \rangle \langle +y | + i | -y \rangle \langle -y | \quad (1.67)$$

#### Example 6: Rigorous Derivation of Rotation Actions

**Finding  $\hat{X}| +z \rangle$ :**

$$\hat{X}| +z \rangle = (| +x \rangle \langle +x | + i | -x \rangle \langle -x |) | +z \rangle \quad (1.68)$$

$$= | +x \rangle \langle +x | +z \rangle + i | -x \rangle \langle -x | +z \rangle \quad (1.69)$$

Using our basis representations:  $\langle +x | +z \rangle = \langle -x | +z \rangle = \frac{1}{\sqrt{2}}$

$$\hat{X}| +z \rangle = | +x \rangle \frac{1}{\sqrt{2}} + i | -x \rangle \frac{1}{\sqrt{2}} \quad (1.70)$$

$$= \frac{1}{\sqrt{2}} (| +x \rangle + i | -x \rangle) \quad (1.71)$$

Converting to Z-basis using  $| +x \rangle = \frac{1}{\sqrt{2}}(| +z \rangle + | -z \rangle)$  and  $| -x \rangle = \frac{1}{\sqrt{2}}(| +z \rangle - | -z \rangle)$ :

$$\hat{X}| +z \rangle = \frac{1}{\sqrt{2}} \left[ \frac{1}{\sqrt{2}}(| +z \rangle + | -z \rangle) + i \frac{1}{\sqrt{2}}(| +z \rangle - | -z \rangle) \right] \quad (1.72)$$

$$= \frac{1}{2} [(1 + i)| +z \rangle + (1 - i)| -z \rangle] \quad (1.73)$$

Factoring out the phase:  $(1 + i) = \sqrt{2}e^{i\pi/4}$ , so:

$$\hat{X}| +z \rangle = e^{i\pi/4} \frac{1}{\sqrt{2}} (| +z \rangle - i | -z \rangle) = e^{i\pi/4} | -y \rangle \quad (1.74)$$

**Finding  $\hat{Y}| +z \rangle$ :**

Similarly:

$$\hat{Y}|+z\rangle = (|+y\rangle\langle+y| + i|-y\rangle\langle-y|)|+z\rangle \quad (1.75)$$

$$= |+y\rangle\langle+y|+z\rangle + i|-y\rangle\langle-y|+z\rangle \quad (1.76)$$

$$= \frac{1}{\sqrt{2}}(|+y\rangle + i|-y\rangle) \quad (1.77)$$

Converting to Z-basis and following similar algebra:

$$\hat{Y}|+z\rangle = e^{i\pi/4}|+x\rangle \quad (1.78)$$

Now let's trace through both possible orderings with the correct phase factors:

**Order 1:**  $\hat{X}$  then  $\hat{Y}$   $|+z\rangle \xrightarrow{\hat{X}} e^{i\pi/4}|-y\rangle \xrightarrow{\hat{Y}} ?$

To find  $\hat{Y}|-y\rangle$ :

$$\hat{Y}|-y\rangle = (|+y\rangle\langle+y| + i|-y\rangle\langle-y|)|-y\rangle \quad (1.79)$$

$$= |+y\rangle \underbrace{\langle+y|-y\rangle}_{=0} + i|-y\rangle \underbrace{\langle-y|-y\rangle}_{=1} \quad (1.80)$$

$$= i|-y\rangle \quad (1.81)$$

Therefore:  $|+z\rangle \xrightarrow{\hat{X}} e^{i\pi/4}|-y\rangle \xrightarrow{\hat{Y}} ie^{i\pi/4}|-y\rangle = e^{i3\pi/4}|-y\rangle$

**Order 2:**  $\hat{Y}$  then  $\hat{X}$   $|+z\rangle \xrightarrow{\hat{Y}} e^{i\pi/4}|+x\rangle \xrightarrow{\hat{X}} ?$

To find  $\hat{X}|+x\rangle$ :

$$\hat{X}|+x\rangle = (|+x\rangle\langle+x| + i|-x\rangle\langle-x|)|+x\rangle \quad (1.82)$$

$$= |+x\rangle \underbrace{\langle+x|+x\rangle}_{=1} + i|-x\rangle \underbrace{\langle-x|+x\rangle}_{=0} \quad (1.83)$$

$$= |+x\rangle \quad (1.84)$$

Therefore:  $|+z\rangle \xrightarrow{\hat{Y}} e^{i\pi/4}|+x\rangle \xrightarrow{\hat{X}} e^{i\pi/4}|+x\rangle$

**Comparison:**

- $\hat{Y}\hat{X}|+z\rangle = e^{i3\pi/4}|-y\rangle$
- $\hat{X}\hat{Y}|+z\rangle = e^{i\pi/4}|+x\rangle$

The final states differ both in direction ( $|-y\rangle$  vs  $|+x\rangle$ ) and global phase ( $e^{i3\pi/4}$  vs  $e^{i\pi/4}$ ), demonstrating that  $\hat{X}\hat{Y} \neq \hat{Y}\hat{X}$ .

### Key Idea: Phase Factors in Quantum Rotations

The 90° rotation operators naturally introduce phase factors that are often omitted in simplified treatments. These phases are essential for:

- Mathematical consistency with the operator definitions
- Correct calculation of sequential transformations



- Understanding the geometric structure of rotation groups
- Proper treatment of commutation relations

While global phases don't affect measurement probabilities, they are crucial for understanding how operators compose and for advanced topics like geometric phases.

This result can be verified through explicit matrix calculations using the matrix representations of these rotation operators. The non-trivial result provides a concrete demonstration that quantum rotations do not commute, with immediate physical consequences observable on the Bloch sphere.

#### Bloch Cube Activity: 1.4: Exploring Non-Commutativity

Using your Bloch cube:

- Start with the cube showing  $|+z\rangle$
- Perform a  $90^\circ$  rotation about the X-axis, then about the Y-axis
- Record the final state
- Reset to  $|+z\rangle$  and perform the rotations in reverse order
- Compare the final states
- Repeat for different initial states and rotation sequences
- Discuss the physical implications of non-commutativity

## 1.7 The Density Operator Formalism

### 1.7.1 Pure States and Statistical Mixtures

Thus far, we have represented quantum states as vectors in Hilbert space. This representation assumes maximal knowledge about the quantum system—what we call a pure state. However, in many practical situations, our knowledge is incomplete, necessitating a more general formalism.

The density operator provides this generalization. For a pure state  $|\psi\rangle$ , the density operator is simply:

$$\rho = |\psi\rangle\langle\psi| \quad (1.85)$$

For example, the density operator for  $|+x\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + |-z\rangle)$  is:

$$\rho_{+x} = |+x\rangle\langle+x| \quad (1.86)$$

$$= \frac{1}{2}(|+z\rangle + |-z\rangle)(\langle+z| + \langle-z|) \quad (1.87)$$

$$= \frac{1}{2}(|+z\rangle\langle+z| + |+z\rangle\langle-z| + |-z\rangle\langle+z| + |-z\rangle\langle-z|) \quad (1.88)$$

This operator contains all information about the quantum state and can be used to calculate expectation values of observables:  $\langle A \rangle = \text{Tr}(\rho A)$ .

When our knowledge is incomplete, we have a statistical mixture of pure states. If the system is in state  $|\psi_i\rangle$  with classical probability  $p_i$ , the density operator becomes:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i| \quad (1.89)$$

### 1.7.2 Properties and Interpretation

The density operator formalism reveals important distinctions between quantum and classical probability. Consider two preparations:

- (a) A pure state  $|\psi\rangle = \sqrt{0.6}|+z\rangle + \sqrt{0.4}|-z\rangle$
- (b) A mixture with 60% probability of  $|+z\rangle$  and 40% probability of  $|-z\rangle$

For Z-measurements, both preparations yield identical statistics. However, they differ dramatically for X-measurements. The pure state shows quantum interference through the off-diagonal terms  $|+z\rangle\langle-z|$  and  $|-z\rangle\langle+z|$  in its density operator, while the mixture does not have these terms.

This difference is captured by the purity  $\text{Tr}(\rho^2)$ . Pure states always have purity 1, while mixed states have purity less than 1.

#### Example 7: Calculating Expectation Values for a Mixed State Ensemble

Consider the mixed state  $\rho = 0.6|+z\rangle\langle+z| + 0.4|-z\rangle\langle-z|$ . We'll calculate the expectation values  $\langle\sigma_x\rangle$ ,  $\langle\sigma_y\rangle$ , and  $\langle\sigma_z\rangle$  using a Bloch cube ensemble.

##### Step 1: Create the ensemble

- Take 5 Bloch cubes
- Set 3 to show  $|+z\rangle$  and 2 to show  $|-z\rangle$
- This ensemble represents our mixed state

**Step 2: Use pairing to simplify** A key insight: opposite states contribute equally but with opposite signs to any measurement. Therefore, any pair of cubes showing  $|+z\rangle$  and  $|-z\rangle$  contributes zero to all expectation values.

- Pair up 2  $|+z\rangle$  cubes with the 2  $|-z\rangle$  cubes
- These 4 cubes (2 pairs) contribute 0 to all expectation values
- We're left with 1 unpaired  $|+z\rangle$  cube

**Step 3: Calculate expectation values** For the ensemble, the expectation value is the average over all cubes:

##### Z-measurement:

- 4 paired cubes: contribute 0
- 1 unpaired  $|+z\rangle$ : contributes +1
- Average over 5 cubes:  $\langle\sigma_z\rangle = \frac{0+1}{5} = 0.2$

##### X-measurement:

- For  $|+z\rangle$ : equal probability of  $\pm 1$ , so expectation is 0
- All 5 cubes have expectation 0 for X-measurement
- $\langle\sigma_x\rangle = 0$

**Y-measurement:**

- Same reasoning as X-measurement
- $\langle\sigma_y\rangle = 0$

**Verification:** We can check this matches the theoretical calculation:  $\langle\sigma_z\rangle = 0.6(+1) + 0.4(-1) = 0.6 - 0.4 = 0.2$

This example shows how the pairing trick simplifies calculations and provides physical intuition: mixed states with equal amounts of opposite states are "balanced" and give zero expectation values for all measurements.

**Contrast with pure state:** The pure state  $\sqrt{0.6}|+z\rangle + \sqrt{0.4}|-z\rangle$  would also give  $\langle\sigma_z\rangle = 0.2$ , but would give non-zero values for  $\langle\sigma_x\rangle$  and  $\langle\sigma_y\rangle$  due to quantum interference. This mixture cannot show such interference because it's just a classical probability distribution over definite states.

**Bloch Cube Activity: 1.5: The Pairing Principle for Mixed States**

Explore the pairing principle with different mixed states:

- Create an ensemble with 4 cubes: 2 showing  $|+x\rangle$  and 2 showing  $|-x\rangle$
- Verify that all expectation values are zero by pairing opposite states
- Now add a 5th cube showing  $|+x\rangle$ . Calculate  $\langle\sigma_x\rangle$ ,  $\langle\sigma_y\rangle$ ,  $\langle\sigma_z\rangle$
- Create a more complex ensemble: 2  $|+z\rangle$ , 2  $|-z\rangle$ , 1  $|+x\rangle$ , 1  $|-x\rangle$
- Use pairing to simplify the calculation of expectation values
- Discuss: Can you always use pairing? What happens with ensembles that have states from all three axes?

## 1.8 Advanced Topics: Beyond the Cube

### 1.8.1 The Full Bloch Sphere

While the Bloch cube restricts us to six special states, the full Bloch sphere encompasses all possible states of a two-level system. Any point on the unit sphere corresponds to a valid quantum state:

$$|\psi(\theta, \phi)\rangle = \cos(\theta/2)|+z\rangle + e^{i\phi}\sin(\theta/2)|-z\rangle \quad (1.90)$$

The Bloch vector  $\vec{r} = (r_x, r_y, r_z)$  provides an alternative parameterization:

$$\rho = \frac{1}{2}(\hat{I} + \vec{r} \cdot \vec{\sigma}) \quad (1.91)$$

For pure states,  $|\vec{r}| = 1$  (on the sphere surface). For mixed states,  $|\vec{r}| < 1$  (inside the sphere). The completely mixed state  $\rho = \frac{1}{2}\hat{I}$  sits at the origin with  $\vec{r} = 0$ .

### 1.8.2 Quantum Process Tomography

Given an unknown quantum state, how can we determine it experimentally? This is the problem of quantum state tomography. For a two-level system, we need to measure in three different bases to fully characterize the state.

If we measure many copies of the state in the X, Y, and Z bases, obtaining expectation values:

$$\langle\sigma_x\rangle = \text{Tr}(\rho\sigma_x) = r_x \quad (1.92)$$

$$\langle\sigma_y\rangle = \text{Tr}(\rho\sigma_y) = r_y \quad (1.93)$$

$$\langle\sigma_z\rangle = \text{Tr}(\rho\sigma_z) = r_z \quad (1.94)$$

We can reconstruct the density operator from these Bloch vector components. This provides a complete experimental characterization of the quantum state.

## 1.9 The Postulates of Quantum Mechanics

Having developed the mathematical and physical tools, we can now state the fundamental postulates of quantum mechanics in a form particularly suited to our Bloch cube approach:

**Postulate 1 (State Space):** The state of a quantum system is represented by a normalized vector  $|\psi\rangle$  in a complex Hilbert space. For our two-state system, this space is two-dimensional.

**Postulate 2 (Observables):** Physical quantities are represented by operators constructed from measurement procedures. The eigenvalues of these operators correspond to possible measurement outcomes. Since measurement outcomes must be real numbers, these operators are necessarily Hermitian.

**Postulate 3 (Born Rule):** Upon measuring an observable  $\hat{A}$  for a system in state  $|\psi\rangle$ , the probability of obtaining eigenvalue  $a_i$  is  $P(|\psi\rangle \rightarrow |a_i\rangle) = |\langle a_i|\psi\rangle|^2$ , where  $|a_i\rangle$  is the corresponding eigenstate.

**Postulate 4 (State Collapse):** After obtaining measurement result  $a_i$ , the system's state becomes  $|a_i\rangle$ .

**Postulate 5 (Unitary Evolution):** The time evolution of an isolated quantum system is governed by the unitary operator  $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$ , where  $\hat{H}$  is the Hamiltonian.

These postulates, while abstractly stated, become concrete through Bloch cube manipulations. Each postulate corresponds to physical operations: preparation, measurement, and rotation of the cube.

### Bloch Cube Activity: 1.6: Verifying the Postulates

Using your Bloch cube, verify each postulate:

- (a) **State Space:** Show that any face can be expressed as a normalized superposition
- (b) **Observables:** Identify the three measurement types with cube orientations
- (c) **Born Rule:** Prepare  $|+x\rangle$  and measure Z many times to verify 50-50 statistics
- (d) **Collapse:** After measuring Z on  $|+x\rangle$ , verify the state is now  $|\pm z\rangle$

(e) **Evolution:** Show that rotations preserve normalization and orthogonality

## 1.10 Matrix Representations

In the computational basis, quantum states are represented as column vectors and operators as matrices. This section formally introduces the matrix representations that complement our Bloch cube approach.

### 1.10.1 Single-Qubit Matrix Representations

For a single qubit, the computational (Z) basis states have the matrix representations:

$$|+z\rangle = |0\rangle \doteq_z \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |-z\rangle = |1\rangle \doteq_z \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (1.95)$$

The projection operators from earlier in this chapter can now be written explicitly as matrices. For example:

$$|0\rangle\langle 0| \doteq_z \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad (1.96)$$

$$|1\rangle\langle 1| \doteq_z \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad (1.97)$$

$$|0\rangle\langle 1| \doteq_z \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad (1.98)$$

$$|1\rangle\langle 0| \doteq_z \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad (1.99)$$

The Pauli matrices, which represent measurements along the three coordinate axes, have the explicit forms:

$$\sigma_x \doteq_z \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (1.100)$$

$$\sigma_y \doteq_z \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (1.101)$$

$$\sigma_z \doteq_z \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (1.102)$$

The rotation operators from earlier also have matrix representations. For example:

$$\hat{R}_z(\theta) = e^{-i\theta\sigma_z/2} \doteq_z \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix} \quad (1.103)$$

$$\hat{Z} = \hat{R}_z(\pi/2) \doteq_z \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad (1.104)$$

Similarly, the discrete rotation operators have the matrix forms:

$$\hat{X} \doteq_z \frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix} \quad (1.105)$$

$$\hat{Y} \doteq_z \frac{1}{2} \begin{pmatrix} 1+i & -1-i \\ 1+i & 1+i \end{pmatrix} \quad (1.106)$$

## 1.11 Comprehensive Basis Representations

Before moving on to more advanced topics, we establish a comprehensive reference for the six Bloch cube states expressed in all three orthonormal bases. This reference will prove invaluable throughout our study of quantum mechanics, as it allows us to quickly read off the representation of any state in any basis.

### 1.11.1 The Six Bloch Cube States in All Bases

Master Reference: All Bloch Cube States in All Bases

**Z-basis representations:**  $\{|+z\rangle, |-z\rangle\}$

$$|+x\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + |-z\rangle) \doteq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1.107)$$

$$|-x\rangle = \frac{1}{\sqrt{2}}(|+z\rangle - |-z\rangle) \doteq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (1.108)$$

$$|+y\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + i|-z\rangle) \doteq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \quad (1.109)$$

$$|-y\rangle = \frac{1}{\sqrt{2}}(|+z\rangle - i|-z\rangle) \doteq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \quad (1.110)$$

$$|+z\rangle = |+z\rangle \doteq \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (1.111)$$

$$|-z\rangle = |-z\rangle \doteq \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (1.112)$$

**X-basis representations:**  $\{|+x\rangle, |-x\rangle\}$

$$|+x\rangle = |+x\rangle \doteq \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (1.113)$$

$$|-x\rangle = |-x\rangle \doteq \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (1.114)$$

$$|+y\rangle = \frac{(1+i)|+x\rangle + (1-i)|-x\rangle}{2} \doteq \frac{1}{2} \begin{pmatrix} 1+i \\ 1-i \end{pmatrix} \quad (1.115)$$

$$|-y\rangle = \frac{(1-i)|+x\rangle + (1+i)|-x\rangle}{2} \doteq \frac{1}{2} \begin{pmatrix} 1-i \\ 1+i \end{pmatrix} \quad (1.116)$$

$$|+z\rangle = \frac{1}{\sqrt{2}}(|+x\rangle + |-x\rangle) \doteq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1.117)$$

$$|-z\rangle = \frac{1}{\sqrt{2}}(|+x\rangle - |-x\rangle) \doteq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (1.118)$$

**Y-basis representations:**  $\{|+y\rangle, |-y\rangle\}$

$$|+x\rangle = \frac{(1-i)|+y\rangle + (1+i)|-y\rangle}{2} \doteq \frac{1}{y} \frac{1}{2} \begin{pmatrix} 1-i \\ 1+i \end{pmatrix} \quad (1.119)$$

$$|-x\rangle = \frac{(1+i)|+y\rangle + (1-i)|-y\rangle}{2} \doteq \frac{1}{y} \frac{1}{2} \begin{pmatrix} 1+i \\ 1-i \end{pmatrix} \quad (1.120)$$

$$|+y\rangle = | + y \rangle \doteq \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (1.121)$$

$$|-y\rangle = | - y \rangle \doteq \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (1.122)$$

$$|+z\rangle = \frac{1}{\sqrt{2}}(|+y\rangle + |-y\rangle) \doteq \frac{1}{y} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1.123)$$

$$|-z\rangle = \frac{1}{\sqrt{2}}(-i|+y\rangle + i|-y\rangle) \doteq \frac{1}{y} \frac{1}{\sqrt{2}} \begin{pmatrix} -i \\ i \end{pmatrix} \quad (1.124)$$

### 1.11.2 Using the Reference Table

This comprehensive table enables several powerful calculations:

#### Example 8: Cross-Basis Inner Products

Calculate  $\langle +x|+y\rangle$  using different basis representations.

**Method 1: Using Z-basis**

$$\langle +x|+y\rangle = \left( \frac{1}{\sqrt{2}}(\langle +z| + \langle -z|) \right) \left( \frac{1}{\sqrt{2}}(|+z\rangle + i|-z\rangle) \right) \quad (1.125)$$

$$= \frac{1}{2}(\langle +z|+z\rangle + i\langle +z|-z\rangle + \langle -z|+z\rangle + i\langle -z|-z\rangle) \quad (1.126)$$

$$= \frac{1}{2}(1 + 0 + 0 + i) = \frac{1+i}{2} \quad (1.127)$$

**Method 2: Using X-basis** From the table:  $|+y\rangle = \frac{(1+i)|+x\rangle + (1-i)|-x\rangle}{2}$

$$\langle +x|+y\rangle = \langle +x| \left( \frac{(1+i)|+x\rangle + (1-i)|-x\rangle}{2} \right) \quad (1.128)$$

$$= \frac{1}{2}[(1+i)\langle +x|+x\rangle + (1-i)\langle +x|-x\rangle] \quad (1.129)$$

$$= \frac{1}{2}[(1+i) \cdot 1 + (1-i) \cdot 0] \quad (1.130)$$

$$= \frac{1+i}{2} \quad (1.131)$$

Both methods give the same result, confirming our calculation.

**Magnitude:**  $|\langle +x|+y\rangle| = \left| \frac{1+i}{2} \right| = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$

**Phase:**  $\arg(\langle +x|+y\rangle) = \arg(1+i) = \pi/4$

### 1.11.3 Transformation Matrices Between Bases

The basis transformations are most clearly understood through their basis-independent operator form, which then yields matrix representations in any chosen basis.

#### Basis-Independent Form

The transformation operators map basis states from one representation to another:

##### Basis Transformation Operators

###### Z to X transformation:

$$U_{xz} = | +x \rangle \langle +z | + | -x \rangle \langle -z | \quad (1.132)$$

This operator maps:  $| +z \rangle \mapsto | +x \rangle$  and  $| -z \rangle \mapsto | -x \rangle$

###### Z to Y transformation:

$$U_{yz} = | +y \rangle \langle +z | + | -y \rangle \langle -z | \quad (1.133)$$

This operator maps:  $| +z \rangle \mapsto | +y \rangle$  and  $| -z \rangle \mapsto | -y \rangle$

###### X to Y transformation:

$$U_{yx} = | +y \rangle \langle +x | + | -y \rangle \langle -x | \quad (1.134)$$

This operator maps:  $| +x \rangle \mapsto | +y \rangle$  and  $| -x \rangle \mapsto | -y \rangle$

These operators transform states according to their definition. To find the matrix representation of any operator  $U$  in a given basis, we construct the matrix by computing all matrix elements in that basis.

#### Deriving Matrix Representations

To find the matrix representation of an operator  $U$  in any basis, we compute the matrix elements  $\langle i | U | j \rangle$  where  $| i \rangle$  and  $| j \rangle$  are the basis states. For example, in the Z-basis, the matrix elements are:

$$U \doteq_z \begin{pmatrix} \langle +z | U | +z \rangle & \langle +z | U | -z \rangle \\ \langle -z | U | +z \rangle & \langle -z | U | -z \rangle \end{pmatrix} \quad (1.135)$$

##### Example 9: Deriving $U_{xz}$ in the Z-basis

The operator  $U_{xz} = | +x \rangle \langle +z | + | -x \rangle \langle -z |$  has the following matrix elements in the Z-basis:

$$\langle +z | U_{xz} | +z \rangle = \langle +z | (| +x \rangle \langle +z | + | -x \rangle \langle -z |) | +z \rangle \quad (1.136)$$

$$= \langle +z | +x \rangle \underbrace{\langle +z | +z \rangle}_{=1} + \langle +z | -x \rangle \underbrace{\langle -z | +z \rangle}_{=0} \quad (1.137)$$

$$= \langle +z | +x \rangle = \frac{1}{\sqrt{2}} \quad (1.138)$$



$$\langle +z|U_{xz}| - z \rangle = \langle +z| + x \rangle \underbrace{\langle +z| - z \rangle}_{=0} + \langle +z| - x \rangle \underbrace{\langle -z| - z \rangle}_{=1} \quad (1.139)$$

$$= \langle +z| - x \rangle = \frac{1}{\sqrt{2}} \quad (1.140)$$

$$\langle -z|U_{xz}| + z \rangle = \langle -z| + x \rangle \underbrace{\langle +z| + z \rangle}_{=1} + \langle -z| - x \rangle \underbrace{\langle -z| + z \rangle}_{=0} \quad (1.141)$$

$$= \langle -z| + x \rangle = \frac{1}{\sqrt{2}} \quad (1.142)$$

$$\langle -z|U_{xz}| - z \rangle = \langle -z| + x \rangle \underbrace{\langle +z| - z \rangle}_{=0} + \langle -z| - x \rangle \underbrace{\langle -z| - z \rangle}_{=1} \quad (1.143)$$

$$= \langle -z| - x \rangle = \frac{-1}{\sqrt{2}} \quad (1.144)$$

Therefore:

$$U_{xz} \doteq_z \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (1.145)$$

**Verification:** This operator transforms the Z-basis states to X-basis states:

$$U_{xz}| + z \rangle = | + x \rangle = \frac{1}{\sqrt{2}}(| + z \rangle + | - z \rangle) \quad (1.146)$$

$$U_{xz}| - z \rangle = | - x \rangle = \frac{1}{\sqrt{2}}(| + z \rangle - | - z \rangle) \quad (1.147)$$

We can verify this using the matrix representation:

$$U_{xz}| + z \rangle \doteq_z \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = | + x \rangle \checkmark \quad (1.148)$$

### Key Idea: Matrix Representations of Operators

The matrix representation of any operator  $U$  in a given basis is constructed from its matrix elements:

- In the Z-basis:  $(U)_{ij} = \langle i|U|j \rangle$  where  $i, j \in \{+z, -z\}$
- The operator equation  $U|\psi\rangle = |\phi\rangle$  becomes a matrix equation when expressed in any basis
- Different bases give different matrix representations of the same operator
- The physical content (eigenvalues, trace, determinant) remains invariant

**Example 10: Deriving  $U_{yz}$  in the Z-basis**

The operator  $U_{yz} = | + y \rangle \langle + z | + | - y \rangle \langle - z |$  has the following matrix elements in the Z-basis:

$$\langle +z | U_{yz} | +z \rangle = \langle +z | (| + y \rangle \langle +z | + | - y \rangle \langle -z |) | +z \rangle \quad (1.149)$$

$$= \langle +z | + y \rangle \underbrace{\langle +z | +z \rangle}_{=1} + \langle +z | - y \rangle \underbrace{\langle -z | +z \rangle}_{=0} \quad (1.150)$$

$$= \langle +z | + y \rangle = \frac{1}{\sqrt{2}} \quad (1.151)$$

$$\langle +z | U_{yz} | -z \rangle = \langle +z | + y \rangle \underbrace{\langle +z | -z \rangle}_{=0} + \langle +z | - y \rangle \underbrace{\langle -z | -z \rangle}_{=1} \quad (1.152)$$

$$= \langle +z | - y \rangle = \frac{1}{\sqrt{2}} \quad (1.153)$$

$$\langle -z | U_{yz} | +z \rangle = \langle -z | + y \rangle \underbrace{\langle +z | +z \rangle}_{=1} + \langle -z | - y \rangle \underbrace{\langle -z | +z \rangle}_{=0} \quad (1.154)$$

$$= \langle -z | + y \rangle = \frac{i}{\sqrt{2}} \quad (1.155)$$

$$\langle -z | U_{yz} | -z \rangle = \langle -z | + y \rangle \underbrace{\langle +z | -z \rangle}_{=0} + \langle -z | - y \rangle \underbrace{\langle -z | -z \rangle}_{=1} \quad (1.156)$$

$$= \langle -z | - y \rangle = \frac{-i}{\sqrt{2}} \quad (1.157)$$

Therefore:

$$U_{yz} \doteq_z \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \quad (1.158)$$

**Verification:** This operator transforms:

$$U_{yz} | +z \rangle = | + y \rangle = \frac{1}{\sqrt{2}} (| + z \rangle + i | - z \rangle) \quad (1.159)$$

$$U_{yz} | -z \rangle = | - y \rangle = \frac{1}{\sqrt{2}} (| + z \rangle - i | - z \rangle) \quad (1.160)$$

**Example 11: Deriving  $U_{yx}$  in the X-basis**

The operator  $U_{yx} = | + y \rangle \langle + x | + | - y \rangle \langle - x |$  has the following matrix elements in the X-basis:

$$\langle +x|U_{yx}|+x\rangle = \langle +x|(|+y\rangle\langle +x| + |-y\rangle\langle -x|)|+x\rangle \quad (1.161)$$

$$= \langle +x|+y\rangle \underbrace{\langle +x|+x\rangle}_{=1} + \langle +x|-y\rangle \underbrace{\langle -x|+x\rangle}_{=0} \quad (1.162)$$

$$= \langle +x|+y\rangle \quad (1.163)$$

From the correct expression  $|+y\rangle = \frac{(1+i)|+x\rangle + (1-i)|-x\rangle}{2}$ , we have  $\langle +x|+y\rangle = \frac{1+i}{2}$ , so:

$$\langle +x|U_{yx}|+x\rangle = \frac{1+i}{2} \quad (1.164)$$

Similarly:

$$\langle +x|U_{yx}|-x\rangle = \langle +x|+y\rangle \underbrace{\langle +x|-x\rangle}_{=0} + \langle +x|-y\rangle \underbrace{\langle -x|-x\rangle}_{=1} = \langle +x|-y\rangle = \frac{1-i}{2} \quad (1.165)$$

$$\langle -x|U_{yx}|+x\rangle = \langle -x|+y\rangle \underbrace{\langle +x|+x\rangle}_{=1} + \langle -x|-y\rangle \underbrace{\langle -x|+x\rangle}_{=0} = \langle -x|+y\rangle = \frac{1-i}{2} \quad (1.166)$$

$$\langle -x|U_{yx}|-x\rangle = \langle -x|+y\rangle \underbrace{\langle +x|-x\rangle}_{=0} + \langle -x|-y\rangle \underbrace{\langle -x|-x\rangle}_{=1} = \langle -x|-y\rangle = \frac{1+i}{2} \quad (1.167)$$

Therefore:

$$U_{yx} \doteq \frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix} \quad (1.168)$$

### Summary of Transformation Matrices

The basis transformation operators have the following matrix representations:

$$U_{xz} \doteq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (\text{transforms } |+z\rangle \rightarrow |+x\rangle, |-z\rangle \rightarrow |-x\rangle) \quad (1.169)$$

$$U_{yz} \doteq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \quad (\text{transforms } |+z\rangle \rightarrow |+y\rangle, |-z\rangle \rightarrow |-y\rangle) \quad (1.170)$$

$$U_{yx} \doteq \frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix} \quad (\text{transforms } |+x\rangle \rightarrow |+y\rangle, |-x\rangle \rightarrow |-y\rangle) \quad (1.171)$$

These matrices satisfy important properties:

- Unitarity:  $U^\dagger U = U U^\dagger = I$
- They implement the basis transformations  $U_{xz}$ ,  $U_{yz}$ , and  $U_{yx}$  defined above
- Each matrix is constructed from the matrix elements  $\langle i|U|j\rangle$  in the respective basis

## 1.12 Chapter Summary

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This chapter has established the mathematical and conceptual foundations of quantum mechanics through the study of two-state systems. Beginning with the familiar mathematics of two-dimensional vectors, we developed the dot-vector notation as a bridge to Dirac's formalism. The introduction of the Bloch cube provided a tangible representation of quantum states, making abstract concepts physically manipulable.

We explored the fundamental features of quantum mechanics: superposition of states, the probabilistic nature of measurement described by the Born rule, state collapse following measurement, and unitary evolution. The non-commutativity of quantum operations emerged naturally from the non-commutativity of rotations. We provided complete basis representations for all six Bloch cube states, enabling calculations in any basis. Finally, we introduced the density operator formalism to handle situations of incomplete knowledge.

This foundation, built on the simplest non-trivial quantum system, contains all the essential features of quantum mechanics. The complete basis representation table will serve as a crucial reference throughout the course. In subsequent chapters, we will extend these concepts to multi-qubit systems, continuous variables, and ultimately to the wave mechanics of particles in space. The physical intuition developed through Bloch cube manipulations will guide us through increasingly abstract mathematical territories.

## 1.13 Problems

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### Problem 1.1 (*Dot-Vec Notation and Basic Operations*)

Using dot-vec notation and the Bloch cube:

- Express the operator that exchanges  $x$  and  $y$  components of any vectors using operators composed of vectors and dot-vectors
- Apply this operator to  $\hat{x}$  and verify the result
- Translate your result to Dirac notation for  $|+z\rangle$  and  $|-z\rangle$ . That is, find an operator that swaps these two states
- Use your Bloch cube to verify this transformation physically

### Problem 1.2 (*Inner Products and Basis Representations*)

For the quantum state  $|+y\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + i|-z\rangle)$ :

- Calculate the inner products  $\langle \pm x | +y \rangle$ ,  $\langle \pm z | +y \rangle$  using the Z-basis
- Verify orthogonality:  $\langle -y | +y \rangle = 0$  using the z-basis
- Determine measurement probabilities for X, Y, and Z measurements
- Express  $|+y\rangle$  in the X-basis using the spectral resolution of identity

### Problem 1.3 (*Bloch Cube Rotations*)

Perform the following sequence on your Bloch cube:

- (a) Start with  $|+z\rangle$  (top face up)
- (b) Apply a  $90^\circ$  rotation about the X-axis
- (c) Apply a  $90^\circ$  rotation about the Y-axis
- (d) Identify the final state and express it in the Z-basis
- (e) Calculate the expectation values for  $\hat{X}$ ,  $\hat{Y}$ , and  $\hat{Z}$  for your final state

**Problem 1.4** (*The Born Rule and Measurement Probabilities*)

The Born rule states that for a system in state  $|\psi\rangle$  and measurement eigenstate  $|\phi_i\rangle$ , the probability is  $P(|\psi\rangle \rightarrow |\phi_i\rangle) = |\langle\phi_i|\psi\rangle|^2$ .

- (a) Consider the Bloch cube state  $|+x\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + |-z\rangle)$ . Verify it is normalized, then use the Born rule to calculate the probabilities for obtaining  $|+z\rangle$  and  $|-z\rangle$  in a Z-measurement.
- (b) For the state  $|+y\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + i|-z\rangle)$ , calculate all six probabilities:  $P(|+y\rangle \rightarrow |\pm z\rangle)$ ,  $P(|+y\rangle \rightarrow |\pm x\rangle)$ , and  $P(|+y\rangle \rightarrow |\pm y\rangle)$ .
- (c) Using your Bloch cube: Create an ensemble of 8 cubes to represent the mixed state  $\rho = 0.5|+x\rangle\langle+x| + 0.5|-x\rangle\langle-x|$ . How many cubes should show each face? Calculate  $\langle\sigma_z\rangle$  for this ensemble and compare to the pure state  $|+y\rangle$ .
- (d) Conceptual: Explain in your own words why the Born rule gives probabilities (not deterministic outcomes) even when we have complete knowledge of the quantum state. How does this differ from classical probability?

**Problem 1.5** (*Properties of Bloch Cube States*)

Analyze the six Bloch cube face states:

- (a) Verify that each state is normalized:  $\langle\psi|\psi\rangle = 1$
- (b) Show that opposite faces are orthogonal (verify for  $|\pm z\rangle$  pair)
- (c) Calculate the overlap  $|\langle+x|+y\rangle|^2$  and interpret physically
- (d) Starting with  $|+y\rangle$ , compare final probability distributions for measuring Z then X versus measuring X then Z

**Problem 1.6** (*Born Rule Applications: Sequential Measurements and Expectation Values*)

Explore deeper applications of the Born rule using only Bloch cube states:

- (a) For each of the six Bloch cube states, calculate  $\langle\sigma_z\rangle$  using the Born rule:
  - (i)  $|+z\rangle$  and  $|-z\rangle$
  - (ii)  $|+x\rangle$  and  $|-x\rangle$
  - (iii)  $|+y\rangle$  and  $|-y\rangle$
  - (iv) What pattern do you observe for opposite faces?

- (b) Using your Bloch cube to simulate: Start with  $|+x\rangle$ . Perform 20 Z-measurements (using coin flips for the 50-50 probability). Record +1 for  $|+z\rangle$  and -1 for  $|-z\rangle$ . Calculate the average and compare to the theoretical  $\langle\sigma_z\rangle = 0$ .
- (c) Sequential measurements: Consider starting with  $|+y\rangle$ 
  - (i) Calculate the probability of measuring  $|+x\rangle$  then  $|+z\rangle$  in sequence
  - (ii) Calculate the probability of measuring  $|+z\rangle$  then  $|+x\rangle$  in sequence
  - (iii) Explain why these probabilities differ despite involving the same measurements

**Problem 1.7** (*Pauli Matrix Operations and Quantum States*)

Working with the Pauli matrices and quantum states:

- (a) Calculate  $\sigma_x^2$ ,  $\sigma_y^2$ , and  $\sigma_z^2$  explicitly
- (b) Verify the commutation relation  $[\sigma_x, \sigma_y] = 2i\sigma_z$
- (c) Find how  $\sigma_x$  acts on  $|+z\rangle$  and  $|+y\rangle$
- (d) Compare the mixed state  $\rho = 0.5|+z\rangle\langle+z| + 0.5|-z\rangle\langle-z|$  with the pure state  $|+x\rangle$  by calculating  $\langle\sigma_z\rangle$  for both

**Problem 1.8** (*Density Operator and Pure vs Mixed States*)

The density operator  $\hat{\rho}$  fully characterizes a quantum state. For a pure state  $|\psi\rangle$ , we have  $\hat{\rho} = |\psi\rangle\langle\psi|$  with  $\text{Tr}(\hat{\rho}^2) = 1$ . Mixed states have  $\text{Tr}(\hat{\rho}^2) < 1$ .

- (a) Construct the density matrices for the six Bloch cube face states
- (b) For each, verify that  $\text{Tr}(\hat{\rho}) = 1$  and  $\text{Tr}(\hat{\rho}^2) = 1$  (confirming purity)
- (c) Consider the thermal state  $\hat{\rho}_T = \frac{1}{2}\hat{I}$ . Calculate  $\text{Tr}(\hat{\rho}_T^2)$  and explain what this purity means physically
- (d) Show that for any density matrix, the eigenvalues  $p_i$  satisfy  $0 \leq p_i \leq 1$  and  $\sum_i p_i = 1$
- (e) Construct a mixed state as a 50-50 mixture of  $|+z\rangle$  and  $|+x\rangle$ :  $\hat{\rho} = 0.5|+z\rangle\langle+z| + 0.5|+x\rangle\langle+x|$ . Calculate its purity

**Problem 1.9** (*Matrix Exponential and Rotations on the Bloch Sphere*)

Single-qubit rotations can be expressed as  $U(\theta) = \exp(-i\theta\hat{n} \cdot \vec{\sigma}/2)$  where  $\hat{n}$  is the rotation axis.

- (a) Calculate  $\exp(-i\theta\sigma_z/2)$  explicitly using the Taylor series
- (b) Verify that your result matches the Z-rotation matrix  $R_z(\theta)$
- (c) Show that  $\exp(-i\theta\sigma_x/2)|+z\rangle\exp(i\theta\sigma_x/2)$  rotates  $|+z\rangle$  around the X-axis by angle  $\theta$
- (d) For  $\theta = \pi/2$ , compute the action of this operator on  $|+z\rangle$  and identify the final state

**Problem 1.10** (*Commutation Relations and Observables*)

Pauli matrices generate all of spin-1/2 quantum mechanics through their (anti)commutation relations:

- (a) Verify all three commutation relations:  $[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k$
- (b) Calculate the anticommutators  $\{\sigma_i, \sigma_j\}$  for all pairs
- (c) Show that  $[\sigma_x, \sigma_z] = -[\sigma_z, \sigma_x]$  (anticommutativity of different axes)
- (d) Use these relations to prove that  $\sigma_x\sigma_y\sigma_z = i\hat{I}$
- (e) Interpret: Why does the fact that  $[\sigma_x, \sigma_z] \neq 0$  mean that X and Z measurements cannot be simultaneously sharp?

**Problem 1.11** (*Basis Transformations and Change of Representation*)

The choice of basis is arbitrary in quantum mechanics. Different choices reveal different information:

- (a) Write the six Bloch cube face states in all three basis choices: Z-basis, X-basis, and Y-basis
- (b) For the state  $|+y\rangle$ , express it in the X-basis by finding its components in  $\{|+x\rangle, |-x\rangle\}$
- (c) Calculate the change-of-basis matrices:  $U_{X \rightarrow Z}$ ,  $U_{Y \rightarrow Z}$ , and  $U_{X \rightarrow Y}$
- (d) Verify that these are unitary:  $U^\dagger U = \hat{I}$
- (e) Consider an operator  $\hat{O} = \sigma_x$  in the Z-basis. What is its matrix representation in the X-basis?

**Problem 1.12** (*Expectation Values and Variance*)

The expectation value  $\langle \hat{A} \rangle = \text{Tr}(\hat{\rho}\hat{A})$  encodes measurement statistics:

- (a) For the state  $|\psi\rangle = \frac{1}{\sqrt{3}}(|+z\rangle + |+x\rangle + |+y\rangle)$ , calculate  $\langle \sigma_x \rangle$ ,  $\langle \sigma_y \rangle$ , and  $\langle \sigma_z \rangle$
- (b) Define variance as  $\Delta A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}$ . For the state  $|+x\rangle$ , calculate  $\Delta \sigma_z$
- (c) Show that for eigenstate measurements, the variance is zero
- (d) For the equal superposition above, calculate the variances in all three directions
- (e) Explain the uncertainty principle relationship between variances of non-commuting observables

**Problem 1.13** (*Quantum Interference and Superposition*)

Quantum superposition exhibits interference effects unlike classical probabilities:

- (a) Consider the state  $|\psi\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + e^{i\phi}|-z\rangle)$  for arbitrary phase  $\phi$ . Show that the probability of measuring  $|+z\rangle$  depends on  $\phi$
- (b) Calculate  $P(\psi \rightarrow +z)$  for  $\phi = 0$ ,  $\phi = \pi/2$ , and  $\phi = \pi$  explicitly
- (c) For  $\phi = \pi$ , simplify the state. What measurement basis makes this state an eigenstate?
- (d) Show that the classical mixture  $\rho = 0.5|+z\rangle\langle +z| + 0.5|-z\rangle\langle -z|$  gives the same probability for all phases  $\phi$  (no interference)
- (e) Use your Bloch cube to visualize why the superposition  $|+z\rangle + e^{i\phi}|-z\rangle$  traces different paths on the sphere as  $\phi$  varies, whereas the classical mixture is always at the center

**Problem 1.14** (*Projectors and Spectral Decomposition*)

Projection operators  $\hat{P}_i = |\phi_i\rangle\langle\phi_i|$  extract components of quantum states:

- For the state  $|\psi\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + |+x\rangle)$ , construct the projection operators  $\hat{P}_{+z}$  and  $\hat{P}_{+x}$
- Apply these operators to  $|\psi\rangle$  and interpret the results as measurement
- Show that  $\hat{P}_{+z} + \hat{P}_{-z} = \hat{I}$  (completeness)
- For a basis  $\{|\phi_i\rangle\}$ , the spectral decomposition of an observable is  $\hat{A} = \sum_i a_i \hat{P}_i$ . Write  $\sigma_z$  in this form using the Z-basis eigenstates
- Express the same operator in the X-basis and verify that the spectrum (eigenvalues) is unchanged

**Problem 1.15** (*Operator Polar Decomposition*)

Any operator can be decomposed as  $\hat{A} = \hat{U}\hat{P}$  where  $\hat{U}$  is unitary and  $\hat{P}$  is positive semidefinite:

- Show that  $\sigma_x$  can be written in polar form (hint:  $\sigma_x$  is both unitary and Hermitian)
- For the operator  $\hat{A} = \hat{I} + i\sigma_y$  (not Hermitian), find its polar decomposition
- Construct the operator  $\hat{A}^\dagger \hat{A}$  and verify that it is positive semidefinite
- For a general density matrix  $\hat{\rho}$ , explain why the polar decomposition naturally emerges from its spectral decomposition with  $p_i \geq 0$
- Consider the operator  $\hat{b} = \frac{1}{\sqrt{2}}(\sigma_x + i\sigma_y) = |-z\rangle\langle +z|$ . Compute  $\hat{b}^\dagger \hat{b}$  and interpret as a projection

**Problem 1.16** (*Two-Level System Kinematics: State Space Geometry*)

The space of all quantum states forms a geometric object—the Bloch ball:

- Show that any single-qubit density matrix can be written as  $\hat{\rho} = \frac{1}{2}(\hat{I} + \vec{r} \cdot \vec{\sigma})$  with  $|\vec{r}| \leq 1$
- For the pure states (surface of the Bloch ball), what is the magnitude of  $\vec{r}$ ?
- For the completely mixed state  $\hat{I}/2$  (center of the Bloch ball), what is  $\vec{r}$ ?
- Show that the constraint  $\text{Tr}(\hat{\rho}^2) \leq 1$  is equivalent to  $|\vec{r}| \leq 1$
- For a state with  $\vec{r} = (1/2, 0, 1/2)$  (normalized), construct the density matrix and verify its purity

## References and Further Reading

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## Chapter 2

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# Tensor Products and Multi-Qubit Systems

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### 2.1 Introduction: The Mathematical Structure of Composite Systems

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#### Key Idea: Exponential Scaling in Quantum Mechanics

The tensor product structure of quantum mechanics leads to an exponential growth in the dimension of Hilbert space as systems are combined. For  $n$  two-level systems, the composite Hilbert space has dimension  $2^n$ . This exponential scaling underlies both the computational power of quantum systems and the difficulty of simulating them classically.

Having established the quantum mechanics of single two-level systems in Chapter 1, we now face a fundamental question: how do we describe multiple quantum systems? The answer lies in the tensor product structure of quantum mechanics, a mathematical framework that governs how quantum systems combine while preserving their essential quantum properties.

This chapter develops the mathematical machinery needed to describe composite quantum systems. We emphasize that our focus remains purely on the kinematical structure—how to build larger Hilbert spaces from smaller ones. The rich physics of entanglement, which makes full use of this structure, is reserved for future study. Our goal here is to establish fluency with tensor products, multi-qubit states, and the operations that act upon them.

The journey from single qubits to multi-qubit systems represents more than a simple generalization. It reveals new phenomena that have no classical analog and opens the door to quantum information processing. By the end of this chapter, you will understand how to construct and manipulate quantum states of arbitrary numbers of qubits, setting the stage for quantum algorithms and quantum many-body physics.

### 2.2 The Physical Setting: Multiple Distinguishable Systems

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### 2.2.1 Combining Independent Quantum Systems

Consider two Bloch cubes placed on a table, labeled “1” and “2” to distinguish them. Each cube represents an independent two-level quantum system, and each can be in any of the six face states we studied in Chapter 1. The fundamental question is: how do we mathematically describe the combined state of both cubes?

The answer employs the tensor product, denoted by  $\otimes$ . If cube 1 is in state  $|\psi\rangle_1$  and cube 2 is in state  $|\phi\rangle_2$ , the combined system is described by the product state:

$$|\Psi\rangle = |\psi\rangle_1 \otimes |\phi\rangle_2 \quad (2.1)$$

This mathematical operation captures a profound physical principle: the state of a composite system contains complete information about all its constituents. The tensor product preserves the individual quantum properties of each subsystem while creating a unified description of the whole.

Several notational conventions exist for tensor products, all representing the same mathematical object. For the specific case where cube 1 shows  $|+z\rangle$  and cube 2 shows  $|-x\rangle$ , we may write:

$$|+z\rangle_1 \otimes |-x\rangle_2 = |+z\rangle|-x\rangle \quad (2.2)$$

$$= |+z\rangle \otimes |-x\rangle \quad (2.3)$$

The subscripts indicate which subsystem each state refers to, though these are often omitted when the context is clear. The convention of writing states adjacently without the explicit  $\otimes$  symbol is particularly common in quantum information, where brevity is valued.

Let us examine a concrete example to solidify these concepts. Suppose cube 1 is in the state  $|+x\rangle_1 = \frac{1}{\sqrt{2}}(|+z\rangle_1 + |-z\rangle_1)$  and cube 2 is in the state  $|+y\rangle_2 = \frac{1}{\sqrt{2}}(|+z\rangle_2 + i|-z\rangle_2)$ . The composite state is:

$$|\Psi\rangle = |+x\rangle_1 \otimes |+y\rangle_2 \quad (2.4)$$

$$= \frac{1}{\sqrt{2}}(|+z\rangle_1 + |-z\rangle_1) \otimes \frac{1}{\sqrt{2}}(|+z\rangle_2 + i|-z\rangle_2) \quad (2.5)$$

$$= \frac{1}{2}(|+z\rangle_1 \otimes |+z\rangle_2 + i|+z\rangle_1 \otimes |-z\rangle_2 + |-z\rangle_1 \otimes |+z\rangle_2 + i|-z\rangle_1 \otimes |-z\rangle_2) \quad (2.6)$$

$$= \frac{1}{2}(|+z\rangle|+z\rangle + i|+z\rangle|-z\rangle + |-z\rangle|+z\rangle + i|-z\rangle|-z\rangle) \quad (2.7)$$

This expansion reveals that the composite system exists in a superposition of four different configurations, with specific phase relationships between them.

### 2.2.2 The Computational Basis for Multiple Qubits

For quantum information processing, we adopt the standard notation where  $|0\rangle \equiv |+z\rangle$  and  $|1\rangle \equiv |-z\rangle$ . This choice, while arbitrary, has become universal in quantum computing. The historical reason for this convention relates to early implementations where the ground state (lower energy) was labeled  $|0\rangle$  and the excited state (higher energy) was labeled  $|1\rangle$ .

For two qubits, the four computational basis states are:

$$|00\rangle = |0\rangle_1 \otimes |0\rangle_2 = |+z\rangle|+z\rangle \quad (2.8)$$

$$|01\rangle = |0\rangle_1 \otimes |1\rangle_2 = |+z\rangle|-z\rangle \quad (2.9)$$

$$|10\rangle = |1\rangle_1 \otimes |0\rangle_2 = |-z\rangle|+z\rangle \quad (2.10)$$

$$|11\rangle = |1\rangle_1 \otimes |1\rangle_2 = |-z\rangle|-z\rangle \quad (2.11)$$

These four states form a complete orthonormal basis for the two-qubit Hilbert space. The orthonormality can be verified using the fundamental property that tensor products preserve inner products:

$$\langle ij|kl\rangle = \langle i|k\rangle_1 \langle j|l\rangle_2 = \delta_{ik}\delta_{jl} \quad (2.12)$$

For instance,  $\langle 01|10\rangle = \langle 0|1\rangle_1 \langle 1|0\rangle_2 = 0 \cdot 0 = 0$ , confirming orthogonality.

The completeness of this basis means that any two-qubit state can be expressed as a linear combination:

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle \quad (2.13)$$

where the complex coefficients satisfy the normalization condition  $|\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2 = 1$ .

### Bloch Cube Activity: 2.1: Exploring Two-Qubit Basis States

Using two Bloch cubes labeled 1 and 2:

- Arrange the cubes to represent each of the four computational basis states
- For each arrangement, verify that you can identify which qubit is in which state
- Count the total number of possible face-state combinations ( $36 = 6 \times 6$ )
- Create the state  $|+x\rangle|-y\rangle$  and express it in the computational basis
- Calculate the probability of finding the system in each computational basis state
- Verify that your probabilities sum to 1

### 2.2.3 Complete Basis Representations for Two-Qubit States

Building on the single-qubit basis representations from Chapter 1, we now provide comprehensive tables for expressing common two-qubit states in different bases. This is essential for cross-basis calculations and understanding how measurements in different bases affect composite systems.

#### Two-Qubit Computational Basis States in Product Bases

**In the ZZ basis (computational basis):**

$$|00\rangle = |00\rangle \quad (2.14)$$

$$|01\rangle = |01\rangle \quad (2.15)$$

$$|10\rangle = |10\rangle \quad (2.16)$$

$$|11\rangle = |11\rangle \quad (2.17)$$

**In the XX basis:**

$$|00\rangle = \frac{1}{2}(|+x\rangle|+x\rangle + |+x\rangle|-x\rangle + |-x\rangle|+x\rangle + |-x\rangle|-x\rangle) \quad (2.18)$$

$$|01\rangle = \frac{1}{2}(|+x\rangle|+x\rangle - |+x\rangle|-x\rangle + |-x\rangle|+x\rangle - |-x\rangle|-x\rangle) \quad (2.19)$$

$$|10\rangle = \frac{1}{2}(|+x\rangle|+x\rangle + |+x\rangle|-x\rangle - |-x\rangle|+x\rangle - |-x\rangle|-x\rangle) \quad (2.20)$$

$$|11\rangle = \frac{1}{2}(|+x\rangle|+x\rangle - |+x\rangle|-x\rangle - |-x\rangle|+x\rangle + |-x\rangle|-x\rangle) \quad (2.21)$$

**In the YY basis:**

$$|00\rangle = \frac{1}{2}(|+\rangle|+\rangle - i|+\rangle|-\rangle - i|-\rangle|+\rangle - |-\rangle|-\rangle) \quad (2.22)$$

$$|01\rangle = \frac{1}{2}(|+\rangle|+\rangle + i|+\rangle|-\rangle - i|-\rangle|+\rangle + |-\rangle|-\rangle) \quad (2.23)$$

$$|10\rangle = \frac{1}{2}(|+\rangle|+\rangle - i|+\rangle|-\rangle + i|-\rangle|+\rangle + |-\rangle|-\rangle) \quad (2.24)$$

$$|11\rangle = \frac{1}{2}(|+\rangle|+\rangle + i|+\rangle|-\rangle + i|-\rangle|+\rangle - |-\rangle|-\rangle) \quad (2.25)$$

**In mixed bases (e.g., XY basis):**

$$|00\rangle = \frac{1}{2}(|+x\rangle|+y\rangle - i|+x\rangle|-y\rangle + |-x\rangle|+y\rangle - i|-x\rangle|-y\rangle) \quad (2.26)$$

$$|01\rangle = \frac{1}{2}(|+x\rangle|+y\rangle + i|+x\rangle|-y\rangle + |-x\rangle|+y\rangle + i|-x\rangle|-y\rangle) \quad (2.27)$$

$$|10\rangle = \frac{1}{2}(|+x\rangle|+y\rangle - i|+x\rangle|-y\rangle - |-x\rangle|+y\rangle + i|-x\rangle|-y\rangle) \quad (2.28)$$

$$|11\rangle = \frac{1}{2}(|+x\rangle|+y\rangle + i|+x\rangle|-y\rangle - |-x\rangle|+y\rangle - i|-x\rangle|-y\rangle) \quad (2.29)$$

These representations enable us to calculate measurement probabilities in any basis. For example, if we have the state  $|00\rangle$  and measure both qubits in the X basis, we can immediately see that all four outcomes  $(\pm x, \pm x)$  occur with equal probability  $1/4$ .

### Example 1: Cross-Basis Measurement Calculation

Consider the state  $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ . Calculate the probability of measuring  $|+x\rangle|+x\rangle$ . Using the XX basis expansions:

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left[ \frac{1}{2}(|+x\rangle|+x\rangle + |+x\rangle|-x\rangle + |-x\rangle|+x\rangle + |-x\rangle|-x\rangle) \right. \quad (2.30)$$

$$\left. + \frac{1}{2}(|+x\rangle|+x\rangle - |+x\rangle|-x\rangle - |-x\rangle|+x\rangle + |-x\rangle|-x\rangle) \right] \quad (2.31)$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{1}{2} [2|+x\rangle|+x\rangle + 2|-x\rangle|-x\rangle] \quad (2.32)$$

$$= \frac{1}{\sqrt{2}} (|+x\rangle|+x\rangle + |-x\rangle|-x\rangle) \quad (2.33)$$

Therefore,  $P(|\psi\rangle \rightarrow |+x\rangle|+x\rangle) = |(\langle +x|\langle +x|)|\psi\rangle|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$ .

We can verify this using the direct inner product:

$$(\langle +x|\langle +x|)|\psi\rangle = \frac{1}{\sqrt{2}} [(\langle +x|\langle +x|)(|+z\rangle|+z\rangle) + (\langle +x|\langle +x|)(|-z\rangle|-z\rangle)] \quad (2.34)$$

$$= \frac{1}{\sqrt{2}} \left[ \frac{1}{2} + \frac{1}{2} \right] \quad (2.35)$$

$$= \frac{1}{\sqrt{2}} \quad (2.36)$$

## 2.3 Mathematical Structure of Tensor Products

### 2.3.1 Vector Space Properties

The tensor product of two vector spaces creates a new vector space whose dimension equals the product of the individual dimensions. For our two-qubit system, each qubit lives in a two-dimensional complex vector space  $\mathcal{H}_1 \cong \mathcal{H}_2 \cong \mathbb{C}^2$ , so the composite system inhabits a four-dimensional space  $\mathcal{H}_1 \otimes \mathcal{H}_2 \cong \mathbb{C}^4$ .

The tensor product operation satisfies several crucial properties that ensure mathematical consistency. First, it exhibits bilinearity:

$$(a|\psi\rangle + b|\phi\rangle) \otimes |\chi\rangle = a|\psi\rangle \otimes |\chi\rangle + b|\phi\rangle \otimes |\chi\rangle \quad (2.37)$$

$$|\psi\rangle \otimes (c|\chi\rangle + d|\xi\rangle) = c|\psi\rangle \otimes |\chi\rangle + d|\psi\rangle \otimes |\xi\rangle \quad (2.38)$$

This bilinearity is not merely a mathematical convenience—it encodes the principle of quantum superposition for composite systems. If a subsystem can exist in a superposition of states, then the composite system inherits this superposition structure.

Second, the tensor product is associative:

$$(|\psi\rangle \otimes |\phi\rangle) \otimes |\chi\rangle = |\psi\rangle \otimes (|\phi\rangle \otimes |\chi\rangle) \quad (2.39)$$

This property ensures that the order of grouping doesn't matter when combining three or more systems, though we must maintain the sequential ordering of the systems themselves. For three qubits, we can write  $|\psi\rangle_1 \otimes |\phi\rangle_2 \otimes |\chi\rangle_3$  without ambiguity about how to parse this expression.

Importantly, the tensor product is not commutative:

$$|\psi\rangle \otimes |\phi\rangle \neq |\phi\rangle \otimes |\psi\rangle \quad (2.40)$$

The order matters because we're describing distinguishable systems. Swapping the order would mean exchanging which physical system is in which state. This non-commutativity reflects the physical reality that qubit 1 and qubit 2 are different objects, perhaps located at different positions in space or implemented using different physical degrees of freedom.

### 2.3.2 Expanding Superposition States

The bilinearity of tensor products becomes particularly important when dealing with superposition states. Consider the case where the first qubit is in the superposition state  $|+\rangle_1 = \frac{1}{\sqrt{2}}(|0\rangle_1 + |1\rangle_1)$  while the second qubit is in the definite state  $|0\rangle_2$ . The combined state is:

$$|\Psi\rangle = |+\rangle_1 \otimes |0\rangle_2 \quad (2.41)$$

$$= \frac{1}{\sqrt{2}}(|0\rangle_1 + |1\rangle_1) \otimes |0\rangle_2 \quad (2.42)$$

$$= \frac{1}{\sqrt{2}}(|0\rangle_1 \otimes |0\rangle_2 + |1\rangle_1 \otimes |0\rangle_2) \quad (2.43)$$

$$= \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \quad (2.44)$$

This expansion demonstrates how single-qubit superpositions lead to multi-qubit superpositions. The resulting state is a coherent superposition of two computational basis states, with the relative phase preserved from the original single-qubit superposition.

Let's examine a more complex example where both qubits are in superposition:

**Example 2: Both Qubits in Superposition**

Consider  $|\psi\rangle_1 = \frac{1}{\sqrt{2}}(|0\rangle_1 + i|1\rangle_1) = | + y \rangle_1$  and  $|\phi\rangle_2 = \frac{1}{\sqrt{2}}(|0\rangle_2 - |1\rangle_2) = | - x \rangle_2$ .  
The composite state is:

$$|\Psi\rangle = | + y \rangle_1 \otimes | - x \rangle_2 \quad (2.45)$$

$$= \frac{1}{\sqrt{2}}(|0\rangle_1 + i|1\rangle_1) \otimes \frac{1}{\sqrt{2}}(|0\rangle_2 - |1\rangle_2) \quad (2.46)$$

$$= \frac{1}{2}[(|0\rangle_1 + i|1\rangle_1) \otimes (|0\rangle_2 - |1\rangle_2)] \quad (2.47)$$

$$= \frac{1}{2}[|0\rangle_1 \otimes |0\rangle_2 - |0\rangle_1 \otimes |1\rangle_2 + i|1\rangle_1 \otimes |0\rangle_2 - i|1\rangle_1 \otimes |1\rangle_2] \quad (2.48)$$

$$= \frac{1}{2}(|00\rangle - |01\rangle + i|10\rangle - i|11\rangle) \quad (2.49)$$

Note how the phases from both individual superpositions combine in the final expression. The phase structure becomes richer as we combine more complex superpositions.

**2.3.3 Inner Products in Composite Systems**

The inner product in the tensor product space inherits its structure from the inner products of the constituent spaces. For product states, we have:

$$\langle \psi_1 \otimes \phi_2 | \chi_1 \otimes \xi_2 \rangle = \langle \psi_1 | \chi_1 \rangle_1 \langle \phi_2 | \xi_2 \rangle_2 \quad (2.50)$$

This factorization property is fundamental to many calculations in quantum information. It allows us to compute inner products efficiently by handling each subsystem separately.

For general states that are not simple tensor products, we must expand them in a common basis:

**Example 3: Inner Product of General States**

Let  $|\Psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$  and  $|\Phi\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ .  
Calculate  $\langle \Phi | \Psi \rangle$ :

$$\langle \Phi | \Psi \rangle = \frac{1}{\sqrt{2}}(\langle 01 | + \langle 10 |) \cdot \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (2.51)$$

$$= \frac{1}{2}[\langle 01 | 00 \rangle + \langle 01 | 11 \rangle + \langle 10 | 00 \rangle + \langle 10 | 11 \rangle] \quad (2.52)$$

$$= \frac{1}{2}[0 + 0 + 0 + 0] \quad (2.53)$$

$$= 0 \quad (2.54)$$

The states are orthogonal! This is a special property—these particular superpositions form part of an orthonormal basis for the two-qubit space known as the Bell basis.

**2.4 Operations on Composite Systems**



### 2.4.1 Local Operations

A fundamental class of operations on composite systems consists of those that act on only one subsystem while leaving others unchanged. These local operations are represented mathematically by tensoring the desired operation with identity operators on the untouched subsystems.

For a two-qubit system, an operation  $\hat{U}$  acting only on the first qubit is represented as  $\hat{U}_1 = \hat{U} \otimes \hat{I}$ , where  $\hat{I}$  is the  $2 \times 2$  identity matrix. Similarly, an operation acting only on the second qubit is  $\hat{U}_2 = \hat{I} \otimes \hat{U}$ .

The physical interpretation is straightforward: we perform a quantum operation on one qubit while doing nothing to the other. This might correspond to applying a magnetic field to one qubit while shielding the other, or addressing one qubit with a laser pulse in an optical system.

Consider the Pauli X operation (NOT gate) applied to the first qubit:

$$\hat{X}_1 = \hat{X} \otimes \hat{I} \doteq \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \doteq \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad (2.55)$$

This matrix acts on the four-dimensional two-qubit space, flipping the first qubit while leaving the second unchanged. We can verify this by applying it to basis states:

$$\hat{X}_1|00\rangle = |10\rangle \quad (2.56)$$

$$\hat{X}_1|01\rangle = |11\rangle \quad (2.57)$$

$$\hat{X}_1|10\rangle = |00\rangle \quad (2.58)$$

$$\hat{X}_1|11\rangle = |01\rangle \quad (2.59)$$

Let's examine how local operations affect superposition states:

#### Example 4: Local Operation on Superposition

Apply  $\hat{Y}_2$  (Pauli Y on the second qubit) to the state  $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ . First, construct the operator:

$$\hat{Y}_2 = \hat{I} \otimes \hat{Y} \doteq \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \doteq \begin{pmatrix} 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{pmatrix} \quad (2.60)$$

Now apply it:

$$\hat{Y}_2|\psi\rangle = \hat{Y}_2 \cdot \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (2.61)$$

$$= \frac{1}{\sqrt{2}}(\hat{Y}_2|00\rangle + \hat{Y}_2|11\rangle) \quad (2.62)$$

$$= \frac{1}{\sqrt{2}}(i|01\rangle - i|10\rangle) \quad (2.63)$$

$$= \frac{i}{\sqrt{2}}(|01\rangle - |10\rangle) \quad (2.64)$$

The operation has transformed our initial state into a different superposition with a global phase factor of  $i$ .

### 2.4.2 Multi-Qubit Operations

Beyond local operations, quantum mechanics allows operations that act on multiple qubits simultaneously. These operations cannot be decomposed into tensor products of single-qubit operations and represent genuinely multi-qubit phenomena. They are essential for creating correlations between qubits and form the backbone of quantum algorithms.

The controlled-NOT (CNOT) gate serves as the canonical example of a two-qubit operation. This gate flips the state of the second qubit (target) if and only if the first qubit (control) is in state  $|1\rangle$ . Its action on the computational basis is:

$$\text{CNOT}|00\rangle = |00\rangle \quad (2.65)$$

$$\text{CNOT}|01\rangle = |01\rangle \quad (2.66)$$

$$\text{CNOT}|10\rangle = |11\rangle \quad (2.67)$$

$$\text{CNOT}|11\rangle = |10\rangle \quad (2.68)$$

In matrix form:

$$\text{CNOT} \doteq_z \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (2.69)$$

The CNOT gate can be understood as a coherent version of classical conditional logic. In classical computing, we might write “IF (qubit 1 = 1) THEN flip qubit 2.” The quantum version maintains coherence, so superpositions are preserved:

#### Example 5: CNOT on Superposition States

Apply CNOT to  $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$ .

$$\text{CNOT}|\psi\rangle = \text{CNOT} \cdot \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \quad (2.70)$$

$$= \frac{1}{\sqrt{2}}(\text{CNOT}|00\rangle + \text{CNOT}|10\rangle) \quad (2.71)$$

$$= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (2.72)$$

The result is an entangled state! The two qubits are now correlated—measuring the first qubit determines the state of the second. While we defer detailed discussion of entanglement to later chapters, this example shows how multi-qubit gates create quantum correlations.

The CNOT gate plays a fundamental role in quantum computation, serving as a basic building block for more complex operations. When combined with single-qubit gates, it forms a universal gate set capable of implementing any quantum computation.

Another important two-qubit gate is the controlled-Z (CZ) gate:

$$\text{CZ} \doteq_z \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad (2.73)$$

This gate applies a phase of -1 if and only if both qubits are in state  $|1\rangle$ . Unlike CNOT, the CZ gate is symmetric with respect to its two inputs—there’s no distinction between control and target.

### 2.4.3 Measurement in Composite Systems

Measurements on composite systems can be either local (measuring individual qubits) or global (measuring multi-qubit observables). Local measurements are the most common in practice and have particularly simple descriptions.

When we measure just the first qubit of a two-qubit system in the computational basis, we’re asking: “Is the first qubit in state  $|0\rangle$  or  $|1\rangle$ ?” The measurement operators are:

$$\hat{P}_0^{(1)} = |0\rangle\langle 0| \otimes \hat{I} \doteq_z \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (2.74)$$

$$\hat{P}_1^{(1)} = |1\rangle\langle 1| \otimes \hat{I} \doteq_z \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (2.75)$$

These projectors satisfy  $\hat{P}_0^{(1)} + \hat{P}_1^{(1)} = \hat{I}$  and  $\hat{P}_0^{(1)}\hat{P}_1^{(1)} = 0$ , as required for a complete measurement.

#### Example 6: Partial Measurement

Consider measuring the first qubit of  $|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ . The probability of finding the first qubit in state  $|0\rangle$  is:

$$P(|\psi\rangle \rightarrow |0\rangle_1) = \langle\psi|\hat{P}_0^{(1)}|\psi\rangle \quad (2.76)$$

$$= \left| \frac{1}{2}(|00\rangle + |01\rangle) \right|^2 \quad (2.77)$$

$$= \frac{1}{4}(2)^2 = \frac{1}{2} \quad (2.78)$$

If we obtain outcome 0, the post-measurement state is:

$$|\psi'\rangle = \frac{\hat{P}_0^{(1)}|\psi\rangle}{\sqrt{P(|\psi\rangle \rightarrow |0\rangle_1)}} \quad (2.79)$$

$$= \frac{\frac{1}{2}(|00\rangle + |01\rangle)}{\sqrt{1/2}} \quad (2.80)$$

$$= \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle) \quad (2.81)$$

$$= |0\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad (2.82)$$

The first qubit is now definitely in state  $|0\rangle$ , while the second qubit remains in superposition.

## 2.5 Scaling to Many Qubits

### 2.5.1 The General Multi-Qubit Framework

The tensor product structure extends naturally to systems with more than two qubits. For  $n$  qubits, the computational basis consists of  $2^n$  states, each specified by a binary string of length  $n$ . The basis states are written as  $|i_1 i_2 \dots i_n\rangle$  where each  $i_j \in \{0, 1\}$ .

The dimension of the Hilbert space grows exponentially with the number of qubits:

- 1 qubit: 2 dimensions
- 2 qubits: 4 dimensions
- 3 qubits: 8 dimensions
- 10 qubits: 1,024 dimensions
- 20 qubits: 1,048,576 dimensions
- 50 qubits:  $\approx 10^{15}$  dimensions

This exponential scaling has profound implications. On one hand, it provides the computational power that makes quantum algorithms potentially more powerful than classical ones. On the other hand, it makes the classical simulation of quantum systems extremely challenging, as the amount of information needed to specify a general  $n$ -qubit state grows exponentially.

For three qubits, the eight computational basis states are:

$$|000\rangle, |001\rangle, |010\rangle, |011\rangle, \quad (2.83)$$

$$|100\rangle, |101\rangle, |110\rangle, |111\rangle \quad (2.84)$$

A general three-qubit state requires eight complex amplitudes (subject to normalization), meaning 15 real parameters are needed to fully specify the state:

$$|\psi\rangle = \sum_{i,j,k \in \{0,1\}} \alpha_{ijk} |ijk\rangle \quad (2.85)$$

where  $\sum_{i,j,k} |\alpha_{ijk}|^2 = 1$ .

Let's explore a specific three-qubit state:

#### Example 7: Three-Qubit GHZ State

Consider the three-qubit state  $|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ .

This state exhibits three-party correlations. Let's calculate the probability of various measurement outcomes:

**Measuring all three qubits:**

$$P(|\text{GHZ}\rangle \rightarrow |000\rangle) = |\langle 000|\text{GHZ}\rangle|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad (2.86)$$

$$P(|\text{GHZ}\rangle \rightarrow |111\rangle) = |\langle 111|\text{GHZ}\rangle|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad (2.87)$$

$$P(|\text{GHZ}\rangle \rightarrow |\text{any other}\rangle) = 0 \quad (2.88)$$

**Measuring just the first qubit:**

$$P(|\text{GHZ}\rangle \rightarrow |0\rangle_1) = |\langle 0 * *|\text{GHZ}\rangle|^2 = \left| \frac{1}{\sqrt{2}} |000\rangle \right|^2 = \frac{1}{2} \quad (2.89)$$

$$P(|\text{GHZ}\rangle \rightarrow |1\rangle_1) = |\langle 1 * *|\text{GHZ}\rangle|^2 = \left| \frac{1}{\sqrt{2}} |111\rangle \right|^2 = \frac{1}{2} \quad (2.90)$$

If we measure the first qubit and get 0, the state collapses to  $|000\rangle$ . If we get 1, it collapses to  $|111\rangle$ . This demonstrates strong correlations among all three qubits.

### 2.5.2 Operations on Selected Qubits

In multi-qubit systems, we often need to apply operations to specific qubits while leaving others untouched. For an  $n$ -qubit system, applying a single-qubit operation  $\hat{U}$  to the  $k$ -th qubit is represented as:

$$\hat{U}^{(k)} = \hat{I} \otimes \cdots \otimes \hat{I} \otimes \underbrace{\hat{U}}_{k\text{-th position}} \otimes \hat{I} \otimes \cdots \otimes \hat{I} \quad (2.91)$$

This notation, while cumbersome, precisely specifies which qubit is affected. The operation acts as the identity on all qubits except the  $k$ -th, where it applies the transformation  $\hat{U}$ .

For example, in a three-qubit system, applying a Hadamard gate to the middle qubit:

$$\hat{H}^{(2)} = \hat{I} \otimes \hat{H} \otimes \hat{I} \quad (2.92)$$

where  $\hat{H} \doteq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  is the Hadamard gate.

Multi-qubit gates can similarly be applied to specific subsets of qubits. For instance, a CNOT gate with qubit  $j$  as control and qubit  $k$  as target affects only those two qubits while acting as the identity on all others. The notation  $\text{CNOT}_{j,k}$  specifies which qubits are involved.

**Example 8: Circuit on Three Qubits**

Starting from  $|000\rangle$ , apply the following sequence: 1.  $\hat{H}^{(1)}$  (Hadamard on first qubit) 2.  $\text{CNOT}_{1,2}$  (CNOT from qubit 1 to qubit 2) 3.  $\text{CNOT}_{2,3}$  (CNOT from qubit 2 to qubit 3)  
Step 1:

$$\hat{H}^{(1)}|000\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |00\rangle \quad (2.93)$$

$$= \frac{1}{\sqrt{2}}(|000\rangle + |100\rangle) \quad (2.94)$$

Step 2:

$$\text{CNOT}_{1,2} \frac{1}{\sqrt{2}}(|000\rangle + |100\rangle) = \frac{1}{\sqrt{2}}(|000\rangle + |110\rangle) \quad (2.95)$$

Step 3:

$$\text{CNOT}_{2,3} \frac{1}{\sqrt{2}}(|000\rangle + |110\rangle) = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle) \quad (2.96)$$

We've created the GHZ state! This circuit demonstrates how local operations and two-qubit gates can create complex multi-qubit states.

**2.5.3 The Challenge of Classical Simulation**

The exponential growth of Hilbert space dimension presents both opportunities and challenges. For quantum computation, this exponential scaling potentially allows quantum computers to explore vast solution spaces efficiently. However, it also means that classical simulation of quantum systems quickly becomes intractable.

Consider the memory requirements for storing a general quantum state:

- 10 qubits:  $2^{10} = 1,024$  complex numbers = 16 KB (assuming double precision)
- 20 qubits:  $2^{20} \approx 10^6$  complex numbers = 16 MB
- 30 qubits:  $2^{30} \approx 10^9$  complex numbers = 16 GB
- 40 qubits:  $2^{40} \approx 10^{12}$  complex numbers = 16 TB
- 50 qubits:  $2^{50} \approx 10^{15}$  complex numbers = 16 PB

Beyond memory, the computational cost of simulating quantum operations also scales exponentially. Applying a general  $n$ -qubit gate requires  $O(2^n)$  operations. This exponential barrier motivates the development of actual quantum computers, which naturally handle this complexity.

However, not all quantum states require exponential resources to describe. States with special structure, such as product states or states with limited entanglement, can often be represented more efficiently. This observation underlies various classical simulation techniques for restricted classes of quantum computations.

## 2.6 Basis Transformations in Composite Systems

### 2.6.1 Local Basis Changes

Understanding how basis transformations work in composite systems is crucial for quantum information processing. If we know how to transform between bases for individual qubits, we can determine the transformation for the composite system using the tensor product structure.

Consider expressing the computational basis state  $|00\rangle$  in terms of the X-basis states  $|\pm x\rangle$ . Since  $|0\rangle = \frac{1}{\sqrt{2}}(|+x\rangle + |-x\rangle)$ , we have:

$$|00\rangle = |0\rangle_1 \otimes |0\rangle_2 \quad (2.97)$$

$$= \frac{1}{\sqrt{2}}(|+x\rangle_1 + |-x\rangle_1) \otimes \frac{1}{\sqrt{2}}(|+x\rangle_2 + |-x\rangle_2) \quad (2.98)$$

$$= \frac{1}{2}(|+x\rangle|+x\rangle + |+x\rangle|-x\rangle + |-x\rangle|+x\rangle + |-x\rangle|-x\rangle) \quad (2.99)$$

This expansion shows that measuring both qubits in the X-basis would yield each of the four possible outcomes with equal probability  $1/4$ .

Let's work through a more complex basis transformation:

#### Example 9: Mixed Basis Measurements

Express  $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle - i|11\rangle)$  in the basis where the first qubit is measured in the X-basis and the second in the Y-basis.

Using the single-qubit transformations from Chapter 1:

$$|0\rangle = \frac{1}{\sqrt{2}}(|+x\rangle + |-x\rangle) \quad (2.100)$$

$$|1\rangle = \frac{1}{\sqrt{2}}(|+x\rangle - |-x\rangle) \quad (2.101)$$

$$|0\rangle = \frac{1}{\sqrt{2}}(|+y\rangle + |-y\rangle) \quad (2.102)$$

$$|1\rangle = \frac{1}{\sqrt{2}}(-i|+y\rangle + i|-y\rangle) \quad (2.103)$$

Therefore:

$$|00\rangle = \frac{1}{2}(|+x\rangle + |-x\rangle)(|+y\rangle + |-y\rangle) \quad (2.104)$$

$$= \frac{1}{2}(|+x\rangle|+y\rangle + |+x\rangle|-y\rangle + |-x\rangle|+y\rangle + |-x\rangle|-y\rangle) \quad (2.105)$$

$$|11\rangle = \frac{1}{2}(|+x\rangle - |-x\rangle)(-i|+y\rangle + i|-y\rangle) \quad (2.106)$$

$$= \frac{1}{2}(-i|+x\rangle|+y\rangle + i|+x\rangle|-y\rangle + i|-x\rangle|+y\rangle - i|-x\rangle|-y\rangle) \quad (2.107)$$

Substituting into  $|\psi\rangle$ :

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left[ \frac{1}{2}(|+x\rangle|+y\rangle + |+x\rangle|-y\rangle + |-x\rangle|+y\rangle + |-x\rangle|-y\rangle) \right. \quad (2.108)$$

$$\left. -i \cdot \frac{1}{2}(-i|+x\rangle|+y\rangle + i|+x\rangle|-y\rangle + i|-x\rangle|+y\rangle - i|-x\rangle|-y\rangle) \right] \quad (2.109)$$

$$= \frac{1}{2\sqrt{2}}[(1-1)|+x\rangle|+y\rangle + (1+1)|+x\rangle|-y\rangle + (1+1)|-x\rangle|+y\rangle + (1-1)|-x\rangle|-y\rangle] \quad (2.110)$$

$$= \frac{1}{\sqrt{2}}(|+x\rangle|-y\rangle + |-x\rangle|+y\rangle) \quad (2.111)$$

So measuring in the XY basis yields either  $(+x, -y)$  or  $(-x, +y)$ , each with probability  $1/2$ .

### 2.6.2 The Role of Phase in Multi-Qubit Systems

Phase relationships become increasingly important in multi-qubit systems. Consider the difference between the states:

$$|\psi_1\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \quad (2.112)$$

$$|\psi_2\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle) \quad (2.113)$$

While both give identical probabilities for computational basis measurements (each outcome has probability  $1/4$ ), they behave differently under other measurements or further operations.

To see the difference, let's measure both qubits in the X-basis:

#### Example 10: Phase Effects in Measurement

For  $|\psi_1\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ :

First, note that this can be factored:

$$|\psi_1\rangle = \frac{1}{2}(|0\rangle + |1\rangle) \otimes (|0\rangle + |1\rangle) \quad (2.114)$$

$$= |+x\rangle \otimes |+x\rangle \quad (2.115)$$

So measuring in the X-basis gives  $(+x, +x)$  with certainty.

For  $|\psi_2\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$ :

This cannot be factored as a product state. Using the X-basis expansions:

$$|\psi_2\rangle = \frac{1}{2}[|00\rangle + |01\rangle + |10\rangle - |11\rangle] \quad (2.116)$$



Working through the basis transformation (as in our table), we find:

$$P(|\psi_2\rangle \rightarrow | + x, +x\rangle) = \left| \frac{1}{2}(1 + 1 + 1 - 1) \right|^2 = \frac{1}{2} \quad (2.117)$$

$$P(|\psi_2\rangle \rightarrow | + x, -x\rangle) = \left| \frac{1}{2}(1 - 1 + 1 + 1) \right|^2 = \frac{1}{2} \quad (2.118)$$

$$P(|\psi_2\rangle \rightarrow | - x, +x\rangle) = P(|\psi_2\rangle \rightarrow | - x, -x\rangle) = 0 \quad (2.119)$$

The phase difference dramatically changes the measurement statistics!

These phase relationships are not merely mathematical artifacts but have observable consequences. They determine interference patterns in quantum algorithms and affect the outcomes of measurements in rotated bases. Understanding and controlling these phases is crucial for quantum computation and quantum information processing.

## 2.7 The Reduced Density Matrix and Partial Trace

### 2.7.1 Describing Subsystems of Composite States

#### Key Idea: From Pure to Mixed: The Cost of Ignoring Correlations

When we have access to only part of an entangled system, we lose quantum coherence. The reduced density matrix captures this transition from pure quantum states to mixed statistical states, revealing how quantum information is stored non-locally in correlations.

So far, we have described composite quantum systems using state vectors in the full tensor product space. But what if we only have access to one part of a composite system? How do we describe what we can observe about qubit A when qubit B is inaccessible? The answer involves the reduced density matrix, obtained through an operation called the partial trace.

### 2.7.2 The Density Matrix Formalism

Before introducing the reduced density matrix, we briefly review the density matrix representation. Any quantum state  $|\psi\rangle$  can be represented by its density matrix:

$$\rho = |\psi\rangle\langle\psi| \quad (2.120)$$

For a pure state, the density matrix satisfies  $\rho^2 = \rho$  and  $\text{Tr}(\rho^2) = 1$ . The density matrix formalism naturally extends to mixed states (statistical ensembles of pure states), where  $\text{Tr}(\rho^2) < 1$ .

### 2.7.3 The Partial Trace Operation

For a two-qubit system in state  $|\psi\rangle_{AB}$ , the reduced density matrix for qubit A is obtained by "tracing out" qubit B:

$$\rho_A = \text{Tr}_B(|\psi\rangle\langle\psi|_{AB}) = \sum_j (\mathbb{I}_A \otimes \langle j|_B) |\psi\rangle\langle\psi|_{AB} (\mathbb{I}_A \otimes |j\rangle_B) \quad (2.121)$$

where  $\{|j\rangle_B\}$  is any orthonormal basis for qubit B's Hilbert space.

#### Mathematical Aside: Matrix Construction of Projection Operators

To understand how the partial trace works in matrix form, let's explicitly construct the projection operators. For a two-qubit system with a  $4 \times 4$  density matrix, we need:

**The operators  $\mathbb{I}_A \otimes |j\rangle_B$  ( $4 \times 2$  matrices):**

For  $j = 0$ :

$$\mathbb{I}_A \otimes |0\rangle_B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (2.122)$$

$$= \begin{pmatrix} 1 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} & 0 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 0 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} & 1 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \quad (2.123)$$

For  $j = 1$ :

$$\mathbb{I}_A \otimes |1\rangle_B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (2.124)$$

$$= \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} \quad (2.125)$$

**The operators  $\mathbb{I}_A \otimes \langle j|_B$  ( $2 \times 4$  matrices):**

For  $j = 0$ :

$$\mathbb{I}_A \otimes \langle 0|_B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes (1 \ 0) \quad (2.126)$$

$$= \begin{pmatrix} 1 \cdot (1 \ 0) & 0 \cdot (1 \ 0) \\ 0 \cdot (1 \ 0) & 1 \cdot (1 \ 0) \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (2.127)$$

For  $j = 1$ :

$$\mathbb{I}_A \otimes \langle 1|_B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes (0 \ 1) \quad (2.128)$$

$$= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (2.129)$$

**The complete operation:** The partial trace performs the matrix multiplication:

$$(\mathbb{I}_A \otimes \langle j|_B) \cdot \rho_{AB} \cdot (\mathbb{I}_A \otimes |j\rangle_B) = (2 \times 4) \cdot (4 \times 4) \cdot (4 \times 2) = (2 \times 2)$$

This sequence of multiplications selects and extracts a  $2 \times 2$  submatrix from the original  $4 \times 4$  density matrix, corresponding to the reduced state of qubit A when B is in state  $|j\rangle$ . Summing over all values of  $j$  gives the final reduced density matrix.

**Verification:** Let's verify this works with our example density matrix. For  $j = 0$ :

$$(\mathbb{I}_A \otimes \langle 0|_B) \cdot \rho_{AB} \cdot (\mathbb{I}_A \otimes |0\rangle_B) \quad (2.130)$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0.4 & 0 & 0 & 0.2 \\ 0 & 0.1 & 0.05 & 0 \\ 0 & 0.05 & 0.1 & 0 \\ 0.2 & 0 & 0 & 0.4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \quad (2.131)$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0.4 & 0 \\ 0 & 0 \\ 0 & 0.1 \\ 0.2 & 0 \end{pmatrix} \quad (2.132)$$

$$= \begin{pmatrix} 0.4 & 0 \\ 0 & 0.1 \end{pmatrix} \quad (2.133)$$

This correctly extracts the elements at positions (0,0), (0,2), (2,0), and (2,2) from  $\rho_{AB}$ , arranging them into a  $2 \times 2$  matrix!

**Key insight:** The projection operators  $(\mathbb{I}_A \otimes \langle j|_B)$  and  $(\mathbb{I}_A \otimes |j\rangle_B)$  act as "selection matrices" that extract the relevant  $2 \times 2$  block from the  $4 \times 4$  density matrix. The dimensions work out perfectly:  $(2 \times 4) \cdot (4 \times 4) \cdot (4 \times 2) = (2 \times 2)$ , reducing the composite system's density matrix to the subsystem's density matrix.

**Physical Insight****What "Tracing Out" Physically Means**

Tracing out is NOT measurement! It represents situations where:

- One qubit is spatially inaccessible (e.g., sent to a distant location)
- One qubit is lost to the environment (decoherence)
- Experimental apparatus can only measure one qubit
- We choose to ignore one qubit and ask what we can observe about the other

The reduced density matrix answers: "What measurement statistics can I obtain from qubit A alone, without any access to qubit B?"

**2.7.4 Computing Partial Traces: A Detailed Example**

Let's work through a concrete example that demonstrates the partial trace operation. Consider a two-qubit system with density matrix:

$$\rho_{AB} = \begin{pmatrix} 0.4 & 0 & 0 & 0.2 \\ 0 & 0.1 & 0.05 & 0 \\ 0 & 0.05 & 0.1 & 0 \\ 0.2 & 0 & 0 & 0.4 \end{pmatrix} \quad (2.134)$$

This matrix has several interesting features:

- It is Hermitian:  $\rho_{AB} = \rho_{AB}^\dagger$
- It has unit trace:  $\text{Tr}(\rho_{AB}) = 0.4 + 0.1 + 0.1 + 0.4 = 1$
- Off-diagonal elements at (0,3) and (3,0) indicate quantum correlations between  $|00\rangle$  and  $|11\rangle$
- Off-diagonal elements at (1,2) and (2,1) indicate quantum correlations between  $|01\rangle$  and  $|10\rangle$

**Example 11: Tracing Over Qubit B**

The partial trace over qubit  $B$  is:

$$\rho_A = \text{Tr}_B(\rho_{AB}) = \sum_{j \in \{0,1\}} (\mathbb{I}_A \otimes \langle j|_B) \rho_{AB} (\mathbb{I}_A \otimes |j\rangle_B) \quad (2.135)$$

(See the Mathematical Aside above for the explicit matrix construction of these projection operators.)

**Step 1: Select elements for each value of  $j$** 

For  $j = 0$  (selecting  $|0\rangle_B$ ): We extract the submatrix corresponding to basis states where  $B$  is in state  $|0\rangle$  (i.e.,  $|00\rangle$  and  $|10\rangle$ , which are indices 0 and 2):

$$(\mathbb{I}_A \otimes \langle 0|_B) \rho_{AB} (\mathbb{I}_A \otimes |0\rangle_B) = \begin{pmatrix} \rho_{00,00} & \rho_{00,10} \\ \rho_{10,00} & \rho_{10,10} \end{pmatrix} = \begin{pmatrix} 0.4 & 0 \\ 0 & 0.1 \end{pmatrix}$$

For  $j = 1$  (selecting  $|1\rangle_B$ ): We extract the submatrix corresponding to basis states where  $B$

is in state  $|1\rangle$  (i.e.,  $|01\rangle$  and  $|11\rangle$ , which are indices 1 and 3):

$$(\mathbb{I}_A \otimes \langle 1|_B) \rho_{AB} (\mathbb{I}_A \otimes |1\rangle_B) = \begin{pmatrix} \rho_{01,01} & \rho_{01,11} \\ \rho_{11,01} & \rho_{11,11} \end{pmatrix} = \begin{pmatrix} 0.1 & 0 \\ 0 & 0.4 \end{pmatrix}$$

**Step 2: Sum the contributions**

$$\rho_A = \begin{pmatrix} 0.4 & 0 \\ 0 & 0.1 \end{pmatrix} + \begin{pmatrix} 0.1 & 0 \\ 0 & 0.4 \end{pmatrix} = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix} = \frac{1}{2} \mathbb{I}$$

### Example 12: Tracing Over Qubit A

Similarly, the partial trace over qubit  $A$  is:

$$\rho_B = \text{Tr}_A(\rho_{AB}) = \sum_{i \in \{0,1\}} (\langle i|_A \otimes \mathbb{I}_B) \rho_{AB} (|i\rangle_A \otimes \mathbb{I}_B) \quad (2.136)$$

**Step 1: Identify the block structure**

When tracing over  $A$ , we can write  $\rho_{AB}$  in block form:

$$\rho_{AB} = \begin{pmatrix} 0.4 & 0 & 0 & 0.2 \\ 0 & 0.1 & 0.05 & 0 \\ 0 & 0.05 & 0.1 & 0 \\ 0.2 & 0 & 0 & 0.4 \end{pmatrix}$$

The upper-left block corresponds to  $i = 0$  ( $|0\rangle_A$  subspace), and the lower-right block corresponds to  $i = 1$  ( $|1\rangle_A$  subspace).

**Step 2: Extract and sum the diagonal blocks**

For  $i = 0$ : Upper-left  $2 \times 2$  block =  $\begin{pmatrix} 0.4 & 0 \\ 0 & 0.1 \end{pmatrix}$

For  $i = 1$ : Lower-right  $2 \times 2$  block =  $\begin{pmatrix} 0.1 & 0 \\ 0 & 0.4 \end{pmatrix}$

**Step 3: Sum the contributions**

$$\rho_B = \begin{pmatrix} 0.4 & 0 \\ 0 & 0.1 \end{pmatrix} + \begin{pmatrix} 0.1 & 0 \\ 0 & 0.4 \end{pmatrix} = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix} = \frac{1}{2} \mathbb{I}$$

### Mathematical Aside: Index Notation Method

We can also compute partial traces using index notation. For a density matrix with elements  $\rho_{ij,kl}$  where  $(i, j) \in \{0, 1\}$  label qubit  $A$  and  $(k, l) \in \{0, 1\}$  label qubit  $B$ :

**Tracing over B:**

$$(\rho_A)_{ij} = \sum_{k \in \{0,1\}} \rho_{ik,jk}$$

**Tracing over A:**

$$(\rho_B)_{kl} = \sum_{i \in \{0,1\}} \rho_{ik,il}$$

For our specific matrix, the mapping between basis states and matrix indices is:

$$|00\rangle \leftrightarrow \text{index } 0 \quad (2.137)$$

$$|01\rangle \leftrightarrow \text{index } 1 \quad (2.138)$$

$$|10\rangle \leftrightarrow \text{index } 2 \quad (2.139)$$

$$|11\rangle \leftrightarrow \text{index } 3 \quad (2.140)$$

Using this method for  $\rho_B$ :

- $(\rho_B)_{00} = \rho_{00,00} + \rho_{10,10} = \rho[0,0] + \rho[2,2] = 0.4 + 0.1 = 0.5$
- $(\rho_B)_{01} = \rho_{00,01} + \rho_{10,11} = \rho[0,1] + \rho[2,3] = 0 + 0 = 0$
- $(\rho_B)_{10} = \rho_{01,00} + \rho_{11,10} = \rho[1,0] + \rho[3,2] = 0 + 0 = 0$
- $(\rho_B)_{11} = \rho_{01,01} + \rho_{11,11} = \rho[1,1] + \rho[3,3] = 0.1 + 0.4 = 0.5$

### 2.7.5 The Block Matrix Method

The block matrix method provides an intuitive way to compute partial traces by organizing the density matrix according to the slow and fast indices.

#### Mathematical Aside: Block Matrix Structure and Index Conventions

**Index convention:** For the basis ordering  $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ :

- A is the *slow index* (leftmost bit) - changes every 2 elements
- B is the *fast index* (rightmost bit) - alternates with each element

**Block organization:** We can write any  $4 \times 4$  density matrix in  $2 \times 2$  blocks organized by the A index:

$$\rho_{AB} = \begin{pmatrix} \rho^{00} & \rho^{01} \\ \rho^{10} & \rho^{11} \end{pmatrix}$$

where each  $\rho^{ij}$  is a  $2 \times 2$  matrix:

- $\rho^{00}$  contains elements for  $A=0 \rightarrow A=0$  transitions, B varies (rows/cols 0,1)
- $\rho^{01}$  contains elements for  $A=0 \rightarrow A=1$  transitions, B varies (rows 0,1; cols 2,3)
- $\rho^{10}$  contains elements for  $A=1 \rightarrow A=0$  transitions, B varies (rows 2,3; cols 0,1)
- $\rho^{11}$  contains elements for  $A=1 \rightarrow A=1$  transitions, B varies (rows/cols 2,3)

**The simple rules:**

- **To trace out B (keep A):**  $\rho_A = \begin{pmatrix} \text{Tr}(\rho^{00}) & \text{Tr}(\rho^{01}) \\ \text{Tr}(\rho^{10}) & \text{Tr}(\rho^{11}) \end{pmatrix}$

Take the trace of each  $2 \times 2$  block (summing over B within each A sector)

- **To trace out A (keep B):**  $\rho_B = \rho^{00} + \rho^{11}$

Sum the diagonal blocks (summing over A values while preserving B structure)

**Intuitive explanation:**

- To trace out the fast index (B): Take the trace within each block—this sums over B values while preserving the A structure
- To trace out the slow index (A): Sum the diagonal blocks—this sums over A values while preserving the B structure
- The pattern matches the index structure: trace out fast  $\rightarrow$  trace within blocks; trace out slow  $\rightarrow$  sum across blocks

Let's apply this method to our example density matrix:

**Example 13: Block Matrix Method Application**

Using the same density matrix from Examples 2.12 and 2.13:

$$\rho_{AB} = \begin{pmatrix} 0.4 & 0 & 0 & 0.2 \\ 0 & 0.1 & 0.05 & 0 \\ 0 & 0.05 & 0.1 & 0 \\ 0.2 & 0 & 0 & 0.4 \end{pmatrix}$$

**Step 1: Identify the  $2 \times 2$  blocks**

$$\rho_{AB} = \begin{pmatrix} \rho^{00} & \rho^{01} \\ \rho^{10} & \rho^{11} \end{pmatrix} = \begin{pmatrix} \begin{pmatrix} 0.4 & 0 \\ 0 & 0.1 \end{pmatrix} & \begin{pmatrix} 0 & 0.2 \\ 0.05 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 0.05 \\ 0.2 & 0 \end{pmatrix} & \begin{pmatrix} 0.1 & 0 \\ 0 & 0.4 \end{pmatrix} \end{pmatrix}$$

**Step 2: Trace out B (keep A)**

Take the trace of each block:

$$\text{Tr}(\rho^{00}) = 0.4 + 0.1 = 0.5 \quad (2.141)$$

$$\text{Tr}(\rho^{01}) = 0 + 0 = 0 \quad (2.142)$$

$$\text{Tr}(\rho^{10}) = 0 + 0 = 0 \quad (2.143)$$

$$\text{Tr}(\rho^{11}) = 0.1 + 0.4 = 0.5 \quad (2.144)$$

Therefore:

$$\rho_A = \begin{pmatrix} \text{Tr}(\rho^{00}) & \text{Tr}(\rho^{01}) \\ \text{Tr}(\rho^{10}) & \text{Tr}(\rho^{11}) \end{pmatrix} = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix} = \frac{1}{2} \mathbb{I}$$

**Step 3: Trace out A (keep B)**

Sum the diagonal blocks:

$$\rho_B = \rho^{00} + \rho^{11} = \begin{pmatrix} 0.4 & 0 \\ 0 & 0.1 \end{pmatrix} + \begin{pmatrix} 0.1 & 0 \\ 0 & 0.4 \end{pmatrix} = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix} = \frac{1}{2}\mathbb{I}$$

Both methods yield the same maximally mixed state, as expected!

### Physical Insight

#### Why the Block Matrix Method Works

The block matrix method reflects the hierarchical structure of the tensor product basis:

- The A index determines which 2×2 block we're in
- The B index determines position within each block
- Tracing means summing over the traced index while keeping the other fixed
- For the fast index (B): sum within blocks (trace operation)
- For the slow index (A): sum across blocks (matrix addition)

This pattern generalizes: for  $n$  qubits, organize the matrix hierarchically and trace/sum at the appropriate level!

### Example 14: Visual Method for Partial Trace

Let's visualize the partial trace operation using the same density matrix, highlighting which elements contribute to each reduced density matrix.

$$\rho_{AB} = \begin{pmatrix} 0.4 & 0 & 0 & 0.2 \\ 0 & 0.1 & 0.05 & 0 \\ 0 & 0.05 & 0.1 & 0 \\ 0.2 & 0 & 0 & 0.4 \end{pmatrix}$$

#### Tracing over B (keeping A):

We can visualize which elements contribute by coloring them:

$$\rho_{AB} = \begin{pmatrix} \text{0.4} & 0 & \text{0} & 0.2 \\ 0 & \text{0.1} & 0.05 & \text{0} \\ \text{0} & 0.05 & \text{0.1} & 0 \\ 0.2 & \text{0} & 0 & \text{0.4} \end{pmatrix}$$

- **Red elements** ( $j = 0$ ): Elements where B is in state  $|0\rangle$  - from basis states  $|00\rangle$  and  $|10\rangle$  (matrix positions 0 and 2)
- **Blue elements** ( $j = 1$ ): Elements where B is in state  $|1\rangle$  - from basis states  $|01\rangle$  and  $|11\rangle$  (matrix positions 1 and 3)



The partial trace collects these elements:

$$\text{For } j = 0 : \begin{pmatrix} \text{0.4} & \text{0} \\ \text{0} & \text{0.1} \end{pmatrix} \quad (2.145)$$

$$\text{For } j = 1 : \begin{pmatrix} \text{0.1} & \text{0} \\ \text{0} & \text{0.4} \end{pmatrix} \quad (2.146)$$

Summing:  $\rho_A = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix}$

**Tracing over A (keeping B):**

Alternatively, we can view the matrix in  $2 \times 2$  blocks:

$$\rho_{AB} = \begin{pmatrix} \begin{pmatrix} \text{0.4} & \text{0} \\ \text{0} & \text{0.1} \end{pmatrix} & \begin{pmatrix} 0 & 0.2 \\ 0.05 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 0.05 \\ 0.2 & 0 \end{pmatrix} & \begin{pmatrix} \text{0.1} & \text{0} \\ \text{0} & \text{0.4} \end{pmatrix} \end{pmatrix}$$

- **Red block**: Upper-left block for  $i = 0$  ( $|0\rangle_A$  subspace)
- **Blue block**: Lower-right block for  $i = 1$  ( $|1\rangle_A$  subspace)

The partial trace sums the diagonal blocks:

$$\rho_B = \begin{pmatrix} \text{0.4} & \text{0} \\ \text{0} & \text{0.1} \end{pmatrix} + \begin{pmatrix} \text{0.1} & \text{0} \\ \text{0} & \text{0.4} \end{pmatrix} = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix}$$

**Key Insight:** The visual method shows that tracing over B selects matrix elements based on B's state value, while tracing over A sums diagonal blocks. Both methods yield the same maximally mixed state, confirming that the quantum correlations are hidden when we look at either subsystem alone.

## 2.8 Physical Interpretation and Implementation

### 2.8.1 Spatial Arrangement of Qubits

The tensor product structure naturally accommodates the physical reality that qubits exist at different spatial locations. When we write  $|\psi\rangle_1 \otimes |\phi\rangle_2$ , the subscripts can be interpreted as labeling physical locations in space. This spatial interpretation becomes particularly relevant when considering the implementation of quantum gates.

In most physical implementations of quantum computers, not all pairs of qubits can directly interact. The connectivity constraints depend on the physical platform:

**Superconducting qubits:** Typically arranged in 2D lattices with nearest-neighbor connectivity. Each qubit can directly interact only with 2-4 neighboring qubits.

**Trapped ions:** All-to-all connectivity is possible, but gates between distant ions may be slower or less accurate.

**Quantum dots:** Often arranged in linear or 2D arrays with nearest-neighbor interactions.

**Photonic systems:** Connectivity determined by the optical network topology.

These connectivity constraints influence quantum algorithm design and the compilation of quantum circuits. Operations between non-adjacent qubits must be decomposed into sequences of allowed gates, potentially introducing additional errors.

#### Example 15: SWAP Networks

Consider a linear array of 4 qubits where only nearest-neighbor gates are allowed. To apply CNOT between qubits 1 and 4:

- (a) SWAP qubits 1 and 2
- (b) SWAP qubits 2 and 3 (qubit 1's state is now at position 3)
- (c) Apply CNOT between positions 3 and 4
- (d) SWAP qubits 2 and 3
- (e) SWAP qubits 1 and 2 (returning qubit 1's state to its original position)

This requires 3 SWAP gates (each decomposable into 3 CNOTs) plus 1 CNOT, totaling 10 two-qubit gates instead of just 1.

### 2.8.2 The Bridge to Many-Body Physics

The connection to spatial arrangement also provides a bridge to condensed matter physics. An array of qubits at fixed positions resembles a lattice model, where quantum particles can occupy discrete sites. This correspondence, explored further in Chapter 3, reveals deep connections between quantum information and many-body physics.

Consider the mapping:

- Qubit at site  $i$  in state  $|0\rangle$ : No particle at site  $i$
- Qubit at site  $i$  in state  $|1\rangle$ : One particle at site  $i$

Under this mapping, the state  $|01001\rangle$  represents particles at sites 2 and 5 of a 5-site lattice. The CNOT gate becomes related to particle hopping, and multi-qubit states describe many-body quantum systems.

This connection has proven fruitful in both directions. Quantum information concepts illuminate many-body physics, while condensed matter insights guide quantum algorithm development. The tensor product structure underlies both fields, emphasizing its fundamental importance in quantum mechanics.

## 2.9 Chapter Summary

This chapter has established the mathematical framework for describing composite quantum systems through the tensor product structure. We have seen how individual quantum systems combine to form larger systems, with the dimension of the composite Hilbert space growing exponentially with the number of constituents.

The key concepts developed include:

- The tensor product operation for states and operators, with complete basis representations
- Local and multi-qubit operations, including the fundamental CNOT gate
- Measurement theory for composite systems, including partial measurements
- Basis transformations using the tensor product structure
- The exponential scaling challenge and its implications
- The reduced density matrix and partial trace operation for describing subsystems
- The distinction between tracing out (ignoring) and measuring subsystems
- Physical interpretations related to spatial arrangement and connectivity

We have provided comprehensive tables showing how two-qubit computational basis states appear in different measurement bases, enabling cross-basis calculations. Through numerous examples, we've demonstrated how phase relationships in multi-qubit systems lead to observable differences in measurement statistics.

The reduced density matrix formalism provides a crucial tool for understanding subsystems of composite quantum states. When we trace out part of an entangled system, the resulting mixed state reveals how quantum information is stored non-locally in correlations. This concept illuminates the fundamental difference between quantum and classical correlations and explains the mechanism of decoherence.

While we have touched on entanglement through examples like the Bell states, a full treatment of entanglement—those special quantum states that cannot be written as tensor products of individual qubit states—is reserved for advanced studies. The mathematical foundation developed here, including the reduced density matrix formalism, provides the tools needed to fully appreciate the subtleties of entanglement when encountered later.

The exponential growth of Hilbert space dimension emerges as both the key resource for quantum computation and the primary obstacle to classical simulation. This fundamental tension drives much of quantum information science and motivates the development of quantum technologies.

### 2.9.1 Distinguishing Tracing from Measurement

It's crucial to understand that tracing out is fundamentally different from measurement:

#### Example 16: Trace vs. Measure

Starting with  $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ :

##### Option 1: Trace out B (ignore it)

- Result:  $\rho_A = \frac{1}{2}\mathbb{I}$  (maximally mixed)
- If we measure A in the Z-basis: 50% chance of  $|0\rangle$ , 50% chance of  $|1\rangle$
- Purity:  $\text{Tr}(\rho_A^2) = \frac{1}{2}$

##### Option 2: Measure B first, then look at A

- If B measured as  $|0\rangle$ : A collapses to pure state  $|0\rangle$

- If B measured as  $|1\rangle$ : A collapses to pure state  $|1\rangle$
- Each outcome has purity:  $\text{Tr}(\rho_A^2) = 1$

The key difference: Measurement collapses the entanglement, while tracing out preserves it (we just can't access it).

### 2.9.2 Physical Applications

The reduced density matrix appears naturally in several important contexts:

- 1. Quantum Decoherence:** When a quantum system entangles with an inaccessible environment, tracing out the environment explains the apparent loss of quantum coherence.
- 2. Quantum Communication:** Alice and Bob share entangled qubits. Alice's local measurement statistics are described by her reduced density matrix.
- 3. Subsystem Thermodynamics:** The thermal state of a subsystem in a larger quantum system is given by the reduced density matrix after tracing out the rest.
- 4. Entanglement Witnesses:** The purity  $\text{Tr}(\rho_A^2)$  quantifies entanglement—it equals 1 for product states and less than 1 for entangled states.

### 2.9.3 The Deeper Meaning

The reduced density matrix reveals a profound aspect of quantum mechanics: information in quantum systems can be stored non-locally in correlations. When subsystems are entangled:

- Neither subsystem has a complete quantum state of its own
- The "missing" information is stored in the quantum correlations
- Ignoring one subsystem converts pure quantum superpositions into classical statistical mixtures
- This is fundamentally different from classical correlations, where subsystems always have definite states

This concept bridges quantum information theory and statistical mechanics, explaining how quantum systems appear classical when we lack access to all degrees of freedom. It also quantifies the non-local nature of quantum information—a key resource in quantum computing and communication.

## 2.10 Problems

### Problem 2.1 (*Two-Qubit Product States*)

For two Bloch cubes representing a two-qubit system:

- Write the four computational basis states  $|00\rangle$ ,  $|01\rangle$ ,  $|10\rangle$ ,  $|11\rangle$  using tensor products
- If both qubits are in state  $|+\rangle$ , express the composite state in the computational basis
- Calculate the probability of each measurement outcome for a Z-measurement on both qubits

- (d) Calculate the probability of measuring the first qubit in  $|0\rangle$  (regardless of the second qubit)
- (e) Use your two Bloch cubes to verify these probability predictions

**Problem 2.2** (*Tensor Product Properties*)

Prove the following basic properties:

- (a)  $(A \otimes B)(C \otimes D) = AC \otimes BD$  for  $2 \times 2$  matrices
- (b)  $\text{Tr}(A \otimes B) = \text{Tr}(A) \cdot \text{Tr}(B)$
- (c) If  $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$  and  $|\phi\rangle = \gamma|0\rangle + \delta|1\rangle$ , write  $|\psi\rangle \otimes |\phi\rangle$  in the computational basis
- (d) Calculate  $(\sigma_z \otimes I)|+x\rangle|+x\rangle$  and express the result in the computational basis
- (e) Verify that  $\langle\psi \otimes \phi|\psi \otimes \phi\rangle = \langle\psi|\psi\rangle\langle\phi|\phi\rangle$

**Problem 2.3** (*CNOT Gate*)

Analyze the controlled-NOT (CNOT) gate:

- (a) Write out how CNOT acts on all four computational basis states
- (b) Construct the  $4 \times 4$  matrix representation of CNOT
- (c) Apply CNOT to the state  $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$
- (d) Apply CNOT to the state  $\frac{1}{\sqrt{2}}(|01\rangle + |11\rangle)$
- (e) Show that  $\text{CNOT}^2 = I$  (applying CNOT twice returns to the original state)
- (f) Use two Bloch cubes to physically implement these CNOT operations

**Problem 2.4** (*Bell State Creation*)

Create and analyze Bell states:

- (a) Starting with  $|00\rangle$ , apply  $H \otimes I$  where  $H = \frac{1}{\sqrt{2}}(\sigma_x + \sigma_z)$
- (b) Apply CNOT to the result to create  $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
- (c) Calculate the probability of each computational basis measurement outcome
- (d) Show that measuring the first qubit always gives the same result as measuring the second qubit
- (e) Express  $|\Phi^+\rangle$  in the XX basis:  $\{|+x\rangle|+x\rangle, |+x\rangle|-x\rangle, |-x\rangle|+x\rangle, |-x\rangle|-x\rangle\}$
- (f) Calculate measurement probabilities in the XX basis

**Problem 2.5** (*Three-Qubit Systems*)

Extend to three qubits:

- (a) Write the eight computational basis states for three qubits

- (b) Express the state  $|+x\rangle|+x\rangle|+x\rangle$  in the computational basis
- (c) Apply  $\sigma_z \otimes I \otimes I$  (Z gate on first qubit only, corresponding to a  $\pi$  rotation around the z-axis)
- (d) Calculate the resulting amplitudes and verify normalization
- (e) Find the probability of measuring the first qubit in  $|0\rangle$
- (f) Use three Bloch cubes to verify your calculation

**Problem 2.6** (*Matrix Representations Using Bit String Convention*)

For  $n$ -qubit systems, we label computational basis states using the bit string to number convention:

$|b_{n-1}b_{n-2}\dots b_1b_0\rangle \equiv |j\rangle$  where  $j = \sum_{i=0}^{n-1} b_i \cdot 2^i$ .

For example, in a 2-qubit system:  $|00\rangle \equiv |0\rangle$ ,  $|01\rangle \equiv |1\rangle$ ,  $|10\rangle \equiv |2\rangle$ ,  $|11\rangle \equiv |3\rangle$ .

For a 3-qubit system:  $|000\rangle \equiv |0\rangle$ ,  $|001\rangle \equiv |1\rangle$ , ...,  $|111\rangle \equiv |7\rangle$ .

- (a) Write the state  $|+x\rangle|+x\rangle$  from Problem 2.1(b) as a 4-dimensional column vector using the ordered basis  $\{|0\rangle, |1\rangle, |2\rangle, |3\rangle\}$
- (b) Write the operator  $\sigma_z \otimes I$  as a  $4 \times 4$  matrix in this same basis. Verify your answer by applying it to the column vector from part (a) and checking that you get the result from Problem 2.2(d)
- (c) Write the CNOT gate as a  $4 \times 4$  matrix using the ordering  $\{|0\rangle, |1\rangle, |2\rangle, |3\rangle\}$  and verify it matches your answer from Problem 2.3(b)
- (d) Write the Bell state  $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$  as a 4-dimensional column vector
- (e) For the three-qubit state  $|+x\rangle|+x\rangle|+x\rangle$  from Problem 2.5(b), write it as an 8-dimensional column vector using the ordered basis  $\{|0\rangle, |1\rangle, \dots, |7\rangle\}$
- (f) Write the operator  $\sigma_z \otimes I \otimes I$  as an  $8 \times 8$  matrix. Show explicitly the first two rows and the last two rows of this matrix (you may use “...” to indicate the middle rows)
- (g) Verify that multiplying the  $8 \times 8$  matrix from part (f) by the column vector from part (e) gives the correct result after applying the Z gate to the first qubit

**Problem 2.7** (*Partial Trace Fundamentals*)

Consider a two-qubit density matrix:

$$\rho_{AB} = \begin{pmatrix} 0.3 & 0 & 0 & 0.1 \\ 0 & 0.2 & 0.05 & 0 \\ 0 & 0.05 & 0.2 & 0 \\ 0.1 & 0 & 0 & 0.3 \end{pmatrix} \quad (2.147)$$

- (a) Use the block matrix method to compute  $\rho_A = \text{Tr}_B(\rho_{AB})$
- (b) Use the projection method to compute  $\rho_B = \text{Tr}_A(\rho_{AB})$
- (c) Calculate the purity  $\text{Tr}(\rho_A^2)$  and  $\text{Tr}(\rho_B^2)$
- (d) What do the purity values tell you about entanglement in the original state?

- (e) Using an ensemble of Bloch cubes, can you represent the reduced states  $\rho_A$  and  $\rho_B$ ? Explain how or why not.

**Problem 2.8** (*Pure vs Mixed States through Partial Trace*)

Consider the following two-qubit states:

- (a) Product state:  $|\psi_1\rangle = |+\rangle_A \otimes |0\rangle_B$
- (b) Bell state:  $|\psi_2\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
- (c) Superposition:  $|\psi_3\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

For each state:

- (i) Construct the full density matrix  $\rho_{AB} = |\psi\rangle\langle\psi|$
- (ii) Calculate both reduced density matrices  $\rho_A$  and  $\rho_B$
- (iii) Determine if each reduced state is pure or mixed
- (iv) Calculate measurement probabilities for the Z-basis on qubit A alone
- (v) Compare your results: which states show entanglement and how can you tell?

**Problem 2.9** (*Measurement vs Tracing Out*)

Start with the Bell state  $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ :

- (a) Option 1 (Trace out qubit B): Calculate  $\rho_A = \text{Tr}_B(|\Phi^+\rangle\langle\Phi^+|)$
- (b) What are the probabilities of measuring qubit A in states  $|0\rangle$  and  $|1\rangle$ ?
- (c) Option 2 (Measure qubit B first): If we measure qubit B in the Z-basis and obtain result  $|0\rangle$ , what is the post-measurement state of the full system?
- (d) What is now the state of qubit A alone?
- (e) If we measure qubit B and obtain result  $|1\rangle$ , what is the state of qubit A?
- (f) Compare the measurement probabilities for qubit A in both options
- (g) Option 3 (Measure qubit B in X-basis): Calculate what happens if we measure qubit B in the  $\{|+\rangle, |-\rangle\}$  basis instead
- (h) Explain the key physical difference between tracing out and measuring

**Problem 2.10** (*Schmidt Decomposition of Entangled States*)

The Schmidt decomposition expresses any two-qubit state uniquely as  $|\psi\rangle = \sum_i \sqrt{\lambda_i} |\phi_i\rangle_A \otimes |\chi_i\rangle_B$  where  $\lambda_i$  are the Schmidt coefficients (eigenvalues of  $\rho_A$ ).

- (a) For the Bell state  $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ , find the Schmidt decomposition
- (b) Calculate the Schmidt number (number of non-zero  $\lambda_i$ ) and explain what it tells you about entanglement

- (c) For a product state  $|\psi\rangle = | +x\rangle_A \otimes | -x\rangle_B$ , compute the Schmidt decomposition and Schmidt number
- (d) Show that the Bell state and the product state have different Schmidt numbers, confirming their different entanglement properties
- (e) For the superposition state  $|\psi\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ , find the Schmidt decomposition

**Problem 2.11** (*Controlled-Phase and Two-Qubit Gate Equivalences*)

Two-qubit gates can often be expressed in multiple equivalent forms:

- (a) The controlled-phase gate is  $\hat{U}_{CP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\phi} \end{pmatrix}$ . Express this in terms of projectors on the  $|11\rangle$  state
- (b) Show that CNOT can be decomposed as  $\text{CNOT} = (I \otimes H)\text{CZ}(I \otimes H)$  where CZ is the controlled-Z gate
- (c) Calculate how CNOT and controlled-phase gates differ in their action on the Bell state  $|\Phi^+\rangle$
- (d) For  $\phi = \pi$ , show that the controlled-phase gate creates the  $|\Psi^-\rangle$  Bell state when applied to  $|00\rangle$  after a Hadamard
- (e) Express the  $\sqrt{\text{CNOT}}$  gate in the same matrix format (it should be a  $4 \times 4$  unitary matrix)

**Problem 2.12** (*GHZ vs W State Distinguishability*)

Multi-qubit entangled states can exhibit qualitatively different properties:

- (a) Define the GHZ state (3-qubit):  $|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$  and the W state:  $|W\rangle = \frac{1}{\sqrt{3}}(|100\rangle + |010\rangle + |001\rangle)$
- (b) For the GHZ state, calculate all three single-qubit reduced density matrices
- (c) For the W state, calculate all three single-qubit reduced density matrices
- (d) Show that tracing out any single qubit from GHZ leaves the two remaining qubits in a separable state
- (e) Show that tracing out any single qubit from W leaves the two remaining qubits in an entangled state
- (f) Explain the physical meaning of these differences for quantum information and resilience to particle loss

**Problem 2.13** (*Entanglement Entropy and Information Theory*)

The entanglement entropy quantifies bipartite entanglement:  $S_A = -\text{Tr}(\rho_A \log_2 \rho_A)$ :

- (a) For a product state  $|\psi\rangle = |0\rangle_A |1\rangle_B$ , show that  $S_A = 0$
- (b) For the Bell state  $|\Phi^+\rangle$ , calculate  $S_A$  and interpret the result



- (c) For the separable mixed state  $\rho = 0.5|00\rangle\langle 00| + 0.5|11\rangle\langle 11|$ , calculate  $S_A$
- (d) Show that  $S_A = S_B$  (entanglement entropy is symmetric)
- (e) For a partially entangled state  $|\psi\rangle = \sqrt{0.7}|00\rangle + \sqrt{0.3}|11\rangle$ , calculate  $S_A$
- (f) Compare the entanglement entropy with the purity: which is a better measure of non-classicality?

**Problem 2.14** (*Partial Transpose and Entanglement Criterion*)

The partial transpose (PPT) criterion detects some (but not all) entangled states. For a density matrix  $\rho_{AB}$ , the partial transpose with respect to system B is  $\rho_{AB}^{T_B}$ :

- (a) Define the partial transpose operation:  $\rho_{AB}^{T_B} = \sum_{i,j,k,l} \rho_{ijkl} |i\rangle\langle j| \otimes |l\rangle\langle k|$
- (b) For the Bell state  $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ , construct  $\rho = |\Phi^+\rangle\langle\Phi^+|$  and compute  $\rho^{T_B}$
- (c) Show that  $\rho^{T_B}$  has a negative eigenvalue, confirming entanglement via the PPT criterion
- (d) For a product state  $|+x\rangle_A \otimes |-x\rangle_B$ , compute the partial transpose and verify it is positive semidefinite
- (e) Explain why PPT is necessary but not sufficient for separability in  $2 \times 2$  and  $2 \times 3$  systems

**Problem 2.15** (*Two-Qubit State Tomography*)

Quantum state tomography reconstructs an unknown state from measurements:

- (a) For a general two-qubit state  $\rho_{AB}$ , how many independent measurement bases are needed to fully characterize it?
- (b) For measurements in the bases  $\{Z, X, Y\} \otimes \{Z, X, Y\}$ , show that you obtain 9 measurement settings (for a  $4 \times 4$  density matrix with  $16 - 4 = 12$  independent real parameters)
- (c) Given measurement outcomes for all nine settings on the Bell state  $|\Phi^+\rangle$ , how would you reconstruct  $\rho = |\Phi^+\rangle\langle\Phi^+|$ ?
- (d) For the separable state  $\rho = |+x\rangle\langle+x| \otimes |0\rangle\langle 0|$ , predict the measurement outcomes in all nine bases before explicit calculation
- (e) Discuss how measurement noise affects the reconstructed state and when tomography becomes impractical

## References and Further Reading

**Nielsen, M. A., and Chuang, I. L.** *Quantum Computation and Quantum Information*, 10th anniversary ed. Cambridge University Press, 2010. Sections 2.1–2.5 cover composite systems, density operators, and the partial trace at the level of this chapter; the standard reference for the formalism developed here.

**Preskill, J.** *Lecture Notes on Quantum Computation*. California Institute of Technology. [theory.caltech.edu/~pres](http://theory.caltech.edu/~pres)  
Chapter 2 of the notes treats foundations of quantum theory, density operators, and reduced states; concise and freely available.

**Mermin, N. D.** *Quantum Computer Science: An Introduction*. Cambridge University Press, 2007. Student-friendly treatment of multi-qubit operations, with explicit circuit diagrams; the most accessible companion to the gate-level material in this chapter.

**Wilde, M. M.** *Quantum Information Theory*, 2nd ed. Cambridge University Press, 2017. [doi:10.1017/9781316809](https://doi.org/10.1017/9781316809) Chapter 4 develops density operators, reduced density matrices, and the partial trace with full mathematical rigor; the natural next step for the formalism.

**Watrous, J.** *The Theory of Quantum Information*. Cambridge University Press, 2018. The most mathematically rigorous treatment of tensor products, channels, and operator structures; a long-term reference rather than a first read.

# Chapter 3

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## Quantum Fields on a Lattice

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### 3.1 Introduction: From Spins to Fields

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#### Key Idea: Universal Quantum Evolution and the Spin-Particle Correspondence

Two profound principles unite in this chapter. First, any unitary transformation on  $N$  qubits can be decomposed into single-qubit rotations and two-qubit entangling gates—a generalization of our single Bloch sphere rotations to many-body systems. Second, these abstract qubits can represent physical particle occupation on lattice sites, transforming quantum information into quantum field theory. Together, these principles show that quantum fields can be understood through discrete gate operations.

In the previous chapters, we developed the mathematical machinery for describing systems of qubits—abstract two-level quantum systems. We saw how a single qubit’s evolution could be described by rotations on the Bloch sphere. We now extend this to  $N$  qubits and reveal a fundamental physical interpretation: each qubit can represent a location in space that either contains a quantum particle or is empty. This reinterpretation is not merely a convenient analogy; it establishes an exact mathematical correspondence that bridges quantum information theory and quantum field theory.

#### 3.1.1 Universal Gate Sets

Before diving into the physics, we establish a crucial mathematical fact:

### Universality for N Qubits

Any unitary operator  $U$  acting on  $N$  qubits can be approximated to arbitrary precision using only:

- Single-qubit rotations:  $R_x(\theta)$ ,  $R_y(\theta)$ ,  $R_z(\phi)$
- Two-qubit entangling gates: CNOT or  $\sqrt{\text{SWAP}}$

This is the many-body generalization of the fact that any single-qubit state can be reached by rotations around different axes of the Bloch sphere.

This universality means that complex quantum evolution—whether it represents computation or physical dynamics—can always be broken down into simple building blocks.

### 3.1.2 The Spin-Particle Mapping

Consider a one-dimensional lattice of sites, each capable of being occupied by at most one particle. This "hard-core" constraint emerges naturally in many physical systems where strong repulsive interactions prevent multiple occupation. We establish the correspondence:

$$|\uparrow\rangle_i \equiv |0\rangle_i \leftrightarrow \text{site } i \text{ is empty} \quad (3.1)$$

$$|\downarrow\rangle_i \equiv |1\rangle_i \leftrightarrow \text{site } i \text{ contains a particle} \quad (3.2)$$

This mapping immediately imbues our abstract quantum formalism with physical meaning. A superposition state  $\frac{1}{\sqrt{2}}(|0\rangle_i + |1\rangle_i)$  no longer represents an abstract qubit in a superposition of computational basis states—it represents a lattice site in a quantum superposition of being empty and occupied.

### Bloch Cube Activity: 3.1: Visualizing the Spin-Particle Correspondence

Arrange three Bloch cubes in a line to represent a three-site lattice:

- Orient each cube so that  $|+z\rangle$  (top) represents "empty" and  $|-z\rangle$  (bottom) represents "occupied"
- Create the configuration with particles at sites 1 and 3: point cubes 1 and 3 downward ( $|-z\rangle$ ), cube 2 upward ( $|+z\rangle$ )
- This represents the quantum state  $|101\rangle$  in our occupation number notation
- Now rotate cube 2 to the  $|+x\rangle$  position—this represents site 2 in superposition  $\frac{1}{\sqrt{2}}(|0\rangle_2 + |1\rangle_2)$
- The full state is now  $|1\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |1\rangle$
- Discuss: What does measuring site 2's occupation mean physically? How does this differ from a classical probabilistic mixture?

## 3.2 Quantum Fields from Ladder Operators

### 3.2.1 Two Types of Particles: Hardcore Bosons and Fermions

We can realize this spin-particle correspondence with two different types of quantum statistics. The choice determines the physical system we're modeling:

#### Two Realizations of Hardcore Particles

**1. Hardcore Bosons:** Particles that cannot doubly occupy a site but have NO exchange phase

- Use operators  $\hat{b}_i$  and  $\hat{b}_i^\dagger$
- Commute at different sites:  $[\hat{b}_i, \hat{b}_j^\dagger] = 0$  for  $i \neq j$
- Physical examples: Cold atoms in optical lattices, photons in cavities, Rydberg atoms

**2. Spinless Fermions:** Particles with Pauli exclusion AND exchange antisymmetry

- Use operators  $\hat{c}_i$  and  $\hat{c}_i^\dagger$
- Anticommute at different sites:  $\{\hat{c}_i, \hat{c}_j^\dagger\} = 0$  for  $i \neq j$
- Physical examples: Electrons in quantum dots, cold fermionic atoms

### 3.2.2 Field Operators for Hardcore Bosons

For hardcore bosons, we define the fundamental ladder operators. Let's first establish their matrix representations in the occupation number basis  $\{|0\rangle, |1\rangle\}$ :

#### Matrix Representations of Basic Operators

In the basis where  $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$  (empty) and  $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$  (occupied):

**Projection operators:**

$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad (\text{projects onto empty state}) \quad (3.3)$$

$$|1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad (\text{projects onto occupied state}) \quad (3.4)$$

**Transition operators:**

$$|1\rangle\langle 0| = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad (\text{creates particle}) \quad (3.5)$$

$$|0\rangle\langle 1| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad (\text{destroys particle}) \quad (3.6)$$

Now we can define the hardcore boson operators with their explicit matrix forms:

### Hardcore Boson Operators

**Creation operator at site  $i$ :**

$$\hat{b}_i^\dagger = |1\rangle_i \langle 0|_i = |\downarrow\rangle_i \langle \uparrow|_i = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}_i \quad (3.7)$$

Creates a particle at site  $i$  if empty, gives zero if occupied.

**Annihilation operator at site  $i$ :**

$$\hat{b}_i = |0\rangle_i \langle 1|_i = |\uparrow\rangle_i \langle \downarrow|_i = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}_i \quad (3.8)$$

Removes a particle from site  $i$  if occupied, gives zero if empty.

**Number operator at site  $i$ :**

$$\hat{n}_i = \hat{b}_i^\dagger \hat{b}_i = |1\rangle_i \langle 1|_i = |\downarrow\rangle_i \langle \downarrow|_i = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}_i \quad (3.9)$$

Counts particles at site  $i$ : eigenvalue 0 if empty, 1 if occupied.

The algebra for hardcore bosons:

$$\text{On-site: } \{\hat{b}_i, \hat{b}_i^\dagger\} = \hat{I}_i \quad (\text{anticommutator - hardcore constraint}) \quad (3.10)$$

$$(\hat{b}_i^\dagger)^2 = 0, \quad (\hat{b}_i)^2 = 0 \quad (\text{no double occupation}) \quad (3.11)$$

$$\text{Different sites: } [\hat{b}_i, \hat{b}_j^\dagger] = 0 \quad \text{for } i \neq j \quad (\text{commutator - bosonic}) \quad (3.12)$$

### 3.2.3 Field Operators for Spinless Fermions

For spinless fermions, we define similar operators with the same matrix representations but different inter-site statistics:

#### Spinless Fermion Operators

**Creation operator at site  $i$ :**

$$\hat{c}_i^\dagger = |1\rangle_i \langle 0|_i = |\downarrow\rangle_i \langle \uparrow|_i = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}_i \quad (3.13)$$

**Annihilation operator at site  $i$ :**

$$\hat{c}_i = |0\rangle_i \langle 1|_i = |\uparrow\rangle_i \langle \downarrow|_i = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}_i \quad (3.14)$$

**Number operator at site  $i$ :**

$$\hat{n}_i = \hat{c}_i^\dagger \hat{c}_i = |1\rangle_i \langle 1|_i = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}_i \quad (3.15)$$

Note: The matrix representations are IDENTICAL to bosons for single sites!

The algebra for spinless fermions:

$$\text{On-site: } \{\hat{c}_i, \hat{c}_i^\dagger\} = \hat{I}_i \quad (\text{same as bosons}) \quad (3.16)$$

$$(\hat{c}_i^\dagger)^2 = 0, \quad (\hat{c}_i)^2 = 0 \quad (\text{same as bosons}) \quad (3.17)$$

$$\text{Different sites: } \{\hat{c}_i, \hat{c}_j^\dagger\} = 0 \quad \text{for } i \neq j \quad (\text{anticommutator - fermionic}) \quad (3.18)$$

#### Example 1: Ladder Operator Action on Lattice States

Consider a three-site system. Let's examine how ladder operators act:

**For hardcore bosons:**

$$\hat{b}_2^\dagger |100\rangle = |110\rangle \quad (\text{creates particle at site 2}) \quad (3.19)$$

$$\hat{b}_1^\dagger |100\rangle = 0 \quad (\text{site 1 already occupied}) \quad (3.20)$$

$$\hat{b}_3^\dagger \hat{b}_2^\dagger \hat{b}_1^\dagger |000\rangle = |111\rangle \quad (\text{fills all sites}) \quad (3.21)$$

**For fermions (same local action):**

$$\hat{c}_2^\dagger |100\rangle = |110\rangle \quad (3.22)$$

$$\hat{c}_1^\dagger |100\rangle = 0 \quad (3.23)$$

$$\hat{c}_3^\dagger \hat{c}_2^\dagger \hat{c}_1^\dagger |000\rangle = |111\rangle \quad (3.24)$$

The difference appears when we exchange operator order (see Section 3.6).

#### Bloch Cube Activity: 3.2: Ladder Operations with Bloch Cubes

Using four Bloch cubes in a line:

- (a) Start with all cubes pointing up (empty lattice):  $|0000\rangle$
- (b) Apply  $\hat{b}_2^\dagger$  (or  $\hat{c}_2^\dagger$ ): flip cube 2 to point down. Result:  $|0100\rangle$
- (c) Apply  $\hat{b}_4^\dagger$  (or  $\hat{c}_4^\dagger$ ): flip cube 4 to point down. Result:  $|0101\rangle$
- (d) Apply  $\hat{b}_1^\dagger \hat{b}_2$  (or  $\hat{c}_1^\dagger \hat{c}_2$ ): moves particle from site 2 to site 1
- (e) Show this by flipping cube 2 up and cube 1 down. Result:  $|1001\rangle$
- (f) Try applying  $\hat{b}_1^\dagger$  again—nothing happens (cube 1 already points down)
- (g) For single particles, bosons and fermions behave identically!

## 3.3 Unitary Gates in Second Quantization

### 3.3.1 Building Gates from Field Operators

We can construct unitary gates directly from creation and annihilation operators. The gate forms are the same for both particle types, using either  $\hat{a} = \hat{b}$  (bosons) or  $\hat{a} = \hat{c}$  (fermions). Let's first see

how these gates look in matrix form:

#### Matrix Representations of Two-Qubit Gates

In the two-qubit computational basis  $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ :

**Identity operator:**

$$\hat{I} \otimes \hat{I} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (3.25)$$

**SWAP gate:**

$$\text{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (3.26)$$

Exchanges the states of two qubits:  $|01\rangle \leftrightarrow |10\rangle$

**Controlled-NOT (CNOT) gate:**

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (3.27)$$

Flips target qubit if control is 1:  $|10\rangle \leftrightarrow |11\rangle$

Now we can express these gates using field operators:

#### Fundamental Unitary Gates

**SWAP gate between sites  $i$  and  $j$ :**

$$\text{SWAP}_{ij} = (\hat{I}_i - \hat{n}_i)(\hat{I}_j - \hat{n}_j) + \hat{n}_i\hat{n}_j + \hat{a}_i^\dagger\hat{a}_j + \hat{a}_j^\dagger\hat{a}_i \quad (3.28)$$

**Phase gate (particle-dependent):**

$$\hat{P}_i(\phi) = \hat{I}_i - \hat{n}_i + e^{i\phi}\hat{n}_i = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}_i \quad (3.29)$$

**Controlled-NOT (CNOT) gate:**

$$\text{CNOT}_{ij} = |00\rangle\langle 00| + |01\rangle\langle 01| + |11\rangle\langle 10| + |10\rangle\langle 11| \quad (3.30)$$

Note:  $\hat{a}$  represents either  $\hat{b}$  (bosons) or  $\hat{c}$  (fermions)

These gates act directly on our lattice states without reference to time evolution or Hamiltonians.

### 3.3.2 The SWAP and Square-Root-of-SWAP Gates

Following Loss and DiVincenzo (1998), we recognize that the SWAP gate and its square root are particularly important for quantum computation with quantum dots:



### The $\sqrt{\text{SWAP}}$ Gate

The  $\sqrt{\text{SWAP}}$  gate creates entanglement between neighboring sites:

$$\sqrt{\text{SWAP}}_{ij} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1+i}{2} & \frac{1-i}{2} & 0 \\ 0 & \frac{1-i}{2} & \frac{1+i}{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (3.31)$$

This gate, combined with single-qubit rotations, forms a universal gate set for quantum computation.

### Example 2: $\sqrt{\text{SWAP}}$ Gate Action

For a three-site system with one particle (same for bosons and fermions):

$$\sqrt{\text{SWAP}}_{12}|100\rangle = \frac{1+i}{2}|100\rangle + \frac{1-i}{2}|010\rangle \quad (3.32)$$

$$= \frac{e^{i\pi/4}}{\sqrt{2}}(|100\rangle + e^{-i\pi/2}|010\rangle) \quad (3.33)$$

The particle becomes delocalized across both sites with a specific phase relationship. Applying it twice recovers the full SWAP:

$$(\sqrt{\text{SWAP}}_{12})^2|100\rangle = \text{SWAP}_{12}|100\rangle = |010\rangle \quad (3.34)$$

## 3.4 Building Quantum States from Creation Operators

### 3.4.1 The Vacuum and Fock States

The vacuum state  $|\text{vac}\rangle = |00\dots 0\rangle$  represents the lattice with no particles anywhere. We build configurations by applying creation operators, but the statistics differ:

**For hardcore bosons:**

$$|n_1 n_2 \dots n_N\rangle_{\text{boson}} = \prod_{j:n_j=1} \hat{b}_j^\dagger |\text{vac}\rangle \quad (3.35)$$

Order doesn't matter:  $\hat{b}_i^\dagger \hat{b}_j^\dagger = \hat{b}_j^\dagger \hat{b}_i^\dagger$

**For spinless fermions:**

$$|n_1 n_2 \dots n_N\rangle_{\text{fermion}} = \prod_{j:n_j=1} \hat{c}_j^\dagger |\text{vac}\rangle \quad (3.36)$$

Order matters:  $\hat{c}_i^\dagger \hat{c}_j^\dagger = -\hat{c}_j^\dagger \hat{c}_i^\dagger$

**Example 3: Building Many-Particle States****Single particle localized at site  $j$  (same for both):**

$$\text{Bosons: } |j\rangle = \hat{b}_j^\dagger |\text{vac}\rangle \quad (3.37)$$

$$\text{Fermions: } |j\rangle = \hat{c}_j^\dagger |\text{vac}\rangle \quad (3.38)$$

**Two particles at sites  $i$  and  $j$  (statistics matter!):**

$$\text{Bosons: } |i, j\rangle = \hat{b}_i^\dagger \hat{b}_j^\dagger |\text{vac}\rangle = \hat{b}_j^\dagger \hat{b}_i^\dagger |\text{vac}\rangle \quad (3.39)$$

$$\text{Fermions: } |i, j\rangle = \hat{c}_i^\dagger \hat{c}_j^\dagger |\text{vac}\rangle = -\hat{c}_j^\dagger \hat{c}_i^\dagger |\text{vac}\rangle \quad (3.40)$$

**Delocalized single particle (same form):**

$$\text{Bosons: } |\psi\rangle = \sum_{j=1}^N \psi_j \hat{b}_j^\dagger |\text{vac}\rangle \quad (3.41)$$

$$\text{Fermions: } |\psi\rangle = \sum_{j=1}^N \psi_j \hat{c}_j^\dagger |\text{vac}\rangle \quad (3.42)$$

**Example entangled two-particle states:**

$$\text{Bosons: } |\Phi^+\rangle = \frac{1}{\sqrt{2}} (\hat{b}_1^\dagger \hat{b}_2^\dagger + \hat{b}_3^\dagger \hat{b}_4^\dagger) |\text{vac}\rangle \quad (3.43)$$

$$\text{Fermions: } |\Phi^-\rangle = \frac{1}{\sqrt{2}} (\hat{c}_1^\dagger \hat{c}_2^\dagger - \hat{c}_3^\dagger \hat{c}_4^\dagger) |\text{vac}\rangle \quad (3.44)$$

Note: These are just illustrative examples. Both particle types can form either  $\pm$  combination—the sign is NOT enforced by the operator algebra! The operators themselves already encode the statistics through their (anti)commutation relations.

**3.4.2 The Total Particle Number and Conservation**

The total number operator counts all particles on the lattice (same form for both):

$$\text{Bosons: } \hat{N} = \sum_{j=1}^N \hat{n}_j = \sum_{j=1}^N \hat{b}_j^\dagger \hat{b}_j \quad (3.45)$$

$$\text{Fermions: } \hat{N} = \sum_{j=1}^N \hat{n}_j = \sum_{j=1}^N \hat{c}_j^\dagger \hat{c}_j \quad (3.46)$$

This operator generates a decomposition of the full Hilbert space:

$$\mathcal{H} = \bigoplus_{m=0}^N \mathcal{H}_m \quad (3.47)$$

where  $\mathcal{H}_m$  contains all states with exactly  $m$  particles. (Note: The direct sum  $\oplus$  represents mutually exclusive subspaces—a state is in  $\mathcal{H}_0$  OR  $\mathcal{H}_1$  OR  $\mathcal{H}_2$ , etc., but never in multiple sectors simultaneously.)

**Example 4: Structure of Particle Number Sectors**

For a 4-site lattice (same for bosons and fermions):

**Zero particles** ( $\mathcal{H}_0$ , dimension 1):

$$\{|0000\rangle\} \quad (3.48)$$

**One particle** ( $\mathcal{H}_1$ , dimension 4):

$$\{|1000\rangle, |0100\rangle, |0010\rangle, |0001\rangle\} \quad (3.49)$$

**Two particles** ( $\mathcal{H}_2$ , dimension 6):

$$\{|1100\rangle, |1010\rangle, |1001\rangle, |0110\rangle, |0101\rangle, |0011\rangle\} \quad (3.50)$$

**Three particles** ( $\mathcal{H}_3$ , dimension 4):

$$\{|1110\rangle, |1101\rangle, |1011\rangle, |0111\rangle\} \quad (3.51)$$

**Four particles** ( $\mathcal{H}_4$ , dimension 1):

$$\{|1111\rangle\} \quad (3.52)$$

Total dimension:  $1 + 4 + 6 + 4 + 1 = 16 = 2^4$

The dimensions follow Pascal's triangle:  $\binom{4}{0}, \binom{4}{1}, \binom{4}{2}, \binom{4}{3}, \binom{4}{4}$

## 3.5 Physical Evolution from Gate Sequences

### 3.5.1 Particle Hopping via Gates

Instead of starting with a hopping Hamiltonian, we can directly implement particle motion using gate sequences:

#### Hopping from Gates

Nearest-neighbor particle hopping can be implemented by:

- Full hop: SWAP gate between adjacent sites (exchanges particle positions)
- Partial hop:  $\sqrt{\text{SWAP}}$  gate (creates quantum superposition)
- Controlled hop: CNOT-based conditional movement
- Multi-hop sequence: Chain of SWAP gates to move particles across the lattice

These gates preserve particle number and maintain the hardcore constraint automatically.

### 3.5.2 Universal Quantum Computation on the Lattice

The combination of single-site operations and two-site gates forms a universal gate set:

**Universal Gate Set (Loss-DiVincenzo)**

Any unitary operation on  $N$  qubits can be approximated to arbitrary precision using:

- (a) Single-qubit rotations:  $\hat{R}_x(\theta), \hat{R}_y(\theta), \hat{R}_z(\phi)$
- (b) Two-qubit entangling gate:  $\sqrt{\text{SWAP}}$  or CNOT

For quantum simulation of particles on a lattice, this means any physically realizable evolution can be implemented through sequences of these basic gates.

### 3.6 Hardcore Bosons versus Spinless Fermions: The Key Difference

#### 3.6.1 The Statistical Difference

While hardcore bosons and spinless fermions have identical on-site properties (both limited to occupation 0 or 1), they differ fundamentally in their multi-site behavior:

**Statistical Distinction****Hardcore Bosons:**

$$[\hat{b}_i, \hat{b}_j^\dagger] = 0 \quad \text{for } i \neq j \quad (\text{commute}) \quad (3.53)$$

$$[\hat{b}_i, \hat{b}_j] = 0 \quad \text{for } i \neq j \quad (3.54)$$

$$\hat{b}_i^\dagger \hat{b}_j^\dagger = \hat{b}_j^\dagger \hat{b}_i^\dagger \quad (\text{symmetric}) \quad (3.55)$$

**Spinless Fermions:**

$$\{\hat{c}_i, \hat{c}_j^\dagger\} = 0 \quad \text{for } i \neq j \quad (\text{anticommute}) \quad (3.56)$$

$$\{\hat{c}_i, \hat{c}_j\} = 0 \quad \text{for } i \neq j \quad (3.57)$$

$$\hat{c}_i^\dagger \hat{c}_j^\dagger = -\hat{c}_j^\dagger \hat{c}_i^\dagger \quad (\text{antisymmetric}) \quad (3.58)$$

#### 3.6.2 Jordan-Wigner Transformation

The Jordan-Wigner transformation maps between spin operators and fermionic operators:

## Jordan-Wigner Transformation

**For hardcore bosons:** Direct mapping to spins!

$$\hat{b}_j = \hat{\sigma}_j^+ = \frac{1}{2}(\sigma_{x,j} + i\sigma_{y,j}) \quad (3.59)$$

$$\hat{b}_j^\dagger = \hat{\sigma}_j^- = \frac{1}{2}(\sigma_{x,j} - i\sigma_{y,j}) \quad (3.60)$$

$$\hat{n}_j = \frac{1}{2}(\hat{I}_j - \sigma_{z,j}) \quad (3.61)$$

**For spinless fermions:** Need non-local string operators!

$$\hat{c}_j = \left( \prod_{k < j} e^{i\pi \hat{n}_k} \right) \hat{\sigma}_j^+ = \left( \prod_{k < j} \sigma_{z,k} \right) \hat{\sigma}_j^+ \quad (3.62)$$

$$\hat{c}_j^\dagger = \left( \prod_{k < j} \sigma_{z,k} \right) \hat{\sigma}_j^- \quad (3.63)$$

$$\hat{n}_j = \frac{1}{2}(\hat{I}_j - \sigma_{z,j}) \quad (\text{same as bosons locally}) \quad (3.64)$$

The string operator  $\prod_{k < j} \sigma_{z,k}$  counts particles to the left, introducing the fermionic sign.

This transformation shows that fermions require non-local phase factors to maintain their anti-commutation relations, while hardcore bosons map directly to local spin operators.

### 3.6.3 Physical Realizations

**Hardcore bosons naturally arise in:**

- (a) **Rydberg atoms:** Strong van der Waals interactions create a "blockade"
- (b) **Photons in cavity arrays:** Strong photon-photon interactions
- (c) **Josephson junction arrays:** Cooper pairs with strong charging energy
- (d) **Ultracold atoms in optical lattices:** Rb-87, Na-23 with strong repulsion

**Spinless fermions naturally arise in:**

- (a) **Electrons in quantum dots:** The Loss-DiVincenzo proposal
- (b) **Cold fermionic atoms:** K-40, Li-6 in optical lattices
- (c) **Nuclear spins:** NMR quantum computing with fermionic statistics

## 3.7 From Many Particles to One: Emergence of the Wavefunction

### 3.7.1 The Single-Particle Sector and Traditional Quantum Mechanics

When restricted to states with exactly one particle, our field theory reduces to traditional single-particle quantum mechanics. Since a single particle has no exchange partner, bosons and fermions behave identically:

$$\mathcal{H}_1 = \text{span}\{\hat{a}_j^\dagger|\text{vac}\rangle : j = 1, 2, \dots, N\} \quad (3.65)$$

where  $\hat{a}$  can be either  $\hat{b}$  (bosons) or  $\hat{c}$  (fermions).

Any single-particle state can be written as:

$$|\psi\rangle = \sum_{j=1}^N \psi_j \hat{a}_j^\dagger |\text{vac}\rangle \quad (3.66)$$

where the complex amplitudes  $\psi_j$  form the discrete wavefunction.

#### Connection to the Wavefunction

The coefficients  $\psi_j$  are precisely the values of the wavefunction at lattice sites:

$$\psi_j = \psi(x_j) \quad \text{where} \quad x_j = ja \quad (3.67)$$

Physical observables emerge naturally:

- Probability density:  $|\psi_j|^2 = |\psi(x_j)|^2$
- Normalization:  $\sum_{j=1}^N |\psi_j|^2 = 1$
- Position expectation:  $\langle x \rangle = a \sum_{j=1}^N j |\psi_j|^2$

### 3.7.2 Position and Momentum Representations

The position basis states are simply:

$$|x_j\rangle = \hat{a}_j^\dagger |\text{vac}\rangle = |0\dots 010\dots 0\rangle \quad (3.68)$$

with the particle at site  $j$ .

For a lattice with periodic boundary conditions, we can define momentum eigenstates:

$$|k\rangle = \frac{1}{\sqrt{N}} \sum_{j=1}^N e^{ikx_j} \hat{a}_j^\dagger |\text{vac}\rangle \quad (3.69)$$

where the allowed momenta are quantized:  $k = \frac{2\pi m}{Na}$  with  $m = 0, 1, \dots, N-1$ .

#### Example 5: Fourier Transform on the Lattice

The transformation between position and momentum representations is the discrete Fourier transform:

$$\psi(x_j) = \frac{1}{\sqrt{N}} \sum_{m=0}^{N-1} \tilde{\psi}(k_m) e^{ik_m x_j} \quad (3.70)$$

$$\tilde{\psi}(k_m) = \frac{1}{\sqrt{N}} \sum_{j=1}^N \psi(x_j) e^{-ik_m x_j} \quad (3.71)$$

This can be implemented exactly using quantum gates—the Quantum Fourier Transform (QFT).

### 3.7.3 Entanglement and Subspace Projections: When is a Superposition Entangled?

We close this chapter with a subtle but fundamental observation about the nature of entanglement. Consider a two-site system with the state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle) \quad (3.72)$$

Is this state entangled? The answer reveals a deep truth: *entanglement depends on the Hilbert space context in which we view the state.*

#### The Context-Dependence of Entanglement

**Viewed in the single-particle sector**  $\mathcal{H}_1 = \text{span}\{|10\rangle, |01\rangle\}$ :

- This is a 2-dimensional space describing one particle's position
- We can map:  $|10\rangle \rightarrow |0\rangle_{\text{eff}}$  and  $|01\rangle \rightarrow |1\rangle_{\text{eff}}$
- Then  $|\psi\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle_{\text{eff}} + |1\rangle_{\text{eff}}) = |+\rangle_{\text{eff}}$
- This is NOT entangled—it's just a superposition of one particle at two locations
- Physically: a single particle wavefunction  $\psi(x)$  spread over two sites

**Viewed in the full two-qubit space**  $\mathcal{H} = \text{span}\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ :

- This is a 4-dimensional space with tensor product structure  $\mathcal{H} = \mathcal{H}_{\text{site1}} \otimes \mathcal{H}_{\text{site2}}$
- The state  $|\psi\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$  is the Bell state  $|\Psi^+\rangle$
- This IS entangled—it cannot be written as  $|\alpha\rangle_1 \otimes |\beta\rangle_2$
- Physically: the two sites have perfectly correlated occupations

The key insight is that entanglement is defined relative to a tensor product decomposition. When we restrict to the  $n = 1$  sector, we lose the tensor product structure—we're no longer asking about correlations between sites but about the spatial distribution of a single entity.

#### Example 6: The Resolution of the Paradox

Consider the following apparent contradiction:

- In the  $n = 1$  sector, no state can be entangled (there's only one particle!)
- The same mathematical object  $\frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$  is maximally entangled in the full space

The resolution: We're asking different questions!

- **$n = 1$  sector question:** "Is the particle's wavefunction entangled with itself?"  $\rightarrow$  No, that's meaningless
- **Full space question:** "Are the two sites entangled?"  $\rightarrow$  Yes, their occupations are correlated

This is analogous to the difference between first and second quantization:

- **First quantization:** Single-particle wavefunction  $\psi(x)$ —no notion of entanglement
- **Second quantization:** Occupation number basis—entanglement between modes/sites

This phenomenon generalizes to  $N$  sites:

#### Generalization to $N$ Sites

For  $N$  sites with one particle, any state  $|\psi\rangle = \sum_{j=1}^N \psi_j |j\rangle$  in the  $n = 1$  sector:

- **Within  $\mathcal{H}_1$ :** Never entangled (it's a single-particle wavefunction)
- **In full  $2^N$ -dimensional space:** Can be highly entangled between sites

Example for 3 sites:

$$|W\rangle = \frac{1}{\sqrt{3}}(|100\rangle + |010\rangle + |001\rangle) \quad (3.73)$$

- In  $\mathcal{H}_1$ : Just a uniform superposition (not entangled)
- In full space: Genuine tripartite entanglement (W-state)

#### Mathematical Aside: Why This Matters

This distinction is crucial for understanding:

- (a) **Quantum simulation:** When simulating fermions with qubits, are we creating "real" entanglement or just mimicking single-particle physics?
- (b) **Quantum computation:** Subspace restrictions can make entangled states appear unentangled
- (c) **Many-body physics:** The transition from few to many particles changes the relevant questions about correlations
- (d) **Measurement theory:** What we measure depends on which subspace we probe

The lesson: Quantum properties like entanglement are not absolute but emerge from our choice of description and the constraints we impose on the system.



**Physical Insight**

In the two-particle sector, an even richer structure emerges. Two distinguishable particles can be entangled through their spatial wavefunctions, their spin degrees of freedom, or both. The interplay between spatial and spin entanglement, combined with fermionic antisymmetry, leads to phenomena like the spin-statistics connection and exchange interactions. This topic, while beyond our current scope, forms the foundation of quantum chemistry and condensed matter physics.

For further exploration of entanglement in identical particle systems, see the references on fermionic entanglement and mode entanglement at the end of this chapter.

## 3.8 Chapter Summary

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This chapter has revealed the profound connection between abstract quantum information and concrete physical systems. We began by establishing that any unitary transformation on  $N$  qubits can be decomposed into single-qubit rotations and two-qubit entangling gates—a powerful generalization of single-qubit Bloch sphere rotations to many-body systems.

Through the spin-particle correspondence, we discovered that qubits can represent occupation numbers on a lattice, with spin up as empty sites and spin down as occupied sites. This transforms collections of two-level systems into quantum fields. We introduced two types of particles:

- **Hardcore bosons** ( $\hat{b}_i, \hat{b}_i^\dagger$ ): Commute at different sites, no exchange phase
- **Spinless fermions** ( $\hat{c}_i, \hat{c}_i^\dagger$ ): Anticommute at different sites, acquire phase under exchange

We showed how quantum gates like SWAP and  $\sqrt{\text{SWAP}}$  have direct physical interpretations as particle exchange and coherent tunneling operations. The Loss-DiVincenzo framework demonstrated that natural physical interactions (exchange coupling) can implement universal quantum computation, with the  $\sqrt{\text{SWAP}}$  gate plus single-qubit rotations forming a complete gate set.

The distinction between hardcore bosons and spinless fermions emerged through their commutation relations—identical local physics but different global quantum statistics. The Jordan-Wigner transformation revealed that fermions require non-local string operators to maintain anticommutation, while bosons map directly to local spin operators. When restricted to single particles, both types behave identically, reducing to familiar quantum mechanics with creation operator coefficients forming the discrete wavefunction.

The framework developed here bridges quantum computation and many-body physics. We've seen that quantum fields can be understood through discrete gate operations, where any unitary evolution decomposes into elementary gates. The choice between bosons and fermions depends on the physical system being modeled, with bosons offering simpler mathematics and fermions capturing electronic physics.

In the following chapter, we will extend this framework to continuous time evolution, showing how Hamiltonians emerge as generators of unitary gates and connecting our discrete gate model to traditional quantum dynamics.

### 3.9 Problems

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**Problem 3.1** (*Spin-Particle Correspondence and Ladder Operators*)

Consider a three-site lattice with the spin-particle correspondence:  $|\uparrow\rangle_i \equiv |0\rangle_i$  (empty) and  $|\downarrow\rangle_i \equiv |1\rangle_i$  (occupied).

- Write the matrix representations for the creation operator  $\hat{b}_2^\dagger$ , annihilation operator  $\hat{b}_2$ , and number operator  $\hat{n}_2$  at site 2
- Apply  $\hat{b}_2^\dagger$  to the state  $|100\rangle$  and  $|110\rangle$ . Explain the physical meaning of each result
- Calculate  $(\hat{b}_2^\dagger)^2|100\rangle$  and explain why this result makes sense for hardcore particles
- Show that  $\hat{n}_2 = \hat{b}_2^\dagger \hat{b}_2$  by explicit matrix multiplication
- Using Bloch cubes, demonstrate the action of  $\hat{b}_2^\dagger$  on the state  $|100\rangle$  by showing the cube orientations before and after

**Problem 3.2** (*Gate Operations on Lattice States*)

Consider the SWAP gate between sites 1 and 2:

$$\text{SWAP}_{12} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (3.74)$$

- Apply this gate to the states  $|10\rangle$ ,  $|01\rangle$ , and  $|11\rangle$ . Interpret the results physically in terms of particle positions
- Apply the gate to the superposition state  $\frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$ . What type of state do you get?
- Now consider the  $\sqrt{\text{SWAP}}$  gate. Apply it to  $|10\rangle$  and show that applying it twice gives the full SWAP operation
- Express the action of  $\text{SWAP}_{12}$  using creation and annihilation operators (hint: think about what operators exchange particles between sites)

**Problem 3.3** (*Building Many-Body States*)

Consider a four-site lattice with hardcore particles.

- Write the vacuum state  $|\text{vac}\rangle$  and create the single-particle states at each site using creation operators
- Create a delocalized single-particle state:  $|\psi\rangle = \frac{1}{2}(\hat{a}_1^\dagger + \hat{a}_2^\dagger + \hat{a}_3^\dagger + \hat{a}_4^\dagger)|\text{vac}\rangle$ . What is this state in the occupation number basis?
- Calculate the total particle number operator  $\hat{N} = \sum_{i=1}^4 \hat{n}_i$  and verify that  $|\psi\rangle$  is an eigenstate with eigenvalue 1
- Create a two-particle state with particles at sites 1 and 3:  $|1, 3\rangle = \hat{a}_1^\dagger \hat{a}_3^\dagger |\text{vac}\rangle$

- (e) How many total two-particle states are possible on this four-site lattice? List them all
- (f) Using the binomial coefficient formula, verify that the total dimension of the Hilbert space is  $2^4 = 16$

**Problem 3.4** (*Entanglement in Different Contexts*)

Consider the state  $|\psi\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$  on a two-site lattice.

- (a) Full Hilbert space view: Treat this as a two-qubit state in the 4-dimensional space  $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ . Show that it cannot be written as a product state  $|a\rangle_1 \otimes |b\rangle_2$
- (b) Calculate the reduced density matrices  $\rho_1$  and  $\rho_2$  by tracing out the other site. What is the purity of each?
- (c) Single-particle sector view: Now view this as a single-particle state in the 2-dimensional space  $\{|10\rangle, |01\rangle\}$ . Define  $|L\rangle = |10\rangle$  (particle on left) and  $|R\rangle = |01\rangle$  (particle on right)
- (d) In this new basis, is  $|\psi\rangle = \frac{1}{\sqrt{2}}(|L\rangle + |R\rangle)$  entangled? Explain your reasoning
- (e) Compare the physical interpretations: when is this state describing “entanglement between sites” vs “superposition of a particle’s position”?
- (f) This demonstrates that entanglement depends on the tensor product structure we choose. Discuss the implications for quantum simulation and many-body physics

**Problem 3.5** (*Tight-Binding Dispersion Relation*)

For a one-dimensional lattice with nearest-neighbor hopping amplitude  $t$ , the single-particle Hamiltonian is  $\hat{H} = -t \sum_{\langle i,j \rangle} (\hat{a}_i^\dagger \hat{a}_j + \hat{a}_j^\dagger \hat{a}_i)$ . In momentum space, this gives  $E(k) = -2t \cos(ka)$  where  $a$  is the lattice spacing.

- (a) Plot the dispersion relation  $E(k)$  for  $k \in [-\pi/a, \pi/a]$  (first Brillouin zone)
- (b) Find the minimum and maximum energy and explain their physical significance
- (c) For a finite chain of  $N = 5$  sites with periodic boundary conditions, write the allowed  $k$  values
- (d) Calculate the density of states  $\rho(E) = |dE/dk|^{-1}$  and explain where it diverges
- (e) For a chain with half-filled occupation (2 particles on 4 sites), estimate the ground state energy using the tight-binding model

**Problem 3.6** (*Momentum Eigenstates on a Ring*)

For a particle on a ring of  $N$  sites with periodic boundary conditions, momentum eigenstates are  $|k\rangle = \frac{1}{\sqrt{N}} \sum_{j=1}^N e^{ikx_j} \hat{a}_j^\dagger |\text{vac}\rangle$  with  $k = 2\pi m/Na$  and  $m = 0, 1, \dots, N-1$ .

- (a) Show that  $|k\rangle$  is an eigenstate of the total momentum operator  $\hat{P} = \sum_j k_j \hat{n}_j$  with eigenvalue  $\hbar k$  (for each particle)
- (b) For  $N = 4$  sites and  $a = 1$ , write out the explicit form of the  $k = \pi/2$  eigenstate in real space
- (c) Calculate  $\langle k|k'\rangle$  for two different momentum eigenstates and verify orthonormality

- (d) Show that the momentum space wavefunction  $\tilde{\psi}(k) = \frac{1}{\sqrt{N}} \sum_j e^{-ikx_j} \psi(x_j)$  is the discrete Fourier transform of the real-space wavefunction
- (e) For a Gaussian wavepacket in momentum space centered at  $k_0$ , describe its behavior in real space on the lattice

**Problem 3.7** (*Particle-Hole Symmetry of Half-Filled Lattice*)

For a half-filled lattice with  $N$  sites and  $N/2$  particles, particle-hole symmetry pairs empty sites with occupied sites. The transformation is  $\hat{a}_i \leftrightarrow \hat{a}_i^\dagger$  (up to phases and careful ordering).

- (a) For a two-site system with one particle, list all states and their particle-hole conjugates
- (b) Show that the state  $\frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$  maps to itself under particle-hole transformation
- (c) For the half-filled 4-site tight-binding model with  $E(k) = -2t \cos(ka)$ , show that  $E(k) = -E(\pi - k)$  (particle-hole symmetry of the spectrum)
- (d) Explain why this symmetry implies that the chemical potential at half-filling is always  $\mu = 0$
- (e) For a non-half-filled system, does particle-hole symmetry still hold? If not, what is broken?

**Problem 3.8** (*Jordan-Wigner Transformation for Fermions*)

The Jordan-Wigner (JW) transformation maps spinless fermions to spins using non-local string operators:  $\hat{c}_j = \left( \prod_{k < j} e^{i\pi \hat{n}_k} \right) \hat{\sigma}_j^+ = \left( \prod_{k < j} \sigma_{z,k} \right) \hat{\sigma}_j^+$ .

- (a) For a two-site system, explicitly write out  $\hat{c}_1$  and  $\hat{c}_2$  using Pauli matrices
- (b) Verify that  $\{\hat{c}_j, \hat{c}_k\} = 0$  for  $j \neq k$  (anticommutation at different sites)
- (c) Verify that  $\{\hat{c}_j, \hat{c}_j^\dagger\} = \hat{I}$  (on-site anticommutation)
- (d) Show that the occupation number operator  $\hat{n}_j = \hat{c}_j^\dagger \hat{c}_j$  is local: it equals  $\frac{1}{2}(\hat{I} - \sigma_{z,j})$
- (e) Compute the hopping term  $\hat{c}_i^\dagger \hat{c}_{i+1}$  in the spin language and explain why non-local string operators appear

**Problem 3.9** (*Hopping in 2D Square Lattice*)

Extend tight-binding to a 2D square lattice with sites labeled  $(m, n)$  and nearest-neighbor hopping  $t$ .

- (a) Write the Hamiltonian:  $\hat{H} = -t \sum_{\langle (m,n), (m',n') \rangle} (\hat{a}_{m,n}^\dagger \hat{a}_{m',n'} + \text{h.c.})$
- (b) In momentum space with  $k = (k_x, k_y)$ , show that  $E(\vec{k}) = -2t[\cos(k_x a) + \cos(k_y a)]$
- (c) Plot the Fermi surface (constant energy contour) for a given filling
- (d) For a half-filled 2D system, estimate the ground state energy
- (e) Explain how the density of states differs at the van Hove singularities (where  $dE/dk = 0$ ) compared to the 1D case

**Problem 3.10** (*Current Operator and Particle Flow*)

The current operator represents particle flow between sites. Define  $\hat{I}_{ij} = i\frac{t}{\hbar}(\hat{a}_i^\dagger \hat{a}_j - \hat{a}_j^\dagger \hat{a}_i)$  (current from  $i$  to  $j$ ).

- Show that current is Hermitian:  $\hat{I}_{ij}^\dagger = -\hat{I}_{ij}$  (current reverses direction under conjugation)
- Verify current conservation:  $\frac{d\hat{n}_i}{dt} + \sum_j \hat{I}_{ij} = 0$  (particle number at site  $i$  changes due to net current flow)
- For a superposition  $|\psi\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$  on a two-site lattice, calculate  $\langle \hat{I}_{12} \rangle$
- Explain the relationship between  $\langle \hat{I}_{ij} \rangle$  and quantum coherence between sites
- In the tight-binding ground state, is there net particle current? Explain why or why not

**Problem 3.11** (*Commutator Algebra for Lattice Operators*)

The commutation relations  $[\hat{a}_i, \hat{a}_j^\dagger] = \delta_{ij}$  and  $[\hat{a}_i, \hat{a}_j] = 0$  (for bosons/hardcore bosons) determine all operator algebra on the lattice.

- Show that  $[\hat{n}_i, \hat{n}_j] = 0$  for all  $i, j$  (particle numbers commute)
- Calculate  $[\hat{a}_i^\dagger \hat{a}_j, \hat{a}_k^\dagger \hat{a}_l]$  for general sites (hint: use Jacobi identity and commutation relations)
- Show that hopping operators at different sites commute:  $[\hat{a}_i^\dagger \hat{a}_{i+1}, \hat{a}_{j+1}^\dagger \hat{a}_j] = 0$  for non-overlapping pairs
- For the number operator  $\hat{N} = \sum_i \hat{n}_i$ , verify that  $[\hat{N}, \hat{a}_i^\dagger] = \hat{a}_i^\dagger$  (creation increases particle count)
- Compute  $[\hat{H}_{\text{hop}}, \hat{n}_i]$  for the hopping Hamiltonian and explain what symmetry this reveals

## References and Further Reading

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## Chapter 4

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# Continuous Time Evolution and Dynamics

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### 4.1 Introduction: From Discrete Rotations to Continuous Time Evolution

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Throughout the previous chapters, we have worked exclusively with unitary operators—from the discrete 90-degree rotations of the Bloch cube to the universal gate sets for many-body quantum systems. These unitary transformations represent the fundamental mathematical structure of quantum mechanics. Now we explore how unitary evolution emerges from exponentiation of Hermitian operators, the infinitesimal generators of time translation, connecting our discrete manipulations to the flowing dynamics of physical systems.

This chapter bridges the gap between our discrete quantum toolkit and the continuous time evolution that governs real physical systems. We will see that the discrete rotations we have studied are special cases of a much richer continuous dynamics, where time evolution is generated by Hermitian operators called Hamiltonians. The Schrödinger equation emerges naturally as the differential form of unitary evolution, and the rotation matrices we constructed for the Bloch cube become specific instances of the general exponential formula  $U(t) = e^{-iHt/\hbar}$ .

The transition from discrete to continuous operations reveals new physical phenomena impossible in our discrete framework. Quantum interference patterns arise from precise phase relationships that evolve continuously. Resonance effects depend critically on timing and frequency matching. Conservation laws emerge from symmetries of the Hamiltonian. Most importantly, we will see how the non-commutativity we encountered in the Pauli matrix algebra underlies all quantum dynamics, from simple two-level systems to complex many-body physics.

### 4.2 Unitary Operators and the Schrödinger Equation

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#### 4.2.1 The Fundamental Postulate of Quantum Dynamics

Every quantum mechanical operation we have encountered—from Bloch cube rotations to quantum gates—has been described by a unitary operator. For any quantum system evolving from time  $t = 0$

to time  $t$ , there exists a unitary operator  $\hat{U}(t)$  such that:

$$|\psi(t)\rangle = \hat{U}(t)|\psi(0)\rangle \quad (4.1)$$

The time evolution operator must satisfy several physical requirements that ensure the consistency of quantum mechanics:

First, it must be unitary:  $\hat{U}^\dagger(t)\hat{U}(t) = \hat{I}$ . This guarantees that quantum states remain normalized for all times. If we start with a normalized state  $\langle\psi(0)|\psi(0)\rangle = 1$ , then  $\langle\psi(t)|\psi(t)\rangle = \langle\psi(0)|\hat{U}^\dagger(t)\hat{U}(t)|\psi(0)\rangle = 1$  for all times.

Second, it must satisfy the group property:  $\hat{U}(t_1)\hat{U}(t_2) = \hat{U}(t_1 + t_2)$ . Evolution for time  $t_1$  followed by evolution for time  $t_2$  must equal evolution for the total time  $t_1 + t_2$ . This property is essential for the consistency of quantum mechanics and ensures that the dynamics are Markovian—that is, future evolution depends only on the present state, not on the past history. A process is Markovian when the probability of transitioning to a future state depends solely on the current state, with no memory of how that state was reached.

Third, it must be continuous:  $\hat{U}(0) = \hat{I}$  and  $\hat{U}(t)$  varies smoothly with time. At  $t = 0$ , no evolution has occurred, so the evolution operator is the identity. The smoothness condition ensures that quantum states don't evolve discontinuously.

#### 4.2.2 Hamiltonian: The Generator of Time Evolution

For infinitesimal time evolution, we can expand the unitary operator to first order in  $dt$ :

$$\hat{U}(dt) = \hat{I} + \hat{G}dt + O(dt^2) \quad (4.2)$$

where  $\hat{G}$  is the generator of time evolution. Since  $\hat{U}(dt)$  must be unitary to first order, we have:

$$\hat{U}^\dagger(dt)\hat{U}(dt) = (\hat{I} + \hat{G}^\dagger dt)(\hat{I} + \hat{G}dt) + O(dt^2) \quad (4.3)$$

$$= \hat{I} + (\hat{G}^\dagger + \hat{G})dt + O(dt^2) = \hat{I} \quad (4.4)$$

This requires  $\hat{G}^\dagger + \hat{G} = 0$ , so  $\hat{G}$  must be anti-Hermitian. We conventionally write  $\hat{G} = -i\hat{H}/\hbar$  where  $\hat{H}$  is a Hermitian operator called the Hamiltonian. This gives:

$$\hat{U}(dt) = \hat{I} - i\hat{H}dt/\hbar \quad (4.5)$$

#### 4.2.3 The Schrödinger Equation for Time-Independent Hamiltonians

For finite time evolution with a **time-independent** Hamiltonian, the group property demands that:

$$\hat{U}(t + dt) = \hat{U}(dt)\hat{U}(t) = \left(\hat{I} - i\hat{H}dt/\hbar\right)\hat{U}(t) \quad (4.6)$$

Rearranging and taking the limit  $dt \rightarrow 0$ :

$$\frac{\hat{U}(t + dt) - \hat{U}(t)}{dt} = -i\hat{H}\hat{U}(t)/\hbar \quad (4.7)$$

$$\frac{d\hat{U}(t)}{dt} = -i\hat{H}\hat{U}(t)/\hbar \quad (4.8)$$

This differential equation has the solution:

$$\hat{U}(t) = e^{-i\hat{H}t/\hbar} \quad (4.9)$$



**provided the Hamiltonian is time-independent.** This crucial assumption allows the exponential form and underlies much of elementary quantum mechanics.

Applying this to an arbitrary state gives us the time-dependent Schrödinger equation:

$$|\psi(t)\rangle = \hat{U}(t)|\psi(0)\rangle = e^{-i\hat{H}t/\hbar}|\psi(0)\rangle \quad (4.10)$$

$$\frac{d}{dt}|\psi(t)\rangle = \frac{d}{dt}e^{-i\hat{H}t/\hbar}|\psi(0)\rangle = -i\hat{H}e^{-i\hat{H}t/\hbar}|\psi(0)\rangle/\hbar \quad (4.11)$$

$$= -i\hat{H}|\psi(t)\rangle/\hbar \quad (4.12)$$

Rearranging yields the familiar form:

$$i\hbar \frac{d}{dt}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle \quad (4.13)$$

The Schrödinger equation is thus the differential formulation of unitary quantum evolution. Every discrete unitary operation we have studied can be understood as arising from continuous evolution under an appropriate Hamiltonian for a specific duration.

## 4.3 Series Solution and Temporal Composition

### 4.3.1 Building Finite Evolution from Infinitesimal Steps

The exponential form of the time evolution operator  $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$  has a profound physical interpretation that connects directly to our understanding of infinitesimal evolution. The fundamental definition of the exponential function:

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n \quad (4.14)$$

Applied to our time evolution operator, this becomes:

$$\hat{U}(t) = e^{-i\hat{H}t/\hbar} = \lim_{n \rightarrow \infty} \left(\hat{I} - \frac{i\hat{H}t}{n\hbar}\right)^n \quad (4.15)$$

This formula has immediate physical meaning: finite time evolution emerges from composing many infinitesimal steps. If we divide the total time  $t$  into  $n$  small intervals of duration  $\Delta t = t/n$ , then:

$$\hat{U}(t) = \lim_{n \rightarrow \infty} [\hat{U}(\Delta t)]^n = \lim_{n \rightarrow \infty} \left[\hat{I} - \frac{i\hat{H}\Delta t}{\hbar}\right]^n \quad (4.16)$$

This shows that evolving for time  $t$  is equivalent to applying the infinitesimal evolution operator  $n$  times, where each application evolves the system for time  $t/n$ . As  $n \rightarrow \infty$ , the discrete steps become continuous evolution.

The exponential can also be understood through its series expansion:

$$\hat{U}(t) = \sum_{n=0}^{\infty} \frac{(-it/\hbar)^n}{n!} \hat{H}^n = \hat{I} - \frac{it}{\hbar} \hat{H} + \frac{1}{2!} \left(\frac{-it}{\hbar}\right)^2 \hat{H}^2 + \dots \quad (4.17)$$

This series provides computational insight. The first few terms show how finite time evolution emerges from infinitesimal changes: the zeroth order term  $\hat{I}$  represents no evolution, the first order

term  $-\frac{it}{\hbar}\hat{H}$  gives the instantaneous rate of change, and the second order term provides acceleration effects.

For two-level systems where  $\hat{H} = \vec{h} \cdot \vec{\sigma}$ , the series simplifies dramatically because  $(\vec{h} \cdot \vec{\sigma})^{2n} = |\vec{h}|^{2n} \hat{I}$  and  $(\vec{h} \cdot \vec{\sigma})^{2n+1} = |\vec{h}|^{2n} (\vec{h} \cdot \vec{\sigma})$ . This allows us to separate even and odd powers:

$$\hat{U}(t) = \sum_{n=0}^{\infty} \frac{(-1)^n (|\vec{h}|t/\hbar)^{2n}}{(2n)!} \hat{I} + \sum_{n=0}^{\infty} \frac{(-1)^n (-i) (|\vec{h}|t/\hbar)^{2n+1}}{(2n+1)!} \frac{\vec{h} \cdot \vec{\sigma}}{|\vec{h}|} \quad (4.18)$$

$$= \cos(|\vec{h}|t/\hbar) \hat{I} - i \sin(|\vec{h}|t/\hbar) \frac{\vec{h} \cdot \vec{\sigma}}{|\vec{h}|} \quad (4.19)$$

This recovers our rotation formula and demonstrates how the matrix exponential reduces to familiar trigonometric functions.

### 4.3.2 Temporal Composition and the Trotter Formula

A fundamental property of time evolution is that it composes: evolution for time  $t_1$  followed by evolution for time  $t_2$  equals evolution for total time  $t_1 + t_2$ . Mathematically:

$$\hat{U}(t_1 + t_2) = \hat{U}(t_2) \hat{U}(t_1) = e^{-i\hat{H}(t_1+t_2)/\hbar} \quad (4.20)$$

This composition property allows us to build long-time evolution from many short-time steps, as we saw in the limit definition of the exponential:

$$\hat{U}(T) = \lim_{N \rightarrow \infty} \left[ \hat{U}(T/N) \right]^N = \lim_{N \rightarrow \infty} \left[ e^{-i\hat{H}T/(N\hbar)} \right]^N \quad (4.21)$$

For small time steps  $\Delta t = T/N$ , we can approximate:

$$\hat{U}(\Delta t) \approx \hat{I} - i\hat{H}\Delta t/\hbar \quad (4.22)$$

This leads to the discrete time evolution:

$$\hat{U}(T) \approx \left( \hat{I} - i\hat{H}T/(N\hbar) \right)^N \quad (4.23)$$

In the limit  $N \rightarrow \infty$ , this reproduces the continuous evolution operator, as we established earlier through the exponential limit definition. This is not just a mathematical curiosity—it shows that continuous quantum evolution can be approximated arbitrarily well by discrete time steps, a principle fundamental to quantum simulation.

When the Hamiltonian is a sum of non-commuting terms,  $\hat{H} = \hat{H}_1 + \hat{H}_2$ , the evolution operator cannot be written as a simple product  $e^{-i\hat{H}_1 t/\hbar} e^{-i\hat{H}_2 t/\hbar}$  because the exponentials don't factor when the operators don't commute. However, for small time steps, the Trotter formula provides an approximation:

$$e^{-i(\hat{H}_1 + \hat{H}_2)\Delta t/\hbar} \approx e^{-i\hat{H}_1 \Delta t/\hbar} e^{-i\hat{H}_2 \Delta t/\hbar} + O((\Delta t)^2) \quad (4.24)$$

This allows complex many-body evolution to be decomposed into simpler pieces that can be implemented individually. For quantum simulation, this means that complicated Hamiltonians can be built from elementary quantum gates, each implementing one term in the Hamiltonian decomposition. The Trotter approach bridges the gap between continuous quantum dynamics and discrete quantum computation, showing how any quantum evolution can be efficiently approximated using a quantum computer with sufficient gates and qubits.

## 4.4 Bloch Sphere Rotations from Hamiltonians

### 4.4.1 From Discrete Faces to Continuous Sphere

The six faces of the Bloch cube correspond to the eigenstates of the three Pauli matrices:

$$\sigma_x |\pm x\rangle = \pm |\pm x\rangle \quad (4.25)$$

$$\sigma_y |\pm y\rangle = \pm |\pm y\rangle \quad (4.26)$$

$$\sigma_z |\pm z\rangle = \pm |\pm z\rangle \quad (4.27)$$

These discrete states are special points on the Bloch sphere, but the sphere contains infinitely many other quantum states. Any pure state of a two-level system can be written as:

$$|\psi(\theta, \phi)\rangle = \cos(\theta/2) | + z \rangle + e^{i\phi} \sin(\theta/2) | - z \rangle \quad (4.28)$$

where  $\theta \in [0, \pi]$  and  $\phi \in [0, 2\pi)$  are the polar and azimuthal angles on the unit sphere.

### 4.4.2 Pauli Matrices as Generators of Rotations

The continuous rotations that connect all points on the Bloch sphere are generated by the Pauli matrices through exponentiation. For any unit vector  $\hat{n} = (n_x, n_y, n_z)$  and rotation angle  $\theta$ , the rotation operator is:

$$\hat{R}_{\hat{n}}(\theta) = e^{-i\theta(\hat{n} \cdot \vec{\sigma})/2} \quad (4.29)$$

where  $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$  is the vector of Pauli matrices.

The key mathematical property that makes this exponential tractable is that  $(\hat{n} \cdot \vec{\sigma})^2 = \hat{I}$  for any unit vector  $\hat{n}$ . This allows us to evaluate the exponential using the identity:

$$e^{-i\theta(\hat{n} \cdot \vec{\sigma})/2} = \cos(\theta/2) \hat{I} - i \sin(\theta/2) (\hat{n} \cdot \vec{\sigma}) \quad (4.30)$$

This is the matrix analog of Euler's formula  $e^{i\theta} = \cos \theta + i \sin \theta$ .

For rotation about the z-axis by angle  $\theta$ :

$$\hat{R}_z(\theta) = e^{-i\theta\sigma_z/2} = \cos(\theta/2) \hat{I} - i \sin(\theta/2) \sigma_z \quad (4.31)$$

$$= \begin{pmatrix} \cos(\theta/2) - i \sin(\theta/2) & 0 \\ 0 & \cos(\theta/2) + i \sin(\theta/2) \end{pmatrix} \quad (4.32)$$

$$= \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix} \quad (4.33)$$

For rotation about the x-axis by angle  $\theta$ :

$$\hat{R}_x(\theta) = \cos(\theta/2) \hat{I} - i \sin(\theta/2) \sigma_x \quad (4.34)$$

$$= \begin{pmatrix} \cos(\theta/2) & -i \sin(\theta/2) \\ -i \sin(\theta/2) & \cos(\theta/2) \end{pmatrix} \quad (4.35)$$

The discrete rotations of the Bloch cube correspond to specific angles. For instance, a 90-degree rotation about the z-axis gives  $\hat{Z} = \hat{R}_z(\pi/2)$ , while a 90-degree rotation about the x-axis yields  $\hat{X} = \hat{R}_x(\pi/2)$ .

**Bloch Cube Visualization and Energy Conservation:** An important property of rotations about the z-axis is that they preserve the expectation value of energy when the Hamiltonian is

proportional to  $\sigma_z$ . Consider a state represented on the Bloch cube undergoing rotation about the z-axis. The energy expectation value is given by  $\langle \hat{H} \rangle = \langle \psi | \sigma_z | \psi \rangle$ , which corresponds to the z-component of the Bloch vector. When we rotate the Bloch cube about the z-axis, the z-component remains unchanged—you can visualize this by imagining spinning the cube about its vertical axis. The x and y components change, but the height (z-component) of any point on the cube stays the same. This geometric invariance directly translates to conservation of energy during the rotation.

#### 4.4.3 Physical Interpretation: Spin in a Magnetic Field

The most direct physical realization of these rotations is a spin-1/2 particle in a magnetic field. The interaction Hamiltonian is:

$$\hat{H} = -\vec{\mu} \cdot \vec{B} = -\gamma \vec{S} \cdot \vec{B} = -\frac{\gamma \hbar}{2} \vec{\sigma} \cdot \vec{B} \quad (4.36)$$

where  $\gamma$  is the gyromagnetic ratio and  $\vec{S} = \frac{\hbar}{2} \vec{\sigma}$  is the spin angular momentum.

For a magnetic field  $\vec{B} = B\hat{n}$  in direction  $\hat{n}$ , the time evolution operator becomes:

$$\hat{U}(t) = e^{-i\hat{H}t/\hbar} = e^{i\gamma Bt(\hat{n} \cdot \vec{\sigma})/2} = \hat{R}_{\hat{n}}(\gamma Bt) \quad (4.37)$$

The spin precesses about the magnetic field direction with Larmor frequency  $\omega_L = \gamma B$ . The angle of rotation grows linearly with time, implementing precisely the continuous rotations we derived algebraically.

#### Example 1

**Larmor Precession** Consider a spin initially in the  $|+x\rangle$  state in a magnetic field  $\vec{B} = B\hat{z}$ . The time evolution is:

$$|\psi(t)\rangle = \hat{R}_z(\omega_L t) | + x \rangle \quad (4.38)$$

$$= e^{-i\omega_L t \sigma_z/2} \frac{1}{\sqrt{2}} (| + z \rangle + | - z \rangle) \quad (4.39)$$

$$= \frac{1}{\sqrt{2}} (e^{-i\omega_L t/2} | + z \rangle + e^{i\omega_L t/2} | - z \rangle) \quad (4.40)$$

$$= \frac{e^{-i\omega_L t/2}}{\sqrt{2}} (| + z \rangle + e^{i\omega_L t} | - z \rangle) \quad (4.41)$$

Factoring out the global phase and converting back to Cartesian coordinates:

$$\langle \sigma_x \rangle(t) = \cos(\omega_L t) \quad (4.42)$$

$$\langle \sigma_y \rangle(t) = \sin(\omega_L t) \quad (4.43)$$

$$\langle \sigma_z \rangle(t) = 0 \quad (4.44)$$

The spin precesses in the xy-plane with period  $T = 2\pi/\omega_L$ , tracing a circle of constant latitude on the Bloch sphere.

## 4.5 Diagonalizing the Hamiltonian

### 4.5.1 Energy Eigenstates and Eigenvalues

The exponential form of the time evolution operator  $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$  is most easily evaluated when the Hamiltonian is diagonal. For any Hermitian operator, we can find a basis of eigenstates where the matrix representation is diagonal.

For a general two-level Hamiltonian of the form  $\hat{H} = \vec{h} \cdot \vec{\sigma}$ , the eigenvalue equation is:

$$\hat{H}|E_{\pm}\rangle = E_{\pm}|E_{\pm}\rangle \quad (4.45)$$

The eigenvalues are  $E_{\pm} = \pm|\vec{h}|$ , and the corresponding eigenstates are:

$$|E_{+}\rangle = \cos(\alpha/2)|+z\rangle + e^{i\beta}\sin(\alpha/2)|-z\rangle \quad (4.46)$$

$$|E_{-}\rangle = -e^{-i\beta}\sin(\alpha/2)|+z\rangle + \cos(\alpha/2)|-z\rangle \quad (4.47)$$

where the angles  $\alpha$  and  $\beta$  are determined by the direction of  $\vec{h}$ :

$$\cos \alpha = \frac{h_z}{|\vec{h}|} \quad (4.48)$$

$$e^{i\beta} = \frac{h_x + ih_y}{\sqrt{h_x^2 + h_y^2}} \quad (4.49)$$

### 4.5.2 Similarity Transformations and Basis Changes

The diagonalization process can be understood as a similarity transformation. If we define the unitary matrix  $\hat{U}$  whose columns are the eigenstates, then:

$$\hat{U}^{\dagger}\hat{H}\hat{U} = \begin{pmatrix} E_{+} & 0 \\ 0 & E_{-} \end{pmatrix} \quad (4.50)$$

This similarity transformation rotates our coordinate system so that the Hamiltonian appears diagonal. In this "natural" basis, the time evolution becomes:

$$e^{-i\hat{H}t/\hbar} = \hat{U} \begin{pmatrix} e^{-iE_{+}t/\hbar} & 0 \\ 0 & e^{-iE_{-}t/\hbar} \end{pmatrix} \hat{U}^{\dagger} \quad (4.51)$$

The eigenstates evolve with simple phase factors, while arbitrary superpositions develop more complex dynamics through the interference of these phases.

#### Example 2

Constructing the Diagonalization Matrix for a Bloch Cube State Consider a Hamiltonian representing a magnetic field along the x-direction:

$$\hat{H} = \frac{\hbar\omega}{2}\sigma_x = \frac{\hbar\omega}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (4.52)$$

The eigenvalues are  $E_{\pm} = \pm\frac{\hbar\omega}{2}$ . To find the eigenstates, we solve:

$$\sigma_x|\psi\rangle = \pm|\psi\rangle \quad (4.53)$$

The eigenstates are the  $|\pm x\rangle$  states from the Bloch cube:

$$|+x\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + |-z\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (4.54)$$

$$|-x\rangle = \frac{1}{\sqrt{2}}(|+z\rangle - |-z\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (4.55)$$

Now we construct the unitary matrix  $\hat{U}$  with these eigenstates as columns:

$$\hat{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (4.56)$$

where the first column is  $|+x\rangle$  and the second column is  $|-x\rangle$ .

The adjoint is:

$$\hat{U}^\dagger = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (4.57)$$

Note that  $\hat{U}^\dagger = \hat{U}$  in this case because the matrix is real and symmetric.

Let's verify the diagonalization explicitly:

$$\hat{U}^\dagger \hat{H} \hat{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \frac{\hbar\omega}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (4.58)$$

$$= \frac{\hbar\omega}{4} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \quad (4.59)$$

$$= \frac{\hbar\omega}{4} \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} \quad (4.60)$$

$$= \begin{pmatrix} \frac{\hbar\omega}{2} & 0 \\ 0 & -\frac{\hbar\omega}{2} \end{pmatrix} \quad (4.61)$$

This is indeed diagonal with eigenvalues  $E_+ = \frac{\hbar\omega}{2}$  and  $E_- = -\frac{\hbar\omega}{2}$ .

**Time Evolution in Both Bases:** In the original  $\{|+z\rangle, |-z\rangle\}$  basis:

$$e^{-i\hat{H}t/\hbar} = e^{-i\omega t\sigma_x/2} = \cos(\omega t/2)\hat{I} - i\sin(\omega t/2)\sigma_x = \begin{pmatrix} \cos(\omega t/2) & -i\sin(\omega t/2) \\ -i\sin(\omega t/2) & \cos(\omega t/2) \end{pmatrix} \quad (4.62)$$

Using the diagonalization:

$$e^{-i\hat{H}t/\hbar} = \hat{U} \begin{pmatrix} e^{-i\omega t/2} & 0 \\ 0 & e^{i\omega t/2} \end{pmatrix} \hat{U}^\dagger \quad (4.63)$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} e^{-i\omega t/2} & 0 \\ 0 & e^{i\omega t/2} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (4.64)$$

$$= \frac{1}{2} \begin{pmatrix} e^{-i\omega t/2} + e^{i\omega t/2} & e^{-i\omega t/2} - e^{i\omega t/2} \\ e^{-i\omega t/2} - e^{i\omega t/2} & e^{-i\omega t/2} + e^{i\omega t/2} \end{pmatrix} \quad (4.65)$$

$$= \begin{pmatrix} \cos(\omega t/2) & -i\sin(\omega t/2) \\ -i\sin(\omega t/2) & \cos(\omega t/2) \end{pmatrix} \quad (4.66)$$

Both methods give the same result, confirming our diagonalization. The key insight: in the  $\{|+x\rangle, |-x\rangle\}$  basis (the eigenbasis), evolution is trivial—just phase factors. The complexity arises only when we express this evolution in a different basis like  $\{|+z\rangle, |-z\rangle\}$ .

**Example 3**

Diagonalizing a Tilted Field Hamiltonian Consider  $\hat{H} = \frac{\hbar\omega}{2}(\sigma_x + \sigma_z)/\sqrt{2}$ , representing a magnetic field at 45 degrees to the z-axis.

The field vector is  $\vec{h} = \frac{\hbar\omega}{2}(1, 0, 1)/\sqrt{2}$ , so  $|\vec{h}| = \frac{\hbar\omega}{2}$  and the angles are:

$$\cos \alpha = \frac{1/\sqrt{2}}{1} = \frac{1}{\sqrt{2}} \Rightarrow \alpha = \pi/4 \quad (4.67)$$

$$\beta = 0 \text{ (since } h_y = 0 \text{ and } h_x > 0) \quad (4.68)$$

The eigenvalues are  $E_+ = \frac{\hbar\omega}{2}$  and  $E_- = -\frac{\hbar\omega}{2}$ .

The eigenstates are:

$$|E_+\rangle = \cos(\pi/8)|+z\rangle + \sin(\pi/8)|-z\rangle \quad (4.69)$$

$$|E_-\rangle = -\sin(\pi/8)|+z\rangle + \cos(\pi/8)|-z\rangle \quad (4.70)$$

To construct the diagonalization matrix  $\hat{V}$  whose columns are the eigenstates:

$$\hat{V} = \begin{pmatrix} \cos(\pi/8) & -\sin(\pi/8) \\ \sin(\pi/8) & \cos(\pi/8) \end{pmatrix} \quad (4.71)$$

The time evolution operator in the energy eigenbasis is:

$$\hat{U}_{\text{diag}}(t) = \begin{pmatrix} e^{-i\omega t} & 0 \\ 0 & e^{i\omega t} \end{pmatrix} \quad (4.72)$$

Following the pattern established in equation (4.51), the time evolution operator in the original basis is:

$$e^{-i\hat{H}t/\hbar} = \hat{V}\hat{U}_{\text{diag}}(t)\hat{V}^\dagger = \hat{V} \begin{pmatrix} e^{-i\omega t} & 0 \\ 0 & e^{i\omega t} \end{pmatrix} \hat{V}^\dagger \quad (4.73)$$

Therefore, any initial state evolves as:

$$|\psi(t)\rangle = \hat{V} \begin{pmatrix} e^{-i\omega t} & 0 \\ 0 & e^{i\omega t} \end{pmatrix} \hat{V}^\dagger |\psi(0)\rangle \quad (4.74)$$

This triple matrix product structure shows how the evolution decomposes into: (1) transformation to the eigenbasis via  $\hat{V}^\dagger$ , (2) simple phase evolution in the eigenbasis, and (3) transformation back to the original basis via  $\hat{V}$ . The rotation axis for this Hamiltonian is along  $(1, 0, 1)/\sqrt{2}$ , and states rotate about this axis with frequency  $2\omega$ .

**4.5.3 General Rotation Formula**

The diagonalization procedure leads to a general formula for rotations about an arbitrary axis. For any unit vector  $\hat{n}$  and angle  $\theta$ :

$$\hat{R}_{\hat{n}}(\theta) = \cos(\theta/2)\hat{I} - i\sin(\theta/2)(\hat{n} \cdot \vec{\sigma}) \quad (4.75)$$

This formula encapsulates all possible single-qubit operations and shows how any rotation can be decomposed into rotations about the coordinate axes. It also reveals that rotations about parallel

axes commute:

$$\hat{R}_{\hat{n}}(\theta_1)\hat{R}_{\hat{n}}(\theta_2) = \hat{R}_{\hat{n}}(\theta_1 + \theta_2) \quad (4.76)$$

while rotations about different axes generally do not commute, reflecting the non-Abelian structure of the rotation group.

## 4.6 Conservation Laws and Symmetries

### 4.6.1 The Connection Between Symmetry and Conservation

One of the most profound insights in physics is Noether's theorem, which states that every continuous symmetry corresponds to a conserved quantity. In quantum mechanics, this manifests as the requirement that observables commute with the Hamiltonian to be conserved.

**Simple Proof for Quantum Systems:** Consider a unitary transformation  $\hat{U}(\epsilon) = e^{-i\epsilon\hat{A}}$  generated by an observable  $\hat{A}$ , where  $\epsilon$  is an infinitesimal parameter. If the Hamiltonian is invariant under this transformation for all  $\epsilon$ , then:

$$\hat{U}^\dagger(\epsilon)\hat{H}\hat{U}(\epsilon) = \hat{H} \quad (4.77)$$

Expanding to first order in  $\epsilon$ :

$$(\hat{I} + i\epsilon\hat{A})\hat{H}(\hat{I} - i\epsilon\hat{A}) = \hat{H} \quad (4.78)$$

$$\hat{H} + i\epsilon(\hat{A}\hat{H} - \hat{H}\hat{A}) + O(\epsilon^2) = \hat{H} \quad (4.79)$$

$$i\epsilon[\hat{A}, \hat{H}] = 0 \quad (4.80)$$

Since this must hold for all  $\epsilon$ , we have  $[\hat{H}, \hat{A}] = 0$ . An observable  $\hat{A}$  is conserved if its expectation value remains constant under time evolution:

$$\frac{d}{dt}\langle\hat{A}\rangle = 0 \quad (4.81)$$

Using the Heisenberg equation of motion:

$$\frac{d}{dt}\langle\hat{A}\rangle = \frac{1}{i\hbar}\langle[\hat{A}, \hat{H}]\rangle \quad (4.82)$$

This vanishes if and only if  $[\hat{H}, \hat{A}] = 0$ . Thus, symmetries of the Hamiltonian (transformations leaving  $\hat{H}$  invariant) directly correspond to conserved quantities (observables that commute with  $\hat{H}$ ).

The physical interpretation is that  $\hat{A}$  generates a symmetry transformation that leaves the Hamiltonian invariant.

### 4.6.2 Conservation Laws in Two-Level Systems

For a general two-level Hamiltonian  $\hat{H} = \vec{h} \cdot \vec{\sigma}$ , the conserved quantity is the component of spin along the direction of  $\vec{h}$ :

$$\hat{A}_{\text{conserved}} = (\hat{h} \cdot \vec{\sigma}) = \frac{\vec{h} \cdot \vec{\sigma}}{|\vec{h}|} \quad (4.83)$$



This can be verified by direct calculation:

$$[\hat{H}, \hat{h} \cdot \vec{\sigma}] = \vec{h} \cdot \vec{\sigma} \cdot [\vec{h} \cdot \vec{\sigma}, \hat{h} \cdot \vec{\sigma}] \quad (4.84)$$

$$= |\vec{h}|^2 [\hat{h} \cdot \vec{\sigma}, \hat{h} \cdot \vec{\sigma}] = 0 \quad (4.85)$$

since any operator commutes with itself.

#### Example 4

**Energy and Angular Momentum Conservation** For a spin in a magnetic field  $\vec{B} = B\hat{z}$ , the Hamiltonian is  $\hat{H} = \frac{\hbar\omega}{2}\sigma_z$ .

The conserved quantities are energy ( $\hat{H}$  itself, since  $[\hat{H}, \hat{H}] = 0$ ) and the z-component of angular momentum ( $\hat{S}_z = \hbar\sigma_z/2$ , since  $[\hat{H}, \sigma_z] = 0$ ). The non-conserved quantities are the x-component, where  $[\hat{H}, \sigma_x] = \frac{\hbar\omega}{2}[\sigma_z, \sigma_x] = i\hbar\omega\sigma_y \neq 0$ , and the y-component, where  $[\hat{H}, \sigma_y] = \frac{\hbar\omega}{2}[\sigma_z, \sigma_y] = -i\hbar\omega\sigma_x \neq 0$ .

The x and y components oscillate with frequency  $2\omega$ , while the z-component and total energy remain constant. This reflects the rotational symmetry about the z-axis.

### 4.6.3 Noether's Theorem and Rotational Symmetry

The rotational symmetry of space leads to conservation of angular momentum. For a two-level system, if the Hamiltonian is invariant under rotations about a particular axis  $\hat{n}$ , then the component of angular momentum along that axis is conserved.

Mathematically, if  $\hat{H}$  commutes with the rotation operator  $\hat{R}_{\hat{n}}(\theta)$  for all angles  $\theta$ , then it also commutes with the generator of that rotation:

$$[\hat{H}, \hat{R}_{\hat{n}}(\theta)] = 0 \Rightarrow [\hat{H}, \hat{n} \cdot \vec{\sigma}] = 0 \quad (4.86)$$

This principle extends to larger systems and other symmetries. Time translation symmetry leads to energy conservation, spatial translation symmetry leads to momentum conservation, and gauge symmetries lead to charge conservation. The two-level system serves as a laboratory for understanding these fundamental connections between symmetry and conservation in their simplest manifestation.

## 4.7 Time-Dependence of Expectation Values and the Heisenberg Picture

### 4.7.1 General Formula for Time Evolution of Observables

The time dependence of expectation values is governed by a general formula that encompasses both the intrinsic time dependence of operators and their evolution due to the system dynamics. For any observable  $\hat{A}(t)$  that may have explicit time dependence:

$$\frac{d}{dt}\langle\hat{A}(t)\rangle = \frac{1}{i\hbar}\langle[\hat{A}(t), \hat{H}]\rangle + \left\langle\frac{\partial\hat{A}(t)}{\partial t}\right\rangle \quad (4.87)$$

This master equation shows two sources of time dependence:

- The commutator term  $\frac{1}{i\hbar}\langle[\hat{A}, \hat{H}]\rangle$  represents evolution due to the system dynamics
- The partial derivative term  $\left\langle\frac{\partial\hat{A}}{\partial t}\right\rangle$  accounts for explicit time dependence in the operator itself

When an observable commutes with the Hamiltonian and has no explicit time dependence, its expectation value is conserved. This provides the quantum mechanical foundation for conservation laws.

### 4.7.2 The Heisenberg Picture

In the Heisenberg picture, we transfer the time dependence from the states to the operators. While the Schrödinger picture keeps operators fixed and evolves states:

$$|\psi_S(t)\rangle = e^{-i\hat{H}t/\hbar}|\psi_S(0)\rangle \quad (4.88)$$

The Heisenberg picture keeps states fixed and evolves operators:

$$\hat{A}_H(t) = e^{i\hat{H}t/\hbar}\hat{A}_S e^{-i\hat{H}t/\hbar} \quad (4.89)$$

where subscripts S and H denote Schrödinger and Heisenberg pictures, respectively.

The expectation values remain the same in both pictures:

$$\langle\hat{A}\rangle(t) = \langle\psi_S(t)|\hat{A}_S|\psi_S(t)\rangle = \langle\psi_H|\hat{A}_H(t)|\psi_H\rangle \quad (4.90)$$

The Heisenberg equation of motion for operators is:

$$\frac{d\hat{A}_H(t)}{dt} = \frac{1}{i\hbar}[\hat{A}_H(t), \hat{H}] + \frac{\partial\hat{A}_H(t)}{\partial t} \quad (4.91)$$

This provides an alternative formulation of quantum dynamics where operators evolve while states remain fixed—often more convenient for calculating time-dependent expectation values.

### 4.7.3 Computing Expectation Values During Evolution

For a state evolving under a time-independent Hamiltonian, any observable  $\hat{A}$  has time-dependent expectation value:

$$\langle\hat{A}\rangle(t) = \langle\psi(t)|\hat{A}|\psi(t)\rangle = \langle\psi(0)|e^{i\hat{H}t/\hbar}\hat{A}e^{-i\hat{H}t/\hbar}|\psi(0)\rangle \quad (4.92)$$

This formula shows that the time dependence can be viewed in two equivalent ways: either the state evolves ( $|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}|\psi(0)\rangle$ ) while the operator  $\hat{A}$  remains fixed, or equivalently, the operator evolves (via the sandwich  $e^{i\hat{H}t/\hbar}\hat{A}e^{-i\hat{H}t/\hbar}$ ) while the state remains fixed. This second viewpoint is formalized as the Heisenberg picture, which we define in the next subsection.

#### Example 5

**Spin Evolution in a Magnetic Field** Consider a spin initially in the  $|+x\rangle$  state evolving under  $\hat{H} = \frac{\hbar\omega}{2}\sigma_z$ . Let's compute the expectation values of all three spin components.

Starting state:  $|\psi(0)\rangle = |+x\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + |-z\rangle)$

Time-evolved state:  $|\psi(t)\rangle = e^{-i\omega t\sigma_z/2}|+x\rangle = \frac{1}{\sqrt{2}}(e^{-i\omega t/2}|+z\rangle + e^{i\omega t/2}|-z\rangle)$

For the x-component:

$$\langle \sigma_x \rangle(t) = \langle \psi(t) | \sigma_x | \psi(t) \rangle \quad (4.93)$$

$$= \frac{1}{2} (e^{i\omega t/2} \langle +z | + e^{-i\omega t/2} \langle -z |) \sigma_x (e^{-i\omega t/2} | +z \rangle + e^{i\omega t/2} | -z \rangle) \quad (4.94)$$

$$= \frac{1}{2} (e^{i\omega t/2} \langle +z | + e^{-i\omega t/2} \langle -z |) (e^{i\omega t/2} | -z \rangle + e^{-i\omega t/2} | +z \rangle) \quad (4.95)$$

$$= \frac{1}{2} (e^{i\omega t} + e^{-i\omega t}) = \cos(\omega t) \quad (4.96)$$

For the y-component:

$$\langle \sigma_y \rangle(t) = \frac{1}{2} (e^{i\omega t/2} \langle +z | + e^{-i\omega t/2} \langle -z |) (-i | -z \rangle + i | +z \rangle) (e^{-i\omega t/2} | +z \rangle + e^{i\omega t/2} | -z \rangle) \quad (4.97)$$

$$= \frac{i}{2} (e^{i\omega t} - e^{-i\omega t}) = \sin(\omega t) \quad (4.98)$$

For the z-component:

$$\langle \sigma_z \rangle(t) = \frac{1}{2} (e^{i\omega t/2} \langle +z | + e^{-i\omega t/2} \langle -z |) (e^{-i\omega t/2} | +z \rangle - e^{i\omega t/2} | -z \rangle) \quad (4.99)$$

$$= \frac{1}{2} (1 - 1) = 0 \quad (4.100)$$

The results show that  $\langle \sigma_x \rangle$  and  $\langle \sigma_y \rangle$  oscillate sinusoidally with frequency  $\omega$ , while  $\langle \sigma_z \rangle$  remains constant at zero.

#### 4.7.4 Geometric Interpretation on the Bloch Sphere

The time evolution of expectation values has a beautiful geometric interpretation. The Bloch vector  $\vec{r}(t) = (\langle \sigma_x \rangle(t), \langle \sigma_y \rangle(t), \langle \sigma_z \rangle(t))$  traces a path on the Bloch sphere as the system evolves.

For a Hamiltonian  $\hat{H} = \vec{h} \cdot \vec{\sigma}$ , the Bloch vector rotates about the axis  $\hat{h} = \vec{h}/|\vec{h}|$  with angular frequency  $\omega = 2|\vec{h}|/\hbar$ . States that are eigenstates of the Hamiltonian correspond to fixed points on the rotation axis, while all other states trace circles of constant latitude.

The key insight is that quantum evolution preserves angles on the Bloch sphere. If two states initially make an angle  $\theta$ , they will continue to subtend the same angle throughout their evolution under any unitary transformation. This geometric constraint underlies many conservation laws and provides intuition for understanding complex quantum dynamics.

##### Example 6

**Understanding Quantum Interference Through Geometry** Consider two states  $|\psi_1\rangle$  and  $|\psi_2\rangle$  with Bloch vectors making an angle  $\theta$  with respect to each other. Their overlap is:

$$|\langle \psi_1 | \psi_2 \rangle|^2 = \cos^2(\theta/2) \quad (4.101)$$

This relationship between the geometric angle and quantum overlap explains why orthogonal states (separated by  $180^\circ$  on the Bloch sphere) have zero overlap, antipodal states differ by a relative phase but have the same measurement probabilities, and states separated by  $90^\circ$

have overlap 1/2, leading to maximum interference visibility.

During time evolution, the angle between states is preserved, so their interference pattern depends only on their initial relative orientation and any additional phases accumulated during evolution.

## 4.8 Time-Dependent Hamiltonians and Nuclear Magnetic Resonance

### 4.8.1 The Problem with Time-Dependent Hamiltonians

Up to this point, we have considered only time-independent Hamiltonians, where energy eigenstates evolve with simple exponential phases and the time evolution operator takes the form  $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$ . However, many physical systems involve time-dependent external fields that drive transitions between energy levels.

When the Hamiltonian depends explicitly on time,  $\hat{H}(t)$ , naively writing

$$\hat{U}(t) = \exp\left(-\frac{i}{\hbar} \int_0^t \hat{H}(t') dt'\right) \quad (\text{INCORRECT in general}) \quad (4.102)$$

is **wrong** unless  $[\hat{H}(t_1), \hat{H}(t_2)] = 0$  for all times  $t_1, t_2$ .

**Why?** The exponential of an integral is not generally equal to integrating exponentials when the integrand doesn't commute with itself at different times. To see this, consider that for time-independent  $\hat{H}$ :

$$e^{-i(\hat{H}_1 + \hat{H}_2)t/\hbar} \neq e^{-i\hat{H}_1 t/\hbar} e^{-i\hat{H}_2 t/\hbar} \quad (4.103)$$

unless  $[\hat{H}_1, \hat{H}_2] = 0$ . The same principle applies when comparing  $\hat{H}(t_1)$  and  $\hat{H}(t_2)$  at different times.

### 4.8.2 Derivation of the Time-Ordered Solution

The correct approach starts from the time-dependent Schrödinger equation:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle \quad (4.104)$$

Integrating both sides from  $t_0$  to  $t$ :

$$|\psi(t)\rangle = |\psi(t_0)\rangle - \frac{i}{\hbar} \int_{t_0}^t \hat{H}(t_1) |\psi(t_1)\rangle dt_1 \quad (4.105)$$

This is an integral equation since  $|\psi(t)\rangle$  appears on both sides. We solve it iteratively by substituting the expression for  $|\psi(t_1)\rangle$  back into itself:

$$|\psi(t)\rangle = |\psi(t_0)\rangle - \frac{i}{\hbar} \int_{t_0}^t \hat{H}(t_1) |\psi(t_0)\rangle dt_1 \quad (4.106)$$

$$+ \left(-\frac{i}{\hbar}\right)^2 \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \hat{H}(t_1) \hat{H}(t_2) |\psi(t_0)\rangle + \dots \quad (4.107)$$

This is the **Dyson series**. The complete time evolution operator is:

$$\hat{U}(t, t_0) = \mathbb{1} + \sum_{n=1}^{\infty} \left( -\frac{i}{\hbar} \right)^n \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \cdots \int_{t_0}^{t_{n-1}} dt_n \hat{H}(t_1) \hat{H}(t_2) \cdots \hat{H}(t_n) \quad (4.108)$$

Note the crucial ordering:  $t \geq t_1 \geq t_2 \geq \cdots \geq t_n \geq t_0$ . The operators are naturally **time-ordered** with later times to the left.

### 4.8.3 The Time-Ordering Operator

The time-ordering operator  $\hat{\mathcal{T}}$  (or simply  $\mathcal{T}$ ) arranges operators chronologically with **later times to the left**:

$$\hat{\mathcal{T}}\{\hat{A}(t_1)\hat{B}(t_2)\} = \begin{cases} \hat{A}(t_1)\hat{B}(t_2) & \text{if } t_1 > t_2 \\ \hat{B}(t_2)\hat{A}(t_1) & \text{if } t_2 > t_1 \end{cases} \quad (4.109)$$

More generally, for multiple operators:

$$\hat{\mathcal{T}}\{\hat{A}(t_1)\hat{B}(t_2)\hat{C}(t_3)\cdots\} = \text{product with operators ordered by decreasing time} \quad (4.110)$$

Using the time-ordering operator, the Dyson series becomes:

$$\hat{U}(t, t_0) = \hat{\mathcal{T}} \exp \left( -\frac{i}{\hbar} \int_{t_0}^t \hat{H}(t') dt' \right) \quad (4.111)$$

$$= \hat{\mathcal{T}} \left\{ \sum_{n=0}^{\infty} \frac{1}{n!} \left( -\frac{i}{\hbar} \right)^n \left[ \int_{t_0}^t \hat{H}(t') dt' \right]^n \right\} \quad (4.112)$$

The time-ordering operator  $\hat{\mathcal{T}}$  converts the unrestricted integral over all time orderings into the properly ordered Dyson series.

### 4.8.4 Physical Interpretation: Causality

Time ordering reflects **causality**: an operator at time  $t_1$  can only affect the system before operators at later times  $t_2 > t_1$  act. The time-ordered product ensures:

- (a) **Sequential application**: The Hamiltonian at earlier times acts first
- (b) **Non-commutativity**: When  $[\hat{H}(t_1), \hat{H}(t_2)] \neq 0$ , the order matters
- (c) **Perturbation theory**: Each term in the Dyson series represents  $n$ th-order processes

#### Example: Time Ordering in a Driven Two-Level System

Consider a spin-1/2 system in a time-varying magnetic field  $\vec{B}(t)$ :

$$\hat{H}(t) = -\gamma \vec{B}(t) \cdot \hat{\vec{S}} = -\gamma [B_x(t)\hat{S}_x + B_y(t)\hat{S}_y + B_z(t)\hat{S}_z] \quad (4.113)$$

If  $\vec{B}(t)$  changes direction, then  $[\hat{H}(t_1), \hat{H}(t_2)] \neq 0$  since different spin components don't commute. For example:

$$[\hat{S}_x, \hat{S}_y] = i\hbar \hat{S}_z \neq 0 \quad (4.114)$$

The evolution cannot be written as a simple exponential—time ordering is essential. To first order in perturbation theory:

$$|\psi(t)\rangle \approx \left[ \mathbb{1} - \frac{i}{\hbar} \int_0^t \hat{H}(t_1) dt_1 \right] |\psi(0)\rangle \quad (4.115)$$

To second order, we must respect time ordering:

$$|\psi(t)\rangle \approx \left[ \mathbb{1} - \frac{i}{\hbar} \int_0^t \hat{H}(t_1) dt_1 \right. \quad (4.116)$$

$$\left. + \left( -\frac{i}{\hbar} \right)^2 \int_0^t dt_1 \int_0^{t_1} dt_2 \hat{H}(t_1) \hat{H}(t_2) \right] |\psi(0)\rangle \quad (4.117)$$

The nested integral structure with  $t_1 > t_2$  ensures that the Hamiltonian at time  $t_2$  acts before the Hamiltonian at time  $t_1$ , preserving causality.

#### 4.8.5 Practical Computation and Connection to Perturbation Theory

For weak time-dependence or perturbative calculations, we truncate the Dyson series:

- **Zeroth order:**  $\hat{U} \approx \mathbb{1}$  (no evolution)
- **First order:**  $\hat{U} \approx \mathbb{1} - \frac{i}{\hbar} \int_{t_0}^t \hat{H}(t') dt'$
- **Second order:** Include the double integral term

This perturbative approach forms the foundation of time-dependent perturbation theory and Fermi's Golden Rule.

**Reduction to Time-Independent Case:** When  $\hat{H}(t) = \hat{H}$  is time-independent, all operators at different times commute:  $[\hat{H}(t_1), \hat{H}(t_2)] = [\hat{H}, \hat{H}] = 0$ . Time ordering becomes trivial, and:

$$\hat{U}(t, t_0) = \mathcal{T} \exp \left( -\frac{i}{\hbar} \int_{t_0}^t \hat{H} dt' \right) \quad (4.118)$$

$$= \exp \left( -\frac{i}{\hbar} \hat{H}(t - t_0) \right) \quad (4.119)$$

recovering our familiar exponential form.

#### 4.8.6 Connection to the Interaction Picture

In the interaction picture with  $\hat{H} = \hat{H}_0 + \hat{V}(t)$ , time ordering appears naturally in the evolution operator:

$$\hat{U}_I(t, t_0) = \hat{\mathcal{T}} \exp \left( -\frac{i}{\hbar} \int_{t_0}^t \hat{V}_I(t') dt' \right) \quad (4.120)$$

where  $\hat{V}_I(t) = e^{i\hat{H}_0 t/\hbar} \hat{V}(t) e^{-i\hat{H}_0 t/\hbar}$  is the interaction in the interaction picture. This formulation is essential for quantum field theory and many-body physics.

**Key Takeaway:** Time ordering is not merely a mathematical technicality but a fundamental requirement when dealing with time-dependent quantum systems where the Hamiltonian at different times doesn't commute. It ensures causality and provides the correct perturbative expansion for quantum evolution.

### 4.8.7 The Driven Two-Level System

A fundamental example is a two-level system with a static energy splitting driven by an oscillating field:

$$\hat{H}(t) = \frac{\hbar\omega_0}{2}\sigma_z + \frac{\hbar\Omega}{2}[\sigma_x \cos(\omega t) + \sigma_y \sin(\omega t)] \quad (4.121)$$

This describes numerous physical systems including an atom in a laser field, where  $\omega_0$  is the atomic transition frequency and  $\Omega$  is proportional to the electric field amplitude; a nuclear spin in an oscillating magnetic field (NMR), where  $\omega_0$  is the Larmor frequency and  $\Omega$  is the Rabi frequency; and a superconducting qubit driven by microwave pulses for quantum computation.

### 4.8.8 The Rotating Frame, Interaction Picture, and Resonance

The key insight for solving driven systems is to transform to the interaction picture (rotating frame). This is analogous to moving to a reference frame rotating at the drive frequency.

**The Interaction Picture:** Recall from Section 4.7 that we have two equivalent formulations of quantum dynamics:

- **Schrödinger picture:** States evolve, operators are fixed:  $|\psi_S(t)\rangle = e^{-i\hat{H}t/\hbar}|\psi_S(0)\rangle$
- **Heisenberg picture:** Operators evolve, states are fixed:  $\hat{A}_H(t) = e^{i\hat{H}t/\hbar}\hat{A}_S e^{-i\hat{H}t/\hbar}$

For time-dependent Hamiltonians, we introduce a third picture that combines features of both. When the Hamiltonian can be separated into a time-independent part  $\hat{H}_0$  and a time-dependent perturbation  $\hat{V}(t)$ :

$$\hat{H}(t) = \hat{H}_0 + \hat{V}(t) \quad (4.122)$$

we define the **interaction picture** (also called the **Dirac picture**) as follows:

$$\text{Interaction picture states: } |\psi_I(t)\rangle = e^{i\hat{H}_0 t/\hbar}|\psi_S(t)\rangle \quad (4.123)$$

$$\text{Interaction picture operators: } \hat{A}_I(t) = e^{i\hat{H}_0 t/\hbar}\hat{A}_S e^{-i\hat{H}_0 t/\hbar} \quad (4.124)$$

In this picture, both states and operators evolve with time, but the time evolution of operators is governed only by  $\hat{H}_0$ , while the time evolution of states is governed by the transformed perturbation  $\hat{V}_I(t)$ . The interaction picture Schrödinger equation becomes:

$$i\hbar \frac{d}{dt}|\psi_I(t)\rangle = \hat{V}_I(t)|\psi_I(t)\rangle \quad (4.125)$$

where  $\hat{V}_I(t) = e^{i\hat{H}_0 t/\hbar}\hat{V}(t)e^{-i\hat{H}_0 t/\hbar}$  is the interaction picture perturbation.

**Connection to the Dyson Series:** The interaction picture provides the natural setting for the Dyson series developed in Section 4.8.3. The full Schrödinger picture evolution operator can be written as:

$$\hat{U}_S(t, t_0) = e^{-i\hat{H}_0(t-t_0)/\hbar}\hat{U}_I(t, t_0) \quad (4.126)$$

where the interaction picture evolution operator is given by the Dyson series applied to  $\hat{V}_I(t)$ :

$$\hat{U}_I(t, t_0) = \mathcal{T} \exp \left( -\frac{i}{\hbar} \int_{t_0}^t \hat{V}_I(t') dt' \right) \quad (4.127)$$

This separation is powerful because:

- The evolution due to  $\hat{H}_0$  is trivial (simple exponential)
- Only the perturbation  $\hat{V}_I(t)$  requires time-ordering via the Dyson series
- When  $\hat{V}_I$  is small, the Dyson series converges rapidly, enabling perturbative calculations

For the driven two-level system below, when we work in the rotating frame and apply the rotating wave approximation, the interaction picture Hamiltonian becomes approximately time-independent, allowing us to avoid the Dyson series altogether and use a simple exponential.

**Application to the Driven Two-Level System:** For our system, we separate:

$$\hat{H}_0 = \frac{\hbar\omega_0}{2}\sigma_z \quad (\text{static field}) \quad (4.128)$$

$$\hat{V}(t) = \frac{\hbar\Omega}{2}[\sigma_x \cos(\omega t) + \sigma_y \sin(\omega t)] \quad (\text{rotating field}) \quad (4.129)$$

The transformation to the interaction picture is:

$$|\psi_I(t)\rangle = e^{i\hat{H}_0 t/\hbar} |\psi_S(t)\rangle = e^{i\omega_0 t \sigma_z/2} |\psi_S(t)\rangle \quad (4.130)$$

Applying the Schrödinger equation in the interaction picture:

$$\hat{H}_I(t) = e^{i\hat{H}_0 t/\hbar} \hat{V}(t) e^{-i\hat{H}_0 t/\hbar} \quad (4.131)$$

$$= e^{i\omega_0 t \sigma_z/2} \frac{\hbar\Omega}{2} [\sigma_x \cos(\omega t) + \sigma_y \sin(\omega t)] e^{-i\omega_0 t \sigma_z/2} \quad (4.132)$$

Using the rotation properties  $e^{i\theta\sigma_z/2}\sigma_x e^{-i\theta\sigma_z/2} = \sigma_x \cos \theta + \sigma_y \sin \theta$  and  $e^{i\theta\sigma_z/2}\sigma_y e^{-i\theta\sigma_z/2} = -\sigma_x \sin \theta + \sigma_y \cos \theta$ :

$$\hat{H}_I(t) = \frac{\hbar\Omega}{2} [(\sigma_x \cos \omega_0 t + \sigma_y \sin \omega_0 t) \cos \omega t \quad (4.133)$$

$$+ (-\sigma_x \sin \omega_0 t + \sigma_y \cos \omega_0 t) \sin \omega t] \quad (4.134)$$

$$= \frac{\hbar\Omega}{2} [\sigma_x (\cos \omega_0 t \cos \omega t - \sin \omega_0 t \sin \omega t) \quad (4.135)$$

$$+ \sigma_y (\sin \omega_0 t \cos \omega t + \cos \omega_0 t \sin \omega t)] \quad (4.136)$$

$$= \frac{\hbar\Omega}{2} [\sigma_x \cos((\omega_0 - \omega)t) + \sigma_y \sin((\omega_0 - \omega)t)] \quad (4.137)$$

This interaction picture Hamiltonian still has explicit time dependence, oscillating at the difference frequency  $(\omega_0 - \omega)$ . When the driving frequency  $\omega$  is close to the natural frequency  $\omega_0$ , this oscillation is slow compared to the original oscillation at  $\omega_0$ .

**Transforming to the Doubly Rotating Frame:** To further simplify, we make a second transformation to a frame rotating at the drive frequency  $\omega$ . Define:

$$|\psi_{rot}(t)\rangle = e^{i\omega t \sigma_z/2} |\psi_I(t)\rangle \quad (4.138)$$

The Hamiltonian in this doubly rotating frame becomes:

$$\hat{H}_{rot}(t) = e^{i\omega t \sigma_z/2} \hat{H}_I(t) e^{-i\omega t \sigma_z/2} - \frac{\hbar\omega}{2}\sigma_z \quad (4.139)$$



**Derivation of the  $-\frac{\hbar\omega}{2}\sigma_z$  term:** This term arises from the time derivative of the rotating frame transformation. When we define  $|\psi_{rot}(t)\rangle = e^{i\omega t\sigma_z/2}|\psi_I(t)\rangle$ , we need to find the equation of motion for  $|\psi_{rot}(t)\rangle$ . Taking the time derivative:

$$\begin{aligned}\frac{d}{dt}|\psi_{rot}(t)\rangle &= \frac{d}{dt}\left[e^{i\omega t\sigma_z/2}|\psi_I(t)\rangle\right] \\ &= \frac{d}{dt}\left[e^{i\omega t\sigma_z/2}\right]|\psi_I(t)\rangle + e^{i\omega t\sigma_z/2}\frac{d}{dt}|\psi_I(t)\rangle \\ &= i\frac{\omega}{2}\sigma_z e^{i\omega t\sigma_z/2}|\psi_I(t)\rangle + e^{i\omega t\sigma_z/2}\frac{d}{dt}|\psi_I(t)\rangle\end{aligned}\quad (4.140)$$

The interaction picture Schrödinger equation is:

$$i\hbar\frac{d}{dt}|\psi_I(t)\rangle = \hat{H}_I(t)|\psi_I(t)\rangle \quad (4.141)$$

Substituting this into our expression:

$$\begin{aligned}\frac{d}{dt}|\psi_{rot}(t)\rangle &= i\frac{\omega}{2}\sigma_z e^{i\omega t\sigma_z/2}|\psi_I(t)\rangle + e^{i\omega t\sigma_z/2}\left(-\frac{i}{\hbar}\hat{H}_I(t)|\psi_I(t)\rangle\right) \\ &= i\frac{\omega}{2}\sigma_z|\psi_{rot}(t)\rangle - \frac{i}{\hbar}e^{i\omega t\sigma_z/2}\hat{H}_I(t)e^{-i\omega t\sigma_z/2}|\psi_{rot}(t)\rangle\end{aligned}\quad (4.142)$$

Multiplying both sides by  $i\hbar$ :

$$i\hbar\frac{d}{dt}|\psi_{rot}(t)\rangle = -\frac{\hbar\omega}{2}\sigma_z|\psi_{rot}(t)\rangle + e^{i\omega t\sigma_z/2}\hat{H}_I(t)e^{-i\omega t\sigma_z/2}|\psi_{rot}(t)\rangle \quad (4.143)$$

This shows that the effective Hamiltonian in the rotating frame is:

$$\hat{H}_{rot}(t) = e^{i\omega t\sigma_z/2}\hat{H}_I(t)e^{-i\omega t\sigma_z/2} - \frac{\hbar\omega}{2}\sigma_z \quad (4.144)$$

The  $-\frac{\hbar\omega}{2}\sigma_z$  term represents the "fictitious" energy shift due to the rotating frame, analogous to fictitious forces in classical rotating reference frames. This is why the effective detuning becomes  $\Delta\omega = \omega_0 - \omega$  in the rotating frame.

To evaluate the transformation, we express the driving field in terms of ladder operators. Starting from:

$$\hat{V}(t) = \frac{\hbar\Omega}{2}[\sigma_x \cos(\omega t) + \sigma_y \sin(\omega t)] \quad (4.145)$$

$$= \frac{\hbar\Omega}{4}[(\sigma_x - i\sigma_y)e^{i\omega t} + (\sigma_x + i\sigma_y)e^{-i\omega t}] \quad (4.146)$$

$$= \frac{\hbar\Omega}{4}[\sigma_+ e^{-i\omega t} + \sigma_- e^{i\omega t}] \quad (4.147)$$

In the interaction picture with  $\hat{H}_0 = \frac{\hbar\omega_0}{2}\sigma_z$ :

$$\hat{V}_I(t) = e^{i\omega_0 t\sigma_z/2}\hat{V}(t)e^{-i\omega_0 t\sigma_z/2} \quad (4.148)$$

$$= \frac{\hbar\Omega}{4}[\sigma_+ e^{i(\omega_0 - \omega)t} + \sigma_- e^{-i(\omega_0 - \omega)t}] \quad (4.149)$$

In the doubly rotating frame, the complete Hamiltonian is:

$$\hat{H}_{rot}(t) = e^{i\omega t\sigma_z/2}\left(\frac{\hbar\omega_0}{2}\sigma_z + \hat{V}_I(t)\right)e^{-i\omega t\sigma_z/2} - \frac{\hbar\omega}{2}\sigma_z \quad (4.150)$$

Since  $e^{i\omega t\sigma_z/2}\sigma_z e^{-i\omega t\sigma_z/2} = \sigma_z$ :

$$\hat{H}_{rot}(t) = \frac{\hbar(\omega_0 - \omega)}{2}\sigma_z + e^{i\omega t\sigma_z/2}\hat{V}_I(t)e^{-i\omega t\sigma_z/2} \quad (4.151)$$

For the second term, we use the transformation properties:

$$e^{i\omega t\sigma_z/2}\sigma_+ e^{-i\omega t\sigma_z/2} = \sigma_+ e^{i\omega t} \quad (4.152)$$

$$e^{i\omega t\sigma_z/2}\sigma_- e^{-i\omega t\sigma_z/2} = \sigma_- e^{-i\omega t} \quad (4.153)$$

Therefore:

$$\hat{H}_{rot}(t) = \frac{\hbar(\omega_0 - \omega)}{2}\sigma_z + \frac{\hbar\Omega}{4}[\sigma_+ e^{i(\omega_0 - \omega)t} e^{i\omega t} + \sigma_- e^{-i(\omega_0 - \omega)t} e^{-i\omega t}] \quad (4.154)$$

Defining the detuning  $\Delta\omega = \omega_0 - \omega$  and expanding the exponentials:

$$\hat{H}_{rot}(t) = \frac{\hbar\Delta\omega}{2}\sigma_z + \frac{\hbar\Omega}{4}[\sigma_+ e^{i\Delta\omega t} e^{i\omega t} + \sigma_- e^{-i\Delta\omega t} e^{-i\omega t}] \quad (4.155)$$

Near resonance ( $\Delta\omega \ll \omega$ ), the exponentials  $e^{\pm i\Delta\omega t}$  vary slowly compared to  $e^{\pm i\omega t}$ . We can separate the expression into slowly and rapidly oscillating parts by writing  $e^{i\Delta\omega t} \approx 1$  for the slowly varying terms:

$$\hat{H}_{rot}(t) = \frac{\hbar\Delta\omega}{2}\sigma_z + \frac{\hbar\Omega}{4}[\sigma_+ + \sigma_-] + \frac{\hbar\Omega}{4}[\sigma_+ e^{2i\omega t} + \sigma_- e^{-2i\omega t}] \quad (4.156)$$

**The Rotating Wave Approximation:** The complete Hamiltonian in the doubly rotating frame is:

$$\hat{H}_{rot}(t) = \frac{\hbar\Delta\omega}{2}\sigma_z + \frac{\hbar\Omega}{4}[\sigma_+ + \sigma_-] + \frac{\hbar\Omega}{4}[\sigma_+ e^{2i\omega t} + \sigma_- e^{-2i\omega t}] \quad (4.157)$$

The last term oscillates rapidly at frequency  $2\omega$ . The rotating wave approximation (RWA) consists of neglecting these rapidly oscillating "counter-rotating" terms. The physical justification is that these terms oscillate much faster than any other timescale in the problem, so they average to zero and don't contribute to the long-time dynamics.

Dropping the rapidly oscillating terms and using  $\sigma_+ + \sigma_- = 2\sigma_x$ :

$$\hat{H}_{eff} = \frac{\hbar\Delta\omega}{2}\sigma_z + \frac{\hbar\Omega}{2}\sigma_x \quad (4.158)$$

This is the effective Hamiltonian in the rotating wave approximation. It is now time-independent, allowing us to use simple exponential time evolution instead of the Dyson series.

#### 4.8.9 Rabi Oscillations and Bloch Vector Dynamics

In the rotating frame with the rotating wave approximation, the effective Hamiltonian generates rotations about the axis  $(\Omega, 0, \Delta\omega)$  with frequency  $\Omega_{eff} = \sqrt{\Omega^2 + (\Delta\omega)^2}$ .

For exact resonance ( $\Delta\omega = 0$ ), the effective Hamiltonian becomes  $\hat{H}_{eff} = \hbar\Omega\sigma_x/2$ , generating rotations about the x-axis. Starting from the ground state  $|+z\rangle$ :

$$|\psi(t)\rangle = e^{-i\Omega t\sigma_x/2}|+z\rangle \quad (4.159)$$

$$= \cos(\Omega t/2)|+z\rangle - i\sin(\Omega t/2)|-z\rangle \quad (4.160)$$

The population oscillates between the two levels:

$$P_{excited}(t) = |\langle -z|\psi(t)\rangle|^2 = \sin^2(\Omega t/2) \quad (4.161)$$

These are called Rabi oscillations, and they provide a precise method for controlling quantum states through the duration and amplitude of applied pulses.

## 4.9 Adiabatic Evolution and Geometric Phases

### 4.9.1 Slow Parameter Variation and the Adiabatic Theorem

When the parameters in a quantum system change slowly compared to the energy gaps, the system can remain in an instantaneous eigenstate while accumulating both dynamical and geometric phases. This adiabatic evolution provides a powerful tool for robust quantum state manipulation and reveals deep connections between quantum mechanics and geometry.

Consider a time-dependent Hamiltonian  $\hat{H}(t)$  that changes slowly over time. At each instant  $t$ , we can define instantaneous eigenstates  $|n(t)\rangle$  and eigenvalues  $E_n(t)$ :

$$\hat{H}(t)|n(t)\rangle = E_n(t)|n(t)\rangle \quad (4.162)$$

The adiabatic theorem states that if the system starts in eigenstate  $|n(0)\rangle$  and the Hamiltonian changes sufficiently slowly, the system will remain in the instantaneous eigenstate  $|n(t)\rangle$  but acquire a phase:

$$|\psi_n(t)\rangle = e^{i\phi_n(t)}|n(t)\rangle \quad (4.163)$$

where  $|\psi_n(t)\rangle$  is the actual state of the system (which follows the  $n$ -th eigenstate adiabatically) and  $|n(t)\rangle$  is the instantaneous eigenstate of  $\hat{H}(t)$  at time  $t$ .

The total phase has two contributions:

$$\phi_n(t) = \phi_n^{(dyn)}(t) + \phi_n^{(geom)}(t) \quad (4.164)$$

$$\phi_n^{(dyn)}(t) = -\frac{1}{\hbar} \int_0^t E_n(t') dt' \quad (\text{dynamical}) \quad (4.165)$$

$$\phi_n^{(geom)}(t) = i \int_0^t \langle n(t') | \frac{d}{dt'} | n(t') \rangle dt' \quad (\text{geometric}) \quad (4.166)$$

**Origin of the Geometric Phase:** The geometric phase arises from the requirement that the state remain normalized during evolution. When we write  $|\psi_n(t)\rangle = e^{i\phi_n(t)}|n(t)\rangle$ , taking the time derivative and using the Schrödinger equation gives:

$$i\hbar \frac{d}{dt} |\psi_n(t)\rangle = \hat{H}(t) |\psi_n(t)\rangle \quad (4.167)$$

Expanding the left side using the product rule:

$$i\hbar \left[ i \frac{d\phi_n}{dt} e^{i\phi_n} |n(t)\rangle + e^{i\phi_n} \frac{d|n(t)\rangle}{dt} \right] = \hat{H}(t) e^{i\phi_n} |n(t)\rangle \quad (4.168)$$

Since  $\hat{H}(t)|n(t)\rangle = E_n(t)|n(t)\rangle$ , this becomes:

$$i\hbar \frac{d\phi_n}{dt} e^{i\phi_n} |n(t)\rangle + i\hbar e^{i\phi_n} \frac{d|n(t)\rangle}{dt} = E_n(t) e^{i\phi_n} |n(t)\rangle \quad (4.169)$$

Dividing through by  $e^{i\phi_n}$  (which is just a phase factor):

$$i\hbar \frac{d\phi_n}{dt} |n(t)\rangle + i\hbar \frac{d|n(t)\rangle}{dt} = E_n(t) |n(t)\rangle \quad (4.170)$$

Taking the inner product with  $\langle n(t) |$ :

$$i\hbar \frac{d\phi_n}{dt} + i\hbar \langle n(t) | \frac{d}{dt} | n(t) \rangle = E_n(t) \quad (4.171)$$

Rearranging:

$$\frac{d\phi_n}{dt} = -\frac{E_n(t)}{\hbar} - i\langle n(t)|\frac{d}{dt}|n(t)\rangle \quad (4.172)$$

**Why the second term is purely real:** The inner product  $\langle n(t)|\frac{d}{dt}|n(t)\rangle$  is purely imaginary, not the phase itself. Since  $\langle n(t)|n(t)\rangle = 1$  for all  $t$ , taking the time derivative:

$$\frac{d}{dt}\langle n(t)|n(t)\rangle = \left\langle \frac{d}{dt}n(t) \middle| n(t) \right\rangle + \left\langle n(t) \middle| \frac{d}{dt}n(t) \right\rangle = 0 \quad (4.173)$$

This means:

$$\left\langle n(t) \middle| \frac{d}{dt}n(t) \right\rangle = -\left\langle \frac{d}{dt}n(t) \middle| n(t) \right\rangle = -\left\langle n(t) \middle| \frac{d}{dt}n(t) \right\rangle^* \quad (4.174)$$

So  $\langle n(t)|\frac{d}{dt}|n(t)\rangle$  is purely imaginary. When multiplied by  $-i$  in the equation for  $d\phi_n/dt$ , the second term becomes purely real. Therefore, both terms on the right side contribute real values to the phase.

Integrating yields the two contributions: the first term gives the dynamical phase (energy-dependent), while the second term gives the geometric phase. This geometric phase depends only on how the instantaneous eigenstate  $|n(t)\rangle$  evolves along its path in Hilbert space, not on the rate of evolution or the energy. It is this path-dependence that gives the phase its geometric character.

The dynamical phase depends on the energy, while the geometric phase depends only on the path traced by the state vector in Hilbert space.

#### 4.9.2 Berry Phase Equals Half the Solid Angle for Two-Level Systems

For a two-level system undergoing adiabatic evolution around a closed path, the geometric Berry phase is exactly half the solid angle enclosed by the path on the Bloch sphere. We now derive this remarkable result.

**Setup:** Consider a two-level Hamiltonian where a magnetic field of constant magnitude traces a path on the unit sphere:

$$\hat{H}(t) = \frac{\hbar\omega_0}{2}\vec{B}(t) \cdot \vec{\sigma} \quad (4.175)$$

where  $\vec{B}(t)$  is a unit vector parameterized by spherical coordinates  $(\theta(t), \phi(t))$ :

$$\vec{B}(t) = (\sin\theta(t)\cos\phi(t), \sin\theta(t)\sin\phi(t), \cos\theta(t)) \quad (4.176)$$

The Hamiltonian depends on time through the magnetic field direction. The energy splitting between the two levels is  $\hbar\omega_0$ , with eigenvalues  $\pm\hbar\omega_0/2$ . As time evolves and the field traces a path on the sphere, the parameters  $(\theta(t), \phi(t))$  change, causing the instantaneous eigenstate to evolve.

**Instantaneous Eigenstates:** At each instant, the ground state eigenstate (lower energy) is:

$$|\psi_-(\theta, \phi)\rangle = \cos(\theta/2)|+z\rangle - e^{i\phi}\sin(\theta/2)|-z\rangle \quad (4.177)$$

This eigenstate points opposite to  $\vec{B}(t)$  on the Bloch sphere (spin anti-aligned with the field).

**The Berry Connection:** From equation (4.174), the geometric phase is:

$$\phi^{(geom)} = i \int_0^t \langle \psi_- | \frac{d}{dt'} | \psi_- \rangle dt' \quad (4.178)$$

Using the chain rule:

$$\frac{d}{dt}|\psi_-\rangle = \frac{\partial|\psi_-\rangle}{\partial\theta}\frac{d\theta}{dt} + \frac{\partial|\psi_-\rangle}{\partial\phi}\frac{d\phi}{dt} \quad (4.179)$$

Substituting and changing variables from time to path parameters:

$$\phi^{(geom)} = i \int_{\text{path}} \left[ \langle \psi_- | \frac{\partial}{\partial \theta} | \psi_- \rangle d\theta + \langle \psi_- | \frac{\partial}{\partial \phi} | \psi_- \rangle d\phi \right] \quad (4.180)$$

We define the **Berry connection**:

$$\vec{A}(\theta, \phi) = i \langle \psi_- | \nabla | \psi_- \rangle = (A_\theta, A_\phi) \quad (4.181)$$

where:

$$A_\theta = i \langle \psi_- | \frac{\partial}{\partial \theta} | \psi_- \rangle \quad (4.182)$$

$$A_\phi = i \langle \psi_- | \frac{\partial}{\partial \phi} | \psi_- \rangle \quad (4.183)$$

The geometric phase for a closed path  $\mathcal{C}$  is:

$$\phi^{(geom)} = \oint_{\mathcal{C}} \vec{A} \cdot d\vec{r} = \oint_{\mathcal{C}} (A_\theta d\theta + A_\phi d\phi) \quad (4.184)$$

**Computing the Berry Connection:**

For  $A_\phi$ , taking the  $\phi$ -derivative:

$$\frac{\partial}{\partial \phi} | \psi_- \rangle = -ie^{i\phi} \sin(\theta/2) | -z \rangle \quad (4.185)$$

Therefore:

$$A_\phi = i \langle \psi_- | \frac{\partial}{\partial \phi} | \psi_- \rangle = i \left[ -ie^{-i\phi} \sin(\theta/2) \right] \left[ -ie^{i\phi} \sin(\theta/2) \right] = -\sin^2(\theta/2) = -\frac{1 - \cos \theta}{2} \quad (4.186)$$

For  $A_\theta$ , taking the  $\theta$ -derivative:

$$\frac{\partial}{\partial \theta} | \psi_- \rangle = -\frac{1}{2} \sin(\theta/2) | +z \rangle - \frac{1}{2} e^{i\phi} \cos(\theta/2) | -z \rangle \quad (4.187)$$

Computing the inner product:

$$A_\theta = i \langle \psi_- | \frac{\partial}{\partial \theta} | \psi_- \rangle = i(-i \cos(\theta/2) \sin(\theta/2)) = \frac{1}{2} \sin \theta \quad (4.188)$$

**From Line Integral to Solid Angle:** Using Stokes' theorem, we convert the line integral to a surface integral:

$$\phi^{(geom)} = \oint_{\mathcal{C}} \vec{A} \cdot d\vec{r} = \iint_S (\nabla \times \vec{A}) \cdot d\vec{S} \quad (4.189)$$

The Berry curvature in spherical coordinates is:

$$\nabla \times \vec{A} = \frac{1}{\sin \theta} \left[ \frac{\partial(\sin \theta A_\theta)}{\partial \theta} - \frac{\partial A_\phi}{\partial \theta} \right] \hat{r} \quad (4.190)$$

Substituting our expressions:

$$\nabla \times \vec{A} = \frac{1}{\sin \theta} \left[ \frac{\partial}{\partial \theta} \left( \frac{\sin^2 \theta}{2} \right) - \frac{\partial}{\partial \theta} \left( -\frac{1 - \cos \theta}{2} \right) \right] \hat{r} \quad (4.191)$$

$$= \frac{1}{\sin \theta} \left[ \sin \theta \cos \theta + \frac{\sin \theta}{2} \right] \hat{r} = \frac{1}{2} \hat{r} \quad (4.192)$$

With surface element  $d\vec{S} = \sin \theta d\theta d\phi \hat{r}$ :

$$\phi^{(geom)} = \iint_S \frac{1}{2} \sin \theta d\theta d\phi = \frac{1}{2} \Omega \quad (4.193)$$

where  $\Omega = \iint_S \sin \theta d\theta d\phi$  is the solid angle enclosed by the path.

With the conventional sign:

$$\boxed{\phi_{geom} = -\frac{1}{2} \Omega} \quad (4.194)$$

**Physical Interpretation:** Why half the solid angle? This factor arises from the spinorial nature of spin-1/2 wavefunctions. A  $2\pi$  rotation does not return the wavefunction to itself but picks up a minus sign:

$$e^{-i(2\pi)\hat{n}\cdot\vec{\sigma}/2} = -\hat{I} \quad (4.195)$$

The rotation group  $SO(3)$  is represented projectively as  $SU(2)$  for spin-1/2 systems. The Berry phase reflects this: when the magnetic field traces a closed path on the sphere, the quantum state acquires a phase that is half the enclosed solid angle.

Geometrically, the Bloch sphere represents quantum states, but physically distinct states correspond to rays in Hilbert space (states differing by a phase are equivalent). The solid angle  $\Omega$  measures the path geometry on the Bloch sphere, while the factor of  $1/2$  accounts for the spinorial structure of the quantum state space.

This result unifies geometry (solid angle), topology (Berry phase), and quantum mechanics (spinor nature), providing a direct experimental signature of these fundamental properties.

For a spin-1/2 system in a slowly rotating magnetic field, the geometric phase has a particularly elegant form. Consider a magnetic field of constant magnitude that traces out a closed path on the unit sphere:

$$\vec{B}(t) = B(\sin \theta(t) \cos \phi(t), \sin \theta(t) \sin \phi(t), \cos \theta(t)) \quad (4.196)$$

If the system starts in the ground state and follows the field adiabatically, after one complete cycle it acquires a geometric phase:

$$\phi_{geom} = -\frac{1}{2} \Omega \quad (4.197)$$

where  $\Omega$  is the solid angle enclosed by the path on the unit sphere. This Berry phase depends only on the geometry of the path, not on the rate of traversal.

### Example 7

**Berry Phase Visualization with the Bloch Cube** The Bloch cube provides an excellent visualization tool for understanding the Berry phase. Consider a quantum state that starts at one face of the cube and is adiabatically transported around a closed loop on the cube's surface.

**Triangular Path on the Bloch Cube:** Start with a state at the north pole of the Bloch sphere (top face of cube,  $|+z\rangle$ ). Follow this path: 1. Move to the equator along the positive x-axis ( $|+x\rangle$ ) 2. Move along the equator to the positive y-axis ( $|+y\rangle$ ) 3. Return to the north pole

This triangular path on the Bloch sphere encloses a solid angle of  $\Omega = \pi/2$  steradians (one-eighth of the sphere). The Berry phase acquired is:

$$\phi_{Berry} = -\frac{1}{2} \times \frac{\pi}{2} = -\frac{\pi}{4} \quad (4.198)$$

You can visualize this on the Bloch cube: the path connects three adjacent vertices, forming

a triangular face of the octahedron inscribed in the cube. The Berry phase is proportional to the area of this triangle projected onto the unit sphere.

**Square Path on the Bloch Cube:** For a path that traverses four faces of the cube in sequence (e.g.,  $|+z\rangle \rightarrow |+x\rangle \rightarrow |-z\rangle \rightarrow |-x\rangle \rightarrow |+z\rangle$ ), the enclosed solid angle is  $\Omega = 2\pi$  steradians (a hemisphere), giving:

$$\phi_{\text{Berry}} = -\pi \quad (4.199)$$

This is precisely a sign flip—the state returns with opposite phase, demonstrating the topological nature of the Berry phase.

### Example 8

**Berry Phase for a Spin-1/2 Particle** Consider a magnetic field that traces a **closed path**: starting along the z-axis, tilting to make angle  $\alpha$  with the z-axis, rotating once around the z-axis at this tilt angle, then returning to the z-axis to close the loop.

The Berry phase is defined only for this complete closed path. To calculate the solid angle enclosed by this closed curve on the unit sphere, we can break it into segments:

**Phase 1 (Tilting down):** The path from north pole to latitude  $\alpha$  traces a meridian line (zero width), contributing no area.

**Phase 2 (Rotation at fixed latitude):** The circular path at constant  $\theta = \alpha$  with  $\phi : 0 \rightarrow 2\pi$  forms the "base" of a spherical cap. Together with the paths to and from the north pole, this encloses a spherical cap of solid angle  $\Omega = 2\pi(1 - \cos \alpha)$ .

**Phase 3 (Tilting back up):** The return path along the same meridian closes the loop but adds no additional enclosed area.

The key insight is that the **total solid angle enclosed by the entire closed path** is  $\Omega = 2\pi(1 - \cos \alpha)$ , which comes entirely from the latitude circle at angle  $\alpha$ . The meridian segments are necessary to close the path but don't contribute to the enclosed area (like the sides of a cone that meet at the apex).

The Berry phase for the complete closed path is:

$$\phi_{\text{geom}} = -\frac{1}{2}\Omega = -\pi(1 - \cos \alpha) \quad (4.200)$$

For  $\alpha = \pi/2$  (equatorial loop), the closed path encloses a hemisphere:  $\phi_{\text{geom}} = -\pi$ . For  $\alpha = \pi$  (great circle through south pole), the closed path encloses the entire sphere:  $\phi_{\text{geom}} = -2\pi$ . The phase depends only on the solid angle enclosed by the closed curve, providing a topological characterization of the evolution.

## 4.10 Many-Body Hamiltonians: From Single Particles to Quantum Fields

### 4.10.1 The Hopping Hamiltonian and Translational Invariance

Having established how Hamiltonians generate time evolution for single two-level systems, we now extend this formalism to many-particle systems on a lattice. Recall from Chapter 3 that we can represent particles on a lattice using creation and annihilation operators  $\hat{a}_i^\dagger$  and  $\hat{a}_i$ . The natural

dynamics for such systems involve particles hopping between lattice sites.

The hopping Hamiltonian in second quantization takes the form:

$$\hat{H}_{\text{hop}} = -\Delta \sum_{\langle i,j \rangle} (\hat{a}_i^\dagger \hat{a}_j + \hat{a}_j^\dagger \hat{a}_i) \quad (4.201)$$

where  $\langle i, j \rangle$  denotes nearest neighbors and  $\Delta > 0$  is the hopping amplitude.

For a periodic chain with  $N$  sites (indexed from 0 to  $N-1$ ), this becomes:

$$\hat{H} = -\Delta \sum_{j=0}^{N-1} (\hat{a}_j^\dagger \hat{a}_{j+1} + \hat{a}_{j+1}^\dagger \hat{a}_j) \quad (4.202)$$

where site indices are understood modulo  $N$  (so site  $N$  is identified with site 0). Here and throughout this section, we use indices  $j, k, l$  for position space (lattice sites) and  $\alpha, \beta, \gamma$  for momentum space.

#### 4.10.2 Momentum Eigenstates as the Natural Basis

The key insight is that the hopping Hamiltonian has translational symmetry. Define the translation operator:

$$\hat{T} = \sum_{j=0}^{N-1} \hat{a}_{j+1}^\dagger \hat{a}_j \quad (4.203)$$

Since  $[\hat{H}, \hat{T}] = 0$ , we can find simultaneous eigenstates of both operators. These are the momentum eigenstates, defined in the single-particle sector as:

$$|k_\alpha\rangle = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{ik_\alpha j} \hat{a}_j^\dagger |\text{vac}\rangle \quad (4.204)$$

where the allowed momenta for periodic boundary conditions are:

$$k_\alpha = \frac{2\pi\alpha}{N}, \quad \alpha = 0, 1, 2, \dots, N-1 \quad (4.205)$$

These states satisfy:

$$\hat{T}|k_\alpha\rangle = e^{ik_\alpha}|k_\alpha\rangle \quad (4.206)$$

$$\hat{H}|k_\alpha\rangle = E(k_\alpha)|k_\alpha\rangle \quad (4.207)$$

where the dispersion relation is:

$$E(k_\alpha) = -2\Delta \cos(k_\alpha) \quad (4.208)$$

#### 4.10.3 General Time Evolution for N Sites

Following the pattern established in equation (4.148), we can write the complete time evolution for a particle on an  $N$ -site ring. In the position basis, we expand the wavefunction as:

$$|\psi(t)\rangle = \sum_{j=0}^{N-1} \psi_j(t) |j\rangle \quad (4.209)$$



where  $|j\rangle$  denotes a particle at site  $j$  and  $\psi_j(t)$  is the amplitude at site  $j$  at time  $t$ .

The transformation from position basis to momentum basis is:

$$|k_\alpha\rangle = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{ik_\alpha j} |j\rangle \quad (4.210)$$

where  $k_\alpha = 2\pi\alpha/N$  for  $\alpha = 0, 1, 2, \dots, N-1$ .

The inverse transformation from momentum to position basis is:

$$|j\rangle = \frac{1}{\sqrt{N}} \sum_{\alpha=0}^{N-1} e^{-ik_\alpha j} |k_\alpha\rangle \quad (4.211)$$

In the momentum basis, time evolution is diagonal:

$$\hat{U}(t)|k_\alpha\rangle = e^{-iE(k_\alpha)t/\hbar}|k_\alpha\rangle = e^{i2\Delta \cos(k_\alpha)t/\hbar}|k_\alpha\rangle \quad (4.212)$$

Now, for an initial state with amplitudes  $\psi_k(0)$ :

$$|\psi(0)\rangle = \sum_{k=0}^{N-1} \psi_k(0) |k\rangle \quad (4.213)$$

The time-evolved state is:

$$|\psi(t)\rangle = \hat{U}(t)|\psi(0)\rangle \quad (4.214)$$

$$= \hat{U}(t) \sum_{k=0}^{N-1} \psi_k(0) |k\rangle \quad (4.215)$$

$$= \sum_{k=0}^{N-1} \psi_k(0) \hat{U}(t) |k\rangle \quad (4.216)$$

To find  $\psi_j(t) = \langle j|\psi(t)\rangle$ , we compute:

$$\psi_j(t) = \sum_{k=0}^{N-1} \psi_k(0) \langle j|\hat{U}(t)|k\rangle \quad (4.217)$$

$$= \sum_{k=0}^{N-1} \psi_k(0) [\hat{U}(t)]_{jk} \quad (4.218)$$

The matrix element  $[\hat{U}(t)]_{jk}$  is calculated using the momentum eigenbasis:

$$[\hat{U}(t)]_{jk} = \langle j|\hat{U}(t)|k\rangle \quad (4.219)$$

$$= \sum_{\alpha=0}^{N-1} \langle j|\hat{U}(t)|k_\alpha\rangle \langle k_\alpha|k\rangle \quad (4.220)$$

$$= \sum_{\alpha=0}^{N-1} \langle j|k_\alpha\rangle e^{-iE(k_\alpha)t/\hbar} \langle k_\alpha|k\rangle \quad (4.221)$$

$$= \frac{1}{N} \sum_{\alpha=0}^{N-1} e^{-ik_\alpha j} \cdot e^{i2\Delta \cos(k_\alpha)t/\hbar} \cdot e^{ik_\alpha k} \quad (4.222)$$

$$= \frac{1}{N} \sum_{\alpha=0}^{N-1} e^{ik_\alpha(k-j)} e^{i2\Delta \cos(k_\alpha)t/\hbar} \quad (4.223)$$

Therefore, the complete time evolution in the position basis is:

$$\psi_j(t) = \sum_{k=0}^{N-1} \left[ \frac{1}{N} \sum_{\alpha=0}^{N-1} e^{ik_\alpha(k-j)} e^{i2\Delta \cos(k_\alpha)t/\hbar} \right] \psi_k(0) \quad (4.224)$$

This can be written in matrix form as:

$$\begin{pmatrix} \psi_0(t) \\ \psi_1(t) \\ \vdots \\ \psi_{N-1}(t) \end{pmatrix} = \hat{S} \cdot \hat{U}_{\text{diag}}(t) \cdot \hat{S}^\dagger \begin{pmatrix} \psi_0(0) \\ \psi_1(0) \\ \vdots \\ \psi_{N-1}(0) \end{pmatrix} \quad (4.225)$$

where  $\hat{S}$  is the matrix transforming to the momentum basis,  $\hat{U}_{\text{diag}}(t)$  contains the diagonal phase factors  $e^{i2\Delta \cos(k_\alpha)t/\hbar}$ , and  $\hat{S}^\dagger$  transforms back to the position basis.

This general formula demonstrates the triple product structure:

- First, transform from position to momentum basis via  $\hat{S}^\dagger$
- Then, apply diagonal time evolution via  $\hat{U}_{\text{diag}}(t)$
- Finally, transform back to position basis via  $\hat{S}$

For a particle initially localized at site  $k_0$ , we have  $\psi_k(0) = \delta_{k,k_0}$ , and the formula simplifies to:

$$\psi_j(t) = [\hat{U}(t)]_{j,k_0} = \frac{1}{N} \sum_{\alpha=0}^{N-1} e^{ik_\alpha(k_0-j)} e^{i2\Delta \cos(k_\alpha)t/\hbar} \quad (4.226)$$

This shows how the particle amplitude spreads from site  $k_0$  to all other sites through quantum hopping.

#### 4.10.4 Explicit Calculation for Three Sites

Consider a ring of three sites (indexed 0, 1, 2) with periodic boundaries. The Hamiltonian in the position basis  $\{|100\rangle, |010\rangle, |001\rangle\}$  is:

$$\hat{H} = -\Delta \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \quad (4.227)$$

The momentum eigenstates are:

$$|k_{\alpha=0}\rangle = \frac{1}{\sqrt{3}}(|100\rangle + |010\rangle + |001\rangle), \quad k_0 = 0 \quad (4.228)$$

$$|k_{\alpha=1}\rangle = \frac{1}{\sqrt{3}}(|100\rangle + e^{i2\pi/3}|010\rangle + e^{i4\pi/3}|001\rangle), \quad k_1 = \frac{2\pi}{3} \quad (4.229)$$

$$|k_{\alpha=2}\rangle = \frac{1}{\sqrt{3}}(|100\rangle + e^{i4\pi/3}|010\rangle + e^{i2\pi/3}|001\rangle), \quad k_2 = \frac{4\pi}{3} \quad (4.230)$$

The corresponding energies are:

$$E(k_0) = -2\Delta \cos(k_0) = -2\Delta \quad (4.231)$$

$$E(k_1) = -2\Delta \cos(k_1) = \Delta \quad (4.232)$$

$$E(k_2) = -2\Delta \cos(k_2) = \Delta \quad (4.233)$$

### 4.10.5 Three-Site Example: Explicit Matrix Calculation

In the momentum basis, the Hamiltonian is diagonal:

$$\hat{H}_{\text{diag}} = \begin{pmatrix} -2\Delta & 0 & 0 \\ 0 & \Delta & 0 \\ 0 & 0 & \Delta \end{pmatrix} \quad (4.234)$$

The time evolution operator in the momentum basis is:

$$\hat{U}_{\text{diag}}(t) = e^{-i\hat{H}_{\text{diag}}t/\hbar} = \begin{pmatrix} e^{i2\Delta t/\hbar} & 0 & 0 \\ 0 & e^{-i\Delta t/\hbar} & 0 \\ 0 & 0 & e^{-i\Delta t/\hbar} \end{pmatrix} \quad (4.235)$$

To find the time evolution operator in the position basis, we need the unitary transformation matrix  $\hat{S}$  whose columns are the momentum eigenstates:

$$\hat{S} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & e^{i2\pi/3} & e^{i4\pi/3} \\ 1 & e^{i4\pi/3} & e^{i2\pi/3} \end{pmatrix} \quad (4.236)$$

The inverse transformation is  $\hat{S}^\dagger = \hat{S}^*$  (complex conjugate):

$$\hat{S}^\dagger = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & e^{-i2\pi/3} & e^{-i4\pi/3} \\ 1 & e^{-i4\pi/3} & e^{-i2\pi/3} \end{pmatrix} \quad (4.237)$$

The full time evolution operator in the position basis is:

$$\hat{U}(t) = \hat{S}\hat{U}_{\text{diag}}(t)\hat{S}^\dagger \quad (4.238)$$

Explicitly:

$$\hat{U}(t) = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & e^{i2\pi/3} & e^{i4\pi/3} \\ 1 & e^{i4\pi/3} & e^{i2\pi/3} \end{pmatrix} \begin{pmatrix} e^{i2\Delta t/\hbar} & 0 & 0 \\ 0 & e^{-i\Delta t/\hbar} & 0 \\ 0 & 0 & e^{-i\Delta t/\hbar} \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & e^{-i2\pi/3} & e^{-i4\pi/3} \\ 1 & e^{-i4\pi/3} & e^{-i2\pi/3} \end{pmatrix} \quad (4.239)$$

After matrix multiplication:

$$[\hat{U}(t)]_{jk} = \frac{1}{3} \left[ e^{i2\Delta t/\hbar} + 2e^{-i\Delta t/\hbar} \cos\left(\frac{2\pi(j-k)}{3}\right) \right] \quad (4.240)$$

This gives the complete quantum dynamics for a particle hopping on a three-site ring.

### 4.10.6 Many-Body Dynamics Under the Hopping Hamiltonian

The beauty of second quantization is that the same Hamiltonian describes dynamics for any number of particles:

**Example 9**

Two-Particle Hopping Dynamics For two particles initially at sites 1 and 2:

$$\hat{H}|110\rangle = -\Delta(\hat{a}_1^\dagger\hat{a}_2 + \hat{a}_2^\dagger\hat{a}_1 + \hat{a}_2^\dagger\hat{a}_3 + \hat{a}_3^\dagger\hat{a}_2)|110\rangle \quad (4.241)$$

Let's evaluate term by term:

- $\hat{a}_1^\dagger\hat{a}_2|110\rangle = 0$  (site 1 already occupied)
- $\hat{a}_2^\dagger\hat{a}_1|110\rangle = 0$  (site 2 already occupied)
- $\hat{a}_2^\dagger\hat{a}_3|110\rangle = 0$  (no particle at site 3)
- $\hat{a}_3^\dagger\hat{a}_2|110\rangle = |101\rangle$  (particle hops 2→3)

Result:  $\hat{H}|110\rangle = -\Delta|101\rangle$

The hard-core constraint naturally emerges—particles cannot hop to occupied sites!

#### 4.10.7 Time Evolution of Many-Body States

The time evolution operator for the hopping Hamiltonian is:

$$\hat{U}(t) = e^{-i\hat{H}_{\text{hop}}t/\hbar} \quad (4.242)$$

For states within a fixed particle number sector, this evolution conserves particle number since:

$$[\hat{H}_{\text{hop}}, \hat{N}] = 0 \quad (4.243)$$

where  $\hat{N} = \sum_i \hat{n}_i$  is the total particle number operator.

**Example 10**

Evolution of a Localized Two-Particle State Consider two particles initially localized at adjacent sites on a 4-site ring:

$$|\psi(0)\rangle = |1100\rangle \quad (4.244)$$

The time evolution involves: 1. Both particles can hop to their respective neighboring sites 2. The hard-core constraint prevents them from occupying the same site 3. The resulting dynamics exhibits both single-particle hopping and two-particle correlations

For short times:

$$|\psi(t)\rangle \approx |1100\rangle - \frac{it}{\hbar}\hat{H}|1100\rangle + \mathcal{O}(t^2) \quad (4.245)$$

$$\approx |1100\rangle - \frac{it}{\hbar}(-\Delta)(|0110\rangle + |1001\rangle) \quad (4.246)$$

$$= |1100\rangle + \frac{i\Delta t}{\hbar}(|0110\rangle + |1001\rangle) \quad (4.247)$$

The particles begin to spread while avoiding each other—a purely quantum many-body effect!

### 4.10.8 From Discrete to Continuous: The Continuum Limit

The lattice hopping Hamiltonian connects to continuum physics in the limit of small lattice spacing. For a particle with wave function  $\psi_j$  at site  $j = xa/a$  (where  $a$  is the lattice constant):

$$-\Delta(\psi_{j+1} + \psi_{j-1} - 2\psi_j) \approx -\Delta a^2 \frac{\partial^2 \psi}{\partial x^2} \quad (4.248)$$

This identifies:

$$\Delta = \frac{\hbar^2}{2ma^2} \quad (4.249)$$

The hopping Hamiltonian thus discretizes the kinetic energy operator of continuous quantum mechanics. This connection reveals that:

- Lattice models capture essential physics of continuous systems
- Hopping amplitude  $\Delta$  relates to particle mass and lattice spacing
- Band structure emerges from discrete translational symmetry
- Many-body effects persist in the continuum limit

Diagonalization reveals the natural basis for evolution: energy eigenstates acquire only phase factors  $e^{-iE_n t/\hbar}$  while superpositions develop complex interference. The time evolution operator decomposes as a triple matrix product  $\hat{U}(t) = \hat{V} \hat{U}_{\text{diag}}(t) \hat{V}^\dagger$ , transforming to the eigenbasis, applying diagonal phases, then transforming back.

Conservation laws emerge from commutation relations: observables satisfying  $[\hat{H}, \hat{A}] = 0$  have constant expectation values, connecting symmetries to conserved quantities through Noether's theorem. The Heisenberg picture  $\hat{A}_H(t) = e^{i\hat{H}t/\hbar} \hat{A}_S e^{-i\hat{H}t/\hbar}$  transfers time dependence from states to operators.

Time-dependent Hamiltonians require time-ordered exponentials  $\mathcal{T} \exp\left(-\frac{i}{\hbar} \int \hat{H}(t') dt'\right)$ . The driven two-level system in NMR demonstrates how the rotating frame and rotating wave approximation reduce time-dependent problems to effective time-independent ones, revealing universal Rabi oscillations. Adiabatic evolution accumulates geometric Berry phases that depend only on the path through parameter space, not the rate of traversal.

The hopping Hamiltonian  $\hat{H} = -\Delta \sum_{\langle i,j \rangle} (\hat{a}_i^\dagger \hat{a}_j + \hat{a}_j^\dagger \hat{a}_i)$  extends these principles to many-body systems. Momentum eigenstates  $|k_n\rangle$  with dispersion  $E(k_n) = -2\Delta \cos(k_n)$  diagonalize the evolution. The general solution  $\psi_j(t) = \sum_i [\hat{U}(t)]_{ji} \psi_i(0)$  maintains the triple product structure, showing how single-particle and many-body dynamics share the same mathematical framework of unitary evolution generated by Hermitian operators.

## 4.11 Problems

### Problem 4.1 (Verifying Unitarity and the Group Property)

Consider the time evolution operator  $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$  for a time-independent Hamiltonian  $\hat{H}$ .

- Show that if  $\hat{H}$  is Hermitian ( $\hat{H}^\dagger = \hat{H}$ ), then  $\hat{U}(t)$  is unitary for all real times  $t$
- Verify the group property:  $\hat{U}(t_1)\hat{U}(t_2) = \hat{U}(t_1 + t_2)$

- (c) Show that  $\hat{U}(-t) = \hat{U}^\dagger(t)$ . What is the physical meaning of this relationship?
- (d) For the Hamiltonian  $\hat{H} = \hbar\omega\sigma_z/2$ , explicitly calculate  $\hat{U}(\pi/\omega)$  and verify it is unitary
- (e) Starting from the infinitesimal form  $\hat{U}(dt) = \hat{I} - i\hat{H}dt/\hbar$ , derive the differential equation for  $\hat{U}(t)$

**Problem 4.2** (*From Discrete Gates to Continuous Evolution*)

The Hadamard gate  $\hat{H}$  and the  $\hat{T}$  gate are fundamental quantum gates:

$$\hat{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \hat{T} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \quad (4.250)$$

- (a) Find a Hamiltonian  $\hat{H}_1$  such that  $e^{-i\hat{H}_1 t_1/\hbar} = \hat{H}$  for some time  $t_1$
- (b) Find a Hamiltonian  $\hat{H}_2$  such that  $e^{-i\hat{H}_2 t_2/\hbar} = \hat{T}$  for some time  $t_2$
- (c) Are your Hamiltonians unique? If not, find another valid choice for each
- (d) What is the minimum time needed to implement each gate given a constraint  $\|\hat{H}\| \leq E_{max}$ ?
- (e) Show that any single-qubit unitary can be generated by some Hamiltonian evolution

**Problem 4.3** (*Series Expansion and Approximation*)

Consider the Hamiltonian  $\hat{H} = \hbar\omega(\sigma_x + \epsilon\sigma_z)$  where  $\epsilon \ll 1$ .

- (a) Calculate the first three terms of the series expansion for  $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$
- (b) For  $\epsilon = 0.1$  and  $\omega t = \pi/4$ , compare your series approximation (using 3 terms) to the exact result
- (c) Show that for the special case  $\epsilon = 0$ , the series simplifies to trigonometric functions
- (d) Use the Trotter formula to approximate  $\hat{U}(t)$  as a product of simpler exponentials
- (e) Calculate the error in the Trotter approximation for small time steps  $\Delta t = t/N$

**Problem 4.4** (*Rotations on the Bloch Sphere*)

A spin-1/2 particle is subject to magnetic fields in different directions.

- (a) For  $\vec{B} = B_0\hat{x}$ , calculate the rotation operator  $\hat{R}_x(\theta)$  where  $\theta = \gamma B_0 t$
- (b) Starting from  $|+z\rangle$ , find the state after rotating by angle  $\pi/3$  about the x-axis
- (c) Calculate the expectation values  $\langle\sigma_x\rangle$ ,  $\langle\sigma_y\rangle$ , and  $\langle\sigma_z\rangle$  for this rotated state
- (d) Now apply a subsequent rotation of  $\pi/2$  about the z-axis. What is the final state?
- (e) Show that rotations about different axes do not commute by computing  $\hat{R}_x(\pi/2)\hat{R}_z(\pi/2)$  and  $\hat{R}_z(\pi/2)\hat{R}_x(\pi/2)$
- (f) Visualize these operations on the Bloch cube, showing how the cube orientation changes with each rotation

**Problem 4.5** (*Compound Rotations and the General Rotation Formula*)

The general rotation formula for a spin-1/2 system is  $\hat{R}_{\hat{n}}(\theta) = \cos(\theta/2)\hat{I} - i\sin(\theta/2)(\hat{n} \cdot \vec{\sigma})$ .

- Show that rotations about parallel axes commute:  $\hat{R}_{\hat{n}}(\theta_1)\hat{R}_{\hat{n}}(\theta_2) = \hat{R}_{\hat{n}}(\theta_1 + \theta_2)$
- For an arbitrary axis  $\hat{n} = (\sin\phi\cos\lambda, \sin\phi\sin\lambda, \cos\phi)$ , write out the explicit matrix form of  $\hat{R}_{\hat{n}}(\theta)$
- Show that the Hadamard gate can be written as a rotation about the axis  $\hat{n} = (\hat{x} + \hat{z})/\sqrt{2}$ . What is the rotation angle?
- Verify that  $\hat{R}_{\hat{n}}(\pi)$  implements a reflection through the axis  $\hat{n}$  on the Bloch sphere
- Starting from the rotation formula, derive the time evolution operator for  $\hat{H} = \vec{h} \cdot \vec{\sigma}$  in terms of rotations
- Show that the composition of two rotations about different axes can be written as a single rotation about a third axis (derive the axis and angle)

**Problem 4.6** (*Diagonalization and Time Evolution*)

Consider the Hamiltonian  $\hat{H} = \hbar\omega \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$ .

- Find the eigenvalues and normalized eigenvectors of  $\hat{H}$
- Construct the diagonalization matrix  $\hat{V}$  and verify that  $\hat{V}^\dagger \hat{H} \hat{V}$  is diagonal
- Using the diagonal form, calculate  $e^{-i\hat{H}t/\hbar}$
- Transform back to the original basis to get the time evolution operator
- If the system starts in state  $|\psi(0)\rangle = |0\rangle$ , find  $|\psi(t)\rangle$
- Calculate the probability of measuring the system in state  $|1\rangle$  as a function of time
- What is the period of oscillation between the two states?

**Problem 4.7** (*Conservation Laws in Exchange-Symmetric Systems*)

Consider two spin-1/2 particles with Hamiltonian  $\hat{H} = J(\vec{S}_1 \cdot \vec{S}_2) + B(S_{1z} + S_{2z})$ , where  $\vec{S}_i = \frac{\hbar}{2}\vec{\sigma}_i$  and the system is invariant under particle exchange.

- Show that the total spin operators  $\vec{S}_{total} = \vec{S}_1 + \vec{S}_2$  commute with  $\hat{H}$
- Express  $\hat{H}$  in terms of  $S_{total}^2$  and  $S_{total,z}$  and show both are conserved quantities
- Find the energy eigenvalues for the triplet ( $S = 1$ ) and singlet ( $S = 0$ ) subspaces
- Construct the four energy eigenstates in the  $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$  basis
- For the initial state  $|\psi(0)\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ , calculate  $\langle S_{total}^2 \rangle$  and  $\langle S_{total,z} \rangle$  and verify they remain constant
- Identify the physical symmetry (particle exchange) and show how it leads to these conservation laws using Noether's theorem concepts

**Problem 4.8** (*Heisenberg vs Schrodinger Pictures*)

For the Hamiltonian  $\hat{H} = \hbar\omega\sigma_y$ , compare time evolution in both pictures.

- In the Schrodinger picture, find  $|\psi_S(t)\rangle$  for initial state  $|\psi_S(0)\rangle = | + z \rangle$
- Calculate  $\langle\sigma_x\rangle(t)$  using the Schrodinger picture
- In the Heisenberg picture, find the time-evolved operator  $\sigma_{x,H}(t)$
- Verify that  $\langle\psi_S(t)|\sigma_x|\psi_S(t)\rangle = \langle\psi_H|\sigma_{x,H}(t)|\psi_H\rangle$
- Find the Heisenberg equations of motion for all three Pauli operators
- Show that these equations form a closed system and solve them directly

**Problem 4.9** (*Driven Two-Level System and Rabi Oscillations*)

A two-level atom interacts with a classical electromagnetic field:

$$\hat{H}(t) = \frac{\hbar\omega_0}{2}\sigma_z + \hbar\Omega\cos(\omega t)\sigma_x \quad (4.251)$$

- Transform to the rotating frame at frequency  $\omega$  and derive the effective Hamiltonian
- For the resonant case ( $\omega = \omega_0$ ), find the time evolution of a state initially in  $|0\rangle$
- Calculate the population inversion  $\langle\sigma_z\rangle(t)$  and identify the Rabi frequency
- What is the minimum time to perform a complete population inversion (a  $\pi$ -pulse)?
- For small detuning  $\Delta = \omega_0 - \omega$ , find the modified Rabi frequency
- Plot the population of the excited state vs time for  $\Delta = 0$ ,  $\Delta = \Omega/2$ , and  $\Delta = 2\Omega$

**Problem 4.10** (*Geometric Phase and Adiabatic Evolution*)

A spin-1/2 particle is in a magnetic field that rotates slowly:

$$\vec{B}(t) = B(\sin\alpha\cos(\omega t), \sin\alpha\sin(\omega t), \cos\alpha) \quad (4.252)$$

where  $\omega \ll \gamma B$  (adiabatic condition) and  $\alpha$  is a fixed tilt angle.

- Write the Hamiltonian  $\hat{H}(t) = -\gamma\vec{B}(t) \cdot \vec{S}$  in terms of Pauli matrices
- Find the instantaneous eigenstates and eigenvalues of  $\hat{H}(t)$
- After one complete rotation (period  $T = 2\pi/\omega$ ), what is the solid angle traced by the magnetic field on the unit sphere?
- Using the formula  $\phi_{geom} = -\frac{1}{2}\Omega$  from the chapter, calculate the Berry phase
- If the system starts in the ground state and follows adiabatically, describe the final state after one period
- Using the Bloch cube visualization from the chapter, explain why a triangular path connecting three adjacent vertices (e.g.,  $| + z \rangle \rightarrow | + x \rangle \rightarrow | + y \rangle \rightarrow | + z \rangle$ ) gives a Berry phase of  $-\pi/4$



**Problem 4.11** (*Hopping on a Four-Site Ring*)

Consider four sites arranged in a ring with nearest-neighbor hopping:

$$\hat{H} = -\Delta \sum_{j=0}^3 (\hat{a}_j^\dagger \hat{a}_{j+1} + \hat{a}_{j+1}^\dagger \hat{a}_j) \quad (4.253)$$

where site 4 is identified with site 0.

- Find all momentum eigenstates  $|k_n\rangle$  and their energies
- Write the time evolution operator in the momentum basis
- If a particle starts localized at site 0, find the probability amplitude at each site as a function of time
- Calculate the probability of finding the particle at site 2 (opposite side) at time  $t = \pi/(2\Delta)$
- Show that the particle returns to site 0 at certain revival times and find the shortest such time
- Add a second particle initially at site 2. How does the hard-core constraint affect the dynamics?

**Problem 4.12** (*Density Matrix Evolution*)

Consider a pure state  $|\psi(0)\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$  evolving under  $\hat{H} = \hbar\omega\sigma_z/2$ .

- Construct the initial density matrix  $\rho(0) = |\psi(0)\rangle\langle\psi(0)|$
- Using  $\rho(t) = \hat{U}(t)\rho(0)\hat{U}^\dagger(t)$ , derive the time-evolved density matrix  $\rho(t)$
- Calculate the trace  $\text{Tr}[\rho(t)]$  and show it remains unity for all times
- Compute the expectation values  $\langle\sigma_x\rangle(t)$ ,  $\langle\sigma_y\rangle(t)$ ,  $\langle\sigma_z\rangle(t)$  from  $\rho(t)$
- Verify that  $\text{Tr}[\rho^2(t)] = 1$  for all times, confirming the state remains pure

**Problem 4.13** (*Heisenberg Equations for Ladder Operators*)

For a quantum harmonic oscillator with  $\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + 1/2)$ , find the Heisenberg equations of motion.

- Derive  $\frac{d\hat{a}}{dt}$  using  $\frac{d\hat{A}_H}{dt} = \frac{1}{i\hbar}[\hat{A}_H, \hat{H}]$
- Solve this equation for  $\hat{a}_H(t)$
- Verify that  $\langle\hat{a}\rangle(t)$  oscillates with frequency  $\omega$
- Show that the uncertainty principle is preserved:  $\Delta x \Delta p \geq \hbar/2$

**Problem 4.14** (*Spin Echo and CPMG Sequences*)

In a two-level system with a strong dephasing field, a sequence of pulses can restore coherence.

- Consider a state  $|\psi(0)\rangle = |+\rangle$  in a field  $\vec{B} = B_0\hat{z} + B_1(t)\hat{x}$
- For a simple echo:  $\pi/2$  pulse, wait time  $\tau$ ,  $\pi$  pulse, wait time  $\tau$ . Show that the accumulated phase is eliminated

- (c) For a Carr-Purcell-Meiboom-Gill (CPMG) sequence with  $N$  pulses, estimate the effective dephasing time versus unrefocused decay
- (d) Plot the echo amplitude as a function of the number of pulses for given dephasing rate

**Problem 4.15** (*Trotter Error Scaling*)

The Trotter formula decomposes time evolution into small steps:  $e^{-i(\hat{H}_1+\hat{H}_2)t/\hbar} \approx [e^{-i\hat{H}_1\Delta t/\hbar}e^{-i\hat{H}_2\Delta t/\hbar}]^N$  where  $\Delta t = t/N$ .

- (a) For  $\hat{H}_1 = \hbar\omega\sigma_x/2$  and  $\hat{H}_2 = \hbar\omega\sigma_z/2$ , compute the exact commutator  $[\hat{H}_1, \hat{H}_2]$
- (b) The Trotter error is given by  $\epsilon_N = O(\Delta t^2) = O(t^2/N^2)$ . For a fixed total evolution time  $t$ , how does the error scale with the number of Trotter steps  $N$ ?
- (c) Estimate the number of steps needed to achieve error less than  $10^{-6}$  for  $t = 1\mu s$  and  $|\hat{H}| \sim 1GHz$

**Problem 4.16** (*Magnus Expansion*)

For a time-dependent Hamiltonian, the time evolution operator can be expanded as  $\hat{U}(t) = e^{\hat{S}(t)}$  where  $\hat{S}(t)$  is called the Magnus expansion.

- (a) Write the first two terms of the Magnus expansion:  $\hat{S}(t) = \hat{S}_1(t) + \hat{S}_2(t) + \dots$
- (b) Apply this to  $\hat{H}(t) = \hbar(\omega_0/2)\sigma_z + \hbar\Omega\cos(\omega t)\sigma_x$  in the rotating frame
- (c) Show that averaging effects suppress certain error terms, improving the effective evolution

**Problem 4.17** (*Quantum Zeno Effect*)

Rapid measurements can freeze quantum evolution. Consider measuring  $|0\rangle$  at intervals  $\Delta t$ .

- (a) For  $\hat{H} = \hbar\omega\sigma_x/2$  starting in  $|0\rangle$ , the unchecked transition probability to  $|1\rangle$  in time  $t$  is approximately  $P(t) \approx (\omega t/2)^2$  for small  $t$
- (b) If we measure  $n$  times at intervals  $\Delta t = t/n$ , the survival probability is approximately  $[1 - (\omega\Delta t/2)^2]^n$
- (c) Show that in the limit  $n \rightarrow \infty$ ,  $[1 - (\omega t/(2n))^2]^n \rightarrow 1$ , demonstrating the quantum Zeno effect
- (d) Discuss the physical interpretation: what prevents transitions when measurements are frequent?

## References and Further Reading

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**Rabi, I. I.** “Space quantization in a gyrating magnetic field.” *Physical Review* **51**, 652–654 (1937). [doi:10.1103/PhysRev.51.652](https://doi.org/10.1103/PhysRev.51.652). The original analysis of the driven two-level system; the source of the Rabi oscillation formula used in the NMR section.

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**Landau, L. D., and Lifshitz, E. M.** *Quantum Mechanics: Non-Relativistic Theory*, 3rd ed. Pergamon Press, 1977. Chapter III on Schrödinger’s equation is concise and physically motivated; recommended for the conservation-law and symmetry discussion.



# Chapter 5

## Continuous Space and the Quantum Harmonic Oscillator

### 5.1 Introduction

#### Key Idea: From Discrete Lattice to Continuous Wave Function

As the lattice spacing approaches zero, the discrete amplitude at each site transforms into a continuous wave function  $\psi(x)$ . Traditional wave mechanics—particles in boxes, harmonic oscillators, quantum tunneling—emerges from the discrete qubit framework developed in previous chapters.

Chapter 3 showed that a single particle on  $n$  lattice sites occupies an  $n$ -dimensional subspace within the full  $2^n$ -dimensional Hilbert space. We now pass to the continuum limit: the lattice spacing  $a \rightarrow 0$ , the number of sites  $n \rightarrow \infty$ , while the total system size  $L = na$  remains fixed.

The limit transforms:

- Discrete position states  $|x_i\rangle$  into continuous states  $|x\rangle$
- Sums into integrals
- Finite-difference operators into differential operators

Differential operators come from finite differences, boundary conditions from lattice edges, and wave functions from continuous limits of discrete amplitudes.

#### 5.1.1 Discrete vs Continuous: Two Perspectives

The discrete-continuous connection is important theoretically and practically:

- **Computational:** Numerical solutions require discretization. Understanding the lattice-continuum connection improves numerical accuracy.

- **Physical:** Many systems (quantum dots, optical lattices, qubit arrays) are inherently discrete. The discrete formulation is exact for these systems.
- **Fundamental:** Whether space is discrete at the Planck scale remains an open question in quantum gravity.

The formulations are complementary.

## 5.2 From Discrete Lattice to Continuous Space

### 5.2.1 Position States in the Discrete and Continuous Pictures

On a one-dimensional lattice with spacing  $a$ , the discrete positions are labeled  $x_j = ja$  for  $j = 1, 2, \dots, n$ . A particle localized at site  $j$  is described by the state  $|x_j\rangle = |0\dots 010\dots 0\rangle$ , where the single 1 appears at position  $j$ . These  $n$  states form an orthonormal basis for the single-particle sector:

$$\langle x_j | x_k \rangle = \delta_{jk} \quad (5.1)$$

The completeness relation for the discrete position basis reads:

$$\sum_{j=1}^n |x_j\rangle \langle x_j| = \hat{I}_{\text{single-particle}} \quad (5.2)$$

In the continuum limit, the discrete index  $j$  becomes the continuous variable  $x$ , the lattice spacing  $a$  approaches zero, and the number of sites  $n$  approaches infinity such that the system size  $L = na$  remains fixed.

The transformations are:

- Kronecker delta  $\delta_{jk}$  becomes Dirac delta  $\delta(x - x')$
- Discrete sums  $\sum_i$  become integrals  $\int dx$
- Finite-dimensional Hilbert space becomes infinite-dimensional

The orthonormality relation becomes:

$$\langle x | x' \rangle = \delta(x - x') \quad (5.3)$$

Discrete states are normalizable ( $\langle x_j | x_j \rangle = 1$ ), but continuous position eigenstates are distributions, not functions—reflecting that perfect localization at a point is physically impossible.

#### Example 1: From Discrete Delta to Dirac Delta

Consider the relationship between localized states on a lattice and in the continuum.

**On the lattice:** A particle perfectly localized at site  $j_0$  (where  $x_{j_0} = j_0 a$ ) has the wave function:

$$\psi_j = \delta_{j,j_0} \quad (5.4)$$

Normalization:  $\sum_j |\psi_j|^2 = 1$ .

**Continuum correspondence:** To connect to the continuum, we must account for the den-

sity of states. The continuum wave function  $\psi(x)$  relates to the discrete amplitudes via:

$$\psi(x_j) = \frac{\psi_j}{\sqrt{a}} \quad (5.5)$$

This scaling ensures that in the continuum limit:

$$\sum_j |\psi_j|^2 = \sum_j a \cdot \left| \frac{\psi_j}{\sqrt{a}} \right|^2 \rightarrow \int dx |\psi(x)|^2 \quad (5.6)$$

For perfect localization at  $x_0$ :

- Discrete:  $\psi_j = \delta_{j,j_0}$
- Continuum:  $\psi(x) = \delta(x - x_0)$

The  $1/\sqrt{a}$  scaling ensures proper normalization in the transition from sums to integrals.

### 5.2.2 The Wave Function as Continuous Amplitude

A general single-particle state on the lattice takes the form:

$$|\psi\rangle = \sum_{j=1}^n \psi_j |x_j\rangle \quad (5.7)$$

where  $\psi_j$  represents the probability amplitude for finding the particle at site  $j$ .

The transformation requires normalization:

#### Example 2: Normalization in the Continuum Limit

Define the continuum wave function as  $\psi(x) = \psi_j/\sqrt{a}$  where  $x = ja$ . Then:

$$\sum_j |\psi_j|^2 = \sum_j a |\psi(x_j)|^2 \quad (5.8)$$

$$\rightarrow \int dx |\psi(x)|^2 \quad (5.9)$$

The factor  $1/\sqrt{a}$  ensures the integral converges to the sum as  $a \rightarrow 0$ .

In the continuum limit, the state becomes:

$$|\psi\rangle = \int_{-\infty}^{\infty} dx \psi(x) |x\rangle \quad (5.10)$$

The coefficient function  $\psi(x) = \langle x|\psi\rangle$  is the wave function—the continuous probability amplitude for finding the particle at position  $x$ .

### 5.2.3 The Mathematical Structure of Wave Functions

Wave functions belong to the Hilbert space  $L^2(\mathbb{R})$  of square-integrable functions:

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx < \infty \quad (5.11)$$

This requirement ensures that the total probability is finite and can be normalized to unity. The inner product in this space is:

$$\langle \phi | \psi \rangle = \int_{-\infty}^{\infty} \phi^*(x) \psi(x) dx \quad (5.12)$$

Properties of the wave function space:

### Example 3: Properties of the Wave Function Space

**1. Linearity:** If  $\psi_1(x)$  and  $\psi_2(x)$  are valid wave functions, so is any linear combination:

$$\psi(x) = c_1 \psi_1(x) + c_2 \psi_2(x) \quad (5.13)$$

where  $|c_1|^2 + |c_2|^2 = 1$  for normalization.

**2. Completeness:** Any square-integrable function can be expanded in a complete basis:

$$\psi(x) = \sum_n c_n \phi_n(x) \quad (5.14)$$

where  $\{\phi_n(x)\}$  forms an orthonormal basis.

**3. Continuity:** Physical wave functions and their derivatives must be continuous except at infinite potential jumps. This ensures finite momentum uncertainty.

**4. Boundary behavior:** For bound states,  $\psi(x) \rightarrow 0$  as  $|x| \rightarrow \infty$  fast enough to be square-integrable.

#### 5.2.4 Inner Products and Probability

The inner product between states transforms from a discrete sum to a continuous integral:

### Example 4: Inner Product Evolution

Discrete lattice:

$$\langle \phi | \psi \rangle = \sum_{j=1}^n \phi_j^* \psi_j \quad (5.15)$$

Continuum limit (with proper normalization):

$$\langle \phi | \psi \rangle = \int_{-\infty}^{\infty} dx \phi^*(x) \psi(x) \quad (5.16)$$

For orthogonal states on a three-site lattice:

$$\psi_1 = \frac{1}{\sqrt{2}}(1, 0, -1) \quad (5.17)$$

$$\psi_2 = \frac{1}{\sqrt{3}}(1, -2, 1) \quad (5.18)$$



We have  $\sum_j \psi_1^*[j]\psi_2[j] = \frac{1}{\sqrt{6}}(1 - 0 - 1) = 0$ .

The continuum analogs are orthogonal functions like  $\sin(x)$  and  $\cos(x)$ .

### 5.2.5 The Emergence of Differential Operators

The nearest-neighbor hopping Hamiltonian on the lattice was:

$$\hat{H}_{\text{hop}} = -t \sum_{j=1}^{n-1} (|x_{j+1}\rangle\langle x_j| + |x_j\rangle\langle x_{j+1}|) \quad (5.19)$$

The hopping operator becomes a differential operator as follows:

**Example 5: Detailed Derivation of the Kinetic Energy Operator**

The hopping Hamiltonian acts on a lattice wave function as:

$$(\hat{H}_{\text{hop}}\psi)_j = -t(\psi_{j+1} + \psi_{j-1}) \quad (5.20)$$

Using Taylor expansion around  $x_j = ja$ :

$$\psi_{j+1} = \psi(x_j + a) = \psi(x_j) + a\psi'(x_j) + \frac{a^2}{2}\psi''(x_j) + \frac{a^3}{6}\psi'''(x_j) + O(a^4) \quad (5.21)$$

$$\psi_{j-1} = \psi(x_j - a) = \psi(x_j) - a\psi'(x_j) + \frac{a^2}{2}\psi''(x_j) - \frac{a^3}{6}\psi'''(x_j) + O(a^4) \quad (5.22)$$

Adding these:

$$\psi_{j+1} + \psi_{j-1} = 2\psi(x_j) + a^2\psi''(x_j) + O(a^4) \quad (5.23)$$

Therefore:

$$(\hat{H}_{\text{hop}}\psi)_j = -t[2\psi(x_j) + a^2\psi''(x_j)] + O(a^4) \quad (5.24)$$

Setting  $t = \hbar^2/(2ma^2)$  and shifting the energy zero to remove the constant term:

$$\hat{H}_{\text{hop}} \rightarrow -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} = \hat{T} \quad (5.25)$$

The parameter  $t = \hbar^2/(2ma^2)$  where:

- $t$ : hopping amplitude (energy scale for tunneling)
- $m$ : particle mass (inertia)
- $a$ : lattice spacing
- $\hbar$ : action scale

### 5.2.6 Momentum States and Dispersion Relations

The continuum limit also transforms how we understand momentum states and their energy-momentum relationships:

**Example 6: Continuum Limit of Momentum States**

The discrete momentum states on a lattice with periodic boundary conditions are:

$$|k_m\rangle = \frac{1}{\sqrt{n}} \sum_{j=1}^n e^{ik_m j} |ja\rangle \quad (5.26)$$

where  $k_m = 2\pi m/L$  with  $m = 0, 1, \dots, n-1$ .

In the continuum limit, these become plane waves:

$$\psi_k(x) = \frac{1}{\sqrt{L}} e^{ikx} \quad (5.27)$$

The dispersion relation transforms from On the lattice:

$$E_m = -2t \cos(k_m a) \quad (5.28)$$

For small  $k_m a$  (long wavelengths compared to lattice spacing):

$$E_m = -2t \cos(k_m a) \quad (5.29)$$

$$\approx -2t \left( 1 - \frac{(k_m a)^2}{2} \right) \quad \text{for } k_m a \ll 1 \quad (5.30)$$

$$= -2t + \frac{\hbar^2 k_m^2}{2m} \quad (5.31)$$

Dropping the constant energy shift, we recover the free particle dispersion:

$$E = \frac{\hbar^2 k^2}{2m} = \frac{p^2}{2m} \quad (5.32)$$

The quadratic free-particle dispersion emerges from the lattice cosine dispersion at long wavelengths.

**5.2.7 The Position Representation of Quantum Mechanics**

In the position representation, quantum mechanical operators become differential operators acting on wave functions.

**Operator Dictionary: Discrete to Continuous**

| Physical Quantity  | Discrete (Lattice)                                            | Continuous (Position)                            |
|--------------------|---------------------------------------------------------------|--------------------------------------------------|
| Position           | $\hat{x} = \sum_j x_j  x_j\rangle\langle x_j $                | $\hat{x} = x$ (multiplication)                   |
| Translation by $a$ | $\hat{T}_a = \sum_j  x_{j+1}\rangle\langle x_j $              | $\hat{T}_a = e^{-i\hat{p}a/\hbar}$               |
| Momentum           | $\hat{p} \approx -i\hbar \frac{\hat{T}_a - \hat{T}_{-a}}{2a}$ | $\hat{p} = -i\hbar \frac{d}{dx}$                 |
| Kinetic Energy     | $\hat{T} = -t(\hat{T}_a + \hat{T}_{-a} - 2\hat{I})$           | $\hat{T} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$ |
| Number at site     | $\hat{n}_j =  x_j\rangle\langle x_j $                         | $\delta(x - x_j)$                                |

The momentum operator emerges from considering infinitesimal translations:

**Example 7: Deriving the Momentum Operator**

For a small translation by distance  $\epsilon$ :

$$\psi(x - \epsilon) = \psi(x) - \epsilon \frac{d\psi}{dx} + O(\epsilon^2) \quad (5.33)$$

The translation operator is:

$$\hat{T}_\epsilon = 1 - \epsilon \frac{d}{dx} = 1 - \frac{i\epsilon}{\hbar} (-i\hbar \frac{d}{dx}) \quad (5.34)$$

Identifying the generator of translations as momentum:

$$\hat{T}_\epsilon = 1 - \frac{i\epsilon}{\hbar} \hat{p} \quad \Rightarrow \quad \hat{p} = -i\hbar \frac{d}{dx} \quad (5.35)$$

For finite translations:  $\hat{T}_a = e^{-i\hat{p}a/\hbar}$ .

**5.2.8 Commutation Relations**

The fundamental commutation relation emerges naturally:

**Example 8: Position-Momentum Commutator**

$$[\hat{x}, \hat{p}]\psi(x) = x(-i\hbar \frac{d\psi}{dx}) - (-i\hbar \frac{d}{dx})(x\psi) \quad (5.36)$$

$$= -i\hbar x \frac{d\psi}{dx} - (-i\hbar) \left( \psi + x \frac{d\psi}{dx} \right) \quad (5.37)$$

$$= -i\hbar x \frac{d\psi}{dx} + i\hbar \psi + i\hbar x \frac{d\psi}{dx} \quad (5.38)$$

$$= i\hbar \psi \quad (5.39)$$

Therefore:  $[\hat{x}, \hat{p}] = i\hbar$ .

This can be verified on the lattice for large  $n$ :

$$[\hat{x}_j, \hat{p}_k]_{\text{lattice}} \approx i\hbar \delta_{jk} + O(a) \quad (5.40)$$

**Verification on the lattice:** On the lattice, position acts as multiplication by  $x_j = ja$ , while momentum implements a finite difference:

$$(\hat{x}\psi)_j = x_j \psi_j \quad (5.41)$$

$$(\hat{p}\psi)_j = -i\hbar \frac{\psi_{j+1} - \psi_{j-1}}{2a} \quad (5.42)$$

Computing the commutator:

$$[\hat{x}, \hat{p}]\psi_j = x_j \cdot \left( -i\hbar \frac{\psi_{j+1} - \psi_{j-1}}{2a} \right) - \hat{p}(x_j \psi_j) \quad (5.43)$$

For the second term, noting that  $x_{j\pm 1} = x_j \pm a$ :

$$\hat{p}(x\psi)_j = -i\hbar \frac{x_{j+1}\psi_{j+1} - x_{j-1}\psi_{j-1}}{2a} \quad (5.44)$$

$$= -i\hbar \left[ \frac{x_j(\psi_{j+1} - \psi_{j-1})}{2a} + \frac{\psi_{j+1} + \psi_{j-1}}{2} \right] \quad (5.45)$$

Therefore:

$$[\hat{x}, \hat{p}]\psi_j = i\hbar \frac{\psi_{j+1} + \psi_{j-1}}{2} \quad (5.46)$$

Using Taylor expansion  $\psi_{j\pm 1} = \psi_j + O(a)$ , we get  $[\hat{x}, \hat{p}] = i\hbar \cdot \mathbb{I} + O(a^2)$  on the lattice, confirming that the canonical commutation relation emerges exactly in the continuum limit.

### 5.2.9 The Baker-Campbell-Hausdorff Formula

The commutation relation  $[\hat{x}, \hat{p}] = i\hbar$  has profound consequences that extend far beyond the uncertainty principle. Before proceeding to multiple representations and specific examples, we introduce a powerful mathematical tool that will be essential throughout our study of continuous quantum systems.

**Physical Intuition: Commutators as Derivatives**

The Baker-Campbell-Hausdorff (BCH) formula describes how operators transform under conjugation by exponentials. This is deeply connected to time evolution in quantum mechanics.

Consider the Heisenberg picture, where operators evolve in time while states remain constant. The key equation is:

$$\frac{d\hat{B}}{dt} = \frac{i}{\hbar}[\hat{H}, \hat{B}] \quad (5.47)$$

The commutator  $[\hat{H}, \cdot]$  acts like a derivative operator in the space of operators!

**Formal Solution: From Schrödinger to Heisenberg Picture**

We know from the Schrödinger picture that states evolve as:

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}|\psi(0)\rangle \quad (5.48)$$

Expectation values must be independent of which picture we use. In the Schrödinger picture:

$$\langle \hat{B} \rangle = \langle \psi(t) | \hat{B} | \psi(t) \rangle = \langle \psi(0) | e^{i\hat{H}t/\hbar} \hat{B} e^{-i\hat{H}t/\hbar} | \psi(0) \rangle \quad (5.49)$$

This suggests defining a time-evolved operator in the Heisenberg picture:

$$\boxed{\hat{B}(t) = e^{i\hat{H}t/\hbar} \hat{B}(0) e^{-i\hat{H}t/\hbar}} \quad (5.50)$$

This is the formal solution to the operator equation of motion.

**Verification:** Let us verify that this exponential form satisfies  $\frac{d\hat{B}}{dt} = \frac{i}{\hbar}[\hat{H}, \hat{B}]$ .

Taking the time derivative using the product rule:

$$\begin{aligned} \frac{d\hat{B}}{dt} &= \frac{d}{dt} \left( e^{i\hat{H}t/\hbar} \hat{B}(0) e^{-i\hat{H}t/\hbar} \right) \\ &= \frac{i\hat{H}}{\hbar} e^{i\hat{H}t/\hbar} \hat{B}(0) e^{-i\hat{H}t/\hbar} + e^{i\hat{H}t/\hbar} \hat{B}(0) \left( -\frac{i\hat{H}}{\hbar} \right) e^{-i\hat{H}t/\hbar} \\ &= \frac{i}{\hbar} \hat{H} \hat{B}(t) - \frac{i}{\hbar} \hat{B}(t) \hat{H} \\ &= \frac{i}{\hbar} [\hat{H}, \hat{B}(t)] \end{aligned} \quad (5.51)$$

The exponential form indeed satisfies the operator equation of motion. This formal solution is the foundation of the BCH formula.

**Taylor Series Expansion with Nested Commutators**

We can expand the formal solution as a Taylor series. Just as for ordinary functions:

$$f(t) = f(0) + tf'(0) + \frac{t^2}{2!}f''(0) + \frac{t^3}{3!}f'''(0) + \dots \quad (5.52)$$

we have for operators:

$$\hat{B}(t) = \hat{B}(0) + t \left. \frac{d\hat{B}}{dt} \right|_{t=0} + \frac{t^2}{2!} \left. \frac{d^2\hat{B}}{dt^2} \right|_{t=0} + \frac{t^3}{3!} \left. \frac{d^3\hat{B}}{dt^3} \right|_{t=0} + \dots \quad (5.53)$$

Now substitute the expressions for the time derivatives:

$$\begin{aligned}\frac{d\hat{B}}{dt} &= \frac{i}{\hbar}[\hat{H}, \hat{B}] \\ \frac{d^2\hat{B}}{dt^2} &= \frac{i}{\hbar}[\hat{H}, \frac{d\hat{B}}{dt}] = \left(\frac{i}{\hbar}\right)^2 [\hat{H}, [\hat{H}, \hat{B}]] \\ \frac{d^3\hat{B}}{dt^3} &= \left(\frac{i}{\hbar}\right)^3 [\hat{H}, [\hat{H}, [\hat{H}, \hat{B}]]]\end{aligned}\tag{5.54}$$

This gives:

$$\hat{B}(t) = \hat{B} + \frac{it}{\hbar}[\hat{H}, \hat{B}] + \frac{(it/\hbar)^2}{2!}[\hat{H}, [\hat{H}, \hat{B}]] + \frac{(it/\hbar)^3}{3!}[\hat{H}, [\hat{H}, [\hat{H}, \hat{B}]]] + \dots\tag{5.55}$$

### Connection to the General BCH Formula

Setting  $\hat{A} = i\hat{H}t/\hbar$ , the series becomes:

$$\hat{B}(t) = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!}[\hat{A}, [\hat{A}, \hat{B}]] + \frac{1}{3!}[\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]] + \dots\tag{5.56}$$

But we also showed that  $\hat{B}(t) = e^{\hat{A}}\hat{B}e^{-\hat{A}}$ . Therefore, we have discovered:

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!}[\hat{A}, [\hat{A}, \hat{B}]] + \frac{1}{3!}[\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]] + \dots\tag{5.57}$$

This is the BCH conjugation formula! The operator  $e^{\hat{A}}$  "flows"  $\hat{B}$  along the direction generated by repeated commutators with  $\hat{A}$ . Each term involves one more nested commutator, exactly as each term in a Taylor series involves one more derivative.

### The General Conjugation Formula

More generally, for any operators  $\hat{A}$  and  $\hat{B}$ :

$$\boxed{e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!}[\hat{A}, [\hat{A}, \hat{B}]] + \frac{1}{3!}[\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]] + \dots}\tag{5.58}$$

Think of this as: the operator  $e^{\hat{A}}$  "flows"  $\hat{B}$  along the direction generated by repeated commutators with  $\hat{A}$ . Each term involves one more nested commutator, just as each term in a Taylor series involves one more derivative. The commutator  $[\hat{A}, \cdot]$  is the "derivative" in operator space.

#### Example 9: BCH for Position Under Momentum Translation

Consider translating position by momentum:  $\hat{A} = -i\epsilon\hat{p}/\hbar$ ,  $\hat{B} = \hat{x}$ .

The first commutator:

$$[\hat{A}, \hat{B}] = [-i\epsilon\hat{p}/\hbar, \hat{x}] = -i\epsilon[\hat{p}, \hat{x}]/\hbar = -i\epsilon(-i\hbar)/\hbar = -\epsilon\tag{5.59}$$

This is a c-number! Therefore all higher commutators vanish:

$$[\hat{A}, [\hat{A}, \hat{B}]] = [\hat{A}, -\epsilon] = 0\tag{5.60}$$

The BCH series terminates:

$$e^{-i\epsilon\hat{p}/\hbar}\hat{x}e^{i\epsilon\hat{p}/\hbar} = \hat{x} - \epsilon \quad (5.61)$$

Momentum translation shifts position by  $\epsilon$ , exactly as expected classically!

### The Key Simplification: C-Number Commutators

When  $[\hat{A}, \hat{B}]$  is a c-number (a multiple of the identity), all higher commutators automatically vanish:

$$[\hat{A}, [\hat{A}, \hat{B}]] = [\hat{A}, c\hat{I}] = c[\hat{A}, \hat{I}] = 0 \quad (5.62)$$

The infinite series becomes just two terms:

$$\boxed{e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}]} \quad (\text{when } [\hat{A}, \hat{B}] \text{ is a c-number}) \quad (5.63)$$

This dramatic simplification is why position and momentum operators have such simple transformation properties under displacement: their commutator  $[\hat{x}, \hat{p}] = i\hbar$  is a c-number. We will exploit this repeatedly when constructing coherent states.

### The Exponential Product Formula

A related BCH result concerns products of exponentials. For two operators  $\hat{A}$  and  $\hat{B}$  that don't commute:

$$e^{\hat{A}+\hat{B}} \neq e^{\hat{A}}e^{\hat{B}} \quad (5.64)$$

The general BCH formula for exponential products is an infinite series involving nested commutators. However, when  $[\hat{A}, \hat{B}]$  commutes with both  $\hat{A}$  and  $\hat{B}$  (automatically satisfied if  $[\hat{A}, \hat{B}]$  is a c-number), the formula simplifies to:

$$\boxed{e^{\hat{A}+\hat{B}} = e^{\hat{A}}e^{\hat{B}}e^{-[\hat{A}, \hat{B}]/2}} \quad (5.65)$$

This form is particularly useful when constructing coherent states of the harmonic oscillator.

#### Example 10: BCH for Harmonic Oscillator Displacement Operator

For the harmonic oscillator displacement operator  $\hat{D}(\alpha) = e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}$ , we have  $\hat{A} = \alpha\hat{a}^\dagger$  and  $\hat{B} = -\alpha^*\hat{a}$ .

The commutator is:

$$[\hat{A}, \hat{B}] = [\alpha\hat{a}^\dagger, -\alpha^*\hat{a}] = -\alpha\alpha^*[\hat{a}^\dagger, \hat{a}] = |\alpha|^2 \quad (5.66)$$

Since this is a c-number, we can apply the exponential product formula:

$$e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}} = e^{\alpha\hat{a}^\dagger}e^{-\alpha^*\hat{a}}e^{-|\alpha|^2/2} = e^{-|\alpha|^2/2}e^{\alpha\hat{a}^\dagger}e^{-\alpha^*\hat{a}} \quad (5.67)$$

This factorization is essential for constructing and analyzing coherent states.



**Summary: Why BCH Matters**

The Baker-Campbell-Hausdorff formula is not just a mathematical curiosity—it reveals the fundamental algebraic structure of quantum mechanics:

- **Commutators are derivatives:** The commutator  $[\hat{A}, \cdot]$  acts like a directional derivative in operator space
- **Exponentials generate flows:**  $e^{\hat{A}}$  generates a "flow" along the direction defined by  $\hat{A}$
- **C-numbers simplify everything:** When commutators yield c-numbers, infinite series truncate to simple expressions
- **Classical-quantum connection:** BCH shows how quantum operators reduce to classical transformations in appropriate limits

We will use BCH extensively in constructing coherent states, analyzing displacement operators, and understanding squeezed states. The formula provides both computational power and conceptual insight into the structure of quantum theory.

**5.2.10 Connections Between Discrete and Continuous****Finite Differences and Differential Operators**

The continuum limit can be reversed to recover discrete approximations:

**Example 11: Finite Difference Approximations**

Second derivative:

$$\frac{d^2\psi}{dx^2} \approx \frac{\psi(x+a) - 2\psi(x) + \psi(x-a)}{a^2} + O(a^2) \quad (5.68)$$

First derivative (symmetric):

$$\frac{d\psi}{dx} \approx \frac{\psi(x+a) - \psi(x-a)}{2a} + O(a^2) \quad (5.69)$$

The discrete Schrödinger equation on a lattice:

$$-\frac{\hbar^2}{2ma^2}(\psi_{j+1} - 2\psi_j + \psi_{j-1}) + V_j\psi_j = E\psi_j \quad (5.70)$$

This can be written as a matrix eigenvalue problem:

$$\begin{pmatrix} V_1 + 2t & -t & 0 & \cdots \\ -t & V_2 + 2t & -t & \cdots \\ 0 & -t & V_3 + 2t & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \vdots \end{pmatrix} = E \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \vdots \end{pmatrix} \quad (5.71)$$

where  $t = \hbar^2/(2ma^2)$ .

## The Deep Connection

The discrete-continuous relationship:

### Physical Insight

The discrete and continuous formulations represent complementary perspectives, neither more fundamental than the other. Each approach has its natural domain of application. Some quantum gravity theories propose discrete spacetime at the Planck scale, while others maintain continuity down to arbitrarily small distances. Condensed matter systems naturally exhibit both discrete descriptions at the atomic level and continuous effective field theories at larger scales. Mathematical physics moves fluidly between discrete algebraic methods and continuous analytic techniques, choosing the approach best suited to each problem.

The practical implications are profound. Lattice models can capture essential physics with finite computational resources, making them ideal for numerical simulations. Continuous formulations provide analytical insights and exact solutions that reveal general principles. Some phenomena like band structure and Wannier states in crystals are naturally discrete, while others like scattering theory and field propagation are naturally continuous. Modern quantum simulation exploits this duality, using discrete quantum systems to model both discrete and continuous physics.

The conceptual unity between these pictures is striking. Wave functions and discrete amplitudes transform into each other via the continuum limit or discretization. Differential operators and difference operators are mutually derivable through systematic expansion procedures. Boundary conditions in the continuum correspond to edge effects on the lattice. Analytical solutions complement numerical methods, each illuminating aspects the other cannot easily capture. This deep connection between discrete and continuous descriptions lies at the heart of modern quantum theory.

## 5.3 Examples of Continuous Hilbert Spaces

### 5.3.1 Multiple Representations

#### The Power of Multiple Perspectives

Quantum systems can be described in multiple representations. Position and momentum space, related by the Fourier transform, are the most fundamental. Problems intractable in one representation often become simple in the other.

#### Position and Momentum Space

Different representations offer complementary insights:

## Position vs Momentum Representation

| Property            | Position Space                                   | Momentum Space                  |
|---------------------|--------------------------------------------------|---------------------------------|
| State               | $\psi(x)$                                        | $\tilde{\psi}(p)$               |
| Position operator   | $\hat{x} = x$                                    | $\hat{x} = i\hbar \frac{d}{dp}$ |
| Momentum operator   | $\hat{p} = -i\hbar \frac{d}{dx}$                 | $\hat{p} = p$                   |
| Kinetic energy      | $\hat{T} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$ | $\hat{T} = \frac{p^2}{2m}$      |
| Free particle       | Differential equation                            | Algebraic equation              |
| Harmonic oscillator | Differential equation                            | Differential equation           |

The relationship between representations is given by the Fourier transform:

## Master Reference: Wave Function Transformations

## Fundamental Transformations:

## Position to Momentum:

$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx e^{-ipx/\hbar} \psi(x) \quad (5.72)$$

## Momentum to Position:

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp e^{ipx/\hbar} \tilde{\psi}(p) \quad (5.73)$$

## Key States in Both Representations:

| State               | Position Space                                            | Momentum Space                                                            |
|---------------------|-----------------------------------------------------------|---------------------------------------------------------------------------|
| Position eigenstate | $\delta(x - x_0)$                                         | $\frac{1}{\sqrt{2\pi\hbar}} e^{-ipx_0/\hbar}$                             |
| Momentum eigenstate | $\frac{1}{\sqrt{2\pi\hbar}} e^{ip_0x/\hbar}$              | $\delta(p - p_0)$                                                         |
| Gaussian            | $(\frac{1}{2\pi\sigma_x^2})^{1/4} e^{-x^2/(4\sigma_x^2)}$ | $(\frac{2\sigma_x^2}{\pi\hbar^2})^{1/4} e^{-\sigma_x^2 p^2/\hbar^2}$      |
| Square well ground  | $\sqrt{\frac{2}{L}} \sin(\frac{\pi x}{L})$                | $\frac{\sqrt{2L\pi}}{p^2 L^2 - \pi^2 \hbar^2} [(-1)^n e^{ipL/\hbar} - 1]$ |
| Harmonic ground     | $(\frac{1}{\pi x_0^2})^{1/4} e^{-x^2/(2x_0^2)}$           | $(\frac{1}{\pi p_0^2})^{1/4} e^{-p^2/(2p_0^2)}$                           |

## Operator Actions:

| Operator                         | Position Space                         | Momentum Space                                    |
|----------------------------------|----------------------------------------|---------------------------------------------------|
| $\hat{x}$                        | $x$ (multiply)                         | $i\hbar \frac{d}{dp}$                             |
| $\hat{p}$                        | $-i\hbar \frac{d}{dx}$                 | $p$ (multiply)                                    |
| $\hat{x}^2$                      | $x^2$                                  | $-\hbar^2 \frac{d^2}{dp^2} + i\hbar \frac{d}{dp}$ |
| $\hat{p}^2$                      | $-\hbar^2 \frac{d^2}{dx^2}$            | $p^2$                                             |
| $\hat{T} = \frac{\hat{p}^2}{2m}$ | $-\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$ | $\frac{p^2}{2m}$                                  |
| $V(\hat{x})$                     | $V(x)$                                 | $V(i\hbar \frac{d}{dp})$                          |

## The Uncertainty Principle Revisited

The Fourier transform relationship provides deep insight into the uncertainty principle:

**Example 12: Gaussian Wave Packet Analysis**

Position space Gaussian:

$$\psi(x) = \left( \frac{1}{2\pi\sigma_x^2} \right)^{1/4} e^{-x^2/(4\sigma_x^2)} \quad (5.74)$$

Fourier transform gives momentum space:

$$\tilde{\psi}(p) = \left( \frac{2\sigma_x^2}{\pi\hbar^2} \right)^{1/4} e^{-\sigma_x^2 p^2/\hbar^2} \quad (5.75)$$

Widths:

$$\Delta x = \sqrt{\langle x^2 \rangle} = \sigma_x \quad (5.76)$$

$$\Delta p = \sqrt{\langle p^2 \rangle} = \frac{\hbar}{2\sigma_x} \quad (5.77)$$

Uncertainty product:

$$\Delta x \Delta p = \sigma_x \cdot \frac{\hbar}{2\sigma_x} = \frac{\hbar}{2} \quad (5.78)$$

The Gaussian saturates the uncertainty bound. For other wave functions:

- Narrower in  $x \rightarrow$  Broader in  $p$
- Cannot be arbitrarily narrow in both
- $\Delta x \Delta p \geq \hbar/2$  always

**Basis Transformations**

Different representations are connected by basis transformations:

**Example 13: Energy Representation**

For a system with energy eigenstates  $\{\psi_n(x)\}$  and eigenvalues  $\{E_n\}$ :

Position representation:

$$\psi(x, t) = \sum_n c_n e^{-iE_n t/\hbar} \psi_n(x) \quad (5.79)$$

Energy representation:

$$c_n(t) = c_n(0) e^{-iE_n t/\hbar} \quad (5.80)$$

The transformation:

$$c_n = \int dx \psi_n^*(x) \psi(x, 0) \quad (\text{analysis}) \quad (5.81)$$

$$\psi(x, 0) = \sum_n c_n \psi_n(x) \quad (\text{synthesis}) \quad (5.82)$$

In energy representation, the Hamiltonian becomes diagonal with matrix elements  $H_{nm} = E_n \delta_{nm}$ . This diagonality makes time evolution trivial, reducing to simple phase factors for

each energy eigenstate. However, the position operator becomes complicated in this basis, with matrix elements  $x_{nm} = \int \psi_n^* x \psi_m dx$  that mix different energy levels.

### 5.3.2 The Time-Independent Schrödinger Equation

#### Energy Eigenstates in Position Space

For a particle in potential  $V(x)$ :

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}) = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \quad (5.83)$$

Energy eigenstates satisfy  $\hat{H}|\psi\rangle = E|\psi\rangle$ , which in position representation becomes:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x) \quad (5.84)$$

This can be rewritten as:

#### Example 14: Local de Broglie Wavelength

Rearranging the Schrödinger equation:

$$\frac{d^2\psi}{dx^2} = -\frac{2m}{\hbar^2} [E - V(x)]\psi = -k^2(x)\psi \quad (5.85)$$

where  $k(x) = \sqrt{2m[E - V(x)]}/\hbar$  is the local wave number.

In classically allowed regions where  $E > V(x)$ , the wave number  $k(x)$  is real and solutions oscillate with local wavelength  $\lambda(x) = 2\pi/k(x)$ . The wavelength decreases as the kinetic energy increases, reflecting the particle's higher local momentum. In classically forbidden regions where  $E < V(x)$ , the wave number becomes imaginary:  $k(x) = i\kappa(x)$  where  $\kappa(x) = \sqrt{2m[V(x) - E]}/\hbar$ . Here solutions decay exponentially as  $\psi \sim e^{-\kappa x}$  with a characteristic penetration depth of order  $1/\kappa$ .

#### General Properties of Solutions

General constraints on wave functions:

#### Example 15: Node Theorem

For bound states in one dimension with a single-well potential, there is a direct relationship between the excitation level and the number of nodes. The ground state ( $n = 0$ ) has no nodes, the first excited state ( $n = 1$ ) has one node, and in general the  $n$ -th excited state has exactly  $n$  nodes.

The reasoning is simple: each node represents a change of sign, and between nodes the wave function maintains definite sign. The kinetic energy  $\int |\psi'|^2 dx$  increases with the number of sign changes because more oscillations require larger derivatives. This increased kinetic energy leads to higher total energy. The node theorem thus provides a way to identify excited states

without solving the full equation—simply count the zeros of the wave function.

### 5.3.3 The Free Particle

The free particle ( $V(x) = 0$ ) illustrates continuous spectrum states, the energy-momentum relationship, boundary conditions, and wave packets.

The free particle Schrödinger equation:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi \quad (5.86)$$

#### Example 16: Free Particle Solutions in Position Space

Setting  $k = \sqrt{2mE}/\hbar$ , the equation becomes:

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0 \quad (5.87)$$

This has the general solution:

$$\psi(x) = Ae^{ikx} + Be^{-ikx} \quad (5.88)$$

Alternatively, using real functions:

$$\psi(x) = C \cos(kx) + D \sin(kx) \quad (5.89)$$

The complex exponential form has clear physical interpretation. The term  $e^{ikx}$  represents a right-moving wave with momentum  $p = \hbar k$ , while  $e^{-ikx}$  represents a left-moving wave with momentum  $p = -\hbar k$ . The energy-momentum relation is  $E = p^2/(2m) = \hbar^2 k^2/(2m)$ , and the de Broglie wavelength is  $\lambda = 2\pi/k = h/p$ .

**The normalization problem:** These plane wave states extend over all space with constant amplitude, making the integral  $\int |\psi|^2 dx$  diverge. This reflects the unphysical nature of states with perfectly defined momentum—they are completely delocalized in position.

#### Example 17: Free Particle in Momentum Space

In momentum space, the free particle problem becomes trivial. The Hamiltonian is:

$$\hat{H} = \frac{\hat{p}^2}{2m} = \frac{p^2}{2m} \quad (5.90)$$

This is already diagonal! The momentum eigenstates are:

$$\tilde{\psi}_p(p') = \delta(p' - p) \quad (5.91)$$

with eigenvalues:

$$E_p = \frac{p^2}{2m} \quad (5.92)$$

Choice of representation affects problem difficulty.

**Connection to position space:** The position space wave function is obtained by inverse Fourier transform:

$$\psi_p(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp' e^{ip'x/\hbar} \delta(p' - p) \quad (5.93)$$

$$= \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar} \quad (5.94)$$

This recovers our plane wave solution.

Free particle states can be normalized in different ways:

#### Example 18: Box Normalization vs Delta Normalization

**Box normalization** (periodic boundary conditions on  $[0, L]$ ):

$$\psi_k(x) = \frac{1}{\sqrt{L}} e^{ikx}, \quad k = \frac{2\pi n}{L}, \quad n \in \mathbb{Z} \quad (5.95)$$

**Delta function normalization** (continuum):

$$\psi_k(x) = \frac{1}{\sqrt{2\pi}} e^{ikx}, \quad \int_{-\infty}^{\infty} \psi_k^*(x) \psi_{k'}(x) dx = \delta(k - k') \quad (5.96)$$

The box normalization approaches delta normalization as  $L \rightarrow \infty$ .

### Wave Packet Dynamics

Physical free particle states are wave packets:

#### Example 19: Wave Packets and Physical States

Physical free particle states are wave packets—superpositions of different momentum components:

$$\psi(x, t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp A(p) e^{i(px - E_p t)/\hbar} \quad (5.97)$$

where  $A(p)$  is the momentum amplitude distribution and  $E_p = p^2/(2m)$ .

For a Gaussian momentum distribution centered at  $p_0$  with width  $\sigma_p$ :

$$A(p) = \left( \frac{2\pi\sigma_p^2}{\pi} \right)^{1/4} \exp \left[ -\frac{(p - p_0)^2}{2\sigma_p^2} \right] \quad (5.98)$$

The resulting position space wave packet at  $t = 0$  is:

$$\psi(x, 0) = \left( \frac{2\sigma_p^2}{\pi\hbar^2} \right)^{1/4} \exp \left[ -\frac{\sigma_p^2 x^2}{\hbar^2} + \frac{ip_0 x}{\hbar} \right] \quad (5.99)$$

This is a Gaussian with width  $\sigma_x = \hbar/(2\sigma_p)$ , demonstrating the uncertainty relation  $\sigma_x\sigma_p = \hbar/2$ .

### Example 20: Gaussian Wave Packet Evolution

Initial Gaussian centered at  $x_0$  with average momentum  $p_0$ :

$$\psi(x, 0) = \left( \frac{1}{2\pi\sigma_0^2} \right)^{1/4} \exp \left[ -\frac{(x - x_0)^2}{4\sigma_0^2} + \frac{ip_0x}{\hbar} \right] \quad (5.100)$$

For a free particle, the time-evolved state is:

$$\psi(x, t) = \left( \frac{1}{2\pi\sigma^2(t)} \right)^{1/4} \exp \left[ -\frac{(x - x_0 - v_0t)^2}{4\sigma^2(t)} + i\phi(x, t) \right] \quad (5.101)$$

where:

$$v_0 = \frac{p_0}{m} \quad (\text{group velocity}) \quad (5.102)$$

$$\sigma^2(t) = \sigma_0^2 \left( 1 + \frac{\hbar^2 t^2}{4m^2\sigma_0^4} \right) \quad (\text{spreading}) \quad (5.103)$$

$$\phi(x, t) = \frac{p_0}{\hbar} \left( x - \frac{p_0 t}{2m} \right) + \text{other terms} \quad (5.104)$$

Key features:

- Center moves classically:  $\langle x \rangle = x_0 + v_0 t$
- Width increases: quantum spreading
- Spreading rate:  $\frac{d\sigma}{dt} \sim \frac{\hbar}{2m\sigma_0}$  for  $t \gg \frac{2m\sigma_0^2}{\hbar}$
- Narrower initial packet spreads faster

### Ehrenfest's Theorem

Expectation values follow classical laws:

### Example 21: Expectation Values Follow Classical Laws

For any quantum state:

$$\frac{d\langle x \rangle}{dt} = \frac{\langle p \rangle}{m} \quad (5.105)$$

$$\frac{d\langle p \rangle}{dt} = - \left\langle \frac{dV}{dx} \right\rangle \quad (5.106)$$



For a harmonic oscillator where  $V = \frac{1}{2}m\omega^2 x^2$ :

$$\frac{d^2 \langle x \rangle}{dt^2} = -\omega^2 \langle x \rangle \quad (5.107)$$

The expectation value oscillates exactly like a classical oscillator!

However, for general potentials:

$$\left\langle \frac{dV}{dx} \right\rangle \neq \left. \frac{dV}{dx} \right|_{\langle x \rangle} \quad (5.108)$$

The difference depends on the wave packet width and potential nonlinearity.

#### 5.3.4 The Infinite Square Well

The infinite square well demonstrates bound states and energy quantization.

The potential is defined as:

$$V(x) = \begin{cases} 0 & \text{for } 0 < x < L \\ \infty & \text{otherwise} \end{cases} \quad (5.109)$$

**Example 22: Solving the Infinite Square Well in Position Space**

Inside the well ( $0 < x < L$ ), the Schrödinger equation is:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi \quad (5.110)$$

This has the general solution:

$$\psi(x) = A \sin(kx) + B \cos(kx) \quad (5.111)$$

where  $k = \sqrt{2mE}/\hbar$ .

Boundary conditions require the wave function to vanish at the walls:

$$\psi(0) = 0 \Rightarrow B = 0 \quad (5.112)$$

$$\psi(L) = 0 \Rightarrow A \sin(kL) = 0 \quad (5.113)$$

The condition  $\sin(kL) = 0$  requires  $kL = n\pi$ :

$$k_n = \frac{n\pi}{L} \quad (5.114)$$

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \quad (5.115)$$

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad (5.116)$$

The normalization factor  $\sqrt{2/L}$  ensures  $\int_0^L |\psi_n|^2 dx = 1$ .

**Key physical features:**

The energy levels scale as  $n^2$ , reflecting the kinetic energy of a wave with  $n$  half-wavelengths fitting in the box. The ground state energy  $E_1 = \pi^2 \hbar^2 / (2mL^2)$  is non-zero, demonstrating zero-point energy—a purely quantum effect with no classical analog. The level spacing  $E_{n+1} - E_n = (2n+1)\pi^2 \hbar^2 / (2mL^2)$  increases linearly with  $n$ , unlike the harmonic oscillator where spacing is constant. Each eigenfunction has exactly  $n - 1$  nodes inside the well, consistent with the node theorem.

**Example 23: Orthogonality and Completeness**

The infinite well eigenfunctions form a complete orthonormal set on  $[0, L]$ :

**Orthogonality:**

$$\int_0^L \psi_n^*(x) \psi_m(x) dx = \frac{2}{L} \int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx \quad (5.117)$$

$$= \frac{2}{L} \cdot \frac{L}{2} \delta_{nm} = \delta_{nm} \quad (5.118)$$

**Completeness:** Any function  $f(x)$  defined on  $[0, L]$  with  $f(0) = f(L) = 0$  can be expanded as:

$$f(x) = \sum_{n=1}^{\infty} c_n \psi_n(x) \quad (5.119)$$

where the coefficients are:

$$c_n = \int_0^L \psi_n^*(x) f(x) dx = \sqrt{\frac{2}{L}} \int_0^L \sin\left(\frac{n\pi x}{L}\right) f(x) dx \quad (5.120)$$

This is precisely the Fourier sine series, showing the deep connection between quantum mechanics and Fourier analysis.

Connection to the discrete lattice:

**Example 24: Lattice Approximation of the Infinite Well**

For an  $N$ -site lattice with hard-wall boundaries:

$$\psi_n^{\text{lattice}}(j) = \sqrt{\frac{2}{N+1}} \sin\left(\frac{n\pi j}{N+1}\right) \quad (5.121)$$

for  $j = 1, 2, \dots, N$  and  $n = 1, 2, \dots, N$ .

As  $N \rightarrow \infty$  with  $L = Na$  fixed, the discrete variables transform continuously: the discrete ratio  $j/N$  becomes the continuous variable  $x/L$ , the lattice wave function  $\psi_n^{\text{lattice}}(j)$  converges to the continuum wave function  $\psi_n(x)$ , and the lowest  $N$  modes of the lattice converge to the corresponding continuum results. This explicit construction demonstrates how the continuum emerges from the discrete.

## 5.4 The Quantum Harmonic Oscillator

The harmonic oscillator appears in molecular vibrations, electromagnetic field modes, phonons, and quantum field theory. It serves as a paradigm for understanding energy quantization, coherent states, and the classical-quantum transition.

$$V(x) = \frac{1}{2}m\omega^2 x^2 \quad (5.122)$$

### 5.4.1 The Operator Method: Creation and Annihilation Operators

The harmonic oscillator Hamiltonian can be written as:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2 \hat{x}^2 \quad (5.123)$$

Define characteristic scales and ladder operators:

$$x_0 = \sqrt{\frac{\hbar}{m\omega}} \quad (\text{oscillator length scale}) \quad (5.124)$$

$$p_0 = \sqrt{m\omega\hbar} \quad (\text{oscillator momentum scale}) \quad (5.125)$$

The position and momentum operators can be written as:

$$\hat{x} = x_0 \frac{\hat{a} + \hat{a}^\dagger}{\sqrt{2}} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger) \quad (5.126)$$

$$\hat{p} = p_0 \frac{\hat{a} - \hat{a}^\dagger}{i\sqrt{2}} = -i\sqrt{\frac{m\omega\hbar}{2}} (\hat{a} - \hat{a}^\dagger) \quad (5.127)$$

Inverting these relations gives the ladder operators:

$$\hat{a} = \frac{1}{\sqrt{2}} \left( \frac{\hat{x}}{x_0} + i \frac{\hat{p}}{p_0} \right) = \sqrt{\frac{m\omega}{2\hbar}} \hat{x} + \frac{i}{\sqrt{2m\omega\hbar}} \hat{p} \quad (5.128)$$

$$\hat{a}^\dagger = \frac{1}{\sqrt{2}} \left( \frac{\hat{x}}{x_0} - i \frac{\hat{p}}{p_0} \right) = \sqrt{\frac{m\omega}{2\hbar}} \hat{x} - \frac{i}{\sqrt{2m\omega\hbar}} \hat{p} \quad (5.129)$$

**Example 25: Properties of Ladder Operators**

The ladder operators satisfy the fundamental commutation relation:

$$[\hat{a}, \hat{a}^\dagger] = 1 \quad (5.130)$$

Using  $[\hat{x}, \hat{p}] = i\hbar$ :

$$[\hat{a}, \hat{a}^\dagger] = \frac{1}{2} \left[ \frac{\hat{x}}{x_0} + i \frac{\hat{p}}{p_0}, \frac{\hat{x}}{x_0} - i \frac{\hat{p}}{p_0} \right] \quad (5.131)$$

$$= \frac{1}{2} \left( -\frac{2i}{x_0 p_0} [\hat{x}, \hat{p}] \right) \quad (5.132)$$

$$= \frac{1}{2} \left( -\frac{2i}{x_0 p_0} \cdot i\hbar \right) \quad (5.133)$$

$$= \frac{\hbar}{x_0 p_0} = \frac{\hbar}{\sqrt{\hbar/(m\omega)} \cdot \sqrt{m\omega\hbar}} = 1 \quad (5.134)$$

Expressing the Hamiltonian in ladder operators:

$$\hat{x}^2 = x_0^2 \frac{(\hat{a} + \hat{a}^\dagger)^2}{2} = \frac{\hbar}{2m\omega} (\hat{a}^2 + \hat{a}^\dagger 2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger \hat{a}) \quad (5.135)$$

$$\hat{p}^2 = p_0^2 \frac{(\hat{a} - \hat{a}^\dagger)^2}{-2} = \frac{m\omega\hbar}{2} (\hat{a}^\dagger \hat{a} + \hat{a}\hat{a}^\dagger - \hat{a}^2 - \hat{a}^\dagger 2) \quad (5.136)$$

Substituting into the Hamiltonian:

$$\hat{H} = \frac{1}{2m} \cdot \frac{m\omega\hbar}{2} (\hat{a}^\dagger \hat{a} + \hat{a}\hat{a}^\dagger - \hat{a}^2 - \hat{a}^\dagger 2) \quad (5.137)$$

$$+ \frac{m\omega^2}{2} \cdot \frac{\hbar}{2m\omega} (\hat{a}^2 + \hat{a}^\dagger 2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger \hat{a}) \quad (5.138)$$

$$= \frac{\hbar\omega}{4} (\hat{a}^\dagger \hat{a} + \hat{a}\hat{a}^\dagger - \hat{a}^2 - \hat{a}^\dagger 2 + \hat{a}^2 + \hat{a}^\dagger 2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger \hat{a}) \quad (5.139)$$

$$= \frac{\hbar\omega}{4} (2\hat{a}^\dagger \hat{a} + 2\hat{a}\hat{a}^\dagger) \quad (5.140)$$

$$= \frac{\hbar\omega}{2} (\hat{a}^\dagger \hat{a} + \hat{a}\hat{a}^\dagger) \quad (5.141)$$

Using  $\hat{a}\hat{a}^\dagger = \hat{a}^\dagger \hat{a} + 1$ , we get:

$$\hat{H} = \hbar\omega \left( \hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \quad (5.142)$$

Defining the number operator  $\hat{n} = \hat{a}^\dagger \hat{a}$ :

$$\hat{H} = \hbar\omega \left( \hat{n} + \frac{1}{2} \right) \quad (5.143)$$

The eigenstates of the number operator  $\hat{n}$  are denoted  $|n\rangle$  with eigenvalues  $n = 0, 1, 2, \dots$ :

$$\hat{n}|n\rangle = n|n\rangle \quad (5.144)$$

$$\hat{H}|n\rangle = \hbar\omega \left(n + \frac{1}{2}\right) |n\rangle \quad (5.145)$$

### Example 26: Action of Ladder Operators

The operators act as:

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle \quad (\text{raises by one quantum}) \quad (5.146)$$

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle \quad (\text{lowers by one quantum}) \quad (5.147)$$

$$\hat{a}|0\rangle = 0 \quad (\text{defines ground state}) \quad (5.148)$$

Any excited state can be built from the ground state:

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle \quad (5.149)$$

The position representation of these states involves Hermite polynomials:

$$\psi_n(x) = \left(\frac{1}{\pi x_0^2}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n\left(\frac{x}{x_0}\right) e^{-x^2/(2x_0^2)} \quad (5.150)$$

where  $H_n$  are Hermite polynomials and  $x_0 = \sqrt{\hbar/(m\omega)}$  is the oscillator length scale.

### Example 27: Harmonic Oscillator Ground State in Position Representation

The ground state is defined by  $\hat{a}|0\rangle = 0$ . In position representation:

$$\hat{a}\psi_0(x) = \frac{1}{\sqrt{2}} \left(\frac{x}{x_0} + \frac{x_0}{\hbar} \hat{p}\right) \psi_0(x) = 0 \quad (5.151)$$

Substituting  $\hat{p} = -i\hbar d/dx$ :

$$\left(\frac{x}{x_0} - ix_0 \frac{d}{dx}\right) \psi_0(x) = 0 \quad (5.152)$$

This gives:

$$\frac{d\psi_0}{dx} = -\frac{x}{x_0^2} \psi_0 = -\frac{m\omega}{\hbar} x \psi_0 \quad (5.153)$$

Solving:  $\psi_0(x) = A e^{-x^2/(2x_0^2)}$

Normalizing:  $\psi_0(x) = \left(\frac{1}{\pi x_0^2}\right)^{1/4} e^{-x^2/(2x_0^2)}$

The ground state is indeed a minimum uncertainty Gaussian with width  $x_0/\sqrt{2}$ .

### 5.4.2 Coherent States: Quantum States Most Like Classical Motion

Coherent states are eigenstates of the annihilation operator and represent the quantum states that most closely resemble classical harmonic oscillator motion.

**Definition 5.4.1.** A coherent state  $|\alpha\rangle$  is defined by:

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle \quad (5.154)$$

where  $\alpha$  is a complex number.

#### Construction and Properties of Coherent States

Coherent states can be generated from the vacuum state by the displacement operator:

$$|\alpha\rangle = \hat{D}(\alpha)|0\rangle = e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}|0\rangle \quad (5.155)$$

The complex parameter  $\alpha$  encodes the displacement in phase space. We can parameterize it as:

$$\alpha = x_\alpha + ip_\alpha \quad (5.156)$$

where  $x_\alpha$  and  $p_\alpha$  are real dimensionless parameters that determine the position and momentum displacements.

The displacement operator can be expressed in terms of position and momentum operators. Substituting the ladder operators:

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\hat{x} + \frac{i}{\sqrt{2m\omega\hbar}}\hat{p} \quad (5.157)$$

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\hat{x} - \frac{i}{\sqrt{2m\omega\hbar}}\hat{p} \quad (5.158)$$

into the displacement operator and working through the algebra, one finds that the displacement operator performs a phase space translation. The action of this operator shifts both position and momentum expectation values:

$$\langle\hat{x}\rangle = x_0\sqrt{2}\text{Re}[\alpha] \quad (5.159)$$

$$\langle\hat{p}\rangle = p_0\sqrt{2}\text{Im}[\alpha] \quad (5.160)$$

where  $x_0 = \sqrt{\hbar/(m\omega)}$  and  $p_0 = \sqrt{m\omega\hbar}$  are the characteristic oscillator scales. The complex parameter  $\alpha = x_\alpha + ip_\alpha$  thus encodes the classical phase space coordinates in dimensionless form.

**Example 28: How Displacement Acts on Wave Functions**

The operator  $e^{-ix_\alpha \hat{p}/\hbar}$  is the translation operator that shifts any wave function by  $x_\alpha$ :

$$e^{-ix_\alpha \hat{p}/\hbar} \psi(x) = \psi(x - x_\alpha) \quad (5.161)$$

To verify this, expand the exponential:

$$e^{-ix_\alpha \hat{p}/\hbar} \psi(x) = \sum_{n=0}^{\infty} \frac{(-ix_\alpha/\hbar)^n}{n!} \hat{p}^n \psi(x) \quad (5.162)$$

$$= \sum_{n=0}^{\infty} \frac{(-ix_\alpha)^n}{n!} \left( -i\hbar \frac{d}{dx} \right)^n \psi(x) \quad (5.163)$$

$$= \sum_{n=0}^{\infty} \frac{(x_\alpha)^n}{n!} \left( -\frac{d}{dx} \right)^n \psi(x) \quad (5.164)$$

$$= \psi(x - x_\alpha) \quad (5.165)$$

This is the Taylor series for  $\psi(x - x_\alpha)$ ! The momentum operator generates spatial translations. Similarly,  $e^{ip_\alpha \hat{x}/\hbar}$  acts as a momentum boost. The momentum boost adds a phase to the wave function:

$$e^{ip_\alpha \hat{x}/\hbar} \psi(x) = e^{ip_\alpha x/\hbar} \psi(x) \quad (5.166)$$

This multiplies the wave function by a plane wave, effectively "kicking" it with momentum  $p_\alpha$ .

**Decomposition of the Displacement Operator**

To better understand the physical meaning of the displacement operator, we can decompose it using the Baker-Campbell-Hausdorff formula. With  $\alpha = x_\alpha + ip_\alpha$ , where  $x_\alpha$  and  $p_\alpha$  are real dimensionless parameters, the displacement operator can be written as:

$$\hat{D}(\alpha) = \exp \left[ \frac{i}{\hbar} (p_\alpha \hat{x} - x_\alpha \hat{p}) \right] = e^{ip_\alpha \hat{x}/\hbar} e^{-ix_\alpha \hat{p}/\hbar} e^{-ip_\alpha x_\alpha/(2\hbar)} \quad (5.167)$$

The last term is merely a c-number phase factor. Up to this phase, the displacement operator factorizes into two simple operations:

$$\hat{D}(\alpha) \approx e^{ip_\alpha \hat{x}/\hbar} \cdot e^{-ix_\alpha \hat{p}/\hbar} \quad (5.168)$$

This factorization reveals that the displacement operator performs two sequential operations:

- (a) First, it translates the state by  $x_\alpha$  in position:  $e^{-ix_\alpha \hat{p}/\hbar} \psi(x) = \psi(x - x_\alpha)$
- (b) Then, it boosts the momentum by  $p_\alpha$ :  $e^{ip_\alpha \hat{x}/\hbar} \psi(x) = e^{ip_\alpha x/\hbar} \psi(x)$

The displacement operator thus performs a complete phase space translation. We can verify this explicitly by computing how it transforms the position and momentum operators:

$$\hat{D}^\dagger(\alpha) \hat{x} \hat{D}(\alpha) = \hat{x} + x_\alpha \quad (5.169)$$

$$\hat{D}^\dagger(\alpha) \hat{p} \hat{D}(\alpha) = \hat{p} + p_\alpha \quad (5.170)$$



To verify these transformations, we use the commutation relations:

$$[\hat{x}, p_\alpha \hat{x}/\hbar] = 0 \quad (5.171)$$

$$[\hat{x}, -x_\alpha \hat{p}/\hbar] = -x_\alpha [\hat{x}, \hat{p}]/\hbar = -ix_\alpha \quad (5.172)$$

$$[\hat{p}, p_\alpha \hat{x}/\hbar] = p_\alpha [\hat{p}, \hat{x}]/\hbar = -ip_\alpha \quad (5.173)$$

$$[\hat{p}, -x_\alpha \hat{p}/\hbar] = 0 \quad (5.174)$$

Using the operator identity  $e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!} [\hat{A}, [\hat{A}, \hat{B}]] + \dots$ , and noting that higher commutators vanish in this case:

$$\hat{D}^\dagger \hat{x} \hat{D} = e^{-i(p_\alpha \hat{x} - x_\alpha \hat{p})/\hbar} \hat{x} e^{i(p_\alpha \hat{x} - x_\alpha \hat{p})/\hbar} \quad (5.175)$$

$$= \hat{x} + \frac{i}{\hbar} [p_\alpha \hat{x} - x_\alpha \hat{p}, \hat{x}] \quad (5.176)$$

$$= \hat{x} + \frac{i}{\hbar} (-x_\alpha [\hat{p}, \hat{x}]) \quad (5.177)$$

$$= \hat{x} + x_\alpha \quad (5.178)$$

Similarly, for the momentum operator:

$$\hat{D}^\dagger \hat{p} \hat{D} = \hat{p} + \frac{i}{\hbar} [p_\alpha \hat{x} - x_\alpha \hat{p}, \hat{p}] \quad (5.179)$$

$$= \hat{p} + \frac{i}{\hbar} (p_\alpha [\hat{x}, \hat{p}]) \quad (5.180)$$

$$= \hat{p} + p_\alpha \quad (5.181)$$

This decomposition reveals the fundamental nature of coherent states: they are quantum states that have been displaced from the vacuum state in phase space. The complex parameter  $\alpha = x_\alpha + ip_\alpha$  encodes this displacement, with the real part determining the position shift and the imaginary part determining the momentum shift. This dual action—combining position displacement with momentum boost—makes coherent states the quantum analog of classical phase space points, bridging the quantum and classical descriptions of harmonic oscillator motion.

Using the Baker-Campbell-Hausdorff formula:

$$|\alpha\rangle = e^{-|\alpha|^2/2} e^{\alpha \hat{a}^\dagger} |0\rangle \quad (5.182)$$

Expanding in the number state basis:

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle \quad (5.183)$$

The probability of finding  $n$  quanta is Poisson distributed:

$$P(n) = |\langle n|\alpha\rangle|^2 = e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!} \quad (5.184)$$

with mean  $\bar{n} = |\alpha|^2$  and variance  $\Delta n^2 = |\alpha|^2$ .

#### Overlap of coherent states:

The amplitude for overlap between two coherent states is

$$\langle \alpha|\beta\rangle = e^{-|\alpha|^2/2} e^{-|\beta|^2/2} e^{\alpha^* \beta}, \quad (5.185)$$

which is generally complex. The gauge-invariant magnitude takes a particularly transparent form:

$$|\langle\alpha|\beta\rangle|^2 = e^{-|\alpha-\beta|^2} \quad (5.186)$$

**Derivation.** Using  $|\alpha - \beta|^2 = |\alpha|^2 + |\beta|^2 - \alpha\beta^* - \alpha^*\beta$  and  $\alpha\beta^* + \alpha^*\beta = 2\operatorname{Re}(\alpha^*\beta)$ ,

$$\begin{aligned} |\langle\alpha|\beta\rangle|^2 &= e^{-|\alpha|^2} e^{-|\beta|^2} e^{\alpha^*\beta + \alpha\beta^*} \\ &= e^{-|\alpha|^2 - |\beta|^2 + 2\operatorname{Re}(\alpha^*\beta)} = e^{-|\alpha-\beta|^2}. \end{aligned} \quad (5.187)$$

The full amplitude differs from  $e^{-|\alpha-\beta|^2/2}$  by a phase  $e^{i\operatorname{Im}(\alpha^*\beta)}$ , but the measurable overlap probability is given by the magnitude squared above.

The form  $|\langle\alpha|\beta\rangle|^2 = e^{-|\alpha-\beta|^2}$  shows that:

- Coherent states are never orthogonal unless  $|\alpha - \beta| \rightarrow \infty$
- The overlap probability decreases as a Gaussian in phase-space distance
- Unlike number states, coherent states form an over-complete basis

### Example 29: Physical Properties of Coherent States

Writing  $\alpha = |\alpha|e^{i\phi}$ , the coherent state has:

**Position expectation:**

$$\langle\hat{x}\rangle = \langle\alpha|\hat{x}|\alpha\rangle = x_0 \frac{\langle\alpha|(\hat{a} + \hat{a}^\dagger)|\alpha\rangle}{\sqrt{2}} \quad (5.188)$$

$$= x_0 \frac{\alpha + \alpha^*}{\sqrt{2}} = x_0 \sqrt{2} |\alpha| \cos(\phi) \quad (5.189)$$

**Momentum expectation:**

$$\langle\hat{p}\rangle = \langle\alpha|\hat{p}|\alpha\rangle = p_0 \frac{\langle\alpha|(\hat{a} - \hat{a}^\dagger)|\alpha\rangle}{i\sqrt{2}} \quad (5.190)$$

$$= p_0 \frac{\alpha - \alpha^*}{i\sqrt{2}} = p_0 \sqrt{2} |\alpha| \sin(\phi) \quad (5.191)$$

**Time evolution:**

$$|\alpha(t)\rangle = e^{-i\hat{H}t/\hbar} |\alpha\rangle = e^{-i\omega t/2} |\alpha e^{-i\omega t}\rangle \quad (5.192)$$

The state remains coherent with  $\alpha(t) = \alpha e^{-i\omega t}$ . The expectation values oscillate:

$$\langle\hat{x}\rangle_t = x_0 \sqrt{2} |\alpha| \cos(\omega t - \phi) \quad (5.193)$$

$$\langle\hat{p}\rangle_t = -p_0 \sqrt{2} |\alpha| \sin(\omega t - \phi) \quad (5.194)$$

This is exactly classical motion! The wave packet maintains its shape (minimum uncertainty) while oscillating.

**Example 30: Wave Function of a Coherent State**

The position space wave function is:

$$\psi_\alpha(x) = \left(\frac{1}{\pi x_0^2}\right)^{1/4} \exp \left[ -\frac{(x - \langle x \rangle)^2}{2x_0^2} + \frac{i\langle p \rangle x}{\hbar} - \frac{i(\langle p \rangle \langle x \rangle + \phi)}{2\hbar} \right] \quad (5.195)$$

where  $\langle x \rangle = x_0\sqrt{2}|\alpha| \cos \phi$  and  $\langle p \rangle = p_0\sqrt{2}|\alpha| \sin \phi$ .

This is a Gaussian wave packet with position width  $\Delta x = x_0/\sqrt{2}$  (identical to the ground state), momentum width  $\Delta p = p_0/\sqrt{2}$ , and uncertainty product  $\Delta x \Delta p = x_0 p_0/2 = \hbar/2$  achieving the minimum uncertainty bound.

**Time Dependence of Coherent States**

Coherent states remain coherent under time evolution, executing classical harmonic motion.

**Example 31: Equation of Motion for Coherent States**

Time evolution from the Schrödinger equation:

$$i\hbar \frac{d}{dt} |\alpha(t)\rangle = \hat{H} |\alpha(t)\rangle \quad (5.196)$$

For the harmonic oscillator Hamiltonian  $\hat{H} = \hbar\omega(\hat{a}^\dagger \hat{a} + 1/2)$ , we can write the formal solution:

$$|\alpha(t)\rangle = e^{-i\hat{H}t/\hbar} |\alpha(0)\rangle \quad (5.197)$$

Expanding the initial coherent state in the number basis:

$$|\alpha(0)\rangle = e^{-|\alpha(0)|^2/2} \sum_{n=0}^{\infty} \frac{[\alpha(0)]^n}{\sqrt{n!}} |n\rangle \quad (5.198)$$

The time evolution operator acts on each number state as:

$$e^{-i\hat{H}t/\hbar} |n\rangle = e^{-i\omega(n+1/2)t} |n\rangle \quad (5.199)$$

Therefore:

$$|\alpha(t)\rangle = e^{-|\alpha(0)|^2/2} \sum_{n=0}^{\infty} \frac{[\alpha(0)]^n}{\sqrt{n!}} e^{-i\omega(n+1/2)t} |n\rangle \quad (5.200)$$

$$= e^{-i\omega t/2} e^{-|\alpha(0)|^2/2} \sum_{n=0}^{\infty} \frac{[\alpha(0)e^{-i\omega t}]^n}{\sqrt{n!}} |n\rangle \quad (5.201)$$

$$= e^{-i\omega t/2} |\alpha(0)e^{-i\omega t}\rangle \quad (5.202)$$

**Key result:** The coherent state remains a coherent state with a time-dependent complex amplitude:

$$\alpha(t) = \alpha(0)e^{-i\omega t} \quad (5.203)$$

The global phase  $e^{-i\omega t/2}$  is physically unobservable and is often dropped. The complex parameter  $\alpha$  rotates in the complex plane with angular frequency  $\omega$ —the same frequency as the classical oscillator!

**Example 32: Position and Momentum Expectation Values**

To find  $\langle x(t) \rangle$  and  $\langle p(t) \rangle$ , we use the relations:

$$\hat{x} = x_0 \frac{\hat{a} + \hat{a}^\dagger}{\sqrt{2}} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger) \quad (5.204)$$

$$\hat{p} = p_0 \frac{\hat{a} - \hat{a}^\dagger}{i\sqrt{2}} = -i\sqrt{\frac{m\omega\hbar}{2}} (\hat{a} - \hat{a}^\dagger) \quad (5.205)$$

Since  $\hat{a}|\alpha(t)\rangle = \alpha(t)|\alpha(t)\rangle$  and  $\langle\alpha(t)|\hat{a}^\dagger = \langle\alpha(t)|\alpha^*(t)$ , we have:

$$\langle x(t) \rangle = \langle\alpha(t)|\hat{x}|\alpha(t)\rangle \quad (5.206)$$

$$= x_0 \frac{\alpha(t) + \alpha^*(t)}{\sqrt{2}} \quad (5.207)$$

$$= x_0 \frac{\alpha(0)e^{-i\omega t} + \alpha^*(0)e^{i\omega t}}{\sqrt{2}} \quad (5.208)$$

Writing  $\alpha(0) = |\alpha_0|e^{i\phi_0}$  where  $|\alpha_0|$  and  $\phi_0$  are real:

$$\langle x(t) \rangle = x_0 \frac{|\alpha_0|[e^{i(\phi_0 - \omega t)} + e^{-i(\phi_0 - \omega t)}]}{\sqrt{2}} \quad (5.209)$$

$$= x_0\sqrt{2}|\alpha_0|\cos(\omega t - \phi_0) \quad (5.210)$$

Similarly for momentum:

$$\langle p(t) \rangle = \langle\alpha(t)|\hat{p}|\alpha(t)\rangle \quad (5.211)$$

$$= p_0 \frac{\alpha(t) - \alpha^*(t)}{i\sqrt{2}} \quad (5.212)$$

$$= p_0 \frac{|\alpha_0|[e^{i(\phi_0 - \omega t)} - e^{-i(\phi_0 - \omega t)}]}{i\sqrt{2}} \quad (5.213)$$

$$= -p_0\sqrt{2}|\alpha_0|\sin(\omega t - \phi_0) \quad (5.214)$$

**Final results:**

$$\boxed{\langle x(t) \rangle = x_0\sqrt{2}|\alpha_0|\cos(\omega t - \phi_0)} \quad (5.215)$$

$$\boxed{\langle p(t) \rangle = -p_0\sqrt{2}|\alpha_0|\sin(\omega t - \phi_0)} \quad (5.216)$$

where  $x_0 = \sqrt{\hbar/(m\omega)}$  and  $p_0 = \sqrt{m\omega\hbar}$ .

**Example 33: Classical Correspondence**

The expectation values satisfy the classical equations of motion. Taking time derivatives:

$$\frac{d\langle x \rangle}{dt} = -x_0\sqrt{2}|\alpha_0|\omega \sin(\omega t - \phi_0) = \frac{\langle p \rangle}{m} \quad (5.217)$$

$$\frac{d\langle p \rangle}{dt} = -p_0\sqrt{2}|\alpha_0|\omega \cos(\omega t - \phi_0) = -m\omega^2\langle x \rangle \quad (5.218)$$

These are exactly the equations of motion for a classical harmonic oscillator! We can verify:

$$\frac{d^2\langle x \rangle}{dt^2} = -\omega^2\langle x \rangle \quad (5.219)$$

Furthermore, the total energy expectation value is:

$$\langle E(t) \rangle = \frac{\langle p^2 \rangle}{2m} + \frac{1}{2}m\omega^2\langle x^2 \rangle \quad (5.220)$$

$$= \hbar\omega \left( |\alpha_0|^2 + \frac{1}{2} \right) \quad (5.221)$$

This is constant in time, as expected for energy conservation. The amplitude  $|\alpha_0|$  determines the classical oscillation amplitude:

$$A_x = x_0\sqrt{2}|\alpha_0| = \sqrt{\frac{2\hbar}{m\omega}}|\alpha_0| \quad (5.222)$$

$$A_p = p_0\sqrt{2}|\alpha_0| = \sqrt{2m\omega\hbar}|\alpha_0| \quad (5.223)$$

The phase  $\phi_0$  sets the initial phase of oscillation.

**Physical Insight**

Coherent states provide the most direct connection between quantum and classical mechanics for the harmonic oscillator. Their expectation values follow classical trajectories exactly—not approximately, but exactly. The quantum state maintains its Gaussian shape throughout the motion, with constant uncertainties  $\Delta x = x_0/\sqrt{2}$  and  $\Delta p = p_0/\sqrt{2}$  that satisfy the minimum uncertainty relation.

This behavior explains why laser light, which consists of coherent photon states, exhibits classical-like electromagnetic field oscillations despite being fundamentally quantum mechanical. The coherent state bridges the quantum-classical divide by encoding classical phase space coordinates  $(x, p)$  into the complex parameter  $\alpha$  through:

$$\text{Re}[\alpha\sqrt{2}] = \frac{\langle x \rangle}{x_0} \quad (5.224)$$

$$\text{Im}[\alpha\sqrt{2}] = \frac{\langle p \rangle}{p_0} \quad (5.225)$$

The rotation of  $\alpha(t)$  in the complex plane directly corresponds to circular motion in classical phase space. This geometric picture—where quantum evolution becomes rotation in a complex plane—extends far beyond the harmonic oscillator and underlies much of modern quantum optics and quantum information theory.

### 5.4.3 Squeezed States: Beyond Classical Limits

While coherent states maintain the symmetric uncertainty of the ground state, squeezed states redistribute the uncertainty between position and momentum while preserving the minimum uncertainty product.

**Definition 5.4.2.** A squeezed state is generated by the squeeze operator:

$$\hat{S}(z) = \exp \left[ \frac{z^*}{2} \hat{a}^2 - \frac{z}{2} (\hat{a}^\dagger)^2 \right] \quad (5.226)$$

where  $z = re^{i\theta}$  is the squeeze parameter.

#### Operator Transformations Under Squeezing

To understand how squeezing affects position and momentum, we need to compute how the squeeze operator transforms  $\hat{x}$  and  $\hat{p}$ . For real squeeze parameter  $r$ , we use the Baker-Campbell-Hausdorff formula with  $\hat{A} = \frac{r}{2}(\hat{a}^2 - (\hat{a}^\dagger)^2)$ .

#### Example 34: Deriving the Position Transformation

Since  $\hat{S}^\dagger(r) = e^{-\hat{A}}$ , we compute  $e^{-\hat{A}}\hat{a}e^{\hat{A}}$  using BCH:

$$e^{-\hat{A}}\hat{a}e^{\hat{A}} = \hat{a} + [-\hat{A}, \hat{a}] + \frac{1}{2!}[-\hat{A}, [-\hat{A}, \hat{a}]] + \dots \quad (5.227)$$

**Key commutators:**

$$\begin{aligned} [\hat{a}^2, \hat{a}] &= 0 \\ [(\hat{a}^\dagger)^2, \hat{a}] &= -2\hat{a}^\dagger \\ [\hat{a}^2, \hat{a}^\dagger] &= 2\hat{a} \end{aligned} \quad (5.228)$$

Computing the nested commutators with  $-\hat{A}$ :

$$\begin{aligned} [-\hat{A}, \hat{a}] &= -\frac{r}{2}(0 - (-2\hat{a}^\dagger)) = -r\hat{a}^\dagger \\ [-\hat{A}, [-\hat{A}, \hat{a}]] &= [-\hat{A}, -r\hat{a}^\dagger] = -r[-\hat{A}, \hat{a}^\dagger] \end{aligned}$$

where  $[-\hat{A}, \hat{a}^\dagger] = -\frac{r}{2}(2\hat{a} - 0) = -r\hat{a}$ , giving:

$$[-\hat{A}, [-\hat{A}, \hat{a}]] = r^2\hat{a} \quad (5.229)$$

Third commutator:

$$[-\hat{A}, [-\hat{A}, [-\hat{A}, \hat{a}]]] = r^2[-\hat{A}, \hat{a}] = -r^3\hat{a}^\dagger \quad (5.230)$$

The pattern alternates with signs! The BCH series becomes:

$$\begin{aligned}\hat{S}^\dagger \hat{a} \hat{S} &= \hat{a} - r \hat{a}^\dagger + \frac{r^2}{2!} \hat{a} - \frac{r^3}{3!} \hat{a}^\dagger + \dots \\ &= \hat{a} \left( 1 + \frac{r^2}{2!} + \frac{r^4}{4!} + \dots \right) - \hat{a}^\dagger \left( r + \frac{r^3}{3!} + \dots \right)\end{aligned}\quad (5.231)$$

Recognizing the Taylor series for hyperbolic functions:

$$\boxed{\hat{S}^\dagger \hat{a} \hat{S} = \hat{a} \cosh r - \hat{a}^\dagger \sinh r} \quad (5.232)$$

Similarly:

$$\boxed{\hat{S}^\dagger \hat{a}^\dagger \hat{S} = \hat{a}^\dagger \cosh r - \hat{a} \sinh r} \quad (5.233)$$

Therefore, for position  $\hat{x} = x_0(\hat{a} + \hat{a}^\dagger)/\sqrt{2}$ :

$$\begin{aligned}\hat{S}^\dagger \hat{x} \hat{S} &= \frac{x_0}{\sqrt{2}} \hat{S}^\dagger (\hat{a} + \hat{a}^\dagger) \hat{S} \\ &= \frac{x_0}{\sqrt{2}} [(\hat{a} + \hat{a}^\dagger) \cosh r - (\hat{a}^\dagger + \hat{a}) \sinh r] \\ &= \frac{x_0}{\sqrt{2}} (\hat{a} + \hat{a}^\dagger) (\cosh r - \sinh r)\end{aligned}\quad (5.234)$$

Using  $\cosh r - \sinh r = e^{-r}$ :

$$\boxed{\hat{S}^\dagger(r) \hat{x} \hat{S}(r) = e^{-r} \hat{x}} \quad (5.235)$$

Similarly for momentum  $\hat{p} = p_0(\hat{a} - \hat{a}^\dagger)/(i\sqrt{2})$ :

$$\begin{aligned}\hat{S}^\dagger \hat{p} \hat{S} &= \frac{p_0}{i\sqrt{2}} [(\hat{a} - \hat{a}^\dagger) \cosh r - (\hat{a}^\dagger - \hat{a}) \sinh r] \\ &= \frac{p_0}{i\sqrt{2}} (\hat{a} - \hat{a}^\dagger) (\cosh r + \sinh r)\end{aligned}\quad (5.236)$$

Using  $\cosh r + \sinh r = e^r$ :

$$\boxed{\hat{S}^\dagger(r) \hat{p} \hat{S}(r) = e^r \hat{p}} \quad (5.237)$$

For  $r > 0$ , position is squeezed by factor  $e^{-r}$  while momentum is anti-squeezed by  $e^r$ .

### Example 35: Properties of Squeezed States

The squeezed vacuum state  $|z\rangle = \hat{S}(z)|0\rangle$  has remarkable properties:

**Position variance:**

$$(\Delta x)^2 = \frac{x_0^2}{2} e^{-2r} \quad (5.238)$$

**Momentum variance:**

$$(\Delta p)^2 = \frac{p_0^2}{2} e^{2r} \quad (5.239)$$

**Uncertainty product:**

$$\Delta x \Delta p = \frac{x_0 p_0}{2} = \frac{\hbar}{2} \quad (5.240)$$

For  $r > 0$ : position uncertainty is "squeezed" below the ground state value while momentum uncertainty increases. The converse occurs for  $r < 0$ .

**Number state expansion:**

$$|z\rangle = \frac{1}{\sqrt{\cosh r}} \sum_{n=0}^{\infty} \frac{\sqrt{(2n)!}}{2^n n!} (-e^{i\theta} \tanh r)^n |2n\rangle \quad (5.241)$$

Only even number states appear due to the parity symmetry of the squeeze operator.

**Example 36: Displaced Squeezed States**

The most general Gaussian state is a displaced squeezed state:

$$|\alpha, z\rangle = \hat{D}(\alpha) \hat{S}(z) |0\rangle \quad (5.242)$$

This state has:

- Displaced mean position and momentum (from  $\alpha$ )
- Squeezed uncertainties (from  $z$ )
- Minimum uncertainty product  $\Delta x \Delta p = \hbar/2$

The wave function is:

$$\psi_{\alpha, z}(x) = \left( \frac{1}{\pi \sigma_x^2} \right)^{1/4} \exp \left[ -\frac{(x - \langle x \rangle)^2}{2\sigma_x^2} + \frac{i\langle p \rangle x}{\hbar} + i\phi(x, t) \right] \quad (5.243)$$

where  $\sigma_x^2 = x_0^2 e^{-2r}$  and  $(\langle x \rangle, \langle p \rangle)$  are determined by  $\alpha$ .

**Physical Insight**

The different quantum states of the harmonic oscillator form a hierarchy of increasing sophistication. Number states  $|n\rangle$  are the energy eigenstates with quantized energy levels  $E_n = \hbar\omega(n + 1/2)$ . These are highly non-classical states with definite energy but completely uncertain phase. Coherent states  $|\alpha\rangle$  are the most classical-like quantum states, maintaining minimum uncertainty while their expectation values oscillate exactly as a classical particle would. Squeezed states  $|z\rangle$  exhibit non-classical uncertainty distributions where position and momentum uncertainties can be redistributed while preserving the minimum uncertainty product. Finally, displaced squeezed states  $|\alpha, z\rangle$  represent the most general Gaussian states, combining both displacement and squeezing.

Coherent states appear throughout physics. Laser light consists of coherent photon states, and driven classical oscillators are well-described by coherent states in the quantum limit. They provide the foundation for semiclassical approximations and are central to quantum



optics and cavity quantum electrodynamics. Squeezed states have become essential tools in precision measurement. The LIGO gravitational wave detector uses squeezed light to reduce quantum noise below the standard quantum limit. Squeezed states enable sub-shot-noise measurements in quantum metrology and provide the basis for continuous variable quantum computing. They also serve as testbeds for quantum mechanics at increasingly macroscopic scales.

#### 5.4.4 Physical Implementation of Squeezing

How can we physically create squeezed states in the laboratory? The key insight is that we need an interaction Hamiltonian that creates or destroys photons in pairs, not individually. This naturally leads to terms proportional to  $(\hat{a}^\dagger)^2$  and  $\hat{a}^2$ .

##### The Parametric Interaction

In quantum optics experiments, squeezed states are typically created using nonlinear crystals where a strong classical "pump" field at frequency  $2\omega$  drives the creation of photon pairs at frequency  $\omega$ . The interaction between the pump field and the quantum field mode can be described by the Hamiltonian:

$$\hat{H}_{\text{int}} = i\hbar g \left[ (\hat{a}^\dagger)^2 e^{-2i\omega t} - \hat{a}^2 e^{2i\omega t} \right] \quad (5.244)$$

where  $g$  is a coupling constant proportional to the pump field amplitude and the nonlinear susceptibility of the crystal.

To understand this Hamiltonian, consider that the pump field can be approximated as a classical field  $E_{\text{pump}} \propto e^{-2i\omega t}$ . The nonlinear crystal mediates an interaction where this pump can create two photons at frequency  $\omega$  (energy conservation requires  $2\omega = \omega + \omega$ ). The term  $(\hat{a}^\dagger)^2$  creates two photons simultaneously, while  $\hat{a}^2$  annihilates two photons.

##### Connection to the Squeeze Operator

Moving to an interaction picture where we remove the fast time dependence, the effective Hamiltonian becomes:

$$\hat{H}_{\text{eff}} = i\hbar g \left[ (\hat{a}^\dagger)^2 - \hat{a}^2 \right] \quad (5.245)$$

The time evolution under this Hamiltonian for a time  $\tau$  gives:

$$\hat{U}(\tau) = \exp \left( -\frac{i\hat{H}_{\text{eff}}\tau}{\hbar} \right) = \exp \left[ g\tau \left( \hat{a}^2 - (\hat{a}^\dagger)^2 \right) \right] \quad (5.246)$$

Comparing with our squeeze operator  $\hat{S}(z) = \exp \left[ \frac{z^*}{2} \hat{a}^2 - \frac{z}{2} (\hat{a}^\dagger)^2 \right]$ , we see that:

$$\hat{U}(\tau) = \hat{S}(z) \quad \text{with} \quad z = -2g\tau \quad (5.247)$$

The interaction time  $\tau$  and coupling strength  $g$  determine the squeeze parameter. For real  $g$  and  $\tau$ , we get  $z = r$  (real), corresponding to squeezing along the position or momentum quadrature.

### Why Only Even Number States?

A crucial feature of squeezed states is that when starting from vacuum, only even number states appear. We can understand this from the structure of the Hamiltonian. Consider the action of the evolution operator on the vacuum state:

$$\hat{S}(r)|0\rangle = \exp\left[\frac{r}{2}\left(\hat{a}^2 - (\hat{a}^\dagger)^2\right)\right]|0\rangle \quad (5.248)$$

Expanding the exponential:

$$\hat{S}(r)|0\rangle = \sum_{n=0}^{\infty} \frac{1}{n!} \left[\frac{r}{2}\left(\hat{a}^2 - (\hat{a}^\dagger)^2\right)\right]^n |0\rangle \quad (5.249)$$

$$= |0\rangle + \frac{r}{2}\left(\hat{a}^2 - (\hat{a}^\dagger)^2\right)|0\rangle + \frac{r^2}{8}\left(\hat{a}^2 - (\hat{a}^\dagger)^2\right)^2|0\rangle + \dots \quad (5.250)$$

The key observation is that  $\hat{a}^2|0\rangle = 0$  (you cannot annihilate two photons from vacuum), so only the  $(\hat{a}^\dagger)^2$  terms survive when acting on  $|0\rangle$ . Each application of  $(\hat{a}^\dagger)^2$  creates two photons:

$$(\hat{a}^\dagger)^2|0\rangle = \sqrt{2!}|2\rangle \quad (5.251)$$

$$[(\hat{a}^\dagger)^2]^2|0\rangle = \sqrt{4!}|4\rangle \quad (5.252)$$

$$[(\hat{a}^\dagger)^2]^3|0\rangle = \sqrt{6!}|6\rangle \quad (5.253)$$

Therefore, the squeezed vacuum state contains only even photon numbers:

$$|r\rangle = \hat{S}(r)|0\rangle = \frac{1}{\sqrt{\cosh r}} \sum_{n=0}^{\infty} \frac{\sqrt{(2n)!}}{2^n n!} (\tanh r)^n |2n\rangle \quad (5.254)$$

This even-number-only structure is a direct consequence of the pairwise creation of photons. The physics is clear: the parametric process creates photons in pairs to conserve energy (two photons at  $\omega$  from one at  $2\omega$ ), leading to only even photon numbers in the output state.

Experimentally, this process is realized in nonlinear crystals pumped by strong laser fields. The squeezed light emerges at half the pump frequency, with the amount of squeezing controlled by the pump power and interaction length. Typical experiments achieve 10-15 dB of squeezing, meaning the noise in one quadrature is reduced by a factor of 3-5 below the vacuum level. These squeezed states are now routinely used in gravitational wave detectors like LIGO to surpass the standard quantum limit for position measurements.

## 5.5 Chapter Summary

This chapter developed the continuum limit of discrete lattice models. Taking lattice spacing  $a \rightarrow 0$  while keeping system size fixed transforms discrete amplitudes into the continuous wave function  $\psi(x)$ .

The harmonic oscillator emerged as a paradigmatic system bridging classical and quantum mechanics through coherent states—eigenstates of the annihilation operator that maintain minimum uncertainty while following classical trajectories. Squeezed states demonstrate quantum control of uncertainty beyond classical limits, with applications in precision measurement.

This connection provides the foundation for numerical methods and applies directly to quantum dots, optical lattices, and qubit arrays. The framework extends to three dimensions, many-particle systems, and quantum field theory.

## 5.6 Problems

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### Problem 5.1 (*Continuum Limit and Commutation Relations*)

- (a) Starting from a discrete lattice with spacing  $a$ , derive the continuum momentum operator  $\hat{p} = -i\hbar \frac{d}{dx}$  by taking the limit  $a \rightarrow 0$  of the discrete momentum.
- (b) Verify the canonical commutation relation  $[\hat{x}, \hat{p}] = i\hbar$  using the differential form of  $\hat{p}$ .
- (c) For a general function  $f(\hat{x})$ , prove that  $[\hat{p}, f(\hat{x})] = -i\hbar \frac{df}{dx}$ .
- (d) Apply this result to find  $[\hat{p}, \hat{x}^3]$  and  $[\hat{x}^2, \hat{p}]$ .

### Problem 5.2 (*Standard Quantum Systems*)

- (a) **Infinite Square Well:** For a particle in a box with  $V(x) = 0$  for  $0 < x < L$  and infinite elsewhere, find the normalized energy eigenstates  $\psi_n(x)$  and eigenvalues  $E_n$ . Calculate  $\langle x \rangle$  and  $\Delta x$  for the ground state.
- (b) **Harmonic Oscillator Ground State:** Verify that  $\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-m\omega x^2/(2\hbar)}$  satisfies the time-independent Schrodinger equation with  $V(x) = \frac{1}{2}m\omega^2 x^2$ . Find the ground state energy  $E_0$  and verify that  $\Delta x \Delta p = \hbar/2$ .

### Problem 5.3 (*Gaussian Wave Packet*)

Consider the wave packet  $\psi(x, 0) = \left(\frac{1}{2\pi\sigma^2}\right)^{1/4} e^{-x^2/(4\sigma^2)} e^{ik_0 x}$ .

- (a) Verify normalization and calculate  $\Delta x$  at  $t = 0$ .
- (b) Find the momentum space wave function  $\tilde{\psi}(p, 0)$  via Fourier transform.
- (c) Calculate  $\langle p \rangle$  and  $\Delta p$ , then verify the uncertainty relation.
- (d) What is the physical interpretation of the parameter  $k_0$ ?

### Problem 5.4 (*Coherent State Overlap*)

Coherent states  $|\alpha\rangle$  satisfy  $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$  and can be expanded as:

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

Calculate the inner product  $\langle \alpha' | \alpha \rangle$  between two coherent states and show that:

$$\langle \alpha' | \alpha \rangle = \exp \left[ -\frac{1}{2} (|\alpha|^2 + |\alpha'|^2 - 2\alpha'^* \alpha) \right]$$

What is the probability  $P(\alpha', \alpha) = |\langle \alpha' | \alpha \rangle|^2$  of finding the system in state  $|\alpha'\rangle$  if it was prepared in state  $|\alpha\rangle$ ? Express your answer in terms of  $|\alpha - \alpha'|$ .

### Problem 5.5 (*Phase Space Distance and Distinguishability*)

- (a) For two coherent states with the same amplitude but different phases,  $\alpha = re^{i\theta}$  and  $\alpha' = re^{i\theta'}$ , calculate the overlap probability  $|\langle\alpha'|\alpha\rangle|^2$  as a function of  $r$  and  $\Delta\theta = \theta' - \theta$ .
- (b) At what phase space distance  $|\alpha - \alpha'|$  does the overlap probability drop below 0.5? Below 0.1?
- (c) Discuss the physical significance: When can two coherent states be considered "approximately orthogonal" and what does this mean for the classical limit?

**Problem 5.6** (*Position Space Wave Functions of Coherent States*)

The position-space wave function of a coherent state centered at phase space point  $\alpha$  is:

$$\psi_\alpha(x) = \left(\frac{1}{\pi x_0^2}\right)^{1/4} \exp\left[-\frac{(x - \langle x \rangle_\alpha)^2}{2x_0^2} + \frac{i\langle p \rangle_\alpha x}{\hbar}\right]$$

where  $\langle x \rangle_\alpha = x_0\sqrt{2}\text{Re}(\alpha)$  and  $\langle p \rangle_\alpha = p_0\sqrt{2}\text{Im}(\alpha)$ .

- (a) Calculate the position space overlap integral  $\int_{-\infty}^{\infty} \psi_{\alpha'}^*(x)\psi_\alpha(x)dx$  between two coherent states.
- (b) Verify that this gives the same result as the abstract calculation  $\langle\alpha'|\alpha\rangle$  from Problem 5.4.
- (c) What is the physical interpretation of the real and imaginary parts of  $\alpha$  in terms of position and momentum?

**Problem 5.7** (*Overcompleteness of Coherent States*)

Unlike energy eigenstates, coherent states form an overcomplete basis. Show that the resolution of the identity for coherent states is:

$$\frac{1}{\pi} \int d^2\alpha |\alpha\rangle\langle\alpha| = \hat{I}$$

where  $d^2\alpha = d(\text{Re } \alpha) d(\text{Im } \alpha)$ .

**Hint:** Work in the number basis and use the integral:

$$\frac{1}{\pi} \int d^2\alpha e^{-|\alpha|^2} \alpha^n (\alpha^*)^m = \delta_{nm} n!$$

**Problem 5.8** (*Coherent State Time Evolution*)

A coherent state  $|\alpha\rangle$  evolves under the harmonic oscillator Hamiltonian  $\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + 1/2)$ .

- (a) Show that the time-evolved state  $|\alpha(t)\rangle = e^{-i\hat{H}t/\hbar}|\alpha\rangle$  is also a coherent state
- (b) Find the time-evolved parameter  $\alpha(t)$  as a function of the initial  $\alpha$  and time  $t$
- (c) Verify that the minimum uncertainty property  $\Delta x \Delta p = \hbar/2$  is preserved
- (d) Describe the motion in phase space: does the coherent state trace out a classical trajectory?

**Problem 5.9** (*Squeezed States*)

Squeezed states provide reduced uncertainty in one quadrature at the expense of increased uncertainty in the other. The squeeze operator is  $\hat{S}(r) = \exp\left(\frac{r}{2}(\hat{a}^2 - \hat{a}^{\dagger 2})\right)$ .

- (a) Show that squeezed states transform position and momentum uncertainties as  $\Delta x \rightarrow e^{-r} \Delta x_0$  and  $\Delta p \rightarrow e^r \Delta p_0$
- (b) For a squeezed ground state  $|\psi_s\rangle = \hat{S}(r)|0\rangle$ , calculate  $\Delta x$  and  $\Delta p$  as functions of  $r$
- (c) Verify that the product  $\Delta x \Delta p = \hbar/2$  remains constant (minimum uncertainty states)
- (d) What is the classical analog: can you achieve one-directional uncertainty reduction classically?

**Problem 5.10** (*Harmonic Oscillator Matrix Elements*)

Calculate the matrix elements of position and momentum in the energy eigenstate basis  $\{|n\rangle\}$  for the harmonic oscillator.

- (a) Using ladder operators  $\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger)$ , find  $\langle n|\hat{x}|m\rangle$
- (b) Similarly, find  $\langle n|\hat{p}|m\rangle$
- (c) Calculate  $\langle n|\hat{x}^2|m\rangle$  and show that  $\langle n|\hat{x}^2|n\rangle = \frac{\hbar}{2m\omega}(2n+1)$
- (d) For the ground state, verify that  $\langle 0|\hat{x}^2|0\rangle$  and  $\langle 0|\hat{p}^2|0\rangle$  are consistent with the uncertainty principle

**Problem 5.11** (*Uncertainty in Number-Phase Operators*)

For a harmonic oscillator with  $\hat{N} = \hat{a}^\dagger \hat{a}$ , consider a phase-like operator  $\hat{\Phi}$  where  $\hat{a} = \sqrt{\hat{N}}e^{i\hat{\Phi}}$ .

- (a) For a Fock state  $|n\rangle$  with  $n \gg 1$ , estimate the phase uncertainty  $\Delta\Phi$
- (b) For a coherent state  $|\alpha\rangle$ , show that the number uncertainty is  $\Delta N = |\alpha|$
- (c) Discuss why the number-phase uncertainty relation  $\Delta N \Delta\Phi \gtrsim 1/2$  is consistent with quantum mechanics
- (d) For a squeezed state, which uncertainty is reduced: number or phase?

**Problem 5.12** (*Finite Potential Well*)

A particle is confined by a finite square well:  $V(x) = 0$  for  $|x| < a$  and  $V(x) = V_0$  for  $|x| \geq a$ .

- (a) Write the bound state wave function, applying appropriate matching conditions at  $x = \pm a$
- (b) Define the dimensionless parameter  $\xi = a\sqrt{2mV_0}/\hbar$ . How many bound states exist for  $\xi = 1$ ,  $\xi = 3$ , and  $\xi = 5$ ?
- (c) For the ground state of a deep well ( $V_0 \rightarrow \infty$ ), show that it approaches the infinite well ground state
- (d) Calculate the penetration depth of the wave function outside the well for the ground state

**Problem 5.13** (*Wave Packet Reflection and Transmission*)

A Gaussian wave packet with  $\langle p \rangle = p_0 > 0$  is incident on a potential step  $V(x) = 0$  for  $x < 0$  and  $V(x) = V_s$  for  $x \geq 0$ , with  $V_s < E = p_0^2/(2m)$ .

- (a) Show that the wave function in each region consists of plane waves with definite momentum
- (b) Using the potential step boundary conditions, derive the reflection and transmission coefficients  $R$  and  $T$  as functions of the incident energy
- (c) Calculate the transmitted momentum  $p_t$  in terms of the incident momentum  $p_0$  and potential height  $V_s$
- (d) A Gaussian packet has width  $\sigma_x$ . How does dispersion affect the transmitted and reflected packets as time progresses?

**Problem 5.14** (*Wave Packet Spreading*)

A free particle wave packet with initial width  $\sigma_x(0)$  spreads due to quantum dispersion.

- (a) Derive the spreading formula:  $\sigma_x^2(t) = \sigma_x^2(0) + \frac{\hbar^2 t^2}{4m^2 \sigma_x^2(0)}$
- (b) Define the dispersion time  $t_d$  as the time at which the packet width doubles. Express  $t_d$  in terms of the initial width and particle mass
- (c) For an electron initially localized to  $\sigma_x(0) = 1 \text{ nm}$ , calculate  $t_d$  in seconds
- (d) Compare the dispersion times for an electron, a proton, and a macroscopic grain of sand ( $1 \mu\text{g}$ ) with the same initial width

## References and Further Reading

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**Griffiths, D. J., and Schroeter, D. F.** *Introduction to Quantum Mechanics*, 3rd ed. Cambridge University Press, 2018. Chapters 2–3 give the standard undergraduate-level entry into the wave equation, the infinite well, and the harmonic oscillator; the gentlest companion to this chapter.

**Cohen-Tannoudji, C., Diu, B., and Laloë, F.** *Quantum Mechanics*, Vol. 1. Wiley-VCH, 1977. Chapters I–V cover the continuum formalism, position and momentum representations, and the harmonic oscillator with worked examples; the most thorough single textbook treatment.

**Shankar, R.** *Principles of Quantum Mechanics*, 2nd ed. Springer, 1994. Chapters 1 and 7 develop the discrete-to-continuum limit, Fourier analysis, and the harmonic oscillator with unusual clarity; particularly recommended for the structure of the position-momentum Fourier relationship.

**Walls, D. F., and Milburn, G. J.** *Quantum Optics*, 2nd ed. Springer, 2008. Chapters 2–3 develop coherent and squeezed states with the formalism extended at the end of this chapter; the natural next reference for the optical applications.

**Aasi, J. et al. (LIGO Scientific Collaboration)** “Enhanced sensitivity of the LIGO gravitational wave detector by using squeezed states of light.” *Nature Photonics* **7**, 613–619 (2013). [doi:10.1038/nphoton.2013.177](https://doi.org/10.1038/nphoton.2013.177). The squeezed states of this chapter, used to beat the standard quantum limit at LIGO; the most concrete demonstration that the formalism here is experimentally consequential.

## Chapter 6

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# Three-Dimensional Space and Angular Momentum

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### 6.1 Introduction: The Rich Structure of Three-Dimensional Quantum Mechanics

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#### Key Idea: Angular Momentum and Rotational Symmetry

The extension from one to three spatial dimensions introduces fundamentally new physics through rotational symmetry. Angular momentum emerges as the generator of rotations, leading to its quantization and the characteristic structure of atomic orbitals. This chapter develops the mathematical framework of three-dimensional quantum mechanics, building the tools needed for understanding atomic and molecular systems.

Throughout our development thus far, we have restricted attention to one-dimensional systems. While this simplification allowed us to develop the essential concepts of quantum mechanics, nature is inherently three-dimensional. The transition to three dimensions is not merely a technical extension—it introduces qualitatively new phenomena arising from rotational symmetry.

The most profound consequence of three-dimensional space is the emergence of angular momentum as a fundamental quantity. Unlike linear momentum, which exists even in one dimension, angular momentum requires the full three-dimensional stage. Its quantization leads directly to the shell structure of atoms, the shapes of molecular orbitals, and ultimately to the periodic table of elements. This chapter develops the mathematical framework of angular momentum and spherical harmonics, establishing the foundation for solving central potential problems in atomic physics.

#### 6.1.1 The Profound Role of Symmetry

Symmetry principles provide powerful constraints on physical systems, and rotational symmetry in three dimensions is among the most consequential. The invariance of physical laws under rotations implies conservation of angular momentum through Noether's theorem. More subtly, the group

structure of rotations—specifically the non-commutativity of rotations about different axes—leads directly to the quantization of angular momentum and the peculiar algebra that governs it.

The mathematical framework we develop here extends far beyond atomic physics. The same angular momentum algebra appears in nuclear physics, particle physics (as isospin and other internal symmetries), and even in the study of rotating molecules and nuclei. The techniques of ladder operators and spherical harmonics find applications in fields ranging from computer graphics to seismology. Understanding three-dimensional quantum mechanics thus opens doors to vast areas of modern physics and technology.

### 6.1.2 From Discrete Spins to Continuous Rotations

Our journey began with discrete spin-1/2 systems and Bloch cubes, where angular momentum appeared in its simplest form. The Pauli matrices satisfied the fundamental commutation relations that define angular momentum, but only two states were possible: spin up and spin down. As we transition to orbital angular momentum in three dimensions, we encounter an infinite tower of possible angular momentum values, each with its multiplet of magnetic substates.

This progression from discrete to continuous, from finite to infinite, mirrors our earlier transition from lattice models to continuous wave functions. Just as wave functions emerged from discrete amplitudes, the continuous rotations of three-dimensional space emerge from the discrete operations we studied with Bloch cubes. The deep unity of quantum mechanics reveals itself in these connections between seemingly disparate formulations.

## 6.2 Wave Functions in Three Dimensions

### 6.2.1 The Three-Dimensional Position Basis

In three dimensions, position eigenstates are labeled by the vector  $\vec{r} = (x, y, z)$ . The fundamental orthonormality relation generalizes to:

$$\langle \vec{r} | \vec{r}' \rangle = \delta^3(\vec{r} - \vec{r}') = \delta(x - x')\delta(y - y')\delta(z - z') \quad (6.1)$$

This three-dimensional Dirac delta function vanishes unless all three coordinates coincide. A general quantum state in three dimensions is expressed as:

$$|\psi\rangle = \int d^3r \psi(\vec{r}) |\vec{r}\rangle \quad (6.2)$$

The wave function  $\psi(\vec{r}) = \langle \vec{r} | \psi \rangle$  now depends on three spatial coordinates. The normalization condition becomes a triple integral:

$$\int d^3r |\psi(\vec{r})|^2 = \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy \int_{-\infty}^{\infty} dz |\psi(x, y, z)|^2 = 1 \quad (6.3)$$

The interpretation remains probabilistic:  $|\psi(\vec{r})|^2 d^3r$  gives the probability of finding the particle in an infinitesimal volume  $d^3r$  around position  $\vec{r}$ . However, the three-dimensional nature introduces new features. The probability current density becomes a vector field  $\vec{j}(\vec{r})$ , and the continuity equation  $\partial\rho/\partial t + \nabla \cdot \vec{j} = 0$  expresses local conservation of probability.



### 6.2.2 Momentum States and the Three-Dimensional Fourier Transform

The momentum basis in three dimensions consists of plane wave states:

$$|\vec{p}\rangle = |p_x, p_y, p_z\rangle \quad (6.4)$$

In position representation:

$$\langle \vec{r} | \vec{p} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\vec{p}\cdot\vec{r}/\hbar} \quad (6.5)$$

The three-dimensional Fourier transform relates position and momentum representations:

$$\tilde{\psi}(\vec{p}) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3r e^{-i\vec{p}\cdot\vec{r}/\hbar} \psi(\vec{r}) \quad (6.6)$$

$$\psi(\vec{r}) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3p e^{i\vec{p}\cdot\vec{r}/\hbar} \tilde{\psi}(\vec{p}) \quad (6.7)$$

The momentum operator in position representation becomes:

$$\hat{\vec{p}} = -i\hbar\nabla = -i\hbar \left( \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \right) \quad (6.8)$$

where  $\hat{x}$ ,  $\hat{y}$ , and  $\hat{z}$  are unit vectors in Cartesian coordinates.

### 6.2.3 The Three-Dimensional Schrödinger Equation

The kinetic energy operator in three dimensions involves derivatives with respect to all three coordinates. Using the Laplacian operator  $\nabla^2$ :

$$\hat{T} = -\frac{\hbar^2}{2m} \nabla^2 = -\frac{\hbar^2}{2m} \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \quad (6.9)$$

The time-independent Schrödinger equation for a particle in a potential  $V(\vec{r})$  becomes:

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{r}) + V(\vec{r})\psi(\vec{r}) = E\psi(\vec{r}) \quad (6.10)$$

This partial differential equation in three variables is considerably more complex than its one-dimensional counterpart. However, symmetries of the potential often allow simplifications through separation of variables.

## 6.3 Separation of Variables and Coordinate Systems

### 6.3.1 The Power of Symmetry-Adapted Coordinates

The choice of coordinate system can dramatically simplify quantum mechanical problems. While the physics is coordinate-independent, the mathematical complexity varies greatly with the choice of coordinates. The guiding principle is to match the coordinate system to the symmetry of the problem: Cartesian coordinates for rectangular symmetries, cylindrical for axial symmetry, and spherical for full rotational symmetry.

This choice affects not just computational convenience but also physical insight. In the appropriate coordinates, conserved quantities become manifest, quantum numbers acquire clear geometric meaning, and the connection between classical and quantum mechanics becomes transparent.

### 6.3.2 Cartesian Coordinates and Separable Potentials

When the potential energy can be written as a sum of functions of individual coordinates:

$$V(x, y, z) = V_x(x) + V_y(y) + V_z(z) \quad (6.11)$$

the Schrödinger equation separates into three independent one-dimensional problems. We seek solutions of the product form:

$$\psi(x, y, z) = \psi_x(x)\psi_y(y)\psi_z(z) \quad (6.12)$$

Substituting into the Schrödinger equation and dividing by  $\psi$ :

$$-\frac{\hbar^2}{2m} \left( \frac{1}{\psi_x} \frac{d^2\psi_x}{dx^2} + \frac{1}{\psi_y} \frac{d^2\psi_y}{dy^2} + \frac{1}{\psi_z} \frac{d^2\psi_z}{dz^2} \right) + V_x + V_y + V_z = E \quad (6.13)$$

Since each term depends on only one variable, they must separately equal constants:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi_x}{dx^2} + V_x(x)\psi_x = E_x\psi_x \quad (6.14)$$

$$-\frac{\hbar^2}{2m} \frac{d^2\psi_y}{dy^2} + V_y(y)\psi_y = E_y\psi_y \quad (6.15)$$

$$-\frac{\hbar^2}{2m} \frac{d^2\psi_z}{dz^2} + V_z(z)\psi_z = E_z\psi_z \quad (6.16)$$

with  $E = E_x + E_y + E_z$ .

#### Example 1

**The Three-Dimensional Particle in a Box** For a rectangular box with sides of length  $L_x$ ,  $L_y$ , and  $L_z$ , the potential is separable with  $V = 0$  inside and  $V = \infty$  at the walls. Each direction yields the familiar one-dimensional box solutions:

$$\psi_{n_x}(x) = \sqrt{\frac{2}{L_x}} \sin\left(\frac{n_x\pi x}{L_x}\right) \quad (6.17)$$

$$\psi_{n_y}(y) = \sqrt{\frac{2}{L_y}} \sin\left(\frac{n_y\pi y}{L_y}\right) \quad (6.18)$$

$$\psi_{n_z}(z) = \sqrt{\frac{2}{L_z}} \sin\left(\frac{n_z\pi z}{L_z}\right) \quad (6.19)$$

The complete wave function:

$$\psi_{n_x, n_y, n_z}(x, y, z) = \sqrt{\frac{8}{L_x L_y L_z}} \sin\left(\frac{n_x\pi x}{L_x}\right) \sin\left(\frac{n_y\pi y}{L_y}\right) \sin\left(\frac{n_z\pi z}{L_z}\right) \quad (6.20)$$

with energy:

$$E_{n_x, n_y, n_z} = \frac{\pi^2 \hbar^2}{2m} \left( \frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2} \right) \quad (6.21)$$

**Degeneracy:** For a cubic box ( $L_x = L_y = L_z = L$ ), states with different quantum number combinations but the same  $n_x^2 + n_y^2 + n_z^2$  have the same energy. For example,  $(2, 1, 1)$ ,  $(1, 2, 1)$ , and  $(1, 1, 2)$  all have  $E = 6\pi^2 \hbar^2 / (2mL^2)$ . This accidental degeneracy reflects the enhanced symmetry of the cube.

### 6.3.3 Spherical Coordinates and Central Potentials

For systems with spherical symmetry, where  $V(\vec{r}) = V(r)$  depends only on the distance from the origin, spherical coordinates prove natural:

$$x = r \sin \theta \cos \phi \quad (6.22)$$

$$y = r \sin \theta \sin \phi \quad (6.23)$$

$$z = r \cos \theta \quad (6.24)$$

Here  $r \in [0, \infty)$  is the radial distance,  $\theta \in [0, \pi]$  is the polar angle from the  $z$ -axis, and  $\phi \in [0, 2\pi)$  is the azimuthal angle in the  $xy$ -plane.

In spherical coordinates, the Laplacian takes a more complex but highly symmetric form:

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \quad (6.25)$$

For central potentials, the wave function separates into radial and angular parts:

$$\psi(r, \theta, \phi) = R(r)Y(\theta, \phi) \quad (6.26)$$

This separation is possible because the angular momentum operators commute with any central potential Hamiltonian.

## 6.4 Angular Momentum in Quantum Mechanics

### 6.4.1 From Classical to Quantum Angular Momentum

Classical angular momentum is defined as  $\vec{L} = \vec{r} \times \vec{p}$ . The quantum mechanical operators are obtained by the standard prescription of replacing classical variables with their operator counterparts:

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y = -i\hbar \left( y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right) \quad (6.27)$$

$$\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z = -i\hbar \left( z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right) \quad (6.28)$$

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x = -i\hbar \left( x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \quad (6.29)$$

These operators generate infinitesimal rotations about their respective axes. A finite rotation by angle  $\alpha$  about the  $z$ -axis, for instance, is implemented by the unitary operator  $e^{-i\alpha\hat{L}_z/\hbar}$ . This connection between angular momentum and rotations is fundamental: angular momentum is the generator of rotations in quantum mechanics, just as linear momentum generates translations.

**Angular Momentum Components in Different Bases** In the  $|\ell, m\rangle$  basis (eigenbasis of  $\hat{L}^2$  and  $\hat{L}_z$ ):

$$\hat{L}_z|\ell, m\rangle = \hbar m|\ell, m\rangle \quad (6.30)$$

$$\hat{L}_+|\ell, m\rangle = \hbar\sqrt{(\ell - m)(\ell + m + 1)}|\ell, m + 1\rangle \quad (6.31)$$

$$\hat{L}_-|\ell, m\rangle = \hbar\sqrt{(\ell + m)(\ell - m + 1)}|\ell, m - 1\rangle \quad (6.32)$$

Table 6.1: Master Reference: Angular Momentum States and Representations

| Quantum Numbers | Eigenvalues                                                 | States                                      | Multiplicity |
|-----------------|-------------------------------------------------------------|---------------------------------------------|--------------|
| $\ell, m$       | $\hat{L}^2: \hbar^2 \ell(\ell + 1)$<br>$\hat{L}_z: \hbar m$ | $ \ell, m\rangle$                           | $2\ell + 1$  |
| $\ell = 0$ (s)  | $\hat{L}^2: 0$                                              | $ 0, 0\rangle$                              | 1            |
| $\ell = 1$ (p)  | $\hat{L}^2: 2\hbar^2$                                       | $ 1, 1\rangle,  1, 0\rangle,  1, -1\rangle$ | 3            |
| $\ell = 2$ (d)  | $\hat{L}^2: 6\hbar^2$                                       | $ 2, 2\rangle, \dots,  2, -2\rangle$        | 5            |

In Cartesian representation:

$$\hat{L}_x = \frac{1}{2}(\hat{L}_+ + \hat{L}_-) \quad (6.33)$$

$$\hat{L}_y = \frac{1}{2i}(\hat{L}_+ - \hat{L}_-) \quad (6.34)$$

Matrix representations for  $\ell = 1$ :

$$\hat{L}_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}_{\ell, m} \quad (6.35)$$

$$\hat{L}_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}_{\ell, m} \quad (6.36)$$

$$\hat{L}_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}_{\ell, m} \quad (6.37)$$

#### 6.4.2 The Fundamental Commutation Relations

The defining feature of angular momentum in quantum mechanics is the non-commutativity of its components:

$$[\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z \quad (6.38)$$

$$[\hat{L}_y, \hat{L}_z] = i\hbar \hat{L}_x \quad (6.39)$$

$$[\hat{L}_z, \hat{L}_x] = i\hbar \hat{L}_y \quad (6.40)$$

These relations can be written compactly as  $[\hat{L}_i, \hat{L}_j] = i\hbar \epsilon_{ijk} \hat{L}_k$ , where  $\epsilon_{ijk}$  is the Levi-Civita symbol.

The non-commutativity has profound physical consequences. It implies that we cannot simultaneously specify all three components of angular momentum with arbitrary precision. This is the rotational analog of the position-momentum uncertainty principle and reflects the fact that rotations about different axes do not commute.

#### 6.4.3 Total Angular Momentum and Simultaneous Eigenstates

While individual components of angular momentum are incompatible observables, the total angular momentum squared:

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 \quad (6.41)$$

commutes with each component:  $[\hat{L}^2, \hat{L}_i] = 0$  for  $i = x, y, z$ . This crucial property allows us to find simultaneous eigenstates of  $\hat{L}^2$  and one component, conventionally chosen as  $\hat{L}_z$ .

### Physical Insight

The ability to know both the magnitude of angular momentum and one component, but not all three components, is analogous to knowing a vector lies on a cone about one axis. The vector precesses around this cone, with the other two components fluctuating while maintaining the fixed magnitude and  $z$ -component. This picture, while classical, provides useful intuition for the quantum mechanical situation.

## 6.5 Angular Momentum in Spherical Coordinates

### 6.5.1 Explicit Forms of the Operators

In spherical coordinates, the angular momentum operators take particularly elegant forms:

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi} \quad (6.42)$$

$$\hat{L}_{\pm} = \hat{L}_x \pm i\hat{L}_y = \hbar e^{\pm i\phi} \left( \pm \frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right) \quad (6.43)$$

$$\hat{L}^2 = -\hbar^2 \left[ \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \quad (6.44)$$

Note that these operators involve only the angular coordinates  $\theta$  and  $\phi$ , not the radial coordinate  $r$ . This is why angular momentum is conserved for any central potential.

### 6.5.2 The Eigenvalue Problem

The simultaneous eigenfunctions of  $\hat{L}^2$  and  $\hat{L}_z$  are the spherical harmonics  $Y_{\ell}^m(\theta, \phi)$ :

$$\hat{L}^2 Y_{\ell}^m(\theta, \phi) = \hbar^2 \ell(\ell + 1) Y_{\ell}^m(\theta, \phi) \quad (6.45)$$

$$\hat{L}_z Y_{\ell}^m(\theta, \phi) = \hbar m Y_{\ell}^m(\theta, \phi) \quad (6.46)$$

The quantum numbers have restricted ranges:  $\ell = 0, 1, 2, \dots$  (the orbital angular momentum quantum number) and  $m = -\ell, -\ell + 1, \dots, \ell - 1, \ell$  (the magnetic quantum number). For each value of  $\ell$ , there are  $2\ell + 1$  possible values of  $m$ .

The quantization of angular momentum magnitude as  $\sqrt{\ell(\ell + 1)}\hbar$  rather than simply  $\ell\hbar$  reflects quantum mechanical subtleties. The “extra”  $\sqrt{\ell(\ell + 1)}$  factor arises from the non-commutativity of angular momentum components and the resulting quantum fluctuations.

## 6.6 Spherical Harmonics: The Angular Wave Functions

### 6.6.1 Explicit Forms and Properties

The spherical harmonics are given by:

$$Y_{\ell}^m(\theta, \phi) = \sqrt{\frac{2\ell + 1}{4\pi} \frac{(\ell - |m|)!}{(\ell + |m|)!}} P_{\ell}^{|m|}(\cos \theta) e^{im\phi} \quad (6.47)$$

where  $P_\ell^{|m|}$  are the associated Legendre polynomials. The first few spherical harmonics are:

$$Y_0^0 = \frac{1}{\sqrt{4\pi}} \quad (6.48)$$

$$Y_1^0 = \sqrt{\frac{3}{4\pi}} \cos \theta \quad (6.49)$$

$$Y_1^{\pm 1} = \mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i\phi} \quad (6.50)$$

$$Y_2^0 = \sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1) \quad (6.51)$$

These functions form a complete orthonormal set on the unit sphere:

$$\int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta Y_{\ell'}^{m'*}(\theta, \phi) Y_\ell^m(\theta, \phi) = \delta_{\ell\ell'} \delta_{mm'} \quad (6.52)$$

### 6.6.2 Physical Interpretation and Orbital Shapes

The spherical harmonics determine the angular distribution of probability in atomic orbitals. The familiar orbital shapes arise from plotting surfaces of constant  $|Y_\ell^m(\theta, \phi)|^2$ :

For  $\ell = 0$  (s orbitals),  $Y_0^0$  is constant, giving spherical symmetry. For  $\ell = 1$  (p orbitals), the three functions correspond to the familiar dumbbell shapes oriented along the coordinate axes. For  $\ell = 2$  (d orbitals), five distinct angular distributions emerge, including the characteristic four-lobed patterns.

The parity of spherical harmonics under inversion ( $\vec{r} \rightarrow -\vec{r}$ , or  $\theta \rightarrow \pi - \theta$ ,  $\phi \rightarrow \phi + \pi$ ) is  $(-1)^\ell$ . This property plays a crucial role in selection rules for atomic transitions.

## 6.7 Ladder Operators and the Algebra of Angular Momentum

### 6.7.1 Raising and Lowering Operators

The ladder operators for angular momentum are defined as:

$$\hat{L}_+ = \hat{L}_x + i\hat{L}_y \quad (6.53)$$

$$\hat{L}_- = \hat{L}_x - i\hat{L}_y \quad (6.54)$$

These operators satisfy the commutation relations:

$$[\hat{L}_z, \hat{L}_\pm] = \pm \hbar \hat{L}_\pm \quad (6.55)$$

$$[\hat{L}_+, \hat{L}_-] = 2\hbar \hat{L}_z \quad (6.56)$$

$$[\hat{L}^2, \hat{L}_\pm] = 0 \quad (6.57)$$

The action on spherical harmonics is:

$$\hat{L}_+ Y_\ell^m = \hbar \sqrt{(\ell - m)(\ell + m + 1)} Y_\ell^{m+1} \quad (6.58)$$

$$\hat{L}_- Y_\ell^m = \hbar \sqrt{(\ell + m)(\ell - m + 1)} Y_\ell^{m-1} \quad (6.59)$$

These operators raise or lower the magnetic quantum number  $m$  while preserving  $\ell$ . They vanish when acting on the extremal states  $Y_\ell^\ell$  (for  $\hat{L}_+$ ) or  $Y_\ell^{-\ell}$  (for  $\hat{L}_-$ ).

### 6.7.2 Constructing Angular Momentum Eigenstates

The ladder operators provide a systematic way to construct all spherical harmonics. Starting from the highest weight state  $Y_\ell^\ell$ , which satisfies  $\hat{L}_+ Y_\ell^\ell = 0$ , we can generate all states with the same  $\ell$  by repeated application of  $\hat{L}_-$ :

$$Y_\ell^m \propto (\hat{L}_-)^{\ell-m} Y_\ell^\ell \quad (6.60)$$

This algebraic approach, based solely on commutation relations, is more general than the differential equation approach and extends to other types of angular momentum, including spin.

## 6.8 Separation in Central Potentials

For any central potential  $V(r)$  depending only on distance from the origin, the Schrödinger equation separates in spherical coordinates:

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_\ell^m(\theta, \phi) \quad (6.61)$$

The angular part is universal, determined entirely by rotational symmetry through the spherical harmonics we have studied. The radial function  $R_{n\ell}(r)$  satisfies:

$$-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{d}{dr} \left( r^2 \frac{dR}{dr} \right) + \left[ \frac{\hbar^2 \ell(\ell+1)}{2mr^2} + V(r) \right] R = ER \quad (6.62)$$

The term  $\frac{\hbar^2 \ell(\ell+1)}{2mr^2}$  acts as an effective centrifugal potential, preventing particles with non-zero angular momentum from approaching the origin. The specific form of  $V(r)$  determines the radial wave functions and energy eigenvalues, but the angular dependence is always given by the spherical harmonics.

This separation is the key to solving atomic and molecular problems in three dimensions. The mathematical tools developed in this chapter—spherical harmonics, ladder operators, and the algebra of angular momentum—provide the framework for understanding the quantum mechanical description of atoms and molecules.

### Optional: Matrix Representations and Rotational Transformation Properties

#### Prerequisites and Scope

This optional section develops the matrix representation of angular momentum operators and establishes how rotations in physical space transform spherical harmonics. These concepts provide crucial preparation for the group theoretical treatment of angular momentum in PHYS 2566. The material is self-contained but not essential for completing this course.

#### Matrix Representations in the Angular Momentum Basis

For each value of  $\ell$ , the  $(2\ell+1)$  states  $|\ell, m\rangle$  with  $m = -\ell, -\ell+1, \dots, \ell-1, \ell$  form a complete basis for the angular momentum subspace. Any operator acting within this subspace can be represented as a  $(2\ell+1) \times (2\ell+1)$  matrix. The matrix elements are defined by:

$$(\hat{A})_{m'm} = \langle \ell, m' | \hat{A} | \ell, m \rangle \quad (6.63)$$

where the row index  $m'$  and column index  $m$  both run from  $-\ell$  to  $\ell$ .

For the diagonal operators  $\hat{L}^2$  and  $\hat{L}_z$ , the matrix representations are particularly simple. Since  $\hat{L}^2|\ell, m\rangle = \hbar^2\ell(\ell+1)|\ell, m\rangle$ , we have:

$$(\hat{L}^2)_{m'm} = \hbar^2\ell(\ell+1)\delta_{m'm} = \hbar^2\ell(\ell+1)I \quad (6.64)$$

where  $I$  is the  $(2\ell+1) \times (2\ell+1)$  identity matrix. Similarly, since  $\hat{L}_z|\ell, m\rangle = \hbar m|\ell, m\rangle$ :

$$(\hat{L}_z)_{m'm} = \hbar m\delta_{m'm} \quad (6.65)$$

which is a diagonal matrix with entries  $\hbar m$  along the diagonal.

### Example 2

**Matrix Representations for  $\ell = 1$**  For the  $\ell = 1$  subspace, the basis states are  $|1, 1\rangle, |1, 0\rangle, |1, -1\rangle$  (in that order). The operators  $\hat{L}^2$  and  $\hat{L}_z$  have matrix representations:

$$\hat{L}^2 = 2\hbar^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 2\hbar^2 I, \quad \hat{L}_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}_{\ell, m} \quad (6.66)$$

For the non-diagonal operators, we use the ladder operator action. Recall:

$$\hat{L}_+|\ell, m\rangle = \hbar\sqrt{(\ell-m)(\ell+m+1)}|\ell, m+1\rangle \quad (6.67)$$

$$\hat{L}_-|\ell, m\rangle = \hbar\sqrt{(\ell+m)(\ell-m+1)}|\ell, m-1\rangle \quad (6.68)$$

For  $\ell = 1$ :

$$\hat{L}_+|1, 1\rangle = 0, \quad \hat{L}_+|1, 0\rangle = \sqrt{2}\hbar|1, 1\rangle, \quad \hat{L}_+|1, -1\rangle = \sqrt{2}\hbar|1, 0\rangle \quad (6.69)$$

$$\hat{L}_-|1, 1\rangle = \sqrt{2}\hbar|1, 0\rangle, \quad \hat{L}_-|1, 0\rangle = \sqrt{2}\hbar|1, -1\rangle, \quad \hat{L}_-|1, -1\rangle = 0 \quad (6.70)$$

This gives:

$$\hat{L}_+ = \hbar\sqrt{2} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad \hat{L}_- = \hbar\sqrt{2} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}_{\ell, m} \quad (6.71)$$

Using  $\hat{L}_x = (\hat{L}_+ + \hat{L}_-)/2$  and  $\hat{L}_y = (\hat{L}_+ - \hat{L}_-)/(2i)$ :

$$\hat{L}_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}_{\ell, m} \quad (6.72)$$

$$\hat{L}_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}_{\ell, m} \quad (6.73)$$

These are Hermitian matrices (as required for observables) with eigenvalues  $\hbar, 0, -\hbar$ , corresponding to the possible  $z$ -components of angular momentum after measurement along the  $x$  or  $y$  axes.



**Example 3**

**Block Diagonal Structure of Angular Momentum Operators** The complete Hilbert space of a particle in three dimensions decomposes into angular momentum subspaces labeled by  $\ell = 0, 1, 2, 3, \dots$ . Any angular momentum operator, when represented in the full basis  $\{|\ell, m\rangle\}$  ordered systematically, exhibits block diagonal structure.

Consider the basis ordering:  $|0, 0\rangle, |1, 1\rangle, |1, 0\rangle, |1, -1\rangle, |2, 2\rangle, |2, 1\rangle, |2, 0\rangle, |2, -1\rangle, |2, -2\rangle, |3, 3\rangle, \dots$ . The operator  $\hat{L}_z$  has the block diagonal form:

$$\hat{L}_z = \begin{pmatrix} 0 & & & & & & & & & & \\ & \hbar & & & & & & & & & \\ & & 0 & & & & & & & & \\ & & & -\hbar & & & & & & & \\ & & & & 2\hbar & & & & & & \\ & & & & & \hbar & & & & & \\ & & & & & & 0 & & & & \\ & & & & & & & -\hbar & & & \\ & & & & & & & & -2\hbar & & \\ & & & & & & & & & 3\hbar & \\ & & & & & & & & & & \ddots \end{pmatrix} \quad (6.74)$$

The structure becomes even clearer when we indicate the block structure explicitly:

$$\hat{L}_z = \begin{pmatrix} \boxed{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \dots \\ \mathbf{0} & \boxed{\begin{matrix} \hbar & & \\ & 0 & \\ & & -\hbar \end{matrix}} & \mathbf{0} & \mathbf{0} & \dots \\ \mathbf{0} & \mathbf{0} & \boxed{\begin{matrix} 2\hbar & & & \\ & \hbar & & \\ & & 0 & \\ & & & -\hbar \\ & & & & -2\hbar \end{matrix}} & \mathbf{0} & \dots \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \boxed{\begin{matrix} 3\hbar & & & & \\ & 2\hbar & & & \\ & & \hbar & & \\ & & & 0 & \\ & & & & -\hbar \\ & & & & & -2\hbar \\ & & & & & & -3\hbar \end{matrix}} & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \quad (6.75)$$

where  $\mathbf{0}$  denotes appropriately-sized zero blocks and the boxed regions show the action on  $\ell = 0$  (dimension 1),  $\ell = 1$  (dimension 3),  $\ell = 2$  (dimension 5), and  $\ell = 3$  (dimension 7) subspaces.

Similarly,  $\hat{L}_x$  has block diagonal structure, but each block contains off-diagonal elements:

$$\hat{L}_x = \begin{pmatrix} \boxed{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots \\ \mathbf{0} & \boxed{\frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}} & \mathbf{0} & \mathbf{0} & \cdots \\ \mathbf{0} & \mathbf{0} & \boxed{\hat{L}_x^{(\ell=2)}} & \mathbf{0} & \cdots \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \boxed{\hat{L}_x^{(\ell=3)}} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \quad (6.76)$$

This block diagonal structure reflects the fundamental fact that angular momentum operators cannot connect states of different  $\ell$ . Each  $\ell$  subspace is an invariant subspace under the action of any angular momentum operator—states with definite total angular momentum  $\ell$  can only be transformed into superpositions of states with the same  $\ell$ . In group theory language, each block is an irreducible representation of the rotation group  $SO(3)$ .

### Angular Momentum Components as a Basis for Operators

A profound fact emerges from examining the  $\ell = 1$  matrices: any  $3 \times 3$  Hermitian matrix can be expressed as a linear combination of  $\hat{L}_x$ ,  $\hat{L}_y$ ,  $\hat{L}_z$ , and the identity  $I$ . To see this, note that a general  $3 \times 3$  Hermitian matrix has 9 real parameters (3 diagonal real entries and 3 complex off-diagonal entries with their conjugates). The identity provides 1 parameter, while  $\hat{L}_x$ ,  $\hat{L}_y$ ,  $\hat{L}_z$  provide 3 more, giving 4 parameters total—but this understates the richness.

More precisely, the space of traceless Hermitian  $3 \times 3$  matrices is 8-dimensional, and the three angular momentum operators  $\hat{L}_x$ ,  $\hat{L}_y$ ,  $\hat{L}_z$  span a 3-dimensional subspace. Any operator acting within the  $\ell = 1$  subspace can be decomposed as:

$$\hat{A} = a_0 I + a_x \hat{L}_x + a_y \hat{L}_y + a_z \hat{L}_z + (\text{other terms}) \quad (6.77)$$

For observables that are rotationally invariant (commuting with all components of  $\vec{L}$ ), only the identity term survives:  $\hat{A} = a_0 I$ . For observables that are vector quantities under rotations, we have  $\hat{A} = \vec{a} \cdot \vec{L}$  for some vector  $\vec{a}$ .

This structure generalizes to all  $\ell$  values. For a general  $\ell$ , the  $(2\ell+1)$  states  $|\ell, m\rangle$  span a  $(2\ell+1)$ -dimensional Hilbert space, and the angular momentum operators provide a natural coordinate system for describing operators within this space. This perspective becomes central in group theory, where the  $\ell$  subspaces are recognized as irreducible representations of the rotation group  $SO(3)$ .

### Rotations in Physical Space and Their Action on Wave Functions

A rotation of the physical coordinate system induces a transformation of quantum states. Consider a rotation  $R$  that transforms coordinates  $(x, y, z) \rightarrow (x', y', z')$  according to the standard rotation matrix:

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = R \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad (6.78)$$

where  $R$  is a  $3 \times 3$  orthogonal matrix with  $\det(R) = +1$  (proper rotations).

In quantum mechanics, rotations are implemented by unitary operators acting on the Hilbert space. For an infinitesimal rotation by angle  $\delta\theta$  about an axis  $\hat{n}$ , the unitary operator is:

$$\hat{U}_R \approx I - \frac{i}{\hbar} \delta\theta (\hat{n} \cdot \vec{L}) \quad (6.79)$$

For finite rotations:

$$\hat{U}_R = e^{-i\theta(\hat{n} \cdot \vec{L})/\hbar} \quad (6.80)$$

This operator transforms states according to  $|\psi'\rangle = \hat{U}_R|\psi\rangle$ . In the position representation, if the original state has wave function  $\psi(\vec{r}) = \langle \vec{r} | \psi \rangle$ , the rotated state has wave function:

$$\psi'(\vec{r}) = \langle \vec{r} | \psi' \rangle = \langle \vec{r} | \hat{U}_R | \psi \rangle = \psi(R^{-1}\vec{r}) \quad (6.81)$$

where  $R^{-1}\vec{r}$  denotes the inverse rotation applied to the position vector.

### Physical Insight

The relationship  $\psi'(\vec{r}) = \psi(R^{-1}\vec{r})$  has a subtle sign: to find the wave function value at point  $\vec{r}$  in the rotated system, we evaluate the original wave function at the point that would rotate to  $\vec{r}$ , which is  $R^{-1}\vec{r}$ . This is an active transformation: the wave function itself rotates, while the coordinate grid remains fixed.

### Transformation of Spherical Harmonics Under Rotations

The spherical harmonics  $Y_\ell^m(\theta, \phi)$  have definite transformation properties under rotations. When we rotate the coordinate system, a spherical harmonic of degree  $\ell$  transforms into a linear combination of spherical harmonics of the same  $\ell$ :

$$\hat{U}_R Y_\ell^m(\theta, \phi) = \sum_{m'=-\ell}^{\ell} D_{m'm}^{(\ell)}(R) Y_\ell^{m'}(\theta, \phi) \quad (6.82)$$

The coefficients  $D_{m'm}^{(\ell)}(R)$  form the Wigner  $D$ -matrices, which provide matrix representations of the rotation group in the basis of angular momentum eigenstates. These matrices satisfy:

$$D_{m'm}^{(\ell)}(R) = \langle \ell, m' | \hat{U}_R | \ell, m \rangle \quad (6.83)$$

For a rotation by angle  $\alpha$  about the  $z$ -axis, the  $D$ -matrix is diagonal:

$$D_{m'm}^{(\ell)}(R_z(\alpha)) = e^{-im\alpha} \delta_{m'm} \quad (6.84)$$

reflecting the simple phase transformation under rotations about the quantization axis.

For more general rotations, the  $D$ -matrices become more complex. Euler angles  $(\alpha, \beta, \gamma)$  parametrize a general rotation as:

$$R(\alpha, \beta, \gamma) = R_z(\alpha) R_y(\beta) R_z(\gamma) \quad (6.85)$$

and the corresponding  $D$ -matrix is:

$$D_{m'm}^{(\ell)}(\alpha, \beta, \gamma) = e^{-im'\alpha} d_{m'm}^{(\ell)}(\beta) e^{-im\gamma} \quad (6.86)$$

where  $d_{m'm}^{(\ell)}(\beta)$  are the reduced rotation matrices (Wigner small- $d$  matrices).

**Example 4**

Rotation of  $\ell = 1$  States For  $\ell = 1$ , consider a rotation by angle  $\beta$  about the  $y$ -axis. The small- $d$  matrix elements are:

$$d^{(1)}(\beta) = \begin{pmatrix} \frac{1+\cos\beta}{2} & -\frac{\sin\beta}{\sqrt{2}} & \frac{1-\cos\beta}{2} \\ \frac{\sin\beta}{\sqrt{2}} & \cos\beta & -\frac{\sin\beta}{\sqrt{2}} \\ \frac{1-\cos\beta}{2} & \frac{\sin\beta}{\sqrt{2}} & \frac{1+\cos\beta}{2} \end{pmatrix}_{m',m} \quad (6.87)$$

For a  $90^\circ$  rotation ( $\beta = \pi/2$ ):

$$d^{(1)}(\pi/2) = \begin{pmatrix} 1/2 & -1/\sqrt{2} & 1/2 \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/2 & 1/\sqrt{2} & 1/2 \end{pmatrix} \quad (6.88)$$

This shows, for instance, that rotating  $Y_1^0$  (the  $p_z$  orbital) by  $90^\circ$  about the  $y$ -axis transforms it into the  $p_x$  orbital:

$$\hat{U}_{R_y(\pi/2)} Y_1^0 = \frac{1}{\sqrt{2}}(Y_1^1 + Y_1^{-1}) \propto \sin\theta \cos\phi \quad (6.89)$$

which is indeed proportional to  $x/r$ , the expected  $p_x$  orbital.

**Example 5**

Explicit Transformation of  $\ell = 1$  Wavefunctions Under Rotation We now demonstrate explicitly how  $p$ -orbital wavefunctions transform under rotation from coordinates  $(x, y, z)$  to new coordinates  $(x', y', z')$ .

**Setup:** Consider the real  $p$ -orbitals commonly used in chemistry. These are related to the complex spherical harmonics by:

$$p_x = \frac{1}{\sqrt{2}}(Y_1^{-1} - Y_1^1) = \sqrt{\frac{3}{4\pi}} \frac{x}{r} = \sqrt{\frac{3}{4\pi}} \sin\theta \cos\phi \quad (6.90)$$

$$p_y = \frac{i}{\sqrt{2}}(Y_1^{-1} + Y_1^1) = \sqrt{\frac{3}{4\pi}} \frac{y}{r} = \sqrt{\frac{3}{4\pi}} \sin\theta \sin\phi \quad (6.91)$$

$$p_z = Y_1^0 = \sqrt{\frac{3}{4\pi}} \frac{z}{r} = \sqrt{\frac{3}{4\pi}} \cos\theta \quad (6.92)$$

In the complex basis  $\{Y_1^1, Y_1^0, Y_1^{-1}\}$ , any  $\ell = 1$  wavefunction can be written as:

$$\psi(\vec{r}) = c_1 Y_1^1(\theta, \phi) + c_0 Y_1^0(\theta, \phi) + c_{-1} Y_1^{-1}(\theta, \phi) \quad (6.93)$$

**Specific rotation:** Consider a rotation by  $\beta = 45^\circ$  about the  $y$ -axis. The Wigner small- $d$  matrix is:

$$d^{(1)}(45^\circ) = \begin{pmatrix} \frac{1+1/\sqrt{2}}{2} & -\frac{1}{2} & \frac{1-1/\sqrt{2}}{2} \\ \frac{1}{2} & \frac{1}{\sqrt{2}} & -\frac{1}{2} \\ \frac{1-1/\sqrt{2}}{2} & \frac{1}{2} & \frac{1+1/\sqrt{2}}{2} \end{pmatrix} = \begin{pmatrix} 0.8536 & -0.5 & 0.1464 \\ 0.5 & 0.7071 & -0.5 \\ 0.1464 & 0.5 & 0.8536 \end{pmatrix}_{m',m} \quad (6.94)$$

where indices run  $(m', m) \in \{1, 0, -1\} \times \{1, 0, -1\}$ .

**Case 1: Transforming the  $p_z$  orbital.**

Start with the  $p_z$  orbital, which in the complex basis is simply  $Y_1^0$  (so  $c_1 = 0$ ,  $c_0 = 1$ ,  $c_{-1} = 0$ ). The coefficient vector is:

$$\vec{c} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad (6.95)$$

After rotation, the new coefficients are:

$$\vec{c}' = d^{(1)}(45^\circ)\vec{c} = \begin{pmatrix} 0.8536 & -0.5 & 0.1464 \\ 0.5 & 0.7071 & -0.5 \\ 0.1464 & 0.5 & 0.8536 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -0.5 \\ 0.7071 \\ 0.5 \end{pmatrix} \quad (6.96)$$

Thus the rotated wavefunction is:

$$\psi'(\vec{r}) = -0.5Y_1^1 + 0.7071Y_1^0 + 0.5Y_1^{-1} \quad (6.97)$$

**Physical interpretation in the rotated frame:** In the new coordinate system  $(x', y', z')$  obtained by rotating the original system by  $45^\circ$  about the  $y$ -axis, this wavefunction represents:

$$\psi'(\vec{r}) = \sqrt{\frac{3}{4\pi}} \frac{z'}{r} \quad (6.98)$$

where  $z'$  is the  $z$ -coordinate in the rotated frame. We can verify this using the coordinate transformation:

$$x' = x \cos(45^\circ) + z \sin(45^\circ) = \frac{x + z}{\sqrt{2}} \quad (6.99)$$

$$y' = y \quad (6.100)$$

$$z' = -x \sin(45^\circ) + z \cos(45^\circ) = \frac{-x + z}{\sqrt{2}} \quad (6.101)$$

Substituting  $z' = (z - x)/\sqrt{2}$  into the expected form:

$$\psi'(\vec{r}) = \sqrt{\frac{3}{4\pi}} \frac{z'}{r} = \sqrt{\frac{3}{4\pi}} \frac{z - x}{r\sqrt{2}} = \frac{1}{\sqrt{2}} \left( \sqrt{\frac{3}{4\pi}} \frac{z}{r} - \sqrt{\frac{3}{4\pi}} \frac{x}{r} \right) = \frac{1}{\sqrt{2}} (Y_1^0 - p_x) \quad (6.102)$$

$$= \frac{1}{\sqrt{2}} \left( Y_1^0 - \frac{1}{\sqrt{2}} (Y_1^{-1} - Y_1^1) \right) = \frac{1}{2} Y_1^1 + \frac{1}{\sqrt{2}} Y_1^0 + \frac{1}{2} Y_1^{-1} \quad (6.103)$$

which matches our  $D$ -matrix result (within normalization)!

**Case 2: Transforming the  $p_x$  orbital.**

Now start with  $p_x = \frac{1}{\sqrt{2}}(Y_1^{-1} - Y_1^1)$ , giving coefficient vector:

$$\vec{c} = \begin{pmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{pmatrix} \quad (6.104)$$

After rotation:

$$\vec{c}' = d^{(1)}(45^\circ)\vec{c} = \begin{pmatrix} 0.8536 & -0.5 & 0.1464 \\ 0.5 & 0.7071 & -0.5 \\ 0.1464 & 0.5 & 0.8536 \end{pmatrix} \begin{pmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{pmatrix} = \begin{pmatrix} -0.5 \\ -0.7071 \\ 0.5 \end{pmatrix} \quad (6.105)$$

The rotated wavefunction is:

$$\psi'(\vec{r}) = -0.5Y_1^1 - 0.7071Y_1^0 + 0.5Y_1^{-1} \quad (6.106)$$

This represents:

$$\psi'(\vec{r}) = \sqrt{\frac{3}{4\pi}} \frac{x'}{r} \quad (6.107)$$

the  $p_x$  orbital in the rotated coordinate system, where  $x' = (x + z)/\sqrt{2}$ .

**Summary of transformation rules:** For a rotation specified by Euler angles  $(\alpha, \beta, \gamma)$ , the complete transformation of an  $\ell = 1$  wavefunction is:

$$\begin{pmatrix} c'_1 \\ c'_0 \\ c'_{-1} \end{pmatrix} = D^{(1)}(\alpha, \beta, \gamma) \begin{pmatrix} c_1 \\ c_0 \\ c_{-1} \end{pmatrix} = e^{-i\alpha M_z} d^{(1)}(\beta) e^{-i\gamma M_z} \begin{pmatrix} c_1 \\ c_0 \\ c_{-1} \end{pmatrix} \quad (6.108)$$

where  $M_z = \text{diag}(1, 0, -1)$ .

This procedure generalizes to arbitrary  $\ell$ : Wigner  $D$ -matrices provide the complete machinery for transforming angular momentum eigenstates under arbitrary rotations, forming the foundation for understanding molecular symmetries, crystal field theory, and group theoretical methods in quantum mechanics.

### Connection to Representation Theory

The transformation properties of spherical harmonics under rotations reveal the deep connection between angular momentum and group theory. Each  $\ell$  subspace carries an irreducible representation of the rotation group  $SO(3)$ . The dimension of this representation is  $2\ell + 1$ , and the  $D$ -matrices provide explicit realizations of group elements as  $(2\ell + 1) \times (2\ell + 1)$  unitary matrices.

Several key properties follow from group theory:

(a) **Composition:** The  $D$ -matrices respect the group multiplication law:

$$D^{(\ell)}(R_1 R_2) = D^{(\ell)}(R_1) D^{(\ell)}(R_2) \quad (6.109)$$

(b) **Unitarity:** The  $D$ -matrices are unitary:

$$\sum_{m''=-\ell}^{\ell} D_{m'm''}^{(\ell)*}(R) D_{m''m}^{(\ell)}(R) = \delta_{m'm} \quad (6.110)$$

(c) **Orthogonality:** Integration over all rotations gives:

$$\int dR D_{m'_1 m_1}^{(\ell_1)*}(R) D_{m'_2 m_2}^{(\ell_2)}(R) = \frac{8\pi^2}{2\ell_1 + 1} \delta_{\ell_1 \ell_2} \delta_{m'_1 m'_2} \delta_{m_1 m_2} \quad (6.111)$$

These properties become essential in analyzing systems with rotational symmetry, computing angular momentum coupling coefficients (Clebsch-Gordan coefficients), and understanding selection rules for transitions.

### Physical Applications and Looking Forward

The matrix representation framework developed here connects to numerous physical phenomena. The rotational states of linear molecules transform according to angular momentum representations, with energy levels  $E_J = BJ(J+1)$  where  $J$  is the rotational quantum number. Electric dipole transitions between atomic states require  $\Delta\ell = \pm 1$  and  $\Delta m = 0, \pm 1$ , consequences of how the position operator transforms under rotations. External magnetic fields break rotational symmetry in the Zeeman effect, splitting energy levels according to their  $m$  quantum numbers. The coupling of orbital and spin angular momentum involves Clebsch-Gordan coefficients, which arise from tensor products of angular momentum representations.

In PHYS 2566, we will develop the full group theoretical framework, treating angular momentum as arising from the symmetry group of spatial rotations. The representation theory of  $SU(2)$  and  $SO(3)$  provides powerful tools for classifying quantum states, deriving selection rules, and understanding the addition of angular momenta. The matrix representations developed here provide the concrete foundation upon which that abstract framework builds.

## 6.9 Connections to Spin and Total Angular Momentum

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### 6.9.1 The Universal Algebra of Angular Momentum

The algebra of angular momentum developed in this chapter applies to any type of angular momentum, not just orbital. The spin-1/2 systems studied in Chapter 1 satisfy the same commutation relations, with spin operators  $\hat{S}_i = \frac{\hbar}{2}\sigma_i$ .

For spin-1/2, we have  $s = 1/2$  and  $m_s = \pm 1/2$ , giving just two states. For orbital angular momentum,  $\ell$  can be any non-negative integer, giving  $2\ell + 1$  states for each  $\ell$ . Despite these differences, both types of angular momentum share the same algebraic structure.

### 6.9.2 Coupling Angular Momenta

When particles possess both orbital and spin angular momentum, or when multiple particles are present, angular momenta must be coupled. The total angular momentum:

$$\vec{J} = \vec{L} + \vec{S} \quad (6.112)$$

satisfies the same commutation relations as its constituents.

The coupling of angular momenta leads to fine structure in atomic spectra, as spin-orbit interaction splits levels with the same  $n$  and  $\ell$  but different total angular momentum  $j = \ell \pm 1/2$ . This physics, while beyond our current scope, builds directly on the foundations established here.

## 6.10 Chapter Summary

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This chapter has extended quantum mechanics from one to three spatial dimensions, revealing the profound consequences of rotational symmetry. Beginning with the familiar wave function formalism, we introduced three-dimensional position states  $|\vec{r}\rangle$  and showed how wave functions now depend on three spatial coordinates:  $\psi(\vec{r}) = \psi(x, y, z)$ . The normalization condition extends to a triple integral over all space, and the probabilistic interpretation remains intact with  $|\psi(\vec{r})|^2 d^3r$  giving the probability of finding the particle in volume element  $d^3r$ .

The transition to three dimensions introduced fundamentally new physics through rotational symmetry and angular momentum. We derived the angular momentum operators  $\hat{L}_x$ ,  $\hat{L}_y$ , and

$\hat{L}_z$  from the classical definition  $\vec{L} = \vec{r} \times \vec{p}$  using the standard prescription of replacing classical variables with quantum operators. The crucial discovery was that these operators do not commute:  $[\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k$ . This non-commutativity, reflecting the fact that rotations about different axes do not commute, has profound consequences. It prevents simultaneous precise knowledge of all three angular momentum components and leads directly to the quantization of angular momentum.

While individual components cannot be simultaneously specified, the total angular momentum squared  $\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$  commutes with each component. This allows us to find simultaneous eigenstates of  $\hat{L}^2$  and one component, conventionally chosen as  $\hat{L}_z$ . The eigenvalue problem, solved using ladder operators  $\hat{L}_{\pm} = \hat{L}_x \pm i\hat{L}_y$ , reveals that angular momentum magnitude is quantized as  $\sqrt{\ell(\ell+1)}\hbar$  where  $\ell = 0, 1, 2, \dots$ , and the  $z$ -component takes values  $m\hbar$  where  $m = -\ell, -\ell + 1, \dots, \ell - 1, \ell$ . The eigenfunctions are the spherical harmonics  $Y_{\ell}^m(\theta, \phi)$ , which form a complete orthonormal basis for functions on the unit sphere.

The power of symmetry-adapted coordinates became evident when we considered central potentials  $V(r)$  depending only on distance from the origin. In spherical coordinates  $(r, \theta, \phi)$ , the Schrödinger equation separates into radial and angular parts:  $\psi(r, \theta, \phi) = R(r)Y_{\ell}^m(\theta, \phi)$ . The angular part is universal, determined entirely by rotational symmetry through the spherical harmonics, while the radial equation depends on the specific potential. An effective centrifugal barrier  $\hbar^2\ell(\ell+1)/(2mr^2)$  emerges, preventing particles with non-zero angular momentum from reaching the origin.

The shapes of atomic orbitals—spherical s orbitals ( $\ell = 0$ ), dumbbell-shaped p orbitals ( $\ell = 1$ ), and more complex d ( $\ell = 2$ ) and f ( $\ell = 3$ ) orbitals—emerge directly from the angular dependence encoded in the spherical harmonics. These angular distributions, combined with specific radial functions for particular potentials, determine the complete structure of atomic wave functions.

Throughout this development, we've seen how the abstract algebra of angular momentum connects our initial study of discrete spin-1/2 systems to the continuous rotations of three-dimensional space. The same commutation relations that governed Pauli matrices for spin now apply to orbital angular momentum, revealing a deep unity in quantum mechanics. The ladder operator formalism, purely algebraic in nature, works equally well for spin and orbital angular momentum, differing only in the allowed values of the quantum numbers.

This chapter establishes the framework for understanding atomic physics, molecular structure, and the coupling of different angular momenta. The progression from one-dimensional problems to the full three-dimensional world, from discrete spins to continuous rotations, and from abstract algebra to concrete orbital shapes illustrates how quantum mechanics builds complexity from simple principles while maintaining its fundamental mathematical coherence.

## 6.11 Problems

### Problem 6.1 (*Angular Momentum Commutation Relations*)

The angular momentum operators are defined by  $\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y$  and cyclic permutations.

- Starting from the canonical commutation relations  $[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$ , derive the fundamental angular momentum commutator  $[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z$  by explicit calculation.
- Verify the cyclic relations  $[\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x$  and  $[\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y$ .
- Define  $\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$ . Show that  $[\hat{L}^2, \hat{L}_z] = 0$  by using the commutation relations.



- (d) Explain physically why the result of part (c) implies that  $\hat{L}^2$  and  $\hat{L}_z$  can have simultaneous eigenstates.
- (e) Define the ladder operators  $\hat{L}_{\pm} = \hat{L}_x \pm i\hat{L}_y$ . Calculate  $[\hat{L}_z, \hat{L}_+]$  and  $[\hat{L}_z, \hat{L}_-]$ .
- (f) Show that  $[\hat{L}^2, \hat{L}_+] = 0$ , demonstrating that ladder operators preserve the total angular momentum.

**Problem 6.2** (*Spherical Harmonics and Ladder Operators*)

The spherical harmonics  $Y_{\ell}^m(\theta, \phi)$  are the eigenfunctions of  $\hat{L}^2$  and  $\hat{L}_z$ .

- (a) Verify by direct calculation that  $Y_0^0 = \frac{1}{\sqrt{4\pi}}$  is normalized over the unit sphere:  $\int_0^{2\pi} d\phi \int_0^{\pi} \sin\theta d\theta |Y_0^0|^2 = 1$ .
- (b) For  $Y_1^0 = \sqrt{\frac{3}{4\pi}} \cos\theta$ , verify that  $\hat{L}_z Y_1^0 = 0$  using  $\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$ .
- (c) Apply the raising operator  $\hat{L}_+ = \hbar e^{i\phi} \left( \frac{\partial}{\partial \theta} + i \cot\theta \frac{\partial}{\partial \phi} \right)$  to  $Y_1^0$  and show that you obtain  $Y_1^1 = -\sqrt{\frac{3}{8\pi}} \sin\theta e^{i\phi}$  up to normalization.
- (d) Apply the lowering operator  $\hat{L}_-$  to  $Y_1^0$  to find  $Y_1^{-1}$ . Verify the normalization.
- (e) Verify orthogonality between  $Y_1^0$  and  $Y_1^1$  by explicit integration.
- (f) The ladder operators satisfy  $\hat{L}_{\pm} Y_{\ell}^m = \hbar \sqrt{(\ell \mp m)(\ell \pm m + 1)} Y_{\ell}^{m \pm 1}$ . Use this to generate all three  $\ell = 1$  states starting from  $Y_1^1$  (given that  $\hat{L}_+ Y_1^1 = 0$ ).

**Problem 6.3** (*Three-Dimensional Particle in a Box*)

Consider a particle of mass  $m$  confined to a rectangular box with sides  $L_x, L_y, L_z$ .

- (a) Argue that the potential  $V(x, y, z) = 0$  inside the box and  $V = \infty$  at the walls is separable. Write the wave function as  $\psi(x, y, z) = \psi_x(x)\psi_y(y)\psi_z(z)$ .
- (b) Show that the energy eigenvalues are  $E_{n_x, n_y, n_z} = \frac{\pi^2 \hbar^2}{2m} \left( \frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2} \right)$  with  $n_x, n_y, n_z = 1, 2, 3, \dots$ . Write the normalized wave function.
- (c) For a cubic box ( $L_x = L_y = L_z = L$ ), find the first five distinct energy levels. For each level, list all quantum number combinations  $(n_x, n_y, n_z)$  and determine the degeneracy.
- (d) For the ground state  $(1, 1, 1)$  of the cubic box, calculate the expectation values  $\langle x \rangle$ ,  $\langle p_x \rangle$ , and verify that the momentum is purely quantum mechanical (zero average).

**Problem 6.4** (*Angular Momentum Quantization and Uncertainty*)

Explore the structure of angular momentum eigenvalues and the uncertainty principle.

- (a) For a state with quantum numbers  $\ell$  and  $m$ , the magnitude of angular momentum is  $|\vec{L}| = \hbar \sqrt{\ell(\ell+1)}$  while the z-component is  $L_z = \hbar m$ . For  $\ell = 2$ , calculate  $|\vec{L}|$  and compare it to the maximum possible z-component  $\hbar m_{\max}$ . Explain why  $|\vec{L}| > |L_z|_{\max}$ .
- (b) For the state  $|2, 1\rangle$ , calculate the uncertainties  $\Delta L_x$  and  $\Delta L_y$  using  $\langle L_x \rangle = \langle L_y \rangle = 0$  and  $\langle L_x^2 \rangle + \langle L_y^2 \rangle = \langle L^2 \rangle - \langle L_z^2 \rangle$ .

- (c) Verify that  $\Delta L_x \cdot \Delta L_y \geq \frac{\hbar}{2} |\langle L_z \rangle|$  is satisfied for the state  $|2, 1\rangle$ .
- (d) Show that for the maximum  $m$  state ( $m = \ell$ ), the uncertainty product  $\Delta L_x \cdot \Delta L_y$  reaches its minimum allowed value, meaning the angular momentum vector is as well-localized as quantum mechanics permits.

**Problem 6.5** (*Separation of Variables in Central Potentials*)

For any spherically symmetric potential  $V(r)$ , the Schrodinger equation separates.

- (a) Starting from  $-\frac{\hbar^2}{2m}\nabla^2\psi + V(r)\psi = E\psi$  in spherical coordinates, substitute  $\psi(r, \theta, \phi) = R(r)Y_\ell^m(\theta, \phi)$  and derive the radial equation for  $R(r)$ .
- (b) Show that the radial equation can be written as  $-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{d}{dr} \left( r^2 \frac{dR}{dr} \right) + V_{\text{eff}}(r)R = ER$  where  $V_{\text{eff}}(r) = V(r) + \frac{\hbar^2 \ell(\ell+1)}{2mr^2}$ .
- (c) Identify the centrifugal barrier term and explain its physical origin. Why does it prevent particles with  $\ell > 0$  from reaching  $r = 0$ ?
- (d) For a three-dimensional isotropic harmonic oscillator  $V(r) = \frac{1}{2}m\omega^2 r^2$ , write the radial equation for the  $\ell = 0$  case. Make the substitution  $u(r) = rR(r)$  and show that  $u$  satisfies a one-dimensional Schrodinger equation.

**Problem 6.6** (*Hydrogen-like Atoms: Angular Momentum Coupling*)

In a hydrogen atom, the electron experiences a Coulomb potential  $V(r) = -ke^2/r$ .

- (a) For the ground state with quantum numbers  $n = 1, \ell = 0, m = 0$ , write the wave function  $\psi_{100}(r, \theta, \phi) = R_{10}(r)Y_0^0(\theta, \phi)$
- (b) Calculate the expectation values  $\langle r \rangle$  and  $\langle r^2 \rangle$  for the ground state
- (c) For the first excited state  $n = 2$ , which values of  $\ell$  are possible? How many distinct states  $|n, \ell, m\rangle$  exist for  $n = 2$ ?
- (d) For  $n = 2, \ell = 1$ , show that there are three possible values of  $m$  and that all three have the same energy (degeneracy)

**Problem 6.7** (*Rigid Rotor*)

A rigid rotor is a particle constrained to move on a sphere of fixed radius  $r = R$ .

- (a) For a rigid rotor, the kinetic energy is  $\hat{T} = \hat{L}^2/(2I)$  where  $I = mR^2$  is the moment of inertia. Show that the energy eigenvalues are  $E_\ell = \hbar^2 \ell(\ell+1)/(2I)$
- (b) Write the normalized eigenfunctions as  $\psi_{\ell,m}(\theta, \phi) = Y_\ell^m(\theta, \phi)$
- (c) Calculate the rotational constant  $B = \hbar^2/(2I)$  for a diatomic molecule with bond length  $R = 1 \text{ \AA}$  and reduced mass  $\mu = 1 \text{ amu}$
- (d) For the first excited rotational state ( $\ell = 1$ ), how many degenerate states exist? What is the energy spacing between  $\ell = 0$  and  $\ell = 1$  states?

**Problem 6.8** (*Matrix Elements of Angular Momentum*)

Calculate matrix elements of angular momentum operators in the  $|\ell, m\rangle$  basis.

- Using  $\hat{L}_z|\ell, m\rangle = \hbar m|\ell, m\rangle$ , write the matrix representation of  $\hat{L}_z$  for  $\ell = 1$
- Using ladder operators, show that  $\langle \ell, m | \hat{L}_x | \ell, m \pm 1 \rangle = \frac{\hbar}{2} \sqrt{(\ell \mp m)(\ell \pm m + 1)}$
- Construct the  $3 \times 3$  matrix for  $\hat{L}_x$  in the  $\ell = 1$  basis:  $\{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle\}$
- Similarly, construct the matrix for  $\hat{L}_y$
- Verify that  $[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z$  by multiplying these matrices

**Problem 6.9** (*Classical Limit of Angular Momentum*)

In the classical limit, large quantum numbers  $\ell \gg 1$  approximate classical behavior.

- For large  $\ell$ , show that  $|\vec{L}| \approx \hbar\ell$  and the minimum uncertainty in the transverse components is  $\Delta L_{\perp} \sim \hbar/\sqrt{\ell}$
- The angular momentum precesses about the z-axis with expectation value  $\langle L_z \rangle = \hbar m$ . For  $m = \ell$ , what is the "opening angle" of the cone traced by the angular momentum vector?
- Estimate the angle: for  $\ell = 100$ , what is the opening angle in degrees?
- In the limit  $\ell \rightarrow \infty$ , show that the angular momentum becomes increasingly well-defined in direction

**Problem 6.10** (*Three-Dimensional Harmonic Oscillator*)

A particle in a 3D isotropic harmonic oscillator potential  $V(r) = \frac{1}{2}m\omega^2 r^2$  can be solved in spherical coordinates.

- Show that the energy levels are  $E_{n_r, \ell} = \hbar\omega(2n_r + \ell + 3/2)$  where  $n_r = 0, 1, 2, \dots$  and  $\ell = 0, 1, 2, \dots$
- Compare the energy level structure to a cubic box: which system has higher degeneracy for the first excited level?
- For the ground state ( $n_r = 0, \ell = 0$ ), write the wave function and calculate  $\langle r^2 \rangle$
- For the first excited level, what values of  $\ell$  contribute and what is the total degeneracy?

**Problem 6.11** (*Three-Dimensional Infinite Spherical Well*)

A particle is confined inside a sphere of radius  $a$ :  $V(r) = 0$  for  $r < a$  and  $V(r) = \infty$  for  $r \geq a$ .

- Write the boundary condition at  $r = a$ : the radial wave function must vanish
- For  $\ell = 0$  (s-wave), the radial equation becomes one-dimensional:  $\frac{d^2 u}{dr^2} + k^2 u = 0$  where  $u(r) = rR(r)$ . Find the normalized ground state wave function
- Calculate the ground state energy  $E_0$
- For  $\ell = 1$  (p-wave), construct the radial equation and compare the first energy eigenvalue to the  $\ell = 0$  case

- (e) Show that the  $\ell = 0$  ground state energy is lower than any  $\ell > 0$  state

**Problem 6.12** (*Angular Momentum Precession in Magnetic Field*)

A system with angular momentum  $\vec{L}$  is placed in a magnetic field  $\vec{B} = B\hat{z}$ .

- The Hamiltonian is  $\hat{H} = -\gamma\vec{L} \cdot \vec{B} = -\gamma B\hat{L}_z$  where  $\gamma$  is the gyromagnetic ratio
- Write the energy eigenstates and eigenvalues for this system
- For initial state  $|\psi(0)\rangle = |\ell, m_0\rangle$ , show that  $|\psi(t)\rangle = e^{i\gamma B t m_0} |\ell, m_0\rangle$  (up to global phase)
- Prepare an initial state  $|\psi(0)\rangle = \frac{1}{\sqrt{2}}(|\ell, \ell\rangle + |\ell, -\ell\rangle)$  and describe the time evolution of  $\langle L_x \rangle$  and  $\langle L_y \rangle$
- What is the precession frequency in terms of  $\gamma$ ,  $B$ , and  $\ell$ ?

**Problem 6.13** (*Spin-Orbit Coupling*)

In heavy atoms, spin-orbit coupling becomes important. The interaction Hamiltonian is  $\hat{H}_{so} = \lambda \vec{L} \cdot \vec{S} / \hbar^2$ .

- Show that  $\vec{L} \cdot \vec{S} = L_z S_z + \frac{1}{2}(L_+ S_- + L_- S_+)$  where  $L_{\pm}$  and  $S_{\pm}$  are ladder operators
- For a state with fixed  $m_\ell$  and  $m_s$ , calculate the first-order energy correction due to spin-orbit coupling
- Discuss why spin-orbit coupling leads to fine structure splitting of atomic levels

## References and Further Reading

- Griffiths, D. J., and Schroeter, D. F.** *Introduction to Quantum Mechanics*, 3rd ed. Cambridge University Press, 2018. Chapter 4 develops three-dimensional quantum mechanics, angular momentum, and the spherical harmonics at the standard undergraduate level.
- Sakurai, J. J., and Napolitano, J.** *Modern Quantum Mechanics*, 3rd ed. Cambridge University Press, 2017. Chapter 3 builds the algebraic theory of angular momentum and the rotation group; the natural graduate-level companion to this chapter and the prerequisite for Chapter 7.
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- Edmonds, A. R.** *Angular Momentum in Quantum Mechanics*. Princeton University Press, 1957. The classic monograph; tightly written and the standard source for representation matrices and conventions used throughout atomic and molecular physics.
- Biedenharn, L. C., and Louck, J. D.** *Angular Momentum in Quantum Physics: Theory and Application*. Cambridge University Press, 1981. Comprehensive reference treating angular momentum from the group-theoretic perspective; the next step after Edmonds for representation theory of  $SU(2)$  and  $SO(3)$ .

# Chapter 7

## Addition of Angular Momentum

### 7.1 Introduction: Addition is Multiplication

In Chapter 6, we developed the quantum theory of angular momentum for a single particle, discovering that angular momentum operators satisfy the fundamental commutation relations  $[\hat{J}_i, \hat{J}_j] = i\hbar\epsilon_{ijk}\hat{J}_k$ . We found that states can be labeled by quantum numbers  $j$  and  $m$ , with eigenvalues  $\hbar^2 j(j+1)$  for  $\hat{J}^2$  and  $\hbar m$  for  $\hat{J}_z$ . The Hilbert space for a single angular momentum  $j$  has dimension  $2j+1$ , corresponding to the  $2j+1$  possible values of  $m$ .

Physical systems often involve multiple sources of angular momentum: orbital and spin angular momenta of a single particle, angular momenta of multiple particles in atoms or molecules, or nuclear spins coupled to electronic angular momentum. How do we describe such composite systems?

**A note on group theory:** Throughout this chapter, we will discuss several results from the representation theory of rotation groups. These results - including the decomposition of tensor products, the relationship between SU(2) and SO(3), and the structure of Wigner symbols - can be understood and applied without rigorous proof. We are using group theory as a framework to organize and understand quantum angular momentum, but we will not provide full mathematical justifications for the group-theoretic statements. Readers interested in the underlying proofs should consult texts on group representation theory.

#### Key Idea: Addition of Angular Momentum is Multiplication of Hilbert Spaces

When we "add" two angular momenta  $\vec{J}_1$  and  $\vec{J}_2$ , the underlying mathematical structure is the **tensor product** of Hilbert spaces:

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \quad (7.1)$$

If angular momentum  $j_1$  has Hilbert space dimension  $2j_1+1$  and angular momentum  $j_2$  has dimension  $2j_2+1$ , the combined system has dimension:

$$\dim(\mathcal{H}) = (2j_1+1)(2j_2+1) \quad (7.2)$$

This is multiplication, not addition. The "addition" of angular momenta refers to the vector sum  $\vec{J} = \vec{J}_1 + \vec{J}_2$ , but the state space structure is multiplicative.

### 7.1.1 Why Two Descriptions of the Same Space?

The tensor product space  $\mathcal{H}_1 \otimes \mathcal{H}_2$  can be spanned by different bases. The choice of basis reflects which physical quantities we consider most natural to specify:

**The uncoupled basis** is expressed as a tensor product of angular momentum ket states:  $|j_1, m_1\rangle \otimes |j_2, m_2\rangle$ . For notational convenience, we compress this into a single ket  $|j_1, m_1; j_2, m_2\rangle$ , but it is essential to remember these are product states from two different Hilbert spaces. This is the natural product basis constructed from eigenstates of  $\hat{J}_1^2$ ,  $\hat{J}_{1z}$ ,  $\hat{J}_2^2$ , and  $\hat{J}_{2z}$ . There are  $(2j_1+1)(2j_2+1)$  such states.

**The coupled basis**  $|j_1, j_2; j, m\rangle$  instead uses the total angular momentum  $\vec{J} = \vec{J}_1 + \vec{J}_2$ . These are eigenstates of  $\hat{J}_1^2$ ,  $\hat{J}_2^2$ ,  $\hat{J}^2$ , and  $\hat{J}_z$ . Remarkably, the same  $(2j_1+1)(2j_2+1)$ -dimensional space can be organized as a direct sum of subspaces with definite total angular momentum:

$$\mathcal{H}_1 \otimes \mathcal{H}_2 = \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} \mathcal{H}_j = \mathcal{H}_{|j_1-j_2|} \oplus \mathcal{H}_{|j_1-j_2|+1} \oplus \cdots \oplus \mathcal{H}_{j_1+j_2-1} \oplus \mathcal{H}_{j_1+j_2} \quad (7.3)$$

where each  $\mathcal{H}_j$  has dimension  $2j+1$ .

This decomposition reflects a deep mathematical structure from group theory that we'll explore in the next section. The key point is that the same Hilbert space admits two complementary descriptions.

### 7.1.2 Physical Motivation: Why Total Angular Momentum?

Why do we care about reorganizing our basis to use total angular momentum? The answer lies in the symmetries and interactions that govern quantum systems.

First, consider conservation laws. If the Hamiltonian of a system is rotationally invariant, meaning the physics doesn't depend on the orientation of our coordinate system, then total angular momentum  $\vec{J}$  is a conserved quantity. This has profound implications for time evolution. States with definite total angular momentum quantum number  $j$  evolve without mixing with states of different  $j$ . The energy eigenstates can be labeled by  $j$ , and transitions between energy levels must satisfy selection rules involving  $j$ . This makes the coupled basis the natural language for describing dynamics in rotationally symmetric systems.

Second, many fundamental interactions depend explicitly on how the constituent angular momenta are oriented relative to each other. The canonical example is spin-orbit coupling, where the interaction energy depends on  $\vec{L} \cdot \vec{S}$ , the dot product of orbital and spin angular momenta. This interaction can be rewritten using the total angular momentum:

$$\vec{L} \cdot \vec{S} = \frac{1}{2}(\hat{J}^2 - \hat{L}^2 - \hat{S}^2) \quad (7.4)$$

In this form, the interaction is diagonal in the coupled basis: each state  $|\ell, s; j, m\rangle$  has a definite value of  $\vec{L} \cdot \vec{S}$  determined entirely by the quantum numbers  $\ell$ ,  $s$ , and  $j$ . In the uncoupled basis  $|\ell, m_\ell; s, m_s\rangle$ , by contrast, the same interaction mixes different states, requiring diagonalization to find energy eigenstates. The coupled basis thus diagonalizes the physically relevant Hamiltonian.

Third, spectroscopy provides the empirical foundation for atomic physics, and experimentally observed atomic energy levels are labeled using total angular momentum quantum numbers. Spectroscopic notation like  $^2P_{3/2}$  specifies  $n$  (principal quantum number),  $\ell$  (orbital angular momentum), total electronic spin  $S$ , and total angular momentum  $j = \ell \pm s$ . This notation reflects the physical

reality that spin-orbit coupling splits energy levels according to  $j$ , not according to the individual projections  $m_\ell$  and  $m_s$ . Reading and interpreting experimental spectra requires fluency in the coupled basis.

### Physical Insight

The choice between uncoupled and coupled bases reflects the physics of the problem. Use the uncoupled basis when  $\vec{J}_1$  and  $\vec{J}_2$  are nearly independent (weak coupling). Use the coupled basis when interactions couple them strongly or when total angular momentum is a good quantum number. The Clebsch-Gordan coefficients allow us to translate between these descriptions as needed.

## 7.2 The Rotation Group: $SU(2)$ and $SO(3)$

Before diving into the detailed mechanics of angular momentum addition, we need to understand the mathematical structure underlying rotations in quantum mechanics. This involves concepts from group theory - the mathematical theory of symmetry.

### 7.2.1 A Minimal Introduction to Group Theory

A **group** is a set of elements together with an operation (often called "multiplication") that combines any two elements to produce a third element in the set. The operation must satisfy four properties: closure (the product of any two elements is in the group), associativity, existence of an identity element, and existence of inverses.

The rotations in three-dimensional space form a group: the product of two rotations is another rotation, there is an identity rotation (do nothing), each rotation has an inverse (rotate by the opposite angle), and composition of rotations is associative.

A **representation** of a group assigns to each group element a matrix (or more generally, a linear operator) in such a way that the group multiplication corresponds to matrix multiplication. If group elements  $g_1$  and  $g_2$  are represented by matrices  $D(g_1)$  and  $D(g_2)$ , then the product  $g_1 g_2$  must be represented by  $D(g_1)D(g_2)$ .

Different representations can have different dimensions. For rotations, the dimension tells us how many components an object has. Scalars have one component (dimension 1), vectors have three components (dimension 3), and more complex objects can have higher dimensions.

A representation is **irreducible** if it cannot be decomposed into smaller representations. When we write a large matrix in block-diagonal form, each independent block is an irreducible representation. Finding irreducible representations is central to understanding how objects transform under symmetries.

These concepts provide the framework for understanding angular momentum in quantum mechanics. Angular momentum quantum numbers label irreducible representations of the rotation group.

### 7.2.2 $SO(3)$ : The Rotation Group in Three Dimensions

The group  $SO(3)$  consists of all rotations in three-dimensional space. Each rotation can be specified by an axis  $\hat{n}$  and an angle  $\theta$ , giving a rotation operator:

$$R(\hat{n}, \theta) = e^{-i\theta \hat{n} \cdot \vec{J}/\hbar} \quad (7.5)$$

These rotations form a group under composition: the product of two rotations is another rotation, there is an identity rotation ( $\theta = 0$ ), and each rotation has an inverse (rotation by  $-\theta$ ).

A complete rotation by  $2\pi$  about any axis returns to the identity:

$$R(\hat{n}, 2\pi) = e^{-2\pi i \hat{n} \cdot \vec{J}/\hbar} = I \quad (7.6)$$

This holds for all classical vectors and orbital angular momentum states, which transform according to representations of  $SO(3)$ . The allowed values of angular momentum are integers:  $\ell = 0, 1, 2, \dots$

### 7.2.3 $SU(2)$ : The Covering Group

Quantum mechanics requires a more subtle structure. The group  $SU(2)$  consists of  $2 \times 2$  unitary matrices with determinant one. Every  $SU(2)$  matrix can be written as:

$$U = e^{-i\theta \hat{n} \cdot \vec{\sigma}/2} \quad (7.7)$$

where  $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$  are the Pauli matrices.

The crucial difference from  $SO(3)$  is that a rotation by  $2\pi$  gives:

$$U(\hat{n}, 2\pi) = e^{-2\pi i \hat{n} \cdot \vec{\sigma}/2} = -I \quad (7.8)$$

A  $2\pi$  rotation produces a minus sign, not the identity. Only a rotation by  $4\pi$  returns to the identity:

$$U(\hat{n}, 4\pi) = I \quad (7.9)$$

This is the defining property of a spinor: an object that picks up a minus sign under a  $2\pi$  rotation.

### 7.2.4 The Double Cover

$SU(2)$  is the double cover of  $SO(3)$ . There is a two-to-one homomorphism from  $SU(2)$  to  $SO(3)$ :

$$\phi : SU(2) \rightarrow SO(3) \quad (7.10)$$

Both  $U$  and  $-U$  in  $SU(2)$  map to the same rotation  $R$  in  $SO(3)$ . This means that for every physical rotation in three-dimensional space, there are two quantum mechanical phase factors that could represent it, differing by a sign.

This mathematical structure has profound physical consequences.  $SU(2)$  admits representations labeled by half-integers  $j = 0, 1/2, 1, 3/2, 2, \dots$ , while  $SO(3)$  allows only integers  $\ell = 0, 1, 2, \dots$ . The half-integer representations of  $SU(2)$  cannot be represented as rotations of vectors in three-dimensional space. They are intrinsically quantum mechanical objects.

### 7.2.5 Physical Implications for Spin

Electrons, protons, neutrons, and many other fundamental particles have spin-1/2. Their quantum states transform according to the spin-1/2 representation of  $SU(2)$ , not  $SO(3)$ . This has several profound consequences. A spin-1/2 particle rotated by  $2\pi$  acquires a phase factor of  $-1$ , not  $+1$ . Two identical spin-1/2 particles (fermions) must have antisymmetric total wavefunctions. The Pauli exclusion principle holds for half-integer spin particles. Spinors require special mathematical treatment distinct from vectors.

The existence of half-integer angular momentum is not an accident or a peculiarity of quantum mechanics. It is a direct consequence of the group structure of rotations in quantum theory, which requires  $SU(2)$  rather than  $SO(3)$ .



### 7.2.6 Connection to Angular Momentum Addition

The decomposition of tensor products of angular momentum representations,

$$j_1 \otimes j_2 = \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} j \quad (7.11)$$

is a statement about how representations of  $SU(2)$  combine. The Clebsch-Gordan coefficients are precisely the matrix elements that accomplish the change of basis between the product representation and the decomposed direct sum.

When we add two half-integer angular momenta, we can obtain either integer or half-integer results. For example,  $1/2 \otimes 1/2 = 0 \oplus 1$  gives both the singlet (integer) and triplet (integer). But  $1/2 \otimes 1 = 1/2 \oplus 3/2$  gives half-integer results. This mixing of integer and half-integer representations under tensor products is a feature of  $SU(2)$  that has no analog in the purely integer representations of  $SO(3)$ .

## 7.3 Cartesian and Spherical Tensors

Having established the group-theoretic foundation, we now examine how composite objects transform under rotations. This connects directly to the spherical harmonics we studied in Chapter 6 and provides concrete examples of how tensor products decompose into irreducible representations.

### 7.3.1 Cartesian Tensors

A Cartesian tensor of rank  $n$  has  $3^n$  components  $T_{i_1 i_2 \dots i_n}$  where each index runs over  $x, y, z$ . Under a rotation  $R$ , the components transform as:

$$T'_{i_1 i_2 \dots i_n} = R_{i_1 j_1} R_{i_2 j_2} \cdots R_{i_n j_n} T_{j_1 j_2 \dots j_n} \quad (7.12)$$

Examples include rank 0 (scalars with 1 component such as mass, charge, and temperature), rank 1 (vectors with 3 components such as position, momentum, and electric field), and rank 2 ( $3 \times 3$  matrices with 9 components such as the stress tensor, moment of inertia, and polarizability).

### 7.3.2 Rotations of Vector Components: A Concrete Example

From Chapter 6, we know that a rotation by angle  $\theta$  about the  $z$ -axis transforms a vector according to:

$$R_z(\theta) \stackrel{c}{=} \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (7.13)$$

Here we use the notation introduced in Chapter 1: the symbol  $\stackrel{c}{=}$  with a dot over the equals sign denotes a matrix representation, while the label underneath indicates which basis we're using. The label  $c$  denotes the Cartesian basis  $(x, y, z)$ . This is a  $3 \times 3$  matrix acting on the three components of a vector. The vector components mix under rotation, but they mix only among themselves - a rotated vector is still a vector.

### 7.3.3 Connection to Spherical Harmonics

In Chapter 6, we encountered the spherical harmonics  $Y_{\ell m}(\theta, \phi)$  as eigenfunctions of  $\hat{L}^2$  and  $\hat{L}_z$  for orbital angular momentum. The simplest spherical harmonics are the  $\ell = 1$  functions, which are directly related to the Cartesian coordinates.

For  $\ell = 1$ , the three spherical harmonics  $Y_{1,m}$  with  $m = -1, 0, +1$  can be expressed in terms of the Cartesian unit vector components  $x/r$ ,  $y/r$ , and  $z/r$ :

$$Y_{1,0}(\theta, \phi) \propto \frac{z}{r} = \cos \theta \quad (7.14)$$

$$Y_{1,\pm 1}(\theta, \phi) \propto \frac{x \pm iy}{r} = \sin \theta e^{\pm i\phi} \quad (7.15)$$

The  $x$ ,  $y$ , and  $z$  coordinates (divided by  $r$  to make them dimensionless) transform as a rank-1 Cartesian tensor - that is, as a vector. But these same three components can be reorganized into the spherical basis  $(Y_{1,+1}, Y_{1,0}, Y_{1,-1})$ , which forms an irreducible representation of the rotation group with  $\ell = 1$ .

This connection is fundamental: the spherical harmonics  $Y_{\ell m}$  provide the spherical tensor basis for objects with angular momentum  $\ell$ . The Cartesian components  $(x, y, z)$  and spherical components  $(Y_{1,+1}, Y_{1,0}, Y_{1,-1})$  describe the same vector space, just in different bases.

### 7.3.4 Tensor Products Under Rotations

Now consider the tensor product of two vectors  $\vec{A}$  and  $\vec{B}$ , which could represent position vectors  $\vec{r}_1$  and  $\vec{r}_2$  of two particles. The 9 components  $T_{ij} = A_i B_j$  form a rank-2 Cartesian tensor. How does this object transform under rotations?

Under a rotation  $R$ , each vector transforms:

$$A'_i = R_{ij} A_j \quad (7.16)$$

$$B'_k = R_{k\ell} B_\ell \quad (7.17)$$

The tensor product components therefore transform as:

$$T'_{ij} = A'_i B'_j = R_{im} A_m R_{jn} B_n = R_{im} R_{jn} T_{mn} \quad (7.18)$$

We can rewrite this as a matrix equation by treating the 9 components as a vector. Label the components as  $(T_{xx}, T_{xy}, T_{xz}, T_{yx}, T_{yy}, T_{yz}, T_{zx}, T_{zy}, T_{zz})$ . The transformation becomes:

$$\vec{T}' \stackrel{c}{=} (R \otimes R) \vec{T} \quad (7.19)$$

where  $R \otimes R$  is the tensor product of the  $3 \times 3$  rotation matrix with itself, giving a  $9 \times 9$  matrix. The label  $c$  again indicates we're working in the Cartesian basis.

### 7.3.5 The $9 \times 9$ Rotation Matrix

For a rotation about the  $z$ -axis, the  $9 \times 9$  matrix  $R_z(\theta) \otimes R_z(\theta)$  explicitly shows how the 9 tensor components mix. This matrix cannot be diagonalized in general - rotations don't have 9 independent eigenvectors with eigenvalue 1.

However, the matrix can be block diagonalized. The key observation is that certain combinations of the 9 components transform only among themselves, forming invariant subspaces. The  $9 \times 9$  matrix breaks into blocks:

$$R_z(\theta) \otimes R_z(\theta) \stackrel{s}{=} \begin{pmatrix} [5 \times 5] & 0 & 0 \\ 0 & [3 \times 3] & 0 \\ 0 & 0 & [1 \times 1] \end{pmatrix} \quad (7.20)$$

where the label  $s$  now denotes the spherical tensor basis. This change of basis transforms the matrix representation from the Cartesian form (label  $c$ ) to the spherical form (label  $s$ ), which reveals the block diagonal structure.

These blocks correspond to:

$$5 \times 5 \text{ block: Symmetric traceless combinations } (\ell = 2) \quad (7.21)$$

$$3 \times 3 \text{ block: Antisymmetric combinations } (\ell = 1) \quad (7.22)$$

$$1 \times 1 \text{ block: Trace } (\ell = 0) \quad (7.23)$$

This block structure is not an accident. It reflects the decomposition  $1 \otimes 1 = 2 \oplus 1 \oplus 0$  from angular momentum addition.

### 7.3.6 Irreducible Representations and Block Diagonalization

A representation of the rotation group is called **irreducible** if it cannot be further decomposed into smaller blocks. The  $3 \times 3$  vector representation is irreducible - you cannot find a change of basis that makes rotation matrices block diagonal for vectors.

The  $9 \times 9$  tensor product representation is **reducible**. By choosing the right basis (symmetric traceless, antisymmetric, and trace combinations), we decompose it into three irreducible blocks. Each block transforms independently under rotations. The 5-component symmetric traceless tensor transforms by a  $5 \times 5$  matrix (the  $\ell = 2$  representation), the 3-component antisymmetric tensor transforms by a  $3 \times 3$  matrix (the  $\ell = 1$  representation), and the 1-component trace transforms by a  $1 \times 1$  matrix, meaning it's invariant (the  $\ell = 0$  representation).

This is the essence of irreducible decomposition: finding the natural building blocks that don't mix under rotations.

### 7.3.7 Example: Product of Position Vectors

Let's be concrete. Consider two position vectors  $\vec{r}_1 = (x_1, y_1, z_1)$  and  $\vec{r}_2 = (x_2, y_2, z_2)$ . Their tensor product gives 9 terms:

$$x_1x_2, \quad x_1y_2, \quad x_1z_2, \quad y_1x_2, \quad y_1y_2, \quad y_1z_2, \quad z_1x_2, \quad z_1y_2, \quad z_1z_2 \quad (7.24)$$

How do these 9 objects decompose under rotations?

**The  $\ell = 0$  piece: Rotationally invariant (1 component)**

The scalar product is rotationally invariant:

$$\vec{r}_1 \cdot \vec{r}_2 = x_1x_2 + y_1y_2 + z_1z_2 \quad (7.25)$$

Under any rotation, this combination remains unchanged. This is the unique  $\ell = 0$  component.

**The  $\ell = 1$  piece: Transforms like a vector (3 components)**

The antisymmetric combinations give a pseudovector (the cross product):

$$(\vec{r}_1 \times \vec{r}_2)_x = y_1 z_2 - z_1 y_2 \quad (7.26)$$

$$(\vec{r}_1 \times \vec{r}_2)_y = z_1 x_2 - x_1 z_2 \quad (7.27)$$

$$(\vec{r}_1 \times \vec{r}_2)_z = x_1 y_2 - y_1 x_2 \quad (7.28)$$

These three components transform like a vector under rotations, forming the  $\ell = 1$  representation.

**The  $\ell = 2$  piece: Quadrupole tensor (5 components)**

The remaining 5 independent components form a symmetric traceless tensor. Examples include:

$$x_1 x_2 - \frac{1}{3}(\vec{r}_1 \cdot \vec{r}_2) \quad (7.29)$$

$$y_1 y_2 - \frac{1}{3}(\vec{r}_1 \cdot \vec{r}_2) \quad (7.30)$$

$$\frac{1}{2}(x_1 y_2 + y_1 x_2) \quad (7.31)$$

$$\frac{1}{2}(x_1 z_2 + z_1 x_2) \quad (7.32)$$

$$\frac{1}{2}(y_1 z_2 + z_1 y_2) \quad (7.33)$$

These transform according to the  $\ell = 2$  representation - the same symmetry as  $d$ -orbitals in atomic physics.

### Reading the Decomposition from Spherical Harmonics

The decomposition of tensor products can be understood directly from the Cartesian expressions of spherical harmonics. The spherical harmonics  $Y_{\ell m}(\theta, \phi)$  for low  $\ell$  have simple Cartesian forms that reveal how products of coordinates naturally organize into irreducible representations.

For a single position vector  $\vec{r} = (x, y, z)$ , the  $\ell = 1$  spherical harmonics in Cartesian form are:

$$Y_{1,0} \propto \frac{z}{r} \quad (7.34)$$

$$Y_{1,+1} \propto -\frac{x + iy}{r} \quad (7.35)$$

$$Y_{1,-1} \propto \frac{x - iy}{r} \quad (7.36)$$

For the product of two position vectors  $\vec{r}_1$  and  $\vec{r}_2$ , the nine terms  $x_1 x_2$ ,  $x_1 y_2$ , etc., reorganize into combinations that transform like spherical harmonics with  $\ell = 0, 1, 2$ . The table below shows this correspondence explicitly:

| $\ell$     | Spherical Harmonic Structure                                                                                                                                    | Tensor Product Combination                                                                                                                                                                                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\ell = 0$ | $Y_{0,0} \propto 1$ (constant)                                                                                                                                  | $x_1x_2 + y_1y_2 + z_1z_2$<br>(rotationally invariant)                                                                                                                                                                           |
| $\ell = 1$ | $Y_{1,0} \propto z$<br>$Y_{1,+1} \propto -(x + iy)$<br>$Y_{1,-1} \propto (x - iy)$                                                                              | $x_1y_2 - y_1x_2$<br>$y_1z_2 - z_1y_2 - i(z_1x_2 - x_1z_2)$<br>$y_1z_2 - z_1y_2 + i(z_1x_2 - x_1z_2)$                                                                                                                            |
| $\ell = 2$ | $Y_{2,0} \propto 3z^2 - r^2$<br>$Y_{2,+1} \propto -(x + iy)z$<br>$Y_{2,-1} \propto (x - iy)z$<br>$Y_{2,+2} \propto (x + iy)^2$<br>$Y_{2,-2} \propto (x - iy)^2$ | $z_1z_2 - \frac{1}{3}(x_1x_2 + y_1y_2 + z_1z_2)$<br>$(x_1z_2 + z_1x_2) + i(y_1z_2 + z_1y_2)$<br>$(x_1z_2 + z_1x_2) - i(y_1z_2 + z_1y_2)$<br>$(x_1x_2 - y_1y_2) + i(x_1y_2 + y_1x_2)$<br>$(x_1x_2 - y_1y_2) - i(x_1y_2 + y_1x_2)$ |

The pattern is clear: products of coordinates organize into the same mathematical structure as spherical harmonics. For  $\ell = 0$ , we get the unique rotationally invariant combination (the dot product). For  $\ell = 1$ , the three components correspond to the cross product written in complex spherical basis. For  $\ell = 2$ , the five symmetric traceless combinations have the same transformation properties as the  $d$ -orbitals.

This correspondence is not accidental. When we form the tensor product  $\vec{r}_1 \otimes \vec{r}_2$ , we are combining two  $\ell = 1$  objects. The decomposition  $1 \otimes 1 = 0 \oplus 1 \oplus 2$  tells us that the resulting object can be reorganized into pieces that transform like  $\ell = 0$ ,  $\ell = 1$ , and  $\ell = 2$  spherical harmonics. The Clebsch-Gordan coefficients provide the precise linear combinations needed to construct these irreducible components from the original Cartesian products.

This decomposition  $3 \otimes 3 = 5 \oplus 3 \oplus 1$  corresponds precisely to the angular momentum addition formula  $1 \otimes 1 = 2 \oplus 1 \oplus 0$ . The Cartesian tensor product decomposes into spherical tensors of definite angular momentum.

## 7.4 The Tensor Product Structure

With this group-theoretic and geometric foundation in place, we now turn to the detailed structure of composite angular momentum systems in quantum mechanics.

### 7.4.1 Hilbert Space Dimensions

Consider two angular momentum systems with quantum numbers  $j_1$  and  $j_2$ . Each lives in its own Hilbert space:

$$\mathcal{H}_1 : \dim = 2j_1 + 1 \quad (\text{states with } m_1 = -j_1, \dots, +j_1) \quad (7.37)$$

$$\mathcal{H}_2 : \dim = 2j_2 + 1 \quad (\text{states with } m_2 = -j_2, \dots, +j_2) \quad (7.38)$$

The composite system lives in the tensor product space:

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \quad (7.39)$$

with dimension:

$$\dim(\mathcal{H}) = (2j_1 + 1) \times (2j_2 + 1) \quad (7.40)$$

**Example:** Two spin-1/2 particles have  $j_1 = j_2 = 1/2$ , so each has a 2-dimensional Hilbert space. The combined system has dimension  $2 \times 2 = 4$ , with basis states  $|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$ .

### 7.4.2 Decomposition into Total Angular Momentum Subspaces

The  $(2j_1 + 1)(2j_2 + 1)$ -dimensional tensor product space can be decomposed into subspaces of definite total angular momentum, as stated in Eq. (7.3). We can verify that the dimensions match by summing over all allowed  $j$  values:

$$\sum_{j=|j_1-j_2|}^{j_1+j_2} (2j+1) = (2j_1+1)(2j_2+1) \quad (7.41)$$

Comparing Eq. (7.41) with Eq. (7.40), we see both expressions give the same total dimension, confirming that the direct sum decomposition in Eq. (7.3) accounts for all states in the tensor product space.

## 7.5 The Uncoupled and Coupled Bases

### 7.5.1 The Uncoupled Basis

Consider a system with two angular momenta  $\vec{J}_1$  and  $\vec{J}_2$ . These could represent physically distinct scenarios. They might describe the orbital angular momentum  $\vec{L}$  and spin angular momentum  $\vec{S}$  of a single electron, where the orbital motion and intrinsic spin represent independent degrees of freedom that can be coupled by spin-orbit interaction. Alternatively, they could represent the angular momenta of two spatially separated particles, for instance, the spins of two electrons in different atoms, or the orbital angular momenta of two nucleons in a nucleus. More generally,  $\vec{J}_1$  and  $\vec{J}_2$  can be any two angular momentum operators that commute with each other, meaning they act on different parts of the Hilbert space or represent different physical degrees of freedom.

The key property is that operators acting on different subsystems commute:

$$[\hat{J}_{1i}, \hat{J}_{2j}] = 0 \quad \text{for all } i, j \quad (7.42)$$

Within each subsystem, the standard angular momentum commutation relations hold:

$$[\hat{J}_{1i}, \hat{J}_{1j}] = i\hbar\epsilon_{ijk}\hat{J}_{1k} \quad (7.43)$$

$$[\hat{J}_{2i}, \hat{J}_{2j}] = i\hbar\epsilon_{ijk}\hat{J}_{2k} \quad (7.44)$$

Since the operators  $\hat{J}_1^2, \hat{J}_{1z}, \hat{J}_2^2$ , and  $\hat{J}_{2z}$  all mutually commute, they form a complete set of commuting observables. We can construct simultaneous eigenstates as tensor products:

$$|j_1, m_1; j_2, m_2\rangle = |j_1, m_1\rangle \otimes |j_2, m_2\rangle \quad (7.45)$$

Often the tensor product symbol  $\otimes$  is suppressed, but it's important to remember that these are product states from two different Hilbert spaces.

These states satisfy:

$$\hat{J}_1^2 |j_1, m_1; j_2, m_2\rangle = \hbar^2 j_1(j_1 + 1) |j_1, m_1; j_2, m_2\rangle \quad (7.46)$$

$$\hat{J}_{1z} |j_1, m_1; j_2, m_2\rangle = \hbar m_1 |j_1, m_1; j_2, m_2\rangle \quad (7.47)$$

$$\hat{J}_2^2 |j_1, m_1; j_2, m_2\rangle = \hbar^2 j_2(j_2 + 1) |j_1, m_1; j_2, m_2\rangle \quad (7.48)$$

$$\hat{J}_{2z} |j_1, m_1; j_2, m_2\rangle = \hbar m_2 |j_1, m_1; j_2, m_2\rangle \quad (7.49)$$

The uncoupled basis provides  $(2j_1 + 1)(2j_2 + 1)$  orthonormal product states that span the full tensor product space  $\mathcal{H}_1 \otimes \mathcal{H}_2$ . This basis is particularly natural when the two angular momenta are nearly independent, such as in weak spin-orbit coupling or widely separated particles.

### 7.5.2 The Total Angular Momentum

The total angular momentum is defined as the vector sum:

$$\vec{J} = \vec{J}_1 + \vec{J}_2 \quad (7.50)$$

or in components:

$$\hat{J}_i = \hat{J}_{1i} + \hat{J}_{2i} \quad \text{for } i = x, y, z \quad (7.51)$$

#### Physical Insight

The total angular momentum  $\vec{J}$  is itself an angular momentum operator, satisfying the standard commutation relations  $[\hat{J}_i, \hat{J}_j] = i\hbar\epsilon_{ijk}\hat{J}_k$ . This can be verified using the individual commutation relations of  $\vec{J}_1$  and  $\vec{J}_2$  and the fact that operators from different subsystems commute.

### 7.5.3 The Coupled Basis

An alternative description uses the total angular momentum. We can show that:

$$[\hat{J}^2, \hat{J}_1^2] = 0 \quad (7.52)$$

$$[\hat{J}^2, \hat{J}_2^2] = 0 \quad (7.53)$$

$$[\hat{J}^2, \hat{J}_z] = 0 \quad (7.54)$$

Therefore,  $\hat{J}_1^2$ ,  $\hat{J}_2^2$ ,  $\hat{J}^2$ , and  $\hat{J}_z$  form another complete set of commuting observables. The corresponding eigenstates are:

$$|j_1, j_2; j, m\rangle \quad (7.55)$$

These states live in the **same** tensor product space  $\mathcal{H}_1 \otimes \mathcal{H}_2$  as the uncoupled basis, but organized differently. They satisfy:

$$\hat{J}_1^2 |j_1, j_2; j, m\rangle = \hbar^2 j_1(j_1 + 1) |j_1, j_2; j, m\rangle \quad (7.56)$$

$$\hat{J}_2^2 |j_1, j_2; j, m\rangle = \hbar^2 j_2(j_2 + 1) |j_1, j_2; j, m\rangle \quad (7.57)$$

$$\hat{J}^2 |j_1, j_2; j, m\rangle = \hbar^2 j(j + 1) |j_1, j_2; j, m\rangle \quad (7.58)$$

$$\hat{J}_z |j_1, j_2; j, m\rangle = \hbar m |j_1, j_2; j, m\rangle \quad (7.59)$$

The coupled basis groups the  $(2j_1 + 1)(2j_2 + 1)$  states according to their total angular momentum  $j$ . Each value of  $j$  gives a  $(2j + 1)$ -dimensional subspace of states with  $m = -j, \dots, +j$ . States within each  $j$  subspace transform among themselves under rotations but never mix with other  $j$  values.

### 7.5.4 Constraints on Total Angular Momentum

Given fixed  $j_1$  and  $j_2$ , what values can the total angular momentum  $j$  take? Since  $\hat{J}_z = \hat{J}_{1z} + \hat{J}_{2z}$ :

$$m = m_1 + m_2 \quad (7.60)$$

The maximum value of  $m$  occurs when both components are aligned:  $m_{\max} = j_1 + j_2$ . Since  $m$  ranges from  $-j$  to  $+j$ , we must have:

$$j_{\max} = j_1 + j_2 \quad (7.61)$$

The minimum value arises when the angular momenta are anti-aligned. The allowed values of  $j$  follow the triangle inequality:

$$|j_1 - j_2| \leq j \leq j_1 + j_2 \quad (7.62)$$

with  $j$  taking integer steps.

The complete list of allowed values is:

$$j \in \{|j_1 - j_2|, |j_1 - j_2| + 1, |j_1 - j_2| + 2, \dots, j_1 + j_2 - 1, j_1 + j_2\} \quad (7.63)$$

This gives a total of:

$$N_j = 2 \min(j_1, j_2) + 1 \quad (7.64)$$

different values of  $j$ .

## 7.6 Clebsch-Gordan Coefficients

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### 7.6.1 Definition and Notation

The uncoupled and coupled bases are related by a unitary transformation. We can expand coupled states in terms of uncoupled states:

$$|j_1, j_2; j, m\rangle = \sum_{m_1, m_2} \langle j_1, m_1; j_2, m_2 | j_1, j_2; j, m \rangle |j_1, m_1; j_2, m_2\rangle \quad (7.65)$$

The expansion coefficients are called Clebsch-Gordan (CG) coefficients:

$$\langle j_1, m_1; j_2, m_2 | j_1, j_2; j, m \rangle \equiv \langle j_1, m_1; j_2, m_2 | j, m \rangle \quad (7.66)$$

An alternative notation uses the symbol:

$$C_{j_1, m_1; j_2, m_2}^{j, m} = \langle j_1, m_1; j_2, m_2 | j, m \rangle \quad (7.67)$$

Yet another common notation is:

$$\langle j_1, m_1; j_2, m_2 | j, m \rangle = \begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & m \end{pmatrix} \quad (7.68)$$



### 7.6.2 Properties of Clebsch-Gordan Coefficients

The CG coefficients have several important properties:

**1. Conservation of  $m$ :**

$$\langle j_1, m_1; j_2, m_2 | j, m \rangle = 0 \quad \text{unless } m = m_1 + m_2 \quad (7.69)$$

**2. Triangle inequality:**

$$\langle j_1, m_1; j_2, m_2 | j, m \rangle = 0 \quad \text{unless } |j_1 - j_2| \leq j \leq j_1 + j_2 \quad (7.70)$$

**3. Orthonormality:**

$$\sum_{m_1, m_2} \langle j_1, m_1; j_2, m_2 | j, m \rangle \langle j_1, m_1; j_2, m_2 | j', m' \rangle = \delta_{jj'} \delta_{mm'} \quad (7.71)$$

$$\sum_{j, m} \langle j_1, m_1; j_2, m_2 | j, m \rangle \langle j_1, m'_1; j_2, m'_2 | j, m \rangle = \delta_{m_1 m'_1} \delta_{m_2 m'_2} \quad (7.72)$$

**4. Symmetry relations:**

$$\langle j_1, m_1; j_2, m_2 | j, m \rangle = (-1)^{j_1+j_2-j} \langle j_2, m_2; j_1, m_1 | j, m \rangle \quad (7.73)$$

$$\langle j_1, m_1; j_2, m_2 | j, m \rangle = (-1)^{j_1-m_1} \sqrt{\frac{2j+1}{2j_2+1}} \langle j_1, -m_1; j, m | j_2, -m_2 \rangle \quad (7.74)$$

**5. Reality (phase convention):** With the Condon-Shortley phase convention, CG coefficients are real:

$$\langle j_1, m_1; j_2, m_2 | j, m \rangle \in \mathbb{R} \quad (7.75)$$

### 7.6.3 Computing Clebsch-Gordan Coefficients

#### The Ladder Operator Method

The most systematic approach uses angular momentum ladder operators. Start from the state with maximum  $m$ :

$$|j_1, j_1; j_2, j_2; j, j\rangle = |j_1, j_1; j_2, j_2\rangle \quad (7.76)$$

where  $j = j_1 + j_2$  and  $m = j$ . This gives:

$$\langle j_1, j_1; j_2, j_2 | j_1 + j_2, j_1 + j_2 \rangle = 1 \quad (7.77)$$

Apply the lowering operator  $\hat{J}_- = \hat{J}_{1-} + \hat{J}_{2-}$  repeatedly to generate states with smaller  $m$ . The action of ladder operators:

$$\hat{J}_\pm |j, m\rangle = \hbar \sqrt{j(j+1) - m(m \pm 1)} |j, m \pm 1\rangle \quad (7.78)$$

$$= \hbar \sqrt{(j \mp m)(j \pm m + 1)} |j, m \pm 1\rangle \quad (7.79)$$

This generates all CG coefficients for a given  $j$  systematically. States with different  $j$  values are found using orthogonality.

### 7.6.4 Reading Clebsch-Gordan Tables

In practice, CG coefficients are typically looked up in standard tables rather than computed from scratch. Understanding how to read these tables is an essential skill.

### Standard Table Format

CG tables are organized by the values of  $j_1$  and  $j_2$  being combined. For each pair  $(j_1, j_2)$ , the table shows the coefficients for all allowed combinations of  $m_1$ ,  $m_2$ , and resulting  $j$ ,  $m$  values.

A typical table entry looks like:

| $j_1 = 1/2, j_2 = 1/2$ |                |                      |                       |
|------------------------|----------------|----------------------|-----------------------|
| $m_1$                  | $m_2$          | $j = 1$              | $j = 0$               |
| $+\frac{1}{2}$         | $+\frac{1}{2}$ | 1                    |                       |
| $+\frac{1}{2}$         | $-\frac{1}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{\sqrt{2}}$  |
| $-\frac{1}{2}$         | $+\frac{1}{2}$ | $\frac{1}{\sqrt{2}}$ | $-\frac{1}{\sqrt{2}}$ |
| $-\frac{1}{2}$         | $-\frac{1}{2}$ | 1                    |                       |

To read this table: the entry in row  $(m_1, m_2)$  and column  $j$  gives the CG coefficient  $\langle j_1, m_1; j_2, m_2 | j, m \rangle$  where  $m = m_1 + m_2$ . Empty entries correspond to forbidden combinations (where  $m \neq m_1 + m_2$  or  $|m| > j$ ).

### Example: Using the Table

From the table above, we can read off the coupled states:

$$|j = 1, m = 1\rangle = |+\frac{1}{2}, +\frac{1}{2}\rangle \quad (7.80)$$

$$|j = 1, m = 0\rangle = \frac{1}{\sqrt{2}}|+\frac{1}{2}, -\frac{1}{2}\rangle + \frac{1}{\sqrt{2}}|-\frac{1}{2}, +\frac{1}{2}\rangle \quad (7.81)$$

$$|j = 0, m = 0\rangle = \frac{1}{\sqrt{2}}|+\frac{1}{2}, -\frac{1}{2}\rangle - \frac{1}{\sqrt{2}}|-\frac{1}{2}, +\frac{1}{2}\rangle \quad (7.82)$$

**Table for  $j_1 = 1, j_2 = 1/2$**

| $j_1 = 1, j_2 = 1/2$ |                |                      |                       |
|----------------------|----------------|----------------------|-----------------------|
| $m_1$                | $m_2$          | $j = 3/2$            | $j = 1/2$             |
| +1                   | $+\frac{1}{2}$ | 1                    |                       |
| +1                   | $-\frac{1}{2}$ | $\sqrt{\frac{1}{3}}$ | $\sqrt{\frac{2}{3}}$  |
| 0                    | $+\frac{1}{2}$ | $\sqrt{\frac{2}{3}}$ | $-\sqrt{\frac{1}{3}}$ |
| 0                    | $-\frac{1}{2}$ | $\sqrt{\frac{2}{3}}$ | $\sqrt{\frac{1}{3}}$  |
| -1                   | $+\frac{1}{2}$ | $\sqrt{\frac{1}{3}}$ | $-\sqrt{\frac{2}{3}}$ |
| -1                   | $-\frac{1}{2}$ | 1                    |                       |

This table is essential for spin-orbit coupling calculations. For example, to find  $|j = 3/2, m = 1/2\rangle$ :

$$|\frac{3}{2}, \frac{1}{2}\rangle = \sqrt{\frac{1}{3}}|1, -\frac{1}{2}\rangle + \sqrt{\frac{2}{3}}|0, +\frac{1}{2}\rangle \quad (7.83)$$

where we use shorthand  $|m_1, m_2\rangle$  for the uncoupled states.

### Square Root Convention

Many tables display entries as fractions under a square root sign. For example,  $\sqrt{2/3}$  might appear as simply  $2/3$  with the understanding that the square root is implied. Some tables show the sign separately, with the magnitude squared in the table. Always check the conventions used in a particular reference.

#### 7.6.5 Explicit Examples

##### Adding $j_1 = 1/2$ and $j_2 = 1/2$

For two spin-1/2 systems, the uncoupled basis has 4 states:

$$|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle \quad (7.84)$$

where we use shorthand  $|\uparrow\rangle = |1/2, 1/2\rangle$  and  $|\downarrow\rangle = |1/2, -1/2\rangle$ .

**Triplet states ( $j = 1$ ):**

$$|1, 1\rangle = |\uparrow\uparrow\rangle \quad (7.85)$$

$$|1, 0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \quad (7.86)$$

$$|1, -1\rangle = |\downarrow\downarrow\rangle \quad (7.87)$$

**Singlet state ( $j = 0$ ):**

$$|0, 0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \quad (7.88)$$

The non-zero CG coefficients are:

$$\langle 1/2, 1/2; 1/2, 1/2 | 1, 1 \rangle = 1 \quad (7.89)$$

$$\langle 1/2, 1/2; 1/2, -1/2 | 1, 0 \rangle = \frac{1}{\sqrt{2}} \quad (7.90)$$

$$\langle 1/2, -1/2; 1/2, 1/2 | 1, 0 \rangle = \frac{1}{\sqrt{2}} \quad (7.91)$$

$$\langle 1/2, -1/2; 1/2, -1/2 | 1, -1 \rangle = 1 \quad (7.92)$$

$$\langle 1/2, 1/2; 1/2, -1/2 | 0, 0 \rangle = \frac{1}{\sqrt{2}} \quad (7.93)$$

$$\langle 1/2, -1/2; 1/2, 1/2 | 0, 0 \rangle = -\frac{1}{\sqrt{2}} \quad (7.94)$$

##### Adding $j_1 = 1$ and $j_2 = 1/2$

This case is relevant for adding orbital angular momentum  $\ell = 1$  to spin  $s = 1/2$ , giving  $j \in \{1/2, 3/2\}$ .

For  $j = 3/2, m = 3/2$ :

$$|3/2, 3/2\rangle = |1, 1; 1/2, 1/2\rangle \quad (7.95)$$

For  $j = 3/2, m = 1/2$ :

$$|3/2, 1/2\rangle = \sqrt{\frac{2}{3}} |1, 1; 1/2, -1/2\rangle + \sqrt{\frac{1}{3}} |1, 0; 1/2, 1/2\rangle \quad (7.96)$$

For  $j = 1/2$ ,  $m = 1/2$  (orthogonal to above):

$$|1/2, 1/2\rangle = \sqrt{\frac{1}{3}}|1, 1; 1/2, -1/2\rangle - \sqrt{\frac{2}{3}}|1, 0; 1/2, 1/2\rangle \quad (7.97)$$

The complete table of CG coefficients for this case appears above.

### Adding Two Integer Angular Momenta: $j_1 = 1$ and $j_2 = 1$

For two  $\ell = 1$  systems, we get  $j \in \{0, 1, 2\}$ , giving 9 states total.

For  $j = 2$  (quintet):

$$|2, 2\rangle = |1, 1; 1, 1\rangle \quad (7.98)$$

$$|2, 1\rangle = \frac{1}{\sqrt{2}}(|1, 1; 1, 0\rangle + |1, 0; 1, 1\rangle) \quad (7.99)$$

$$|2, 0\rangle = \frac{1}{\sqrt{6}}(|1, 1; 1, -1\rangle + 2|1, 0; 1, 0\rangle + |1, -1; 1, 1\rangle) \quad (7.100)$$

For  $j = 1$  (triplet):

$$|1, 0\rangle = \frac{1}{\sqrt{2}}(|1, 1; 1, -1\rangle - |1, -1; 1, 1\rangle) \quad (7.101)$$

For  $j = 0$  (singlet):

$$|0, 0\rangle = \frac{1}{\sqrt{3}}(|1, 1; 1, -1\rangle - |1, 0; 1, 0\rangle + |1, -1; 1, 1\rangle) \quad (7.102)$$

## 7.7 Physical Applications

### 7.7.1 Spin-Orbit Coupling

For a single electron with orbital angular momentum  $\vec{L}$  and spin  $\vec{S}$ , the total angular momentum is:

$$\vec{J} = \vec{L} + \vec{S} \quad (7.103)$$

With  $s = 1/2$ , the possible values of  $j$  are:

$$j = \ell \pm \frac{1}{2} \quad (7.104)$$

This seemingly simple addition has profound consequences for atomic spectra. The spin-orbit interaction energy depends on  $\vec{L} \cdot \vec{S}$ , which as we showed earlier can be written as  $(\hat{J}^2 - \hat{L}^2 - \hat{S}^2)/2$ . This means states with different  $j$  have different energies even when they share the same  $n$  and  $\ell$ . The result is fine structure splitting: what would be a single energy level in the absence of spin splits into two levels separated by the spin-orbit coupling energy.

Consider the hydrogen  $n = 2$ ,  $\ell = 1$  level as a concrete example. Without spin-orbit coupling, this is the  $2p$  level with six degenerate states (three from  $m_\ell = -1, 0, 1$  times two from  $m_s = \pm 1/2$ ). With spin-orbit coupling, these six states reorganize into two multiplets. The  $j = 3/2$  level contains four states with magnetic quantum numbers  $m = -3/2, -1/2, 1/2, 3/2$ . The  $j = 1/2$  level contains two states with  $m = -1/2, 1/2$ . These two levels have slightly different energies, producing the famous sodium D-line doublet at 589 nm, which corresponds to transitions from the  $3s$  level to the  $3p_{1/2}$  and  $3p_{3/2}$  levels.

We will study spin-orbit coupling in detail when we analyze the hydrogen atom in the next chapter.

### 7.7.2 Nuclear Hyperfine Structure

Nuclear spin  $\vec{I}$  couples to electronic angular momentum  $\vec{J}$ :

$$\vec{F} = \vec{J} + \vec{I} \quad (7.105)$$

This produces hyperfine splitting in atomic spectra, used in atomic clocks and precision measurements.

### 7.7.3 Multi-Electron Atoms

In atoms with multiple electrons, the situation becomes more complex because we must couple several angular momenta. The appropriate coupling scheme depends on the relative strength of different interactions. In light atoms, LS coupling (Russell-Saunders coupling) applies when electron-electron repulsion dominates over spin-orbit effects. In heavy atoms, jj coupling becomes more appropriate when spin-orbit coupling is strong.

We will return to multi-electron atoms after developing the theory of the hydrogen atom, which will provide the foundation for understanding atomic structure more generally.

## 7.8 Wigner 3j and 6j Symbols

When working with angular momentum in more advanced contexts, alternative notations for CG coefficients prove useful. These symbols have superior symmetry properties and appear naturally in certain physical calculations. We also introduce the general formalism of spherical tensors and their transformation properties under rotations.

### 7.8.1 The 3j Symbols

The Wigner 3j symbols are related to CG coefficients by:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = \frac{(-1)^{j_1-j_2-m_3}}{\sqrt{2j_3+1}} \langle j_1, m_1; j_2, m_2 | j_3, -m_3 \rangle \quad (7.106)$$

Note the sign of  $m_3$  on the right-hand side: it appears as  $-m_3$ , not  $+m_3$ . This ensures the 3j symbol has enhanced symmetry.

### Symmetry Properties

The 3j symbols have remarkable symmetry properties that make them preferable for theoretical work:

**1. Even permutations of columns:** Any even permutation of the three columns leaves the 3j symbol unchanged:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = \begin{pmatrix} j_2 & j_3 & j_1 \\ m_2 & m_3 & m_1 \end{pmatrix} = \begin{pmatrix} j_3 & j_1 & j_2 \\ m_3 & m_1 & m_2 \end{pmatrix} \quad (7.107)$$

**2. Odd permutations introduce a phase:** Odd permutations multiply the 3j symbol by  $(-1)^{j_1+j_2+j_3}$ :

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_2 & j_1 & j_3 \\ m_2 & m_1 & m_3 \end{pmatrix} \quad (7.108)$$

### 3. Sign reversal of all $m$ values:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{pmatrix} \quad (7.109)$$

**4. Selection rules:** The  $3j$  symbol vanishes unless three conditions are satisfied:  $m_1 + m_2 + m_3 = 0$  (which differs from CG coefficients where  $m = m_1 + m_2$ ), the triangle inequality  $|j_1 - j_2| \leq j_3 \leq j_1 + j_2$  holds, and  $|m_i| \leq j_i$  for all  $i$ .

### Physical Interpretation

The  $3j$  symbols describe the coupling of three angular momenta to give zero total angular momentum:

$$\vec{J}_1 + \vec{J}_2 + \vec{J}_3 = 0 \quad (7.110)$$

This symmetry in treating all three angular momenta equally (rather than singling out two to be added) makes  $3j$  symbols natural for describing processes with three-fold symmetry, such as matrix elements of operators between angular momentum states, radiation transition probabilities in atoms, nuclear beta decay angular correlations, and scattering processes with three participating particles.

### Relation to Physical Matrix Elements

Matrix elements of spherical tensor operators  $T_q^{(k)}$  between angular momentum states have the form:

$$\langle j', m' | T_q^{(k)} | j, m \rangle \propto \begin{pmatrix} j & k & j' \\ m & q & -m' \end{pmatrix} \quad (7.111)$$

The  $3j$  symbol automatically encodes the selection rules for such transitions:  $\Delta m = m' - m = q$  and  $|j - j'| \leq k \leq j + j'$ . This makes  $3j$  symbols essential for calculating electromagnetic transition rates and selection rules in atomic physics.

### 7.8.2 The $6j$ Symbols

When coupling three angular momenta, different coupling orders can be used. Consider  $\vec{J}_1$ ,  $\vec{J}_2$ , and  $\vec{J}_3$ . We could form  $(\vec{J}_1 + \vec{J}_2) + \vec{J}_3$  with intermediate  $\vec{J}_{12} = \vec{J}_1 + \vec{J}_2$ , or alternatively  $\vec{J}_1 + (\vec{J}_2 + \vec{J}_3)$  with intermediate  $\vec{J}_{23} = \vec{J}_2 + \vec{J}_3$ .

The  $6j$  symbols relate these different coupling schemes:

$$|(j_1 j_2) j_{12}; j_3; j m\rangle = \sum_{j_{23}} \sqrt{(2j_{12} + 1)(2j_{23} + 1)} \begin{Bmatrix} j_1 & j_2 & j_{12} \\ j_3 & j & j_{23} \end{Bmatrix} |j_1; (j_2 j_3) j_{23}; j m\rangle \quad (7.112)$$

These arise in recoupling problems and are essential for complex atomic structure calculations involving more than two angular momenta. Examples include LS coupling versus jj coupling in multi-electron atoms and multi-photon processes in quantum optics.

### 7.8.3 Spherical Tensors: The General Picture

Spherical tensors provide a powerful framework for understanding how composite objects transform under rotations, generalizing the concrete examples we saw in the Cartesian tensor section.

A spherical tensor of rank  $k$  has  $2k + 1$  components  $T_q^{(k)}$  where  $q = -k, -k + 1, \dots, k - 1, k$ . These components transform among themselves according to:

$$T_q'^{(k)} \stackrel{s}{=} \sum_{q'=-k}^k D_{qq'}^{(k)}(R) T_{q'}^{(k)} \quad (7.113)$$

where  $D_{qq'}^{(k)}(R)$  is a  $(2k+1) \times (2k+1)$  matrix called the Wigner D-matrix for the  $\ell = k$  representation. The label  $s$  indicates this transformation law is written in the spherical basis. The D-matrices are the matrix representations of rotations in the basis of spherical harmonics or, more generally, in the basis that diagonalizes  $\hat{J}^2$  and  $\hat{J}_z$ .

The key advantage is that spherical tensors transform according to irreducible representations. Each rank  $k$  forms a separate, unmixed representation. Components of different  $k$  never mix under rotations, just as we saw in the block diagonal structure of tensor products.

### 7.8.4 Wigner D-Matrices: Explicit Examples

The Wigner D-matrices provide the explicit rotation matrices for each angular momentum representation. For a general rotation specified by Euler angles  $(\alpha, \beta, \gamma)$ , the D-matrix elements are:

$$D_{m'm}^{(j)}(\alpha, \beta, \gamma) = e^{-im'\alpha} d_{m'm}^{(j)}(\beta) e^{-im\gamma} \quad (7.114)$$

where  $d_{m'm}^{(j)}(\beta)$  are the Wigner small-d matrices that depend only on the rotation angle  $\beta$  about the  $y$ -axis.

For rotations about the  $z$ -axis by angle  $\theta$ , the D-matrices simplify dramatically:

$$D_{m'm}^{(j)}(R_z(\theta)) = \delta_{m'm} e^{-im\theta} \quad (7.115)$$

The matrix is diagonal, with each diagonal element a phase factor.

#### Example: $j = 1/2$ (Spin-1/2)

For spin-1/2, the  $2 \times 2$  D-matrix for rotation about the  $z$ -axis is:

$$D^{(1/2)}(R_z(\theta)) \stackrel{s}{=} \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{+i\theta/2} \end{pmatrix} \quad (7.116)$$

For rotation about the  $y$ -axis by angle  $\beta$ , the small-d matrix is:

$$d^{(1/2)}(\beta) = \begin{pmatrix} \cos(\beta/2) & -\sin(\beta/2) \\ \sin(\beta/2) & \cos(\beta/2) \end{pmatrix} \quad (7.117)$$

This is precisely the matrix representation of rotations acting on spinors.

**Example:  $j = 1$  (Vectors)**

For  $j = 1$ , the  $3 \times 3$  D-matrix for rotation about the  $z$ -axis is:

$$D^{(1)}(R_z(\theta)) \doteq_s \begin{pmatrix} e^{-i\theta} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{+i\theta} \end{pmatrix} \quad (7.118)$$

Note that this is written in the spherical basis with components labeled by  $m = +1, 0, -1$  (in that order). This differs from the Cartesian rotation matrix we saw in Eq. (7.13).

For rotation about the  $y$ -axis, the small-d matrix elements are:

$$d^{(1)}(\beta) = \begin{pmatrix} \frac{1+\cos\beta}{2} & -\frac{\sin\beta}{\sqrt{2}} & \frac{1-\cos\beta}{2} \\ \frac{\sin\beta}{\sqrt{2}} & \cos\beta & -\frac{\sin\beta}{\sqrt{2}} \\ \frac{1-\cos\beta}{2} & \frac{\sin\beta}{\sqrt{2}} & \frac{1+\cos\beta}{2} \end{pmatrix} \quad (7.119)$$

**Example:  $j = 2$  (Quadrupole)**

For  $j = 2$ , the  $5 \times 5$  D-matrix for rotation about the  $z$ -axis is:

$$D^{(2)}(R_z(\theta)) \doteq_s \begin{pmatrix} e^{-2i\theta} & 0 & 0 & 0 & 0 \\ 0 & e^{-i\theta} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & e^{+i\theta} & 0 \\ 0 & 0 & 0 & 0 & e^{+2i\theta} \end{pmatrix} \quad (7.120)$$

The components are ordered by  $m = +2, +1, 0, -1, -2$ .

**7.8.5 Using Wigner D-Matrix Tables**

Wigner D-matrices are tabulated in the same references that provide Clebsch-Gordan coefficients. Standard references include Edmonds (*Angular Momentum in Quantum Mechanics*, Princeton, 1957), Varshalovich, Moskalev, and Khersonskii (*Quantum Theory of Angular Momentum*, World Scientific, 1988), Rose (*Elementary Theory of Angular Momentum*, Wiley, 1957), and the online NIST Digital Library of Mathematical Functions (<https://dlmf.nist.gov/>).

**Reading D-Matrix Tables**

D-matrix tables typically provide the small-d matrix elements  $d_{m'm}^{(j)}(\beta)$  for rotation about the  $y$ -axis. A typical table entry looks like:

| $d^{(2)}(\beta)$ matrix elements |                                            |                                      |                                            |         |         |
|----------------------------------|--------------------------------------------|--------------------------------------|--------------------------------------------|---------|---------|
| $m' \backslash m$                | +2                                         | +1                                   | 0                                          | -1      | -2      |
| +2                               | $\cos^4(\beta/2)$                          | $-2 \cos^3(\beta/2) \sin(\beta/2)$   | $\sqrt{6} \cos^2(\beta/2) \sin^2(\beta/2)$ | $\dots$ | $\dots$ |
| +1                               | $2 \cos^3(\beta/2) \sin(\beta/2)$          | $\cos(\beta)(2 \cos^2(\beta/2) - 1)$ | $\dots$                                    | $\dots$ | $\dots$ |
| 0                                | $\sqrt{6} \cos^2(\beta/2) \sin^2(\beta/2)$ | $\dots$                              | $\frac{3 \cos(2\beta)+1}{4}$               | $\dots$ | $\dots$ |

To read this table: the entry in row  $m'$  and column  $m$  gives  $d_{m'm}^{(j)}(\beta)$ . Due to symmetry properties, many tables only show half the entries.



### Symmetry Properties

The d-matrices satisfy several useful symmetries:

$$d_{m'm}^{(j)}(\beta) = (-1)^{m'-m} d_{mm'}^{(j)}(\beta) \quad (\text{transpose symmetry}) \quad (7.121)$$

$$d_{m'm}^{(j)}(\beta) = (-1)^{m'-m} d_{-m,-m'}^{(j)}(\beta) \quad (\text{reflection symmetry}) \quad (7.122)$$

$$d_{m'm}^{(j)}(0) = \delta_{m'm} \quad (\text{identity at } \beta = 0) \quad (7.123)$$

$$d_{m'm}^{(j)}(\pi) = (-1)^{j+m} \delta_{m',-m} \quad (\text{at } \beta = \pi) \quad (7.124)$$

These properties allow you to compute missing table entries from the ones provided.

### Practical Usage

When calculating how a spherical tensor component transforms under a rotation:

- (a) Identify the rank  $j$  of your tensor
- (b) Determine the Euler angles  $(\alpha, \beta, \gamma)$  for your rotation
- (c) Look up the  $d_{m'm}^{(j)}(\beta)$  elements from tables
- (d) Apply the full D-matrix formula:  $D_{m'm}^{(j)}(\alpha, \beta, \gamma) = e^{-im'\alpha} d_{m'm}^{(j)}(\beta) e^{-im\gamma}$
- (e) Sum over  $m$  according to Eq. (7.113)

For simple rotations about coordinate axes, use the simplified forms we derived above.

#### 7.8.6 Tensor Products and Angular Momentum

The tensor product of two spherical tensors decomposes into spherical tensors of definite rank:

$$T^{(k_1)} \otimes T^{(k_2)} = \bigoplus_{k=|k_1-k_2|}^{k_1+k_2} T^{(k)} \quad (7.125)$$

This is precisely the angular momentum addition formula. The rank  $k$  of a spherical tensor is its angular momentum quantum number. The  $(2k_1+1)(2k_2+1)$ -dimensional reducible representation decomposes into irreducible blocks of dimensions  $2k+1$  for  $k = |k_1 - k_2|, \dots, k_1 + k_2$ .

#### 7.8.7 Example: Tensor Product of Two Vectors ( $\ell = 1$ )

Consider two vectors  $\vec{A}$  and  $\vec{B}$ . In Cartesian coordinates, their tensor product gives 9 components:

$$T_{ij} = A_i B_j \quad \text{for } i, j \in \{x, y, z\} \quad (7.126)$$

This is a rank-2 Cartesian tensor with 9 components. However, not all these components transform independently under rotations. We can decompose this tensor into three irreducible parts.

**Decomposition:**  $3 \otimes 3 = 5 \oplus 3 \oplus 1$

In terms of angular momentum:  $1 \otimes 1 = 2 \oplus 1 \oplus 0$ . The 9 Cartesian components decompose into three pieces:  $\ell = 2$  gives the symmetric traceless tensor (5 components),  $\ell = 1$  gives the antisymmetric tensor (3 components), and  $\ell = 0$  gives the trace (1 component).

**The  $\ell = 0$  piece: Dot product (1 component)**

The trace of the tensor gives a scalar:

$$T^{(0)} = \frac{1}{3}(A_x B_x + A_y B_y + A_z B_z) = \frac{1}{3} \vec{A} \cdot \vec{B} \quad (7.127)$$

This is the unique rotationally invariant combination of two vectors. Under rotations,  $\vec{A} \cdot \vec{B}$  remains unchanged, as expected for an  $\ell = 0$  object.

**The  $\ell = 1$  piece: Cross product (3 components)**

The antisymmetric part of the tensor corresponds to the cross product:

$$T_i^{(1)} = \frac{1}{2}(A_j B_k - A_k B_j) = \frac{1}{2}(\vec{A} \times \vec{B})_i \quad (7.128)$$

where  $(i, j, k)$  is a cyclic permutation of  $(x, y, z)$ . This gives three independent components:

$$T_x^{(1)} = A_y B_z - A_z B_y \quad (7.129)$$

$$T_y^{(1)} = A_z B_x - A_x B_z \quad (7.130)$$

$$T_z^{(1)} = A_x B_y - A_y B_x \quad (7.131)$$

These three components form a vector (an  $\ell = 1$  spherical tensor) that transforms like  $\vec{A} \times \vec{B}$  under rotations.

**The  $\ell = 2$  piece: Symmetric traceless tensor (5 components)**

The symmetric traceless part is:

$$T_{ij}^{(2)} = \frac{1}{2}(A_i B_j + A_j B_i) - \frac{1}{3} \delta_{ij} \vec{A} \cdot \vec{B} \quad (7.132)$$

This tensor is symmetric,  $T_{ij}^{(2)} = T_{ji}^{(2)}$ , which reduces 9 components to 6, and traceless,  $T_{xx}^{(2)} + T_{yy}^{(2)} + T_{zz}^{(2)} = 0$ , which further reduces 6 components to 5. The 5 independent components correspond to the  $2\ell + 1 = 5$  components of a spherical tensor with  $\ell = 2$ . Examples include the quadrupole moment tensor and the  $d$ -orbitals in atomic physics.

## Physical Examples

This decomposition appears throughout physics. For a charge distribution  $\rho(\vec{r})$ , the components  $x_i x_j$  form a rank-2 tensor whose symmetric traceless part gives the electric quadrupole moment tensor  $Q_{ij}$  with  $\ell = 2$ , creating quadrupole electric fields. In atomic physics, combining two  $p$ -orbitals ( $\ell = 1$ ) gives  $s$  ( $\ell = 0$ ),  $p$  ( $\ell = 1$ ), and  $d$  ( $\ell = 2$ ) orbitals, where the  $s$ -orbital is spherically symmetric (the dot product),  $p$ -orbitals are dipolar (related to the cross product structure), and  $d$ -orbitals have quadrupole symmetry. The interaction of light (photon angular momentum  $\ell = 1$ ) with matter involves the tensor product of the electromagnetic field with atomic transition operators, and the selection rules for electric dipole ( $\ell = 1$ ), quadrupole ( $\ell = 2$ ), and higher multipole transitions follow from this tensor decomposition. The polarizability tensor  $\alpha_{ij}$  relates the induced dipole moment to an applied electric field, where its symmetric traceless part describes anisotropic polarizability (different response along different axes), while its trace gives the average isotropic polarizability.

The block diagonal structure we found for  $\ell = 1 \otimes \ell = 1$  generalizes to all tensor products. The Wigner D-matrices provide the explicit rotation matrices for each irreducible block, and Clebsch-Gordan coefficients provide the change of basis that achieves the block diagonalization.

### 7.8.8 Connection to Clebsch-Gordan Coefficients

The decomposition of tensor products is accomplished by Clebsch-Gordan coefficients. For two vectors in spherical tensor form:

$$T_q^{(k)} = \sum_{q_1, q_2} \langle 1, q_1; 1, q_2 | k, q \rangle A_{q_1}^{(1)} B_{q_2}^{(1)} \quad (7.133)$$

For  $k = 0$  (scalar),  $k = 1$  (vector), and  $k = 2$  (rank-2 tensor), the CG coefficients automatically project out the dot product, cross product, and symmetric traceless combinations, respectively.

## 7.9 Chapter Summary

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The addition of angular momenta is a central problem in quantum mechanics, arising whenever a system possesses multiple sources of angular momentum. We developed two complementary descriptions:

The **uncoupled basis**  $|j_1, m_1; j_2, m_2\rangle$  specifies each angular momentum separately and is natural when the subsystems are weakly interacting. The **coupled basis**  $|j_1, j_2; j, m\rangle$  specifies the total angular momentum and is preferred when interactions couple the subsystems or when total angular momentum is conserved.

The tensor product of two angular momentum spaces decomposes into a direct sum of subspaces with definite total  $j$ :

$$\mathcal{H}_1 \otimes \mathcal{H}_2 = \mathcal{H}_{|j_1-j_2|} \oplus \mathcal{H}_{|j_1-j_2|+1} \oplus \cdots \oplus \mathcal{H}_{j_1+j_2} \quad (7.134)$$

The transformation between bases is accomplished by **Clebsch-Gordan coefficients**, which encode the overlap between uncoupled and coupled states. These coefficients satisfy conservation of  $m$ , requiring  $m = m_1 + m_2$ , reflecting the additivity of angular momentum projections. They obey the triangle inequality  $|j_1 - j_2| \leq j \leq j_1 + j_2$ , which determines the allowed values of total angular momentum for given constituent angular momenta. The coefficients also satisfy orthonormality relations that ensure the transformation between bases is unitary, along with various symmetry relations that reflect the underlying rotational symmetry of the problem.

We computed explicit examples for common cases: two spin-1/2 particles (singlet and triplet), adding  $\ell = 1$  to  $s = 1/2$  (fine structure), and two  $\ell = 1$  systems. Standard CG tables provide a practical resource for looking up these coefficients.

Physical applications include spin-orbit coupling, which produces fine structure splitting in atomic spectra, and nuclear hyperfine structure arising from the coupling of nuclear spin to electronic angular momentum. The 3j and 6j symbols provide alternative notations with enhanced symmetry properties, useful when combining more than two angular momenta.

The next chapter applies these angular momentum addition techniques to the hydrogen atom, where we'll see how spin-orbit coupling produces fine structure splitting of energy levels and how the coupled basis emerges as the natural description for atomic states.

## 7.10 Problems

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**Problem 7.1** (*Hilbert Space Dimensions and Tensor Products*)

Consider two angular momentum systems with quantum numbers  $j_1$  and  $j_2$ .

- A spin-1/2 particle (electron) and a spin-1 particle are combined into a composite system. What is the dimension of the total Hilbert space  $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$ ? Write out all basis states in the uncoupled basis  $|j_1, m_1; j_2, m_2\rangle$  explicitly.
- Using the decomposition formula  $\mathcal{H}_1 \otimes \mathcal{H}_2 = \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} \mathcal{H}_j$ , determine which total angular momentum values  $j$  are allowed for this system. For each value of  $j$ , state the dimension of the corresponding subspace  $\mathcal{H}_j$  and verify that the total dimension matches your answer from part (a).
- Two spin-1 particles are combined. Determine the allowed values of total angular momentum  $j$  and their multiplicities. Identify which  $j$  value corresponds to a symmetric state under particle exchange, which to an antisymmetric state, and which are mixed symmetry.

**Problem 7.2** (*Uncoupled versus Coupled Bases*)

For a system of two spin-1/2 particles, we can use either the uncoupled basis  $|m_1, m_2\rangle$  or the coupled basis  $|j, m\rangle$ .

- Write out the four uncoupled basis states explicitly. For each state, determine the eigenvalues of  $\hat{J}_{1z}$ ,  $\hat{J}_{2z}$ , and  $\hat{J}_z = \hat{J}_{1z} + \hat{J}_{2z}$ .
- The state with maximum  $m$  value and the state with minimum  $m$  value can be immediately identified as coupled basis eigenstates. Identify these two states and explain why no calculation is needed.
- Consider the operator  $\vec{J}_1 \cdot \vec{J}_2 = \frac{1}{2}(\hat{J}^2 - \hat{J}_1^2 - \hat{J}_2^2)$ . Using this expression, determine the eigenvalue of  $\vec{J}_1 \cdot \vec{J}_2$  for each coupled basis state  $|j, m\rangle$ . What are the numerical values (in units of  $\hbar^2$ ) for the singlet and triplet states?

**Problem 7.3** ( *$SU(2)$ ,  $SO(3)$ , and Spinors*)

The distinction between  $SU(2)$  and  $SO(3)$  leads to fundamental differences between integer and half-integer angular momentum.

- A rotation by angle  $\theta$  about the  $z$ -axis acts on a spin-1/2 state as  $\hat{R}_z(\theta) = e^{-i\theta\hat{J}_z/\hbar} = e^{-i\theta\sigma_z/2}$ . Calculate the matrix representation of  $\hat{R}_z(2\pi)$  and  $\hat{R}_z(4\pi)$  in the basis  $\{|\uparrow\rangle, |\downarrow\rangle\}$ . What is the physical interpretation?
- For orbital angular momentum with  $\ell = 1$ , show that  $\hat{R}_z(2\pi) = \mathbb{I}$  by considering how  $Y_{\ell m}(\theta, \phi) \propto e^{im\phi}$  transforms under  $\phi \rightarrow \phi + 2\pi$ . Compare this result with part (a).
- The tensor product  $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$  gives both a singlet ( $j = 0$ , integer) and a triplet ( $j = 1$ , integer). Verify the dimension formula  $(2j_1 + 1)(2j_2 + 1) = \sum_j (2j + 1)$  for  $\frac{1}{2} \otimes 1 = \frac{1}{2} \oplus \frac{3}{2}$ . What pattern governs whether tensor products of half-integer angular momenta give integer or half-integer results?

**Problem 7.4** (*Cartesian and Spherical Tensor Decomposition*)

Consider two position vectors  $\vec{r}_1 = (x_1, y_1, z_1)$  and  $\vec{r}_2 = (x_2, y_2, z_2)$ .

- The tensor product  $\vec{r}_1 \otimes \vec{r}_2$  has 9 components that decompose into irreducible parts with  $\ell = 0, 1, 2$ . Write explicit expressions for: (i) the  $\ell = 0$  scalar, (ii) the three components of the  $\ell = 1$  antisymmetric tensor, (iii) one example component from the  $\ell = 2$  symmetric traceless tensor. Verify that  $1 + 3 + 5 = 9$ .
- Show that the  $\ell = 0$  scalar combination  $\vec{r}_1 \cdot \vec{r}_2$  is invariant under rotation about the  $z$ -axis by angle  $\theta$ .
- The spherical harmonics for  $\ell = 1$  are proportional to:  $Y_{1,0} \propto \cos \theta$ ,  $Y_{1,\pm 1} \propto \sin \theta e^{\pm i\phi}$ . The Cartesian coordinates can be expressed as linear combinations of  $(Y_{1,+1}, Y_{1,0}, Y_{1,-1})$ . Invert these relations to express  $x/r$ ,  $y/r$ , and  $z/r$  in terms of the  $Y_{1,m}$  and identify the coefficients.

**Problem 7.5** (*Triangle Inequality and Selection Rules*)

The allowed values of total angular momentum must satisfy  $|j_1 - j_2| \leq j \leq j_1 + j_2$ .

- Consider the addition of orbital angular momentum  $\ell = 2$  (a  $d$ -orbital) and spin  $s = 1/2$ . List all allowed values of total angular momentum  $j$ . For each allowed  $j$ , determine how many states (values of  $m$ ) belong to that multiplet. Verify the total count matches  $(2\ell + 1)(2s + 1)$ .
- In spectroscopic notation  $^{2S+1}L_J$ , for a single electron in a  $d$ -orbital ( $\ell = 2$ ) with spin  $s = 1/2$ , the term symbol is  $^2D_J$ . What values can  $J$  take? Write out the complete term symbols.
- The spin-orbit Hamiltonian is  $H_{\text{SO}} = \xi(\vec{r})\vec{L} \cdot \vec{S}$ . Using the identity  $\vec{L} \cdot \vec{S} = \frac{1}{2}(\hat{J}^2 - \hat{L}^2 - \hat{S}^2)$ , show that the coupled basis states  $|\ell, s; j, m\rangle$  are eigenstates of  $H_{\text{SO}}$  (assuming  $\xi$  depends only on  $r$ ). For the  $^2D$  states, determine the energy splitting pattern if  $\xi$  is constant. Which state has lower energy for  $\xi > 0$ ?

**Problem 7.6** (*Clebsch-Gordan Structure and Orthogonality*)

Although we have not yet calculated Clebsch-Gordan coefficients explicitly, we can understand their structure.

- The coupled and uncoupled bases are related by  $|j_1, j_2; j, m\rangle = \sum_{m_1, m_2} C_{j_1 m_1, j_2 m_2}^{j m} |j_1, m_1; j_2, m_2\rangle$ . The constraint  $m = m_1 + m_2$  follows from conservation of the  $z$ -component of angular momentum. Explain why the Clebsch-Gordan coefficient  $C_{j_1 m_1, j_2 m_2}^{j m} = 0$  whenever  $m \neq m_1 + m_2$ .
- For two spin-1/2 particles, the triplet state with  $m = 0$  is  $|j = 1, m = 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$ . The singlet state must be orthogonal to this. Write down the singlet state  $|j = 0, m = 0\rangle$  and verify orthogonality explicitly.
- The coupled basis forms a complete orthonormal set. For two spin-1/2 particles, verify the completeness relation  $\sum_{j, m} |j, m\rangle\langle j, m| = \mathbb{I}$  by expanding both sides in the uncoupled basis and recovering the  $4 \times 4$  identity matrix.

**Problem 7.7** (*Construction of Clebsch-Gordan Coefficients for Spin-1/2 + Spin-1/2*)

Construct the Clebsch-Gordan coefficients for  $1/2 \otimes 1/2$  using ladder operators.

- The highest state  $|1, +1\rangle$  (triplet,  $j = 1, m = 1$ ) is the unique state with maximum  $m$ . Write  $|1, +1\rangle = |\uparrow\uparrow\rangle$ .

- (b) Apply the lowering operator  $\hat{J}_- = \hat{J}_{1-} + \hat{J}_{2-}$  to  $|1, +1\rangle$  to obtain  $|1, 0\rangle$  (triplet with  $m = 0$ ). Normalize correctly.
- (c) Apply  $\hat{J}_-$  again to obtain  $|1, -1\rangle$  (triplet with  $m = -1$ ).
- (d) The fourth state with  $m = 0$  must be the singlet  $|0, 0\rangle$ . Using orthogonality with the triplet state from part (b), write down  $|0, 0\rangle$  explicitly.
- (e) Verify that your four coupled states form an orthonormal basis by computing inner products.

**Problem 7.8** (*Addition of Three Spin-1/2 Particles*)

Three spin-1/2 particles are combined into a composite system.

- (a) First, couple particles 1 and 2:  $\mathcal{H}_{12} = \mathcal{H}_1 \otimes \mathcal{H}_2 = 0 \oplus 1$ . Then couple the result to particle 3. Show that the total decomposition is  $(0 \oplus 1) \otimes \frac{1}{2} = \frac{1}{2} \oplus \frac{3}{2}$ .
- (b) Alternatively, first couple particles 1 and 3:  $\mathcal{H}_{13} = 0 \oplus 1$ . Then couple to particle 2, giving the same decomposition. Why must both orderings give the same result?
- (c) How many total states are there? Verify that  $2 \times 2 \times 2 = 8 = 2 + 6$  (two doublets and one sextet).
- (d) Identify the state with maximum total spin projection  $m_{\max} = 3/2$ . What is the total angular momentum quantum number for this state?
- (e) Construct the fully antisymmetric three-spin state (the Fermi sea state for spin-3/2 fermions). This state is completely antisymmetric under all particle exchanges.

**Problem 7.9** (*Wigner-Eckart Theorem and Reduced Matrix Elements*)

The Wigner-Eckart theorem relates matrix elements to reduced matrix elements and Clebsch-Gordan coefficients.

- (a) For a spherical tensor operator  $T_q^{(k)}$  acting between  $\ell \rightarrow \ell'$  states, the matrix element is  $\langle \ell', m' | T_q^{(k)} | \ell, m \rangle = C_{\ell m, k q}^{\ell' m'} \langle \ell' || T^{(k)} || \ell \rangle$ . For the dipole operator  $z \propto T_0^{(1)} \propto Y_{1,0}$ , verify the selection rules  $\Delta\ell = \pm 1$ ,  $\Delta m = 0$  by noting which Clebsch-Gordan coefficients are nonzero.
- (b) In hydrogen, the selection rule for allowed transitions is  $\Delta\ell = \pm 1$ . Using the Wigner-Eckart theorem, explain why transitions  $2s \rightarrow 1s$  and  $3d \rightarrow 2d$  are forbidden, while  $2p \rightarrow 1s$  is allowed.
- (c) For the magnetic moment operator  $\mu_z = -g\mu_B S_z/\hbar$  acting on spin-1/2, compute the reduced matrix element  $\langle 1/2 || \mu_z || 1/2 \rangle$  and verify it is independent of  $m$ .

**Problem 7.10** (*Spin-Orbit Coupling and the Lande g-Factor*)

The spin-orbit interaction for a single electron is  $H_{\text{SO}} = \lambda \vec{L} \cdot \vec{S}$ .

- (a) Using  $\vec{L} \cdot \vec{S} = \frac{1}{2}(\hat{J}^2 - \hat{L}^2 - \hat{S}^2)$ , show that the energy shifts for a state  $|\ell, s; j, m\rangle$  are  $E_{\text{SO}} = \frac{\lambda\hbar^2}{2}[j(j+1) - \ell(\ell+1) - s(s+1)]$ .
- (b) For the  $^2P$  term ( $\ell = 1$ ,  $s = 1/2$ ), determine the two levels  $j = 1/2$  and  $j = 3/2$  and their energy splitting.

- (c) The magnetic moment in a magnetic field is  $\mu_z = -g_J \mu_B m_j$  where  $g_J$  is the Lande g-factor. Show that:

$$g_J = 1 + \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)}$$

- (d) For the  $^2P_{3/2}$  state, calculate  $g_J$  numerically. What is the energy shift in a magnetic field  $B = 1$  Tesla (in eV)?

**Problem 7.11** (*Isospin Symmetry and Nucleon Doublets*)

In nuclear physics, the nucleon (proton and neutron) forms an isospin doublet with  $T = 1/2$ .

- Write the nucleon isospin states as  $|1/2, +1/2\rangle$  (proton) and  $|1/2, -1/2\rangle$  (neutron). Under isospin rotations, these transform as a spin-1/2 system. What is the physical analog of the spin raising and lowering operators?
- For two nucleons, construct the coupled isospin states: singlet  $|0, 0\rangle$  and triplet  $|1, T_z\rangle$  with  $T_z = -1, 0, +1$ . Identify which states correspond to  $(pp)$ ,  $(pn)$ ,  $(np)$ , and  $(nn)$ .
- The strong nuclear force treats protons and neutrons identically and is invariant under isospin rotations. Under the Pauli exclusion principle for identical fermions, which two-nucleon states are allowed: even isospin, odd isospin, or both? Explain.
- The electric dipole operator breaks isospin symmetry. It transforms as  $T = 1$  (like a vector). Which isospin transitions does it allow between states of different nuclei?

**Problem 7.12** (*Tensor Operators and Magnetic Quadrupole Moments*)

The electric quadrupole moment is a rank-2 spherical tensor.

- A rank-2 tensor has  $2k + 1 = 5$  components. The electric quadrupole moment tensor is  $Q_q^{(2)} \propto (3z_i z_j - \delta_{ij} r^2)$  in Cartesian form. What are the five spherical components?
- For a state with  $j = 1$  (like  $^3P$ ), compute the expectation value  $\langle j = 1 || Q^{(2)} || j = 1 \rangle$  using the Wigner-Eckart theorem.
- The second-order Stark effect in hydrogen involves the polarizability tensor, which is rank-2. Can this tensor mix states with  $\ell = 0$  to  $\ell = 0$ ? Why or why not?

## References and Further Reading

**Sakurai, J. J., and Napolitano, J.** *Modern Quantum Mechanics*, 3rd ed. Cambridge University Press, 2017. Sections 3.7–3.10 give the canonical graduate treatment of angular momentum addition, Clebsch–Gordan coefficients, and tensor operators; the natural next stop after this chapter.

**Edmonds, A. R.** *Angular Momentum in Quantum Mechanics*. Princeton University Press, 1957. The standard monograph; tightly written and the source most often cited for the conventions and symmetry relations used here.

**Varshalovich, D. A., Moskalev, A. N., and Khersonskii, V. K.** *Quantum Theory of Angular Momentum*. World Scientific, 1988. Comprehensive reference with explicit tables of 3-j, 6-j, and 9-j symbols; the first place to look when a CG coefficient is needed in practice.

- Wigner, E. P.** *Group Theory and Its Application to the Quantum Mechanics of Atomic Spectra*. Academic Press, 1959. Translation of the 1931 original. The historical foundation: angular momentum addition derived from the representation theory of  $SU(2)$ .
- Zare, R. N.** *Angular Momentum: Understanding Spatial Aspects in Chemistry and Physics*. Wiley, 1988. Particularly clear on spherical tensor operators and the Wigner–Eckart theorem; recommended for the rank-2 tensor decomposition discussed at the end of this chapter.



# Chapter 8

## The Hydrogen Atom

### 8.1 Introduction

The hydrogen atom represents one of the greatest triumphs of quantum mechanics. As the simplest atom with only one electron bound to a proton, it serves as the perfect testing ground for quantum mechanical principles while being exactly solvable. In this chapter, we solve the Schrödinger equation for the hydrogen atom using the techniques developed thus far, particularly the angular momentum formalism from Chapters 6 and 7.

#### Key Idea: Central Force Problems in Quantum Mechanics

The hydrogen atom is the prototypical central force problem in quantum mechanics. The spherical symmetry allows us to separate the radial and angular parts of the wave function, leading to the famous quantum numbers  $n$ ,  $\ell$ , and  $m$ .

Before diving into the solution, we pause to discuss unit systems. The choice of units dramatically affects the appearance of our equations, and graduate students must be fluent in multiple conventions.

### 8.2 Units in Electromagnetic and Atomic Physics

#### 8.2.1 Why Units Matter

The choice of units in physics is more than bookkeeping—it reflects what we consider fundamental and can dramatically simplify or complicate our equations. Consider the hydrogen atom Hamiltonian. In SI units:

$$\hat{H} = -\frac{\hbar^2}{2m_e}\nabla^2 - \frac{e^2}{4\pi\epsilon_0 r} \quad (8.1)$$

In atomic units:

$$\hat{H} = -\frac{1}{2}\nabla^2 - \frac{1}{r} \quad (8.2)$$

The physics is identical, but the second form lets you focus on the structure without the constant clutter. Different unit systems are optimized for different purposes.

### 8.2.2 SI Units

SI units (Système International), also called MKS units, are the modern international standard. The base units are the meter, kilogram, second, and ampere. The permittivity of free space  $\epsilon_0$  and permeability  $\mu_0$  appear explicitly in electromagnetic equations.

Coulomb's law reads:

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \quad (8.3)$$

The factors of  $4\pi\epsilon_0$  appear throughout, which can obscure the physics but makes dimensional analysis straightforward. SI is standard in engineering, experimental physics, and undergraduate courses. Griffiths' textbooks use SI.

### 8.2.3 Gaussian Units

Gaussian units (CGS) were the standard in theoretical physics for most of the 20th century. The defining choice is that Coulomb's law takes its simplest form:

$$F = \frac{q_1 q_2}{r^2} \quad (8.4)$$

with no  $4\pi\epsilon_0$  factor. The unit of charge, the *statcoulomb* or *esu*, is defined to make this work.

The trade-off is that factors of  $c$  appear explicitly in Maxwell's equations and in relationships between electric and magnetic quantities. Gaussian units remain standard in plasma physics, astrophysics, and much of theoretical physics. Jackson's *Classical Electrodynamics*, Landau & Lifshitz, and most graduate quantum mechanics texts (Sakurai, Shankar) use Gaussian conventions.

### 8.2.4 Heaviside-Lorentz Units

Heaviside-Lorentz units modify Gaussian units by absorbing factors of  $4\pi$  into the definitions of fields and charges. This "rationalizes" Maxwell's equations: the differential forms contain no  $4\pi$  factors, while they appear in Coulomb's law:

$$F = \frac{1}{4\pi} \frac{q_1 q_2}{r^2} \quad (8.5)$$

This system is standard in quantum field theory and particle physics, where the clean form of Maxwell's equations is advantageous.

### 8.2.5 Atomic Units

Atomic units are constructed by setting the fundamental constants of atomic physics equal to unity:

$$\hbar = m_e = e = 1 \quad (8.6)$$

(building on Gaussian conventions for electromagnetic quantities). This is not merely a notational trick—it defines a complete system of units with natural scales for atomic phenomena.

**The Bohr radius.** The **Bohr radius**  $a_0$  is defined as:

$$a_0 = \frac{\hbar^2}{m_e e^2} = 1 \quad (\text{atomic units}) \quad (8.7)$$

In SI,  $a_0 = 5.29 \times 10^{-11} \text{ m} = 0.529 \text{ \AA}$ . This is the natural length scale for atoms: the most probable electron-proton distance in the hydrogen ground state is exactly one Bohr radius.

The Bohr radius emerges from balancing kinetic and potential energy. The uncertainty principle requires  $\Delta p \sim \hbar/r$  for confinement to radius  $r$ , giving kinetic energy  $\sim \hbar^2/(m_e r^2)$ . The potential energy is  $\sim -e^2/r$ . Minimizing the total energy yields  $r \sim \hbar^2/(m_e e^2) = a_0$ .

**The Hartree.** The **Hartree**  $E_h$  (named after Douglas Hartree of Hartree-Fock fame) is the atomic unit of energy, defined as the potential energy of two elementary charges separated by one Bohr radius:

$$E_h = \frac{e^2}{a_0} = 1 \quad (\text{atomic units}) \quad (8.8)$$

In SI,  $1 E_h = 27.2 \text{ eV} = 4.36 \times 10^{-18} \text{ J}$ . Equivalently,  $E_h = m_e v_0^2$  where  $v_0 = e^2/\hbar$  is the characteristic atomic velocity.

**The Rydberg.** The **Rydberg** (symbol Ry) is defined as half a Hartree:

$$1 \text{ Ry} = \frac{1}{2} E_h = 13.6 \text{ eV} \quad (8.9)$$

This is precisely the ionization energy of hydrogen—the energy required to remove the electron from the ground state. The Rydberg is often more convenient for spectroscopy since hydrogen energy levels are  $E_n = -\text{Ry}/n^2$ .

In atomic units, energies are measured in Hartrees, so the hydrogen ground state energy is  $E_1 = -1/2$  (meaning  $-\frac{1}{2} E_h = -1 \text{ Ry} = -13.6 \text{ eV}$ ).

**Other atomic units.** Setting  $\hbar = m_e = e = 1$  determines units for all other quantities; see Table 8.2 in Section 8.2.7.

**The speed of light in atomic units.** In atomic units,  $c$  is *not* equal to 1. Instead:

$$c = \frac{1}{\alpha} \approx 137 \quad (8.10)$$

where  $\alpha = e^2/(\hbar c) \approx 1/137$  is the fine structure constant. This large value of  $c$  reflects the fact that typical atomic velocities ( $v \sim 1$  in atomic units) are much smaller than the speed of light, justifying non-relativistic quantum mechanics for atoms. Relativistic corrections are suppressed by factors of  $(v/c)^2 = \alpha^2 \sim 10^{-4}$ .

### 8.2.6 Using Atomic Units in Practice

A common source of confusion: if  $a_0 = 1$  in atomic units, what do we write when a distance equals one Bohr radius? Do we write “ $r = 1$ ” or “ $r = a_0$ ”?

The answer depends on context:

**During calculations,** use pure numbers. When working in atomic units, all quantities are dimensionless numbers. The ground state wave function is  $\psi = \pi^{-1/2} e^{-r}$ , the most probable radius is  $r^* = 1$ , and the ground state energy is  $E_1 = -1/2$ . This is the whole point of atomic units: no constants cluttering the algebra.

**When reporting results,** restore the units for clarity. Write “the most probable radius is  $r^* = 1 a_0 = 0.529 \text{ \AA}$ ” or “the ground state energy is  $E_1 = -1/2 E_h = -13.6 \text{ eV}$ .” This makes the physical scale explicit and allows comparison with experiment.

**In equations that define the system,** you can go either way. The hydrogen Hamiltonian can be written as

$$\hat{H} = -\frac{1}{2} \nabla^2 - \frac{1}{r} \quad (8.11)$$

with the understanding that  $r$  is measured in Bohr radii and  $H$  in Hartrees, or equivalently

$$\hat{H} = -\frac{1}{2}\nabla^2 - \frac{1}{r/a_0} \cdot E_h \quad (8.12)$$

but the first form is cleaner and standard.

The key insight is that atomic units work exactly like setting  $c = 1$  in special relativity or  $\hbar = 1$  in quantum mechanics: during calculations you drop the constants, and when comparing to experiment you restore them by dimensional analysis.

### Physical Insight

Think of atomic units as measuring everything in “natural” atomic scales. Saying “ $r = 2$ ” means the distance is twice the Bohr radius. Saying “ $E = -0.5$ ” means the energy is half a Hartree (one Rydberg). The numbers directly tell you the physics in atomic terms.

## 8.2.7 Comparison of Key Formulas

Table 8.1 shows how key formulas appear in different unit systems.

Table 8.1: Formulas in different unit systems.

| Expression              | SI                                                             | Gaussian                                        | Atomic                               |
|-------------------------|----------------------------------------------------------------|-------------------------------------------------|--------------------------------------|
| Coulomb potential       | $-\frac{e^2}{4\pi\epsilon_0 r}$                                | $-\frac{e^2}{r}$                                | $-\frac{1}{r}$                       |
| H Hamiltonian           | $-\frac{\hbar^2}{2m_e}\nabla^2 - \frac{e^2}{4\pi\epsilon_0 r}$ | $-\frac{\hbar^2}{2m_e}\nabla^2 - \frac{e^2}{r}$ | $-\frac{1}{2}\nabla^2 - \frac{1}{r}$ |
| Fine structure constant | $\frac{e^2}{4\pi\epsilon_0 \hbar c}$                           | $\frac{e^2}{\hbar c}$                           | $\frac{1}{c}$                        |

Table 8.2 lists the fundamental atomic constants.

Table 8.2: Fundamental atomic constants.

| Constant        | Symbol | Definition          | Atomic value           | SI value               |
|-----------------|--------|---------------------|------------------------|------------------------|
| Bohr radius     | $a_0$  | $\hbar^2/(m_e e^2)$ | 1                      | 0.529 Å                |
| Hartree         | $E_h$  | $e^2/a_0$           | 1                      | 27.2 eV                |
| Rydberg         | Ry     | $E_h/2$             | 1/2                    | 13.6 eV                |
| Atomic time     | $t_0$  | $\hbar/E_h$         | 1                      | 24.2 as                |
| Atomic velocity | $v_0$  | $a_0 E_h/\hbar$     | 1                      | $2.19 \times 10^6$ m/s |
| Speed of light  | $c$    | —                   | $1/\alpha \approx 137$ | $3 \times 10^8$ m/s    |

### 8.2.8 Maxwell's Equations in Different Units

For completeness, Tables 8.3 and 8.4 show Maxwell's equations in all four unit systems, illustrating why different communities prefer different conventions.

Table 8.3: Maxwell's equations in SI and Gaussian units.

| Equation       | SI                                                                                             | Gaussian                                                                                           |
|----------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Gauss (E)      | $\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$                                               | $\nabla \cdot \vec{E} = 4\pi\rho$                                                                  |
| Gauss (B)      | $\nabla \cdot \vec{B} = 0$                                                                     | $\nabla \cdot \vec{B} = 0$                                                                         |
| Faraday        | $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$                                 | $\nabla \times \vec{E} = -\frac{1}{c} \frac{\partial \vec{B}}{\partial t}$                         |
| Ampère-Maxwell | $\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$ | $\nabla \times \vec{B} = \frac{4\pi}{c} \vec{J} + \frac{1}{c} \frac{\partial \vec{E}}{\partial t}$ |

Table 8.4: Maxwell's equations in Heaviside-Lorentz and atomic units.

| Equation       | Heaviside-Lorentz                                                                               | Atomic                                                                                    |
|----------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Gauss (E)      | $\nabla \cdot \vec{E} = \rho$                                                                   | $\nabla \cdot \vec{E} = 4\pi\rho$                                                         |
| Gauss (B)      | $\nabla \cdot \vec{B} = 0$                                                                      | $\nabla \cdot \vec{B} = 0$                                                                |
| Faraday        | $\nabla \times \vec{E} = -\frac{1}{c} \frac{\partial \vec{B}}{\partial t}$                      | $\nabla \times \vec{E} = -\alpha \frac{\partial \vec{B}}{\partial t}$                     |
| Ampère-Maxwell | $\nabla \times \vec{B} = \frac{1}{c} \vec{J} + \frac{1}{c} \frac{\partial \vec{E}}{\partial t}$ | $\nabla \times \vec{B} = 4\pi\alpha \vec{J} + \alpha \frac{\partial \vec{E}}{\partial t}$ |

Heaviside-Lorentz has the cleanest form: no  $4\pi$  factors in the differential equations. Atomic units are awkward for electrodynamics because  $\alpha \approx 1/137$  appears where  $1/c$  would be in Gaussian units—atomic units are optimized for non-relativistic quantum mechanics, where  $c$  doesn't naturally appear.

### 8.2.9 Converting Between Systems

The key conversion from SI to Gaussian is:

$$\frac{e^2}{4\pi\epsilon_0} \quad (\text{SI}) \longleftrightarrow \quad e^2 \quad (\text{Gaussian}) \quad (8.13)$$

To convert any formula, replace every  $e^2/(4\pi\epsilon_0)$  with  $e^2$ .

To convert from Gaussian to atomic units, set  $\hbar = m_e = e = 1$ . Lengths become pure numbers (measured in units of  $a_0$ ); energies become pure numbers (measured in units of  $E_h$ ).

The fine structure constant  $\alpha$  is dimensionless and therefore identical in all unit systems:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} = \frac{e^2}{\hbar c} = \frac{1}{c} \approx \frac{1}{137} \quad (8.14)$$

### 8.2.10 Which System to Use?

**SI units:** Engineering, experimental physics, undergraduate teaching, numerical comparisons with experiment.

**Gaussian units:** Classic texts (Jackson, Landau), plasma physics, astrophysics, condensed matter theory, avoiding  $4\pi\epsilon_0$  clutter.

**Heaviside-Lorentz units:** Quantum field theory, particle physics, when you want clean Maxwell equations.

**Atomic units:** Atomic/molecular/optical physics, quantum chemistry, whenever you want the simplest Schrödinger equation.

#### Key Idea: The Fine Structure Constant

The fine structure constant  $\alpha \approx 1/137$  is the one truly dimensionless number characterizing electromagnetism. It measures the strength of the electromagnetic interaction:  $\alpha = e^2/(\hbar c)$  in Gaussian units. Its smallness ( $\alpha \ll 1$ ) is why perturbation theory works for atomic physics and why atoms exist at all. In atomic units,  $\alpha = 1/c$ , making the non-relativistic nature of atomic physics manifest.

For this chapter and most atomic physics applications, we use **atomic units**:  $\hbar = m_e = e = 1$ . Lengths are measured in Bohr radii ( $a_0 = 1$ ), energies in Hartrees ( $E_h = 1$ ), and the Coulomb potential is simply  $V(r) = -1/r$ .

## 8.3 The Schrödinger Equation in Spherical Coordinates

For a particle of mass  $\mu$  (in units of  $m_e$ ) in a central potential  $V(r)$ , the time-independent Schrödinger equation is:

$$-\frac{1}{2\mu}\nabla^2\psi + V(r)\psi = E\psi \quad (8.15)$$

For hydrogen with an infinitely heavy nucleus,  $\mu = 1$ . In spherical coordinates  $(r, \theta, \phi)$ , the Laplacian takes the form:

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \quad (8.16)$$

### 8.3.1 Separation of Variables

We seek solutions of the form:

$$\psi(r, \theta, \phi) = R(r)Y_\ell^m(\theta, \phi) \quad (8.17)$$

where  $Y_\ell^m(\theta, \phi)$  are the spherical harmonics we studied in Chapter 6.

## 8.4 The Radial Equation

After separation, the radial equation becomes:

$$\frac{1}{r^2} \frac{d}{dr} \left( r^2 \frac{dR}{dr} \right) + \left[ 2\mu(E - V(r)) - \frac{\ell(\ell+1)}{r^2} \right] R = 0 \quad (8.18)$$

For the Coulomb potential  $V(r) = -1/r$ , this becomes the hydrogen radial equation.

### 8.4.1 The Effective Potential

#### Physical Insight

The term  $\frac{\ell(\ell+1)}{2\mu r^2}$  acts as a centrifugal barrier, preventing the electron from approaching the nucleus too closely for  $\ell > 0$ . This is a purely quantum mechanical effect arising from the angular momentum of the electron.

## 8.5 Energy Levels and Wave Functions

### 8.5.1 The Hydrogen Hamiltonian

In atomic units, the hydrogen Hamiltonian takes its simplest possible form:

$$\hat{H} = -\frac{1}{2}\nabla^2 - \frac{1}{r} \quad (8.19)$$

This elegant expression contains no fundamental constants at all—just the kinetic energy operator and the Coulomb attraction.

### 8.5.2 Energy Eigenvalues

The energy eigenvalues are:

$$E_n = -\frac{1}{2n^2} \quad (8.20)$$

where  $n = 1, 2, 3, \dots$  is the principal quantum number. These energies are measured in Hartrees.

In terms of Rydbergs ( $1 \text{ Ry} = \frac{1}{2}E_h = 13.6 \text{ eV}$ ):

$$E_n = -\frac{\text{Ry}}{n^2} \quad (8.21)$$

The ground state energy is exactly  $E_1 = -\frac{1}{2}E_h = -1 \text{ Ry} = -13.6 \text{ eV}$ .

### 8.5.3 Radial Wave Functions

The normalized radial wave functions are:

$$R_{n\ell}(r) = \sqrt{\left(\frac{2}{n}\right)^3 \frac{(n-\ell-1)!}{2n(n+\ell)!}} e^{-r/n} \left(\frac{2r}{n}\right)^\ell L_{n-\ell-1}^{2\ell+1}\left(\frac{2r}{n}\right) \quad (8.22)$$

where  $r$  is measured in Bohr radii and  $L_k^p(x)$  are the associated Laguerre polynomials.

**Explicit forms:**

Ground state (1s):

$$R_{10}(r) = 2e^{-r} \quad (8.23)$$

2s state:

$$R_{20}(r) = \frac{1}{2\sqrt{2}} \left(1 - \frac{r}{2}\right) e^{-r/2} \quad (8.24)$$

2p state:

$$R_{21}(r) = \frac{1}{2\sqrt{6}} r e^{-r/2} \quad (8.25)$$

3s state:

$$R_{30}(r) = \frac{2}{81\sqrt{3}} (27 - 18r + 2r^2) e^{-r/3} \quad (8.26)$$

## 8.6 Quantum Numbers and Degeneracy

### Key Idea: Quantum Numbers of Hydrogen

The hydrogen atom wave functions are characterized by three quantum numbers: the principal quantum number  $n = 1, 2, 3, \dots$  (which determines energy), the orbital angular momentum  $\ell = 0, 1, 2, \dots, n-1$ , and the magnetic quantum number  $m = -\ell, -\ell+1, \dots, \ell-1, \ell$ . The degeneracy of energy level  $n$  is  $n^2$ .

The total degeneracy is:

$$g_n = \sum_{\ell=0}^{n-1} (2\ell+1) = n^2 \quad (8.27)$$

Including electron spin:  $g_n = 2n^2$ .

## 8.7 Probability Densities and Orbitals

### 8.7.1 Radial Probability Distribution

The probability of finding the electron between  $r$  and  $r + dr$  is:

$$P(r)dr = |R_{n\ell}(r)|^2 r^2 dr \quad (8.28)$$

For the ground state:

$$P_{1s}(r) = 4r^2 e^{-2r} \quad (8.29)$$

The most probable radius (where  $dP/dr = 0$ ) is  $r^* = 1$  (one Bohr radius).The expectation value is  $\langle r \rangle = 3/2$  (in units of  $a_0$ ).

### 8.7.2 Angular Distributions and Orbital Shapes

The angular part determines the shape of orbitals. The  $s$  orbitals ( $\ell = 0$ ) are spherically symmetric,  $p$  orbitals ( $\ell = 1$ ) are dumbbell shaped, and  $d$  orbitals ( $\ell = 2$ ) have more complex four-lobed shapes.



## 8.8 The Effective Potential

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The effective potential in atomic units is:

$$V_{\text{eff}}(r) = -\frac{1}{r} + \frac{\ell(\ell+1)}{2r^2} \quad (8.30)$$

For  $\ell > 0$ , this has a minimum at:

$$r_{\text{min}} = \ell(\ell+1) \quad (8.31)$$

with value:

$$V_{\text{eff}}(r_{\text{min}}) = -\frac{1}{2\ell(\ell+1)} \quad (8.32)$$

## 8.9 Hydrogen-like Atoms

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For ions with nuclear charge  $Z$  ( $\text{He}^+$ ,  $\text{Li}^{2+}$ , etc.), the potential becomes  $V(r) = -Z/r$ . The energy levels and wave functions scale simply:

$$E_n^{(Z)} = -\frac{Z^2}{2n^2} \quad (8.33)$$

$$R_{n\ell}^{(Z)}(r) = Z^{3/2} R_{n\ell}(Zr) \quad (8.34)$$

The effective Bohr radius shrinks:  $a_0^{(Z)} = 1/Z$ .

**Examples:**

| Ion              | $Z$ | $E_1$ (in $E_h$ ) | $E_1$ (in eV) |
|------------------|-----|-------------------|---------------|
| H                | 1   | $-1/2$            | $-13.6$       |
| $\text{He}^+$    | 2   | $-2$              | $-54.4$       |
| $\text{Li}^{2+}$ | 3   | $-9/2$            | $-122.4$      |

## 8.10 Beyond the Bohr Model: Outlook

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The energy levels we have derived represent the leading-order solution to the hydrogen atom. Several small corrections modify these energies.

In atomic units, the speed of light is  $c = 1/\alpha \approx 137$ . This large value reflects the fact that typical atomic velocities ( $v \sim 1$  in atomic units) are much smaller than  $c$ , justifying our non-relativistic treatment. Relativistic corrections scale as  $(v/c)^2 \sim \alpha^2 \sim 10^{-4}$ .

**Fine structure** ( $\sim \alpha^2 E_n$ ): relativistic kinetic energy, spin-orbit coupling, Darwin term.

**Hyperfine structure** ( $\sim (m_e/m_p)\alpha^2 E_n$ ): electron-nuclear spin interaction.

**Lamb shift** ( $\sim \alpha^3 \ln \alpha \cdot E_n$ ): quantum electrodynamic vacuum fluctuations.

These corrections will be treated systematically using perturbation theory in Chapter 9.

## 8.11 Chapter Summary

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In atomic units, the hydrogen atom achieves maximal simplicity:

| Quantity                   | Atomic Units                                   |
|----------------------------|------------------------------------------------|
| Hamiltonian                | $\hat{H} = -\frac{1}{2}\nabla^2 - \frac{1}{r}$ |
| Energy levels              | $E_n = -\frac{1}{2n^2}$                        |
| Ground state energy        | $E_1 = -\frac{1}{2}E_h = -1 \text{ Ry}$        |
| Ground state wave function | $\psi_{100} = \pi^{-1/2}e^{-r}$                |
| Most probable radius       | $r^* = 1 a_0$                                  |
| Mean radius (ground state) | $\langle r \rangle = \frac{3}{2} a_0$          |
| Degeneracy                 | $g_n = n^2$                                    |
| Hydrogen-like ( $Z$ )      | $E_n = -Z^2/(2n^2)$                            |

The absence of fundamental constants focuses attention on the physics: the competition between kinetic energy (which favors spreading out) and potential energy (which favors localization), resolved by the uncertainty principle into discrete bound states.

In the next chapter, we develop perturbation theory to handle atoms and systems that cannot be solved exactly like hydrogen.

## 8.12 Problems

**Note:** Throughout these problems, we use atomic units ( $\hbar = m_e = e = 1$ ) unless otherwise specified.

### Problem 8.1 (*The Schrödinger Equation in Spherical Coordinates*)

The time-independent Schrödinger equation for a particle of mass  $\mu$  in a central potential  $V(r)$  is (in atomic units):

$$-\frac{1}{2\mu}\nabla^2\psi + V(r)\psi = E\psi$$

- Starting from the Laplacian in spherical coordinates, show that the angular part can be written as  $-\hat{L}^2/r^2$ .
- By substituting  $\psi(r, \theta, \phi) = R(r)Y_\ell^m(\theta, \phi)$  into the Schrödinger equation, derive the radial equation:

$$\frac{1}{r^2} \frac{d}{dr} \left( r^2 \frac{dR}{dr} \right) + \left[ 2\mu(E - V(r)) - \frac{\ell(\ell+1)}{r^2} \right] R = 0$$

- Define  $u(r) = rR(r)$  and show that the radial equation becomes:

$$-\frac{1}{2\mu} \frac{d^2u}{dr^2} + \left[ V(r) + \frac{\ell(\ell+1)}{2\mu r^2} \right] u = Eu$$

### Problem 8.2 (*Atomic Units and the Hydrogen Atom*)

In atomic units, the hydrogen atom Hamiltonian is  $\hat{H} = -\frac{1}{2}\nabla^2 - \frac{1}{r}$ , and the energy eigenvalues are  $E_n = -1/(2n^2)$ .

- (a) The Bohr radius is  $a_0 = \hbar^2/(m_e e^2)$  in Gaussian units. Verify this has dimensions of length and calculate its numerical value.
- (b) The Hartree is  $E_h = e^2/a_0$ . Express this in SI units and calculate its numerical value in eV.
- (c) Express the hydrogen ground state energy  $E_1 = -1/2$  in: (i) Hartrees, (ii) Rydbergs, (iii) electron volts.
- (d) For hydrogen-like ions with nuclear charge  $Z$ , show that  $E_n^{(Z)} = -Z^2/(2n^2)$ . Calculate the ionization energy of  $\text{He}^+$  in eV.

**Problem 8.3** (*Quantum Numbers and Degeneracy*)

The hydrogen atom wave functions are characterized by three quantum numbers:  $n$ ,  $\ell$ , and  $m$ .

- (a) For a given principal quantum number  $n$ , list all allowed values of  $\ell$  and explain why these are the only allowed values.
- (b) For a given  $\ell$ , list all allowed values of  $m$  and explain the physical origin of this constraint.
- (c) Calculate the total degeneracy  $g_n = n^2$  by counting all allowed combinations of  $\ell$  and  $m$  for a given  $n$ .
- (d) Including electron spin ( $s = 1/2$ ), what is the total degeneracy of energy level  $E_n$ ? For  $n = 3$ , enumerate all states using the notation  $|n, \ell, m_\ell, m_s\rangle$ .

**Problem 8.4** (*Radial Wave Functions and Probability Distributions*)

In atomic units, the normalized radial wave functions for hydrogen are:

$$R_{n\ell}(r) = \sqrt{\left(\frac{2}{n}\right)^3 \frac{(n-\ell-1)!}{2n(n+\ell)!}} e^{-r/n} \left(\frac{2r}{n}\right)^\ell L_{n-\ell-1}^{2\ell+1}\left(\frac{2r}{n}\right)$$

- (a) Write out the explicit form of  $R_{10}(r)$  for the ground state ( $n = 1$ ,  $\ell = 0$ ).
- (b) Write out the explicit form of  $R_{21}(r)$  for the  $2p$  state.
- (c) The radial probability density is  $P(r) = |R_{n\ell}(r)|^2 r^2$ . For the ground state, find the most probable radius  $r^*$  by solving  $dP/dr = 0$ .
- (d) Calculate the expectation value  $\langle r \rangle$  for the ground state and compare it to  $r^*$ . Why are these different?

**Problem 8.5** (*The Effective Potential and Classical Turning Points*)

In atomic units, the effective potential for the hydrogen radial equation is:

$$V_{\text{eff}}(r) = -\frac{1}{r} + \frac{\ell(\ell+1)}{2r^2}$$

- (a) Sketch  $V_{\text{eff}}(r)$  for  $\ell = 0, 1, 2$  on the same axes. Describe how the centrifugal barrier affects the behavior near  $r = 0$ .
- (b) For  $\ell > 0$ , find the radius  $r_{\min}$  where  $V_{\text{eff}}$  has its minimum.

- (c) Calculate  $V_{\text{eff}}(r_{\min})$  and show that it equals  $-1/[2\ell(\ell+1)]$ .
- (d) For the  $2p$  state ( $n = 2$ ,  $\ell = 1$ ), find the inner and outer classical turning points where  $E_2 = V_{\text{eff}}(r)$ .

**Problem 8.6** (*Angular Momentum and Orbital Shapes*)

The angular part of the hydrogen wave function is given by the spherical harmonics  $Y_\ell^m(\theta, \phi)$ .

- (a) The  $s$  orbitals ( $\ell = 0$ ) have  $Y_0^0 = 1/\sqrt{4\pi}$ . Explain why  $s$  orbitals are spherically symmetric.
- (b) The  $p$  orbitals ( $\ell = 1$ ) have complex spherical harmonics  $Y_1^0 \propto \cos\theta$  and  $Y_1^{\pm 1} \propto \sin\theta e^{\pm i\phi}$ . The real combinations are  $p_z \propto \cos\theta$ ,  $p_x \propto \sin\theta \cos\phi$ ,  $p_y \propto \sin\theta \sin\phi$ . Show that  $p_x$  and  $p_y$  are linear combinations of  $Y_1^{+1}$  and  $Y_1^{-1}$ .
- (c) Calculate  $\langle \hat{L}_z \rangle$  for the state  $p_x$ . Is  $p_x$  an eigenstate of  $\hat{L}_z$ ?
- (d) For a  $p_z$  orbital, the probability density is  $|\psi|^2 \propto |R_{n1}(r)|^2 \cos^2\theta$ . In which directions is the probability maximum? Where is it zero? Describe the orbital shape.

**Problem 8.7** (*Hydrogen Expectation Values and Virial Theorem*)

- (a) For the hydrogen ground state  $|1s\rangle$ , calculate  $\langle r \rangle$ ,  $\langle r^2 \rangle$ , and  $\langle 1/r \rangle$  in atomic units. Use the radial wave function  $R_{10}(r) = 2e^{-r}$ .
- (b) The virial theorem for a power-law potential  $V \propto r^k$  states:  $2\langle T \rangle = k\langle V \rangle$ . For the Coulomb potential ( $k = -1$ ), show that  $\langle T \rangle = -E$  and  $\langle V \rangle = 2E$ .
- (c) Verify the virial theorem numerically for the  $n = 2$  state using  $E_2 = -1/8$ .
- (d) What is the ratio  $\langle 1/r \rangle / \langle r \rangle^{-1}$  for the ground state? Interpret this result.

**Problem 8.8** (*Relativistic Corrections and the Darwin Term*)

The relativistic correction to the kinetic energy is  $T_{\text{rel}} = -p^4/(8m^3c^2)$  in leading order.

- (a) In atomic units with  $c \approx 137$ , the relativistic correction to energy is  $E_{\text{rel}} = -\langle p^4 \rangle / (8c^2)$ . For the ground state, use  $\langle p^2 \rangle = 1$  to estimate  $\langle p^4 \rangle$  by assuming  $\langle p^4 \rangle \sim \langle p^2 \rangle^2$ .
- (b) The exact result is  $E_{\text{rel}} = -\alpha^2 E_n / (2n^3) [n/\ell + 1/2]^{-1} - 3/4]$  where  $\alpha = 1/c$ . For  $n = 1$ ,  $\ell = 0$ , calculate the fractional shift in energy.
- (c) The Darwin term (QED effect) contributes  $\Delta E_{\text{Darwin}} = \alpha^2 \pi \delta^3(\vec{r})/2$  for  $s$  states. Using  $\psi_{1s}(0) = 1/\sqrt{\pi}$ , calculate this contribution.
- (d) The total fine structure splitting between  $2S_{1/2}$  and  $2P_{1/2}$  (historically the Lamb shift) is about 1058 MHz. Convert to eV and compare to the splitting predicted by Dirac theory alone.

**Problem 8.9** (*Helium Ground State and Pauli Exclusion*)

Two electrons in helium interact via Coulomb repulsion. In the simplest approximation, each electron is in a hydrogen-like orbital with effective nuclear charge  $Z_{\text{eff}}$ .

- (a) The ground state has both electrons in the  $1s$  orbital with opposite spins:  $\psi(1, 2) = |1s\rangle_1 |1s\rangle_2 \times$  singlet spin. The total energy is  $E_{\text{tot}} = 2E_1(Z_{\text{eff}}) + \langle V_{ee} \rangle$  where  $V_{ee} = 1/r_{12}$  is the electron-electron repulsion.
- (b) Using the variational method with trial wave function  $\psi_{1s}(r) = (Z^3/\pi)^{1/2} e^{-Zr}$  for both electrons, show that the total energy is  $E(Z) = -2Z^2 + (5/8)Z$  (in atomic units).
- (c) Minimize  $E(Z)$  to find the optimal  $Z_{\text{eff}}$ . What is the resulting ground state energy and how does it compare to the experimental value of  $-5.81$  eV?
- (d) Explain why  $Z_{\text{eff}} < 2$ . What is the physical origin of the screening effect?

**Problem 8.10** (*Spectroscopic Transitions and Selection Rules*)

Selection rules for electric dipole transitions are  $\Delta\ell = \pm 1$ ,  $\Delta m = 0, \pm 1$  (from  $z$  and  $x \pm iy$  components of the dipole operator).

- (a) Using the selection rules, list all allowed transitions from the  $n = 3$  level of hydrogen to the  $n = 2$  level. Identify the emitted photon wavelengths using  $E = -1/(2n^2)$ .
- (b) Why is the  $2s \rightarrow 1s$  transition forbidden? Why is the  $2p \rightarrow 1s$  transition allowed?
- (c) The transition  $1s \rightarrow 2p$  requires absorption of a photon with energy  $\Delta E = E_2 - E_1 = 3/8$  (in atomic units). Calculate the wavelength in nm and identify which spectral line this is (Lyman-alpha).
- (d) For hydrogen-like atoms with nuclear charge  $Z$ , the transition frequency scales as  $\omega \propto Z^2$ . Estimate the wavelength of the Lyman-alpha line for  $\text{He}^+$ .

**Problem 8.11** (*Fine Structure and Lamb Shift*)

The fine structure arises from spin-orbit coupling and relativistic corrections. The Lamb shift is a QED effect that splits the  $2S_{1/2}$  and  $2P_{1/2}$  levels.

- (a) The spin-orbit energy is  $E_{\text{SO}} = \alpha^2/(2n^3) \times f(j, \ell, s)$  where  $f$  depends on quantum numbers. For the  $2P$  level ( $\ell = 1$ ,  $s = 1/2$ ), the two levels are  $2P_{1/2}$  and  $2P_{3/2}$ . Calculate the spin-orbit splitting.
- (b) The relativistic mass correction is  $\Delta E_{\text{rel}} = -\alpha^2/(2n^3) \times [3/4 - n/(j + 1/2)]$ . Calculate this for  $2S_{1/2}$  and  $2P_{1/2}$ .
- (c) The Lamb shift (QED) adds about  $+1.6$  to the  $2S_{1/2}$  energy (in units of  $10^{-3}$  eV relative to  $n = 2$ ) but not to  $2P_{1/2}$ . Estimate the total  $2S_{1/2} - 2P_{1/2}$  splitting and compare to the measured 1058 MHz.

**Problem 8.12** (*Finite Nuclear Size Effects*)

Real nuclei are not point charges. The nuclear charge distribution affects the  $1s$  orbital significantly.

- (a) For a uniformly charged sphere of radius  $R$  (nuclear radius), the potential is  $V(r) = -Z/r$  for  $r > R$  and  $V(r) = -Z[3/(2R) - r^2/(2R^3)]$  for  $r < R$ .
- (b) For hydrogen,  $R \sim 10^{-5}$  in atomic units. The shift in  $1s$  energy due to finite size is  $\Delta E_{1s} \sim (Z\alpha)^2(R/a_0)^2|E_1|$ . Estimate this numerically.

- (c) The shift in the muonic hydrogen  $1s$  level (muon mass  $207m_e$ ) is about 200 times larger. Explain why.
- (d) How could atomic spectroscopy of muonic atoms be used to measure the nuclear radius?

## References and Further Reading

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- Cohen-Tannoudji, C., Diu, B., and Laloë, F.** *Quantum Mechanics*, Vol. 1. Wiley-VCH, 1977. Chapter VII treats the hydrogen atom in conventional units; useful for cross-checking the atomic-units formulas of this chapter against the SI versions.

# Chapter 9

## Perturbation Theory

### 9.1 Introduction

While the hydrogen atom represents a pinnacle of exact solvability in quantum mechanics, most physical systems cannot be solved analytically. Perturbation theory provides a systematic approach to finding approximate solutions when the Hamiltonian can be written as a sum of a solvable part and a small perturbation.

#### Key Idea: The Perturbative Approach

Perturbation theory treats a complicated problem as a simple, solvable problem plus a small correction. We expand solutions in powers of a small parameter, obtaining increasingly accurate approximations.

### 9.2 Time-Independent Perturbation Theory

#### 9.2.1 Non-Degenerate Perturbation Theory

Consider a Hamiltonian of the form:

$$H = H_0 + \lambda V \quad (9.1)$$

where  $H_0$  is the unperturbed Hamiltonian with known eigenstates  $|n^{(0)}\rangle$  and eigenvalues  $E_n^{(0)}$ , and  $\lambda V$  is a small perturbation.

We expand the perturbed eigenstates and energies as power series in  $\lambda$ :

$$|n\rangle = |n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + \dots \quad (9.2)$$

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots \quad (9.3)$$

#### 9.2.2 Derivation of First-Order Corrections

Substituting the expansions (9.2) and (9.3) into the eigenvalue equation  $(H_0 + \lambda V)|n\rangle = E_n|n\rangle$ :

$$(H_0 + \lambda V)(|n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \dots) = (E_n^{(0)} + \lambda E_n^{(1)} + \dots)(|n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \dots) \quad (9.4)$$

Expanding and collecting terms by powers of  $\lambda$ :

$$\lambda^0 : H_0|n^{(0)}\rangle = E_n^{(0)}|n^{(0)}\rangle \quad (9.5)$$

$$\lambda^1 : H_0|n^{(1)}\rangle + V|n^{(0)}\rangle = E_n^{(0)}|n^{(1)}\rangle + E_n^{(1)}|n^{(0)}\rangle \quad (9.6)$$

Equation (9.5) is just the unperturbed eigenvalue equation, which is satisfied by assumption.

**First-order energy correction:** To find  $E_n^{(1)}$ , take the inner product of equation (9.6) with  $\langle n^{(0)}|$ :

$$\langle n^{(0)}|H_0|n^{(1)}\rangle + \langle n^{(0)}|V|n^{(0)}\rangle = E_n^{(0)}\langle n^{(0)}|n^{(1)}\rangle + E_n^{(1)}\langle n^{(0)}|n^{(0)}\rangle \quad (9.7)$$

Since  $H_0$  is Hermitian,  $\langle n^{(0)}|H_0 = E_n^{(0)}\langle n^{(0)}|$ , so the first term on the left equals  $E_n^{(0)}\langle n^{(0)}|n^{(1)}\rangle$ . This cancels the first term on the right, leaving:

$$\boxed{E_n^{(1)} = \langle n^{(0)}|V|n^{(0)}\rangle} \quad (9.8)$$

**First-order state correction:** To find  $|n^{(1)}\rangle$ , expand it in the complete basis of unperturbed states:

$$|n^{(1)}\rangle = \sum_k c_k^{(1)}|k^{(0)}\rangle \quad (9.9)$$

Take the inner product of equation (9.6) with  $\langle m^{(0)}|$  for  $m \neq n$ :

$$\langle m^{(0)}|H_0|n^{(1)}\rangle + \langle m^{(0)}|V|n^{(0)}\rangle = E_n^{(0)}\langle m^{(0)}|n^{(1)}\rangle + E_n^{(1)}\underbrace{\langle m^{(0)}|n^{(0)}\rangle}_{=0} \quad (9.10)$$

Using  $\langle m^{(0)}|H_0 = E_m^{(0)}\langle m^{(0)}|$  and  $\langle m^{(0)}|n^{(1)}\rangle = c_m^{(1)}$ :

$$E_m^{(0)}c_m^{(1)} + \langle m^{(0)}|V|n^{(0)}\rangle = E_n^{(0)}c_m^{(1)} \quad (9.11)$$

Solving for  $c_m^{(1)}$ :

$$c_m^{(1)} = \frac{\langle m^{(0)}|V|n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}} \quad (m \neq n) \quad (9.12)$$

Therefore:

$$\boxed{|n^{(1)}\rangle = \sum_{m \neq n} \frac{\langle m^{(0)}|V|n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}}|m^{(0)}\rangle} \quad (9.13)$$

### Physical Insight

The first-order energy shift is just the expectation value of the perturbation in the unperturbed state. States mix in proportion to their coupling strength divided by their energy separation.

### Example 1

**Spin-1/2 in a Tilted Magnetic Field** Consider a spin-1/2 particle in a strong magnetic field along  $z$  with a weak transverse component along  $x$ . The Hamiltonian is:

$$H = -\gamma B_z S_z - \gamma B_x S_x = \underbrace{-\gamma B_z S_z}_{H_0} + \underbrace{(-\gamma B_x S_x)}_V \quad (9.14)$$



where  $B_x \ll B_z$ .

The unperturbed eigenstates are  $|\uparrow\rangle$  and  $|\downarrow\rangle$  with energies  $E_{\uparrow}^{(0)} = -\frac{1}{2}\gamma B_z$  and  $E_{\downarrow}^{(0)} = +\frac{1}{2}\gamma B_z$ . For the first-order energy correction to the spin-up state:

$$E_{\uparrow}^{(1)} = \langle \uparrow | (-\gamma B_x S_x) | \uparrow \rangle = -\gamma B_x \langle \uparrow | S_x | \uparrow \rangle = 0 \quad (9.15)$$

since  $S_x$  has no diagonal matrix elements in the  $S_z$  basis. Similarly,  $E_{\downarrow}^{(1)} = 0$ . The first-order state correction is:

$$|\uparrow^{(1)}\rangle = \frac{\langle \downarrow | (-\gamma B_x S_x) | \uparrow \rangle}{E_{\uparrow}^{(0)} - E_{\downarrow}^{(0)}} |\downarrow\rangle = \frac{-\gamma B_x (1/2)}{-\gamma B_z} |\downarrow\rangle = \frac{B_x}{2B_z} |\downarrow\rangle \quad (9.16)$$

The perturbed spin-up state tilts toward the  $x$ -direction:

$$|\uparrow'\rangle \approx |\uparrow\rangle + \frac{B_x}{2B_z} |\downarrow\rangle \quad (9.17)$$

This makes physical sense: the spin aligns with the total field direction.

### Example 2

**Zeeman Effect on Hydrogen Ground State** Consider a hydrogen atom in its ground state ( $1s$ , with  $\ell = 0$ ) placed in a uniform magnetic field  $\vec{B} = B\hat{z}$ . The perturbation is:

$$V = -\vec{\mu} \cdot \vec{B} = \frac{e}{2m_e} (\vec{L} + 2\vec{S}) \cdot \vec{B} = \frac{eB}{2m_e} (L_z + 2S_z) \quad (9.18)$$

For the  $1s$  state,  $\ell = 0$  so  $L_z|1s\rangle = 0$ . The unperturbed ground state is two-fold degenerate due to spin:  $|1s, \uparrow\rangle$  and  $|1s, \downarrow\rangle$ .

However, these states are eigenstates of  $S_z$ , so the perturbation is already diagonal:

$$E_{1s,\uparrow}^{(1)} = \frac{eB}{2m_e} \langle 1s, \uparrow | 2S_z | 1s, \uparrow \rangle = \frac{eB}{2m_e} = \mu_B B \quad (9.19)$$

$$E_{1s,\downarrow}^{(1)} = \frac{eB}{2m_e} \langle 1s, \downarrow | 2S_z | 1s, \downarrow \rangle = -\mu_B B \quad (9.20)$$

where  $\mu_B = \frac{e}{2m_e} = \frac{1}{2}$  in atomic units is the Bohr magneton.

The magnetic field splits the ground state by  $\Delta E = 2\mu_B B$ . For a 1 Tesla field, this is about  $1.16 \times 10^{-4}$  eV.

*Context:* Magnetic field control of atomic states is foundational to atomic clocks, MRI, and quantum sensing. Trapped ion qubits often use Zeeman sublevels as computational basis states, where precise control of this splitting enables gate operations.

### 9.2.3 Second-Order Corrections

The second-order energy correction involves virtual transitions:

$$E_n^{(2)} = \sum_{m \neq n} \frac{|\langle m^{(0)} | V | n^{(0)} \rangle|^2}{E_n^{(0)} - E_m^{(0)}} \quad (9.21)$$

#### Physical Insight

Second-order corrections arise from “virtual transitions” to intermediate states. The ground state always shifts down in energy (since all denominators are negative), reflecting the system’s ability to lower its energy by mixing with excited states.

#### Example 3

**Anharmonic Oscillator: Quartic Perturbation** Consider a harmonic oscillator with a quartic perturbation:

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 + \lambda x^4 \quad (9.22)$$

The unperturbed energies are  $E_n^{(0)} = \omega(n + \frac{1}{2})$ . For the ground state ( $n = 0$ ), we need  $\langle 0 | x^4 | 0 \rangle$ . Using the ladder operator representation  $x = \sqrt{\frac{1}{2\omega}}(a + a^\dagger)$  (with  $m = 1$  in atomic units):

$$x^4 = \left(\frac{1}{2\omega}\right)^2 (a + a^\dagger)^4 \quad (9.23)$$

The first-order correction requires terms with equal numbers of  $a$  and  $a^\dagger$ :

$$E_0^{(1)} = \lambda \langle 0 | x^4 | 0 \rangle = \lambda \left(\frac{1}{2\omega}\right)^2 \langle 0 | (a + a^\dagger)^4 | 0 \rangle \quad (9.24)$$

Expanding and keeping terms that return to  $|0\rangle$ :

$$\langle 0 | (a + a^\dagger)^4 | 0 \rangle = \langle 0 | (aa^\dagger aa^\dagger + aa^\dagger a^\dagger a + a^\dagger a a a^\dagger + 2 \text{ more}) | 0 \rangle = 3 \quad (9.25)$$

Therefore:

$$E_0^{(1)} = \frac{3\lambda}{4\omega^2} \quad (9.26)$$

For the second-order correction,  $x^4$  connects  $|0\rangle$  to  $|2\rangle$  and  $|4\rangle$ :

$$\langle 2 | x^4 | 0 \rangle = \left(\frac{1}{2\omega}\right)^2 \sqrt{2} \cdot 3 = \frac{3\sqrt{2}}{4\omega^2} \quad (9.27)$$

$$\langle 4 | x^4 | 0 \rangle = \left(\frac{1}{2\omega}\right)^2 \sqrt{24} = \frac{\sqrt{24}}{4\omega^2} \quad (9.28)$$

The second-order correction is:

$$E_0^{(2)} = \frac{|\langle 2 | x^4 | 0 \rangle|^2}{E_0^{(0)} - E_2^{(0)}} + \frac{|\langle 4 | x^4 | 0 \rangle|^2}{E_0^{(0)} - E_4^{(0)}} = -\frac{21\lambda^2}{8\omega^5} \quad (9.29)$$

The negative second-order shift reflects the general principle that the ground state energy is pushed down by coupling to excited states.

**Example 4**

Quadratic Stark Effect in Hydrogen Working in atomic units, for a hydrogen atom in a weak electric field  $\mathcal{E}$  along the  $z$ -axis, the perturbation is:

$$V = \mathcal{E}z = \mathcal{E}r \cos \theta \quad (9.30)$$

For the ground state  $|1s\rangle$ , which has definite parity (even under  $\vec{r} \rightarrow -\vec{r}$ ), the first-order shift vanishes:

$$E_1^{(1)} = \mathcal{E}\langle 1s|z|1s\rangle = 0 \quad (9.31)$$

since  $z$  is odd under parity. Physically, the spherically symmetric  $1s$  state has no permanent electric dipole moment.

The second-order correction requires summing over all intermediate states:

$$E_1^{(2)} = \sum_{n\ell m \neq 1s} \frac{|\mathcal{E}\langle n\ell m|z|1s\rangle|^2}{E_1 - E_n} \quad (9.32)$$

The selection rule  $\Delta\ell = \pm 1$  from the angular integration means only  $p$ -states contribute. After detailed calculation:

$$E_1^{(2)} = -\frac{9}{4}\mathcal{E}^2 = -\frac{1}{2}\alpha_H\mathcal{E}^2 \quad (9.33)$$

where  $\alpha_H = 9/2$  is the ground state polarizability of hydrogen in atomic units.

The quadratic dependence on field strength is characteristic of induced (rather than permanent) dipole moments. The polarizability  $\alpha_H = 4.5$  a.u. is quite small, but for Rydberg states with  $n \gg 1$ , the polarizability scales as  $n^7$ , making electric field control extremely effective.

*Context:* Electric field control of atoms underlies optical tweezers, atom traps, and Rydberg atom quantum simulators. The enormous polarizability of Rydberg atoms makes Stark shifts a primary tool for control and readout in Rydberg-based quantum computing. In precision atomic clocks, understanding and compensating for Stark shifts from blackbody radiation is essential for reaching  $10^{-18}$  fractional accuracy.

## 9.3 Degenerate Perturbation Theory

When unperturbed states are degenerate, standard perturbation theory fails because the denominators  $E_n^{(0)} - E_m^{(0)}$  vanish for states within the degenerate subspace. We must use degenerate perturbation theory.

### 9.3.1 The Degenerate Subspace

For a  $g$ -fold degenerate level with energy  $E^{(0)}$ , we form the perturbation matrix:

$$W_{ij} = \langle \psi_i^{(0)} | V | \psi_j^{(0)} \rangle \quad (9.34)$$

within the degenerate subspace. The first-order energy corrections are the eigenvalues of this  $g \times g$  matrix.

### 9.3.2 Good Quantum Numbers

The eigenvectors of  $W$  give the correct zeroth-order states:

$$|n, \alpha\rangle = \sum_{i=1}^g c_i^{(\alpha)} |\psi_i^{(0)}\rangle \quad (9.35)$$

where  $\alpha$  labels the eigenstates within the formerly degenerate subspace. These eigenvectors correspond to “good quantum numbers” that remain well-defined in the presence of the perturbation.

#### Key Idea: Lifting Degeneracy

Perturbations often break symmetries, lifting degeneracies. The perturbation selects a preferred basis within the degenerate subspace, namely the basis that diagonalizes the perturbation.

#### Example 5

**Two Coupled Spins with Symmetry-Breaking Perturbation** Consider two spin-1/2 particles with exchange coupling and a perturbation that treats them asymmetrically:

$$H = J\vec{S}_1 \cdot \vec{S}_2 + \epsilon(S_{1z} - S_{2z}) \quad (9.36)$$

The unperturbed Hamiltonian  $H_0 = J\vec{S}_1 \cdot \vec{S}_2$  has eigenstates in the coupled basis: the singlet  $|0, 0\rangle$  with  $E = -\frac{3}{4}J\hbar^2$  and the triplet  $|1, m\rangle$  with  $E = \frac{1}{4}J\hbar^2$ .

The triplet is three-fold degenerate. The perturbation  $V = \epsilon(S_{1z} - S_{2z})$  breaks the exchange symmetry. In the triplet subspace:

$$|1, 1\rangle = |\uparrow\uparrow\rangle \quad (9.37)$$

$$|1, 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \quad (9.38)$$

$$|1, -1\rangle = |\downarrow\downarrow\rangle \quad (9.39)$$

Computing the perturbation matrix elements:

$$\langle 1, 1|V|1, 1\rangle = \epsilon\langle\uparrow\uparrow|(S_{1z} - S_{2z})|\uparrow\uparrow\rangle = \epsilon(\frac{\hbar}{2} - \frac{\hbar}{2}) = 0 \quad (9.40)$$

$$\langle 1, 0|V|1, 0\rangle = \frac{\epsilon}{2}[\langle\uparrow\downarrow| + \langle\downarrow\uparrow|](S_{1z} - S_{2z})[|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle] = 0 \quad (9.41)$$

$$\langle 1, -1|V|1, -1\rangle = 0 \quad (9.42)$$

All diagonal elements vanish! The off-diagonal elements:

$$\langle 1, 1|V|1, 0\rangle = \langle 1, 0|V|1, -1\rangle = 0 \quad (9.43)$$

but

$$\langle\uparrow\downarrow|(S_{1z} - S_{2z})|\uparrow\downarrow\rangle = \hbar, \quad \langle\downarrow\uparrow|(S_{1z} - S_{2z})|\downarrow\uparrow\rangle = -\hbar \quad (9.44)$$

The perturbation matrix in the  $\{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle\}$  basis is zero, but this is because these are already the good states. The perturbation does not mix states with different  $m$ , and within each  $m$  subspace the matrix is diagonal.

However, if we had worked in the uncoupled basis  $\{|\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle\}$  for  $m = 0$ :

$$W = \epsilon\hbar \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (9.45)$$

The eigenstates are  $|\uparrow\downarrow\rangle$  and  $|\downarrow\uparrow\rangle$  with energies  $\pm\epsilon\hbar$ . The perturbation breaks the triplet-singlet structure and makes the uncoupled basis the “good” basis.

### Example 6

**Linear Stark Effect in Hydrogen ( $n = 2$ )** Working in atomic units, the  $n = 2$  level of hydrogen is four-fold degenerate (ignoring spin): one  $2s$  state and three  $2p$  states. An electric field  $\mathcal{E}$  along  $z$  introduces the perturbation  $V = \mathcal{E}z$ .

The states are  $|2, 0, 0\rangle$  ( $2s$ ),  $|2, 1, 0\rangle$  ( $2p_z$ ),  $|2, 1, 1\rangle$  ( $2p_+$ ), and  $|2, 1, -1\rangle$  ( $2p_-$ ).

Since  $[L_z, z] = 0$ , the perturbation conserves  $m$ . The  $m = \pm 1$  states cannot mix with  $m = 0$  states, so the  $4 \times 4$  problem separates:

- $m = +1$ :  $\langle 2, 1, 1|V|2, 1, 1\rangle = 0$  (no shift)
- $m = -1$ :  $\langle 2, 1, -1|V|2, 1, -1\rangle = 0$  (no shift)
- $m = 0$ :  $2 \times 2$  subspace of  $|2s\rangle$  and  $|2p_z\rangle$

For the  $m = 0$  subspace, diagonal elements vanish by parity:

$$\langle 2s|z|2s\rangle = \langle 2p_z|z|2p_z\rangle = 0 \quad (9.46)$$

The crucial off-diagonal element:

$$\langle 2s|z|2p_z\rangle = \int_0^\infty r^2 dr \int d\Omega R_{20}(r)Y_0^0 \cdot r \cos\theta \cdot R_{21}(r)Y_1^0 \quad (9.47)$$

Using explicit wave functions (in atomic units where  $a_0 = 1$ ):

$$R_{20} = \frac{1}{4\sqrt{2}}(2-r)e^{-r/2} \quad (9.48)$$

$$R_{21} = \frac{1}{4\sqrt{6}}r e^{-r/2} \quad (9.49)$$

After integration:  $\langle 2s|z|2p_z\rangle = -3$ .

The perturbation matrix in the  $\{|2s\rangle, |2p_z\rangle\}$  basis:

$$W = \mathcal{E} \begin{pmatrix} 0 & -3 \\ -3 & 0 \end{pmatrix} \quad (9.50)$$

Eigenvalues:  $E^{(1)} = \pm 3\mathcal{E}$ . Eigenstates:

$$|\pm\rangle = \frac{1}{\sqrt{2}}(|2s\rangle \mp |2p_z\rangle) \quad (9.51)$$

The linear Stark effect gives energy shifts proportional to  $\mathcal{E}$ , not  $\mathcal{E}^2$ ! This is because the degenerate  $2s$  and  $2p$  states can hybridize to form states with permanent electric dipole moments.

*Context:* The linear Stark effect demonstrates how degeneracy enables qualitatively different physics. Rydberg atoms, with their near-degenerate high- $n$  manifolds, exhibit enormous electric dipole moments ( $\sim n^2$  a.u.), making them exquisitely sensitive to electric fields. This sensitivity is exploited in Rydberg-atom-based electric field sensors and in creating strong dipole-dipole interactions for quantum simulation.

### 9.3.3 Choice of Basis: Coupled vs. Uncoupled vs. Full Diagonalization

A recurring theme in perturbation theory is the choice of basis. When multiple quantum numbers could label states, choosing the right basis can dramatically simplify calculations.

#### Key Idea: Choosing the Right Basis

The “right” basis for perturbation theory is one in which the perturbation is diagonal or block-diagonal. This basis corresponds to the quantum numbers that remain good (conserved) in the presence of the perturbation.

Consider a system with two angular momenta  $\vec{L}$  and  $\vec{S}$ . We can work in:

- **Uncoupled basis:**  $|\ell, m_\ell, s, m_s\rangle$ , eigenstates of  $L^2, L_z, S^2, S_z$
- **Coupled basis:**  $|\ell, s, j, m_j\rangle$ , eigenstates of  $L^2, S^2, J^2, J_z$

The choice depends on which operators commute with the perturbation:

- If  $[V, J_z] = 0$  but  $[V, L_z] \neq 0$ : use coupled basis
- If  $[V, L_z] = 0$  and  $[V, S_z] = 0$ : use uncoupled basis
- If neither is diagonal: must diagonalize explicitly

#### Example 7

**Hydrogen Fine Structure: Comparing Basis Choices** The fine structure Hamiltonian for hydrogen includes the spin-orbit term:

$$H_{SO} = \frac{1}{2m_e^2 c^2} \frac{1}{r} \frac{dV}{dr} \vec{L} \cdot \vec{S} = \xi(r) \vec{L} \cdot \vec{S} \quad (9.52)$$

**Attempt 1: Uncoupled basis**  $|n, \ell, m_\ell, m_s\rangle$

In this basis,  $\vec{L} \cdot \vec{S} = L_x S_x + L_y S_y + L_z S_z$  has off-diagonal elements. For example,  $L_+ S_-$  connects  $|m_\ell, m_s\rangle$  to  $|m_\ell + 1, m_s - 1\rangle$ . The perturbation matrix is not diagonal, and we would need to diagonalize a  $(2\ell + 1) \times 2$  dimensional matrix for each  $\ell$ .

**Attempt 2: Coupled basis**  $|n, \ell, j, m_j\rangle$

Using  $\vec{J} = \vec{L} + \vec{S}$ :

$$\vec{L} \cdot \vec{S} = \frac{1}{2}(J^2 - L^2 - S^2) \quad (9.53)$$

In the coupled basis, this is diagonal:

$$\langle n, \ell, j, m_j | \vec{L} \cdot \vec{S} | n, \ell, j, m_j \rangle = \frac{\hbar^2}{2} [j(j+1) - \ell(\ell+1) - s(s+1)] \quad (9.54)$$

No diagonalization needed! The first-order energy shift is:

$$E_{SO}^{(1)} = \frac{\hbar^2}{2} [j(j+1) - \ell(\ell+1) - \frac{3}{4}] \langle \xi(r) \rangle_{n\ell} \quad (9.55)$$

For  $\ell > 0$ ,  $j = \ell \pm \frac{1}{2}$ :

$$j = \ell + \frac{1}{2} : \quad E_{SO}^{(1)} = \frac{\ell \hbar^2}{2} \langle \xi \rangle \quad (9.56)$$

$$j = \ell - \frac{1}{2} : \quad E_{SO}^{(1)} = -\frac{(\ell+1) \hbar^2}{2} \langle \xi \rangle \quad (9.57)$$

The coupled basis is the natural choice because  $J^2$  and  $J_z$  commute with  $\vec{L} \cdot \vec{S}$ , making  $j$  and  $m_j$  good quantum numbers.

*Context:* Basis choice is not just mathematical convenience; it reflects physical insight about which quantities are conserved. The same principle governs LS-coupling versus jj-coupling in multi-electron atoms, molecular orbital versus valence bond descriptions in chemistry, and product versus entangled bases in quantum information. In quantum computing, gate design requires choosing bases adapted to the available control Hamiltonians.

## 9.4 Time-Dependent Perturbation Theory

For time-dependent Hamiltonians  $H = H_0 + V(t)$ , we need different techniques. In Chapter 4, we developed the general framework for time-dependent evolution, including the Dyson series (Section 4.8) and the interaction picture (Section 4.9). Here we apply these tools in the perturbative regime, where the time-dependent part  $V(t)$  is small compared to  $H_0$ .

### 9.4.1 The Interaction Picture

Recall from Section 4.9 that the interaction picture provides a hybrid between the Schrödinger and Heisenberg pictures, transferring the “trivial” evolution under  $H_0$  to the operators while states evolve only due to the perturbation. In the interaction picture, states evolve according to:

$$i \frac{\partial}{\partial t} |\psi_I(t)\rangle = V_I(t) |\psi_I(t)\rangle \quad (9.58)$$

with  $V_I(t) = e^{iH_0 t} V(t) e^{-iH_0 t}$ .

As we showed in Chapter 4, the formal solution involves the time-ordered exponential (Dyson series):

$$|\psi_I(t)\rangle = \mathcal{T} \exp \left( -i \int_0^t V_I(t') dt' \right) |\psi_I(0)\rangle \quad (9.59)$$

Time ordering is essential when  $[V_I(t_1), V_I(t_2)] \neq 0$  for different times, reflecting the causal structure of quantum evolution. In perturbation theory, we truncate this series at low order.

### 9.4.2 Transition Probabilities

To first order, the amplitude for transition from state  $|i\rangle$  to state  $|f\rangle$  is:

$$c_{fi}(t) = -i \int_0^t dt' \langle f | V_I(t') | i \rangle = -i \int_0^t dt' e^{i\omega_{fi}t'} \langle f | V(t') | i \rangle \quad (9.60)$$

where  $\omega_{fi} = E_f - E_i$ .

#### Example 8

**Rabi Oscillations in a Driven Two-Level System** Consider a spin-1/2 in a static field  $B_0 \hat{z}$  driven by an oscillating transverse field:

$$H = -\gamma B_0 S_z - \gamma B_1 \cos(\omega t) S_x \quad (9.61)$$

This is precisely the NMR system we analyzed exactly in Section 4.9 of Chapter 4. There, we transformed to the rotating frame and applied the rotating wave approximation to obtain the effective time-independent Hamiltonian. Here we take a complementary approach: we treat the oscillating field perturbatively, valid when  $\Omega_1 \ll |\omega_0 - \omega|$  (far from resonance) or for short times.

Let  $\omega_0 = \gamma B_0$  (Larmor frequency) and  $\Omega_1 = \gamma B_1$  (Rabi frequency). The unperturbed states are  $|\uparrow\rangle$  and  $|\downarrow\rangle$  with energy splitting  $\omega_0$ .

Starting in  $|\downarrow\rangle$ , the transition amplitude to  $|\uparrow\rangle$  is:

$$c_{\uparrow}(t) = -i \int_0^t dt' e^{i\omega_0 t'} (-\gamma B_1 \cos \omega t') \langle \uparrow | S_x | \downarrow \rangle \quad (9.62)$$

Using  $\langle \uparrow | S_x | \downarrow \rangle = 1/2$  and  $\cos \omega t' = \frac{1}{2}(e^{i\omega t'} + e^{-i\omega t'})$ :

$$c_{\uparrow}(t) = \frac{i\Omega_1}{4} \int_0^t dt' \left( e^{i(\omega_0 + \omega)t'} + e^{i(\omega_0 - \omega)t'} \right) \quad (9.63)$$

Near resonance ( $\omega \approx \omega_0$ ), the first term oscillates rapidly and averages to zero (rotating wave approximation). The second term gives:

$$c_{\uparrow}(t) \approx \frac{i\Omega_1}{4} \frac{e^{i(\omega_0 - \omega)t} - 1}{i(\omega_0 - \omega)} = \frac{\Omega_1}{2(\omega_0 - \omega)} \sin\left(\frac{(\omega_0 - \omega)t}{2}\right) e^{i(\omega_0 - \omega)t/2} \quad (9.64)$$

The transition probability:

$$P_{\uparrow}(t) = |c_{\uparrow}(t)|^2 = \frac{\Omega_1^2}{(\omega_0 - \omega)^2} \sin^2\left(\frac{(\omega_0 - \omega)t}{2}\right) \quad (9.65)$$

At exact resonance ( $\omega = \omega_0$ ), this formula breaks down, but the exact solution gives:

$$P_{\uparrow}(t) = \sin^2\left(\frac{\Omega_1 t}{2}\right) \quad (9.66)$$

The system oscillates coherently between  $|\downarrow\rangle$  and  $|\uparrow\rangle$  at the Rabi frequency  $\Omega_1$ . Perturbation theory is valid when  $\Omega_1 \ll |\omega_0 - \omega|$  or for short times  $t \ll 2\pi/\Omega_1$ .

Compare this perturbative result with the exact rotating-frame solution from Section 4.9. The exact on-resonance result  $P_{\uparrow}(t) = \sin^2(\Omega_1 t/2)$  agrees with what we would obtain by summing the perturbation series to all orders. The perturbative approach is particularly valuable for understanding the off-resonant response and for systems where the driving is genuinely weak.



**Example 9**

**Electric Dipole Transitions in Hydrogen** Consider a hydrogen atom interacting with an oscillating electric field  $\vec{\mathcal{E}}(t) = \mathcal{E}_0 \hat{z} \cos(\omega t)$ . The perturbation is:

$$V(t) = e\mathcal{E}_0 z \cos(\omega t) \quad (9.67)$$

The transition amplitude from state  $|i\rangle = |n_i, \ell_i, m_i\rangle$  to  $|f\rangle = |n_f, \ell_f, m_f\rangle$  is:

$$c_{fi}(t) \propto \langle f|z|i\rangle \int_0^t dt' e^{i\omega_{fi}t'} \cos(\omega t') \quad (9.68)$$

The matrix element  $\langle f|z|i\rangle$  imposes **selection rules**:

$$z = r \cos \theta = r \sqrt{\frac{4\pi}{3}} Y_1^0(\theta, \phi) \quad (9.69)$$

The angular integration:

$$\int d\Omega Y_{\ell_f}^{m_f*} Y_1^0 Y_{\ell_i}^{m_i} \quad (9.70)$$

vanishes unless:

$$\Delta m = m_f - m_i = 0 \quad (9.71)$$

$$\Delta \ell = \ell_f - \ell_i = \pm 1 \quad (9.72)$$

These are the **electric dipole selection rules**. Additionally, parity must change since  $z$  is odd.

For the Lyman- $\alpha$  transition  $2p \rightarrow 1s$ :

$$\langle 1s|z|2p_z\rangle = \langle 1, 0, 0|r \cos \theta|2, 1, 0\rangle = \frac{2^7}{3^5} a_0 \approx 0.74 a_0 \quad (9.73)$$

*Context:* Selection rules determine which transitions can be driven by lasers, the foundation of laser cooling, optical pumping, and quantum state preparation. “Forbidden” transitions (electric quadrupole, magnetic dipole) have much smaller matrix elements but are used for ultra-narrow clock transitions because their slower decay rates give longer coherence times. The matrix elements directly determine oscillator strengths used throughout astrophysical spectroscopy.

## 9.5 Fermi's Golden Rule

### 9.5.1 Transition Rates to a Continuum

When transitions occur to a continuum of final states, we compute the transition rate rather than the transition probability. Starting from time-dependent perturbation theory with a periodic perturbation  $V(t) = V e^{-i\omega t} + V^\dagger e^{i\omega t}$ , the transition rate to a continuum of final states is:

$$\Gamma_{i \rightarrow f} = 2\pi |\langle f|V|i\rangle|^2 \rho(E_f) \quad (9.74)$$

where  $\rho(E_f)$  is the density of final states at the energy determined by energy conservation:  $E_f = E_i + \omega$  (absorption) or  $E_f = E_i - \omega$  (emission).

#### Mathematical Aside: Derivation of Fermi's Golden Rule

The transition amplitude at time  $t$  is  $c_f(t) = -i\langle f|V|i\rangle \frac{e^{i(\omega_{fi}-\omega)t}-1}{i(\omega_{fi}-\omega)}$ . The probability  $|c_f(t)|^2 \propto \frac{\sin^2[(\omega_{fi}-\omega)t/2]}{(\omega_{fi}-\omega)^2}$  becomes sharply peaked at  $\omega_{fi} = \omega$  as  $t$  increases. In the limit  $t \rightarrow \infty$ , this approaches  $\pi t \delta(\omega_{fi} - \omega)$ , giving a constant transition rate.

#### Example 10

**Ionization from a Bound State** Consider a particle initially bound in a one-dimensional well, perturbed by an oscillating electric field that can ionize it into continuum states. For a particle in the ground state of an infinite square well of width  $L$ :

$$\psi_1(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right), \quad E_1 = \frac{\pi^2}{2L^2} \quad (9.75)$$

An oscillating field  $V(t) = E_0 x \cos(\omega t)$  can eject the particle into continuum states. For final states with energy  $E_f = k_f^2/2$ , energy conservation requires:

$$\omega = E_f - E_1 = \frac{k_f^2}{2} - \frac{\pi^2}{2L^2} \quad (9.76)$$

The matrix element  $\langle k_f|x|1\rangle$  couples the bound state to the continuum. The density of states for a free particle is  $\rho(E) = \frac{L}{2\pi} \sqrt{\frac{2}{E}}$ , and Fermi's golden rule gives:

$$\Gamma = 2\pi |E_0|^2 |\langle k_f|x|1\rangle|^2 \rho(E_f) \quad (9.77)$$

This ionization rate increases with field strength as  $E_0^2$  and depends on the overlap between bound and continuum wave functions.

#### Example 11

**Spontaneous Emission: Hydrogen Lyman- $\alpha$**  For spontaneous emission, an excited atom decays by emitting a photon into the electromagnetic vacuum. The transition rate for electric dipole radiation is:

$$A_{fi} = \frac{\omega_{fi}^3}{3\pi\epsilon_0\hbar c^3} |\langle f|\vec{d}|i\rangle|^2 \quad (9.78)$$

where  $\vec{d} = e\vec{r}$  is the electric dipole operator.

For the Lyman- $\alpha$  transition ( $2p \rightarrow 1s$ ) in hydrogen:

- Transition energy:  $\hbar\omega = 10.2$  eV (wavelength 121.6 nm)
- Dipole matrix element:  $|\langle 1s|er|2p\rangle|^2 = e^2 \left(\frac{2^7}{3^5} a_0\right)^2$

The spontaneous emission rate:

$$A_{2p \rightarrow 1s} = \frac{\omega^3}{3\pi\epsilon_0\hbar c^3} \cdot \frac{e^2 2^{15} a_0^2}{3^{10}} = 6.27 \times 10^8 \text{ s}^{-1} \quad (9.79)$$

The lifetime of the  $2p$  state is  $\tau = 1/A = 1.6 \text{ ns}$ .

For comparison, Rydberg states with principal quantum number  $n$  have:

$$A_n \propto \frac{1}{n^3} \quad (9.80)$$

A Rydberg state with  $n = 50$  has a lifetime of order milliseconds, a factor of  $10^6$  longer than low-lying states. This suppressed spontaneous emission enables long coherence times in Rydberg-atom quantum computing.

*Context:* Spontaneous emission rates determine atomic lifetimes, laser threshold conditions, and quantum efficiency of light sources. The Lyman- $\alpha$  line is the most important spectral line in astrophysics, used for mapping cosmic hydrogen distribution and studying the epoch of reionization. In cavity QED, the Purcell effect modifies spontaneous emission rates, enabling single-photon sources for quantum communication.

## 9.6 The Sudden Approximation

When a perturbation is applied suddenly (faster than characteristic system timescales), the wave function has no time to adjust. We use the sudden approximation.

### 9.6.1 Overlap Probabilities

If the Hamiltonian suddenly changes from  $H_0$  to  $H'$  at  $t = 0$ , the system's state  $|\psi\rangle$  is unchanged, but it is no longer an eigenstate of the new Hamiltonian. The probability of finding the system in eigenstate  $|\phi'_n\rangle$  of  $H'$  is:

$$P_n = |\langle \phi'_n | \psi \rangle|^2 \quad (9.81)$$

#### Example 12

**Sudden Change of Harmonic Oscillator Frequency** Consider a particle in the ground state of a harmonic oscillator with frequency  $\omega_1$ . At  $t = 0$ , the frequency suddenly changes to  $\omega_2$ .

The initial state is:

$$\psi_0^{(1)}(x) = \left(\frac{m\omega_1}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega_1 x^2}{2\hbar}\right) \quad (9.82)$$

The new ground state is:

$$\psi_0^{(2)}(x) = \left(\frac{m\omega_2}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega_2 x^2}{2\hbar}\right) \quad (9.83)$$

The probability of remaining in the ground state is:

$$P_0 = |\langle \psi_0^{(2)} | \psi_0^{(1)} \rangle|^2 = \left| \int_{-\infty}^{\infty} dx \psi_0^{(2)*}(x) \psi_0^{(1)}(x) \right|^2 \quad (9.84)$$

The integral of two Gaussians:

$$\langle \psi_0^{(2)} | \psi_0^{(1)} \rangle = \left( \frac{m\omega_1}{\pi\hbar} \right)^{1/4} \left( \frac{m\omega_2}{\pi\hbar} \right)^{1/4} \sqrt{\frac{2\pi\hbar}{m(\omega_1 + \omega_2)}} = \frac{2\sqrt{\omega_1\omega_2}}{\omega_1 + \omega_2} \quad (9.85)$$

Therefore:

$$P_0 = \frac{4\omega_1\omega_2}{(\omega_1 + \omega_2)^2} \quad (9.86)$$

For a small change  $\omega_2 = \omega_1(1 + \epsilon)$ :

$$P_0 \approx 1 - \frac{\epsilon^2}{4} + O(\epsilon^3) \quad (9.87)$$

For a doubling of frequency ( $\omega_2 = 2\omega_1$ ):  $P_0 = 8/9 \approx 0.89$ .

The particle is most likely to remain in the ground state, but there is a finite probability of excitation to higher states (only even states, by parity).

### Example 13

**Sudden Nuclear Charge Change: Beta Decay** In nuclear  $\beta^-$  decay, a neutron converts to a proton, suddenly increasing the nuclear charge from  $Z$  to  $Z + 1$ . Consider an electron initially in the ground state of a hydrogen-like atom.

The initial wave function is:

$$\psi_{1s}^{(Z)}(r) = \frac{1}{\sqrt{\pi}} \left( \frac{Z}{a_0} \right)^{3/2} e^{-Zr/a_0} \quad (9.88)$$

After the decay, the electron finds itself in the Coulomb potential of charge  $Z + 1$ . The probability of remaining in the  $1s$  state of the new atom:

$$P_{1s} = |\langle \psi_{1s}^{(Z+1)} | \psi_{1s}^{(Z)} \rangle|^2 \quad (9.89)$$

The overlap integral:

$$\langle \psi_{1s}^{(Z+1)} | \psi_{1s}^{(Z)} \rangle = \frac{1}{\pi} \left( \frac{Z(Z+1)}{a_0^2} \right)^{3/2} \int_0^\infty 4\pi r^2 dr e^{-(2Z+1)r/a_0} \quad (9.90)$$

$$= 4 \left( \frac{Z(Z+1)}{a_0^2} \right)^{3/2} \frac{2a_0^3}{(2Z+1)^3} = \frac{8[Z(Z+1)]^{3/2}}{(2Z+1)^3} \quad (9.91)$$

For tritium decay ( $Z = 1 \rightarrow Z = 2$ ):

$$P_{1s} = \left( \frac{8(1)(2)^{3/2}}{27} \right)^2 = \left( \frac{16\sqrt{2}}{27} \right)^2 \approx 0.70 \quad (9.92)$$

There is a 30% probability that the electron is shaken into an excited state or ionized by the sudden change in nuclear charge.

*Context:* The sudden approximation applies to fast quenches in ultracold atomic gases, where trap parameters can be changed faster than atomic motion timescales. In condensed matter, sudden quenches are used to study non-equilibrium dynamics and thermalization. The beta decay calculation was an early confirmation of quantum mechanical overlap predictions.

## 9.7 The Adiabatic Approximation

When changes occur slowly compared to system dynamics, the system remains in an instantaneous eigenstate. We developed the adiabatic theorem in detail in Section 4.10 of Chapter 4, including the derivation of the Berry phase for two-level systems. Here we summarize the key results and place them in the context of perturbation theory.

### 9.7.1 The Adiabatic Theorem

Recall from Section 4.10 that if a system starts in the  $n$ th eigenstate of  $H(0)$  and  $H(t)$  changes slowly enough that transitions to other states are negligible, the system remains in the  $n$ th instantaneous eigenstate of  $H(t)$ :

$$|\psi(t)\rangle = e^{i\gamma_n(t)} e^{-\frac{i}{\hbar} \int_0^t E_n(t') dt'} |n(t)\rangle \quad (9.93)$$

The adiabatic condition requires that the rate of change be slow compared to level spacings:

$$\hbar \left| \frac{\langle m | \dot{H} | n \rangle}{(E_n - E_m)^2} \right| \ll 1 \quad \text{for all } m \neq n \quad (9.94)$$

This condition can be understood through time-dependent perturbation theory: the transition amplitude to state  $|m\rangle$  involves the matrix element  $\langle m | \dot{H} | n \rangle$  divided by the energy denominator  $(E_n - E_m)$ . When the adiabatic condition is satisfied, these transition amplitudes remain small for all times.

### 9.7.2 Berry Phase

As derived in Section 4.10.2, the geometric phase  $\gamma_n(t)$  acquired during adiabatic evolution depends only on the path taken through parameter space, not on the rate of traversal. The key result for two-level systems is that the Berry phase equals half the solid angle subtended by the path on the Bloch sphere:

$$\gamma_{\text{geom}} = -\frac{1}{2}\Omega \quad (9.95)$$

where  $\Omega$  is the solid angle enclosed by the closed path traced by the Hamiltonian's parameters.

For a general cyclic evolution in parameter space:

$$\gamma_n = i \oint \langle n(R) | \nabla_R | n(R) \rangle \cdot dR \quad (9.96)$$

This Berry connection  $\vec{A}_n(R) = i \langle n(R) | \nabla_R | n(R) \rangle$  plays a role analogous to the vector potential in electromagnetism, with the Berry curvature acting like a magnetic field in parameter space.

**Physical Insight**

The Berry phase is topological in nature: it depends on the geometry of the parameter space and the structure of eigenstate degeneracies. As we showed in Chapter 4 through the Bloch cube visualization, the factor of  $1/2$  arises from the spinorial nature of spin-1/2 wavefunctions. The Berry phase underlies phenomena from the Aharonov-Bohm effect to topological insulators.

## 9.8 Variational Methods

The variational principle provides an alternative approach to approximation that does not require a small parameter.

### 9.8.1 The Variational Principle

For any normalized trial wave function  $|\psi_{\text{trial}}\rangle$ :

$$E_0 \leq \frac{\langle \psi_{\text{trial}} | H | \psi_{\text{trial}} \rangle}{\langle \psi_{\text{trial}} | \psi_{\text{trial}} \rangle} \quad (9.97)$$

where  $E_0$  is the true ground state energy. The equality holds only when  $|\psi_{\text{trial}}\rangle$  is the exact ground state.

### 9.8.2 The Rayleigh-Ritz Method

We parameterize trial functions and minimize the energy expectation value with respect to the parameters.

**Example 14**

**Gaussian Trial Function for Harmonic Oscillator** Consider the harmonic oscillator  $H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$  with a Gaussian trial function:

$$\psi_{\alpha}(x) = \left(\frac{2\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2} \quad (9.98)$$

where  $\alpha > 0$  is the variational parameter.

The kinetic energy:

$$\langle T \rangle = -\frac{\hbar^2}{2m} \int_{-\infty}^{\infty} dx \psi_{\alpha}^* \frac{d^2 \psi_{\alpha}}{dx^2} \quad (9.99)$$

Using  $\frac{d^2 \psi_{\alpha}}{dx^2} = (4\alpha^2 x^2 - 2\alpha)\psi_{\alpha}$ :

$$\langle T \rangle = \frac{\hbar^2}{2m} (2\alpha - 4\alpha^2 \langle x^2 \rangle) = \frac{\hbar^2}{2m} \left( 2\alpha - 4\alpha^2 \cdot \frac{1}{4\alpha} \right) = \frac{\hbar^2 \alpha}{2m} \quad (9.100)$$

The potential energy:

$$\langle V \rangle = \frac{1}{2} m \omega^2 \langle x^2 \rangle = \frac{1}{2} m \omega^2 \cdot \frac{1}{4\alpha} = \frac{m \omega^2}{8\alpha} \quad (9.101)$$

Total energy:

$$E(\alpha) = \frac{\hbar^2 \alpha}{2m} + \frac{m\omega^2}{8\alpha} \quad (9.102)$$

Minimizing:  $\frac{dE}{d\alpha} = \frac{\hbar^2}{2m} - \frac{m\omega^2}{8\alpha^2} = 0$ , giving:

$$\alpha_{\min} = \frac{m\omega}{2\hbar} \quad (9.103)$$

The minimum energy:

$$E_{\min} = \frac{\hbar^2}{2m} \cdot \frac{m\omega}{2\hbar} + \frac{m\omega^2}{8} \cdot \frac{2\hbar}{m\omega} = \frac{\hbar\omega}{4} + \frac{\hbar\omega}{4} = \frac{\hbar\omega}{2} \quad (9.104)$$

This is exactly the ground state energy! The Gaussian trial function with optimal  $\alpha$  is the exact ground state wave function. This is a special case where the variational method gives the exact answer.

### Example 15

**Helium Ground State** The helium atom has two electrons in the Coulomb field of a  $Z = 2$  nucleus:

$$H = -\frac{\hbar^2}{2m}(\nabla_1^2 + \nabla_2^2) - \frac{2e^2}{4\pi\epsilon_0} \left( \frac{1}{r_1} + \frac{1}{r_2} \right) + \frac{e^2}{4\pi\epsilon_0} \frac{1}{r_{12}} \quad (9.105)$$

The electron-electron repulsion makes this problem unsolvable analytically. We use a trial function where each electron occupies a hydrogen-like  $1s$  orbital with effective charge  $Z_{\text{eff}}$ :

$$\psi(r_1, r_2) = \frac{Z_{\text{eff}}^3}{\pi a_0^3} e^{-Z_{\text{eff}}(r_1+r_2)/a_0} \quad (9.106)$$

The effective charge accounts for screening: each electron partially shields the other from the nucleus.

The energy expectation value can be computed analytically:

$$E(Z_{\text{eff}}) = \left( Z_{\text{eff}}^2 - 2Z \cdot Z_{\text{eff}} + \frac{5}{8} Z_{\text{eff}} \right) \frac{e^2}{4\pi\epsilon_0 a_0} \quad (9.107)$$

For helium ( $Z = 2$ ):

$$E(Z_{\text{eff}}) = \left( Z_{\text{eff}}^2 - 4Z_{\text{eff}} + \frac{5}{8} Z_{\text{eff}} \right) \text{Ry} = \left( Z_{\text{eff}}^2 - \frac{27}{8} Z_{\text{eff}} \right) \text{Ry} \quad (9.108)$$

Minimizing:  $\frac{dE}{dZ_{\text{eff}}} = 2Z_{\text{eff}} - \frac{27}{8} = 0$ , giving  $Z_{\text{eff}} = \frac{27}{16} = 1.6875$ .

The variational energy:

$$E_{\min} = - \left( \frac{27}{16} \right)^2 \text{Ry} = -2.848 \text{ Ry} = -77.5 \text{ eV} \quad (9.109)$$

The experimental value is  $-78.975 \text{ eV}$ , so the simple one-parameter variational calculation gives 98% of the correct binding energy.

*Context:* Variational methods are the foundation of computational quantum chemistry. Hartree-Fock, density functional theory (DFT), and coupled-cluster methods all build on variational principles. Variational quantum eigensolvers (VQE) are leading algorithms for near-term quantum computers, using parameterized quantum circuits as trial wave functions. Tensor network methods in condensed matter physics also rely on variational optimization.

## 9.9 WKB Approximation

The WKB (Wentzel-Kramers-Brillouin) approximation provides semiclassical wave functions valid when the potential varies slowly on the scale of the de Broglie wavelength.

### 9.9.1 The WKB Wave Function

In classically allowed regions where  $E > V(x)$ , the local momentum is  $p(x) = \sqrt{2m(E - V(x))}$ . The WKB wave function is:

$$\psi(x) \approx \frac{C}{\sqrt{p(x)}} \exp\left(\pm \frac{i}{\hbar} \int^x p(x') dx'\right) \quad (9.110)$$

In classically forbidden regions where  $E < V(x)$ ,  $p(x)$  becomes imaginary:

$$\psi(x) \approx \frac{C}{\sqrt{|p(x)|}} \exp\left(\pm \frac{1}{\hbar} \int^x |p(x')| dx'\right) \quad (9.111)$$

### 9.9.2 Bohr-Sommerfeld Quantization

For bound states between classical turning points  $a$  and  $b$ , the quantization condition is:

$$\oint p dx = 2 \int_a^b p(x) dx = 2\pi\hbar \left(n + \frac{1}{2}\right) \quad (9.112)$$

The  $\frac{1}{2}$  arises from connection formulas at the turning points.

#### Example 16

**WKB for the Harmonic Oscillator** For  $V(x) = \frac{1}{2}m\omega^2 x^2$ , the classical turning points at energy  $E$  are  $x = \pm\sqrt{2E/m\omega^2}$ .

The local momentum:  $p(x) = \sqrt{2m(E - \frac{1}{2}m\omega^2 x^2)} = \sqrt{2mE} \sqrt{1 - \frac{m\omega^2 x^2}{2E}}$ .

The quantization integral:

$$\int_{-a}^a p(x) dx = \sqrt{2mE} \int_{-a}^a \sqrt{1 - x^2/a^2} dx = \sqrt{2mE} \cdot \frac{\pi a}{2} \quad (9.113)$$

where  $a = \sqrt{2E/m\omega^2}$ .

Substituting:

$$\sqrt{2mE} \cdot \frac{\pi}{2} \sqrt{\frac{2E}{m\omega^2}} = \frac{\pi E}{\omega} \quad (9.114)$$



The quantization condition  $\frac{2\pi E}{\omega} = 2\pi\hbar(n + \frac{1}{2})$  gives:

$$E_n = \hbar\omega \left(n + \frac{1}{2}\right) \quad (9.115)$$

WKB gives the exact harmonic oscillator spectrum! This is another special case where the semiclassical approximation is exact.

### Example 17

**WKB for Hydrogen: Radial Quantization** Working in atomic units, the radial Schrödinger equation for hydrogen can be written as a one-dimensional problem:

$$-\frac{1}{2} \frac{d^2 u}{dr^2} + V_{\text{eff}}(r)u = Eu \quad (9.116)$$

where  $u = rR(r)$  and

$$V_{\text{eff}}(r) = -\frac{1}{r} + \frac{\ell(\ell+1)}{2r^2} \quad (9.117)$$

The classical turning points  $r_1$  and  $r_2$  satisfy  $E = V_{\text{eff}}(r)$ . The WKB quantization:

$$\int_{r_1}^{r_2} \sqrt{2(E - V_{\text{eff}}(r))} dr = \pi \left(n_r + \frac{1}{2}\right) \quad (9.118)$$

where  $n_r = 0, 1, 2, \dots$  is the radial quantum number.

This integral can be evaluated using contour integration. The result is:

$$\frac{1}{\sqrt{-2E}} - \left(\ell + \frac{1}{2}\right) = \left(n_r + \frac{1}{2}\right) \quad (9.119)$$

Defining  $n = n_r + \ell + 1$ :

$$E_n = -\frac{1}{2n^2} \quad (9.120)$$

which is the exact hydrogen spectrum in atomic units (where the ground state energy is  $-1/2$  Hartree  $= -1$  Rydberg).

WKB reproduces the exact hydrogen spectrum.

*Context:* Semiclassical methods bridge quantum and classical descriptions, essential for understanding the quantum-classical correspondence. WKB tunneling rates govern alpha decay, scanning tunneling microscopy, and Josephson junction dynamics. The method extends to classically chaotic systems through trace formulas connecting periodic orbits to quantum spectra.

## 9.10 The Complete Hydrogen Atom

Having developed perturbation theory techniques, we now apply them systematically to understand the real hydrogen atom beyond the idealized Coulomb solution. This synthesis demonstrates how perturbation theory connects textbook quantum mechanics to precision atomic physics.

### 9.10.1 The Hierarchy of Energy Scales

The hydrogen atom exhibits a beautiful hierarchy of energy scales, each probing different physics:

| Effect              | Energy Scale                                   | Origin                  |
|---------------------|------------------------------------------------|-------------------------|
| Gross structure     | $Ry \approx 13.6 \text{ eV}$                   | Coulomb interaction     |
| Fine structure      | $\alpha^2 Ry \sim 10^{-4} \text{ eV}$          | Relativity + spin-orbit |
| Lamb shift          | $\alpha^3 Ry \sim 10^{-6} \text{ eV}$          | QED vacuum fluctuations |
| Hyperfine structure | $(m_e/m_p)\alpha^2 Ry \sim 10^{-7} \text{ eV}$ | Nuclear magnetic moment |

Here  $\alpha = e^2/(4\pi\epsilon_0\hbar c) \approx 1/137$  is the fine structure constant.

#### Physical Insight

The hierarchy arises because different interactions involve different combinations of fundamental constants. The fine structure constant  $\alpha$  controls relativistic corrections, the electron-to-proton mass ratio governs nuclear effects, and higher powers of  $\alpha$  indicate quantum electrodynamic corrections. This pattern of hierarchical scales, where each level of structure involves smaller energy scales, recurs throughout physics and underlies the effective field theory approach to modern particle physics.

### 9.10.2 Fine Structure in Detail

The fine structure Hamiltonian consists of three terms:

$$H_{\text{fine}} = H_{\text{rel}} + H_{SO} + H_{\text{Darwin}} \quad (9.121)$$

#### 1. Relativistic kinetic energy correction:

The relativistic kinetic energy  $T = \sqrt{p^2 c^2 + m^2 c^4} - mc^2$  expands as:

$$T = \frac{p^2}{2m} - \frac{p^4}{8m^3 c^2} + \dots \quad (9.122)$$

The first-order perturbation:

$$E_{\text{rel}}^{(1)} = -\frac{1}{2mc^2} \langle \psi_{n\ell m} | \frac{p^4}{4m^2} | \psi_{n\ell m} \rangle \quad (9.123)$$

Using  $p^2/2m = E_n - V(r)$  and the known expectation values  $\langle r^{-1} \rangle$  and  $\langle r^{-2} \rangle$ :

$$E_{\text{rel}}^{(1)} = -\frac{E_n^2}{2mc^2} \left( \frac{4n}{\ell + 1/2} - 3 \right) \quad (9.124)$$

#### 2. Spin-orbit coupling:

From the electron's rest frame, the nucleus appears to orbit, creating a magnetic field that couples to the electron spin:

$$H_{SO} = \frac{1}{2m_e^2 c^2} \frac{1}{r} \frac{dV}{dr} \vec{L} \cdot \vec{S} = \frac{e^2}{8\pi\epsilon_0 m_e^2 c^2 r^3} \vec{L} \cdot \vec{S} \quad (9.125)$$

Using the coupled basis  $|n, \ell, j, m_j\rangle$  where  $\vec{L} \cdot \vec{S} = \frac{1}{2}(J^2 - L^2 - S^2)$ :

$$E_{SO}^{(1)} = \frac{e^2}{16\pi\epsilon_0 m_e^2 c^2} \langle r^{-3} \rangle_{n\ell} [j(j+1) - \ell(\ell+1) - \frac{3}{4}] \hbar^2 \quad (9.126)$$

For  $\ell > 0$ , this gives different energies for  $j = \ell + \frac{1}{2}$  and  $j = \ell - \frac{1}{2}$ .

### 3. Darwin term:

The Darwin term arises from the “zitterbewegung” of the electron (rapid oscillation at the Compton wavelength scale) and affects only  $s$ -states:

$$H_{\text{Darwin}} = \frac{\pi \hbar^2}{2m_e^2 c^2} \frac{e^2}{4\pi \epsilon_0} \delta^3(\vec{r}) \quad (9.127)$$

For  $s$ -states:  $E_{\text{Darwin}}^{(1)} = \frac{\pi \hbar^2 e^2}{8\pi \epsilon_0 m_e^2 c^2} |\psi_{n00}(0)|^2$ .

### Combined fine structure:

Remarkably, all three contributions combine to give a simple formula that depends only on  $n$  and  $j$ :

$$E_{\text{fine}} = -\frac{\alpha^2 \text{Ry}}{n^3} \left( \frac{1}{j + 1/2} - \frac{3}{4n} \right) \quad (9.128)$$

where  $\text{Ry} \approx 13.6 \text{ eV}$  is the Rydberg energy. This is the Sommerfeld fine structure formula. Using  $E_n = -\text{Ry}/n^2$ , it can equivalently be written  $E_{\text{fine}} = (\alpha^2 E_n/n) [1/(j + 1/2) - 3/(4n)]$ , which is negative because  $E_n < 0$ .

For  $n = 2$ :

- $2s_{1/2}$  ( $\ell = 0, j = 1/2$ ):  $E_{\text{fine}} = -\frac{5\alpha^2}{64} \text{Ry}$
- $2p_{1/2}$  ( $\ell = 1, j = 1/2$ ):  $E_{\text{fine}} = -\frac{5\alpha^2}{64} \text{Ry}$
- $2p_{3/2}$  ( $\ell = 1, j = 3/2$ ):  $E_{\text{fine}} = -\frac{\alpha^2}{64} \text{Ry}$

The  $2s_{1/2}$  and  $2p_{1/2}$  states are predicted to be degenerate, a degeneracy broken only by the Lamb shift, which requires quantum electrodynamics.

*Context:* The fine structure constant  $\alpha$  governs the strength of electromagnetic interactions and appears throughout atomic physics. Spin-orbit coupling is the mechanism underlying spintronic devices and topological insulators. In artificial atoms (quantum dots), the “fine structure” splitting is tunable, enabling engineered spin-photon interfaces for quantum networks.

### 9.10.3 The Zeeman Effect: Weak, Strong, and Intermediate Fields

When hydrogen is placed in a magnetic field, the choice of perturbative approach depends on the field strength relative to fine structure splitting.

#### Weak-field Zeeman effect (Anomalous Zeeman effect):

When  $\mu_B B \ll E_{\text{fine}}$ , the fine structure dominates. We treat  $H_{\text{Zeeman}} = \frac{\mu_B}{\hbar} (L_z + 2S_z) B$  as a perturbation on states labeled by  $|n, \ell, j, m_j\rangle$ .

Since  $J_z = L_z + S_z$  but not  $L_z$  and  $S_z$  individually are good quantum numbers, we use:

$$\langle L_z + 2S_z \rangle = \langle J_z \rangle + \langle S_z \rangle = \hbar m_j + \langle S_z \rangle \quad (9.129)$$

The projection theorem gives  $\langle S_z \rangle = m_j \hbar \cdot \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)}$ .

The energy shift:

$$E_{\text{Zeeman}}^{(1)} = \mu_B B m_j g_j \quad (9.130)$$

where the Landé  $g$ -factor is:

$$g_j = 1 + \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)} \quad (9.131)$$

For  $j = \ell + \frac{1}{2}$ :  $g_j = 1 + \frac{1}{2\ell+1}$ . For  $j = \ell - \frac{1}{2}$ :  $g_j = 1 - \frac{1}{2\ell+1}$ .

**Strong-field Zeeman effect (Paschen-Back effect):**

When  $\mu_B B \gg E_{\text{fine}}$ , the magnetic field dominates. The good quantum numbers are  $m_\ell$  and  $m_s$  (uncoupled basis), and fine structure becomes a perturbation:

$$E = E_n + \mu_B B(m_\ell + 2m_s) + E_{\text{fine}}^{(1)}(m_\ell, m_s) \quad (9.132)$$

**Intermediate fields:**

When  $\mu_B B \sim E_{\text{fine}}$ , neither basis diagonalizes the total perturbation. We must diagonalize the combined  $H_{\text{fine}} + H_{\text{Zeeman}}$  within the degenerate subspace.

**Example 18**

$n = 2$  Hydrogen in an Intermediate Magnetic Field Consider the  $n = 2$  states (ignoring the  $2p_{3/2}$  for simplicity). The  $2s_{1/2}$  and  $2p_{1/2}$  states each have  $m_j = \pm 1/2$ .

In the basis  $\{|2s_{1/2}, +\frac{1}{2}\rangle, |2s_{1/2}, -\frac{1}{2}\rangle, |2p_{1/2}, +\frac{1}{2}\rangle, |2p_{1/2}, -\frac{1}{2}\rangle\}$ , the total perturbation matrix is:

$$H' = \begin{pmatrix} E_s + \mu_B B g_s \cdot \frac{1}{2} & 0 & 0 & 0 \\ 0 & E_s - \mu_B B g_s \cdot \frac{1}{2} & 0 & 0 \\ 0 & 0 & E_p + \mu_B B g_p \cdot \frac{1}{2} & 0 \\ 0 & 0 & 0 & E_p - \mu_B B g_p \cdot \frac{1}{2} \end{pmatrix} \quad (9.133)$$

where  $g_s = 2$  (pure spin) and  $g_p = 2/3$  for  $2p_{1/2}$ .

Without the Lamb shift,  $E_s = E_p$ , and states with the same  $m_j$  would cross. The Lamb shift ( $E_s > E_p$  by about 1058 MHz) creates an avoided crossing, and at intermediate fields, states are superpositions of  $s$  and  $p$  character.

*Context:* Magneto-optical traps (MOTs) exploit Zeeman shifts combined with laser detuning to create position-dependent forces that cool and trap atoms. This technique launched the ultracold atom revolution. Zeeman slows decelerate atomic beams for loading MOTs. In astrophysics, the Paschen-Back regime is relevant for white dwarf and neutron star atmospheres, where magnetic fields reach  $10^8$ – $10^{11}$  T.

#### 9.10.4 Hyperfine Structure

The proton has a magnetic moment  $\vec{\mu}_p = g_p \frac{e}{2m_p} \vec{I}$  where  $\vec{I}$  is the nuclear spin ( $I = 1/2$  for hydrogen) and  $g_p \approx 5.59$ . This creates a magnetic field that couples to the electron spin.

The hyperfine Hamiltonian is:

$$H_{\text{hf}} = A \vec{I} \cdot \vec{S} \quad (9.134)$$

where  $A$  depends on the electron wave function at the nucleus.

For the ground state:

$$A_{1s} = \frac{2}{3} \mu_0 g_p \frac{e^2}{4\pi m_e m_p} |\psi_{1s}(0)|^2 = \frac{8}{3} g_p \frac{m_e}{m_p} \alpha^2 E_1 \quad (9.135)$$

Coupling  $\vec{I}$  and  $\vec{S}$  gives total spin  $\vec{F} = \vec{I} + \vec{S}$ :

$$\vec{I} \cdot \vec{S} = \frac{1}{2}(F^2 - I^2 - S^2) = \frac{\hbar^2}{2}[F(F+1) - I(I+1) - S(S+1)] \quad (9.136)$$

For hydrogen ( $I = S = 1/2$ ),  $F = 0$  or  $F = 1$ :

$$F = 1 : \quad E_{\text{hf}} = +\frac{A\hbar^2}{4} \quad (9.137)$$

$$F = 0 : \quad E_{\text{hf}} = -\frac{3A\hbar^2}{4} \quad (9.138)$$

The hyperfine splitting of the ground state:

$$\Delta E_{\text{hf}} = A\hbar^2 = 5.87 \times 10^{-6} \text{ eV} \quad \Leftrightarrow \quad \nu = 1420.4 \text{ MHz} \quad \Leftrightarrow \quad \lambda = 21 \text{ cm} \quad (9.139)$$

*Context:* The 21 cm hydrogen line is radio astronomy’s most important spectral line. It maps the distribution of neutral hydrogen throughout the universe, probes the cosmic “dark ages” before the first stars, and tests cosmological models. Closer to home, the hyperfine transition in cesium-133 defines the SI second, and hyperfine states in trapped ions (such as  $^{171}\text{Yb}^+$  and  $^{133}\text{Ba}^+$ ) serve as qubit states in leading quantum computing platforms. The hydrogen maser, based on stimulated emission of the 21 cm transition, remains one of the most stable frequency references available.

### 9.10.5 Summary: When Different Methods Apply

The hydrogen atom illustrates how to choose among perturbative approaches:

| Situation                                        | Method                             |
|--------------------------------------------------|------------------------------------|
| Non-degenerate level, small perturbation         | Standard perturbation theory       |
| Degenerate level, perturbation breaks degeneracy | Degenerate perturbation theory     |
| Competing perturbations of similar size          | Full matrix diagonalization        |
| Perturbation commutes with some $H_0$ eigenbasis | Choose that basis                  |
| Oscillating perturbation, transitions            | Time-dependent perturbation theory |
| Transition to continuum                          | Fermi’s golden rule                |
| Fast change                                      | Sudden approximation               |
| Slow change                                      | Adiabatic theorem                  |
| No small parameter                               | Variational method                 |

The art of perturbation theory lies in recognizing which regime applies and choosing the appropriate basis to simplify calculations.

## 9.11 Chapter Summary

Perturbation theory provides powerful tools for approximate solutions throughout quantum mechanics and beyond. Time-independent perturbation theory handles energy shifts and state mixing in static systems, with first-order shifts determined by expectation values and second-order shifts by virtual transitions to intermediate states. Degenerate perturbation theory addresses situations when symmetries are broken, requiring diagonalization within the degenerate subspace to find the appropriate zeroth-order states and good quantum numbers. Time-dependent perturbation theory describes transitions between states driven by external fields. Fermi’s golden rule gives transition rates to continua of final states, underlying phenomena from spontaneous emission to scattering. The sudden and adiabatic approximations handle the limiting cases of fast and slow parameter changes. Variational methods provide upper bounds on ground state energies without requiring a small expansion parameter. The WKB approximation treats semiclassical systems with slowly varying potentials, connecting quantum mechanics to classical trajectories.

The hydrogen atom, with its hierarchy of corrections from gross structure through fine structure to hyperfine structure, exemplifies how these methods work together. Choice of basis (coupled versus uncoupled, weak-field versus strong-field) reflects physical insight about which quantum numbers remain good under the perturbation at hand. These same principles extend to multi-electron atoms, molecules, solids, and nuclei, making perturbation theory one of the most widely used tools in theoretical physics.

## 9.12 Problems

**Note:** Throughout these problems, we use atomic units ( $\hbar = m_e = e = 1$ ) unless otherwise specified.

### Problem 9.1 (*Non-Degenerate Perturbation Theory: Infinite Square Well*)

A particle in the ground state of an infinite square well of width  $L$  (walls at  $x = 0$  and  $x = L$ ) experiences a perturbation  $V(x) = V_0 \sin(\pi x/L)$ .

- Calculate the first-order energy correction  $E_1^{(1)}$  for the ground state.
- Calculate the first-order energy correction  $E_2^{(1)}$  for the first excited state. Why is this qualitatively different from part (a)?
- Calculate the second-order energy correction to the ground state energy. Which excited state contributes most?
- For what value of  $V_0$  relative to  $E_1^{(0)}$  does perturbation theory break down? Give a quantitative criterion.

### Problem 9.2 (*Non-Degenerate Perturbation Theory: Anharmonic Oscillator*)

A harmonic oscillator has a cubic perturbation:  $H = p^2/(2m) + m\omega^2 x^2/2 + \lambda x^3$ . Use  $x = (x_0/\sqrt{2})(a + a^\dagger)$  with  $x_0 = \sqrt{\hbar/(m\omega)}$ .

- Show that  $E_n^{(1)} = \lambda \langle n|x^3|n \rangle$  vanishes for every  $n$ .
- Calculate  $(a + a^\dagger)^3|0\rangle$  iteratively.
- Compute the matrix elements  $\langle 1|x^3|0 \rangle$  and  $\langle 3|x^3|0 \rangle$ .
- Calculate the second-order ground state energy correction  $E_0^{(2)}$  and express it in terms of  $\lambda$ ,  $x_0$ , and  $\hbar\omega$ .
- Although the potential is unbounded below for large  $|x|$ , the perturbation series gives useful results for small  $\lambda$ . Explain why physically.

### Problem 9.3 (*Quadratic Stark Effect in Hydrogen*)

The hydrogen ground state  $|1s\rangle$  (energy  $E_1 = -1/2$  in atomic units) is placed in a uniform electric field  $\mathcal{E}$  along  $z$ , giving perturbation  $V = \mathcal{E}z$ .

- Show that  $E_1^{(1)} = \mathcal{E} \langle 1s|z|1s \rangle = 0$ . What symmetry is responsible?
- The second-order correction is  $E_1^{(2)} = \sum_{n\ell m} |\mathcal{E} \langle n, \ell, m|z|1s \rangle|^2 / (E_1 - E_n)$ . Explain why only  $p$ -states ( $\ell = 1, m = 0$ ) contribute. Cite two selection rules.

- (c) The exact result is  $E_1^{(2)} = -\frac{9}{4}\mathcal{E}^2$ , giving polarizability  $\alpha_H = 9/2$  in atomic units. Express  $\alpha_H$  in SI units and calculate its numerical value.
- (d) Show that replacing all  $E_n$  in the denominator with  $E_2$  gives an upper bound on  $|E_1^{(2)}|$ . What polarizability does this yield, and how does it compare to the exact value?

**Problem 9.4** (*Degenerate Perturbation Theory: Four-State System*)

A system with four states and unperturbed energies  $E_1^{(0)} = E_3^{(0)} = 0$  and  $E_2^{(0)} = E_4^{(0)} = 10$  has perturbation matrix:

$$V \approx \begin{pmatrix} 2 & 0.3 & 1 & 0 \\ 0.3 & -1 & 0 & 2 \\ 1 & 0 & -2 & 0.5 \\ 0 & 2 & 0.5 & 3 \end{pmatrix}$$

- (a) Identify the two degenerate subspaces and extract the  $2 \times 2$  perturbation submatrices.
- (b) Diagonalize the  $E^{(0)} = 0$  subspace to find  $E_{\pm}^{(1)}$  and the correct zeroth-order eigenstates.
- (c) Diagonalize the  $E^{(0)} = 10$  subspace to find its first-order corrections.
- (d) Sketch the energy level diagram showing how the two degenerate pairs split. Are any levels still degenerate?
- (e) Estimate whether the second-order corrections from off-diagonal elements between subspaces are negligible.

**Problem 9.5** (*Two Coupled Spins with Perturbation*)

Two spin-1/2 particles interact via  $H = J\vec{S}_1 \cdot \vec{S}_2 + \epsilon S_{1z}$  where  $\epsilon \ll J$  is a perturbation.

- (a) Show that the unperturbed energies are  $E_{\text{singlet}} = -3J/4$  and  $E_{\text{triplet}} = J/4$  (three-fold degenerate).
- (b) Construct the  $3 \times 3$  perturbation matrix for the triplet subspace using the Clebsch-Gordan decomposition of triplet states.
- (c) Diagonalize this matrix to find the first-order splittings of the three triplet states.
- (d) Does the perturbation mix singlet and triplet? Calculate  $\langle 0,0|S_{1z}|1,0\rangle$  to check. At what order in perturbation theory does this mixing enter?

**Problem 9.6** (*Linear Stark Effect in Hydrogen  $n = 2$* )

The  $n = 2$  level of hydrogen (ignoring spin) is 4-fold degenerate:  $|2s\rangle$ ,  $|2p_z\rangle$ ,  $|2p_+\rangle$ ,  $|2p_-\rangle$ . An electric field  $\mathcal{E}$  along  $z$  gives perturbation  $V = \mathcal{E}z$ .

- (a) Using selection rules  $\Delta m = 0$  and  $\Delta \ell = \pm 1$ , show that the only nonzero matrix element within  $n = 2$  is  $\langle 2s|z|2p_z\rangle$ .
- (b) Verify that  $\langle 2s|z|2p_z\rangle = -3$  (atomic units) using the given radial wave functions  $R_{20}(r) = \frac{1}{2\sqrt{2}}(2-r)e^{-r/2}$  and  $R_{21}(r) = \frac{1}{2\sqrt{6}}re^{-r/2}$ .
- (c) Write out the full  $4 \times 4$  perturbation matrix. Show that it is block-diagonal in  $m$ .

- (d) Find the four first-order energy corrections and the correct zeroth-order eigenstates.
- (e) The states  $|\pm\rangle = (|2s\rangle \mp |2p_z\rangle)/\sqrt{2}$  are  $sp$ -hybrid orbitals with permanent dipole moments. Calculate  $\langle \pm | z | \pm \rangle$ . How is this possible for parity eigenstates?

**Problem 9.7** (*Deriving Fermi's Golden Rule*)

For a constant perturbation  $V$  suddenly switched on at  $t = 0$ , the first-order transition probability is:

$$P_{i \rightarrow f}(t) = \frac{|\langle f | V | i \rangle|^2}{\hbar^2} \frac{\sin^2(\omega_{fi}t/2)}{(\omega_{fi}/2)^2}$$

where  $\omega_{fi} = (E_f - E_i)/\hbar$ .

- (a) Sketch  $P_{i \rightarrow f}$  as a function of  $\omega_{fi}$  for fixed  $t$ . What is the peak height and width? How do these scale with  $t$ ?
- (b) For a continuum of final states with density  $\rho(E_f)$ , show that the total probability is  $P(t) = \frac{2\pi}{\hbar} |\langle f | V | i \rangle|^2 \rho(E_i) t$  by using  $\int_{-\infty}^{\infty} \frac{\sin^2(xt/2)}{(x/2)^2} dx = 2\pi t$ .
- (c) Derive the transition rate  $\Gamma = dP/dt$  (Fermi's golden rule):  $\Gamma = \frac{2\pi}{\hbar} |\langle f | V | i \rangle|^2 \rho(E_i)$ . Explain why a constant rate requires a continuum.
- (d) Show that Fermi's golden rule requires both  $t$  large (sinc-squared acts like delta function) and  $t$  small ( $P \ll 1$ ), but these conditions are compatible when  $\Gamma \ll \Delta E/\hbar$ .

**Problem 9.8** (*Sudden and Adiabatic: Harmonic Oscillator Frequency Change*)

A harmonic oscillator in its ground state with frequency  $\omega_1$  suddenly changes to frequency  $\omega_2 = 4\omega_1$ .

- (a) **Sudden limit:** Calculate the probability of remaining in the ground state and the probability of reaching the first excited state. Use parity to argue that only even- $n$  final states are accessible.
- (b) **Adiabatic limit:** What is the final-state probability and energy? What is the adiabatic phase?
- (c) The energy expectation immediately after the change is  $\langle H' \rangle = \langle \psi_0^{(1)} | H' | \psi_0^{(1)} \rangle$  where  $H' = p^2/(2m) + m\omega_2^2 x^2/2$ . Calculate  $\langle H' \rangle$  in both limits and interpret.
- (d) Prove that in the sudden case, the particle can only be found in even- $n$  states of the new oscillator (parity conservation).

**Problem 9.9** (*Variational Method: Helium Ground State*)

Two electrons in helium interact via  $H = -\nabla_1^2/2 - \nabla_2^2/2 - 2/r_1 - 2/r_2 + 1/r_{12}$  (atomic units).

- (a) Use a trial wave function  $\psi(r_1, r_2) = (Z^3/\pi)^{1/2} e^{-Zr_1} (Z^3/\pi)^{1/2} e^{-Zr_2}$  where  $Z$  is the variational parameter. Calculate  $\langle H \rangle$  as a function of  $Z$ .
- (b) Show that the energy is  $E(Z) = -2Z^2 + (5/8)Z$  (the electron-electron repulsion integral yields  $5Z/8$ ).
- (c) Minimize to find the optimal  $Z_{\text{eff}}$  and the resulting ground state energy. Compare to the experimental value of  $-5.81$  eV.



- (d) Explain physically why  $Z_{\text{eff}} < 2$ . What does the second term in  $E(Z)$  represent?

**Problem 9.10** (*WKB Approximation: Tunneling and Quantization*)

- (a) For a potential  $V(x) = |x|$  (linear potential), find the WKB quantization condition  $\oint p(x)dx = 2\pi n\hbar$  for bound states. What are the allowed energies?
- (b) For potential  $V(x) = |x|^\alpha$  with  $\alpha > 0$ , calculate the leading-order WKB spectrum and identify which potentials give equally spaced levels.
- (c) A particle with energy  $E$  incident on a step potential  $V(x) = V_0$  (for  $x > 0$ ) has WKB transmission coefficient  $T \approx e^{-2\gamma}$  where  $\gamma = \int_0^d \sqrt{2m(V_0 - E)}/\hbar dx$  is the tunneling exponent. For a rectangular barrier of height  $V_0 = 5$  eV and width  $d = 1$  Å, calculate the transmission for  $E = 4$  eV.

**Problem 9.11** (*Born Approximation and Scattering Amplitude*)

The scattering amplitude in the Born approximation is  $f(\vec{q}) = -\frac{m}{4\pi}(2\pi)^3 \tilde{V}(\vec{q})$  where  $\tilde{V}$  is the Fourier transform of the potential and  $\vec{q} = \vec{k}_i - \vec{k}_f$  is the momentum transfer.

- (a) For a Yukawa potential  $V(r) = -\frac{g^2}{4\pi}e^{-\mu r}/r$ , calculate the Fourier transform  $\tilde{V}(\vec{q})$ .
- (b) Find the scattering amplitude  $f(\theta)$  and the differential cross section  $d\sigma/d\Omega = |f(\theta)|^2$  in terms of the scattering angle  $\theta$ .
- (c) For low-energy scattering ( $\mu \ll k$ ), show that the  $s$ -wave phase shift is  $\delta_0 \approx -ka_s$  where  $a_s$  is the scattering length. Relate  $a_s$  to the potential.

**Problem 9.12** (*Rotating Wave Approximation and Rabi Oscillations*)

A two-level atom with transition frequency  $\omega_0$  is driven by an oscillating field  $\vec{E}(t) = E_0 \cos(\omega t)\hat{z}$ , producing perturbation  $V(t) = -dE_0 \cos(\omega t)$  where  $d$  is the dipole moment.

- (a) Decompose  $V(t)$  into rotating and counter-rotating terms:  $V = V^{\text{rot}}e^{-i\omega t} + \text{c.c.}$  plus counter-rotating terms oscillating at  $2\omega$ .
- (b) In the rotating frame with  $U = e^{-i\omega_0 t\sigma_z/2}$ , show that the counter-rotating terms oscillate at  $2\omega$  (far off-resonance) and can be dropped if  $|\omega - \omega_0| \ll \omega$ . This is the rotating wave approximation.
- (c) In the RWA, the Hamiltonian becomes  $H_{\text{RWA}} = \Delta\sigma_z/2 + \Omega\sigma_x$  where  $\Delta = \omega - \omega_0$  is the detuning and  $\Omega = dE_0$  is the Rabi frequency.
- (d) For resonant driving ( $\Delta = 0$ ), show that the population oscillates between states:  $P_e(t) = \sin^2(\Omega t/2)$ . At what time does the system achieve maximum transition probability (a  $\pi$  pulse)?

## References and Further Reading

**Sakurai, J. J., and Napolitano, J.** *Modern Quantum Mechanics*, 3rd ed. Cambridge University Press, 2017. Chapter 5 covers time-independent and time-dependent perturbation theory at the level of this chapter; the standard graduate reference for both the formalism and the hydrogen fine-structure application.

- Cohen-Tannoudji, C., Diu, B., and Laloë, F.** *Quantum Mechanics*, Vol. 2. Wiley-VCH, 1977. Chapters XI–XIII work through stationary, degenerate, and time-dependent perturbation theory with worked examples; the most thorough textbook treatment, particularly clear on degenerate perturbation theory and good quantum numbers.
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- Landau, L. D., and Lifshitz, E. M.** *Quantum Mechanics: Non-Relativistic Theory*, 3rd ed. Pergamon Press, 1977. Chapter VI on perturbation theory and Chapter VII on the WKB approximation are characteristically terse and physically motivated; recommended for the variational and semiclassical methods touched on at the end of this chapter.

## Chapter 10

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# Entanglement and Bell's Theorem

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### 10.1 Introduction

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When a qubit is in the state  $|+x\rangle$  and we measure  $\sigma_z$ , we get  $+1$  or  $-1$  with equal probability. Why? One natural explanation is that the qubit actually *has* a definite  $z$ -spin value; we simply don't know what it is. The state  $|+x\rangle$  would then be an incomplete description, reflecting our ignorance rather than any genuine indeterminacy in nature.

This is not a naive idea. It is a precise physical hypothesis called *realism*: that measurement reveals pre-existing values rather than creating them. Einstein held this view, and it captures a deep intuition about how the physical world works.

In this chapter, we will show that realism works perfectly well for a single qubit. We will construct an explicit model in which hidden Bloch cubes encode definite answers to multiple spin measurements simultaneously. The Bloch cube provides a natural taxonomy of measurement directions—face, edge, and corner—and hidden variable models can be built for each. The quantum mechanical probabilities emerge from our ignorance of which cubes the particle carries.

The model fails only when we consider *entangled* systems and impose *locality*, the requirement that a measurement on one particle cannot instantaneously affect a distant particle. Bell's theorem proves that no model can be simultaneously realistic and local while reproducing the predictions of quantum mechanics. The measurement directions that maximally violate Bell's inequality turn out to be faces and edges of the Bloch cube—a connection that makes the geometry of the violation visible. This is not a philosophical statement; it is a mathematical theorem with experimental consequences, and the experiments have spoken decisively.

### 10.2 Hidden Variables for a Single Qubit

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#### 10.2.1 What Realism Requires

Consider a spin-1/2 particle in state  $|+x\rangle$ . Quantum mechanics tells us nothing definite about the outcome of a  $\sigma_z$  measurement, assigning equal probabilities to  $\pm 1$ . A realist interpretation says that this uncertainty reflects our ignorance: the particle possesses a definite value of  $\sigma_z$ , determined by some variable  $\lambda$  that quantum mechanics fails to track.

More ambitiously, a realist would like every spin component to have a definite value simultaneously. The Bloch cube from Chapter 1 provides a natural language for making this idea precise.

### 10.2.2 Hidden Bloch Cubes

Recall that the Bloch cube has six faces corresponding to the states  $|\pm x\rangle$ ,  $|\pm y\rangle$ , and  $|\pm z\rangle$ . A quantum state like  $|+z\rangle$  specifies a definite outcome for  $\sigma_z$  measurements but leaves  $\sigma_x$  and  $\sigma_y$  undetermined. One Bloch cube, oriented with  $|+z\rangle$  facing up, tells us the answer to the  $Z$  question but says nothing definite about  $X$  or  $Y$ .

Now consider your viewing angle. Looking straight down at the top face, you see  $|+z\rangle$  and nothing else. The four side faces ( $|\pm x\rangle$  and  $|\pm y\rangle$ ) are hidden by perspective. But tilt your line of sight toward one of the four upper corners, and two side faces come into view. The corner where the  $+x$ ,  $+y$ , and  $+z$  faces meet reveals both  $|+x\rangle$  and  $|+y\rangle$ . Rotate the cube  $90^\circ$  about the vertical axis and you see the corner where  $-x$ ,  $+y$ , and  $+z$  meet, revealing  $|-x\rangle$  and  $|+y\rangle$ . Another  $90^\circ$  shows  $|-x\rangle$  and  $|-y\rangle$ ; another shows  $|+x\rangle$  and  $|-y\rangle$ . The four corners of the top face correspond to exactly four combinations of hidden answers:

$$(|+x\rangle, |+y\rangle), \quad (|-x\rangle, |+y\rangle), \quad (|-x\rangle, |-y\rangle), \quad (|+x\rangle, |-y\rangle) \quad (10.1)$$

A hidden variable theory says that the particle's complete state is specified not just by the visible top face but by the particular corner you would see if you tilted your view. Each corner encodes definite answers to the  $X$  and  $Y$  questions that quantum mechanics refuses to provide. The complete description of a single particle is therefore a triple:

$$\underbrace{(|+z\rangle)}_{\text{visible}}, \underbrace{(|\pm x\rangle)}_{\text{hidden}}, \underbrace{(|\pm y\rangle)}_{\text{hidden}} \quad (10.2)$$

When we measure  $\sigma_z$ , we read the visible face and get  $+1$  with certainty. When we measure  $\sigma_x$  or  $\sigma_y$ , we read the corresponding hidden answer encoded in the corner. There is no randomness in any individual measurement. Each outcome is predetermined. The apparent randomness of quantum mechanics arises because we cannot see which corner the cube is oriented toward.

### 10.2.3 Reproducing Quantum Predictions

For this picture to work, the hidden corners must be distributed in a way that reproduces the quantum probabilities. Consider an ensemble of particles all prepared in  $|+z\rangle$ . Every particle has the same visible face, but the corner orientation varies from particle to particle:

| Hidden $X$ answer | Hidden $Y$ answer | Probability |
|-------------------|-------------------|-------------|
| $ +x\rangle$      | $ +y\rangle$      | $p_1$       |
| $ +x\rangle$      | $ -y\rangle$      | $p_2$       |
| $ -x\rangle$      | $ +y\rangle$      | $p_3$       |
| $ -x\rangle$      | $ -y\rangle$      | $p_4$       |

Quantum mechanics predicts that measuring  $\sigma_x$  on  $|+z\rangle$  gives  $\pm 1$  with equal probability, and likewise for  $\sigma_y$ . This requires:

$$P(+x) = p_1 + p_2 = \frac{1}{2} \quad (10.3)$$

$$P(+y) = p_1 + p_3 = \frac{1}{2} \quad (10.4)$$

Together with the normalization  $p_1 + p_2 + p_3 + p_4 = 1$  and the requirement that all  $p_i \geq 0$ , these three equations in four unknowns appear to leave a one-parameter family of solutions. But a symmetry argument pins down the answer completely.

Notice what the marginal constraints already imply. From  $p_1 + p_2 = \frac{1}{2}$  and  $p_1 + p_3 = \frac{1}{2}$  we get  $p_2 = p_3$ . Substituting into the normalization gives  $p_4 = p_1$ . So opposite corners of the top face automatically carry equal probability: the diagonal pair  $(|+x\rangle, |+y\rangle)$  and  $(|-x\rangle, |-y\rangle)$  each have weight  $p_1$ , while  $(|+x\rangle, |-y\rangle)$  and  $(|-x\rangle, |+y\rangle)$  each have weight  $\frac{1}{2} - p_1$ . The two diagonal pairs sum to  $2p_1$  and  $1 - 2p_1$  respectively, and the sole remaining freedom is the value of  $p_1$ .

The symmetry of  $|+z\rangle$  fixes it. The state  $|+z\rangle$  is invariant under rotations about the  $z$ -axis. A  $90^\circ$  rotation cycles the four top-face corners and interchanges the two diagonal pairs. Since the quantum state does not change under this rotation, the probability distribution over hidden corners must not change either: the two diagonal-pair sums must be equal. Setting  $2p_1 = 1 - 2p_1$  gives  $p_1 = \frac{1}{4}$ , and therefore:

$$p_1 = p_2 = p_3 = p_4 = \frac{1}{4} \quad (10.5)$$

The uniform distribution is not merely the simplest solution; it is the unique one consistent with the rotational symmetry of the quantum state. The two hidden answers are independent and uniformly random.

This model is satisfying, but limited: it only handles measurements along the three Bloch cube face directions  $X$ ,  $Y$ , and  $Z$ . Spin can be measured along *any* direction. Can the hidden-variable picture be extended to other axes?

#### 10.2.4 Edge States of the Bloch Cube

The Bloch cube defines a natural hierarchy of measurement directions. The six **face centers**  $\pm\hat{x}$ ,  $\pm\hat{y}$ ,  $\pm\hat{z}$  are the directions we have already used. The eight **body-diagonal** (corner) directions  $(\pm\hat{x} \pm \hat{y} \pm \hat{z})/\sqrt{3}$  admit an analogous hidden-variable construction not pursued here. Between these lie the twelve **edge midpoints**: each edge of the cube connects two adjacent face centers, and its midpoint defines a spin measurement axis at  $45^\circ$  from both faces.

In the  $xz$ -plane, the two edge directions are:

$$\hat{e}_+ = \frac{\hat{z} + \hat{x}}{\sqrt{2}} \quad (45^\circ \text{ from } \hat{z}), \quad \hat{e}_- = \frac{\hat{x} - \hat{z}}{\sqrt{2}} \quad (135^\circ \text{ from } \hat{z}) \quad (10.6)$$

The direction  $\hat{e}_+$  points from the center of the cube toward the midpoint of the edge shared by the  $+z$  and  $+x$  faces. It lies exactly halfway between two face axes. The direction  $\hat{e}_-$  points toward the midpoint of the edge shared by the  $+x$  and  $-z$  faces, diametrically opposite to  $(\hat{z} - \hat{x})/\sqrt{2}$ .

For a particle in the state  $|+z\rangle$ , quantum mechanics predicts (recall that the probability of obtaining  $+1$  along  $\hat{n}$  is  $\cos^2(\theta/2)$ , where  $\theta$  is the angle between  $\hat{n}$  and the prepared state):

$$P(+1 | \hat{e}_+) = \cos^2(22.5^\circ) \approx 0.854 \quad (10.7)$$

$$P(+1 | \hat{e}_-) = \cos^2(67.5^\circ) \approx 0.146 \quad (10.8)$$

The edge direction  $\hat{e}_+$  is close to  $\hat{z}$  (only  $45^\circ$  away), so the particle is very likely to be found spin-up along it. The edge direction  $\hat{e}_-$  is far from  $\hat{z}$  ( $135^\circ$ ), so spin-up is unlikely. These probabilities— $\cos^2(22.5^\circ) \approx 85\%$  and its complement  $\sin^2(22.5^\circ) \approx 15\%$ —will reappear when we test Bell's inequality.

Now try to extend the face model to handle edge measurements. The face model tells us the outcome of a  $\sigma_x$  measurement is simply  $\epsilon_x$ , the hidden face value carried by the particle. But notice that this is the same as computing  $\text{sign}(\hat{x} \cdot \vec{\lambda})$ , where  $\vec{\lambda} = (\epsilon_x, \epsilon_y, \epsilon_z)$  is the hidden triple viewed as a

vector. Indeed,  $\hat{x} \cdot \vec{\lambda} = \epsilon_x = \pm 1$ , whose sign is just  $\epsilon_x$  itself. The same holds for the other two face directions:  $\text{sign}(\hat{y} \cdot \vec{\lambda}) = \epsilon_y$  and  $\text{sign}(\hat{z} \cdot \vec{\lambda}) = \epsilon_z$ . For the three face directions, “read the hidden face value” and “compute the sign of the dot product with the hidden triple” are the same instruction.

This reformulation suggests a natural way to extend the model beyond face directions. For an arbitrary measurement direction  $\hat{n}$ , define the outcome function:

$$A(\hat{n}) = \text{sign}(\hat{n} \cdot \vec{\lambda}) \quad (10.9)$$

This rule agrees with the face model whenever  $\hat{n}$  is a face direction, and it provides an answer—the sign of  $\hat{n} \cdot \vec{\lambda}$ —for any other direction as well. It is the simplest generalization of the face-reading rule to the full unit sphere: project the hidden triple onto the measurement axis, and report the sign.

Applying Eq. (10.9) to the edge direction  $\hat{e}_+$  gives:

$$A(\hat{e}_+) = \text{sign}\left(\frac{\epsilon_x + \epsilon_z}{\sqrt{2}}\right) = \text{sign}(\epsilon_x + \epsilon_z) \quad (10.10)$$

When  $\epsilon_x = \epsilon_z$  (both +1 or both −1), the sign is definite: +1 or −1 respectively. But when  $\epsilon_x = -\epsilon_z$ , the argument is zero and the sign is undefined. The edge direction  $\hat{e}_+$  lies exactly on the boundary between the hemispheres defined by these hidden triples. For half of the four relevant face configurations, the model has no answer.

This is not a problem that can be patched by adopting a convention for  $\text{sign}(0)$ . The deeper issue is geometric: the face model partitions the unit sphere using planes perpendicular to  $\hat{x}$ ,  $\hat{y}$ , and  $\hat{z}$ , and the edge directions sit exactly on these partition boundaries. The discrete model is too coarse to resolve directions halfway between its grid points.

The model that resolves this limitation—Bell’s continuous construction—handles all directions, including edges, faces, corners, and everything in between. But the fact that the face model breaks down *precisely at the edge directions* will prove deeply significant: these are exactly the directions where Bell’s inequality is maximally violated.

### 10.2.5 Bell’s Continuous Model

#### From faces to edges to all directions

The face model assigned a definite  $\pm 1$  answer to each of three face directions. We have just seen that it cannot cleanly handle the twelve edge directions. A more ambitious realist would assign a definite  $\pm 1$  answer to *every* direction on the unit sphere simultaneously—faces, edges, corners, and the continuum of directions between them. Mathematically, such an assignment is a function:

$$f : S^2 \rightarrow \{+1, -1\} \quad (10.11)$$

that maps each measurement direction  $\hat{n}$  to a definite outcome  $f(\hat{n}) = \pm 1$ . The face model is just  $f$  sampled at three points:  $f(\hat{x})$ ,  $f(\hat{y})$ ,  $f(\hat{z})$ . In the continuous limit,  $f$  is defined everywhere.

The complete hidden-variable description of a particle in state  $|+z\rangle$  is the pair  $(|+z\rangle, f)$ , where the quantum state tells us  $f(\hat{z}) = +1$  with certainty and the function  $f$  fills in the rest. A particle carrying a specific  $f$  has no randomness at all: every spin question has a predetermined answer, just as a particle carrying a specific set of hidden face values had definite answers to all three face questions. The apparent randomness of quantum mechanics arises from not knowing which  $f$  the particle carries.

The ensemble is described by a probability distribution  $\rho[f]$  over such functions, constrained so that the average of  $f(\hat{n})$  over the ensemble reproduces the quantum prediction for every direction:

$$P(+1|\hat{n}) = \sum_{f: f(\hat{n})=+1} \rho[f] = \frac{1 + \hat{n} \cdot \hat{s}}{2} \quad (10.12)$$

Here  $\hat{s}$  is the Bloch vector of the prepared quantum state (e.g.,  $\hat{s} = \hat{z}$  for  $|+z\rangle$ ),  $\hat{n}$  is the measurement direction—the question we choose to ask—and  $f(\hat{n}) = \pm 1$  is the predetermined answer that the hidden function  $f$  assigns to that question. The sum runs over all functions  $f$  in the ensemble that happen to answer  $+1$  for the direction  $\hat{n}$ , weighted by their probabilities  $\rho[f]$ . This is a weaker requirement than it might seem. Individual functions can have answer patterns that look nothing like the quantum probability distribution. A particular  $f$  might assign  $+1$  to two nearly parallel directions and  $-1$  to a direction between them. That is perfectly acceptable, because the model only needs to get the statistics right on average, not direction by direction for each particle. Just as in the face model, where one particular hidden triple gave  $\sigma_x = +1$  and  $\sigma_y = +1$  even though the quantum state  $|+z\rangle$  has no preference between these, each  $f$  carries its own idiosyncratic assignment; the quantum probabilities emerge only from averaging over the ensemble.

There is a subtlety here that strengthens the realist position considerably. Many different distributions  $\rho[f]$  over many different kinds of functions can satisfy Eq. (10.12). The situation is exactly analogous to the completely mixed state  $\rho = I/2$  of a single qubit. That density matrix can be decomposed as an equal mixture of  $|+z\rangle$  and  $|-z\rangle$ , or equally well as a mixture of  $|+x\rangle$  and  $|-x\rangle$ , or as a uniform distribution over all pure states on the Bloch sphere. No experiment can distinguish these decompositions; the density matrix captures everything observable, and the choice of decomposition is physically meaningless. In the same way, no single-qubit experiment can distinguish between different hidden variable models that all reproduce the same quantum statistics. The realist posits that each particle carries a definite  $f$ , but the specific  $f$  is forever inaccessible—only the ensemble average  $\rho[f]$  has observable consequences. This means that single-qubit realism is not merely compatible with quantum mechanics; it is *unfalsifiable* by single-qubit experiments. We can neither confirm nor deny the existence of the hidden variables. The realist can always claim they are there, and no measurement on a single particle can prove otherwise.

### Bell's sign construction

The space of all functions  $f : S^2 \rightarrow \{+1, -1\}$  is enormous. Bell's insight was to find a specific, geometrically natural family of such functions that suffices to reproduce all quantum predictions.

The face model motivates the construction. In Section 10.2.4, we showed that the face model's measurement rule can be rewritten as  $A(\hat{n}) = \text{sign}(\hat{n} \cdot \vec{\lambda})$ , where  $\vec{\lambda} = (\epsilon_x, \epsilon_y, \epsilon_z)$  is the hidden triple viewed as a vector. That vector takes only eight discrete values, one for each combination of face signs. Bell's insight was to promote this to a continuous construction: let the hidden variable be a unit vector  $\hat{\lambda}$  that can point anywhere on the sphere, and define:

$$f_{\hat{\lambda}}(\hat{n}) = \text{sign}(\hat{n} \cdot \hat{\lambda}) \quad (10.13)$$

This single vector  $\hat{\lambda}$  compresses the entire function  $f$  into one object. It divides the unit sphere into two hemispheres: every direction  $\hat{n}$  within  $90^\circ$  of  $\hat{\lambda}$  gets  $f = +1$ , and every direction on the opposite side gets  $f = -1$ . One vector, one clean geometric rule, and the answer to every conceivable measurement is determined. Crucially, the hemisphere boundary is now a great circle that can be oriented in any direction, so there is no preferred grid of directions along which the model might fail.



The edge-direction ambiguity of the face model—where  $\text{sign}(0)$  was undefined—is a set of measure zero in the continuous model: for almost every  $\hat{\lambda}$ , the dot product  $\hat{e}_+ \cdot \hat{\lambda}$  is nonzero.

The quantum probabilities emerge from averaging over the ensemble, exactly as in the face model. There, each particle carried a definite triple  $(\epsilon_x, \epsilon_y, \epsilon_z)$ , and the probability  $P(+x) = 1/2$  arose from the uniform distribution over the four possible triples. In the continuous model, each particle carries a definite  $\hat{\lambda}$ , and the ensemble is described by a probability density  $\rho(\hat{\lambda})$  over the unit sphere. The probability of obtaining  $+1$  along direction  $\hat{n}$  is the integral of  $\rho(\hat{\lambda})$  over all  $\hat{\lambda}$  whose hemispheres include  $\hat{n}$ :

$$P(+1|\hat{n}) = \int_{\hat{n} \cdot \hat{\lambda} > 0} \rho(\hat{\lambda}) d\Omega \quad (10.14)$$

In 1966, John Bell showed that for any quantum state with Bloch vector  $\hat{s}$ , there exists a density  $\rho(\hat{\lambda})$  such that this integral reproduces

$$P(+1|\hat{n}) = \frac{1 + \hat{n} \cdot \hat{s}}{2} = \cos^2 \frac{\theta}{2} \quad (10.15)$$

exactly, for every measurement direction  $\hat{n}$ —faces, edges, corners, and all directions in between. The Bloch cube face model is a discrete special case of this construction, restricted to three measurement axes.

### 10.2.6 The Lesson

The hidden variable program for a single qubit is in an extraordinarily strong position. Not only can we construct explicit models that reproduce all quantum predictions—from the discrete face model to Bell's continuous construction—but the realist hypothesis is *unfalsifiable* by any single-particle experiment. Many different hidden variable models, with different  $f$ 's and different  $\rho[f]$ 's, all produce the same quantum statistics. No measurement can reveal which model is correct, or even whether any such model is “really” at work behind the scenes. The realist can always maintain that particles carry hidden variables, and no single-qubit experiment can prove otherwise.

This seems like a decisive victory for realism. If one had only single-particle quantum mechanics, one might reasonably conclude that quantum mechanics is simply an incomplete statistical theory—useful for making predictions, but missing a deeper deterministic layer in which every question has a definite answer. This is precisely what Einstein, Podolsky, and Rosen argued in 1935.

The story does not end here. When we turn to entangled systems and demand locality, the hidden variable program collapses—and it collapses completely. Bell's theorem does not merely rule out one particular model, or one particular choice of  $f$ , or one particular  $\rho[f]$ . It rules out *every* local hidden variable model simultaneously, regardless of the structure of  $\lambda$ , regardless of how the hidden variables are constructed, regardless of the distribution over the ensemble. That is what makes Bell's result so remarkable.

## 10.3 Bipartite Entanglement

### 10.3.1 Product States vs. Entangled States

For a two-qubit system in  $\mathcal{H}_A \otimes \mathcal{H}_B$ , a product state has the form:

$$|\psi\rangle = |\phi\rangle_A \otimes |\chi\rangle_B \quad (10.16)$$

In a product state, each subsystem has a definite quantum state of its own. Measuring subsystem  $A$  tells us nothing about subsystem  $B$  that we didn't already know.



An entangled state cannot be written in product form. The simplest example is:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (10.17)$$

Here, neither qubit has a definite state individually. The state of the whole is more than the sum of its parts.

### 10.3.2 The Four Bell States

The maximally entangled Bell basis consists of four orthonormal states:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (10.18)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \quad (10.19)$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \quad (10.20)$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \quad (10.21)$$

These form a complete orthonormal basis for the two-qubit Hilbert space. We encountered these states in Chapter 2; they will now play a central role in understanding the foundations of quantum mechanics.

#### Physical Insight

Consider the state  $|\Phi^+\rangle$ . If Alice measures her qubit in the computational basis and obtains  $|0\rangle$ , the state collapses to  $|00\rangle$ , so Bob's qubit is certainly in state  $|0\rangle$ . If Alice obtains  $|1\rangle$ , Bob's qubit is certainly in state  $|1\rangle$ . The outcomes are perfectly correlated, even if Alice and Bob are light-years apart.

This correlation is not, by itself, mysterious. One could imagine the qubits carried pre-determined values, each qubit with its own hidden Bloch cube packed at the source, like a pair of gloves shipped to different cities. Bell's theorem shows that no such local explanation can account for all quantum predictions.

## 10.4 The EPR Paradox and Local Realism

### 10.4.1 The EPR Argument

Having seen that realism works perfectly for a single qubit, Einstein, Podolsky, and Rosen (EPR) argued in 1935 that it should work for entangled systems too. Their reasoning was as follows.

Consider two particles prepared in the singlet state:

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \quad (10.22)$$

where  $|0\rangle$  and  $|1\rangle$  represent spin-up and spin-down along the  $z$ -axis. Alice and Bob each take one particle and travel to distant locations.

If Alice measures spin along the  $z$ -axis, she can predict with certainty what Bob will obtain if he also measures along  $z$ : the outcomes are perfectly anti-correlated. But Alice could equally well measure along the  $x$ -axis, and then she could predict Bob's  $x$ -spin with certainty.

EPR introduced a criterion of physical reality: "If, without in any way disturbing a system, we can predict with certainty the value of a physical quantity, then there exists an element of physical reality corresponding to this physical quantity."

Since Alice's measurement cannot instantaneously affect Bob's distant particle (by the principle of locality), and since Alice can predict either Bob's  $z$ -spin or  $x$ -spin with certainty depending on her choice, EPR concluded that both spin components must have definite values prior to measurement. But quantum mechanics does not assign definite values to non-commuting observables like  $S_x$  and  $S_z$ . Therefore, EPR argued, quantum mechanics must be an incomplete description of physical reality.

The argument is compelling. We have just seen that a single qubit *can* carry definite values for all spin components, encoded in a hidden Bloch cube, with quantum mechanics emerging as a statistical description of which hidden values appear in the ensemble. EPR were simply extending this to the entangled case: each particle would carry its own set of hidden cubes, with the additional requirement of locality ensuring that Alice's measurement choice does not disturb Bob's.

The face-cube model makes this extension concrete. Imagine the source prepares a pair of Bloch cubes, one for Alice and one for Bob, with every face anti-correlated:

|               | Alice's cube    | Bob's cube      |
|---------------|-----------------|-----------------|
| $Z$ (visible) | $  - z \rangle$ | $  + z \rangle$ |
| $X$ (hidden)  | $  + x \rangle$ | $  - x \rangle$ |
| $Y$ (hidden)  | $  - y \rangle$ | $  + y \rangle$ |

Both cubes sit with their  $Z$  faces on top. Looking straight down, you see only the  $Z$  answers: Alice has spin-down, Bob has spin-up. The  $X$  and  $Y$  answers are on the side faces—present but hidden by perspective. If Alice measures  $\sigma_z$ , she reads her top face and gets  $-1$ ; if she measures  $\sigma_x$  instead, she reads the hidden side face and gets  $+1$ . Either way, the answer was there before she looked. Bob's cube, being anti-correlated face by face, guarantees that whenever they measure along the same axis, they get opposite results. The source packs each pair with one of the four possible anti-correlated face-value assignments, each with probability  $1/4$ , and the singlet-state statistics emerge from not knowing which pair was packed.

### 10.4.2 Local Realism

The EPR argument rests on two assumptions that together constitute *local realism*:

**Realism:** Physical properties have definite values that exist independent of observation. When we measure an observable, we are revealing a pre-existing value, not creating one. As we showed in Section 10.2.2, the hidden-cube model makes this assumption concrete and reproduces all single-qubit quantum predictions.

**Locality:** An action performed at one location cannot instantaneously influence physical reality at a distant location. Influences propagate at most at the speed of light.

These assumptions seem eminently reasonable. Realism captures the intuition that the moon exists whether or not anyone is looking at it, and we have demonstrated that it is entirely compatible with single-particle quantum mechanics. Locality is built into the structure of special relativity. For nearly thirty years after EPR, most physicists assumed that quantum mechanics could eventually be supplemented by hidden variables that would restore a local realistic description.

### 10.4.3 Local Hidden Variable Theories

A local hidden variable theory for two particles extends the single-qubit model in a natural way. The source prepares *two* Bloch cubes, one for Alice's particle and one for Bob's, each with a definite set of hidden face values  $\vec{\lambda} = (\epsilon_x, \epsilon_y, \epsilon_z)$ , and sends them off in opposite directions. When Alice measures spin along  $\vec{a}$ , she evaluates  $A(\vec{a}) = \text{sign}(\vec{a} \cdot \vec{\lambda}_A)$ . Bob does the same with his hidden triple:  $B(\vec{b}) = \text{sign}(\vec{b} \cdot \vec{\lambda}_B)$ .

The crucial constraint is *locality*: Alice's cube determines her outcome for *any* measurement she might choose, independent of what Bob does with his. Alice's setting  $\vec{a}$  does not appear in Bob's outcome function, and vice versa. Both cubes can be correlated (the source can pack them according to any joint distribution  $\rho(\vec{\lambda}_A, \vec{\lambda}_B)$ ), but once the particles separate, each outcome depends only on the local measurement setting and the local hidden triple.

This is a straightforward generalization of the model that worked so well for a single qubit. The question is whether such a model can reproduce the quantum mechanical predictions for entangled states.

## 10.5 Bell's Theorem

In 1964, John Bell proved a remarkable theorem: no local hidden variable theory can reproduce all the predictions of quantum mechanics. The hidden variable approach that worked flawlessly for a single qubit fails irrecoverably when applied to entangled pairs with a locality constraint.

### 10.5.1 The Scenario

Consider the following experimental setup. A source prepares pairs of spin-1/2 particles in the singlet state  $|\Psi^-\rangle$  and sends one to Alice and one to Bob. Alice chooses to measure spin along one of two directions,  $\vec{a}$  or  $\vec{a}'$ . Bob independently chooses to measure along  $\vec{b}$  or  $\vec{b}'$ . Each measurement yields +1 or -1.

We define the correlation function:

$$E(\vec{a}, \vec{b}) = \langle A(\vec{a}) \cdot B(\vec{b}) \rangle \quad (10.23)$$

This is the average value of the product of outcomes when Alice measures along  $\vec{a}$  and Bob measures along  $\vec{b}$ .

### 10.5.2 The Continuous Hidden Variable Model

Recall that Bell's single-qubit model (Section 10.2.5) used a continuous hidden variable  $\hat{\lambda}$ , a unit vector distributed over the sphere, with outcome rule  $A(\hat{n}) = \text{sign}(\hat{n} \cdot \hat{\lambda})$ . Bell showed that a suitable probability density  $\rho(\hat{\lambda})$  reproduces all single-qubit quantum predictions exactly.

For two entangled particles, we try the natural extension: give each pair a shared hidden variable  $\hat{\lambda}$ , and impose anti-correlation by taking  $A(\hat{a}) = \text{sign}(\hat{a} \cdot \hat{\lambda})$  and  $B(\hat{b}) = -\text{sign}(\hat{b} \cdot \hat{\lambda})$ . For the singlet state, the simplest choice is a uniform distribution  $\rho(\hat{\lambda}) = 1/4\pi$ .

The correlation function is:

$$E(\hat{a}, \hat{b}) = - \int \frac{d\Omega}{4\pi} \text{sign}(\hat{a} \cdot \hat{\lambda}) \text{sign}(\hat{b} \cdot \hat{\lambda}) \quad (10.24)$$

This integral has an elegant geometric interpretation. The planes perpendicular to  $\hat{a}$  and  $\hat{b}$  divide the unit sphere into four regions. In two of them, the signs of  $\hat{a} \cdot \hat{\lambda}$  and  $\hat{b} \cdot \hat{\lambda}$  agree; in the other two,

they disagree. The fraction of the sphere where they *disagree* is  $\theta/\pi$ , where  $\theta$  is the angle between  $\hat{a}$  and  $\hat{b}$ . The fraction where they agree is  $(\pi - \theta)/\pi$ . Therefore:

$$E(\theta) = - \left[ \frac{\pi - \theta}{\pi} - \frac{\theta}{\pi} \right] = -1 + \frac{2\theta}{\pi} \quad (10.25)$$

This is a straight line from  $E = -1$  at  $\theta = 0$  to  $E = +1$  at  $\theta = \pi$ .

Compare the two results:

|                                             | $\theta = 0$ | $\theta = \pi/2$ | $\theta = \pi$ |
|---------------------------------------------|--------------|------------------|----------------|
| Local hidden variable model (straight line) | -1           | 0                | +1             |
| Quantum mechanics ( $-\cos \theta$ )        | -1           | 0                | +1             |

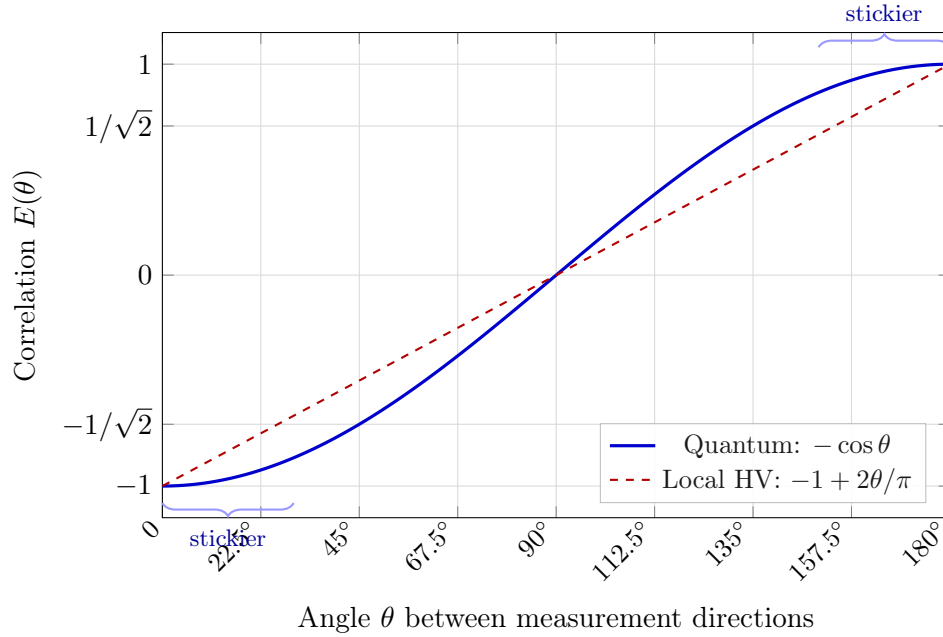


Figure 10.1: Comparison of the correlation function  $E(\theta)$  for the local hidden variable model and quantum mechanics. The quantum prediction  $-\cos \theta$  (solid blue) is *stickier* near perfect anti-correlation ( $\theta = 0$ ) and perfect correlation ( $\theta = 180^\circ$ ) than the local hidden variable model (dashed red), which departs linearly. Both agree at  $\theta = 0^\circ$ ,  $90^\circ$ , and  $180^\circ$ .

Both agree at  $\theta = 0$ ,  $\pi/2$ , and  $\pi$ , as shown in Fig. 10.1. But the discrepancy between them is physically significant.

The crucial difference appears near  $\theta = 0$ . The quantum correlation  $-\cos \theta \approx -1 + \theta^2/2$  is flat at perfect anti-correlation; the curve barely moves away from  $-1$  for small  $\theta$ . The local model  $-1 + 2\theta/\pi$  departs linearly, with a constant slope  $2/\pi \approx 0.64$ . Quantum correlations are *stickier* near perfect correlation than any local model can produce. This is Bell's original observation from his 1964 paper: the quantum cosine clings to  $-1$  at small angles more tightly than the linear function, and clings to  $+1$  at large angles more tightly than the linear function. The straight line is the best a local model with a uniform distribution can do, and it is not good enough.

Recall that for a single qubit, Bell's continuous model reproduced all quantum predictions exactly. For two entangled qubits, the same construction fails. The problem is the locality constraint, which forces the correlation function to be a weighted average of sign products—and no such average can produce the curvature of  $-\cos \theta$ .

### 10.5.3 The CHSH Inequality

The continuous model fails to reproduce the quantum correlation for entangled pairs. But could some cleverer local model succeed, perhaps with a non-uniform distribution or a more complicated hidden variable space? The CHSH inequality shows that the answer is no: *every* local hidden variable theory, regardless of its internal structure, is bounded by the same constraint.

The generality of the argument rests on exactly two properties that any local hidden variable theory must have:

- (a) **Bounded outcomes.** Each measurement yields  $A = \pm 1$  or  $B = \pm 1$ . This is not an assumption about hidden variables; it is a fact about spin-1/2: the eigenvalues of any spin component are  $\pm\hbar/2$ , and we label these  $\pm 1$ .
- (b) **Locality.** Alice's outcome  $A(\vec{a}, \lambda)$  depends on her measurement setting  $\vec{a}$  and the shared hidden variable  $\lambda$ , but *not* on Bob's setting  $\vec{b}$ . Likewise,  $B(\vec{b}, \lambda)$  does not depend on  $\vec{a}$ .

Everything else is unrestricted. The hidden variable  $\lambda$  can be anything: a discrete triple, a point on the unit sphere, the complete works of Shakespeare, or an arbitrarily complicated instruction set. The functions  $A(\vec{a}, \lambda)$  and  $B(\vec{b}, \lambda)$  can depend on their arguments in any way they like, as long as each output is  $\pm 1$ . The distribution  $\rho(\lambda)$  can be anything normalizable. The CHSH argument treats  $A$  and  $B$  as black boxes and never looks inside.

For each particle pair carrying a particular  $\lambda$ , Alice could measure along  $\vec{a}$  or  $\vec{a}'$ , and Bob could measure along  $\vec{b}$  or  $\vec{b}'$ . Consider the combination:

$$S(\lambda) = A(\vec{a}, \lambda)[B(\vec{b}, \lambda) - B(\vec{b}', \lambda)] + A(\vec{a}', \lambda)[B(\vec{b}, \lambda) + B(\vec{b}', \lambda)] \quad (10.26)$$

This is where boundedness does its work. Since  $B(\vec{b}, \lambda)$  and  $B(\vec{b}', \lambda)$  are each exactly  $\pm 1$ , they either agree or disagree. If they agree,  $B(\vec{b}) - B(\vec{b}') = 0$  and  $B(\vec{b}) + B(\vec{b}') = \pm 2$ . If they disagree,  $B(\vec{b}) - B(\vec{b}') = \pm 2$  and  $B(\vec{b}) + B(\vec{b}') = 0$ . In either case, one bracket vanishes and the other has magnitude 2. The surviving bracket is multiplied by an  $A$  value of magnitude 1, so  $|S(\lambda)| = 2$  for every single  $\lambda$ .

Table 10.1 displays all sixteen combinations explicitly. Every row gives  $|S| = 2$ , confirming the per- $\lambda$  bound by exhaustion.

This is the critical step. For any particular  $\lambda$ , the quantity  $S$  is exactly  $\pm 2$ —not approximately, not on average, but for every individual instruction set the source could produce. No cleverness in choosing  $\lambda$  or designing the functions  $A$  and  $B$  can make  $|S(\lambda)|$  exceed 2, because the bound follows from nothing more than  $\pm 1$  arithmetic.

Since  $|S(\lambda)| \leq 2$  holds for every  $\lambda$  individually, averaging over any distribution  $\rho(\lambda)$  preserves the bound:

$$|\langle S \rangle| = \left| \int \rho(\lambda) S(\lambda) d\lambda \right| \leq \int \rho(\lambda) |S(\lambda)| d\lambda \leq 2 \quad (10.27)$$

Writing this in terms of correlation functions:

$$|E(\vec{a}, \vec{b}) - E(\vec{a}, \vec{b}') + E(\vec{a}', \vec{b}) + E(\vec{a}', \vec{b}')| \leq 2 \quad (10.28)$$

This is the **CHSH inequality**, named after Clauser, Horne, Shimony, and Holt. Any local hidden variable theory must satisfy it, regardless of the structure of  $\lambda$ , regardless of the form of  $A$  and  $B$ , and regardless of the distribution  $\rho(\lambda)$ . The entire argument used only two facts: that outcomes are  $\pm 1$ , and that Alice's outcome does not depend on Bob's setting. The sign construction, the hemisphere picture, the uniform distribution—none of these played any role. They were useful for building intuition about specific models, but the inequality transcends every particular model.

Table 10.1: All sixteen combinations of the four  $\pm 1$  outcomes and the resulting CHSH value  $S(\lambda)$ . In each row, exactly one of the two brackets vanishes while the other has magnitude 2, giving  $|S| = 2$  regardless of the combination. The “vanishes/survives” columns show which bracket is zero and which is  $\pm 2$ .

| $A$ | $A'$ | $B$ | $B'$ | $B-B'$          | $B+B'$          | $S$ |
|-----|------|-----|------|-----------------|-----------------|-----|
| +1  | +1   | +1  | +1   | 0               | +2 $\leftarrow$ | +2  |
| +1  | +1   | +1  | -1   | +2 $\leftarrow$ | 0               | +2  |
| +1  | +1   | -1  | +1   | -2 $\leftarrow$ | 0               | -2  |
| +1  | +1   | -1  | -1   | 0               | -2 $\leftarrow$ | -2  |
| +1  | -1   | +1  | +1   | 0               | +2 $\leftarrow$ | -2  |
| +1  | -1   | +1  | -1   | +2 $\leftarrow$ | 0               | +2  |
| +1  | -1   | -1  | +1   | -2 $\leftarrow$ | 0               | -2  |
| +1  | -1   | -1  | -1   | 0               | -2 $\leftarrow$ | +2  |
| -1  | +1   | +1  | +1   | 0               | +2 $\leftarrow$ | +2  |
| -1  | +1   | +1  | -1   | +2 $\leftarrow$ | 0               | -2  |
| -1  | +1   | -1  | +1   | -2 $\leftarrow$ | 0               | +2  |
| -1  | +1   | -1  | -1   | 0               | -2 $\leftarrow$ | -2  |
| -1  | -1   | +1  | +1   | 0               | +2 $\leftarrow$ | -2  |
| -1  | -1   | +1  | -1   | +2 $\leftarrow$ | 0               | -2  |
| -1  | -1   | -1  | +1   | -2 $\leftarrow$ | 0               | +2  |
| -1  | -1   | -1  | -1   | 0               | -2 $\leftarrow$ | +2  |

#### 10.5.4 Quantum Violation: Faces and Edges

For the singlet state  $|\Psi^-\rangle$ , quantum mechanics predicts:

$$E(\vec{a}, \vec{b}) = -\vec{a} \cdot \vec{b} = -\cos \theta_{ab} \quad (10.29)$$

where  $\theta_{ab}$  is the angle between directions  $\vec{a}$  and  $\vec{b}$ .

##### Example 1

**Derivation of the Quantum Correlation** Let Alice measure spin along  $\vec{a} = \hat{z}$  and Bob measure along a direction  $\vec{b}$  that makes angle  $\theta$  with  $\hat{z}$ .

In the singlet state, if Alice obtains +1 (spin up along  $\hat{z}$ ), then Bob's particle is in state  $|1\rangle$  (spin down along  $\hat{z}$ ). Bob's measurement along  $\vec{b}$  then yields +1 with probability  $\sin^2(\theta/2)$  and -1 with probability  $\cos^2(\theta/2)$ , giving expected value  $\sin^2(\theta/2) - \cos^2(\theta/2) = -\cos \theta$ .

Similarly, if Alice obtains -1, Bob's expected value is  $+\cos \theta$ .

Since Alice gets  $\pm 1$  with equal probability:

$$E(\vec{a}, \vec{b}) = \frac{1}{2}(+1)(-\cos \theta) + \frac{1}{2}(-1)(+\cos \theta) = -\cos \theta \quad (10.30)$$

The measurement directions that maximally violate the CHSH inequality have a natural interpretation in the Bloch cube taxonomy introduced in Section 10.2.4. Alice measures along two **face** directions:

$$\hat{a} = \hat{z} \quad (0^\circ), \quad \hat{a}' = \hat{x} \quad (90^\circ) \quad (10.31)$$

Bob measures along two **edge** directions:

$$\hat{b} = \frac{\hat{z} + \hat{x}}{\sqrt{2}} \quad (45^\circ), \quad \hat{b}' = \frac{\hat{x} - \hat{z}}{\sqrt{2}} \quad (135^\circ) \quad (10.32)$$

In the  $xz$ -plane, these four directions alternate face–edge–face–edge at  $0^\circ$ ,  $45^\circ$ ,  $90^\circ$ , and  $135^\circ$  from the  $z$ -axis. Bob's edge directions sit at the midpoints of the edges connecting Alice's two face directions on the Bloch cube.

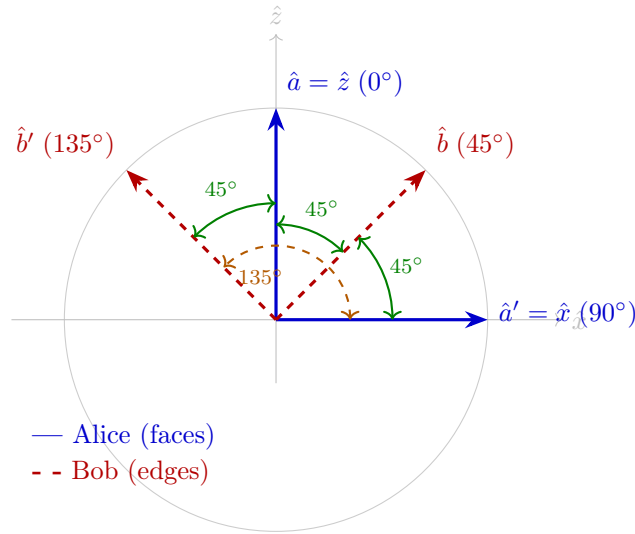


Figure 10.2: The four CHSH measurement directions in the  $xz$ -plane. Alice measures along two face directions of the Bloch cube ( $\hat{a} = \hat{z}$  and  $\hat{a}' = \hat{x}$ , solid blue). Bob measures along two edge directions ( $\hat{b}$  at  $45^\circ$  and  $\hat{b}'$  at  $135^\circ$ , dashed red). Adjacent directions are separated by  $45^\circ$ , producing the three-close-one-far pattern: the pairs  $(\hat{a}, \hat{b})$ ,  $(\hat{a}', \hat{b})$ , and  $(\hat{a}', \hat{b}')$  are each  $45^\circ$  apart, while  $(\hat{a}, \hat{b}')$  is  $135^\circ$  apart.

The geometry produces a crucial asymmetry (Fig. 10.2). Three of the four pairs are neighboring face–edge combinations, each separated by  $45^\circ$ :

$$\theta_{ab} = \theta_{a'b} = \theta_{a'b'} = 45^\circ \quad (10.33)$$

The fourth pair—Alice's face direction  $\hat{z}$  and Bob's far edge direction  $(\hat{x} - \hat{z})/\sqrt{2}$ —are  $135^\circ$  apart:

$$\theta_{ab'} = 135^\circ \quad (10.34)$$

This three-close-one-far pattern is not a choice we impose; it is forced by the face–edge geometry of the Bloch cube.

The quantum correlations follow:

$$E(\vec{a}, \vec{b}) = E(\vec{a}', \vec{b}) = E(\vec{a}', \vec{b}') = -\cos 45^\circ = -\frac{1}{\sqrt{2}}, \quad E(\vec{a}, \vec{b}') = -\cos 135^\circ = +\frac{1}{\sqrt{2}} \quad (10.35)$$

The CHSH combination gives:

$$S = E(\vec{a}, \vec{b}) - E(\vec{a}, \vec{b}') + E(\vec{a}', \vec{b}) + E(\vec{a}', \vec{b}') = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -2\sqrt{2} \quad (10.36)$$

Therefore  $|S| = 2\sqrt{2} \approx 2.83$ , which exceeds the local hidden variable bound of 2 (Fig. 10.3).

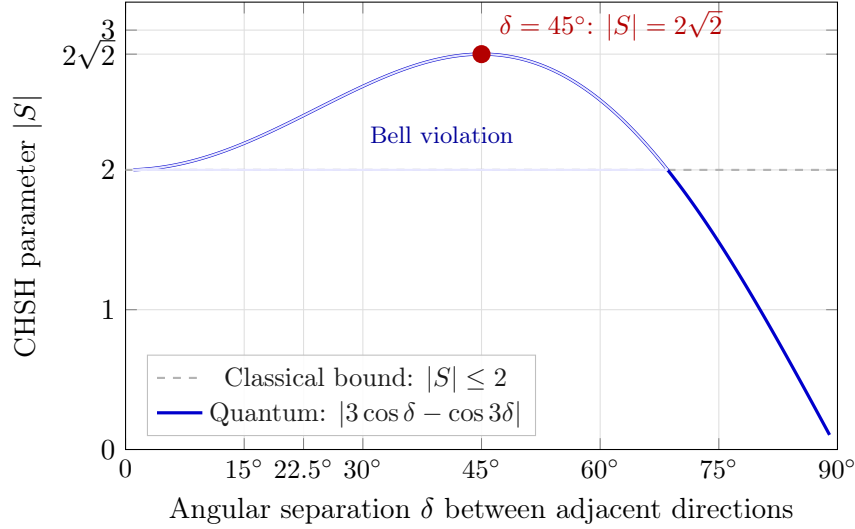


Figure 10.3: The CHSH parameter  $|S|$  as a function of the angular separation  $\delta$  between adjacent measurement directions, assuming four equally spaced directions at angles  $0, \delta, 2\delta, 3\delta$  in the  $xz$ -plane. The quantum prediction (solid blue) exceeds the classical bound of 2 (dashed gray) for a range of angles centered on  $\delta = 45^\circ$ , where the violation is maximal at  $|S| = 2\sqrt{2} \approx 2.83$ . The shaded region marks the Bell violation. The optimal  $45^\circ$  spacing corresponds to the face-edge separation on the Bloch cube.

#### Key Idea: Bell's Theorem

No local hidden variable theory can reproduce all predictions of quantum mechanics. The CHSH inequality, which any local realistic theory must satisfy, is violated by quantum mechanical predictions for entangled states. Experiments confirm the quantum predictions. The Bloch cube makes the geometry of the violation visible. The optimal CHSH measurement directions are two faces and two edges of the Bloch cube, and the violation lives in the face-edge asymmetry: three pairs at  $45^\circ$  and one at  $135^\circ$ . The  $45^\circ$  angle that defines an edge direction is the same angle that drives the maximal quantum violation.

### 10.5.5 Where the Single-Qubit Model Breaks Down

It is instructive to see precisely why the hidden variable strategy that worked for one qubit fails for two.

For a single qubit, Bell's model used a hidden variable  $\hat{\lambda}$  to determine all measurement outcomes through  $A(\hat{n}) = \text{sign}(\hat{n} \cdot \hat{\lambda})$ . One might try to extend this by giving each particle its own hidden variable: Alice's particle carries  $\hat{\lambda}_A$  and Bob's carries  $\hat{\lambda}_B$ , with outcomes  $A(\vec{a}) = \text{sign}(\vec{a} \cdot \hat{\lambda}_A)$  and  $B(\vec{b}) = \text{sign}(\vec{b} \cdot \hat{\lambda}_B)$ , both prepared at the source.



This model is explicitly local: Alice's outcome depends only on  $\vec{a}$  and  $\vec{\lambda}_A$ , and Bob's only on  $\vec{b}$  and  $\vec{\lambda}_B$ . The correlations arise solely from the joint distribution  $\rho(\vec{\lambda}_A, \vec{\lambda}_B)$  established at the source.

The problem is that no distribution  $\rho(\vec{\lambda}_A, \vec{\lambda}_B)$  can produce the quantum correlation  $E(\vec{a}, \vec{b}) = -\cos \theta_{ab}$  for all measurement directions simultaneously. The proof is Bell's theorem: any such local model satisfies  $|S| \leq 2$ , but quantum mechanics demands  $|S| = 2\sqrt{2}$ . The quantum correlations are too strong to arise from shared classical information, no matter how cleverly the hidden variables at the source are prepared.

## 10.6 GHZ States: A Starker Contradiction

Bell's theorem involves statistical predictions: local hidden variables and quantum mechanics give different probabilities. Greenberger, Horne, and Zeilinger (GHZ) discovered an even more dramatic conflict involving definite predictions, not just probabilities.

### 10.6.1 The GHZ State

Consider three qubits in the state:

$$|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle - |111\rangle) \quad (10.37)$$

This is a maximally entangled state of three particles. A source prepares this state and sends one qubit to Alice, one to Bob, and one to Charlie.

### 10.6.2 Quantum Predictions for GHZ

Each experimenter can measure their qubit in either the  $X$  basis (eigenstates of  $\sigma_x$ ) or the  $Y$  basis (eigenstates of  $\sigma_y$ ), obtaining outcomes  $\pm 1$ .

The quantum predictions for specific measurement combinations can be computed using the operator eigenvalue approach. Consider the operator  $X_1X_2X_3 = \sigma_x \otimes \sigma_x \otimes \sigma_x$ . Acting on the GHZ state:

$$X_1X_2X_3|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(\sigma_x|0\rangle \otimes \sigma_x|0\rangle \otimes \sigma_x|0\rangle - \sigma_x|1\rangle \otimes \sigma_x|1\rangle \otimes \sigma_x|1\rangle) = \frac{1}{\sqrt{2}}(|111\rangle - |000\rangle) = -|\text{GHZ}\rangle \quad (10.38)$$

So  $|\text{GHZ}\rangle$  is an eigenstate of  $X_1X_2X_3$  with eigenvalue  $-1$ , meaning the product of the three  $X$  outcomes is  $-1$  on every run, not just on average. By the same method (using  $\sigma_y|0\rangle = -i|1\rangle$  and  $\sigma_y|1\rangle = i|0\rangle$ ), one finds that  $|\text{GHZ}\rangle$  is also an eigenstate of the operators  $X_1Y_2Y_3$ ,  $Y_1X_2Y_3$ , and  $Y_1Y_2X_3$ , each with eigenvalue  $+1$ . The four predictions are:

$$\langle X_1Y_2Y_3 \rangle = +1 \quad (10.39)$$

$$\langle Y_1X_2Y_3 \rangle = +1 \quad (10.40)$$

$$\langle Y_1Y_2X_3 \rangle = +1 \quad (10.41)$$

$$\langle X_1X_2X_3 \rangle = -1 \quad (10.42)$$

The first three equations say that when we measure  $XY Y$ ,  $YXY$ , or  $YYX$ , the product of the three outcomes is always  $+1$ . The fourth says that when we measure  $XXX$ , the product is always  $-1$ . Crucially, all four are eigenvalue equations: the outcomes are definite, not merely statistical.

### 10.6.3 The Contradiction with Local Realism

Suppose local hidden variables govern the outcomes. Each particle carries predetermined values  $x_i = \pm 1$  (its hidden answer to an  $X$  measurement) and  $y_i = \pm 1$  (its hidden answer to a  $Y$  measurement). All six values are determined at the source and carried locally to the three detectors.

The first three quantum predictions require:

$$x_1 y_2 y_3 = +1 \quad (10.43)$$

$$y_1 x_2 y_3 = +1 \quad (10.44)$$

$$y_1 y_2 x_3 = +1 \quad (10.45)$$

Multiply these three equations:

$$(x_1 y_2 y_3)(y_1 x_2 y_3)(y_1 y_2 x_3) = (+1)(+1)(+1) = +1 \quad (10.46)$$

The left side simplifies to:

$$x_1 x_2 x_3 \cdot y_1^2 y_2^2 y_3^2 = x_1 x_2 x_3 \cdot (+1) = x_1 x_2 x_3 \quad (10.47)$$

since each  $y_i = \pm 1$ .

Therefore the hidden variables predict  $x_1 x_2 x_3 = +1$ : the predetermined  $X$  answers from the three particles must multiply to  $+1$ .

But quantum mechanics predicts  $\langle X_1 X_2 X_3 \rangle = -1$ : the product is always  $-1$ .

This is not a statistical disagreement. It is an outright contradiction. No assignment of predetermined values to the three particles can simultaneously satisfy all four quantum predictions. The hidden variable program, which worked flawlessly for one qubit and failed only statistically for two, now fails on every single run of a three-particle experiment.

#### Key Idea: The GHZ Argument

The GHZ state provides an “all-or-nothing” refutation of local hidden variables. Unlike Bell’s original argument, which requires statistical analysis of many runs, the GHZ argument shows a logical contradiction from a single set of measurements. No assignment of predetermined values to the three particles can reproduce all four quantum predictions simultaneously.

## 10.7 Experimental Tests

Bell’s theorem and the GHZ argument are not philosophical speculations—they make quantitative predictions that differ from any local hidden variable theory. Over more than four decades, a series of increasingly sophisticated experiments has confirmed the quantum predictions and ruled out local realism.

### 10.7.1 Bell Inequality Tests with Photon Pairs

The first experimental test of a Bell inequality was performed by Freedman and Clauser in 1972 using polarization-entangled photon pairs produced by an atomic cascade source. The measured correlations violated the CHSH inequality, favoring quantum mechanics over local hidden variables.

The most influential early experiments were those of Aspect, Dalibard, and Roger (1982), who introduced rapid switching of polarizer settings during the flight of the photons. This addressed

the *locality loophole*: the concern that if the measurement settings were chosen in advance, some local signal could communicate Alice’s setting to Bob’s detector in time to influence the result. By switching settings after the photons had separated, the experiment ensured that no light-speed signal could carry the information. The measured correlations violated the CHSH inequality in agreement with quantum predictions.

However, each early experiment left open at least one loophole:

**The locality loophole:** Were the measurement settings truly chosen with spacelike separation, so that no signal could connect Alice’s choice to Bob’s outcome? Aspect’s switching experiment was a major step. Weihs et al. (1998) strengthened this by using quantum random number generators to select settings and separating the detectors by 400 meters, ensuring strict spacelike separation.

**The detection loophole:** Photon detectors are imperfect—typical efficiencies were 10–20% in early experiments. If only a small fraction of emitted pairs are detected, one might worry that the detected subset is not representative of the full ensemble. A sufficiently biased subset could violate the CHSH inequality even if the full ensemble did not. Rowe et al. (2001) closed this loophole using trapped ions, which can be detected with near-unit efficiency, though the ions were not spacelike-separated.

The definitive experiments came in 2015, when three independent groups closed *both* loopholes simultaneously. Hensen et al. used entangled electron spins in nitrogen-vacancy centers in diamond, separated by 1.3 km. Giustina et al. and Shalm et al. used high-efficiency superconducting photon detectors with spacelike-separated measurement stations. All three experiments confirmed the quantum predictions and violated the CHSH inequality with high statistical significance.

The experimental verdict is unambiguous: local hidden variable theories are ruled out.

### 10.7.2 The GHZ Experiment

The GHZ argument is theoretically sharper than CHSH—an all-or-nothing contradiction on every run, rather than a statistical violation accumulated over many runs. But it is experimentally more demanding, requiring entanglement of at least three particles.

The first observation of three-photon GHZ entanglement was achieved by Bouwmeester et al. (1999) in Zeilinger’s group in Vienna. The technique used two pairs of polarization-entangled photons produced by pulsed parametric down-conversion. A polarizing beam splitter combined one photon from each pair, and detection of a “trigger” photon in a way that erased which-pair information projected the remaining three photons into a GHZ-entangled state.

The full nonlocality test was reported by Pan et al. (2000) in *Nature*. The experiment followed the logic of the GHZ argument directly. Three experiments measured the polarization correlations corresponding to  $XYX$ ,  $YXY$ , and  $YYX$  on the three-photon GHZ state. In each case, the product of the three outcomes was found to be +1, consistent with the quantum eigenvalue predictions. Under the assumption of local realism, these three results predict that the  $XXX$  measurement must also yield product +1. The fourth experiment measured  $XXX$  and found the product to be −1—in agreement with quantum mechanics and in direct contradiction with local realism.

The measured error rate was approximately 15% per measurement configuration. The experiment required fourfold coincidence detection (three GHZ photons plus the trigger), and entanglement was confirmed only when all detectors fired, leaving the detection loophole open. Subsequent experiments have extended GHZ tests to four, six, and even ten entangled photons, and have been performed on platforms such as superconducting qubits and trapped ions where detection efficiencies are much higher.

Zeilinger shared the 2022 Nobel Prize in Physics, in part for this experimental program demonstrating multi-photon entanglement and its foundational implications.

### Physical Insight

The progression of experimental difficulty mirrors the theoretical progression. Testing CHSH requires entangling two particles; testing GHZ requires three. Closing all loopholes simultaneously—spacelike separation *and* high detection efficiency—took decades of technological development in source brightness, detector efficiency, and random number generation. In every case, the quantum predictions have been confirmed.

## 10.8 What Bell's Theorem Means

The experiments described in the previous section have spoken decisively: the quantum predictions are correct, and local hidden variable theories are ruled out. What does this imply for our understanding of reality?

### 10.8.1 The Assumptions

Bell's theorem derives a contradiction from three assumptions. **Realism** holds that measurement outcomes are determined by pre-existing properties. **Locality** requires that the outcome at one location cannot depend on the measurement setting at a distant location for spacelike-separated measurements. **Freedom of choice** asserts that the measurement settings can be chosen independently of the hidden variables.

Since experiments violate the Bell inequality, at least one assumption must be false.

### 10.8.2 Giving Up Locality

One response is to accept that quantum mechanics involves genuine nonlocality: what happens at one location can instantaneously influence what happens at a distant location.

This does not permit faster-than-light signaling. Although Alice's measurement instantaneously affects the quantum state at Bob's location, Bob cannot detect this without knowing Alice's result, which must arrive by classical (slower-than-light) communication. Quantum mechanics maintains "peaceful coexistence" with special relativity in this sense.

Nevertheless, the nonlocality is real: the joint probability distribution for Alice's and Bob's outcomes cannot be explained by any local mechanism. The universe exhibits correlations that cannot be accounted for by information traveling from a common source.

### 10.8.3 Giving Up Realism

An alternative is to abandon realism: measurement outcomes are not revealing pre-existing values but are created in the act of measurement. On this view, it is simply meaningless to ask what Bob's spin "really is" before he measures it.

This is the orthodox Copenhagen interpretation. It maintains locality in the sense that no physical influence travels between the particles, because there is no underlying reality to be influenced. The correlations arise from the entangled quantum state, which is not itself a local object.

Note the irony: the single-qubit hidden variable models of Section 2 gave every qubit definite answers to all spin questions, and Bell's continuous model extended this to every direction on the sphere. The Copenhagen interpretation says we should not have taken that success too seriously. What seemed like evidence for an underlying deterministic reality was merely the fact that single-particle statistics happen to be reproducible by hidden variables. The failure only becomes apparent

in the richer correlational structure of entangled systems, where shared hidden variables cannot produce correlations as strong as those quantum mechanics predicts.

#### 10.8.4 Giving Up Freedom of Choice

A more radical option is “superdeterminism”: perhaps the measurement settings are not truly free but are correlated with the hidden variables. If the universe conspired to always choose settings that produce the observed correlations, no violation of locality or realism would be required.

Most physicists reject this option as undermining the foundations of experimental science. If we cannot choose measurement settings independently, we cannot trust any experiment.

#### 10.8.5 Interpretations of Quantum Mechanics

Different interpretations of quantum mechanics respond to Bell's theorem in different ways:

**Copenhagen interpretation:** Accepts that quantum mechanics is complete and that asking about pre-measurement values is meaningless. The “collapse” of the wavefunction is not a physical process but an update of our knowledge.

**Many-worlds interpretation:** All outcomes occur in different branches of a universal wavefunction. There is no collapse and no nonlocality; the appearance of definite outcomes arises from decoherence and the observer's own branching.

**Bohmian mechanics:** A deterministic, nonlocal hidden variable theory. Particles have definite positions at all times, guided by a “pilot wave” that depends on the configuration of all particles. Explicitly nonlocal, it reproduces quantum predictions exactly.

**Collapse theories (GRW):** Physical collapse mechanisms cause spontaneous localization, modifying the Schrödinger equation. The nonlocality is explicit in the collapse dynamics.

Bell's theorem does not adjudicate between these interpretations. They all reproduce the experimental predictions. What it does rule out is any interpretation that is both local and realistic in the senses defined above.

#### 10.8.6 The Significance for Physics

Bell's theorem transforms what had been a philosophical debate into an empirical question. The EPR argument seemed to show that quantum mechanics was incomplete and must be supplemented by hidden variables. Bell showed that such a supplementation, if local, leads to testable predictions that differ from quantum mechanics. Experiments have decided the matter: the quantum predictions are correct.

**On the nature of reality:** The classical picture of a world with definite properties existing independently of observation, with influences propagating locally, cannot be the whole story. Quantum mechanics reveals a world that is, at a fundamental level, stranger than our everyday intuitions suggest.

**On the completeness of quantum mechanics:** In a precise sense, quantum mechanics cannot be “completed” by adding local hidden variables. Whatever deeper theory might underlie quantum mechanics, it cannot restore a classical worldview.

**On the unity of physics:** The tension between quantum nonlocality and relativistic locality remains an open problem. Quantum field theory manages to combine quantum mechanics with special relativity, but the interpretation of this combination, especially regarding entanglement and measurement, is not fully understood.

## 10.9 Chapter Summary

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The Bloch cube defines a natural hierarchy of measurement directions—faces, edges, and corners—that organizes the entire hidden variable story. For a single qubit, hidden variable models succeed completely at every level of this hierarchy: the face model reproduces predictions for the three face directions, an analogous corner model handles the four body diagonals, and Bell's 1966 continuous model extends to all directions on the sphere. Single-particle quantum randomness is fully compatible with realism, and the realist hypothesis is unfalsifiable by any single-particle experiment.

Entangled states cannot be written as products of individual subsystem states. The Bell states form a complete basis for two-qubit systems and exhibit perfect correlations between distant measurements.

The EPR argument, emboldened by the success of single-qubit realism, attempted to extend hidden variables to entangled systems while maintaining locality. Bell's theorem proves this is impossible. The CHSH inequality provides a quantitative test: local theories satisfy  $|S| \leq 2$ , while quantum mechanics predicts  $|S| = 2\sqrt{2}$  for optimal measurements on singlet states. The optimal measurement directions are two faces and two edges of the Bloch cube, arranged in the  $xz$ -plane at  $0^\circ$ ,  $45^\circ$ ,  $90^\circ$ , and  $135^\circ$ . The face-edge geometry produces the asymmetry—three pairs at  $45^\circ$ , one at  $135^\circ$ —that drives the maximal violation.

The  $45^\circ$  angle that defines an edge direction is the same angle that appears in the CHSH game's winning probability  $\cos^2(22.5^\circ) \approx 85\%$ , exceeding the classical limit of 75%.

GHZ states provide an even starker contradiction: three sets of hidden face values make a definite prediction ( $x_1x_2x_3 = +1$ ) that quantum mechanics contradicts ( $-1$ ) in every single run, not just statistically.

Experimental tests confirm quantum mechanical predictions. At least one of the assumptions underlying local realism must be abandoned. How we interpret this result depends on our preferred interpretation of quantum mechanics, but no interpretation can restore a fully local and realistic worldview.

## 10.10 Problems

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### Problem 10.1 (*Hidden Bloch Cubes: The Face Model*)

A particle is prepared in the state  $|+z\rangle$ . In the face model, the complete hidden-variable description is a triple  $(|+z\rangle, |\epsilon_x\rangle, |\epsilon_y\rangle)$  where  $\epsilon_x, \epsilon_y = \pm 1$  are hidden answers to the  $X$  and  $Y$  questions. The four corners of the top face are equally probable:  $p_1 = p_2 = p_3 = p_4 = 1/4$ .

- Verify that this uniform distribution reproduces  $P(+x) = 1/2$  and  $P(+y) = 1/2$ , as quantum mechanics requires.
- The face model assigns a measurement rule  $A(\hat{n}) = \text{sign}(\hat{n} \cdot \vec{\lambda})$ , where  $\vec{\lambda} = (\epsilon_x, \epsilon_y, \epsilon_z)$ . Show that this correctly returns  $\epsilon_x$  when measuring along  $\hat{x}$  and  $\epsilon_z$  when measuring along  $\hat{z}$ .
- Now consider the edge direction  $\hat{e}_+ = (\hat{z} + \hat{x})/\sqrt{2}$ . Show that the face model predicts  $P(+1|\hat{e}_+) = 3/4$  for a particle in  $|+z\rangle$ .
- Quantum mechanics predicts  $P(+1|\hat{e}_+) = \cos^2(22.5^\circ) \approx 0.854$ . Compare this with your answer from (c). Does the face model succeed or fail at edge directions?
- The face model works with four equally-weighted hidden states. Explain physically why a finite discrete model cannot reproduce a continuous function like  $\cos^2(\theta/2)$  for arbitrary angles  $\theta$ .

**Problem 10.2** (*Bell's Continuous Hidden Variable Model*)

Bell's continuous model assigns each particle a hidden unit vector  $\hat{\lambda}$  on the sphere, with the measurement rule  $f_{\hat{\lambda}}(\hat{n}) = \text{sign}(\hat{n} \cdot \hat{\lambda})$ . This divides the sphere into two hemispheres separated by the great circle perpendicular to  $\hat{n}$ .

- For a particle prepared in  $|+z\rangle$  (Bloch vector  $\hat{s} = \hat{z}$ ), the probability of  $+1$  along direction  $\hat{n}$  at polar angle  $\theta$  from  $\hat{z}$  must satisfy  $P(+1 | \hat{n}) = (1 + \cos \theta)/2$ . Show that this equals  $\cos^2(\theta/2)$ .
- Consider  $\hat{n} = \hat{z}$ . The measurement hemisphere is  $\hat{n} \cdot \hat{\lambda} > 0$ , i.e., the upper hemisphere. Show geometrically that  $P(+1 | \hat{z}) = 1$  requires all the probability weight  $\rho(\hat{\lambda})$  to lie in the upper hemisphere.
- Now consider  $\hat{n}$  at angle  $\theta$  from  $\hat{z}$ . The great circle perpendicular to  $\hat{n}$  carves out a fraction of the upper hemisphere that gives  $f = -1$ . Sketch the geometry for  $\theta = 45^\circ$  and explain qualitatively why  $P(+1) < 1$  but  $P(+1) > 1/2$ .
- Bell's model reproduces all single-qubit quantum predictions for any direction  $\hat{n}$ . Does this mean hidden variables are the correct description of a single qubit? Explain what "unfalsifiable" means in this context.

**Problem 10.3** (*Bell States and Entanglement*)

The four Bell states are:

$$\begin{aligned} |\Phi^+\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) & |\Phi^-\rangle &= \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \\ |\Psi^+\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) & |\Psi^-\rangle &= \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \end{aligned}$$

- Show that the four Bell states are orthonormal.
- Compute the reduced density matrix  $\rho_A = \text{Tr}_B(|\Phi^+\rangle\langle\Phi^+|)$ . Repeat for  $|\Psi^-\rangle$ . What do you notice?
- For  $|\Phi^+\rangle$ , suppose Alice measures  $\sigma_z$  and obtains  $+1$ . What is Bob's post-measurement state? Repeat for the case where Alice obtains  $-1$ .
- For  $|\Psi^-\rangle$  (the singlet), suppose Alice measures along an arbitrary direction  $\hat{n}$  and obtains  $+1$ . Show that Bob's post-measurement state is the spin-down state along  $\hat{n}$ .
- The singlet state has perfect anti-correlation in every basis simultaneously. Explain why this does not contradict locality.

**Problem 10.4** (*The EPR Argument and Locality*)

Alice and Bob share the singlet state  $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$  and are separated by a large distance.

- Alice measures  $\sigma_z$  on her qubit. With what probability does she obtain each outcome? For each outcome, what can she predict about Bob's  $\sigma_z$  result with certainty?
- Alice instead measures  $\sigma_x$ . For each outcome, what can she predict about Bob's  $\sigma_x$  result?
- EPR's locality assumption: Alice's choice of measurement cannot instantaneously affect Bob's particle. Using this assumption, argue that Bob's particle must have had definite values of both  $\sigma_z$  and  $\sigma_x$  before either measurement.

- (d) Quantum mechanics says  $\sigma_z$  and  $\sigma_x$  cannot simultaneously have definite values. What logical conclusion did EPR draw?
- (e) Identify the assumption in the EPR argument that Bell's theorem ultimately shows must be abandoned.

**Problem 10.5** (*The CHSH Inequality*)

In a local hidden variable theory, Alice's outcome  $A(\vec{a}, \lambda) = \pm 1$  depends on her setting  $\vec{a}$  and a shared hidden variable  $\lambda$ , but not on Bob's setting  $\vec{b}$ . Similarly,  $B(\vec{b}, \lambda) = \pm 1$  is independent of  $\vec{a}$ .

- (a) For a single value of  $\lambda$ , define  $S(\lambda) = A(\vec{a})[B(\vec{b}) - B(\vec{b}')] + A(\vec{a}')[B(\vec{b}) + B(\vec{b}')]$ . Show that if  $B(\vec{b}) = B(\vec{b}')$ , then  $|S(\lambda)| = 2$ .
- (b) Show that if  $B(\vec{b}) = -B(\vec{b}')$ , then  $|S(\lambda)| = 2$  as well.
- (c) Since  $B(\vec{b})$  and  $B(\vec{b}')$  are each  $\pm 1$ , one of the two cases above must hold. Conclude that  $|S(\lambda)| = 2$  for every  $\lambda$ .
- (d) Averaging over  $\rho(\lambda)$ , derive the CHSH inequality:  $|E(\vec{a}, \vec{b}) - E(\vec{a}, \vec{b}') + E(\vec{a}', \vec{b}) + E(\vec{a}', \vec{b}')| \leq 2$ .
- (e) Could a cleverly chosen distribution  $\rho(\lambda)$  violate this bound? Explain why or why not.

**Problem 10.6** (*Quantum Violation with Faces and Edges*)

For the singlet state  $|\Psi^-\rangle$ , the quantum correlation function is  $E(\vec{a}, \vec{b}) = -\cos \theta_{ab}$ , where  $\theta_{ab}$  is the angle between the measurement directions.

- (a) Alice measures along two face directions  $\hat{a} = \hat{z}$  and  $\hat{a}' = \hat{x}$ . Bob measures along two edge directions  $\hat{b} = (\hat{z} + \hat{x})/\sqrt{2}$  and  $\hat{b}' = (\hat{x} - \hat{z})/\sqrt{2}$ . Compute the four angles  $\theta_{ab}, \theta_{ab'}, \theta_{a'b}, \theta_{a'b'}$ .
- (b) Calculate the CHSH combination  $S = E(\vec{a}, \vec{b}) - E(\vec{a}, \vec{b}') + E(\vec{a}', \vec{b}) + E(\vec{a}', \vec{b}')$  using  $E(\theta) = -\cos \theta$ .
- (c) Verify that  $|S| = 2\sqrt{2}$ , violating the CHSH bound. Explain the role of the face-edge geometry in producing maximal violation.
- (d) For a local hidden variable model with  $\hat{\lambda}$  uniformly distributed on the sphere, show that  $E_{\text{local}}(\theta) = -1 + 2\theta/\pi$ .
- (e) Compare the behaviors of the local and quantum correlations near  $\theta = 0$ . Which departs more slowly from perfect anti-correlation, and what does this tell you about quantum "stickiness"?

**Problem 10.7** (*Concurrence and Entanglement Measures*)

The concurrence  $C(\rho)$  is an entanglement measure for two-qubit states defined by  $C = \max(0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4)$ , where  $\lambda_i$  are the eigenvalues of  $R = \sqrt{\sqrt{\rho} \tilde{\rho} \sqrt{\rho}}$  in decreasing order, and  $\tilde{\rho} = (\sigma_y \otimes \sigma_y) \rho^* (\sigma_y \otimes \sigma_y)$ .

- (a) For a pure state  $|\psi\rangle_{AB}$ , show that the concurrence equals  $C = 2|\alpha\gamma - \beta\delta|$  where  $|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$ .
- (b) Compute the concurrence for the Bell state  $|\Phi^+\rangle$ .



- (c) Compute the concurrence for a separable state  $|\psi\rangle_A \otimes |\phi\rangle_B$ . What does this tell you about separable states?
- (d) For the mixed state  $\rho = p|\Phi^+\rangle\langle\Phi^+| + (1-p)\frac{\mathbb{I}}{4}$ , the concurrence is  $C = \max(0, 2p/2 - 1/2)$ . For what value of  $p$  does entanglement disappear?
- (e) Relate the concurrence to the entanglement of formation  $E_f(\rho) = h\left(\frac{1+\sqrt{1-C^2}}{2}\right)$ , where  $h(x) = -x \log_2 x - (1-x) \log_2 (1-x)$ .

**Problem 10.8** (*Teleportation and Superdense Coding*)

- (a) In quantum teleportation, Alice holds half of a Bell pair and the unknown state  $|\psi\rangle$  on her qubit. She performs a Bell measurement on these two qubits, projecting onto one of four Bell states with equal probability  $1/4$ . If the measurement outcome is  $|\Psi^+\rangle$ , what unitary operation must Bob apply to his qubit to recover  $|\psi\rangle$ ?
- (b) Verify that after Bob applies the corrective unitary, his state is indeed  $|\psi\rangle$  (up to global phase).
- (c) In superdense coding, Alice and Bob share a Bell pair. Alice applies one of four unitary operations to her qubit:  $I$ ,  $\sigma_x$ ,  $\sigma_z$ , or  $i\sigma_y$ . For each operation, determine the resulting two-qubit state. Are they orthogonal?
- (d) Why does superdense coding allow Alice to transmit two classical bits to Bob by sending only one qubit?
- (e) List the two key differences between teleportation and superdense coding in terms of resource use (pre-shared entanglement, classical bits, and qubits sent).

**Problem 10.9** (*Entanglement Entropy of Bipartite Pure States*)

- (a) For a pure bipartite state  $|\psi\rangle_{AB}$  with Schmidt decomposition  $|\psi\rangle = \sum_i \sqrt{\lambda_i} |u_i\rangle_A |v_i\rangle_B$ , define the entanglement entropy as  $S_E = -\sum_i \lambda_i \log_2 \lambda_i$ .
- (b) Show that  $S_E = S_A = S_B$ , where  $S_A = -\text{Tr}(\rho_A \log_2 \rho_A)$  is the von Neumann entropy of the reduced density matrix on subsystem  $A$ .
- (c) Compute  $S_E$  for the Bell state  $|\Phi^+\rangle$ . Is it maximally entangled?
- (d) Compute  $S_E$  for a separable state  $|\psi\rangle_A |\phi\rangle_B$ .
- (e) For the state  $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ , what is the entanglement entropy? Compare it with  $|\psi'\rangle = \frac{1}{\sqrt{2}}|0\rangle(|0\rangle + |1\rangle)$ . Which is more entangled?

**Problem 10.10** (*The GHZ State and Three-Body Correlations*)

The GHZ state for three qubits is:

$$|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle - |111\rangle)$$

- (a) Verify that  $X_1Y_2Y_3|\text{GHZ}\rangle = +|\text{GHZ}\rangle$ ,  $Y_1X_2Y_3|\text{GHZ}\rangle = +|\text{GHZ}\rangle$ , and  $Y_1Y_2X_3|\text{GHZ}\rangle = +|\text{GHZ}\rangle$ .
- (b) Verify that  $X_1X_2X_3|\text{GHZ}\rangle = -|\text{GHZ}\rangle$ .
- (c) Assume local hidden variables with predetermined values  $x_i = \pm 1$  and  $y_i = \pm 1$  for each qubit. Show that the eigenvalue equations  $x_1y_2y_3 = +1$ ,  $y_1x_2y_3 = +1$ ,  $y_1y_2x_3 = +1$ , and  $x_1x_2x_3 = -1$  are mutually incompatible.
- (d) Explain why the GHZ contradiction is stronger than the CHSH inequality: one is deterministic while the other is statistical.
- (e) Can you prepare a state of three qubits such that all three of  $X_1Y_2Y_3$ ,  $Y_1X_2Y_3$ , and  $Y_1Y_2X_3$  measure to  $+1$  while  $X_1X_2X_3$  measures to  $+1$  (not  $-1$ )? Why or why not?

**Problem 10.11** (*Werner States and the Entanglement Threshold*)

A Werner state is a mixture of a Bell pair and white noise:

$$\rho_W(p) = p|\Psi^-\rangle\langle\Psi^-| + (1-p)\frac{\mathbb{I}_4}{4}$$

where  $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$  and  $p \in [0, 1]$ .

- (a) Show that  $\rho_W(p)$  is a valid density matrix for all  $p \in [0, 1]$ .
- (b) Compute the concurrence  $C(p)$  of the Werner state. For what value  $p_c$  does entanglement disappear?
- (c) Compute the CHSH expectation value  $\langle S \rangle$  for measurements along the optimal face-edge geometry used in Problem 6. Show that  $\langle S \rangle = 2\sqrt{2}p$ .
- (d) Determine the critical value  $p_{\text{CHSH}}$  below which the Werner state cannot violate the CHSH inequality, even with optimal measurements.
- (e) Compare  $p_c$  (entanglement threshold) with  $p_{\text{CHSH}}$  (CHSH violation threshold). What does the difference tell you about the relationship between entanglement and nonlocality?

**Problem 10.12** (*No-Cloning Theorem*)

- (a) State the no-cloning theorem: it is impossible to construct a unitary operation that copies an arbitrary unknown quantum state.
- (b) Prove the no-cloning theorem by assuming a hypothetical "cloning" unitary  $U$  satisfying  $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$  for any state  $|\psi\rangle$ . Apply  $U$  to an arbitrary superposition  $|\phi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$  and derive a contradiction with unitarity.
- (c) Explain why classical information can be copied (e.g., one can always copy a bit) but quantum information cannot.
- (d) Show that the no-cloning theorem is consistent with the linearity of quantum mechanics. (Hint: consider whether the cloning operation is linear in the state.)

- (e) If one is allowed to make multiple copies of a state provided one knows in advance what that state is (i.e., the state is classical knowledge, not quantum data), can one clone? Reconcile your answer with the no-cloning theorem.

**Problem 10.13** (*Mermin Inequality for Three Qubits*)

- (a) The Mermin inequality for three qubits states that any local hidden variable model predicts  $|M| \leq 4$ , where  $M = \langle A_1 B_1 C_1 \rangle + \langle A_1 B_2 C_2 \rangle + \langle A_2 B_1 C_2 \rangle - \langle A_2 B_2 C_1 \rangle$  for dichotomic observables  $A_i, B_i, C_i \in \{+1, -1\}$ .
- (b) Derive the Mermin bound by writing  $M = A_1(B_1 C_1 + B_2 C_2) + A_2(B_1 C_2 - B_2 C_1)$  and using the fact that  $(C_1 \pm C_2) \in \{0, \pm 2\}$  for dichotomic observables.
- (c) For the GHZ state  $|\text{GHZ}\rangle$  and measurements  $A_1 = \sigma_z$ ,  $A_2 = \sigma_x$ ,  $B_1 = \sigma_z$ ,  $B_2 = \sigma_x$ ,  $C_1 = \sigma_z$ ,  $C_2 = \sigma_x$ , compute  $\langle A_i B_j C_k \rangle$ .
- (d) Calculate  $M$  for the GHZ state with the measurements in part (c). Does it violate the Mermin inequality?
- (e) Compare the structure of the Mermin inequality to the CHSH inequality: how many measurement choices are involved in each, and how many parties?

## References and Further Reading

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# Chapter 11

## The Dirac Equation

### 11.1 Introduction

**Conventions.** Throughout this chapter we work in natural units  $\hbar = c = 1$  and use the metric signature  $g_{\mu\nu} = \text{diag}(+1, -1, -1, -1)$  (mostly minus), except where indicated otherwise. Greek indices  $\mu, \nu, \dots$  run over 0, 1, 2, 3 and Latin indices  $i, j, \dots$  over the spatial components 1, 2, 3.

The Schrödinger equation  $i\partial_t\psi = H\psi$  is first-order in time but, through  $H = -\nabla^2/2m$ , second-order in space. This asymmetry is fundamentally non-relativistic: special relativity treats time and space on equal footing, so a relativistic wave equation should treat their derivatives symmetrically.

#### 11.1.1 The Klein-Gordon Equation

The most obvious fix is to promote  $E^2 = p^2 + m^2$  directly to an operator equation. Using  $E \rightarrow i\partial_t$  and  $\vec{p} \rightarrow -i\nabla$  gives

$$(\partial_t^2 - \nabla^2 + m^2)\phi = 0 \quad (11.1)$$

This is the Klein-Gordon equation. Its plane wave solutions  $\phi = e^{i(\vec{k}\cdot\vec{x} - \omega t)}$  satisfy  $\omega^2 = k^2 + m^2$ , giving the correct relativistic dispersion. But the equation is second-order in time, which ruins the probability interpretation. To see why, we derive the conserved density by multiplying by  $\phi^*$  and subtracting the complex conjugate:

$$\phi^* \partial_t^2 \phi - \phi \partial_t^2 \phi^* = \phi^* \nabla^2 \phi - \phi \nabla^2 \phi^* \quad (11.2)$$

The left side equals  $\partial_t(\phi^* \partial_t \phi - \phi \partial_t \phi^*)$  and the right side is  $\nabla \cdot (\phi^* \nabla \phi - \phi \nabla \phi^*)$ . This gives the continuity equation  $\partial_t \rho + \nabla \cdot \vec{j} = 0$  with

$$\rho = \frac{i}{2m} (\phi^* \partial_t \phi - \phi \partial_t \phi^*) \quad (11.3)$$

where the factor of  $1/2m$  is chosen so that  $\rho \rightarrow |\phi|^2$  in the non-relativistic limit. For a positive-frequency plane wave with  $\omega = +\sqrt{k^2 + m^2}$ , this gives  $\rho = \omega/m > 0$ . But for a negative-frequency solution with  $\omega \rightarrow -\omega$ , we get  $\rho = -\omega/m < 0$ . Since a general solution superposes both,  $\rho$  can take either sign and cannot be interpreted as a probability density.

The root cause is that a second-order equation requires two pieces of initial data ( $\phi$  and  $\partial_t\phi$ ), so the conserved density necessarily involves  $\partial_t\phi$  and is not positive-definite. (In quantum field theory the Klein-Gordon equation is rehabilitated:  $\rho$  becomes a charge density, which can naturally be negative. But as a single-particle equation, it fails.)

### 11.1.2 Dirac's Insight

Dirac's response (1928) was to go the other direction: instead of promoting the Schrödinger equation to second-order in both derivatives, demote the spatial part to first-order. This means writing  $i\partial_t\psi = H\psi$  with

$$H = -i\alpha\partial_x + \beta m \quad (11.4)$$

where  $\alpha$  and  $\beta$  are constants to be determined. Applying  $H$  twice must recover the relativistic dispersion,  $H^2 = p^2 + m^2$ , and expanding  $H^2 = (-i\alpha\partial_x + \beta m)^2 = \alpha^2 p^2 + \{\alpha, \beta\}mp + \beta^2 m^2$  shows this requires  $\alpha^2 = 1$ ,  $\beta^2 = 1$ , and  $\{\alpha, \beta\} = 0$ . No ordinary numbers satisfy all three conditions. The crucial realization is that  $\alpha$  and  $\beta$  can be *matrices*—the simplest choice being  $2 \times 2$  matrices like the Pauli matrices. This forces  $\psi$  to be a multi-component object (a spinor rather than a scalar), and spin-1/2, antiparticles, and the electron's magnetic moment all emerge as consequences.

Following our earlier approach, we will build up to the full Dirac equation by first considering discrete versions, providing both mathematical insight and connections to condensed matter physics.

#### Key Idea: From Discrete to Continuous

By starting with discrete time and space, we can see how the Dirac equation emerges naturally from requiring a first-order difference equation that respects relativistic symmetries. This approach also reveals deep connections to topological insulators and graphene.

A word on interpretation as we proceed. In the discrete and condensed-matter models—the 1D lattice, the SSH chain, graphene—the two-component spinor encodes *sublattice pseudospin*: the two components label which atom in the unit cell the electron occupies, and the Pauli matrices  $\sigma_x, \sigma_y, \sigma_z$  act on that sublattice degree of freedom. Real electron spin is a separate, spectator degree of freedom suppressed in those models. In the full relativistic Dirac equation, by contrast, the Pauli matrices act on *genuine spin-1/2*, and the four-component spinor additionally accommodates particle and antiparticle solutions. The mathematical structure—anticommuting matrices whose squares and mutual anticommutators are fixed by the metric—is identical in both settings, which is why the condensed-matter analogy illuminates the algebra so cleanly. But the physical meaning of the internal degree of freedom changes at the transition from lattice to relativistic theory, and that distinction should be kept in mind throughout the chapter.

## 11.2 Discrete Time and Space Dirac Equation

### 11.2.1 The Discrete Setting

Consider a particle on a 1D lattice with spacing  $a = 1$  and discrete time steps  $\Delta t = 1$ . We also set  $\hbar = c = 1$ . Position:  $x = n$  and time:  $t = m$ .

The discrete Schrödinger equation is:

$$i(\psi_n(m+1) - \psi_n(m)) = (H\psi)_n(m) \quad (11.5)$$

### 11.2.2 Relativistic Constraints

As discussed in the introduction, we want a wave equation that is first-order in both time and space, so that (1) the probability density  $\rho = \psi^\dagger \psi$  is guaranteed positive, and (2) the equation respects the relativistic symmetry between time and space. Squaring the equation must recover the relativistic dispersion  $E^2 = p^2 + m^2$ , which forces the coefficients of the spatial and mass terms to be anticommuting matrices rather than ordinary numbers. On our discrete lattice, this means the Hamiltonian must involve Pauli matrices acting on a two-component spinor.

### 11.2.3 The Discrete Dirac Hamiltonian

To write the Hamiltonian we work in the position representation, where the state at time  $m$  is specified by a two-component spinor  $\psi_n(m)$  at each site  $n$  (we will usually suppress the time index and write  $\psi_n$ , since the Hamiltonian acts at a single time step). The action of the Hamiltonian on this wavefunction is

$$(H\psi)_n = \frac{1}{2i}\sigma_x(\psi_{n+1} - \psi_{n-1}) + m\sigma_z\psi_n \quad (11.6)$$

The first term is a nearest-neighbor hopping: the value of  $\psi$  at site  $n$  is coupled to its neighbors at  $n \pm 1$  through  $\sigma_x$ . The second term is an on-site mass that acts through  $\sigma_z$ .

#### Physical Insight

The appearance of two components is not accidental; relativity requires doubling the degrees of freedom to accommodate both positive and negative energy solutions. On the lattice, this manifests as the two sublattices.

#### Example 1

**Explicit Spinor Evolution** Consider the discrete Dirac equation with  $m = 0$  (massless case).

Writing the spinor as  $\psi_n = \begin{pmatrix} \psi_n^A \\ \psi_n^B \end{pmatrix}$ , the Hamiltonian becomes

$$H\psi_n = \frac{1}{2i} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \psi_{n+1}^A - \psi_{n-1}^A \\ \psi_{n+1}^B - \psi_{n-1}^B \end{pmatrix} = \frac{1}{2i} \begin{pmatrix} \psi_{n+1}^B - \psi_{n-1}^B \\ \psi_{n+1}^A - \psi_{n-1}^A \end{pmatrix} \quad (11.7)$$

The two components couple to each other through the hopping terms. Component  $A$  at site  $n$  evolves based on the values of component  $B$  at neighboring sites, and vice versa. This mixing between components is the essence of how the Dirac equation generates spin dynamics.

### 11.2.4 Dispersion Relation

To find the energy spectrum, we look for stationary states of the form  $\psi_n = u e^{ikn}$ , where  $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$  is a constant spinor and  $k$  is the lattice momentum. Substituting into the Hamiltonian,

$$H(u e^{ikn}) = \frac{1}{2i}\sigma_x(u e^{ik(n+1)} - u e^{ik(n-1)}) + m\sigma_z u e^{ikn} = e^{ikn} \left[ \frac{e^{ik} - e^{-ik}}{2i}\sigma_x + m\sigma_z \right] u \quad (11.8)$$

Recognizing  $(e^{ik} - e^{-ik})/2i = \sin k$ , the eigenvalue equation  $Hu = Eu$  becomes

$$[\sin(k)\sigma_x + m\sigma_z]u = Eu \quad (11.9)$$

or in matrix form,

$$\begin{pmatrix} m & \sin k \\ \sin k & -m \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = E \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \quad (11.10)$$

Setting  $\det(H - EI) = 0$  gives  $(m - E)(-m - E) - \sin^2 k = 0$ , so

$$E(k) = \pm E_k, \quad E_k \equiv \sqrt{\sin^2(k) + m^2} \quad (11.11)$$

To find the eigenstates, take the first row of the matrix equation:  $(m - E)u_1 + \sin(k)u_2 = 0$ . For the positive-energy branch  $E = +E_k$ , this gives  $u_2 = (E_k - m)u_1 / \sin k$ , so the unnormalized eigenvector is proportional to  $\begin{pmatrix} \sin k \\ E_k - m \end{pmatrix}$ . Equivalently, using the second row to eliminate in the other direction, we can write it as  $\begin{pmatrix} E_k + m \\ \sin k \end{pmatrix}$ , which avoids a singularity at  $k = 0$ . Normalizing,

$$u_+(k) = \frac{1}{\sqrt{2E_k(E_k + m)}} \begin{pmatrix} E_k + m \\ \sin k \end{pmatrix} \quad (11.12)$$

where we used  $|u_+|^2 = (E_k + m)^2 + \sin^2 k = 2E_k(E_k + m)$ . The negative-energy eigenstate is orthogonal to  $u_+$ :

$$u_-(k) = \frac{1}{\sqrt{2E_k(E_k + m)}} \begin{pmatrix} -\sin k \\ E_k + m \end{pmatrix} \quad (11.13)$$

In the non-relativistic limit  $k \ll 1$  and  $m \gg k$ , we have  $E_k \approx m + k^2/2m$  and  $\sin k \approx k$ , so  $u_+$  approaches  $\begin{pmatrix} 1 \\ k/2m \end{pmatrix}$ —predominantly upper component, with a small lower component of order  $v/c$ . In the massless limit  $m = 0$ , we have  $E_k = |\sin k|$  and the eigenstates become  $u_{\pm} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm \text{sgn}(\sin k) \end{pmatrix}$ , which are eigenstates of  $\sigma_x$ —each component propagates independently in one direction. We will see in section 11.3.3 that this property is called chirality.

For  $k \ll 1$ :  $E \approx \pm\sqrt{k^2 + m^2}$ , recovering the relativistic energy-momentum relation.

Here is an [interactive simulation](#) that diagonalizes this Hamiltonian on a finite lattice and visualizes the eigenstates as Bloch vectors.

## 11.3 Discrete Space, Continuous Time

### 11.3.1 The Lattice Dirac Equation

With continuous time but discrete space:

$$i \frac{\partial \psi_n}{\partial t} = \frac{1}{2i} \sigma_x (\psi_{n+1} - \psi_{n-1}) + m \sigma_z \psi_n \quad (11.14)$$

### 11.3.2 Connection to Condensed Matter

This lattice Dirac equation is not merely a mathematical stepping stone; it directly describes real physical systems. In graphene, the honeycomb lattice structure leads to linear band crossings at two inequivalent points in the Brillouin zone. Near these Dirac points, electrons behave as massless Dirac fermions with  $m = 0$ . The effective speed of light is replaced by the Fermi velocity  $v_F \approx 1/300$ .





Figure 11.1: Energy eigenstates of discrete Dirac Hamiltonian.

At the edges of topological insulators, protected surface states arise that obey effective Dirac equations. The bulk of a topological insulator is gapped, but the boundary hosts gapless states that cannot be removed without closing the bulk gap. These edge states are robust against disorder because they are protected by time-reversal symmetry.

The Su-Schrieffer-Heeger (SSH) model provides another realization. This model describes electrons hopping on a dimerized chain with alternating bond strengths. In the topological phase, zero-energy states appear at the ends of the chain, directly analogous to the edge states in the lattice Dirac equation.

### Example 2

**Graphene as a Dirac Material** In graphene's honeycomb lattice, low-energy excitations near the Dirac points obey:

$$H_{\text{graphene}} = v_F(\sigma_x k_x + \sigma_y k_y) \quad (11.15)$$

where  $v_F \approx 1/300$  is the Fermi velocity. This makes graphene a “relativistic” material despite non-relativistic electron speeds.

### 11.3.3 Chiral Symmetry

Working in momentum space, we linearize near  $k = 0$  (setting  $\sin k \approx k$ ), giving the Hamiltonian as the  $2 \times 2$  matrix

$$H(k) = k \sigma_x + m \sigma_z = \begin{pmatrix} m & k \\ k & -m \end{pmatrix} \quad (11.16)$$

The structure is transparent in the  $\sigma_z$  eigenbasis: the kinetic term  $k\sigma_x$  is purely off-diagonal, coupling sublattice  $A$  to  $B$  and  $B$  to  $A$ . The mass term  $m\sigma_z$  is diagonal, assigning opposite on-site energies to the two sublattices.

Chiral symmetry is the statement that  $H(k)$  is purely off-diagonal—equivalently, that  $\sigma_z$  anti-commutes with  $H(k)$ . Computing directly,

$$\{H(k), \sigma_z\} = k\{\sigma_x, \sigma_z\} + m\{\sigma_z, \sigma_z\} = 0 + 2mI \quad (11.17)$$

The first term vanishes because  $\sigma_x$  and  $\sigma_z$  anticommute; the second is zero only when  $m = 0$ . So chiral symmetry holds if and only if  $m = 0$ . When  $m \neq 0$  the mass term is diagonal and commutes with  $\sigma_z$ , breaking the symmetry. Physically, a nonzero mass makes the two sublattices inequivalent.

Chiral symmetry constrains the spectrum. If  $|\chi\rangle$  is an eigenstate of  $H(k)$  with energy  $E$ , then  $\sigma_z|\chi\rangle$  is an eigenstate with energy  $-E$ :

$$H(k)\sigma_z|\chi\rangle = -\sigma_z H(k)|\chi\rangle = -E(\sigma_z|\chi\rangle) \quad (11.18)$$

Every nonzero energy eigenvalue is paired with its negative. This is already visible in the dispersion  $E(k) = \pm\sqrt{k^2 + m^2}$ : the two branches are exactly  $\pm E_k$ . A zero-energy state satisfies  $E = -E = 0$  and needs no partner.

This pairing is what protects zero-energy edge states on a finite chain. Such a state sits at exactly  $E = 0$  by the symmetry of the spectrum. Any perturbation that preserves chiral symmetry must preserve the  $\pm E$  pairing, so it cannot shift a zero mode to nonzero energy without simultaneously producing a partner for it. On a finite chain with boundaries, the number of unpaired zero modes is a topological invariant tied to the bulk winding number, so no chirally symmetric perturbation can remove them. They can only disappear if two zero modes from opposite ends hybridize, or if the chiral symmetry is broken by a mass term.

## 11.4 The Continuous Dirac Equation in 1+1D

### 11.4.1 Taking the Continuum Limit

As  $a \rightarrow 0$ :

$$\frac{\psi_{n+1} - \psi_{n-1}}{2} \rightarrow \frac{\partial\psi}{\partial x} \quad (11.19)$$

The continuum 1+1D Dirac equation becomes:

$$i\frac{\partial\psi}{\partial t} = \left(-i\sigma_x\frac{\partial}{\partial x} + m\sigma_z\right)\psi \quad (11.20)$$

#### Example 3

**Plane Wave Solutions in 1+1D** Consider a plane wave solution  $\psi(x, t) = ue^{i(kx - \omega t)}$  where  $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$  is a constant spinor. Substituting into the 1+1D Dirac equation gives

$$\omega \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} m & k \\ k & -m \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \quad (11.21)$$

The eigenvalue equation  $\det(H - E) = 0$  gives  $(m - E)(-m - E) - k^2 = 0$ , which yields  $E^2 = k^2 + m^2$ . This is exactly the relativistic energy-momentum relation.

### 11.4.2 Covariant Form

The 1+1D Dirac equation as written above treats time and space asymmetrically: the left-hand side has a time derivative while the right-hand side has a spatial derivative, with different matrices ( $I$  and  $\sigma_x$ ) multiplying each. This obscures the fact that special relativity demands time and space be treated on equal footing. To make this symmetry manifest, we want to write the equation in a form where  $\partial_t$  and  $\partial_x$  appear in a single object  $\partial_\mu = (\partial_t, \partial_x)$ , contracted with a set of matrices  $\gamma^\mu$  that encode the algebraic structure.

To do this, multiply both sides of the 1+1D Dirac equation by  $\sigma_z$  from the left:

$$i\sigma_z\partial_t\psi = -i\sigma_z\sigma_x\partial_x\psi + m\sigma_z^2\psi = -i(i\sigma_y)\partial_x\psi + m\psi = \sigma_y\partial_x\psi + m\psi \quad (11.22)$$

where we used  $\sigma_z\sigma_x = i\sigma_y$ . Rearranging into the form  $i\gamma^\mu\partial_\mu\psi - m\psi = 0$ , we read off  $\gamma^0 = \sigma_z$  and  $\gamma^1 = i\sigma_y$ :

$$(i\gamma^\mu\partial_\mu - m)\psi = 0 \quad (11.23)$$

This is the covariant form. By “covariant” we mean that the equation takes the same form in every inertial frame: under a Lorentz transformation the derivative  $\partial_\mu$  and the spinor  $\psi$  each transform, but because  $\gamma^\mu\partial_\mu$  contracts a matrix-valued four-vector with a derivative four-vector, the structure  $i\gamma^\mu\partial_\mu\psi - m\psi = 0$  is preserved. The Hamiltonian form, by contrast, treats  $\partial_t$  and  $\partial_x$  differently, so Lorentz invariance is present but hidden.

But what constraints must the  $\gamma^\mu$  matrices satisfy? The answer comes from demanding the correct dispersion relation. Every solution of the Dirac equation should also satisfy the relativistic energy-momentum relation  $E^2 = p^2 + m^2$ , which in operator form is the Klein-Gordon equation  $(\partial_\mu\partial^\mu + m^2)\psi = 0$ . To check this, act on the Dirac equation with the operator  $(i\gamma^\nu\partial_\nu + m)$ :

$$(i\gamma^\nu\partial_\nu + m)(i\gamma^\mu\partial_\mu - m)\psi = (-\gamma^\nu\gamma^\mu\partial_\nu\partial_\mu - m^2)\psi = 0 \quad (11.24)$$

#### Mathematical Aside: Covariant and Contravariant Indices

In special relativity a four-vector carries either an upper (contravariant) or lower (covariant) index. The spacetime position is  $x^\mu = (t, \vec{x})$  with an upper index; the associated lower-index object is  $x_\mu = (t, -\vec{x})$ . The metric tensor  $g_{\mu\nu} = \text{diag}(+1, -1, -1, -1)$  converts between them:  $v_\mu = g_{\mu\nu}v^\nu$  (lowering), while its inverse  $g^{\mu\nu}$  raises,  $v^\mu = g^{\mu\nu}v_\nu$ . A repeated upper-lower pair is summed by the Einstein convention,  $v^\mu w_\mu \equiv \sum_\mu v^\mu w_\mu$ , and the result is a Lorentz scalar. The structure is directly analogous to bras and kets:  $v^\mu$  is like a ket  $|v\rangle$ , the metric maps it to the covariant dual  $v_\mu$  which is like the bra  $\langle v|$ , and the contraction  $v^\mu w_\mu$  is the inner product.

For this to reduce to  $(\partial_\mu\partial^\mu + m^2)\psi = 0$ , we need  $\gamma^\nu\gamma^\mu\partial_\nu\partial_\mu = g^{\mu\nu}\partial_\mu\partial_\nu$ . Since partial derivatives commute ( $\partial_\nu\partial_\mu = \partial_\mu\partial_\nu$ ), only the symmetric part of  $\gamma^\nu\gamma^\mu$  contributes, so the condition is

$$\frac{1}{2}(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu) = g^{\mu\nu} \quad \text{i.e.} \quad \{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} \quad (11.25)$$

This is the *Clifford algebra*, and it is not an assumption we impose by hand—it is forced on us by the requirement that the first-order Dirac equation be consistent with the second-order Klein-Gordon equation. Any set of matrices satisfying this relation will produce the correct relativistic dispersion. The covariant form also makes Lorentz invariance manifest, since  $\gamma^\mu\partial_\mu$  transforms as a scalar when  $\psi$  transforms appropriately (as we will see in the section on Lorentz covariance). This would be difficult to verify starting from the Hamiltonian form, where time and space play different roles.

### Mathematical Aside: Properties of the Clifford Algebra

The relation  $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$  implies several useful properties.

**Squares and anticommutativity.** Taking  $\mu = \nu$  gives  $(\gamma^\mu)^2 = g^{\mu\mu}I$  (no sum):  $(\gamma^0)^2 = +I$  and  $(\gamma^i)^2 = -I$ . For  $\mu \neq \nu$ , the anticommutator vanishes, so  $\gamma^\mu\gamma^\nu = -\gamma^\nu\gamma^\mu$ : distinct gamma matrices anticommute.

**Independent elements.** In  $d$  spacetime dimensions, products of gamma matrices generate a  $2^d$ -dimensional algebra. In 3+1D ( $d = 4$ ) the 16 independent elements are  $I$ ,  $\gamma^\mu$  (4),  $\gamma^\mu\gamma^\nu$  with  $\mu < \nu$  (6),  $\gamma^\mu\gamma^\nu\gamma^\rho$  (4), and  $\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3$  (1). These correspond precisely to the scalar, vector, antisymmetric tensor, pseudovector, and pseudoscalar bilinears of the Dirac theory.

**Chirality.** The matrix  $\gamma^5$  anticommutes with all  $\gamma^\mu$ :  $\{\gamma^5, \gamma^\mu\} = 0$ . It therefore commutes with  $H$  only when  $m = 0$ , which is the relativistic generalization of the chiral symmetry we found on the lattice. Its eigenvalues  $\pm 1$  define left- and right-handed Weyl spinors.

**Trace identities.** Since  $(\gamma^\mu)^2 = \pm I$  and  $\gamma^\mu = -(\gamma^\nu\gamma^\mu)\gamma^\nu/g^{\nu\nu}$  for any fixed  $\nu \neq \mu$ , one can show  $\text{tr}(\gamma^\mu) = 0$ , and more generally that the trace of any odd number of gamma matrices vanishes. The key even-product traces are  $\text{tr}(\gamma^\mu\gamma^\nu) = 4g^{\mu\nu}$  and  $\text{tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho})$ . These identities are the workhorses of QED Feynman diagram calculations.

**Representation dimension.** In  $d$  dimensions the minimal matrix representation has dimension  $2^{\lfloor d/2 \rfloor}$ . In 1+1D that gives  $2 \times 2$  matrices (the Pauli matrices suffice); in 3+1D it gives  $4 \times 4$  matrices, forcing the Dirac spinor to have four components. Different explicit choices of  $4 \times 4$  matrices satisfying the algebra—the Dirac, Weyl, and Majorana representations—are all related by similarity transformations and describe the same physics.

#### Example 4

Verifying the Gamma Matrix Algebra in 1+1D The gamma matrices must satisfy  $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$  where  $g = \text{diag}(1, -1)$ . Let us check:  $\gamma^0 = \sigma_z$  and  $\gamma^1 = i\sigma_y$ . We need  $(\gamma^0)^2 = 1$ ,  $(\gamma^1)^2 = -1$ , and  $\{\gamma^0, \gamma^1\} = 0$ . Indeed,

$$(\gamma^0)^2 = \sigma_z^2 = I \quad (11.26)$$

$$(\gamma^1)^2 = (i\sigma_y)^2 = -\sigma_y^2 = -I \quad (11.27)$$

$$\gamma^0\gamma^1 + \gamma^1\gamma^0 = \sigma_z(i\sigma_y) + (i\sigma_y)\sigma_z = i(\sigma_z\sigma_y + \sigma_y\sigma_z) = 0 \quad (11.28)$$

using the anticommutation of Pauli matrices.

## 11.5 The Dirac Equation in 3+1D

### 11.5.1 The Need for Four Components

In 3+1D, we need three anticommuting spatial matrices plus one for mass. The Clifford algebra  $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$  requires matrices satisfying  $(\gamma^0)^2 = I$ ,  $(\gamma^i)^2 = -I$  for  $i = 1, 2, 3$ , and  $\gamma^\mu\gamma^\nu = -\gamma^\nu\gamma^\mu$  for  $\mu \neq \nu$ . No set of  $2 \times 2$  matrices can satisfy all these conditions because the Pauli matrices plus identity span the space of  $2 \times 2$  Hermitian matrices, but we need four mutually anticommuting matrices. The minimal representation therefore requires  $4 \times 4$  matrices, giving the Dirac spinor four components.

### 11.5.2 The Standard Representation

The gamma matrices can be written compactly as tensor products of Pauli matrices:

$$\gamma^0 = \sigma_z \otimes I, \quad \gamma^i = i\sigma_y \otimes \sigma^i \quad (11.29)$$

The first (left) tensor factor acts on particle/antiparticle space; the second (right) factor acts on spin space. Expanding the tensor products gives the standard block form:

$$\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} \quad (11.30)$$

This structure also clarifies the connection to the 1+1D case: the 1+1D gamma matrices  $\gamma^0 = \sigma_z$  and  $\gamma^1 = i\sigma_y$  are precisely the left tensor factors, with the spin space being trivial.

The Clifford algebra (equation 11.4.2),  $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}I$ , is straightforward to verify in the tensor product form:  $(\gamma^0)^2 = \sigma_z^2 \otimes I = I$ ,  $(\gamma^i)^2 = (i\sigma_y)^2 \otimes (\sigma^i)^2 = (-I) \otimes I = -I$ , and anticommutativity between distinct  $\gamma^\mu$  follows from the Pauli anticommutation relations in each factor.

#### Example 5

**Explicit Verification of the Clifford Algebra** Let us verify that  $\{\gamma^1, \gamma^2\} = 0$ . We compute

$$\gamma^1 \gamma^2 = \begin{pmatrix} 0 & \sigma^1 \\ -\sigma^1 & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma^2 \\ -\sigma^2 & 0 \end{pmatrix} = \begin{pmatrix} -\sigma^1 \sigma^2 & 0 \\ 0 & -\sigma^1 \sigma^2 \end{pmatrix} \quad (11.31)$$

and similarly

$$\gamma^2 \gamma^1 = \begin{pmatrix} -\sigma^2 \sigma^1 & 0 \\ 0 & -\sigma^2 \sigma^1 \end{pmatrix} \quad (11.32)$$

Since  $\sigma^1 \sigma^2 = i\sigma^3 = -\sigma^2 \sigma^1$ , we have  $\gamma^1 \gamma^2 = -\gamma^2 \gamma^1$ , confirming  $\{\gamma^1, \gamma^2\} = 0$ .

### 11.5.3 The Full Dirac Equation

The Dirac equation in covariant form:

$$(i\gamma^\mu \partial_\mu - m)\psi = 0 \quad (11.33)$$

Or explicitly:

$$i \frac{\partial \psi}{\partial t} = (-i\vec{\alpha} \cdot \nabla + \beta m) \psi \quad (11.34)$$

where  $\alpha^i = \gamma^0 \gamma^i$  and  $\beta = \gamma^0$ .

#### Example 6

**Computing the Alpha Matrices** From the gamma matrices, we find

$$\alpha^i = \gamma^0 \gamma^i = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} = \begin{pmatrix} 0 & \sigma^i \\ \sigma^i & 0 \end{pmatrix} \quad (11.35)$$

These satisfy  $(\alpha^i)^2 = I$  and  $\alpha^i \alpha^j + \alpha^j \alpha^i = 2\delta^{ij}I$  for  $i \neq j$ . Combined with  $\beta = \gamma^0$ , we have  $\alpha^i \beta + \beta \alpha^i = 0$ , which ensures that the Hamiltonian  $H = \vec{\alpha} \cdot \vec{p} + \beta m$  correctly reproduces the

relativistic dispersion when squared.

## 11.6 Solutions of the Dirac Equation

### 11.6.1 Plane Wave Solutions

For a free particle with four-momentum  $p^\mu = (E, \vec{p})$ , we try the plane wave ansatz

$$\psi(x) = u(p)e^{-ip \cdot x} \quad (11.36)$$

where  $p \cdot x = Et - \vec{p} \cdot \vec{x}$  and  $u(p)$  is a constant four-component spinor. Substituting into  $(i\gamma^\mu \partial_\mu - m)\psi = 0$ , the derivative acts only on the exponential:  $\partial_\mu e^{-ip \cdot x} = -ip_\mu e^{-ip \cdot x}$ . The exponential cancels, leaving the algebraic condition

$$(\gamma^\mu p_\mu - m)u(p) = 0 \quad (11.37)$$

This  $4 \times 4$  matrix equation has nontrivial solutions only when  $\det(\gamma^\mu p_\mu - m) = 0$ , which enforces the relativistic dispersion  $E^2 = \vec{p}^2 + m^2$ .

### 11.6.2 Positive and Negative Energy Solutions

For each momentum  $\vec{p}$ , the Dirac equation admits four independent solutions. Two of these correspond to positive energy  $E = +\sqrt{\vec{p}^2 + m^2}$  and describe a particle with spin up or spin down. The remaining two correspond to negative energy  $E = -\sqrt{\vec{p}^2 + m^2}$  and similarly come in spin up and spin down varieties. The existence of negative energy solutions was initially troubling, as it seemed to imply that particles could radiate away unlimited amounts of energy by transitioning to ever more negative energy states.

#### Key Idea: The Dirac Sea

Dirac interpreted negative energy states as a filled “sea.” Holes in this sea behave as antiparticles with positive energy and opposite charge, predicting the positron before its discovery.

#### Example 7

**Energy of a Relativistic Electron** Consider a particle with momentum  $p = m$  (a moderately relativistic case). The energy is

$$E = \pm\sqrt{p^2 + m^2} = \pm\sqrt{2}m \quad (11.38)$$

The positive energy solution describes a free particle with kinetic energy  $(\sqrt{2} - 1)m$ . The negative energy solution, in modern interpretation, describes an antiparticle state.

### 11.6.3 Spinor Solutions

The standard normalization convention is  $u^{(r)\dagger}u^{(s)} = 2E\delta^{rs}$ , chosen because  $2E$  transforms as the time component of a four-vector, making the normalization Lorentz covariant. Equivalently,

$\bar{u}^{(r)}u^{(s)} = 2m\delta^{rs}$  where  $\bar{u} = u^\dagger\gamma^0$ . In the rest frame ( $\vec{p} = 0$ ,  $E = m$ ), this requires  $u^\dagger u = 2m$ , giving

$$u^{(1)} = \sqrt{2m} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad u^{(2)} = \sqrt{2m} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad (11.39)$$

### Example 8

**Boosted Spinors** For a particle moving with momentum  $\vec{p} = p\hat{z}$ , we solve  $(\gamma^\mu p_\mu - m)u = 0$  directly. Using the standard representation with  $p^\mu = (E, 0, 0, p)$  and  $p_\mu = (E, 0, 0, -p)$ ,

$$\gamma^\mu p_\mu = \gamma^0 E - \gamma^3 p = \begin{pmatrix} E & -p\sigma^3 \\ p\sigma^3 & -E \end{pmatrix} \quad (11.40)$$

where we used the block forms  $\gamma^0 = \text{diag}(I, -I)$  and  $\gamma^3 = \begin{pmatrix} 0 & \sigma^3 \\ -\sigma^3 & 0 \end{pmatrix}$ . Writing  $u = \begin{pmatrix} \phi \\ \chi \end{pmatrix}$ , the equation  $(\gamma^\mu p_\mu - m)u = 0$  splits into two coupled equations:

$$\begin{aligned} (E - m)\phi &= p\sigma^3\chi \\ p\sigma^3\phi &= (E + m)\chi \end{aligned}$$

The second equation gives  $\chi = \frac{p\sigma^3}{E+m}\phi$ . For spin up,  $\phi = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ , and since  $\sigma^3\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ , we get  $\chi = \frac{p}{E+m}\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ . Assembling and normalizing gives

$$u^{(1)}(p) = \sqrt{E+m} \begin{pmatrix} 1 \\ 0 \\ \frac{p}{E+m} \\ 0 \end{pmatrix} \quad (11.41)$$

where  $E = \sqrt{p^2 + m^2}$ . In the non-relativistic limit  $p \ll m$ ,  $E \approx m$  and the lower component  $p/(E+m) \approx p/2m \ll 1$ , so the spinor is predominantly upper. In the ultra-relativistic limit  $p \gg m$ ,  $E \approx p$  and  $p/(E+m) \approx 1$ , so upper and lower components are comparable.

## 11.7 Properties of the Dirac Equation

### 11.7.1 Current Conservation

The Dirac equation implies conservation of probability current:

$$\partial_\mu j^\mu = 0 \quad (11.42)$$

where  $j^\mu = \bar{\psi}\gamma^\mu\psi$  and  $\bar{\psi} = \psi^\dagger\gamma^0$ .

### Example 9

**The Probability Current Components:**

The time component of the current is  $j^0 = \bar{\psi}\gamma^0\psi = \psi^\dagger\gamma^0\gamma^0\psi = \psi^\dagger\psi$ , which is manifestly positive. This resolves the problem that plagued the Klein-Gordon equation. The spatial

components are  $j^i = \bar{\psi}\gamma^i\psi = \psi^\dagger\gamma^0\gamma^i\psi = \psi^\dagger\alpha^i\psi$ . For a plane wave solution, the current  $\vec{j}$  points in the direction of the particle's momentum, as expected for probability flow.

### 11.7.2 Spin and Magnetic Moment

#### Spin-1/2 from the Matrix Structure

Ordinary orbital angular momentum  $\vec{L} = \vec{r} \times \vec{p}$  is not conserved for the Dirac Hamiltonian  $H = \vec{\alpha} \cdot \vec{p} + \beta m$ . This is not a defect of the theory; it is a clue. In non-relativistic quantum mechanics,  $\vec{L}$  fails to be conserved precisely when there is an additional angular momentum degree of freedom—spin—that must be included. The question is whether the Dirac equation supplies a natural candidate for that extra piece.

The answer is yes. Each  $4 \times 4$  matrix in the Dirac theory is built from  $2 \times 2$  blocks. Define the spin operator

$$\vec{S} = \frac{1}{2}\vec{\Sigma}, \quad \Sigma^i = \begin{pmatrix} \sigma^i & 0 \\ 0 & \sigma^i \end{pmatrix} \quad (11.43)$$

This is the unique rotationally natural  $4 \times 4$  operator that (a) acts within each  $2 \times 2$  block without mixing upper and lower components, and (b) satisfies the angular momentum algebra  $[S^i, S^j] = i\epsilon^{ijk}S^k$ . Its eigenvalues are  $\pm\frac{1}{2}$ , inherited from  $\sigma^i$ . The total angular momentum is  $\vec{J} = \vec{L} + \vec{S}$ .

#### Example 10

$[H, J^z] = 0$ : Spin-1/2 is Not a Postulate We show that  $[H, L^z] \neq 0$  but  $[H, J^z] = [H, L^z + S^z] = 0$ , establishing that  $\vec{J}$  is the conserved total angular momentum and that the Dirac equation necessarily describes spin-1/2 particles.

**Step 1:**  $[H, L^z]$  is **nonzero**. With  $L^z = xp_y - yp_x$  and using  $[p_i, x_j] = -i\delta_{ij}$ , the mass term  $\beta m$  commutes with  $L^z$ . For the kinetic terms, using  $[p_x, x] = -i$  and  $[p_y, y] = -i$  (all other pairs commute):

$$[p_x, xp_y - yp_x] = [p_x, x]p_y = -ip_y \quad (11.44)$$

$$[p_y, xp_y - yp_x] = -[p_y, y]p_x = +ip_x \quad (11.45)$$

Therefore  $[H, L^z] = \alpha^x(-ip_y) + \alpha^y(ip_x) = i(\alpha^y p_x - \alpha^x p_y) \neq 0$ .

**Step 2:**  $[H, S^z]$  **exactly cancels**. We need  $[\alpha^i, \Sigma^z]$ . Using  $\alpha^i = \begin{pmatrix} 0 & \sigma^i \\ \sigma^i & 0 \end{pmatrix}$ :

$$[\alpha^i, \Sigma^z] = \begin{pmatrix} 0 & [\sigma^i, \sigma^z] \\ [\sigma^i, \sigma^z] & 0 \end{pmatrix} \quad (11.46)$$

From the Pauli commutation relation  $[\sigma^i, \sigma^j] = 2i\epsilon^{ijk}\sigma^k$ :

$$[\sigma^x, \sigma^z] = -2i\sigma^y \quad \Rightarrow \quad [\alpha^x, \Sigma^z] = -2i\alpha^y \quad (11.47)$$

$$[\sigma^y, \sigma^z] = +2i\sigma^x \quad \Rightarrow \quad [\alpha^y, \Sigma^z] = +2i\alpha^x \quad (11.48)$$

Assembling:

$$\begin{aligned} [H, S^z] &= \frac{1}{2}p_x[\alpha^x, \Sigma^z] + \frac{1}{2}p_y[\alpha^y, \Sigma^z] \\ &= \frac{1}{2}p_x(-2i\alpha^y) + \frac{1}{2}p_y(+2i\alpha^x) = -i(\alpha^y p_x - \alpha^x p_y) \end{aligned} \quad (11.49)$$



This is exactly  $-[H, L^z]$ . Therefore  $[H, J^z] = [H, L^z] + [H, S^z] = 0$ . The same argument applies to  $J^x$  and  $J^y$ . Spin-1/2 emerges from the four-component structure of the Dirac equation—it is not put in by hand.

### The Magnetic Moment and $g = 2$

To see how a magnetic moment emerges from the Dirac equation, include an electromagnetic field via minimal coupling  $\partial_\mu \rightarrow \partial_\mu + ieA_\mu$ , or equivalently  $\vec{p} \rightarrow \vec{\Pi} \equiv \vec{p} - e\vec{A}$ . Writing  $\psi = \begin{pmatrix} \phi \\ \chi \end{pmatrix}$  and using the standard-representation Hamiltonian form, the Dirac equation becomes

$$\begin{aligned}(E - m - e\Phi)\phi &= \vec{\sigma} \cdot \vec{\Pi} \chi \\ (E + m - e\Phi)\chi &= \vec{\sigma} \cdot \vec{\Pi} \phi\end{aligned}$$

In the non-relativistic limit, write  $E = m + \varepsilon$  with  $\varepsilon \ll m$ , and neglect  $e\Phi$  relative to  $2m$  in the second equation:  $\chi \approx \frac{\vec{\sigma} \cdot \vec{\Pi}}{2m} \phi$ . Substituting into the first equation gives

$$\varepsilon \phi = \left[ \frac{(\vec{\sigma} \cdot \vec{\Pi})^2}{2m} + e\Phi \right] \phi \quad (11.50)$$

The key step is expanding  $(\vec{\sigma} \cdot \vec{\Pi})^2$  using the identity  $(\vec{\sigma} \cdot \vec{A})(\vec{\sigma} \cdot \vec{B}) = \vec{A} \cdot \vec{B} + i\vec{\sigma} \cdot (\vec{A} \times \vec{B})$ :

$$(\vec{\sigma} \cdot \vec{\Pi})^2 = \vec{\Pi}^2 + i\vec{\sigma} \cdot (\vec{\Pi} \times \vec{\Pi}) \quad (11.51)$$

Since  $\vec{\Pi} = \vec{p} - e\vec{A}$  involves an operator  $\vec{p} = -i\nabla$ , the cross product  $\vec{\Pi} \times \vec{\Pi}$  does not vanish: the commutator  $[\Pi_i, \Pi_j] = -ie(\partial_i A_j - \partial_j A_i) = -ie\epsilon_{ijk}B_k$  gives  $i\vec{\sigma} \cdot (\vec{\Pi} \times \vec{\Pi}) = -e\vec{\sigma} \cdot \vec{B}$ . Therefore

$$\varepsilon \phi = \left[ \frac{(\vec{p} - e\vec{A})^2}{2m} - \frac{e}{2m} \vec{\sigma} \cdot \vec{B} + e\Phi \right] \phi \quad (11.52)$$

This is the Pauli equation. The magnetic interaction term is  $-\frac{e}{2m} \vec{\sigma} \cdot \vec{B} = -2 \cdot \frac{e}{2m} \vec{S} \cdot \vec{B}$  (since  $\vec{S} = \frac{1}{2}\vec{\sigma}$ ), giving the magnetic moment

$$\vec{\mu} = \frac{e}{2m} \vec{\Sigma}, \quad \vec{\Sigma} = \begin{pmatrix} \vec{\sigma} & 0 \\ 0 & \vec{\sigma} \end{pmatrix} \quad (11.53)$$

with  $g = 2$ : the magnetic moment is twice the classical expectation  $e\vec{S}/m$ .

#### Example 11

**The Electron g-Factor from the  $\vec{\sigma}$  Identity** The general form of the magnetic moment interaction is  $-g\frac{e}{2m}\vec{S} \cdot \vec{B}$ , where  $g$  is the g-factor and  $\vec{S} = \frac{1}{2}\vec{\sigma}$ . The  $g = 2$  prediction follows directly from the Dirac algebra. The term  $i\vec{\sigma} \cdot (\vec{\Pi} \times \vec{\Pi})$  arises from the non-commutativity of  $\vec{\Pi}$ : using  $[\Pi_i, \Pi_j] = -ie\epsilon_{ijk}B_k$ ,

$$i\sigma_k(\vec{\Pi} \times \vec{\Pi})_k = i\sigma_k \epsilon_{kij} \frac{1}{2}[\Pi_i, \Pi_j] = i\sigma_k \epsilon_{kij} \frac{1}{2}(-ie\epsilon_{ijk}B_k) = -e\sigma_k B_k \quad (11.54)$$

So  $(\vec{\sigma} \cdot \vec{\Pi})^2 = \vec{\Pi}^2 - e\vec{\sigma} \cdot \vec{B}$ , and the Hamiltonian contains  $-\frac{e}{2m}\vec{\sigma} \cdot \vec{B} = -\frac{e}{m}\vec{S} \cdot \vec{B} = -g\frac{e}{2m}\vec{S} \cdot \vec{B}$  with  $g = 2$ . This factor of 2 is a direct consequence of the first-order structure of the Dirac equation: it would not appear in a naïve second-order theory. Experimentally,  $g \approx 2.002319$ ; the small deviation is explained by QED loop corrections.

## 11.8 Lorentz Covariance

### 11.8.1 Spinor Transformations

Under a Lorentz transformation  $x^\mu \rightarrow \Lambda^\mu_\nu x^\nu$ :

$$\psi(x) \rightarrow S(\Lambda)\psi(\Lambda^{-1}x) \quad (11.55)$$

where  $S(\Lambda)$  is the spinor representation.

#### Example 12

**Rotation of a Dirac Spinor** Under a rotation by angle  $\theta$  about the  $z$ -axis, the spinor transformation matrix is

$$S(R_z(\theta)) = \exp\left(-\frac{i\theta}{2}\Sigma^3\right) = \begin{pmatrix} e^{-i\theta/2} & 0 & 0 & 0 \\ 0 & e^{i\theta/2} & 0 & 0 \\ 0 & 0 & e^{-i\theta/2} & 0 \\ 0 & 0 & 0 & e^{i\theta/2} \end{pmatrix} \quad (11.56)$$

A rotation by  $2\pi$  gives  $S(R_z(2\pi)) = -I$ , so the spinor changes sign. This is the hallmark of spin-1/2: a  $360^\circ$  rotation produces a phase of  $-1$ , and only a  $720^\circ$  rotation returns the spinor to its original value.

### 11.8.2 Bilinear Covariants

From a Dirac spinor  $\psi$ , we can construct quantities that transform in well-defined ways under Lorentz transformations. The scalar  $\bar{\psi}\psi$  is invariant under Lorentz transformations and appears in the mass term of the Lagrangian. The vector  $\bar{\psi}\gamma^\mu\psi$  transforms as a four-vector and gives the probability current. The antisymmetric tensor  $\bar{\psi}\sigma^{\mu\nu}\psi$ , where  $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$ , has six independent components and relates to the electromagnetic field coupling. The pseudovector (or axial vector)  $\bar{\psi}\gamma^\mu\gamma^5\psi$  transforms as a four-vector but picks up an extra sign under parity. The pseudoscalar  $\bar{\psi}\gamma^5\psi$ , where  $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ , is invariant under proper Lorentz transformations but changes sign under parity.

#### Example 13

**The Pseudoscalar and Parity** Under parity, spatial coordinates flip sign:  $(t, \vec{x}) \rightarrow (t, -\vec{x})$ . The spinor transforms as  $\psi \rightarrow \gamma^0\psi$ . Therefore

$$\bar{\psi}\gamma^5\psi \rightarrow (\psi^\dagger\gamma^0\gamma^0)\gamma^5(\gamma^0\psi) = \psi^\dagger\gamma^5\gamma^0\psi = -\psi^\dagger\gamma^0\gamma^5\psi = -\bar{\psi}\gamma^5\psi \quad (11.57)$$

using  $\{\gamma^5, \gamma^0\} = 0$ . The pseudoscalar changes sign under parity, which is why it is called “pseudo.”

## 11.9 The Dirac Equation in Electromagnetic Fields

### 11.9.1 Minimal Coupling

Including electromagnetic fields via minimal coupling:

$$(i\gamma^\mu D_\mu - m)\psi = 0 \quad (11.58)$$

where  $D_\mu = \partial_\mu + ieA_\mu$ .

#### Example 14

**The Dirac Equation in a Uniform Magnetic Field** Consider a magnetic field  $\vec{B} = B\hat{z}$  with vector potential  $\vec{A} = \frac{1}{2}\vec{B} \times \vec{r} = \frac{B}{2}(-y, x, 0)$ . The Dirac Hamiltonian becomes

$$H = \vec{\alpha} \cdot (\vec{p} - e\vec{A}) + \beta m \quad (11.59)$$

In the non-relativistic limit, this reduces to the Pauli Hamiltonian with the interaction term  $-\vec{\mu} \cdot \vec{B} = -g\frac{e}{2m}\vec{S} \cdot \vec{B}$ . With  $g = 2$ , the spin-up and spin-down states split by  $\Delta E = \frac{eB}{m}$ , giving the electron spin resonance frequency.

### 11.9.2 The Hydrogen Atom Revisited

The Dirac equation applied to the hydrogen atom yields the fine structure splitting automatically, without needing to add it as a perturbation. The spin-orbit coupling, which in the non-relativistic treatment must be introduced by hand, emerges naturally from the relativistic formulation. The energy levels depend on both  $n$  and  $j$  (total angular momentum), leading to the fine structure formula

$$E_{n,j} = m \left[ 1 + \left( \frac{\alpha}{n - (j + 1/2) + \sqrt{(j + 1/2)^2 - \alpha^2}} \right)^2 \right]^{-1/2} \quad (11.60)$$

where  $\alpha \approx 1/137$  is the fine structure constant. This formula agrees with experiment to high precision. A small remaining discrepancy, the Lamb shift, is explained by quantum electrodynamic effects involving virtual photon exchange.

#### Example 15

**Fine Structure Splitting in Hydrogen** For the  $n = 2$  level, the Dirac equation predicts that states with  $j = 1/2$  (the  $2S_{1/2}$  and  $2P_{1/2}$  states) are degenerate, while  $j = 3/2$  (the  $2P_{3/2}$  state) has higher energy. The splitting between  $j = 1/2$  and  $j = 3/2$  is approximately  $\frac{\alpha^2}{16} \times 13.6 \text{ eV} \approx 4.5 \times 10^{-5} \text{ eV}$ , corresponding to a wavelength splitting of about 0.016 nm in the Balmer- $\alpha$  line at 656 nm. This prediction matched experimental observations and confirmed the validity of the relativistic treatment.

## 11.10 Modern Applications

### 11.10.1 Topological Insulators

The surface states of 3D topological insulators obey an effective 2D Dirac equation:

$$H_{\text{surface}} = v_F(\vec{\sigma} \times \vec{k})_z \quad (11.61)$$

These surface states have several remarkable properties. The spin direction is locked perpendicular to the momentum, creating a “spin texture” in momentum space. Backscattering is forbidden because reversing momentum would require flipping spin, which is protected by time-reversal symmetry. This leads to robust metallic conduction on the surface even when the bulk is insulating.

#### Example 16

**Spin-Momentum Locking** For a surface state with momentum  $\vec{k} = k(\cos \phi, \sin \phi)$ , the Hamiltonian gives

$$H = v_F(k_x \sigma_y - k_y \sigma_x) = v_F k(-\sin \phi \sigma_x + \cos \phi \sigma_y) \quad (11.62)$$

The eigenstates have spin pointing perpendicular to  $\vec{k}$  in the surface plane. An electron moving in the  $+x$  direction has spin pointing in the  $+y$  direction, while an electron moving in the  $-x$  direction has spin pointing in the  $-y$  direction. To backscatter (reverse  $k_x$ ), the spin would need to flip, but non-magnetic impurities cannot flip spin, so backscattering is suppressed.

### 11.10.2 Weyl Semimetals

Materials with separated Weyl points host chiral fermions obeying:

$$H_{\text{Weyl}} = \pm v_F \vec{\sigma} \cdot \vec{k} \quad (11.63)$$

The  $\pm$  sign defines the chirality of the Weyl node. Unlike Dirac points, which can gap out, Weyl points are topologically stable because they act as monopoles of Berry curvature in momentum space. They can only be removed by annihilating with a Weyl point of opposite chirality.

#### Example 17

**Chiral Anomaly in Weyl Semimetals** In parallel electric and magnetic fields, charge is pumped between Weyl nodes of opposite chirality. When  $\vec{E} \parallel \vec{B}$ , the zeroth Landau level at each Weyl point carries current in a direction determined by the chirality. This leads to a chiral current

$$\frac{\partial n_R}{\partial t} - \frac{\partial n_L}{\partial t} = \frac{e^2}{2\pi^2} \vec{E} \cdot \vec{B} \quad (11.64)$$

where  $n_R$  and  $n_L$  are the densities at right- and left-handed nodes. This chiral anomaly, a quantum field theory phenomenon, manifests as negative magnetoresistance in Weyl semimetals: the resistance decreases when magnetic field is applied parallel to the current.

## 11.11 Chapter Summary

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The Dirac equation unifies quantum mechanics and special relativity in a remarkably elegant framework. We have seen that discrete versions of the equation connect directly to condensed matter physics, describing electrons in graphene and edge states in topological materials. The requirement of relativistic covariance forces the wave function to be a spinor with two components in 1+1 dimensions and four components in 3+1 dimensions.

The equation's predictive power is striking. It naturally incorporates spin-1/2 without any additional assumptions and predicts the electron's magnetic moment with the correct g-factor of 2. The existence of negative energy solutions led Dirac to propose the filled sea interpretation, which correctly predicted the existence of antiparticles before the positron was discovered experimentally.

When applied to the hydrogen atom, the Dirac equation automatically produces the fine structure splitting that must be added perturbatively in non-relativistic treatments. The relativistic framework provides the foundation for quantum electrodynamics and modern quantum field theory.

Contemporary applications extend beyond particle physics to condensed matter systems. Surface states of topological insulators obey effective Dirac equations with spin-momentum locking and topological protection. Weyl semimetals host chiral fermions that exhibit the quantum chiral anomaly. These connections demonstrate that the Dirac equation, nearly a century after its discovery, remains central to our understanding of quantum phenomena across multiple fields of physics.

This relativistic framework prepares us for exploring symmetries and conservation laws in the next chapter.

## 11.12 Problems

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### Problem 11.1 (*Klein-Gordon Probability Current*)

The Klein-Gordon equation is  $(\partial_t^2 - \nabla^2 + m^2)\phi = 0$ .

- Multiply the Klein-Gordon equation by  $\phi^*$  and subtract the complex conjugate to derive the continuity equation  $\partial_t \rho + \nabla \cdot \vec{j} = 0$  with  $\rho = \frac{i}{2m}(\phi^* \partial_t \phi - \phi \partial_t \phi^*)$  and identify  $\vec{j}$ .
- For the positive-frequency plane wave  $\phi = e^{i(\vec{k} \cdot \vec{x} - \omega t)}$  with  $\omega = +\sqrt{k^2 + m^2}$ , show that  $\rho = \omega/m > 0$ .
- For the negative-frequency solution  $\phi = e^{i(\vec{k} \cdot \vec{x} + \omega t)}$ , show that  $\rho < 0$ . Why does this rule out  $\rho$  as a probability density?
- In the non-relativistic limit  $k \ll m$ , show that  $\omega \approx m + k^2/2m$  and that  $\rho \approx |\phi|^2$ .
- Why is the Klein-Gordon equation second-order in time, and how does this lead to the problem with  $\rho$ ? How does the Dirac equation avoid this issue?

### Problem 11.2 (*Discrete Dirac Dispersion*)

The discrete 1D Dirac Hamiltonian acts on two-component spinors via  $(H\psi)_n = \frac{1}{2i}\sigma_x(\psi_{n+1} - \psi_{n-1}) + m\sigma_z \psi_n$ .

- Substitute the Bloch ansatz  $\psi_n = u e^{ikn}$  and show that the eigenvalue equation reduces to  $[\sin(k)\sigma_x + m\sigma_z]u = Eu$ .

- (b) Write this as a  $2 \times 2$  matrix equation and use  $\det(H - EI) = 0$  to derive  $E(k) = \pm \sqrt{\sin^2(k) + m^2}$ .
- (c) Show that the normalized positive-energy eigenstate is  $u_+(k) = \frac{1}{\sqrt{2E_k(E_k+m)}} \begin{pmatrix} E_k + m \\ \sin k \end{pmatrix}$  where  $E_k = \sqrt{\sin^2 k + m^2}$ .
- (d) In the limit  $k \ll 1$  and  $m \gg k$ , expand  $E_k$  and show that  $E_k \approx m + k^2/2m$ .
- (e) For  $m = 0$ , find the two eigenstates and show each is an eigenstate of  $\sigma_x$ . What does this tell you about the two spinor components?

**Problem 11.3** (*Chiral Symmetry and Zero Modes*)

Working in momentum space with the linearized Hamiltonian  $H(k) = k\sigma_x + m\sigma_z$ :

- (a) Compute  $\{\sigma_z, H(k)\}$  and show that it equals  $2mI$ . Conclude that chiral symmetry holds if and only if  $m = 0$ .
- (b) Assuming  $m = 0$ , show that if  $|\chi\rangle$  is an eigenstate of  $H(k)$  with energy  $E$ , then  $\sigma_z|\chi\rangle$  is an eigenstate with energy  $-E$ .
- (c) A zero-energy state satisfies  $E = 0$ . What is its partner under the pairing argument? Why can chiral symmetry accommodate unpaired zero modes?
- (d) On a finite lattice, what boundary condition is required to support a zero mode at one end of the chain?
- (e) Explain why a small mass perturbation breaks chiral symmetry and generically lifts zero modes away from  $E = 0$ .

**Problem 11.4** (*Clifford Algebra in 3+1D*)

In the standard representation, the gamma matrices are  $\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$  and  $\gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}$  for  $i = 1, 2, 3$ .

- (a) Verify directly that  $(\gamma^0)^2 = I$ ,  $(\gamma^i)^2 = -I$ , and  $\{\gamma^0, \gamma^i\} = 0$ .
- (b) Verify that  $\{\gamma^1, \gamma^2\} = 0$  using the block matrix form and the Pauli anticommutation relation.
- (c) Show that the Clifford algebra  $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}I$  is forced by requiring that every solution of the Dirac equation also satisfies the Klein-Gordon equation.
- (d) Define  $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ . Show that  $\{\gamma^5, \gamma^\mu\} = 0$  for any  $\mu$ .
- (e) The Weyl representation diagonalizes  $\gamma^5$ . Given that  $(\gamma^5)^2 = I$ , what are its eigenvalues and multiplicities? What do these define?

**Problem 11.5** (*Spin-1/2 from the Dirac Equation*)

The Dirac Hamiltonian is  $H = \vec{\alpha} \cdot \vec{p} + \beta m$  where  $\alpha^i = \begin{pmatrix} 0 & \sigma^i \\ \sigma^i & 0 \end{pmatrix}$  and  $\beta = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$ . The spin operator is  $\vec{S} = \frac{1}{2}\vec{\Sigma}$  with  $\Sigma^i = \begin{pmatrix} \sigma^i & 0 \\ 0 & \sigma^i \end{pmatrix}$ .

- (a) Show that  $[L^z, H] \neq 0$  by computing the relevant commutators. What does this say about orbital angular momentum conservation?
- (b) Using Pauli commutation relations, compute  $[\alpha^x, \Sigma^z]$  and  $[\alpha^y, \Sigma^z]$ .
- (c) Show that  $[H, S^z] = -[H, L^z]$ . Conclude that  $[H, J^z] = 0$  where  $\vec{J} = \vec{L} + \vec{S}$  is the total angular momentum.
- (d) From  $(\Sigma^i)^2 = I$  and  $\{\Sigma^i, \Sigma^j\} = 2\delta^{ij}I$ , show that  $\vec{S}^2 = \frac{3}{4}I$  and extract the spin quantum number.
- (e) Explain in two sentences why spin-1/2 is a prediction of the Dirac equation rather than an assumption.

**Problem 11.6** (*Lorentz Covariance of the Dirac Equation*)

- (a) Under a Lorentz transformation  $x^\mu \rightarrow \Lambda^\mu_\nu x^\nu$ , the spinor transforms as  $\psi(x) \rightarrow S(\Lambda)\psi(\Lambda^{-1}x)$ . Show that the Dirac equation  $(i\gamma^\mu \partial_\mu - m)\psi = 0$  is covariant, i.e., has the same form in all inertial frames.
- (b) Verify that a boost in the  $z$ -direction by rapidity  $\eta$  gives  $S(\text{boost}) = \exp(-i\eta\gamma^0\gamma^3/2)$ . What is the relationship between  $\eta$  and the velocity  $v$ ?
- (c) Under a rotation by angle  $\theta$  about the  $z$ -axis, the spinor transformation is  $S(R_z(\theta)) = \exp(-i\theta\Sigma^3/2)$ . Show that a  $2\pi$  rotation gives  $S(R_z(2\pi)) = -I$  (the spinor acquires a sign).
- (d) Explain why this sign change is not a problem for physical observables (which are bilinear in  $\psi$ ).
- (e) Define the bilinear covariants: the scalar  $\bar{\psi}\psi$ , the vector  $\bar{\psi}\gamma^\mu\psi$ , and the pseudoscalar  $\bar{\psi}\gamma^5\psi$ . Under parity ( $\vec{x} \rightarrow -\vec{x}$ ), how does each transform?

**Problem 11.7** (*Positive-Energy Projector and Antiparticles*)

- (a) For a free particle with momentum  $\vec{p}$ , the positive-energy projector is  $\Lambda_+(\vec{p}) = \frac{\gamma^\mu p_\mu + m}{2m}$  where  $p^\mu = (E, \vec{p})$  with  $E = \sqrt{\vec{p}^2 + m^2}$ . Show that  $\Lambda_+^2 = \Lambda_+$  (idempotent) and that  $\text{Tr}(\Lambda_+) = 2$  (two-dimensional).
- (b) Verify that  $(\gamma^\mu p_\mu - m)\Lambda_+(\vec{p}) = 0$ , confirming that the positive-energy spinors lie in the range of  $\Lambda_+$ .
- (c) The negative-energy projector is  $\Lambda_-(\vec{p}) = I - \Lambda_+(\vec{p})$ . Verify that it projects onto negative-energy states.
- (d) In the rest frame ( $\vec{p} = 0, E = m$ ), show that  $\Lambda_+ = \frac{\gamma^0 + 1}{2}$  and that the two states in its range are the spin-up and spin-down spinors in the standard representation.
- (e) Define the charge-conjugation operator  $C = -i\gamma^2\gamma^0$  such that  $C^{-1}\gamma^\mu C = -\gamma^{\mu T}$ . Show that  $C^{-1}(\gamma^\mu p_\mu - m)C = -(\gamma^\mu p_\mu + m)$ . What does this mean physically?

**Problem 11.8** (*Gamma Matrices in Different Representations*)

- (a) The standard (Dirac) representation uses  $\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$  and  $\gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}$ . The Weyl representation diagonalizes  $\gamma^5$  as  $\gamma^5 = \begin{pmatrix} -I & 0 \\ 0 & I \end{pmatrix}$ . Show that a change of representation is a similarity transformation:  $\gamma_{\text{Weyl}}^\mu = U^{-1} \gamma_{\text{Dirac}}^\mu U$  for some unitary  $U$ .
- (b) Compute the  $U$  that transforms from the Dirac to the Weyl representation.
- (c) In the Weyl representation, the Dirac spinor decomposes into right- and left-handed Weyl spinors. Write the Weyl representation Dirac equation in block form and interpret the upper and lower blocks.
- (d) Why is the Weyl representation useful for massless particles? What happens when you add a mass term?
- (e) Verify that the Clifford algebra  $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}I$  holds in the Weyl representation.

**Problem 11.9** (*Helicity Eigenstates of Massless Dirac Fermions*)

For a massless Dirac particle, the Hamiltonian is  $H = \vec{\alpha} \cdot \vec{p}$ .

- (a) The helicity operator is  $h = \vec{S} \cdot \hat{p} = \frac{1}{2} \vec{\Sigma} \cdot \hat{p}$  where  $\hat{p} = \vec{p}/|\vec{p}|$ . Show that  $[H, h] = 0$  for massless particles.
- (b) For a particle moving along  $\hat{z}$  with momentum  $p$ , show that the helicity eigenvalue equation reduces to  $\sigma^z u = \pm u$  where  $u$  is a two-component spinor.
- (c) Solve for the helicity eigenstates: the right-handed state  $u_R$  (helicity  $+1/2$ ) and left-handed state  $u_L$  (helicity  $-1/2$ ).
- (d) Verify that  $u_R$  and  $u_L$  are orthogonal and that they are eigenstates of  $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$  in the massless limit.
- (e) Show that the massless Dirac equation  $i\gamma^\mu \partial_\mu \psi = 0$  decouples into separate equations for  $u_R$  and  $u_L$ . What does this tell you about Weyl spinors?

**Problem 11.10** (*Foldy-Wouthuysen Non-Relativistic Limit*)

- (a) In the non-relativistic limit, write  $\psi = e^{-imt} \tilde{\psi}$  where  $\tilde{\psi}$  is slowly varying. Decompose the spinor as  $\tilde{\psi} = \begin{pmatrix} \phi \\ \chi \end{pmatrix}$  where  $\chi = O(v/c)\phi$ .
- (b) From the Dirac equation in the presence of an electromagnetic field, show that in the non-relativistic limit  $\chi \approx \frac{\vec{\alpha} \cdot \vec{\Pi}}{2m} \phi$  where  $\vec{\Pi} = \vec{p} - e\vec{A}$ .
- (c) Substitute this back into the upper component equation to derive the Pauli equation:  $i\partial_t \phi = \left[ \frac{(\vec{p} - e\vec{A})^2}{2m} + e\Phi - \frac{e}{2m} \vec{\sigma} \cdot \vec{B} \right] \phi$ .
- (d) Extract the magnetic moment interaction term and show that the g-factor is  $g = 2$ .



- (e) Explain why the Foldy-Wouthuysen transformation is useful for understanding the connection between relativistic Dirac theory and non-relativistic quantum mechanics.

**Problem 11.11** (*Zitterbewegung: The Electron's Trembling Motion*)

- (a) For a free Dirac electron, the velocity is  $\vec{v} = d\vec{x}/dt = \vec{\alpha}$ . Compute  $d\vec{\alpha}/dt$  using the Heisenberg equation  $\dot{O} = i[H, O]$  where  $H = \vec{\alpha} \cdot \vec{p} + \beta m$ .
- (b) Show that the center-of-mass position  $\vec{x}_{\text{cm}} = \vec{x} - \vec{v}t$  satisfies  $d\vec{x}_{\text{cm}}/dt = \vec{v} - \vec{\alpha} = -i[\vec{v}, H]/E$  where  $E = \sqrt{\vec{p}^2 + m^2}$ .
- (c) The fluctuation of the velocity about its mean is  $\delta\vec{v} = \vec{v}(t) - \langle\vec{v}\rangle$ . Show that this oscillates with frequency  $\omega_z = 2m$  (in natural units) and amplitude  $\sim v/c$ .
- (d) Explain physically why this oscillation (Zitterbewegung) arises and why it is suppressed in the non-relativistic limit.
- (e) Estimate the Compton wavelength  $\lambda_C = h/(mc)$  and the zitterbewegung amplitude. Why is this effect unobservable in non-relativistic systems?

**Problem 11.12** (*Charge Conjugation and the Dirac Sea*)

- (a) The charge conjugation operator  $C$  acting on the Dirac equation satisfies  $C^{-1}\gamma^\mu C = -\gamma^{\mu T}$  (charge-conjugated gamma matrices). Define  $C = -i\gamma^2\gamma^0$  in the standard representation and verify this property.
- (b) Under charge conjugation,  $\psi \rightarrow C\bar{\psi}^T = -i\gamma^2\gamma^0(\psi^\dagger\gamma^0)^T$ . Show that the Dirac equation for  $\psi$  maps to the Dirac equation for the conjugate spinor with opposite charge  $-e$ .
- (c) In Dirac's interpretation, a hole in the filled negative-energy sea is interpreted as a positron (antiparticle with positive energy  $+E$  and charge  $+e$ ). Explain how the hole's energy and charge relate to the missing negative-energy electron.
- (d) Consider a positive-energy electron state  $u_+(\vec{p})$  from problem 5 (boosted spinors). What is the corresponding conjugate spinor  $v_-(-\vec{p})$  for a positron with momentum  $-\vec{p}$ ? Show that  $v_-$  is the charge-conjugated version of  $u_+$ .
- (e) While the Dirac sea interpretation is historically important, modern quantum field theory replaces it with second quantization. Briefly explain why second quantization is more satisfactory.

## References and Further Reading

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# Chapter 12

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## The Renormalization Group

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### 12.1 Introduction: The Problem of Scale

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At the beginning of this course, we started with the simplest quantum system: a single qubit living in a two-dimensional Hilbert space. We then built up systematically: tensor products gave us multiple qubits, lattice systems connected discrete sites, and finally we took continuum limits to recover the familiar Schrödinger equation in continuous space. Throughout this progression, a fundamental question lurked beneath the surface: when we describe physics at different scales, are we looking at the same theory, or do the rules themselves change?

The renormalization group (RG) provides a profound answer: physical theories transform under changes of scale. Rather than asking “what is the true theory,” we should ask “how do theories flow as we coarse-grain our description?” This perspective resolves the discrete-versus-continuous tension that motivated our entire pedagogical arc. The continuum limit is not simply the limit as the lattice spacing goes to zero; it represents a particular fixed point in the space of all possible theories. The RG is not a group in the mathematical sense, but rather a semigroup of scale transformations that maps a theory to an effective theory at longer length scales. By systematically integrating out short-distance degrees of freedom, it reveals which properties of a system are robust under coarse-graining and which are sensitive to microscopic details.

A single example from high-energy physics makes the payoff vivid. The fine-structure constant  $\alpha \approx 1/137$  appears throughout atomic physics and feels like a fixed property of nature. But the RG reveals that  $\alpha$  is a function of the energy scale  $\mu$  at which you probe the interaction: approximately  $1/137$  at atomic energies, but measurably larger ( $\approx 1/128$ ) at the energy of the  $Z$  boson ( $\mu \approx 91$  GeV). The coupling constant *runs*. All three fundamental forces exhibit running couplings, and as Figure 12.1 shows, evolving them upward in energy using the RG flow equations reveals a remarkable convergence near  $\mu \sim 10^{16}$  GeV, suggesting that the electromagnetic, weak, and strong forces may be three low-energy faces of a single unified interaction. This convergence is entirely invisible at any fixed energy scale. It is a statement about flow, about how the effective description of nature changes as you change the resolution at which you examine it. We will not derive this result, as doing so requires quantum field theory, but it sets the ambition: the RG reveals structure in physics that would otherwise be completely hidden.

Before turning to the RG itself, it is worth recalling the two broad classes of phase transition. In

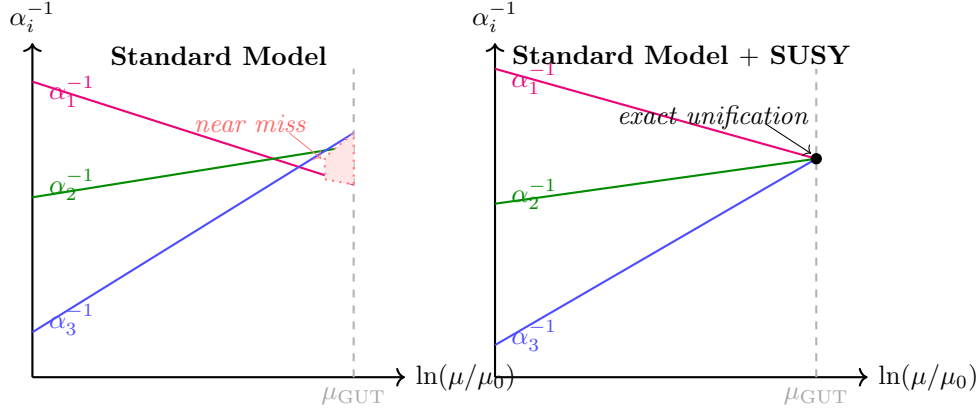


Figure 12.1: Schematic of running coupling constants. Each of the three fundamental forces has a dimensionless coupling constant that changes with the energy scale  $\mu$  at which it is probed. Plotting the inverse couplings  $\alpha_i^{-1}$  against  $\ln \mu$ , each runs as a straight line whose slope is set by the structure of that force. Left: in the Standard Model the three lines converge toward  $\mu \sim 10^{16}$  GeV but fall slightly short of a common point (shaded near-miss region). Right: a supersymmetric extension of the theory modifies the slopes just enough that all three lines meet exactly, suggesting a single unified interaction at high energy.

a *first-order* transition (melting ice, boiling water), the order parameter jumps discontinuously and the two phases coexist at the transition point; the correlation length remains finite. In a *second-order* (or *continuous*) transition, the order parameter vanishes continuously and the correlation length  $\xi$  diverges. It is the second-order transitions that exhibit the scale invariance the RG is designed to exploit.

The correlation length is defined through the connected spin-spin correlation function

$$G(r) = \langle s_i s_{i+r} \rangle - \langle s_i \rangle \langle s_{i+r} \rangle, \quad (12.1)$$

which measures how much the spin at site  $i+r$  “knows about” the spin at site  $i$  beyond any uniform magnetization. Far from the critical point,  $G(r)$  decays exponentially,

$$G(r) \sim e^{-r/\xi}, \quad (12.2)$$

defining the correlation length  $\xi$ : the characteristic distance over which spins are statistically correlated. As the temperature approaches the critical temperature  $T_c$  from either side,  $\xi$  diverges as  $\xi \sim |T - T_c|^{-\nu}$ , where  $\nu$  is the correlation-length critical exponent. At the critical point itself,  $\xi$  diverges, removing the only length scale from the problem. The correlation function must then decay as a power law rather than an exponential, and the question is: what power? The answer begins with the simplest continuum model of a fluctuating order parameter, a free (Gaussian) scalar field  $\phi(\mathbf{x})$  that plays the role of a coarse-grained, continuous version of the discrete spin  $s_i$ . The correlation function for this field is  $G(\mathbf{r}) = \langle \phi(\mathbf{x}) \phi(\mathbf{x} + \mathbf{r}) \rangle - \langle \phi \rangle^2$ , the direct continuum analog of the spin-spin correlator defined above. At the critical point it satisfies

$$-\nabla^2 G(\mathbf{r}) = \delta^d(\mathbf{r}), \quad (12.3)$$

i.e.,  $G(\mathbf{r})$  is the Green’s function of the Laplacian in  $d$  dimensions. Fourier transforming both sides gives  $k^2 \tilde{G}(\mathbf{k}) = 1$ , so  $\tilde{G}(\mathbf{k}) = 1/k^2$ . The real-space correlation function is then

$$G(\mathbf{r}) = \int \frac{d^d k}{(2\pi)^d} \frac{e^{i\mathbf{k} \cdot \mathbf{r}}}{k^2}. \quad (12.4)$$

The  $r$ -dependence can be extracted without evaluating the integral explicitly. Substitute  $\mathbf{k} = \mathbf{q}/r$ , so that  $d^d k = r^{-d} d^d q$ ,  $k^2 = q^2/r^2$ , and  $\mathbf{k} \cdot \mathbf{r} = \mathbf{q} \cdot \hat{r}$ :

$$G(r) = r^{-(d-2)} \int \frac{d^d q}{(2\pi)^d} \frac{e^{i\mathbf{q} \cdot \hat{r}}}{q^2}. \quad (12.5)$$

The remaining integral is a pure number independent of  $r$  (it depends only on the direction  $\hat{r}$ , and after angular averaging becomes a numerical constant), so

$$G(r) \sim r^{-(d-2)} \quad (\text{free field}). \quad (12.6)$$

This is the prediction of mean-field theory, in which fluctuations are treated as non-interacting. In an interacting system the exponent receives a correction:

$$G(r) \sim r^{-(d-2+\eta)}, \quad (12.7)$$

where  $\eta$  is the *anomalous dimension*. It measures how much the true critical correlations deviate from the free-field prediction;  $\eta = 0$  in mean-field theory, but takes the value  $\eta = 1/4$  in the 2D Ising model.

The RG is equally central to classical and quantum statistical mechanics. Near a continuous phase transition, a magnet approaching its Curie temperature, helium approaching its superfluid transition, a binary mixture near its critical point,  $\xi$  diverges and fluctuations exist on all length scales at once. This is the hallmark of an RG fixed point. The striking consequence is *universality*: the critical exponents that govern the power-law singularities of thermodynamic quantities depend only on symmetry and spatial dimensionality, not on microscopic details. A magnet and a liquid-gas system near their respective critical points can obey identical power laws despite having nothing microscopically in common. Kenneth Wilson was awarded the Nobel Prize in Physics in 1982 for making this connection precise.

### Key Idea: Scale Transformations and Effective Theories

The renormalization group is a semigroup of scale transformations that maps theories to effective theories at longer length scales. Whether discrete or continuous descriptions are “more fundamental” depends on which RG fixed point you are near, resolving a question posed in Chapter 1.

This chapter develops the RG through the Ising model, a lattice of binary spins interacting with their neighbors. The Ising Hamiltonian  $H = -J \sum s_i^z s_{i+1}^z$  is a perfectly valid quantum-mechanical operator, but because it is diagonal in the  $z$ -basis, the entire analysis can be carried out using classical statistical mechanics: no superpositions or operator ordering enter. This is deliberate: working in a setting where classical methods suffice isolates the core logic of coarse-graining in the simplest possible way. The procedure, called decimation, integrates out alternating spins exactly and yields a recursion relation for the effective coupling. Fixed points of this recursion correspond to phases of matter, and their stability determines whether a phase transition occurs. The goal is a concrete, fully worked example that builds the intuition underlying both the quantum field theory of running couplings and the statistical mechanics of critical phenomena.

## 12.2 Block Spin Renormalization: The Ising Model

### 12.2.1 The One-Dimensional Ising Model

We begin with the simplest example that captures the essential physics: spins on a one-dimensional lattice. The Hamiltonian is:

$$H = -J \sum_{i=1}^{N-1} s_i s_{i+1} \quad (12.8)$$

where  $s_i = \pm 1$  are classical spins and  $J > 0$  favors alignment. These values are the eigenvalues of  $\hat{\sigma}^z$ , familiar throughout this course, but here each spin simply takes a definite classical value: there is no superposition.

What is new is temperature. The system does not sit in a single configuration; instead it samples all  $2^N$  configurations  $\{s_1, \dots, s_N\}$  with probabilities set by the Boltzmann weight  $e^{-\beta H}$ , where  $\beta \equiv 1/(k_B T)$  is the inverse temperature. Lower-energy configurations are exponentially more probable; at high temperature  $\beta \rightarrow 0$  and all configurations become equally likely. The operator  $e^{-\beta H}$  should look familiar: it has exactly the same structure as the time-evolution operator  $e^{-i\hat{H}t/\hbar}$  from Chapter 4, with the replacement  $it/\hbar \rightarrow \beta$ . This is not a coincidence. The normalization factor that makes this a valid probability distribution is the partition function

$$Z = \text{Tr} \left( e^{-\beta \hat{H}} \right), \quad (12.9)$$

and the thermal density operator

$$\hat{\rho}_{\text{th}} = \frac{e^{-\beta \hat{H}}}{Z} \quad (12.10)$$

assigns probability  $p_n = e^{-\beta E_n}/Z$  to each energy eigenstate  $|n\rangle$  and gives expectation values via  $\langle A \rangle = \text{Tr}(\hat{\rho}_{\text{th}} \hat{A})$ . You have seen the density operator formalism for qubits; here it extends to systems with many degrees of freedom. For the classical Ising model the Hamiltonian is diagonal in the spin-configuration basis, so there are no off-diagonal coherences and the trace is just a sum over configurations:

$$\begin{aligned} Z &= \sum_{\{s_i\}} e^{-\beta H} \\ &= \sum_{\{s_i\}} \exp \left( \beta J \sum_{i=1}^{N-1} s_i s_{i+1} \right) \\ &= \sum_{\{s_i\}} \prod_{i=1}^{N-1} e^{\beta J s_i s_{i+1}}. \end{aligned} \quad (12.11)$$

The last step uses the fact that the exponential of a sum equals the product of exponentials. Because  $H$  is a sum of independent bond terms,  $e^{-\beta H}$  factorizes into one factor per bond. This factorization is what makes decimation possible: each spin  $s_i$  appears in only a finite number of bond factors, so summing over  $s_i$  affects only those factors and leaves the rest of the product unchanged. All thermodynamic quantities follow from  $Z$ : the average energy is  $-\partial \ln Z / \partial \beta$ , the free energy is  $F = -k_B T \ln Z$ , and so on. The decimation procedure below amounts to performing part of this sum analytically.

### 12.2.2 Decimation: Integrating Out Every Other Spin

The central idea is simple: we have a chain of  $N$  spins, and we want to trade it for a shorter chain that captures the same long-distance physics. The way to do this is to perform part of the sum in  $Z$

exactly, specifically to sum over every other spin while leaving the remaining spins untouched. This is called *decimation*, because we are removing (“decimating”) a fraction of the degrees of freedom.

Label the spins by site:  $s_1, s_2, s_3, s_4, \dots$ . We will sum over the even-site spins  $s_2, s_4, \dots$  and keep the odd-site spins  $s_1, s_3, s_5, \dots$ . The question is whether the result of that partial sum can itself be written as an Ising model, just with a different coupling constant and a different lattice spacing. If yes, we have a map from one Ising model to another: that map is the RG transformation.

To see how this works, focus on a single even-site spin  $s_2$  sitting between its two odd-site neighbors  $s_1$  and  $s_3$ . In the full partition function,  $s_2$  appears only in the two bond factors  $e^{\beta J s_1 s_2}$  and  $e^{\beta J s_2 s_3}$ , so summing over  $s_2 = \pm 1$  affects only those two factors:

$$\begin{aligned} \sum_{s_2=\pm 1} e^{\beta J s_1 s_2} e^{\beta J s_2 s_3} &= e^{\beta J (s_1 + s_3)} + e^{-\beta J (s_1 + s_3)} \\ &= 2 \cosh[\beta J (s_1 + s_3)]. \end{aligned} \quad (12.12)$$

Define the dimensionless coupling  $K \equiv \beta J$ ; then  $s_1$  and  $s_3$  each equal  $\pm 1$ , and the argument of cosh takes only two values:

$$s_1 = s_3 : \quad 2 \cosh(2K), \quad (12.13)$$

$$s_1 = -s_3 : \quad 2 \cosh(0) = 2. \quad (12.14)$$

Now ask: can this be written in the form  $A e^{K' s_1 s_3}$  for some constants  $A$  and  $K'$ ? The temperature is fixed throughout,  $\beta = 1/k_B T$  does not change when we decimate. What changes is the effective coupling  $J'$  between surviving spins, and it is the dimensionless combination  $K' \equiv \beta J'$  (same  $\beta$ , new  $J'$ ) that the RG maps. Since  $s_1 s_3 = +1$  when the spins are parallel and  $-1$  when they are antiparallel, the ansatz gives two conditions:

$$A e^{+K'} = 2 \cosh(2K), \quad (12.15)$$

$$A e^{-K'} = 2. \quad (12.16)$$

Multiplying equations (12.15) and (12.16) gives  $A^2 = 4 \cosh(2K)$ , so

$$A = 2[\cosh(2K)]^{1/2}. \quad (12.17)$$

Dividing equation (12.15) by equation (12.16) gives  $e^{2K'} = \cosh(2K)$ , so the RG recursion relation is

$$K' = \frac{1}{2} \ln[\cosh(2K)]. \quad (12.18)$$

The same factor  $A$  appears for every decimated spin, so after integrating out all even sites the partition function becomes

$$Z = A^{N/2} Z', \quad (12.19)$$

where  $Z'$  is the partition function of an Ising model on the remaining  $N/2$  spins with effective coupling  $J' = K'/\beta = k_B T \cdot K'$ . The factor  $A^{N/2}$  contributes an additive constant to the free energy per site but does not affect correlations or critical behavior. The physics we care about is entirely encoded in the recursion relation (12.18), which tells us how the coupling flows when we double the lattice spacing by removing every other spin.

**Physical Insight**

Decimation does not change the temperature. The heat bath is external, and  $\beta = 1/k_B T$  is the same before and after summing over the even-site spins. What changes is the effective coupling  $J' = k_B T \cdot K'$  between the surviving spins. Since  $K' < K$  for any finite  $K$ , coarse-graining always weakens the effective interaction. The fixed point  $K^* = 0$  means the effective coupling has been driven to zero: the long-distance physics looks completely disordered, as a paramagnet would at any temperature. This is not because the temperature rose, but because repeated coarse-graining washes out short-range correlations, leaving no tendency for neighboring block spins to align. The fixed point  $K^* = \infty$  is the only alternative, approached only as  $T \rightarrow 0$  where thermal fluctuations vanish entirely.

Because the partition function of the 1D Ising model depends only on the single dimensionless combination  $K = J/k_B T$ , the flow  $K \rightarrow K'$  can be reinterpreted as holding  $J$  fixed and letting the effective temperature rise,  $T \rightarrow T' = J/(k_B K') > T$ . In this reading, coarse-graining “heats” the system, and the disordered fixed point  $K^* = 0$  is literally the  $T \rightarrow \infty$  limit. The two interpretations, renormalizing  $J$  at fixed  $T$  or renormalizing  $T$  at fixed  $J$ , are equivalent because only the ratio  $K$  is physical.

This equivalence is special to the 1D model and does not generalize. In any system with more than one independent coupling, the RG flow changes each coupling at a different rate and can generate entirely new interactions absent from the original Hamiltonian. Temperature is a single overall scale factor on the Hamiltonian; it cannot account for the differential running of multiple couplings simultaneously. Moreover, in the 2D Ising model the critical fixed point sits at a finite value  $K^* \neq 0$ , so the flow near criticality does not head toward infinite temperature at all. And in quantum systems at  $T = 0$  the RG is perfectly well-defined even though there are no thermal fluctuations whatsoever. In all these cases, RG is a statement about coarse-graining, not about temperature.

**12.2.3 Flow Diagram and Fixed Points**

Define the dimensionless coupling  $K = \beta J$ . The RG equation becomes:

$$K' = \frac{1}{2} \ln[\cosh(2K)] \quad (12.20)$$

For small  $K$  (high temperature):

$$K' \approx \frac{1}{2} \ln \left[ 1 + \frac{(2K)^2}{2} \right] \approx K^2 \quad (12.21)$$

This exhibits two fixed points:  $K^* = 0$  (vanishing effective coupling, fully disordered) where  $K' = 0$  exactly, and  $K^* = \infty$  (zero temperature, perfect order) where  $K' = K$  asymptotically.

The flow diagram shows that all finite temperatures flow toward  $K^* = 0$  under repeated coarse-graining. The one-dimensional Ising model has no phase transition at finite temperature.

**Example 1**

**Why 1D Has No Phase Transition:** The absence of a nontrivial fixed point has a physical interpretation. Consider a 1D chain at temperature  $T$ . A single broken bond (a domain wall) costs energy  $2J$  but gains entropy  $\sim k_B \ln N$  from being anywhere in the chain. For large  $N$ ,



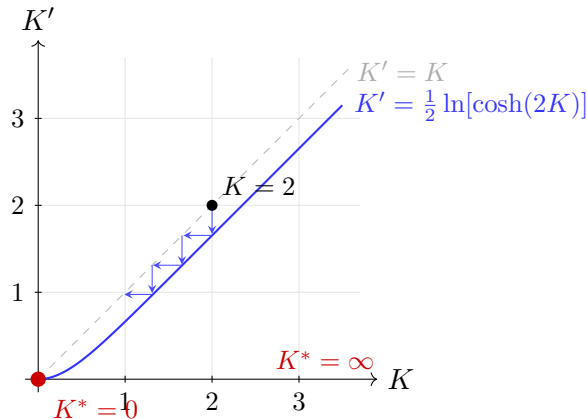


Figure 12.2: RG flow diagram for the 1D Ising model. The blue curve shows  $K' = \frac{1}{2} \ln[\cosh(2K)]$ , which lies strictly below the diagonal  $K' = K$  for all finite  $K > 0$ . The cobweb construction illustrates successive RG steps starting from  $K = 2$ : drop vertically to the curve to read off  $K'$ , then move horizontally back to the diagonal to set the new starting point. The sequence converges to the disordered fixed point  $K^* = 0$ ; the only other fixed point,  $K^* = \infty$ , is approached only at  $T = 0$ .

entropy always wins at any finite  $T$ , destroying long-range order.

The RG makes this manifest: at any finite temperature, coarse-graining eventually averages over domain walls, washing out correlations. Only at  $T = 0$ , where thermal fluctuations vanish, can order persist. This connects to our earlier discussion of finite vs. infinite systems: thermodynamic phases require the limit  $N \rightarrow \infty$ , but 1D systems are “too small” in the transverse direction for this limit to produce phase transitions.

## 12.3 Two-Dimensional Ising Model and Critical Behavior

### 12.3.1 The Square Lattice

For the 2D Ising model on a square lattice:

$$H = -J \sum_{\langle ij \rangle} s_i s_j \quad (12.22)$$

where the sum runs over nearest-neighbor pairs. Unlike the 1D case, this model exhibits a genuine second-order phase transition at a critical temperature  $T_c$  that can be determined by an elegant argument due to Kramers and Wannier (1941). The Kramers-Wannier argument constructs two exact representations of the same partition function, one organized around the low-temperature ordered phase and one around the high-temperature disordered phase, then shows that the two are related by an exact transformation that pins down  $T_c$ .

**Heuristic picture.** Before working through the details, here is the idea in one sentence: both the low-temperature and high-temperature representations of  $Z$  turn out to be sums over closed loops drawn on the lattice, weighted by a per-bond factor that depends on  $K = \beta J$ . The critical temperature is then determined by asking when the two loop gases describe the same system.

At low temperature, the spins are nearly all aligned and excitations above the ground state are islands of flipped spins. What costs energy is the *boundary* of each island: every bond crossing the boundary is a broken bond and pays an energetic price in the Boltzmann factor. These boundaries are necessarily closed curves, because an island is always fully enclosed. The partition function becomes a sum over all possible collections of such closed curves, with each bond on a curve contributing a small weight  $e^{-2K}$ . When  $T$  is low,  $K$  is large,  $e^{-2K}$  is tiny, and long curves are strongly suppressed, so  $Z$  is dominated by the empty configuration plus a few short loops.

At high temperature, the spins are nearly independent and correlations are rare. An exact rewriting of each bond factor (carried out below) expresses  $e^{Ks_i s_j}$  as an uncorrelated piece plus a small correlated piece proportional to  $\tanh K$ . Selecting the correlated piece on some subset of bonds and then averaging over spin configurations, only subsets forming *closed loops* survive the average: an open path has dangling endpoints whose spins average to zero, while a closed loop visits each spin an even number of times and so cannot vanish. The partition function is again a sum over closed loops, but now the per-bond weight is  $\tanh K$ , which is small when  $K$  is small.

So both expansions give  $Z$  as a loop gas on the square lattice, differing only in their per-bond weight:  $e^{-2K}$  at low  $T$ ,  $\tanh K$  at high  $T$ . As  $T$  rises,  $e^{-2K}$  grows from 0 toward 1 while  $\tanh K$  shrinks from 1 toward 0, and the two curves must cross somewhere. Kramers and Wannier observed that at the crossing the two loop gases are mathematically identical: the system maps onto itself. If the Ising model has a single sharp transition between ordered and disordered phases, that transition can only sit at the self-dual point where the two weights coincide,

$$\tanh K_c = e^{-2K_c}, \quad (12.23)$$

and solving this equation (shown near the end of this section) gives  $\sinh(2K_c) = 1$ , so  $k_B T_c/J \approx 2.269$ . The critical temperature drops out of a one-line self-consistency condition, without ever computing a thermodynamic quantity directly. The derivations that follow establish the two loop-gas representations carefully and justify why the self-dual point must be the transition; readers content with the heuristic can skim them and rejoin the argument at equation (12.35).

**Low-temperature expansion.** At low temperature, the dominant configurations are nearly ordered. A square lattice with  $N$  sites has  $2N$  bonds under periodic boundary conditions. Writing  $K = \beta J$ , a configuration with  $b$  broken bonds (bonds where  $s_i s_j = -1$ ) has Boltzmann weight

$$e^{K \sum_{\langle ij \rangle} s_i s_j} = e^{K(2N-b)} e^{-Kb} = e^{2KN} e^{-2Kb}. \quad (12.24)$$

Every spin configuration maps to a collection of broken bonds, and the broken bonds form the boundaries of flipped domains. On the square lattice, these boundaries are closed loops on the dual lattice (a square lattice whose sites sit at the centers of the original faces, with bonds crossing the original bonds). Crucially, global spin flip leaves the loop collection unchanged, so the map from spin configurations to loop collections is exactly 2-to-1. The partition function is therefore

$$Z = 2e^{2KN} \sum_{\mathcal{P}} (e^{-2K})^{|\mathcal{P}|}, \quad (12.25)$$

where the sum runs over all collections  $\mathcal{P}$  of closed loops on the dual lattice and  $|\mathcal{P}|$  is their total bond length. The empty collection (ground state) contributes 1.

**High-temperature expansion.** At high temperature,  $K = \beta J$  is small, and we seek an expansion in powers of  $K$  (equivalently, in powers of  $\tanh K$ , which is also small). The starting point is

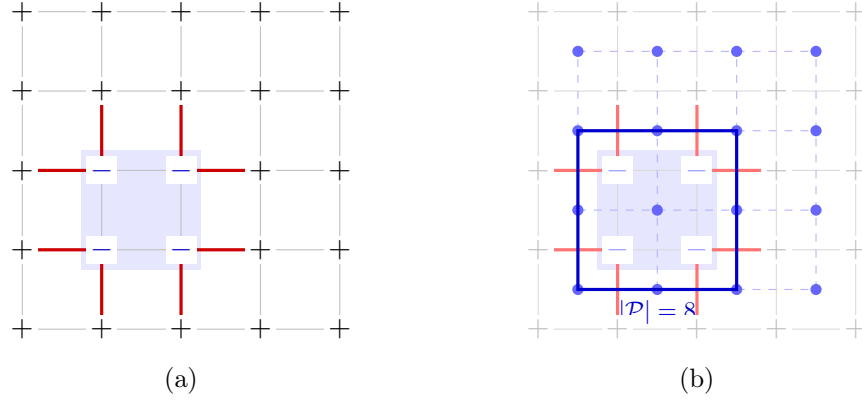


Figure 12.3: Low-temperature expansion: domain walls as closed loops on the dual lattice. (a) A  $2 \times 2$  block of flipped spins ( $-$ , blue, shaded) surrounded by  $+$  spins. The eight broken bonds (thick red) form the boundary of the flipped domain. (b) The dual lattice (dashed lines, sites at half-integer positions shown as filled circles) overlaid on the same configuration. Each broken bond on the original lattice is crossed by exactly one dual bond. The eight broken bonds map to a single closed loop of perimeter  $|\mathcal{P}| = 8$  on the dual lattice (bold blue), contributing weight  $(e^{-2K})^8$  in equation (12.25). Global spin flip ( $+ \leftrightarrow -$  everywhere) produces the same loop collection, accounting for the factor of 2 in equation (12.25).

an exact rewriting of each bond factor  $e^{K s_i s_j}$  that takes advantage of the fact that  $s_i s_j$  takes only the two values  $\pm 1$ . For either sign,

$$e^{\pm K} = \cosh K \pm \sinh K = \cosh K (1 \pm \tanh K), \quad (12.26)$$

and the two cases combine into a single expression by noting that the second term carries the same sign as  $s_i s_j$ :

$$e^{K s_i s_j} = \cosh K (1 + \tanh K \cdot s_i s_j). \quad (12.27)$$

This identity is exact, not a truncated Taylor series; it holds because  $(s_i s_j)^2 = 1$  collapses all higher powers. The partition function is a sum over spin configurations of a product of one such factor per bond:

$$\begin{aligned} Z &= \sum_{\{s_i\}} \prod_{\langle ij \rangle} e^{K s_i s_j} \\ &= (\cosh K)^{2N} \sum_{\{s_i\}} \prod_{\langle ij \rangle} (1 + \tanh K \cdot s_i s_j), \end{aligned} \quad (12.28)$$

where  $(\cosh K)^{2N}$  pulls out one  $\cosh K$  from each of the  $2N$  bonds on the square lattice with periodic boundary conditions.

Now “expanding the product” means multiplying out the  $2N$  factors in parentheses. Each factor offers two choices: either the 1 term or the  $\tanh K \cdot s_i s_j$  term. A term in the fully expanded product is specified by going through all  $2N$  bonds and recording, for each one, which of the two choices was made. Let  $\mathcal{B}$  denote the subset of bonds for which the  $\tanh K \cdot s_i s_j$  term was chosen; the remaining bonds contribute factors of 1. Multiplying everything together, the fully expanded product is

$$\prod_{\langle ij \rangle} (1 + \tanh K \cdot s_i s_j) = \sum_{\mathcal{B}} (\tanh K)^{|\mathcal{B}|} \prod_{\langle ij \rangle \in \mathcal{B}} s_i s_j, \quad (12.29)$$

where  $|\mathcal{B}|$  is the number of bonds in  $\mathcal{B}$  and the sum runs over all  $2^{2N}$  subsets of bonds. The empty subset ( $\mathcal{B} = \emptyset$ ) contributes 1; the full subset contributes  $(\tanh K)^{2N} \prod_{\langle ij \rangle} s_i s_j$ . Every intermediate possibility appears exactly once. This is the same bookkeeping one uses to expand  $(1+a)(1+b)(1+c) = 1 + a + b + c + ab + ac + bc + abc$ : each term is labeled by the subset of factors from which the non-unit piece was taken.

Substituting equation (12.29) back into  $Z$  and exchanging the order of the sum over spin configurations with the sum over bond subsets:

$$Z = (\cosh K)^{2N} \sum_{\mathcal{B}} (\tanh K)^{|\mathcal{B}|} \sum_{\{s_i\}} \prod_{\langle ij \rangle \in \mathcal{B}} s_i s_j. \quad (12.30)$$

Each bond in  $\mathcal{B}$  contributes one factor of  $s_i$  and one factor of  $s_j$  to the product. Collecting these factors by site, the spin product becomes

$$\prod_{\langle ij \rangle \in \mathcal{B}} s_i s_j = \prod_i s_i^{n_i}, \quad (12.31)$$

where  $n_i$  is the number of bonds in  $\mathcal{B}$  incident on site  $i$  (the degree of site  $i$  within  $\mathcal{B}$ ). The spin sum then factorizes over sites because different  $s_i$  are independent:

$$\sum_{\{s_i\}} \prod_i s_i^{n_i} = \prod_i \left( \sum_{s_i = \pm 1} s_i^{n_i} \right). \quad (12.32)$$

Each site sum is elementary:

$$\sum_{s=\pm 1} s^n = \begin{cases} 2 & n \text{ even,} \\ 0 & n \text{ odd.} \end{cases} \quad (12.33)$$

A single odd  $n_i$  anywhere on the lattice kills the entire term. Only subsets  $\mathcal{B}$  in which *every* site has even degree survive, and each surviving term contributes  $2^N$  (one factor of 2 per site). The subsets with all even degrees are exactly the unions of closed loops on the original lattice: a closed loop enters and leaves each site it visits the same number of times, giving even degree at every vertex, while any open path has odd degree at its endpoints and is killed by the spin sum. The partition function becomes

$$Z = 2^N (\cosh K)^{2N} \sum_{\mathcal{P}'} (\tanh K)^{|\mathcal{P}'|}, \quad (12.34)$$

where  $\mathcal{P}'$  runs over all collections of closed loops on the original lattice and  $|\mathcal{P}'|$  is their total bond length. At small  $K$ ,  $\tanh K$  is small, so the sum is dominated by the shortest loops (on the square lattice, the elementary plaquettes of perimeter 4), with longer loops contributing higher-order corrections in  $\tanh K$ .

**Duality and the critical point.** Comparing equations (12.25) and (12.34): both have the form of a prefactor times a weighted sum over closed loops on a square lattice, with per-bond weights  $e^{-2K}$  and  $\tanh K$  respectively. On the square lattice the dual lattice is again a square lattice, so the two loop sums are structurally identical. Define  $K^*$  by

$$\tanh K^* = e^{-2K}. \quad (12.35)$$

The loop sum in (12.25) at coupling  $K$  then equals the loop sum in (12.34) at coupling  $K^*$ , giving the exact relation

$$Z(K) = \frac{2e^{2KN}}{2^N (\cosh K^*)^{2N}} Z(K^*). \quad (12.36)$$

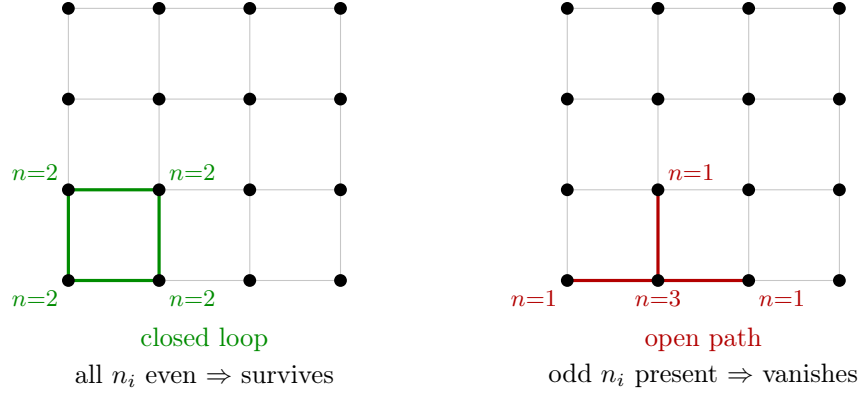


Figure 12.4: High-temperature expansion: the closed-loop condition. A bond subset  $\mathcal{B}$  contributes to  $Z$  only if every site  $i$  has even degree  $n_i$  (the number of bonds in  $\mathcal{B}$  incident on  $i$ ), because  $\sum_{s=\pm 1} s^n = 0$  for  $n$  odd. Left: a unit square (four bonds, green) has  $n_i = 2$  at all four corners; the spin sums all yield 2, contributing  $(\tanh K)^4 \cdot 2^N$  to  $Z$ . Right: an L-shaped path of three bonds (red) has  $n_i = 1$  at its three endpoints and  $n_i = 3$  at the junction; the spin sum at each of these sites vanishes, so this subset contributes zero. Only bond subsets forming unions of closed loops on the original lattice survive, giving equation (12.34).

The prefactor is a smooth function of  $K$ , so any non-analyticity in  $Z(K)$  at  $K_c$  implies a non-analyticity in  $Z(K^*)$  at the dual value  $K_c^*$ . The map (12.35) sends small  $K$  (high temperature, disordered) to large  $K^*$  (low temperature, ordered) and vice versa, so it relates the two sides of the transition. If the model has a unique phase transition, the transition point must be its own image under the duality, i.e.,  $K_c = K_c^*$ . Setting  $K = K^*$  in (12.35) and using  $\tanh K = (e^{2K} - 1)/(e^{2K} + 1)$ :

$$\begin{aligned}
 \frac{e^{2K_c} - 1}{e^{2K_c} + 1} &= e^{-2K_c} \\
 e^{2K_c} - 1 &= 1 + e^{-2K_c} \\
 e^{2K_c} - e^{-2K_c} &= 2 \\
 \sinh(2K_c) &= 1,
 \end{aligned} \tag{12.37}$$

so  $2K_c = \ln(1 + \sqrt{2})$ , giving

$$\frac{k_B T_c}{J} = \frac{1}{K_c} = \frac{2}{\ln(1 + \sqrt{2})} \approx 2.269. \tag{12.38}$$

Kramers and Wannier obtained this result in 1941, three years before Onsager's complete exact solution.

### 12.3.2 Kadanoff Block Spin and the Scaling Hypothesis

The 1D decimation worked because the calculation is exact: summing over every other spin produces a perfect Ising model with a new coupling, and nothing else. The 2D case is more demanding. Integrating out a fraction of spins on a square lattice generates not just a renormalized nearest-neighbor coupling but also next-nearest-neighbor couplings, four-spin couplings, and infinitely many higher-order terms. Tracking all of these exactly is intractable.

Kadanoff (1966) cut through this difficulty with a physical argument rather than a calculation. Near a critical point the system is scale-invariant: fluctuations occur on every length scale simultaneously and there is no characteristic microscopic scale. If we group  $L \times L$  spins into blocks and assign each block a single block spin

$$S_I = \text{sign} \left( \sum_{i \in I} s_i \right), \quad (12.39)$$

then near criticality the system of block spins should look statistically identical to the original system, rescaled in temperature and field. All the complicated higher-order couplings generated by the block-spin procedure shrink to zero under further coarse-graining near the critical point; they shift  $T_c$  slightly but do not change the universality class.

This is Kadanoff's *scaling hypothesis*: the only effect of a block-spin step with rescaling factor  $b = L$  on the two relevant couplings is

$$t' = b^{y_t} t, \quad (12.40)$$

$$h' = b^{y_h} h, \quad (12.41)$$

where  $t = (T - T_c)/T_c$  is the reduced temperature,  $h$  is a dimensionless external field, and  $y_t > 0$ ,  $y_h > 0$  are called the *thermal* and *magnetic scaling exponents*. Both are positive, meaning both  $t$  and  $h$  grow under coarse-graining: they are the relevant operators at the 2D Ising fixed point. The fixed point itself sits at  $t = h = 0$  (exactly at  $T_c$ , zero field), and the scaling hypothesis asserts that near this point the RG acts as a pure rescaling of the two relevant parameters. Kadanoff could not derive this from the microscopic model, but it is consistent with the exact solution and captures the essential physics: a scale-invariant system has no intrinsic length, so scale transformations can only multiply relevant couplings by powers of  $b$ .

More generally, near any RG fixed point the linearized flow assigns an eigenvalue  $b^y$  to each coupling in the Hamiltonian. The sign of  $y$  classifies operators into three categories.

#### Classification of Operators

**Relevant operators** have  $y > 0$  and grow under coarse-graining. They drive the system away from the fixed point and determine which phase the system is in. An example is the temperature deviation  $t = (T - T_c)/T_c$ .

**Irrelevant operators** have  $y < 0$  and shrink under coarse-graining. They flow toward zero at the fixed point and do not affect universal critical behavior. An example is microscopic lattice details.

**Marginal operators** have  $y = 0$  and remain unchanged by RG to leading order. They require higher-order analysis to determine whether they are marginally relevant or marginally irrelevant. Marginal operators are rare and special; they arise, for example, in certain four-dimensional quantum field theories studied in particle physics.

With this classification in hand, Kadanoff's scaling hypothesis asserts that  $t$  and  $h$  are the only relevant operators at the 2D Ising fixed point, and that all the complicated couplings generated by block-spinning are irrelevant.

### 12.3.3 Linearized RG, Scaling Relations, and Critical Exponents

The scaling hypothesis can be written as a linearized RG matrix. Near the fixed point  $(t, h) = (0, 0)$ , the transformation acts as

$$\begin{pmatrix} t' \\ h' \end{pmatrix} = \begin{pmatrix} b^{y_t} & 0 \\ 0 & b^{y_h} \end{pmatrix} \begin{pmatrix} t \\ h \end{pmatrix}. \quad (12.42)$$

The matrix is diagonal because the Ising Hamiltonian is symmetric under  $h \rightarrow -h$ , so temperature deviations and field deviations cannot mix under RG. The eigenvalues  $b^{y_t}$  and  $b^{y_h}$  are both greater than 1, confirming that  $t$  and  $h$  are relevant.

**Correlation length exponent  $\nu$ .** The correlation length  $\xi$  is a physical length. When we coarse-grain by a factor  $b$ , the lattice spacing changes from  $a$  to  $ba$ , but the physical correlation length  $\xi a$  does not change (the RG reorganizes degrees of freedom without altering the physics). Expressed in units of the new lattice spacing, the correlation length is therefore  $\xi' = \xi/b$ . Since  $\xi$  depends on  $t$ , and one RG step maps  $t \rightarrow b^{y_t} t$ , we can write  $\xi(t)/b = \xi(b^{y_t} t)$ , or equivalently

$$\xi(t) = b \xi(b^{y_t} t). \quad (12.43)$$

This is a functional equation for  $\xi(t)$  that we can solve by a judicious choice of  $b$ . Set  $b = |t|^{-1/y_t}$  so that the argument of  $\xi$  on the right becomes  $\pm 1$ . Then  $\xi(t) = |t|^{-1/y_t} \xi(\pm 1) \sim |t|^{-1/y_t}$ , where  $\xi(\pm 1)$  is a nonuniversal constant. Comparing with the definition  $\xi \sim |t|^{-\nu}$  gives

$$\boxed{\nu = \frac{1}{y_t}}. \quad (12.44)$$

**Free energy scaling.** The partition function  $Z$  is invariant under a single RG step: decimation or block-spinning reorganizes the sum over spin configurations, grouping terms together, but it does not change the total. Every original configuration still contributes, just counted through the intermediate (coarse-grained) variables. The total free energy  $F = -k_B T \ln Z$  is therefore also invariant.

Now write  $F$  in two different ways. Before the RG step, the system has  $N$  sites and is described by the reduced couplings  $(t, h)$ . Writing the free energy as  $N$  times a free energy density  $f(t, h)$ :

$$F = N f(t, h). \quad (12.45)$$

After a block-spin step with rescaling factor  $b$  in  $d$  spatial dimensions, each block contains  $b^d$  original sites and is replaced by a single new effective site. The number of effective sites is therefore

$$N' = \frac{N}{b^d}, \quad (12.46)$$

and the new couplings are  $t' = b^{y_t} t$  and  $h' = b^{y_h} h$  from the scaling hypothesis. The same total free energy, now written in terms of the new description, is

$$F = N' f(t', h') = \frac{N}{b^d} f(b^{y_t} t, b^{y_h} h). \quad (12.47)$$

Setting these two expressions for  $F$  equal and canceling the common factor  $N$ :

$$f(t, h) = b^{-d} f(b^{y_t} t, b^{y_h} h). \quad (12.48)$$

This is the central scaling form for the free energy density. It says that  $f$  at couplings  $(t, h)$  equals  $f$  at the renormalized couplings  $(b^{y_t} t, b^{y_h} h)$ , up to the geometric factor  $b^{-d}$  that accounts for each effective site representing  $b^d$  original sites.

**Standard critical exponents.** Before extracting consequences from the scaling form, we introduce the four standard exponents that label the power-law singularities of thermodynamic quantities near  $T_c$ . Each is defined by tuning either the reduced temperature  $t = (T - T_c)/T_c$  at zero field, or the field  $h$  at  $t = 0$ :

$$m(t, 0) \sim |t|^\beta \quad (t \rightarrow 0^-), \quad (12.49)$$

$$\chi(t, 0) \sim |t|^{-\gamma}, \quad (12.50)$$

$$m(0, h) \sim h^{1/\delta}, \quad (12.51)$$

$$C(t, 0) \sim |t|^{-\alpha}. \quad (12.52)$$

Here  $m = -\partial f / \partial h$  is the magnetization (the order parameter),  $\chi = \partial m / \partial h$  is the zero-field susceptibility (the linear response of  $m$  to a small applied field), and  $C \sim -\partial^2 f / \partial t^2$  is the specific heat at zero field. The exponent  $\beta$  governs how the spontaneous magnetization vanishes as  $T \rightarrow T_c^-$ ;  $\gamma$  governs the divergence of the susceptibility, which becomes infinite at  $T_c$  because fluctuations are correlated across all length scales;  $\delta$  fixes the shape of the critical isotherm at  $t = 0$ ; and  $\alpha$  governs the singularity of the specific heat (with  $\alpha = 0$  corresponding to a logarithmic divergence rather than a power law). Together with the correlation-length exponent  $\nu$  already obtained, these are the five standard exponents characterizing a continuous transition.

Equation (12.48), combined with the definitions of the standard critical exponents above, determines all four of  $\beta$ ,  $\gamma$ ,  $\delta$ , and  $\alpha$  in terms of  $y_t$ ,  $y_h$ , and  $d$ . The strategy is the same in every case: differentiate both sides the appropriate number of times, apply the chain rule to pick up factors of  $b^{y_t}$  or  $b^{y_h}$ , then make a clever choice of  $b$  that eliminates one of the two arguments on the right-hand side.

**Magnetization exponent  $\beta$ .** The magnetization is defined by

$$m(t, h) = -\frac{\partial f(t, h)}{\partial h}. \quad (12.53)$$

Differentiate both sides of equation (12.48) with respect to  $h$ . The left-hand side gives  $-m(t, h)$ . On the right-hand side,  $h$  enters only through the rescaled argument  $h' = b^{y_h} h$ , so the chain rule gives

$$\begin{aligned} \frac{\partial}{\partial h} \left[ b^{-d} f(b^{y_t} t, b^{y_h} h) \right] &= b^{-d} \frac{\partial f}{\partial h'} \bigg|_{(b^{y_t} t, b^{y_h} h)} \cdot \frac{\partial h'}{\partial h} \\ &= b^{-d} \cdot b^{y_h} \frac{\partial f}{\partial h'} \bigg|_{(b^{y_t} t, b^{y_h} h)} \\ &= -b^{y_h-d} m(b^{y_t} t, b^{y_h} h). \end{aligned} \quad (12.54)$$

Equating the two sides (the minus signs cancel):

$$m(t, h) = b^{y_h-d} m(b^{y_t} t, b^{y_h} h). \quad (12.55)$$

Now set  $h = 0$  on both sides and choose  $b$  to eliminate  $t$  from the argument on the right. The choice  $b = |t|^{-1/y_t}$  does the job, because then  $b^{y_t} t = |t|^{-1} t = \text{sgn}(t) = \pm 1$ . Substituting:

$$m(t, 0) = (|t|^{-1/y_t})^{y_h-d} m(\pm 1, 0) = |t|^{(d-y_h)/y_t} m(\pm 1, 0). \quad (12.56)$$

The quantity  $m(\pm 1, 0)$  is a nonuniversal number (the magnetization at the reference point), not a function of  $t$ . The  $t$ -dependence is therefore entirely in the prefactor  $|t|^{(d-y_h)/y_t}$ . Comparing with



the conventional definition  $m \sim |t|^\beta$  (below  $T_c$ ):

$$\boxed{\beta = \frac{d - y_h}{y_t} = \nu(d - y_h),} \quad (12.57)$$

where in the last step we used  $\nu = 1/y_t$ .

**Susceptibility exponent  $\gamma$ .** The zero-field susceptibility is

$$\chi(t, h) = \frac{\partial m}{\partial h} = -\frac{\partial^2 f}{\partial h^2}, \quad (12.58)$$

so it requires a second derivative with respect to  $h$ . Differentiate equation (12.55) once more with respect to  $h$ , and apply the chain rule to the  $b^{y_h}h$  argument again. A second factor of  $b^{y_h}$  comes down:

$$\chi(t, h) = b^{y_h-d} \cdot b^{y_h} \chi(b^{y_t}t, b^{y_h}h) = b^{2y_h-d} \chi(b^{y_t}t, b^{y_h}h). \quad (12.59)$$

Set  $h = 0$  and again choose  $b = |t|^{-1/y_t}$ :

$$\chi(t, 0) = (|t|^{-1/y_t})^{2y_h-d} \chi(\pm 1, 0) = |t|^{-(2y_h-d)/y_t} \chi(\pm 1, 0). \quad (12.60)$$

Comparing with the definition  $\chi \sim |t|^{-\gamma}$ :

$$\boxed{\gamma = \frac{2y_h - d}{y_t} = \nu(2y_h - d).} \quad (12.61)$$

**Critical isotherm exponent  $\delta$ .** Exactly at the critical temperature,  $t = 0$ , and the magnetization depends only on  $h$ . The conventional definition is  $m \sim h^{1/\delta}$ , so we extract  $\delta$  by following the  $h$ -dependence of  $m$  at  $t = 0$ .

Return to the magnetization scaling relation (12.55) and set  $t = 0$ :

$$m(0, h) = b^{y_h-d} m(0, b^{y_h}h). \quad (12.62)$$

This time, rather than eliminating  $t$ , we eliminate  $h$  from the right-hand side. Choose  $b = |h|^{-1/y_h}$  so that  $b^{y_h}h = |h|^{-1}h = \pm 1$ :

$$m(0, h) = (|h|^{-1/y_h})^{y_h-d} m(0, \pm 1) = |h|^{(d-y_h)/y_h} m(0, \pm 1). \quad (12.63)$$

Comparing with  $m \sim h^{1/\delta}$  reads off  $1/\delta = (d - y_h)/y_h$ , so

$$\boxed{\delta = \frac{y_h}{d - y_h}.} \quad (12.64)$$

**Specific heat exponent  $\alpha$ .** The specific heat (at fixed field  $h = 0$ ) comes from a second derivative of the free energy density with respect to temperature:

$$C(t) \sim -\frac{\partial^2 f(t, 0)}{\partial t^2}. \quad (12.65)$$

Differentiate equation (12.48) twice with respect to  $t$ . Each derivative acts only through the rescaled argument  $t' = b^{y_t}t$ , and each application of the chain rule pulls down a factor of  $b^{y_t}$ :

$$\frac{\partial f(t, h)}{\partial t} = b^{-d} \cdot b^{y_t} \left. \frac{\partial f}{\partial t'} \right|_{(b^{y_t}t, b^{y_h}h)} = b^{y_t-d} \left. \frac{\partial f}{\partial t'} \right|_{(b^{y_t}t, b^{y_h}h)}, \quad (12.66)$$

$$\frac{\partial^2 f(t, h)}{\partial t^2} = b^{y_t-d} \cdot b^{y_t} \left. \frac{\partial^2 f}{\partial t'^2} \right|_{(b^{y_t}t, b^{y_h}h)} = b^{2y_t-d} \left. \frac{\partial^2 f}{\partial t'^2} \right|_{(b^{y_t}t, b^{y_h}h)}. \quad (12.67)$$

Therefore

$$C(t, h) \sim b^{2y_t - d} C(b^{y_t} t, b^{y_h} h). \quad (12.68)$$

Set  $h = 0$  and choose  $b = |t|^{-1/y_t}$ :

$$C(t, 0) \sim (|t|^{-1/y_t})^{2y_t - d} C(\pm 1, 0) = |t|^{-(2y_t - d)/y_t} C(\pm 1, 0) = |t|^{d/y_t - 2} C(\pm 1, 0). \quad (12.69)$$

Comparing with  $C \sim |t|^{-\alpha}$  gives  $-\alpha = d/y_t - 2$ , or

$$\boxed{\alpha = 2 - \frac{d}{y_t} = 2 - d\nu.} \quad (12.70)$$

This last relation,  $2 - \alpha = d\nu$ , is the *hyperscaling relation*. It connects the specific heat exponent to the correlation length exponent and the spatial dimension, and it is a nontrivial prediction of the scaling hypothesis (it fails above the upper critical dimension, where mean-field exponents take over and fluctuations no longer determine the critical behavior).

These are not independent results. All four exponents ( $\beta$ ,  $\gamma$ ,  $\delta$ ,  $\alpha$ ) follow from just two numbers ( $y_t$  and  $y_h$ ) through the free energy scaling (12.48). This is the power of the RG: instead of calculating each exponent separately, the scaling hypothesis reduces everything to two eigenvalues of the linearized RG transformation.

### Example 2

2D Ising Critical Exponents: Onsager's exact solution (1944) gives  $\nu = 1$  and  $\beta = 1/8$  for  $d = 2$ . From these, the scaling exponents are  $y_t = 1/\nu = 1$  and  $y_h = d - \beta/\nu = 2 - 1/8 = 15/8$ . All remaining exponents follow:

$$\gamma = \nu(2y_h - d) = 1\left(\frac{15}{4} - 2\right) = \frac{7}{4}, \quad (12.71)$$

$$\delta = \frac{y_h}{d - y_h} = \frac{15/8}{1/8} = 15, \quad (12.72)$$

$$\alpha = 2 - d\nu = 0 \quad (\text{logarithmic divergence}). \quad (12.73)$$

These are exact, not approximate. Onsager's solution proceeds via the transfer matrix method: the 2D partition function is rewritten as a trace over products of  $2^L \times 2^L$  matrices (one per row of the lattice), reducing the statistical mechanics problem to finding the largest eigenvalue of this matrix. The calculation is a tour de force of algebra, and we do not reproduce it here, but the key point is that the critical exponents can be extracted analytically. The values happen to be simple fractions, a consequence of the unusually high degree of symmetry at the 2D Ising fixed point, which constrains the exponents far more tightly than in a generic critical system. Kadanoff's framework guarantees that exponents  $y_t$  and  $y_h$  must exist and that all other critical exponents follow from them through scaling relations; the numerical values require either Onsager's exact solution or a more refined calculation such as the  $\varepsilon$ -expansion.

## 12.4 Universality and RG Flows

The classification of operators into relevant, irrelevant, and marginal has a powerful consequence: at criticality, irrelevant operators vanish, leaving only relevant and marginal operators. Systems with the same relevant operators exhibit identical critical behavior regardless of microscopic details.

### Physical Insight

Universality explains why liquid-gas critical points, ferromagnetic transitions, and binary alloy phase separations all have the same critical exponents. They share the same RG fixed point structure despite vastly different microscopic physics.

What does it mean for two different theories to flow to the same fixed point? Consider the space of all possible Hamiltonians as a high-dimensional “theory space,” where each point represents a different set of coupling constants. The RG defines flow lines in this space: trajectories along which theories evolve as we coarse-grain.

A fixed point is a theory that remains unchanged under RG transformations. Near each fixed point, we can linearize the RG flow and identify directions (eigenvectors) that grow (relevant), shrink (irrelevant), or remain unchanged (marginal). The *basin of attraction* of a fixed point consists of all theories that flow toward it under repeated coarse-graining.

### Example 3

Universality in the 2D Ising Model: Consider three different microscopic models:

- (a) Square lattice Ising model:  $H_1 = -J \sum_{\langle ij \rangle_{\text{square}}} s_i s_j$
- (b) Triangular lattice Ising model:  $H_2 = -J \sum_{\langle ij \rangle_{\text{tri}}} s_i s_j$
- (c) Square lattice with next-nearest neighbors:  $H_3 = -J \sum_{\langle ij \rangle} s_i s_j - J_2 \sum_{\langle\langle ij \rangle\rangle} s_i s_j$

All three have different microscopic physics: different coordination numbers, different numbers of neighbors, different short-range structures. Yet at their critical temperatures, they exhibit *identical* critical exponents:  $\beta = 1/8$ ,  $\gamma = 7/4$ ,  $\nu = 1$ ,  $\delta = 15$ .

Why? The lattice geometry and next-nearest neighbor couplings correspond to irrelevant operators. Under RG flow toward the critical point, these differences wash out. All three theories lie in the basin of attraction of the same fixed point: the 2D Ising fixed point. At long distances, they become indistinguishable.

This has profound implications. Once we know the fixed point structure, we can predict critical behavior for entire classes of systems without solving each microscopic model, giving the RG tremendous predictive power. Measurements on completely different physical systems (magnets, fluids, alloys) can test the same theoretical predictions. The theory space perspective shifts our thinking: rather than asking about “the theory” of a system, we think of theories as points in theory space that flow under scale transformations. The long-distance physics depends only on which fixed point you flow toward, not where you started.

Two theories flow to the same fixed point if and only if they have the same set of relevant operators. Irrelevant operators specify *which* microscopic realization you are studying, but they do not affect universal critical behavior. This is why condensed matter physicists can study critical phenomena using simple lattice models even though real materials have enormously complex microscopic structure. The complexity lives in irrelevant operators that vanish at long distances under RG flow.

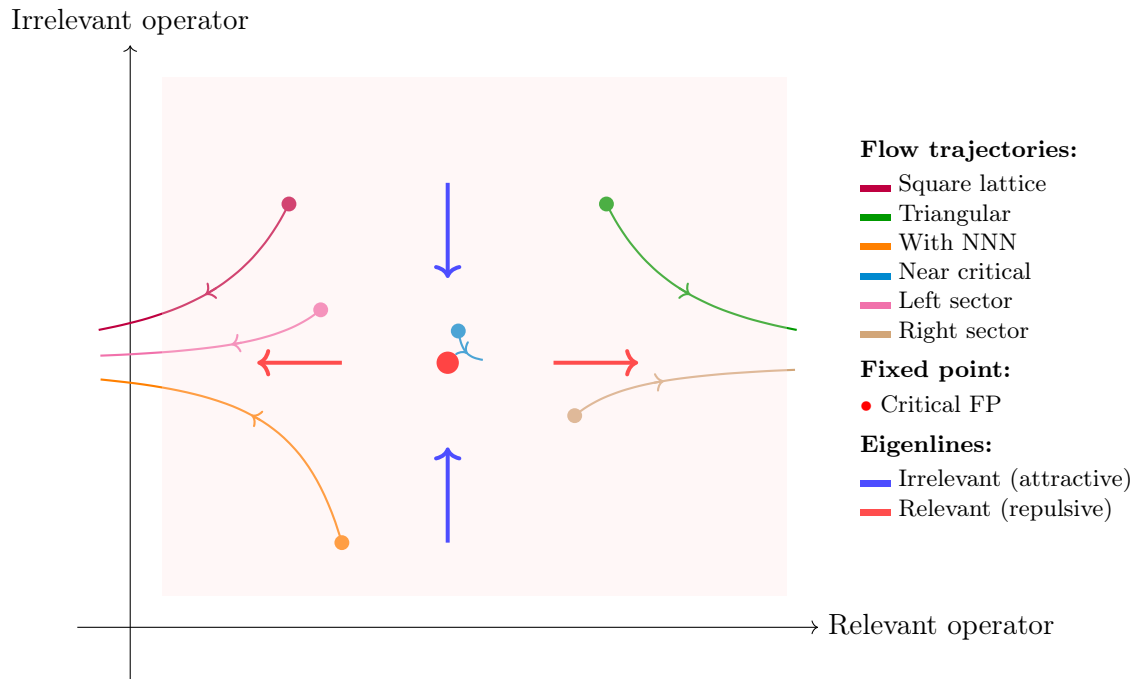


Figure 12.5: Theory space and RG flows near the critical fixed point. Different microscopic models start at different points (colored dots) and flow along curved trajectories computed from the linearized RG equations. The fixed point acts as a saddle: flows are attracted along the irrelevant direction (vertical, blue arrows) but repelled along the relevant direction (horizontal, red arrows). Trajectories veer toward the fixed point as irrelevant operators decay, then diverge along the relevant direction to different phases. All theories in the shaded basin eventually pass through the same critical behavior.

**Example 4**

Identifying Relevant and Irrelevant Operators: Consider the 2D Ising model near criticality. We can write down many possible terms in the Hamiltonian:

$$H = -J \sum_{\langle ij \rangle} s_i s_j - J_2 \sum_{\langle\langle ij \rangle\rangle} s_i s_j - h \sum_i s_i - h_4 \sum_{\langle ijkl \rangle} s_i s_j s_k s_l + \dots \quad (12.74)$$

How important is each term? The answer is a property of the *fixed point*, not something that can be read off from the Hamiltonian by inspection. One must either solve the model exactly or perform an explicit RG calculation to determine the scaling exponents  $y$  for each coupling. For the 2D Ising model, Onsager's exact solution provides the answer. The fixed point has exactly two relevant directions: the temperature deviation  $t \sim (T - T_c)/T_c$  with  $y_t = 1 > 0$ , and the magnetic field  $h$  with  $y_h = 15/8 > 0$ . Since the fixed point has only two relevant directions, every other coupling must have  $y < 0$  and is therefore irrelevant. The next-nearest-neighbor coupling  $J_2$  and the four-spin interaction  $h_4$  both fall into this category: they flow to zero under coarse-graining and do not affect critical behavior.

Physically this is plausible. Under a block-spin step with block size  $b$ , the block spin averages over  $b^d$  microscopic spins. Details that distinguish nearest-neighbor from next-nearest-neighbor geometry, or that correlate four spins around a plaquette, are short-range structures that get washed out by this averaging. But plausibility is not proof: the classification of operators requires a calculation, not an intuitive argument. This is why we can ignore lattice details when computing critical exponents. The microscopic differences between square, triangular, and hexagonal lattices all correspond to irrelevant operators that vanish under RG flow to the critical point.

This is the proper definition of a universality class: it is the set of all theories that flow to the same fixed point under RG. Two theories are equivalent, in the sense of sharing the same long-distance physics, if and only if they belong to the same universality class. The equivalence is one of RG fate, not of microscopic structure. Away from criticality this equivalence is coarser, both theories flow to the same trivial fixed point and agree on which phase they inhabit, but the universality class still leaves its mark in the shared structure of corrections to scaling and in how each theory approaches the transition. The fixed point is the organizing principle; the basin of attraction and the flow geometry are what partition theory space into equivalence classes.

## 12.5 Quantum Phase Transitions

### 12.5.1 Zero-Temperature Criticality

Classical phase transitions are driven by thermal fluctuations. At high temperature, entropy dominates and disordered phases prevail; at low temperature, energy dominates and ordered phases emerge. The critical point separating them occurs at a specific temperature  $T_c$ , determined by the competition between energy and entropy.

Quantum mechanics introduces a fundamentally different mechanism. At absolute zero, thermal fluctuations vanish entirely, yet a system can still undergo a phase transition if a non-thermal parameter, a coupling constant, an applied field, a chemical doping, is tuned through a critical value. These are *quantum phase transitions* (QPTs), driven not by thermal energy but by quantum fluctuations arising from the uncertainty principle. The ground state of a quantum system is not simply

the lowest-energy classical configuration; it is a superposition spread over many configurations by zero-point motion. As a control parameter is varied, this ground-state wave function can change qualitatively, undergoing a sharp, nonanalytic transition at  $T = 0$ .

### 12.5.2 The Transverse-Field Ising Model

The paradigmatic model for a QPT is the transverse-field Ising chain,

$$H = -J \sum_i \sigma_i^z \sigma_{i+1}^z - h \sum_i \sigma_i^x, \quad (12.75)$$

which students of this course will recognize immediately. The first term favors ferromagnetic alignment along  $z$ ; the transverse field along  $x$  introduces quantum fluctuations. Because  $\sigma_i^z$  and  $\sigma_i^x$  do not commute, the two terms are in direct competition and cannot be simultaneously minimized.

The limiting cases are transparent. When  $h = 0$  the ground state is twofold degenerate: all spins aligned along  $+z$  or all along  $-z$ . This is the ferromagnetic (ordered) phase, with  $\langle \sigma_i^z \rangle \neq 0$ . When  $h \gg J$  the transverse field dominates and each spin settles into  $|+x\rangle = (|\uparrow\rangle + |\downarrow\rangle)/\sqrt{2}$ , giving  $\langle \sigma_i^z \rangle = 0$ . This is the paramagnetic (disordered) phase. The terminology refers to the order parameter  $\langle \sigma_i^z \rangle$ : the state is “disordered” in the sense that it has no spontaneous magnetization along  $z$  and respects the  $\mathbb{Z}_2$  symmetry  $\sigma_i^z \rightarrow -\sigma_i^z$  generated by  $\prod_i \sigma_i^x$ . The ground state itself is of course a perfectly definite product state  $|+x\rangle^{\otimes N}$ ; “disorder” here is a statement about correlations of the order parameter, not about the wavefunction. Between these extremes, at a critical value  $h_c = J$  (the exact value follows from mapping to the 2D classical Ising model via the transfer matrix, discussed below), there is a quantum critical point. The spatial correlation length diverges as  $\xi \sim |h - h_c|^{-\nu}$ , exactly as at a classical critical point. But a quantum critical point has a feature absent from classical transitions: the energy gap  $\Delta$  between the ground state and first excited state also closes,

$$\Delta \sim \xi^{-z} \sim |h - h_c|^{z\nu}. \quad (12.76)$$

The relation  $\Delta \sim \xi^{-z}$  defines the *dynamical critical exponent*  $z$ ; we state it here without proof, as its derivation requires the transfer matrix formalism. Figure 12.6 shows the exact gap  $\Delta = 2|J - h|$  obtained from the Jordan-Wigner solution of the 1D transverse-field Ising model, displaying the characteristic V-shape: the gap decreases linearly from  $2J$  at  $h = 0$ , closes at the quantum critical point  $h_c = J$ , and then grows linearly as  $2h$  for  $h > J$ . The linear closing is the signature  $\Delta \sim |h - h_c|^{z\nu}$  with  $z\nu = 1$ , and for this model the exact values are  $z = 1$  and  $\nu = 1$  separately.

A closing gap means a diverging timescale: the inverse gap  $\hbar/\Delta$  sets the characteristic time for quantum fluctuations between the two competing ground states. The dynamical exponent describes how this timescale grows relative to the spatial correlation length, with correlation time  $\xi_\tau \sim \xi^z$ . At a classical critical point, only spatial correlations diverge. At a quantum critical point, spatial and temporal correlations diverge together, and  $z$  controls their relative rate. For the 1D transverse-field Ising model, the exact solution gives  $z = 1$ : space and imaginary time scale identically at the critical point.

### 12.5.3 The Quantum-Classical Correspondence

The deepest structural insight into QPTs is a mapping between quantum systems in  $d$  spatial dimensions and classical systems in  $d + 1$  dimensions, where the extra dimension is imaginary time.

The connection runs through a substitution already noted in this chapter. Chapter 4 introduced the time-evolution operator  $e^{-i\hat{H}t/\hbar}$ , and the Ising model section above observed that the Boltzmann factor  $e^{-\beta\hat{H}}$  has the same mathematical form with  $t$  replaced by  $-i\beta\hbar$ . Make this substitution

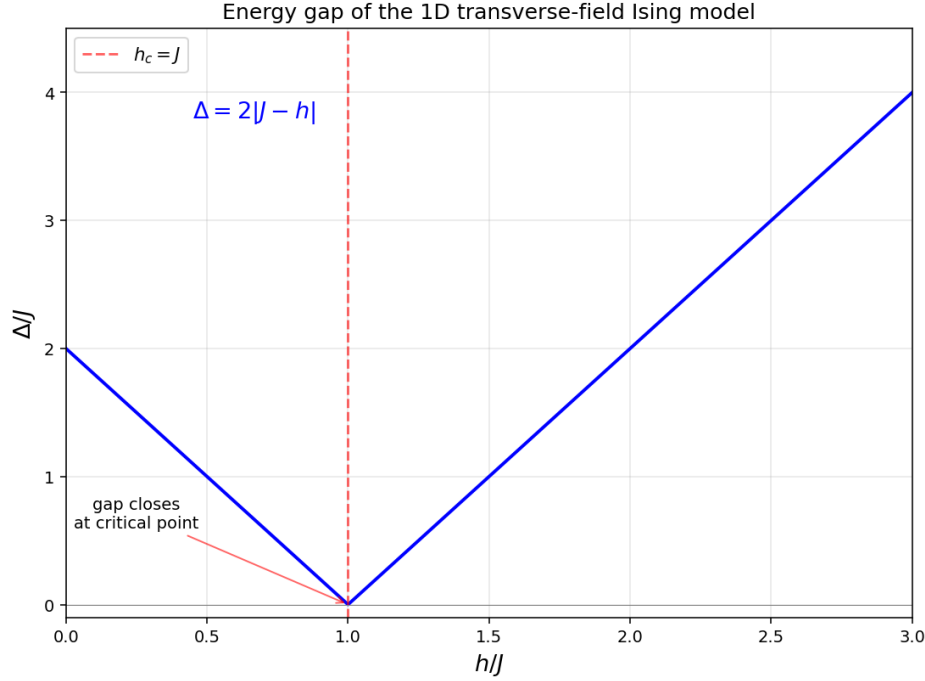


Figure 12.6: Energy gap  $\Delta$  of the 1D transverse-field Ising model as a function of the transverse field  $h$ , in units of the Ising coupling  $J$ . The exact result  $\Delta = 2|J - h|$  takes the value  $2J$  at  $h = 0$  (the cost to flip a spin against its ferromagnetically aligned neighbors), decreases linearly with slope  $-2J$  until it closes at the quantum critical point  $h_c = J$ , and then grows linearly with slope  $+2J$  as  $\Delta = 2(h - J)$  for  $h > J$ , asymptoting to  $2h$  in the deep paramagnetic regime (the Zeeman cost to flip a spin against the transverse field). The linear closing reflects the critical exponent product  $z\nu = 1$ .

explicit: define imaginary time  $\tau$  by  $t = -i\tau$ , where  $\tau$  is real and has units of time. The evolution operator becomes

$$e^{-i\hat{H}t/\hbar}\big|_{t=-i\tau} = e^{-\hat{H}\tau/\hbar}. \quad (12.77)$$

This is identical to the Boltzmann factor when  $\tau = \beta\hbar = \hbar/(k_B T)$ . The partition function is therefore

$$Z = \text{Tr}(e^{-\beta\hat{H}}) = \text{Tr}\left(e^{-\hat{H}\tau/\hbar}\bigg|_{\tau=\beta\hbar}\right), \quad (12.78)$$

which is quantum evolution in imaginary time over a total “length”  $\beta\hbar$ , with periodic boundary conditions (since the trace closes the path). At finite temperature, this imaginary-time direction is compact with circumference  $\hbar/(k_B T)$ . As  $T \rightarrow 0$ ,  $\beta \rightarrow \infty$  and the imaginary-time direction becomes infinitely long, adding a genuine new spatial-like dimension to the problem.

**Why this is legitimate.** The substitution  $t \rightarrow -i\tau$  may look like an arbitrary trick, but no physics is being analytically continued. Both  $e^{-i\hat{H}t/\hbar}$  and  $e^{-\beta\hat{H}}$  are well-defined operators built from the same Hermitian Hamiltonian  $\hat{H}$ , with the same spectrum and the same eigenstates. They are simply different functions of  $\hat{H}$ . The partition function  $Z = \text{Tr}(e^{-\beta\hat{H}})$  is a number we want to compute; the observation is that this number happens to equal  $\text{Tr}(e^{-i\hat{H}t/\hbar})$  evaluated at the formal value  $t = -i\beta\hbar$ . We are identifying two expressions, not deforming time itself.

**Where the extra dimension comes from.** To compute the trace, insert resolutions of the identity. Split  $\beta = N\Delta\tau$  into  $N$  small slices and use  $e^{-\beta\hat{H}} = (e^{-\Delta\tau\hat{H}})^N$ . Between each pair of factors, insert  $\sum_n |n\rangle\langle n| = \mathbb{I}$  in some convenient basis. The trace becomes

$$Z = \sum_{n_1, \dots, n_N} \langle n_1 | e^{-\Delta\tau\hat{H}} | n_N \rangle \langle n_N | e^{-\Delta\tau\hat{H}} | n_{N-1} \rangle \cdots \langle n_2 | e^{-\Delta\tau\hat{H}} | n_1 \rangle. \quad (12.79)$$

Each intermediate index  $n_k$  labels the state at imaginary time  $\tau_k = k\Delta\tau$ ; the original Hilbert space has been replaced by an array of  $N$  copies of itself, indexed by a discrete time slice. The trace forces  $n_{N+1} = n_1$ , giving periodic boundary conditions in  $\tau$ . The “extra dimension” is just the bookkeeping label for these intermediate sums.

**A worked example: one qubit in a transverse field.** Take  $\hat{H} = -h\sigma^x$  on a single qubit. Then  $Z = \text{Tr}(e^{\beta h\sigma^x}) = 2 \cosh(\beta h)$ , which we can also obtain directly from the eigenvalues  $\pm h$ . Now compute it the imaginary-time way. Slice  $\beta = N\Delta\tau$  and insert  $\sigma^z$ -eigenstates  $|s\rangle$  with  $s = \pm 1$  between slices:

$$Z = \sum_{s_1, \dots, s_N} \prod_{k=1}^N \langle s_{k+1} | e^{\Delta\tau h\sigma^x} | s_k \rangle, \quad s_{N+1} = s_1. \quad (12.80)$$

The single-slice matrix element is elementary. Expand  $e^{\Delta\tau h\sigma^x} = \cosh(\Delta\tau h)\mathbb{I} + \sinh(\Delta\tau h)\sigma^x$  and use  $\langle s' | \mathbb{I} | s \rangle = \delta_{s's}$ ,  $\langle s' | \sigma^x | s \rangle = \delta_{s', -s}$ :

$$\langle s' | e^{\Delta\tau h\sigma^x} | s \rangle = \begin{cases} \cosh(\Delta\tau h) & s' = s, \\ \sinh(\Delta\tau h) & s' = -s. \end{cases} \quad (12.81)$$

Both cases can be written in the unified form  $A e^{K s' s}$ , with  $A^2 = \cosh(\Delta\tau h) \sinh(\Delta\tau h) = \frac{1}{2} \sinh(2\Delta\tau h)$  and  $e^{2K} = \coth(\Delta\tau h)$ . Substituting:

$$Z = A^N \sum_{\{s_k\}} \exp \left( K \sum_{k=1}^N s_{k+1} s_k \right). \quad (12.82)$$

This is exactly the partition function of a *classical* 1D Ising chain of  $N$  spins with periodic boundary conditions and coupling  $K$ . A single qubit in 0 spatial dimensions has become a classical Ising model in 1 dimension, with the new dimension being imaginary time. Summing this classical chain by the transfer-matrix method (a standard exercise) reproduces  $Z = 2 \cosh(\beta h)$  exactly in the  $N \rightarrow \infty$  limit; the equivalence is not approximate.

The same construction applied to the 1D transverse-field Ising chain (now with both  $\sigma^z \sigma^z$  and  $\sigma^x$  terms) generates a 2D classical Ising model on a square lattice, where one direction is the original spatial chain and the other is imaginary time.

For the 1D transverse-field Ising model, this Euclidean path-integral representation maps onto the 2D classical Ising model, solved exactly by Onsager in 1944. The QPT at  $h_c$  in the quantum model corresponds precisely to the thermal transition at  $T_c$  in the classical model, and the two systems share the same universality class and the same RG fixed point. The full derivation parallels the single-qubit calculation above but with two basis labels per site (spatial position and time slice) and an additional  $\sigma^z \sigma^z$  factor in each slice; the essential mechanism is the same  $t \rightarrow -i\tau$  substitution that converts quantum dynamics into classical statistical mechanics in one higher dimension. More generally, a  $d$ -dimensional quantum system at a QPT belongs to the universality class of a  $(d+z)$ -dimensional classical system, with  $z$  controlling the anisotropy between spatial and imaginary-time directions.



This correspondence means that the entire RG machinery developed for classical critical phenomena, relevant and irrelevant operators, universality, scaling relations among critical exponents, applies directly to quantum phase transitions. The two subjects are facets of the same mathematical structure.

## 12.6 From Spins to Fields

The RG framework developed so far rests on Kadanoff's scaling hypothesis, which assumes that scaling exponents  $y_t$  and  $y_h$  exist at the critical fixed point and lets all other critical exponents follow from them. But Kadanoff's hypothesis takes those two numbers as inputs, supplied by Onsager's exact solution in 2D, or by experiment and numerical simulation in 3D. The remainder of this chapter closes that gap. A continuum description of the order parameter, combined with a momentum-space implementation of the RG, produces a beta function whose fixed points and eigenvalues are the scaling exponents Kadanoff's hypothesis required. The key insight, due to Wilson and Fisher (1972), is that working in  $d = 4 - \varepsilon$  dimensions converts a degenerate marginal situation into a calculable interacting fixed point. Setting  $\varepsilon = 1$  gives quantitative predictions for three-dimensional critical exponents.

The earlier sections worked with discrete spins  $s_i = \pm 1$  on a lattice. After many rounds of block-spin coarse-graining, the lattice spacing becomes irrelevant and the order parameter varies smoothly over large distances. At that point it is natural to replace the discrete spin at each site with a continuous real number  $\phi(\mathbf{x})$ : the coarse-grained magnetization, averaged over a small region around  $\mathbf{x}$ . In the ordered phase  $\langle \phi \rangle \neq 0$ ; in the disordered phase  $\langle \phi \rangle = 0$ . This is the same field that appeared in the free-Gaussian discussion of the correlation function in the chapter introduction.

### 12.6.1 The Partition Function as a Functional Integral

For a lattice system the partition function sums over all spin configurations weighted by  $e^{-\beta H}$ . For a continuous field the analogous object is

$$Z = \int \mathcal{D}\phi e^{-S_E[\phi]}, \quad (12.83)$$

where  $\int \mathcal{D}\phi$  means “sum over every possible function  $\phi(\mathbf{x})$ ” and  $S_E[\phi]$  is the *Euclidean action*, which plays the role of energy: configurations with small  $S_E$  are exponentially favored, exactly as low-energy spin configurations are favored by  $e^{-\beta H}$ . The subscript  $E$  refers to the quantum-classical correspondence developed earlier in this chapter: the thermal density matrix  $e^{-\beta H}$  is mathematically equivalent to the quantum time-evolution operator evaluated at imaginary time. Expressing the partition function as a functional integral produces weights  $e^{-S_E}$  that are real and positive like Boltzmann factors.

### 12.6.2 The $\phi^4$ Action

The action  $S_E[\phi]$  must respect the same symmetries as the underlying lattice model: spin-flip symmetry  $\phi \rightarrow -\phi$  (so only even powers appear), and translation and rotation invariance. The simplest action with these properties is

$$S_E[\phi] = \int d^d x \left[ \frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} m^2 \phi^2 + \frac{\lambda}{4!} \phi^4 + \cdots \right]. \quad (12.84)$$

Each term has a clear physical meaning. The gradient term  $(\nabla\phi)^2$  penalizes spatial variation: a uniform field costs nothing, rapid fluctuations are suppressed. This is the continuum analog of the Ising nearest-neighbor coupling. The mass term  $\frac{1}{2}m^2\phi^2$  controls ordering: when  $m^2 > 0$  the only minimum is  $\phi = 0$  (disordered); when  $m^2 < 0$  the minimum shifts to  $\phi \neq 0$  (ordered). Near criticality,  $m^2 \propto T - T_c$ , so tuning  $m^2$  through zero is tuning through the phase transition. The quartic term  $\frac{\lambda}{4!}\phi^4$  with  $\lambda > 0$  stabilizes the action when  $m^2 < 0$ , creating the double-well potential that describes the two ordered states. This theory is called  $\phi^4$  theory.

Equation (12.84) is the Landau-Ginzburg free energy familiar from statistical mechanics. The field theory is not a new object: it is the continuum limit of the block-spin description, now equipped with tools to perform the RG explicitly.

## 12.7 Dimensional Analysis and the Upper Critical Dimension

Before doing any calculation, dimensional analysis already reveals the most important qualitative fact about  $\phi^4$  theory. The calculation is short but deserves to be worked out explicitly, because its conclusion drives everything that follows.

**How many dimensionful quantities are there?** Only one. The action (12.84) contains no  $\hbar$ , no  $c$ , no particle mass; the only ingredients are the field  $\phi$  and the couplings  $m^2$  and  $\lambda$ , integrated over space. The sole dimensionful scale in the problem is the short-distance cutoff, the lattice spacing  $a$  (or equivalently the cutoff momentum  $\Lambda = 1/a$ ). Every dimensionful quantity in the theory can therefore be expressed as a pure number times a power of  $a$ . We write  $[X] = a^p$  to mean that  $X$  has dimensions of length to the  $p$ ; equivalently,  $[X] = \text{mass}^{-p}$  since inverse length plays the role of mass here.

This is a weaker convention than the atomic units used throughout the rest of this course. In atomic units one sets  $\hbar = m_e = e = 1$  (with  $4\pi\epsilon_0 = 1$ ), collapsing mechanical problems to a single scale, the Bohr radius  $a_0$ ; note that  $c = 1/\alpha \approx 137$  is *not* unity. In high-energy physics one instead sets  $\hbar = c = 1$ , making length and inverse mass interchangeable. The  $\phi^4$  theory has neither  $\hbar$  nor  $c$  nor  $m_e$  to begin with; the unit of length is whatever cutoff scale you chose when regulating the theory. In all three conventions the structural point is the same: fixing enough fundamental constants to unity leaves exactly one independent dimension, and every coupling and field is labeled by its power of that one dimension.

**The dimension of the field.** Because  $S_E$  appears in  $e^{-S_E}$ , it must be a pure number,  $[S_E] = a^0$ . The integration measure contributes  $[d^d x] = a^d$ , so every term in the integrand must carry dimension  $a^{-d}$  to cancel.

Start with the gradient term, which has the simplest structure. A derivative  $\nabla$  has dimension  $a^{-1}$  (it is  $\partial/\partial x$ ), so

$$[(\nabla\phi)^2] = a^{-2} [\phi]^2. \quad (12.85)$$

Setting this equal to  $a^{-d}$  gives  $[\phi]^2 = a^{-(d-2)}$ , so

$$[\phi] = a^{-(d-2)/2}. \quad (12.86)$$

In  $d = 4$ , for instance,  $[\phi] = a^{-1}$ , i.e.,  $\phi$  has the dimension of inverse length (or of mass). This is the *canonical* dimension of the field, the dimension demanded by requiring the kinetic term alone to be dimensionless.

**The dimension of  $m^2$ .** The mass term has dimension  $[m^2] \cdot [\phi^2] = [m^2] \cdot a^{-(d-2)}$ . Setting this to  $a^{-d}$  gives

$$[m^2] = a^{-2}, \quad [m] = a^{-1}, \quad (12.87)$$

so  $m$  is an inverse length (equivalently, a mass). This matches the intuition from the correlation-function discussion at the start of this chapter:  $m$  is the inverse correlation length away from criticality, and the phase transition is approached by tuning  $m^2 \rightarrow 0$ .

**The dimension of  $\lambda$ .** The quartic term has dimension  $[\lambda] \cdot [\phi^4] = [\lambda] \cdot a^{-2(d-2)}$ . Setting this to  $a^{-d}$ :

$$[\lambda] = a^{-d} \cdot a^{2(d-2)} = a^{d-4}. \quad (12.88)$$

Converting to mass units via  $[X] = \text{mass}^{-p}$  when  $[X] = a^p$ ,

$$[\lambda] = \text{mass}^{4-d} \equiv \text{mass}^\varepsilon, \quad \varepsilon \equiv 4 - d. \quad (12.89)$$

This is the crucial result. From the operator classification developed earlier, a coupling with positive mass dimension grows under coarse-graining (relevant); one with negative mass dimension shrinks (irrelevant). The  $\phi^4$  coupling  $\lambda$  therefore changes character depending on dimension:

| Dimension | $[\lambda]$   | Classification |
|-----------|---------------|----------------|
| $d > 4$   | negative      | irrelevant     |
| $d = 4$   | dimensionless | marginal       |
| $d < 4$   | positive      | relevant       |

Above  $d = 4$  interactions are irrelevant: the long-distance physics is governed by the free-field (Gaussian) fixed point, and mean-field exponents ( $\nu = 1/2$ ,  $\eta = 0$ ) are exact. Below  $d = 4$  interactions are relevant: they grow under coarse-graining and drive the system to a new interacting fixed point with different exponents. At  $d = 4$  exactly,  $\lambda$  is dimensionless and the situation is marginal, with quantum fluctuations determining the outcome. This makes  $d = 4$  the *upper critical dimension* of  $\phi^4$  theory, the boundary below which mean-field theory fails.

#### Key Idea: Upper Critical Dimension

The failure of mean-field theory in three dimensions is not an accident. The interaction coupling  $\lambda$  is classically relevant for  $d < 4$ , meaning fluctuations grow as one coarse-grains, pushing the system away from the mean-field (Gaussian) fixed point toward a new fixed point with different critical exponents. The dimension  $d = 4$  is the boundary: above it, mean-field theory is exact; below it, fluctuations matter.

## 12.8 Aside: Cutoffs in Single-Particle Quantum Mechanics

The preceding dimensional analysis may feel unfamiliar, but the underlying logic appeared much earlier in this course. Chapter 3 constructed the continuum Schrödinger equation as the limit of a tight-binding chain with lattice spacing  $a$ , and the continuum limit was recovered by taking  $a \rightarrow 0$  at fixed  $\hbar^2/(2ma^2)$ . That lattice spacing played exactly the role the cutoff  $a = 1/\Lambda$  plays here in  $\phi^4$  theory: it is the shortest distance the theory can resolve, and physics at scales  $\gg a$  is described by the continuum effective theory. Dimensional analysis in the continuum is, in both cases, bookkeeping for how quantities scale relative to the one microscopic length the regulator supplies.

The shared logic runs deeper than just “there is a lattice somewhere.” Several features transfer directly.

**Continuum theories need cutoffs to be well-defined.** In single-particle QM, this is usually invisible, because the standard problems (harmonic oscillator, hydrogen atom, particle in a box) are finite without any regulator. But the moment one considers singular potentials, the infinities of continuum QM become manifest. A delta-function potential  $V(x) = g\delta(x)$  in one dimension is perfectly integrable and produces a single bound state, but the analogous contact interaction in two or three dimensions requires a short-distance cutoff to define at all; the bare coupling has to be tuned with the cutoff to give a finite physical binding energy. Likewise the attractive inverse-square potential  $V(r) = -\hbar^2 g/(2mr^2)$  exhibits the “fall to the center” pathology for  $g$  above a critical value and requires a self-adjoint extension, a regulator in disguise. Field theory generalizes this: every loop integral  $\int d^d k/(k^2 + m^2)^n$  diverges at large  $k$  unless cut off at  $|k| < \Lambda = 1/a$ . The lattice is implicit in one-body QM and explicit in  $\phi^4$ , but the same principle applies: the continuum is a limit, and the regulator is what makes the limit sensible.

**Dimensional analysis classifies perturbations identically in both settings.** The operator classification of Section 12.7 (relevant, marginal, irrelevant, according to the sign of the mass dimension) is not specific to statistical field theory. It applies, implicitly, to perturbations of single-particle QM as well. Consider a 1D Hamiltonian  $\hat{H}_0 = -\nabla^2/(2m)$  perturbed by a short-range potential  $V(x)$  of characteristic range  $a$  and strength  $g_n$  with  $[g_n] = a^{n-1}$ . The dimensionless coupling at a resolution  $\ell \gg a$  is  $g_n \ell^{1-n}$ , which grows when  $n > 1$ , stays fixed when  $n = 1$ , and shrinks when  $n < 1$ . Translated back to familiar examples:

- A delta function  $V = g\delta(x)$  has  $[g] = a^0$  in  $d = 1$  (the integral  $\int V dx$  must give an energy, and  $\int \delta(x) dx = 1$  is dimensionless in the sense of having no  $a$  dependence once  $\delta(x)$  is written in units of the cutoff). A little more carefully, the dimensionless ratio controlling the physics is  $g(m/\hbar^2)a$ , which *shrinks* as  $a \rightarrow 0$ . The 1D delta function is *irrelevant* in the RG sense.
- An inverse-square potential  $V = -\hbar^2 g/(2mr^2)$  has dimensionless  $g$ . No power of the cutoff appears in the dimensionless coupling; it is *marginal*. This is precisely the potential whose subtle behavior (fall-to-the-center, required self-adjoint extension) reflects marginality: logarithmic running of the effective coupling replaces simple power-law scaling, and the physics depends sensitively on the regulator.
- A  $1/r^4$  potential has  $[g] = a^2$ ; its dimensionless coupling grows as  $a \rightarrow 0$ , so it is *relevant*. The short-distance physics dominates, and the continuum theory is singular without a cutoff.

The trichotomy is exactly the one we found for  $\phi^4$ . Relevant perturbations drive the low-energy physics away from the free Hamiltonian; irrelevant perturbations become progressively less important at long distances; marginal perturbations sit on the boundary and require more careful analysis.

**Where the analogy genuinely breaks.** Two important differences keep the parallel from being complete.

First, in most single-particle problems the continuum limit is painless. A particle in a smooth confining potential is already well-defined in the continuum; the lattice is a numerical convenience, not a conceptual necessity. Finite-difference solvers for the Schrödinger equation use a grid spacing because computers demand it, not because the physics demands it. In  $\phi^4$  theory the situation is qualitatively different: even perturbation theory at one loop diverges without a cutoff. The cutoff is mandatory, and the central question of renormalization is whether one can remove it in a controlled way.

Second, the RG flow in a non-interacting single-particle problem is trivial. Coarse-graining a particle in a smooth potential just produces a coarser grid with renormalized effective-mass and effective-potential parameters; the flow has no fixed points other than the free theory. The entire apparatus of the Wilson-Fisher fixed point, the running coupling, and nontrivial critical exponents exists because  $\phi^4$  is an *interacting* theory whose coupling  $\lambda$  generates a nontrivial flow competing with classical scaling. Single-particle QM is the base case of a theory where everything is classical and the flow is geometric; interacting field theory is where the flow first becomes dynamical.

### Physical Insight

The lattice spacing introduced in Chapter 3 to discretize the Schrödinger equation, and the cutoff  $\Lambda = 1/a$  introduced here to regulate  $\phi^4$  theory, are the same conceptual object. Both set the shortest length scale the theory resolves; both make dimensional analysis well-defined by supplying the one unit of length relative to which couplings are measured; both classify perturbations into relevant, marginal, and irrelevant by how their dimensionless versions behave under coarse-graining. The content of the renormalization group is the extension of this classification from the kinematic setting of single-particle QM, where the classification reduces to dimensional analysis of a potential, to the dynamical setting of interacting field theory, where quantum fluctuations generate nontrivial flow between couplings.

## 12.9 The Renormalization Group Flow

The momentum-space RG is the continuum analog of decimation. Every field configuration can be decomposed into Fourier modes at different wavevectors. The RG proceeds in two steps: first integrate out the short-wavelength (high-momentum) modes between  $\Lambda/b$  and  $\Lambda$ , then rescale momenta by  $k \rightarrow bk$  to restore the original cutoff. Repeating this generates a flow in the space of coupling constants, exactly as for the lattice models considered earlier.

**The RG “time”  $\ell$ .** It is convenient to parametrize the sequence of RG steps continuously. Define

$$\ell \equiv \ln b, \quad (12.90)$$

so that one RG step with rescaling factor  $b$  advances  $\ell$  by  $\ln b$ , and an infinitesimal step advances  $\ell$  by  $d\ell$ . The cutoff after  $\ell$  steps is  $\Lambda(\ell) = \Lambda_0 e^{-\ell}$ :  $\ell$  increases as we coarse-grain to longer distances. Every coupling in the theory becomes a function of  $\ell$ , and the RG flow is specified by a set of differential equations, one per coupling.

**Dimensionless couplings.** The couplings  $m^2$  and  $\lambda$  carry powers of the cutoff, as established in Section 12.7:  $[m^2] = \Lambda^2$  and  $[\lambda] = \Lambda^\varepsilon$  where  $\varepsilon = 4 - d$ . Because the cutoff itself flows under RG, the bare couplings flow trivially just from this dimensional scaling. To isolate the non-trivial, fluctuation-driven part of the flow we work with dimensionless versions:

$$t \equiv m^2/\Lambda^2, \quad (12.91)$$

$$g \equiv \lambda \Lambda^{-\varepsilon} K_d, \quad (12.92)$$

where  $K_d = S_{d-1}/(2\pi)^d$  is a conventional numerical normalization (the surface area of the unit sphere in  $d$  dimensions divided by  $(2\pi)^d$ ) that absorbs the angular integrals appearing in the one-loop calculation below, making the coefficients of the beta function come out as simple integers. With these definitions  $t$  and  $g$  are pure numbers.

**Classical part of the flow.** Part of the flow follows immediately from dimensional analysis and requires no calculation. Holding the bare coupling  $\lambda$  fixed while lowering the cutoff  $\Lambda \rightarrow \Lambda e^{-d\ell}$ , the dimensionless coupling defined in (12.92) changes as

$$\left. \frac{dg}{d\ell} \right|_{\text{classical}} = \varepsilon g, \quad (12.93)$$

simply because  $g \propto \Lambda^{-\varepsilon}$  and  $d \ln \Lambda / d\ell = -1$ . This is the relevant scaling that Kadanoff's hypothesis assigned to  $\lambda$  with exponent  $\varepsilon$ , now derived from the action rather than assumed.

**Fluctuation corrections.** Integrating out the momentum shell  $\Lambda e^{-d\ell} < |k| < \Lambda$  generates additional contributions. The mechanism is the following: in perturbation theory the  $\phi^4$  vertex couples four fields, and contracting two pairs of fields from two different vertices across the momentum shell produces an effective correction to the remaining four-field (coupling) term. The correction is an integral over the shell, and the resulting flow equation at one-loop order is

$$\frac{dg}{d\ell} = \varepsilon g - 3g^2 + \dots, \quad (12.94)$$

which we state without proof: the calculation that produces the coefficient 3 is a standard one-loop exercise in quantum field theory, requiring the Feynman rules for  $\phi^4$  theory and evaluation of an integral over the momentum shell. The factor of 3 is a combinatorial count of the ways the four external legs of two  $\phi^4$  vertices can be paired so that two pairs are contracted across the shell and two pairs remain external. Two features of equation (12.94) deserve comment. The correction is *quadratic* in  $g$  because the diagram involves two vertices (one factor of  $\lambda$  each). It is *negative* because integrating out high-momentum modes reduces the strength of the interaction at long wavelengths: short-wavelength fluctuations screen the bare coupling. The sign is what makes a non-trivial fixed point possible.

**The beta function.** The right-hand side of equation (12.94) is, by definition, the *beta function*:

$$\beta(g) \equiv \frac{dg}{d\ell} = \varepsilon g - 3g^2. \quad (12.95)$$

The name comes from quantum field theory, where  $\beta(g)$  describes how a running coupling evolves with scale. A zero of  $\beta$  is a fixed point of the RG flow, and the sign of  $\beta'(g)$  at a fixed point determines whether the flow is attractive ( $\beta' < 0$ ) or repulsive ( $\beta' > 0$ ). The competition between the classical term  $\varepsilon g$  and the fluctuation correction  $-3g^2$  is what determines the fixed-point structure analyzed next.

## 12.10 The Wilson-Fisher Fixed Point

Setting  $dg/d\ell = 0$  in (12.95) gives two fixed points:  $g^* = 0$  and  $g^* = \varepsilon/3$ . Figure 12.7 shows their structure.

The **Gaussian fixed point** at  $g^* = 0$  is the free, non-interacting theory. Above four dimensions ( $\varepsilon < 0$ ), it is stable: any small interaction is driven back to zero by the  $-3g^2$  term. Below four dimensions ( $\varepsilon > 0$ ), it is unstable: the  $\varepsilon g$  term wins at small  $g$  and interactions grow.

The **Wilson-Fisher fixed point** at  $g^* = \varepsilon/3$  exists only for  $\varepsilon > 0$ . It is stable: the beta function has negative slope there, so any nearby value of  $g$  flows back toward  $g^*$ . This is the fixed point governing critical behavior in  $d < 4$ .

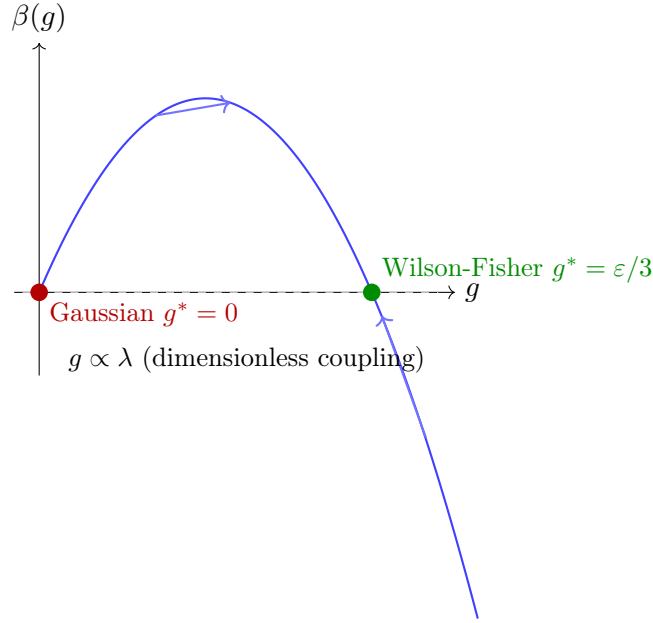


Figure 12.7: The beta function  $\beta(g) = \varepsilon g - 3g^2$  for  $\phi^4$  theory in  $d = 4 - \varepsilon$  (shown for  $\varepsilon = 1$ , i.e.,  $d = 3$ ). The classical term  $\varepsilon g$  drives  $g$  upward; the fluctuation term  $-3g^2$  drives it downward. The Gaussian fixed point is unstable for  $d < 4$ ; the Wilson-Fisher fixed point at  $g^* = \varepsilon/3$  is stable.

### 12.10.1 Critical Exponents From the Fixed Point

Once the Wilson-Fisher fixed point is located, the critical exponents follow by linearizing the RG flow around it, exactly the procedure applied to Kadanoff's scaling hypothesis earlier. The linearized flow has two eigenvalues. The first,

$$y_t = 2 - \frac{\varepsilon}{3} + O(\varepsilon^2), \quad (12.96)$$

is the thermal scaling exponent: it is positive (temperature deviation is relevant), and it is the number that Kadanoff's hypothesis required as input. The correlation length exponent follows directly:

$$\nu = \frac{1}{y_t} \approx \frac{1}{2} + \frac{\varepsilon}{12}. \quad (12.97)$$

At  $\varepsilon = 0$  (four dimensions),  $\nu = 1/2$ : mean-field theory. At  $\varepsilon = 1$  (three dimensions),  $\nu \approx 0.583$ , compared to the exact 3D Ising value  $\nu = 0.630$ . The second eigenvalue  $y_g = -\varepsilon$  is negative, confirming the fixed point is stable in the coupling direction.

The magnetic scaling exponent  $y_h \approx 3 - \varepsilon/2$  follows from how an external magnetic field rescales under the RG. All other critical exponents ( $\beta$ ,  $\gamma$ ,  $\delta$ ,  $\alpha$ ) then follow from  $y_t$  and  $y_h$  through the scaling relations derived earlier in this chapter. The  $\varepsilon$  expansion thus converts Kadanoff's hypothesis into a calculation.

#### Physical Insight

The Wilson-Fisher fixed point exists because of a competition between two tendencies. Classical scaling drives the coupling  $g$  upward (interactions are relevant below four dimensions). Fluctuations drive  $g$  downward (short-wavelength modes, when integrated out, weaken the



effective long-wavelength coupling). These two effects balance at  $g^* = \varepsilon/3$ . The resulting fixed point is not mean-field ( $g^* = 0$ ) but it is close to mean-field when  $\varepsilon$  is small, which is why the  $\varepsilon$  expansion works.

### 12.10.2 Universality from Symmetry and Dimension

The Wilson-Fisher fixed point at  $g^* = \varepsilon/3$  has no free parameters. It depends only on the spatial dimension  $d$  and the  $\mathbb{Z}_2$  symmetry of the  $\phi^4$  action. This is the field-theory explanation of the universality observed earlier: any microscopic model with  $\mathbb{Z}_2$  symmetry in  $d < 4$  flows to the same fixed point under coarse-graining, regardless of lattice type, interaction range, or coupling strength. The fixed point and its eigenvalues are determined by symmetry and dimension alone.

The irrelevant operators discussed earlier in this chapter now have explicit identities. The  $\phi^6$  term has mass dimension  $6 - 2d$ , which is negative for  $d > 3$ : it is irrelevant at the Wilson-Fisher fixed point in three dimensions. Lattice anisotropy, next-neighbor interactions, the precise shape of the lattice all map onto irrelevant operators that flow to zero under repeated coarse-graining. They shift  $T_c$  but leave the critical exponents unchanged.

When the symmetry group changes, so does the universality class. An  $N$ -component order parameter with  $O(N)$  symmetry gives a beta function

$$\frac{dg}{d\ell} = \varepsilon g - \frac{N+8}{3} g^2, \quad (12.98)$$

with fixed point  $g^* = 3\varepsilon/(N+8)$ . The Ising model ( $N = 1$ ), the XY model for superfluids ( $N = 2$ ), and the Heisenberg model for isotropic ferromagnets ( $N = 3$ ) all fit this framework with different values of  $N$ , producing distinct universality classes with different exponents, consistent with what is observed experimentally.

## 12.11 Chapter Summary

The RG resolves the tension between discrete and continuous descriptions that we encountered in Chapter 1. When we started with qubits and built up to the Schrödinger equation, we were implicitly performing an RG flow. The lattice quantum field theory in Chapter 3 provided a regularization, a well-defined theory at all scales. Taking the continuum limit meant flowing to a particular fixed point where the lattice spacing became irrelevant. This perspective transforms how we think about fundamental physics. Rather than asking “is space fundamentally discrete or continuous?”, we should ask “which RG fixed point describes nature at the scales we observe?” Quantum field theory works not because spacetime is truly continuous, but because the continuum fixed point accurately describes physics at length scales much larger than any underlying cutoff.

The shift from fixed theories to flows in theory space represents one of the deepest insights in modern theoretical physics. Rather than asking “what is the theory at all scales?” the RG teaches us to ask “how do theories transform under changes of scale?” Fixed points organize the space of all theories, with a small number of relevant operators determining which fixed point a system flows toward, while the vast majority of microscopic details are washed out as irrelevant.

Three pillars of the RG framework have emerged. First, the scaling hypothesis: near a critical fixed point, the linearized RG acts as a pure rescaling of a small number of relevant couplings, and all critical exponents follow from the corresponding eigenvalues. Second, universality: systems flowing to the same fixed point share identical long-distance physics regardless of microscopic detail,



with the universality class determined by symmetry and dimension alone. Third, the quantum-classical correspondence: a  $d$ -dimensional quantum system at a quantum critical point belongs to the universality class of a  $(d+z)$ -dimensional classical system, so the entire RG machinery applies equally to thermal and quantum transitions.

These pillars are not independent assumptions but consequences of a single calculation. The Landau-Ginzburg free energy is the Euclidean action for a scalar field  $\phi(\mathbf{x})$ , and writing the partition function as a functional integral  $Z = \int \mathcal{D}\phi e^{-S_E[\phi]}$  places it on the same footing as a quantum field theory. Momentum-space RG produces the beta function  $dg/d\ell = \varepsilon g - 3g^2$ , whose Wilson-Fisher fixed point at  $g^* = \varepsilon/3$  governs critical behavior in  $d < 4$ . The scaling exponents  $y_t \approx 2 - \varepsilon/3$  and  $y_h \approx 3 - \varepsilon/2$  are not inputs from experiment but eigenvalues of the linearized flow, and setting  $\varepsilon = 1$  gives quantitative predictions for three-dimensional critical behavior. This unifies statistical mechanics, quantum field theory, and condensed matter physics, and provides a satisfying resolution to questions about the nature of the continuum that motivated our journey from qubits to quantum fields.

## 12.12 Problems

### Problem 12.1 (1D Ising Decimation and RG Flow)

Consider the one-dimensional Ising model with  $H = -J \sum_{i=1}^{N-1} s_i s_{i+1}$  where  $s_i = \pm 1$  and  $J > 0$ . The partition function is  $Z = \sum_{\{s_i\}} e^{-\beta H}$ .

- Show that  $Z = \sum_{\{s_i\}} \prod_{i=1}^{N-1} e^{K s_i s_{i+1}}$  where  $K = \beta J$ .
- Focus on the even-site spin  $s_2$  between  $s_1$  and  $s_3$ . Perform the sum  $\sum_{s_2} e^{K s_1 s_2} e^{K s_2 s_3}$  for the cases  $s_1 = s_3$  and  $s_1 = -s_3$  separately.
- Express the result as  $A e^{K' s_1 s_3}$  and determine the RG recursion:  $K' = \frac{1}{2} \ln[\cosh(2K)]$ .
- Show that  $K^* = 0$  is a stable fixed point by expanding for small  $K$ . Verify that  $K' < K$  for  $0 < K < 1$ .
- Argue that  $K^* = \infty$  is also a fixed point. Is it stable or unstable?
- Starting from  $K = 1.0$ , iterate the recursion numerically to compute  $K', K'', K'''$ . In which direction does the flow proceed?

### Problem 12.2 (Critical Correlations and Anomalous Dimension)

At a critical point, the free-field correlation function in  $d$  spatial dimensions is  $G(\mathbf{r}) = \int \frac{d^d k}{(2\pi)^d} \frac{e^{i\mathbf{k} \cdot \mathbf{r}}}{k^2}$ .

- Use the substitution  $\mathbf{k} = \mathbf{q}/r$  to show that  $G(r) \sim r^{-(d-2)}$ .
- Evaluate the exponent for  $d = 1, 2, 3, 4$ . For which dimension does the correlator grow with distance?
- The interacting critical correlator is  $G(r) \sim r^{-(d-2+\eta)}$  where  $\eta$  is the anomalous dimension. For the 2D Ising model,  $\eta = 1/4$ . What is the power-law exponent?
- Away from criticality, correlations decay as  $G(r) \sim e^{-r/\xi}$  with  $\xi \sim |t|^{-\nu}$ . Explain why power-law decay emerges precisely when  $\xi$  diverges.

**Problem 12.3** (*Scaling Relations and Critical Exponents*)

The scaling hypothesis asserts that under a block-spin transformation with rescaling factor  $b$ , the free energy density satisfies  $f(t, h) = b^{-d} f(b^{y_t} t, b^{y_h} h)$  where  $t = (T - T_c)/T_c$  is the reduced temperature,  $h$  is the field, and  $y_t, y_h$  are RG exponents.

- Set  $b = |t|^{-1/y_t}$  and use  $\xi(t) = b \xi(b^{y_t} t)$  to derive  $\nu = 1/y_t$ .
- Differentiate  $f$  once with respect to  $h$  to obtain the magnetization  $m(t, h) = b^{y_h-d} m(b^{y_t} t, b^{y_h} h)$ . Derive  $\beta = (d - y_h)/y_t$ .
- Differentiate twice to obtain the susceptibility and derive  $\gamma = (2y_h - d)/y_t$ .
- Differentiate twice with respect to  $t$  to derive  $\alpha = 2 - d/y_t$ . Verify the hyperscaling relation  $2 - \alpha = d\nu$ .
- For the 2D Ising model with  $\nu = 1$  and  $\beta = 1/8$ , extract  $y_t$  and  $y_h$ . Compute  $\gamma$ ,  $\delta = y_h/(d - y_h)$ , and  $\alpha$ .
- Verify the Rushbrooke inequality  $\alpha + 2\beta + \gamma \geq 2$ .

**Problem 12.4** (*Kramers-Wannier Duality and the Critical Temperature*)

The 2D Ising partition function can be expanded in low temperature (variable  $u = e^{-2K}$ ) or high temperature (variable  $v = \tanh K$ ) forms, related by duality:  $e^{-2K^*} = \tanh K$ .

- Explain physically what  $u = e^{-2K}$  counts. Why is it small at low temperature?
- Show that the duality maps high temperature (small  $K$ ) to low temperature (large  $K^*$ ) and vice versa.
- Argue that a unique phase transition must occur at the self-dual point  $K_c = K_c^*$ .
- From the self-duality condition  $e^{-2K_c} = \tanh K_c$ , derive  $\sinh(2K_c) = 1$  and show that  $\frac{k_B T_c}{J} = \frac{2}{\ln(1+\sqrt{2})} \approx 2.269$ .
- The 1D Ising model is self-dual at  $K = 0$  (not finite temperature). Explain why this is consistent with no phase transition at finite  $T$ .

**Problem 12.5** (*Quantum Phase Transition in the Transverse-Field Ising Model*)

The transverse-field Ising model has Hamiltonian  $H = -J \sum_i \sigma_i^z \sigma_{i+1}^z - h \sum_i \sigma_i^x$ .

- For  $h = 0$ , describe the ground state(s) and compute  $\langle \sigma_i^z \rangle$ .
- For  $h \gg J$ , find the ground state and show that  $\langle \sigma_i^z \rangle = 0$ .
- Why does the non-commutativity of the two terms in  $H$  lead to a quantum phase transition at  $T = 0$ ?
- The quantum-classical correspondence maps a  $d$ -dimensional quantum system at a QPT onto a  $(d + z)$ -dimensional classical system, where  $z$  is the dynamical critical exponent. For the 1D transverse-field Ising model with  $z = 1$ , what classical model results? What universality class?

- (e) The energy gap closes as  $\Delta \sim |h - h_c|^{z\nu}$ . Using  $z = 1$  and  $\nu = 1$ , compute the gap exponent and explain what this means physically.

**Problem 12.6** (*Ising Magnetization at  $T = 0$  with Transverse Field*)

- (a) For the transverse-field Ising model with  $J = 1$  and varying field  $h$ , the ground state magnetization is  $m(h) = \langle \sigma^z \rangle$ . At  $h = 0$ ,  $m(0) = 1$ . At the quantum critical point  $h_c = 1$ ,  $m(h_c) = 0$ . Show that  $m(h) \sim (1 - h)^\beta$  near  $h_c$  with  $\beta = 1/8$  (matching 2D Ising).
- (b) Compute  $m(h)$  numerically (or analytically for the 1D chain) at  $h = 0.5, 0.8, 0.95, 0.99$  and verify the power-law scaling near  $h_c = 1$ .
- (c) Near the critical point, the magnetization susceptibility  $\chi = \partial m / \partial h$  diverges. Show that  $\chi \sim |h - h_c|^{-\gamma}$  with  $\gamma = 7/4$  (2D Ising).
- (d) Compare the magnetization curves for different values of  $J$  (the coupling strength). Does the critical field  $h_c$  depend on  $J$ ? How does the critical exponent  $\beta$  depend on  $J$ ?
- (e) Explain in physical terms why increasing the transverse field  $h$  reduces the ferromagnetic magnetization.

**Problem 12.7** (*RG Fixed-Point Linearization for Transverse-Field Ising*)

- (a) Near the quantum critical point of the transverse-field Ising model, the RG flow in the  $(h, J)$  plane can be linearized around the critical point. Write the RG transformation as  $h' = b^{y_h} h$ ,  $J' = b^{y_J} J$  (or equivalently,  $K'_h = b^{y_h} K_h$ ,  $K'_J = b^{y_J} K_J$ ).
- (b) At the critical point  $(h_c, J_c) = (1, 1)$  (up to rescaling), the RG scaling exponents for the transverse-field Ising model are related to those of the 2D classical Ising model. Using the quantum-classical correspondence with  $z = 1$ , relate  $y_h$  and  $y_J$  to the classical exponents  $y_t$  and  $y_h$ .
- (c) The correlation length along the  $h$  direction diverges as  $\xi_h \sim (h - h_c)^{-\nu_h}$ . Show that  $\nu_h = 1/y_h$  and relate it to the classical exponents.
- (d) Near the critical point, sketch the RG flow trajectories in the  $(h, J)$  plane. Identify the directions of relevant, irrelevant, and marginal perturbations.
- (e) A perturbation away from criticality (e.g., a small magnetic field  $B\sigma^z$ ) is relevant or irrelevant depending on its sign (which direction it points in coupling space). Explain what this means for the long-distance physics.

**Problem 12.8** (*Correlation Length Critical Exponent from RG*)

- (a) The correlation length  $\xi$  is the scale at which correlations decay. In terms of the lattice spacing  $a$ , after a block-spin transformation with rescaling factor  $b$ , the correlation length becomes  $\xi' = \xi/b$  (measured in units of the new lattice spacing  $ba$ ).

- (b) If the reduced temperature  $t = (T - T_c)/T_c$  transforms as  $t' = b^{y_t} t$ , show that  $\xi(t) = b \xi(b^{y_t} t)$ . By choosing  $b = |t|^{-1/y_t}$ , derive  $\xi \sim |t|^{-\nu}$  with  $\nu = 1/y_t$ .
- (c) For the 2D Ising model,  $y_t = 1$  (the scaling exponent of temperature), so  $\nu = 1$ . Compute the correlation length at  $t = 0.01$  (i.e., 1
- (d) As  $T \rightarrow T_c$ , the correlation length diverges. Explain physically why arbitrarily large spatial structures appear at the critical point and why the system becomes scale-invariant.
- (e) Compute the correlation length for a different universality class (e.g., the Ising model in a different dimension) and compare with the 2D result.

**Problem 12.9** (*Finite-Size Scaling Collapse*)

At a critical point, finite-size systems of linear size  $L$  exhibit scaling:  $m(T, L) = L^{-\beta/\nu} f_m(|T - T_c| L^{1/\nu})$  where  $f_m$  is a universal scaling function.

- (a) Simulate the 1D or 2D Ising model at various temperatures near  $T_c$  and system sizes  $L$ . Compute the magnetization  $m(T, L)$  for each combination.
- (b) Plot  $m(T, L) \cdot L^{\beta/\nu}$  versus  $|T - T_c| L^{1/\nu}$  on a log-log scale (using literature values  $\beta = 1/8$ ,  $\nu = 1$  for the 2D Ising model). Verify that the data from different system sizes collapse onto a single universal curve.
- (c) The quality of the collapse is a test of the critical exponents. If the exponents are incorrect, the data will not collapse. Vary  $\beta$  and  $\nu$  slightly and show how the collapse deteriorates.
- (d) Extract the universal scaling function  $f_m(x)$  from the collapsed data and compare with theoretical predictions or other simulations.
- (e) Use the finite-size scaling method to estimate  $T_c$  from your simulation data (the temperature at which the collapse is best).

**Problem 12.10** (*Kosterlitz-Thouless Transition: Qualitative Picture*)

- (a) The 2D XY model (classical Heisenberg model with spins in the plane) has the Hamiltonian  $H = -J \sum_{\langle i,j \rangle} \cos(\theta_i - \theta_j)$  where  $\theta_i \in [0, 2\pi)$  is the angle of the spin at site  $i$ .
- (b) At low temperature, spins are quasi-aligned, supporting long-range order and power-law correlations. At high temperature, spins are disordered. However, unlike the Ising model, the 2D XY model has a different transition mechanism: the Kosterlitz-Thouless (KT) transition.
- (c) The key role is played by topological defects called vortices, which are point defects where the phase winds by  $2\pi$  around a plaquette. Describe qualitatively what happens as temperature increases: (1) at  $T < T_{KT}$ , vortices are bound in pairs (vortex-antivortex pairs) with opposite winding; (2) at  $T > T_{KT}$ , vortices unbind and proliferate.
- (d) The KT transition is characterized by a different universality class than the Ising model. The correlation length diverges as  $\xi \sim \exp(C|T - T_{KT}|^{-1/2})$  (essential singularity) rather than power-law, and there is no conventional order parameter with a critical exponent.

- (e) Sketch the phase diagram and the vortex density as a function of temperature. Explain why the transition is called "topological."
- (f) How does the 2D XY model differ from the 2D Ising model in terms of symmetry and the nature of the phase transition? What role does the continuous symmetry play?

**Problem 12.11** (*Quantum Hall Plateau: RG Argument*)

- (a) In the quantum Hall regime, electrons in a strong perpendicular magnetic field  $B$  form discrete Landau levels. The quantum Hall conductance is quantized in units of  $e^2/h$ . A plateau at conductance  $\sigma = \nu e^2/h$  (for filling fraction  $\nu$ ) arises from the insulating incompressible state.
- (b) Disorder in the system broadens the Landau levels into bands and can cause transitions between plateaus. A key insight from RG is that the random network of extended states at energy  $E_c$  (separating the localized states of different plateaus) is a critical two-dimensional electron system.
- (c) At the transition (critical point), the localization length diverges:  $\xi \sim |E - E_c|^{-\nu_c}$  with a critical exponent  $\nu_c \approx 2.3$  (for the integer quantum Hall effect).
- (d) The quantum Hall plateau is broad in energy (or magnetic field) because only a small window around  $E_c$  supports extended states. States far from  $E_c$  are localized, contributing nothing to the conductance.
- (e) Sketch the density of states  $\rho(E)$  and the localization length  $\xi(E)$  across a Landau level. Indicate the plateau region (localized,  $\sigma = \nu e^2/h$ ) and the critical point (extended, metallic).
- (f) Explain qualitatively why the quantum Hall conductance is precisely quantized ( $\nu e^2/h$ ) and why it is insensitive to disorder at the plateau centers. What role does the topological robustness play?

**Problem 12.12** (*Conformal Invariance and Operator Scaling Dimensions*)

- (a) At a critical point, the theory is scale-invariant and, in many cases (especially in 2D), has a larger symmetry: conformal invariance. This means the theory is invariant under angle-preserving transformations (conformal maps).
- (b) In conformal field theory (CFT), every operator has a scaling dimension  $\Delta$  such that under a scale transformation  $x \rightarrow \lambda x$ , the operator transforms as  $\mathcal{O}(x) \rightarrow \lambda^{-\Delta} \mathcal{O}(\lambda x)$ .
- (c) For the 2D Ising model at criticality, the primary operators include the identity  $I$  (dimension  $\Delta = 0$ ), the energy density  $\epsilon$  (dimension  $\Delta = 1$ ), the magnetization  $\sigma$  (dimension  $\Delta = 1/8$ ), and the disorder operator  $\mu$  (dimension  $\Delta = 1/8$ ).
- (d) The correlation function of two identical primary operators has the form  $\langle \mathcal{O}(x) \mathcal{O}(0) \rangle \sim |x|^{-2\Delta}$ . Explain how this relates to the power-law exponent in the problem on critical correlations.
- (e) For the 2D Ising model, compute  $\Delta_\sigma = 1/8$  from the known anomalous dimension  $\eta = 1/4$  (using  $\Delta = (d-2+\eta)/2$  for the Ising magnetization in  $d$  dimensions). Verify that the resulting correlation exponent matches the literature.

- (f) Explain why conformal invariance constrains the form of correlators and why it is a powerful tool for exact solutions in 2D.

## References and Further Reading

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- Cardy, J.** *Scaling and Renormalization in Statistical Physics*. Cambridge Lecture Notes in Physics, Cambridge University Press, 1996. The standard graduate introduction to the RG and critical phenomena, emphasizing scaling and field-theoretic methods. Chapters 3, 5, and 7 cover Kadanoff scaling, the  $\varepsilon$ -expansion, and quantum critical phenomena, respectively.
- Goldenfeld, N.** *Lectures on Phase Transitions and the Renormalization Group*. Frontiers in Physics, Addison-Wesley, 1992. A pedagogical companion to Cardy with extensive worked examples; particularly clear on the conceptual foundations of universality and the meaning of fixed points.
- Sachdev, S.** *Quantum Phase Transitions*, 2nd ed. Cambridge University Press, 2011. The definitive reference on quantum critical phenomena. Chapter 5 covers the transverse-field Ising model in detail, including the exact Jordan-Wigner solution used to obtain the gap shown in Figure 12.6.
- Wilson, K. G., and Kogut, J.** “The renormalization group and the  $\varepsilon$ -expansion.” *Physics Reports* **12**, 75–199 (1974). [doi:10.1016/0370-1573\(74\)90023-4](https://doi.org/10.1016/0370-1573(74)90023-4). The classic review by the architect of the modern RG, written shortly after Wilson’s Nobel-winning work. Demanding but rewarding.
- Kadanoff, L. P.** “Scaling laws for Ising models near  $T_c$ .” *Physics Physique Fizika* **2**, 263–272 (1966). [doi:10.1103/PhysicsPhysiqueFizika.2.263](https://doi.org/10.1103/PhysicsPhysiqueFizika.2.263). The original block-spin paper that introduced the scaling hypothesis used in this chapter.
- Kramers, H. A., and Wannier, G. H.; and Onsager, L.** “Statistics of the two-dimensional ferromagnet,” *Physical Review* **60**, 252–262 (1941), [doi:10.1103/PhysRev.60.252](https://doi.org/10.1103/PhysRev.60.252); and “Crystal statistics. I. A two-dimensional model with an order-disorder transition,” *Physical Review* **65**, 117–149 (1944), [doi:10.1103/PhysRev.65.117](https://doi.org/10.1103/PhysRev.65.117). The duality argument that locates  $T_c$  exactly, followed three years later by Onsager’s full solution; together they are the source of the 2D Ising critical exponents quoted in this chapter.